GE Medical Systems

Technical Publications

Direction 2248333

Revision 4

Magnet and System Upgrade to Signa® WideOpen

Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Immediately complete "**Damage Loss Claim Form**" after damage is found. Form is available via MS Exchange Mail at Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/Damage Loss Claim Form. Send completed form to e–mail address listed on form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address:

ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF

Rev. 11/15/2000

GE Medical Systems

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Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST–HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST–HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

警告

・このサービスマニュアルには英語版しかありません。
・ＧＥＭＳ以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
・このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
A	Aug 19, 1999	Preliminary Release
0	Oct 8, 1999	Initial Release
1	Sep 14, 2002	Overview Tab, Sec 1 – Updated catalogs for 9.x and 10.x upgrade configurations. Revised Discarded Material Return Process information by indicating GoldSeal as the main contact source for return of removed equipment. Sec 2 – Moved flow charts to section 2 from section 3. Changed 8.x to 9.x with ACGD/PDU and added 10.x (MGD) flow charts. Sec 3 – Revised return label information with GEMS GoldSeal as the contact source. Sec 4 – Revised cable information for 9.x and 10.x upgrade configurations. Added Cable rating information. Sec 5 – Revised for 9.x and 10.x releases. Tab 1 – Added ACGD and MGD cable installation information. Tab 2 – Revised Fixed Site magnet enclosure installation to include MGD and added Mobile installation. Tab 4 – Updated section to include LX and 10.x penetration panel upgrade instructions. Added note for correct O–ring installation of HN right angle connectors on Pen Panel (CQA 1029849). Tab 8 – Added Direction 2170004 for 8.x/98.x cabinet installation. Added Direction 2338502 for MGD System cabinet. Tab 13 – Updated Operator workspace installation instructions for 9.x and 10.x. Tab 15 – Added ACGD/PDU installation instructons. Tab 16 – Added SRF and SRFD2 installation instructions.
2	Jun 16, 2003	Tab 15 – Added neutral connection for SSM in PDU power cable connections table (SPR# MRIge82869). Revised wording on DIP switch settings. Tab 16 – Revised 1.0T and 1.5T designations for identifying Runs 236/748 and 235/745 (SPR # MRIge83424). Added note for location of UFI Label kit.
3	Jan 18, 2005	Overview Tab, Sec 4 – Added “N” to ACGD cabinet connection for Run 966 (MRIge82868).
4	Nov 10, 2010	Overview Tab, Section 1 – Revised Ferrous Material Handling warning, added 200 gauss plot minimum movement area, and note for required MR training. Tab 1 – Directions 2286976, 2287106, & 2338872: Revised Ferrous Material Handling warnings. Tab 2 – Directions 2223857, and 2223858: Revised Ferrous Material Handling warnings, Dock handling warnings, Rear Pedestal handling warnings, and instructions for installing blower box. (CAPA 1859293)

LIST OF EFFECTIVE PAGES

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* This revision number/letter corresponds to the indicated document’s revision control system.

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SECTION 1 – SYSTEM UPGRADE INFORMATION

1–1 INTRODUCTION

This manual documents the de–installation and installation of the magnet enclosure, cabinets, operator workspace, applicable cables, and other required materials to upgrade an existing site's magnet and system to a Signa 9.x WideOpen or 10.x (MGD) with LCC magnet. The magnet with partial enclosure covers and Sumitomo water chiller are installed via Direction 2226318 or other applicable installation instructions.

Depending on the system being upgraded, there may be items that are not installed. Return these items with the deinstalled parts of the old system.

The Upgrade configurations documented throughout this publication are:

<u>CURRENT SYSTEM CONFIGURATION</u> (w/ Oxford, S1, S2, S3, S4, S5, or CX magnet	➧ <u>UPGRADED SYSTEM CONFIGURATION</u>
Signa Advantage 4.x Signa Advantage 5.x 1.5T, 1.0T, & 0.5T Signa Horizon 5.x Signa LX	Signa 9.x MR/*i* with LCC Magnet & WideOpen Enclosure Signa 10.x (MGD) MR/*i* with LCC Magnet & WideOpen Enclosure

NOTICE:

During this hardware upgrade, the magnet may be ramped to full field strength. Individuals working in the magnet room must be MR safety trained.

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE. THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.

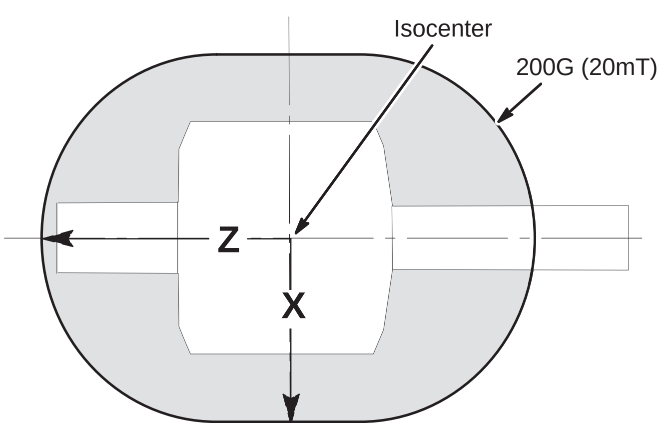


Illustration is not to scale

Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in)[1]	205 cm (81 in)[1]
Unshielded Magnet	Measure [2]	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

1–2 SITE READY CHECK

Direction 2246012 Pre–Sale/Site Survey For Upgrades to CXK4 Magnet with WideOpen Enclosure must be completed and signed–off by Local Customer Team to qualify a site for magnet upgrade. Direction 2246012 requires input supplied by the customer and by GE. Electronic copy of Direction 2246012 is available to GEMS personnel via the GE Intranet at http://3.87.118.28/optec3/common/upgradeinfo/upgrade.htm.

Pre–installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2142460, Signa LX Upgrade Pre–installation,* Pre–installation Checklist tab.

Tools and test equipment to install and calibrate the Signa Horizon 8.x system are listed in *Direction 2142460, Signa LX Upgrade Pre–installation*, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46–301450G1 Fiber Optic Cable Connector Repair kit

1–3 DISCARDED MATERIAL RETURN PROCESS

The components being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

Note

The following equipment or material removed by the upgrade are ***required to be returned*** to GoldSeal:

- SGD or Gradient Cabinet
- System Cabinet (if removed)
- Indigo Computer with all associated boards/cables (includes reflex board, cable kit, and audio cables)
- All other replaced components

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal.

The process for returning material is as follows:

1. Contact either of the following for disposition, with GoldSeal being the primary contact.
   - GoldSeal – Contact the MR Product Specialist at GoldSeal Logistics phone number 877–251–4468. If GoldSeal is accepting the returned equipment then all transportation arrangements and packaging will be processed through them. If GoldSeal is not accepting the return then they will advise you to contact GEMS Renewable Resources.
   - GEMS Renewable Resources – For coordination of transportation arrangements and packaging. Phone (414)–747–6997 or Dial Comm 8*579–6997; Fax (414)–747–6855; Field ASPEN 1–800–525–1516, Box #26041.
2. When calling, the following information is needed.
   - FDO (Future Delivery Order) number under which the upgrade was shipped.
   - Site name, address, and phone numbers.
   - Date of completed upgrade.
   - Name and Beeper number of Field Engineer that could be available when the equipment is picked up.
3. Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.
4. Process product locater information for all serialized components removed from the system according to policy.

1–4 UPGRADE SCHEDULE CONSIDERATIONS

FMIs, periodic maintenance, installation of prerequisite upgrades and/or options (Multi–coil, Spectroscopy, etc.) will directly add to the man–hour and system down–time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1–5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room
- Yellow – Post–installation (Phantoms, Coils).

"Product Delivery Instructions" (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M1090NC is for the Signa Horizon EchoSpeed Upgrade). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summery of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1–6 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans–shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
   http://gein2.med.ge.com/gib
   The FE Site Verification consists of three components that are available on the web from the main menu. They are:
   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page – http://supportcentral.ge.com/15563

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans–shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached. When complete, return to: **Product Locator File, P.O. Box 414, W–523, Milwaukee, WI 53201–0414**

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1–7 UPGRADE SEQUENCE

Follow the Upgrade Flow chart in SECTION 3 for best installation efficiency.

Note

All on–site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

Alternate tasks are noted above the task block to set them aside from parallel tasks.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

  Aside: Removing and installing cables is more efficient and accurate when the rooms are un–cluttered. It is strongly recommended that this be done after equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Functional Checks
- System calibrations
- System performance checks
- Cover replacement
- Complete final tasks and return site to customer.

1–8 SITE CONFIGURATION INFORMATION

Refer to the appropriate section listed below for overview of content when upgrading to system with LCC magnet.

- SECTION 1–9, UPGRADE TO 1.5T SIGNA 10.x MR/*i* WideOpen with EXCITE TECHNOLOGY
- SECTION 1–11, UPGRADE TO SIGNA 9.x MR/*i* WideOpen

1–9 UPGRADES TO 1.5T SIGNA 10.x MR/*i* WideOpen WITH EXCITE TECHNOLOGY

Illustration 1–1 shows the system's major equipment after a 1.5T Signa LX with EXCITE Technology (MGD) upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with EXCITE Technology product configuration. The following sub–sections provide an overview of the various Signa LX with EXCITE Technology upgrade hardware.

Note

At this time upgrades to EXCITE Technology are not available for systems with 1.0T Magnets or systems with CRM Body Coils.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER TO DIRECTION 2142460 – POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER TO DIRECTION 2142460 – POWER REQUIREMENTS.

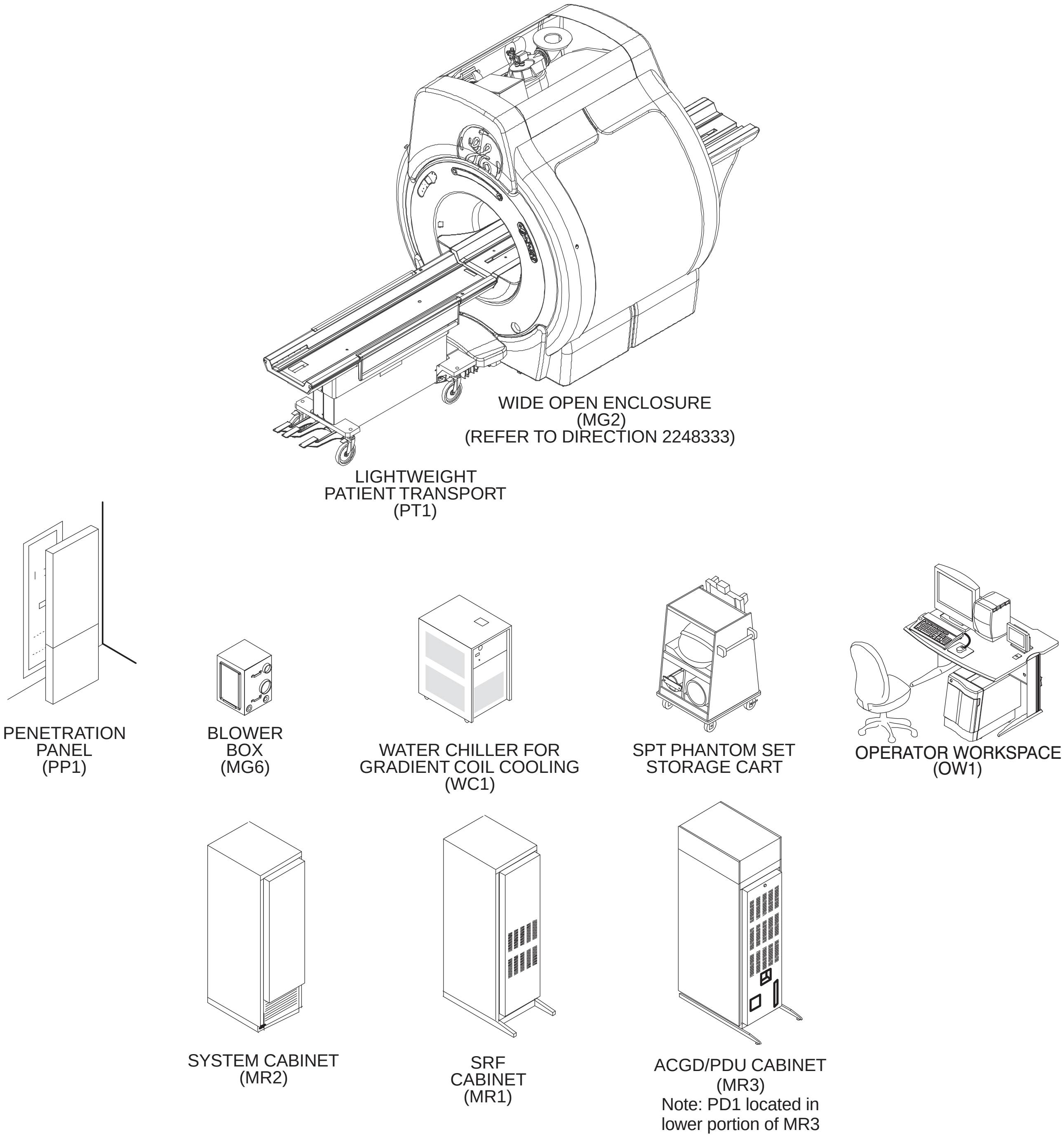
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre–Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating at Release 8.X prior or coincident with the Magnet Upgrade option.

1–9 UPGRADES TO 1.5T SIGNA 10.x MR/*i* WideOpen WITH EXCITE TECHNOLOGY (Continued)



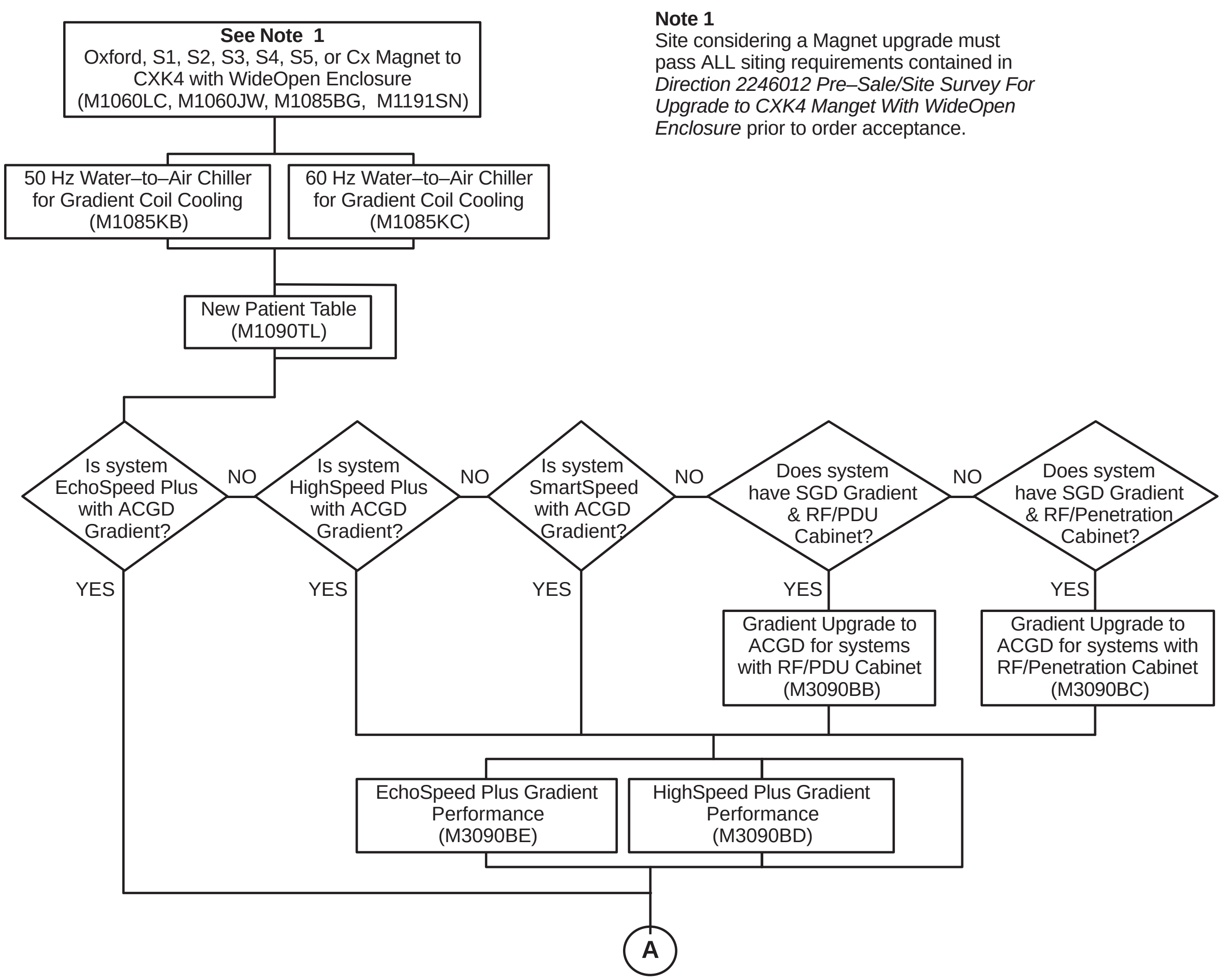
SIGNA LX WITH EXCITE TECHNOLOGY SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1–1

1–9–1 Upgrades from Signa Horizon LX Systems

Illustrations 1–2 and 1–3 lists the upgrade catalogs for Signa LX with ACGD or SGD to Signa LX With EXCITE Technology and LCC magnet. Illustration 1–4 shows the cabinets replaced and modified by the listed catalogs.

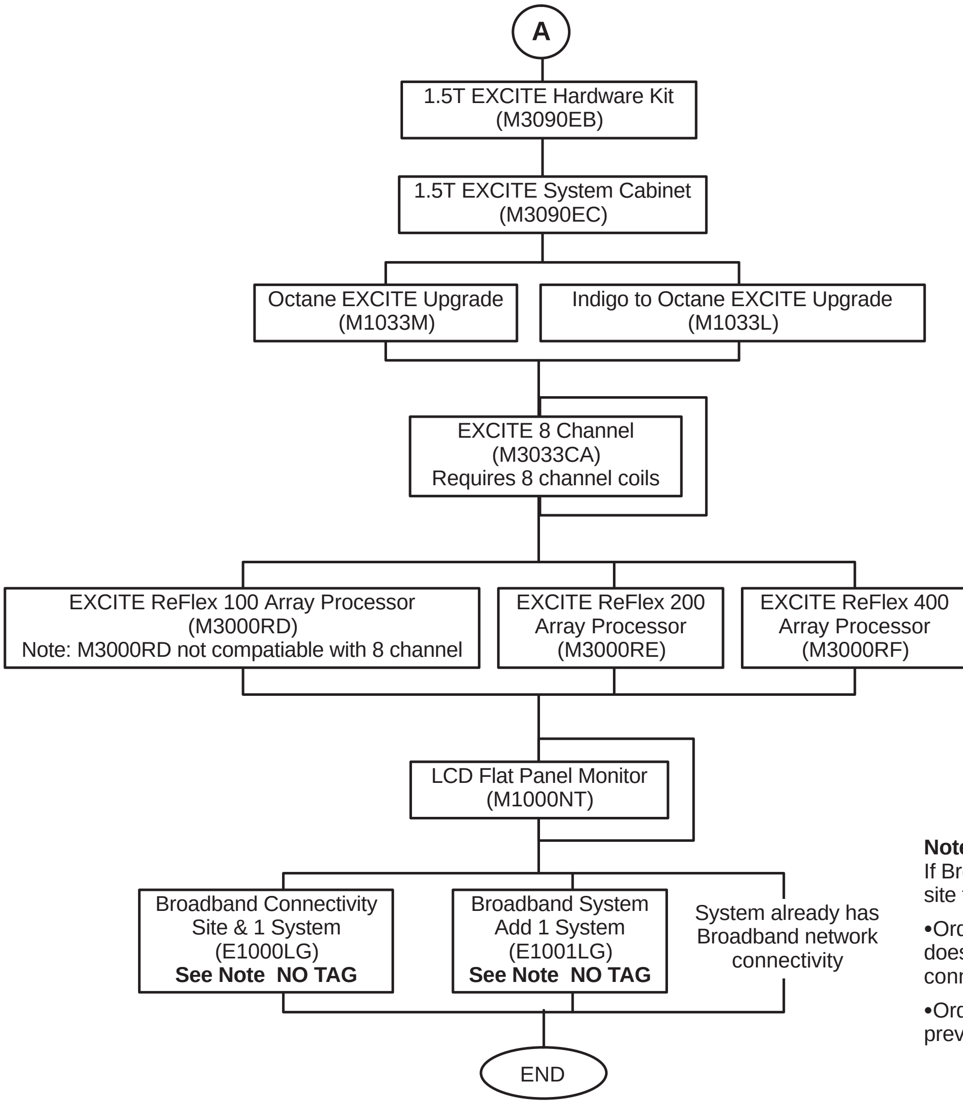
Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA LX TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1 – 2

1–9–1 Upgrades from Signa Horizon LX System (Continued)



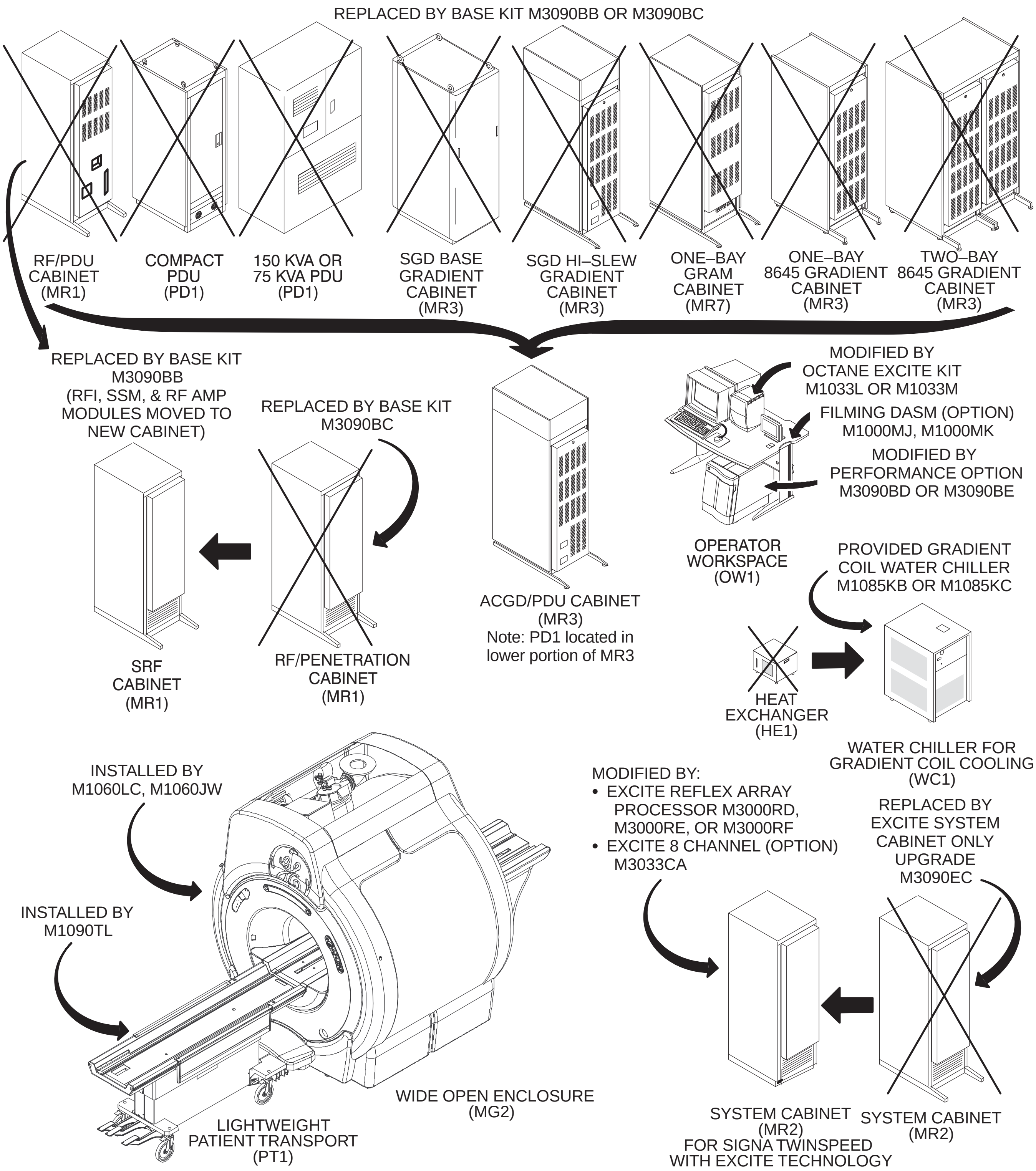
Note 2
If Broadband is available at customer site then choose one of the options:

•Order E1000LG if customer site does not have an existing connection.

•Order E1001LG if customer site has previously ordered E1000LG

1.5T SIGNA LX TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1–3

1–9–1 Upgrades from Signa Horizon LX System (Continued)



SIGNA HORIZON LX SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1–4

1–9–2 Upgrades from Signa Horizon 5.x System

Illustrations 1–5 through 1–6 lists the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX With EXCITE Technology. Illustrations 1–7 and 1–8 how the cabinets replaced or modified and new equipment added by the listed catalogs.

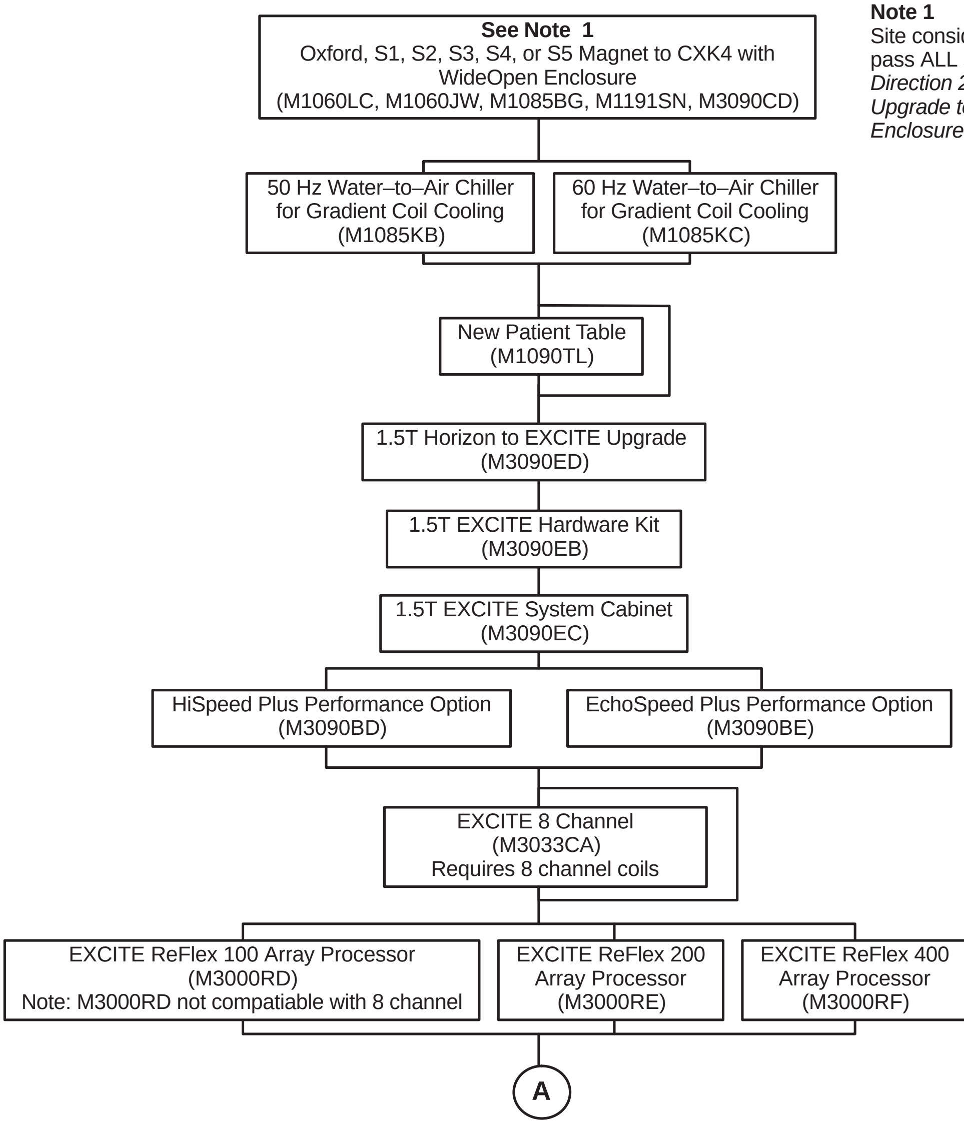
Refer to Section 1–10 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require upgrade catalogs.

Note

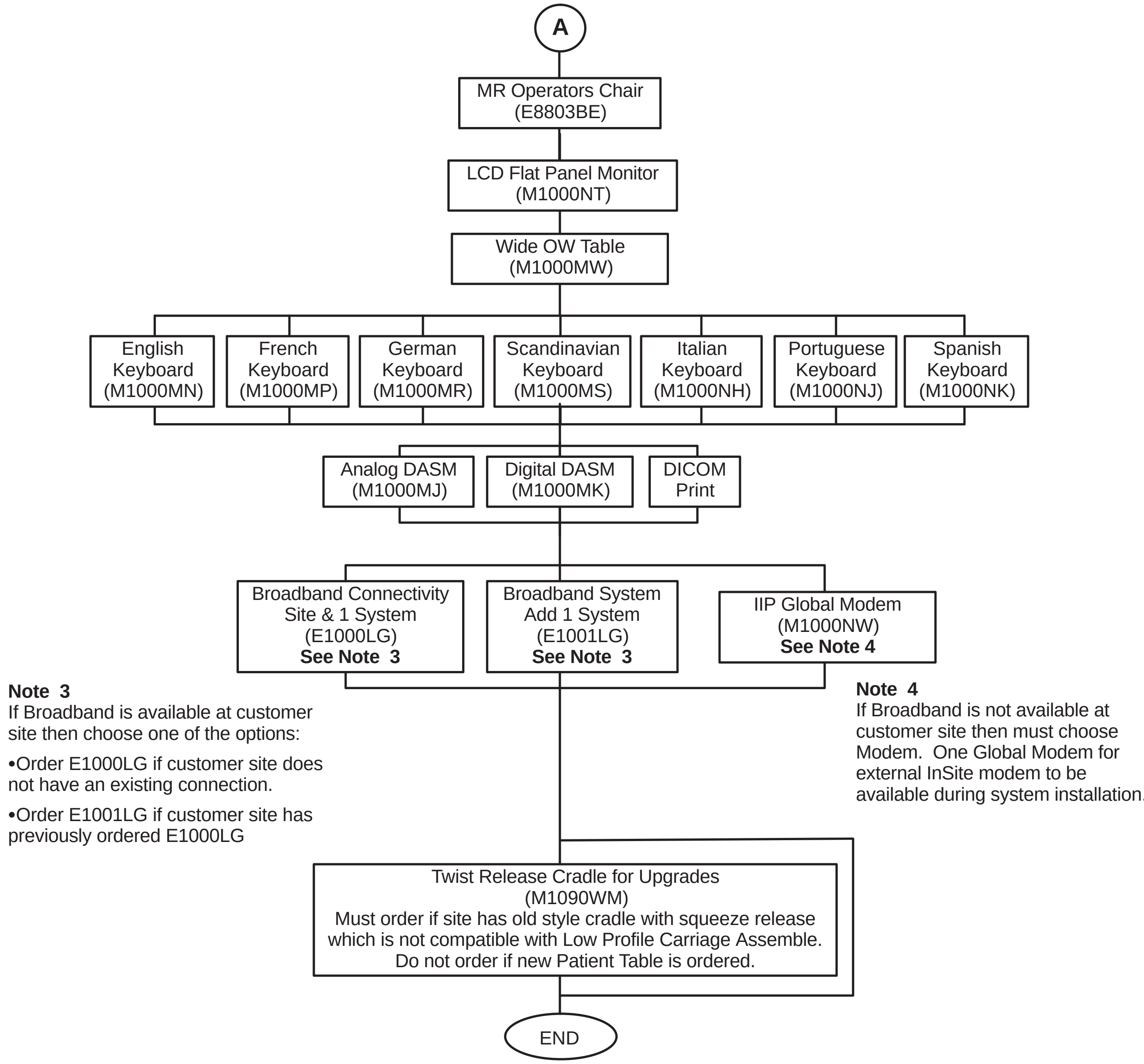
Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



Note 1
Site considering a Magnet upgrade must pass ALL siting requirements contained in *Direction 2246012 Pre–Sale/Site Survey For Upgrade to CXK4 Manget With WideOpen Enclosure* prior to order acceptance.

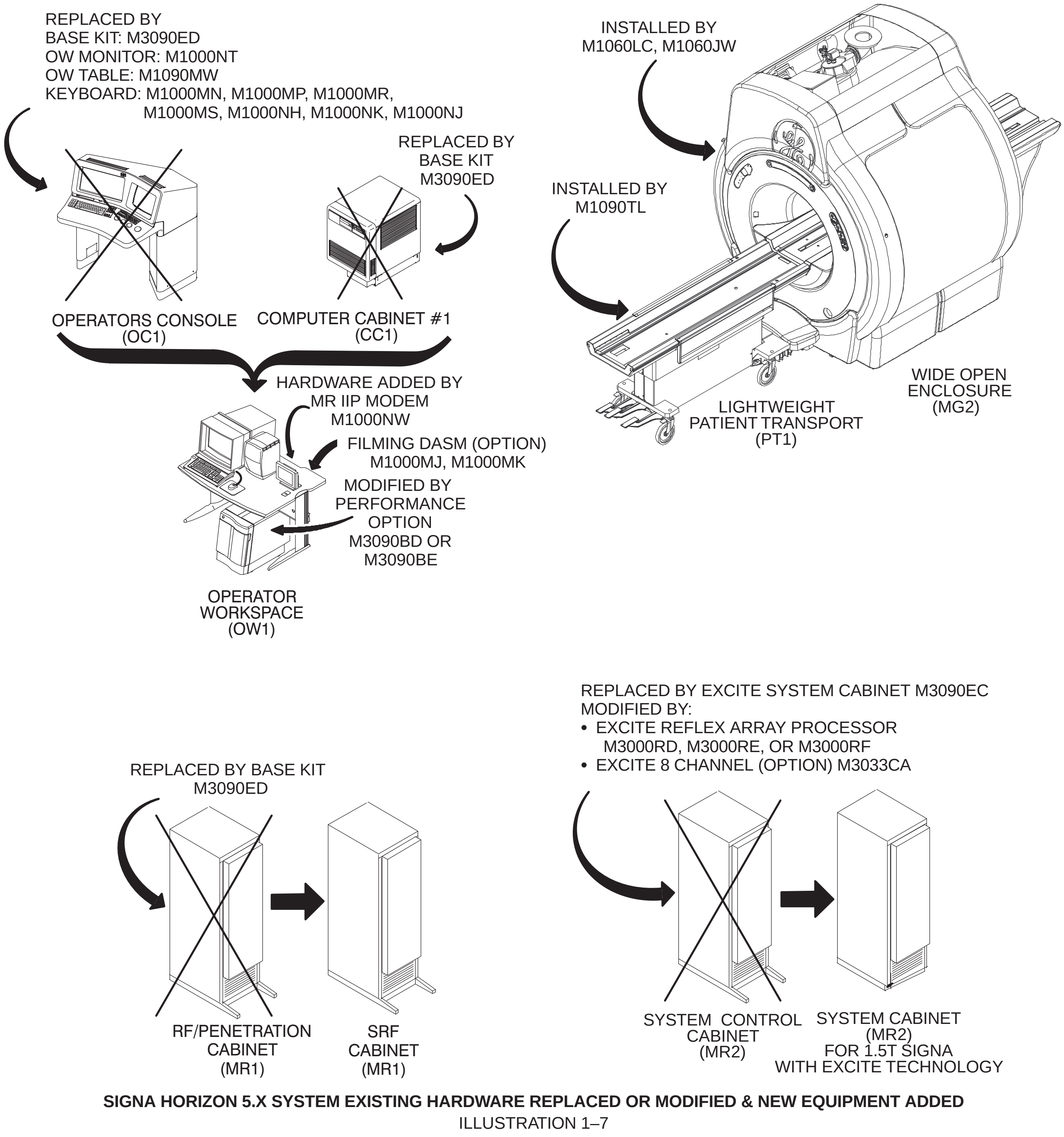
1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1 – 5

1–9–2 Upgrades from Signa Horizon 5.x System (Continued)



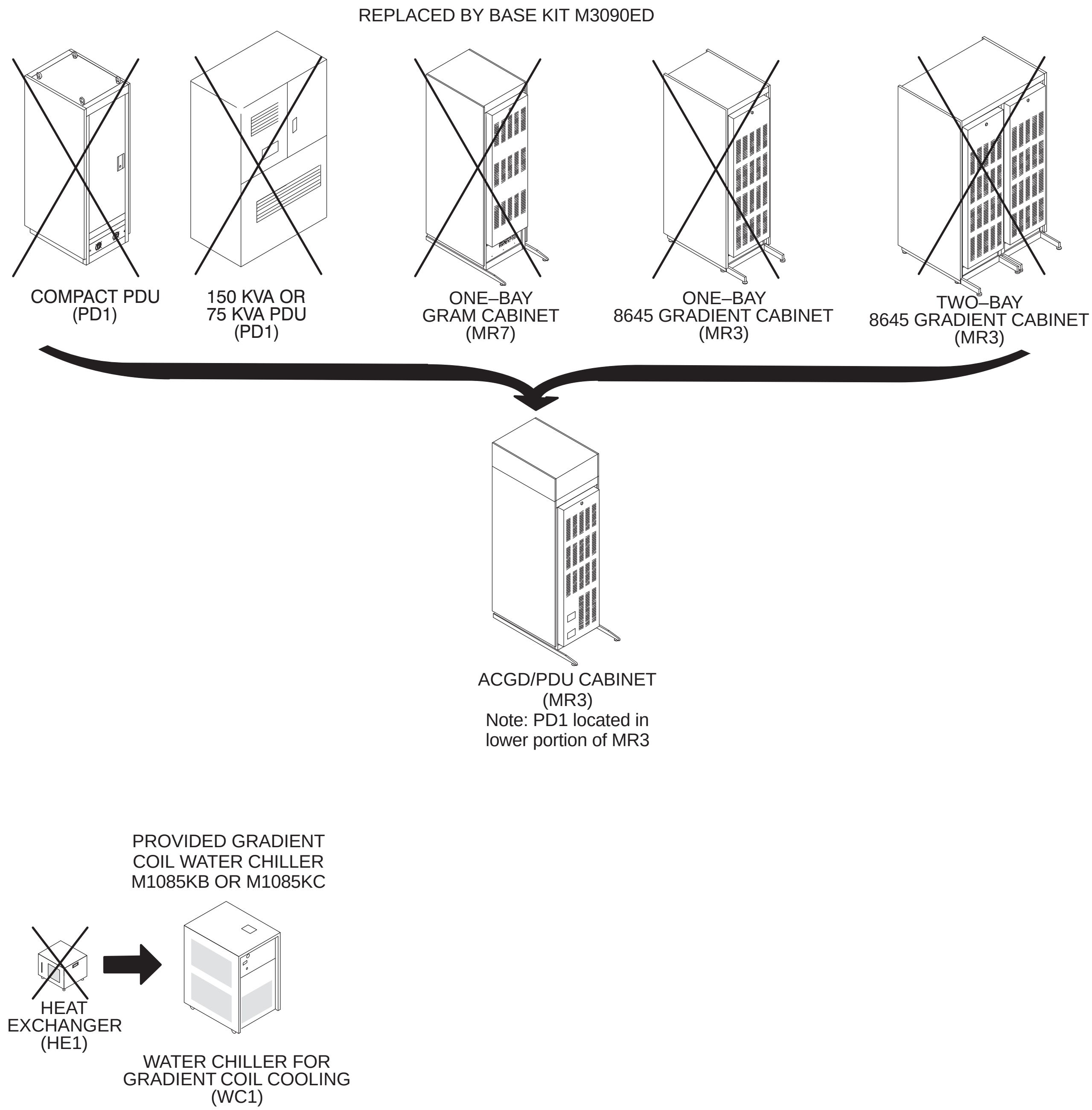
SIGNA HORIZON 5.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1–6

1–9–2 Upgrades from Signa Horizon 5.x System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–7

1–9–2 Upgrades from Signa Horizon 5.x System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–8

1–9–3 Upgrades from Signa Advantage 1.5T 5.x & 4.x

Illustrations 1–15 through 1–10 list the upgrade catalogs for 1.5T Signa Horizon 5.X software only system, 1.5T Signa Advantage 5.X hardware system, and 1.5T Signa 4.X system to 1.5T Signa LX with EXCITE Technology. Signa 5.1–5.4 and 4.X hardware systems are the following configurations:

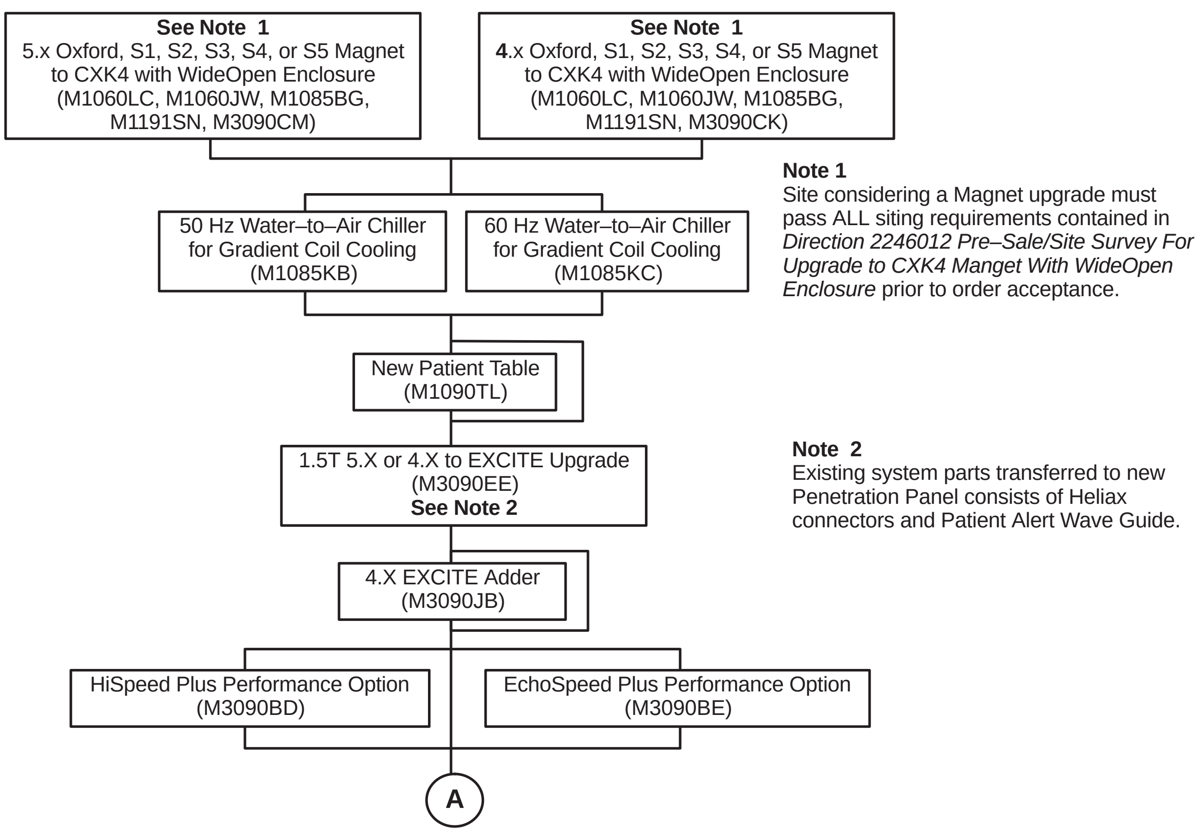
- Signa Horizon 5.X Software only systems which have a 55 cm Body Coil
- Signa Advantage 5.X systems identified by the Genesis Computer, 1–bay System Cabinet, and 8607 Gradient Cabinet.
- Signa Advantage 4.X hardware systems are Signa Advantage 4.5 to 4.8 which can be identified by RF Cabinet model number 46–306900 and Signa Advantage 4.0 to pre–4.5 which can be identified by RF Cabinet model number 46–282900.

The flowchart shown in Illustrations 1–15 through 1–10 lists the Signa Horizon 5.X Software Only, Advantage 5.X and 4.X to 1.5T Signa LX with EXCITE Technolog upgrade catalogs. The equipment replaced and modified by the catalogs listed in the flowchart are shown in the illustrations as note below:

- Signa Horizon 1.5T 5.X Software Only, Advantage 5.X see Illustrations 1–11 and 1–12
- Signa Horizon 1.5T 4.X see Illustrations 1–13 and 1–14.

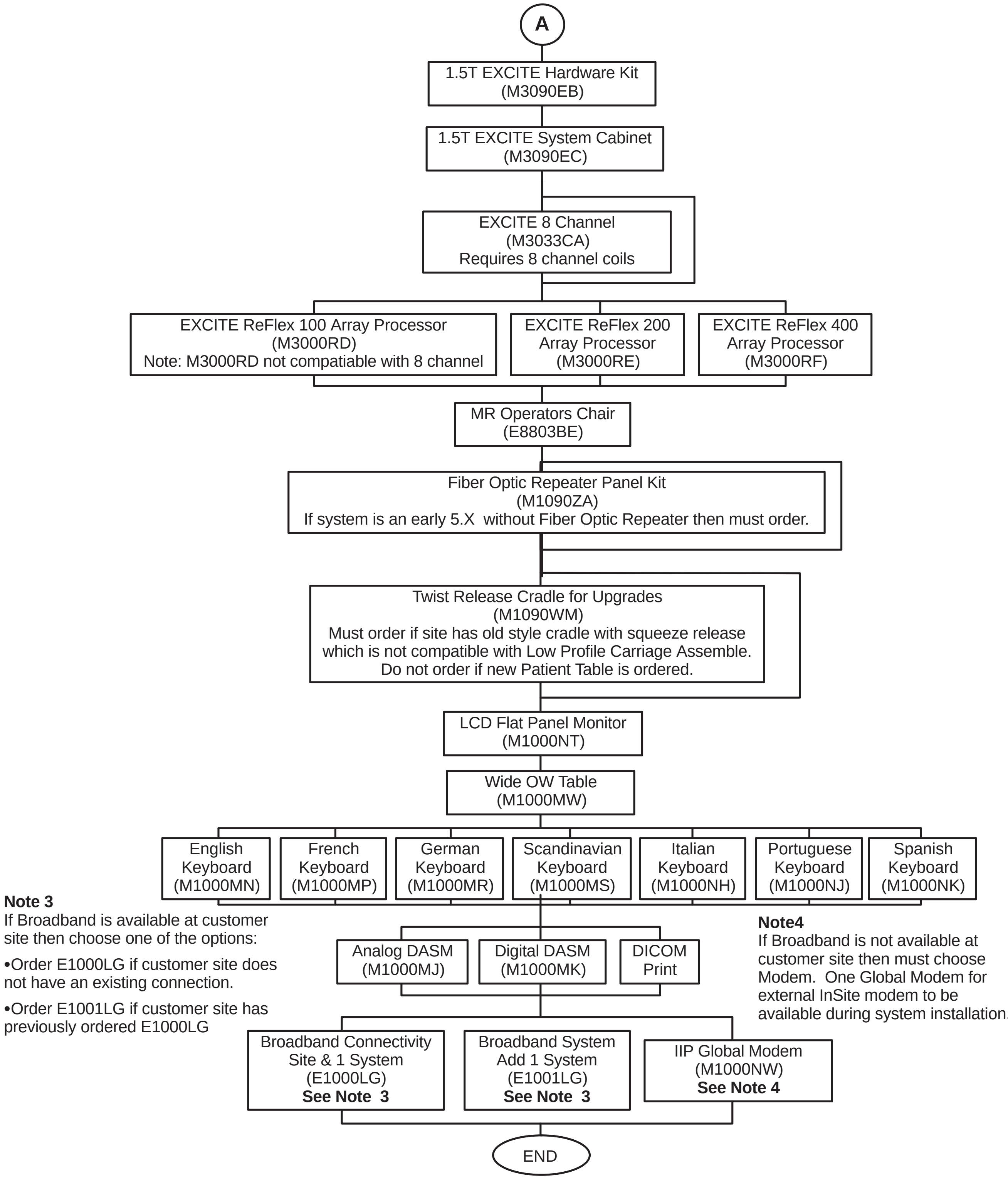
Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



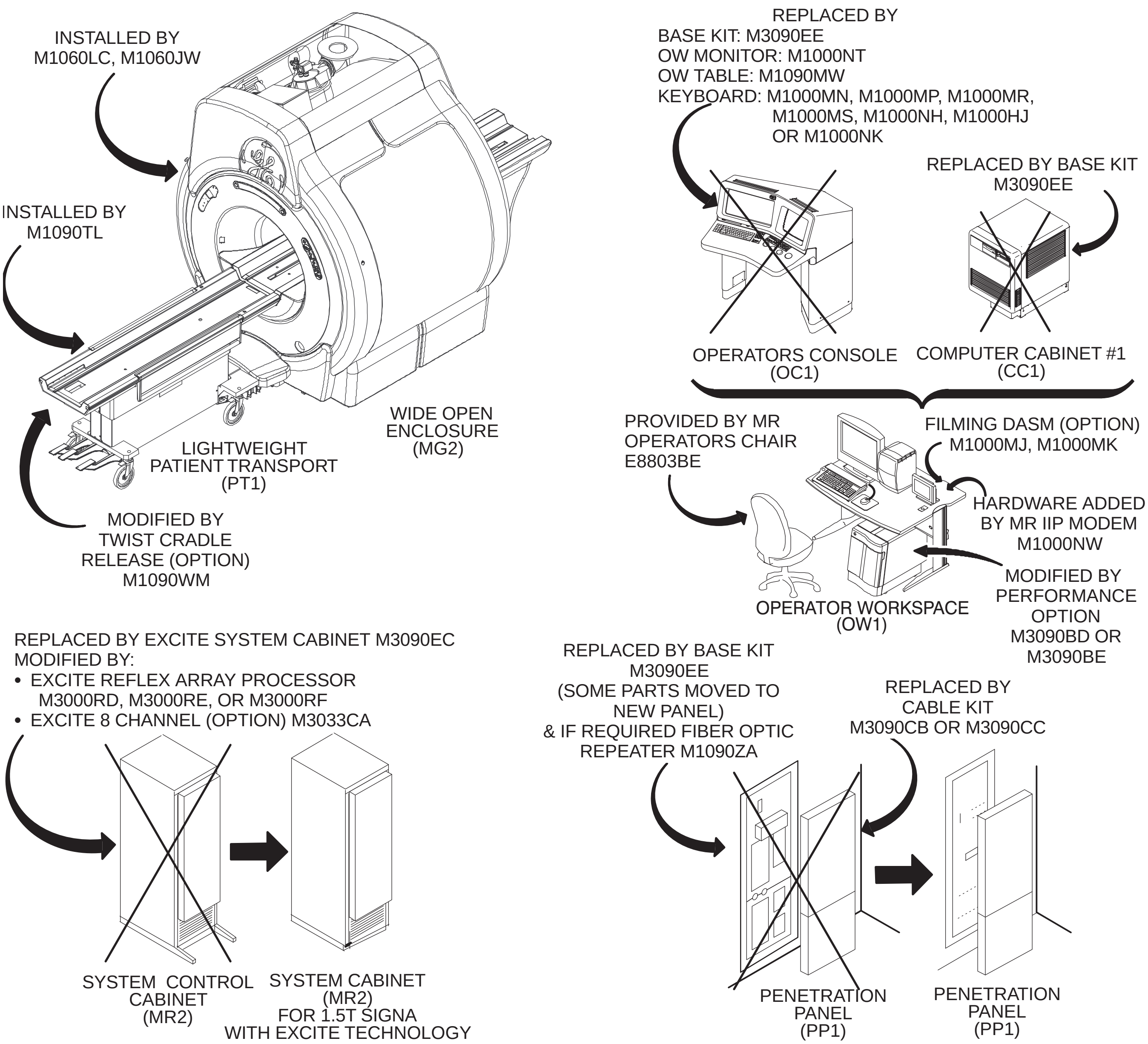
SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X, 4.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1–9

1–9–3 Upgrades from Signa Advantage 1.5T 5.x & 4.x (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X, 4.X TO SIGNA LX WITH EXCITE TECHNOLOGY CATALOGS
ILLUSTRATION 1 – 10

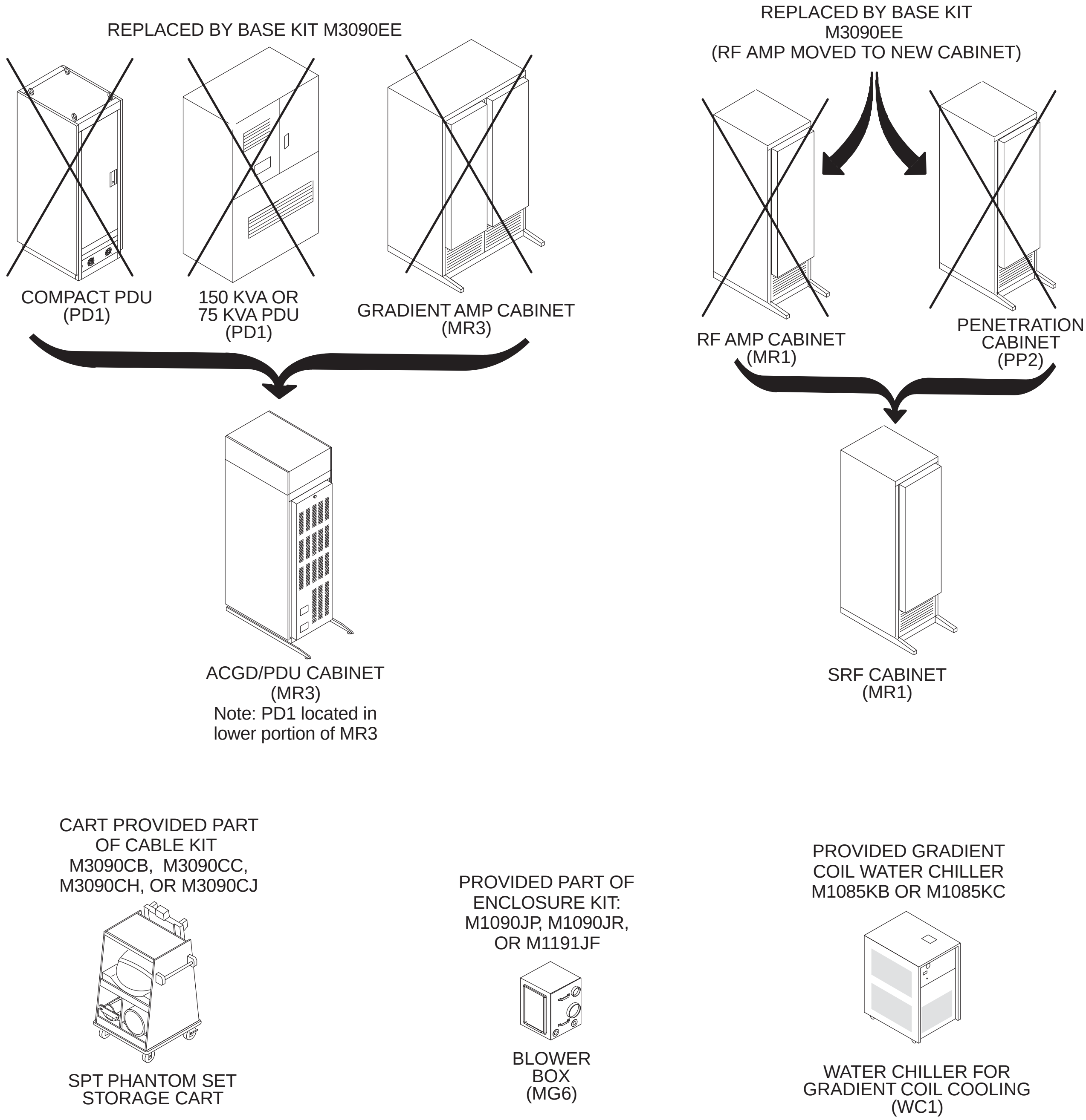
1–9–3 Upgrades from Signa Advantage 1.5T 5.x & 4.x (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

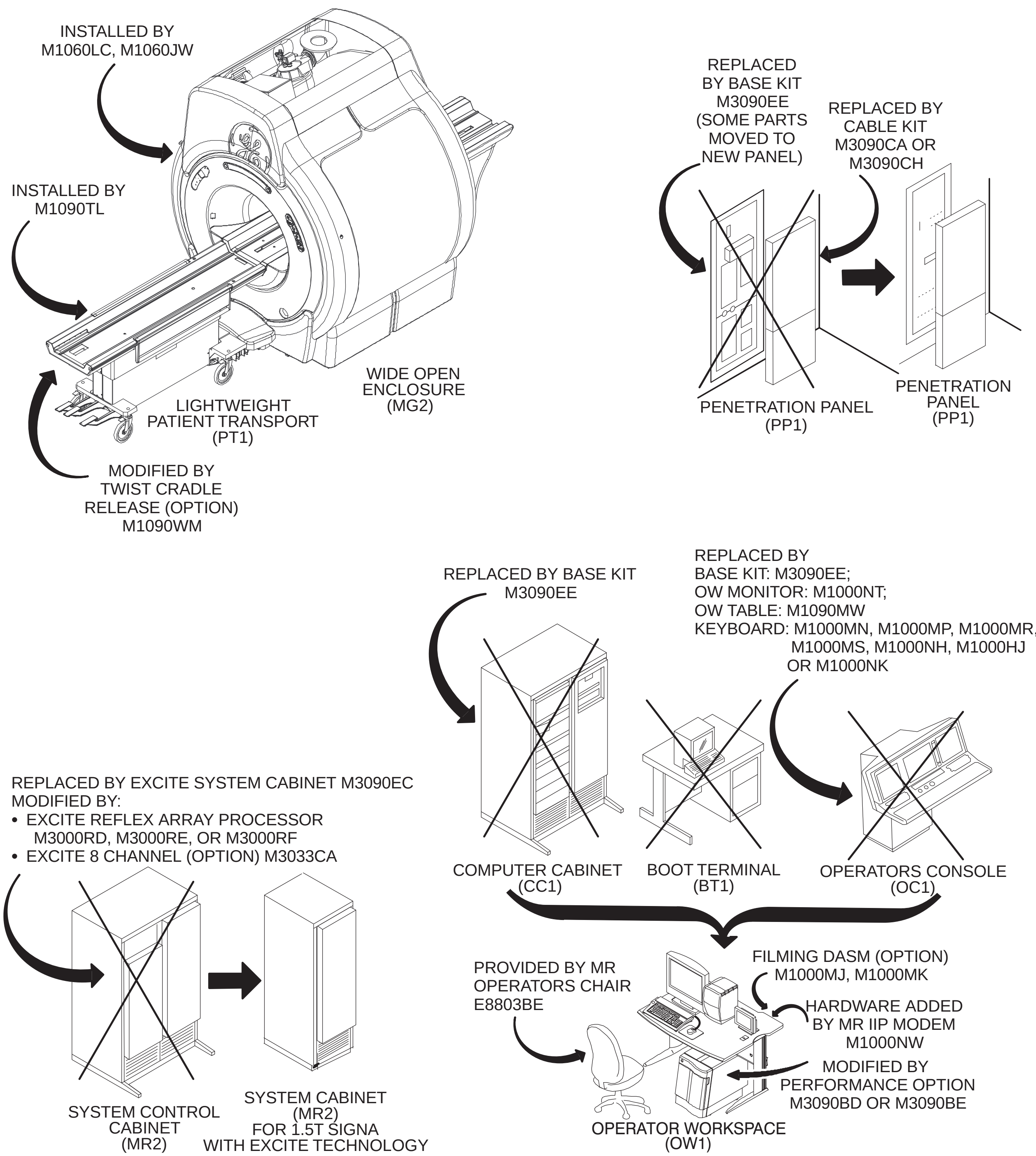
ILLUSTRATION 1–11

1–9–3 Upgrades from Signa Advantage 1.5T 5.x & 4.x (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–12

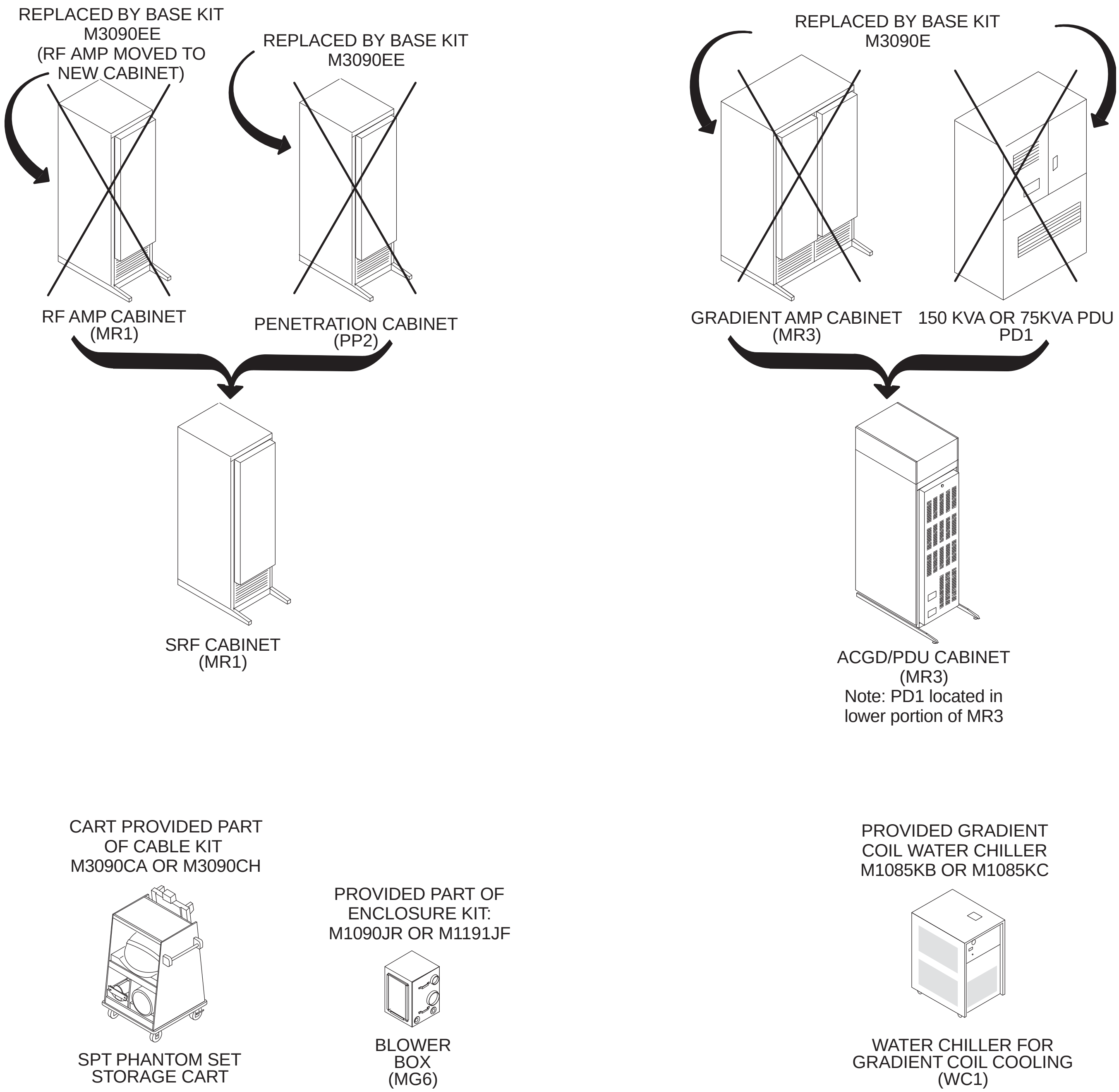
1–9–3 Upgrades from Signa Advantage 1.5T 5.x & 4.x (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1–13

1–9–3 Upgrades from Signa Advantage 1.5T 5.x & 4.x (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1–14

1–10 UPGRADES TO 1.5T SIGNA 9.x MR/*i* WideOpen WITH ACGD

Illustration 1–15 shows the system's major equipment after a Signa LX with ACGD upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with ACGD product configuration. The following sub–sections provide an overview of the various Signa LX with ACGD upgrade hardware.

WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER SECTION NO TAG – POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER SECTION NO TAG – POWER REQUIREMENTS.

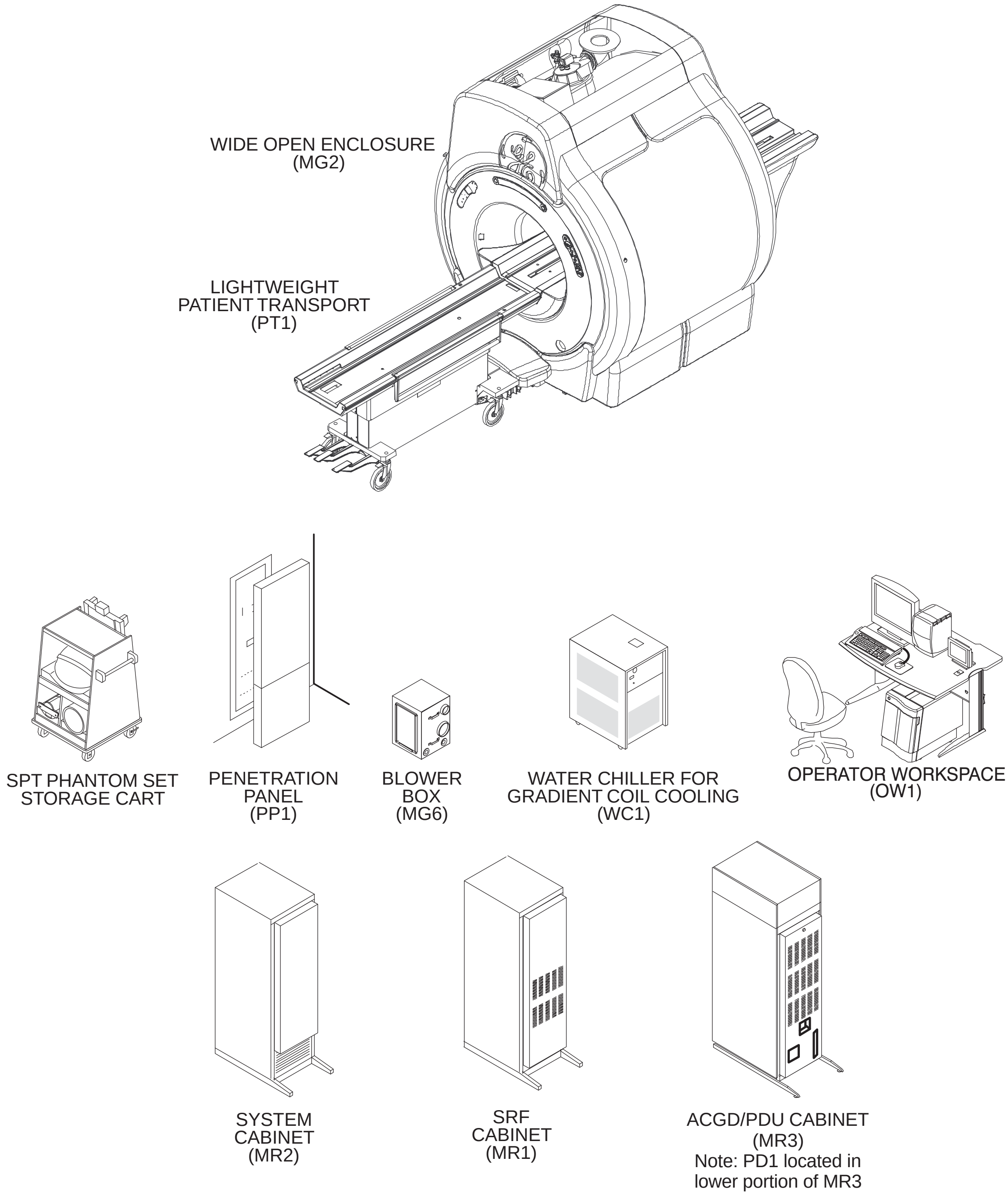
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre–Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating a Release 8.X prior or coincident with the Magnet Upgrade option.

1–10 UPGRADES TO 1.5T SIGNA 9.x MR/*i* WideOpen WITH ACGD (Continued)



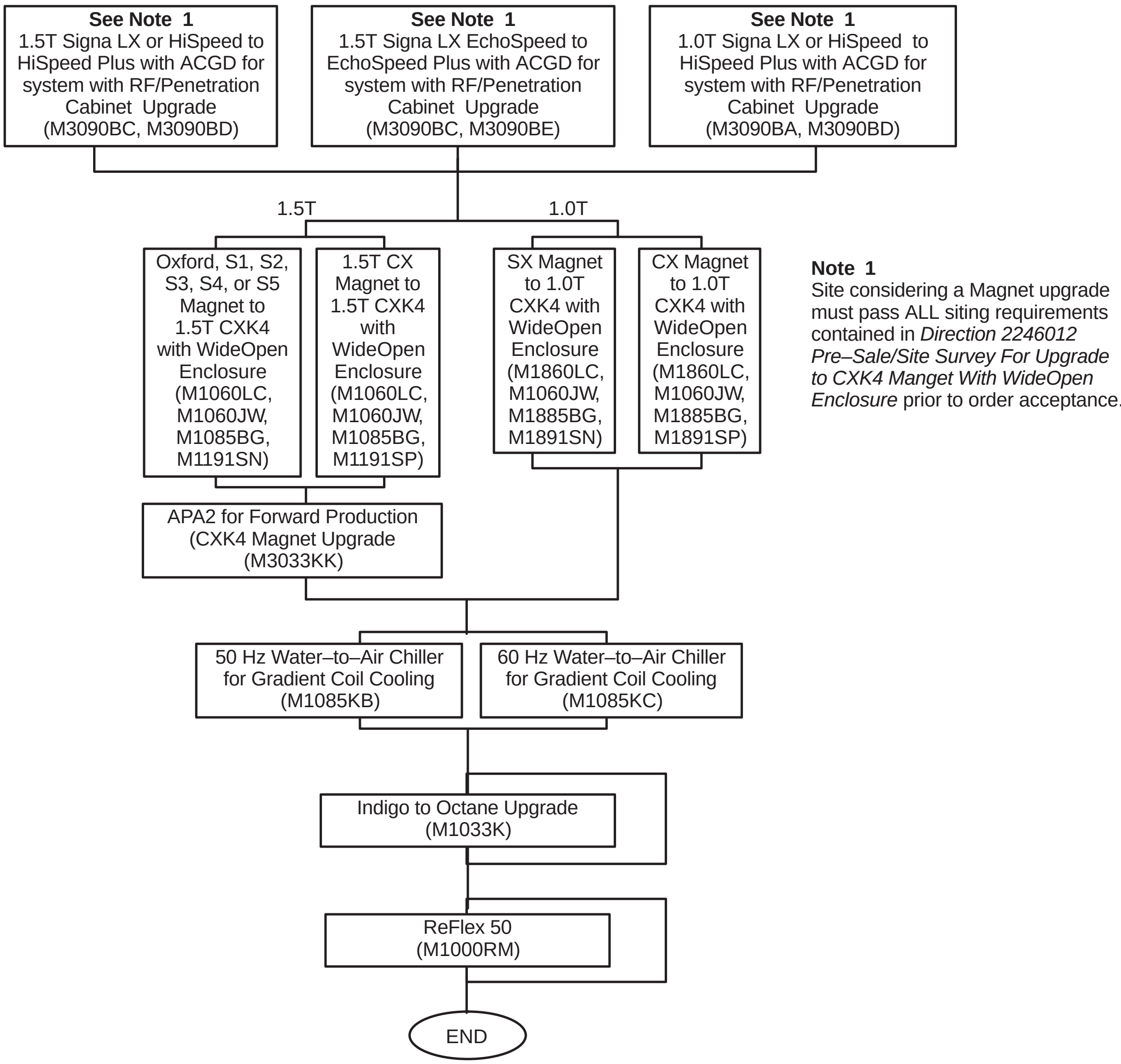
SIGNA LX WITH ACGD SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1–15

1–10–1 Upgrades from Horizon LX with 8645 Gradients

Illustration 1–16 lists the upgrade catalogs for Signa Horizon LX system with 8645 Gradients to Signa Horizon LX with ACGD. Illustration 1–17 shows the cabinets replaced and modified by the listed catalogs.

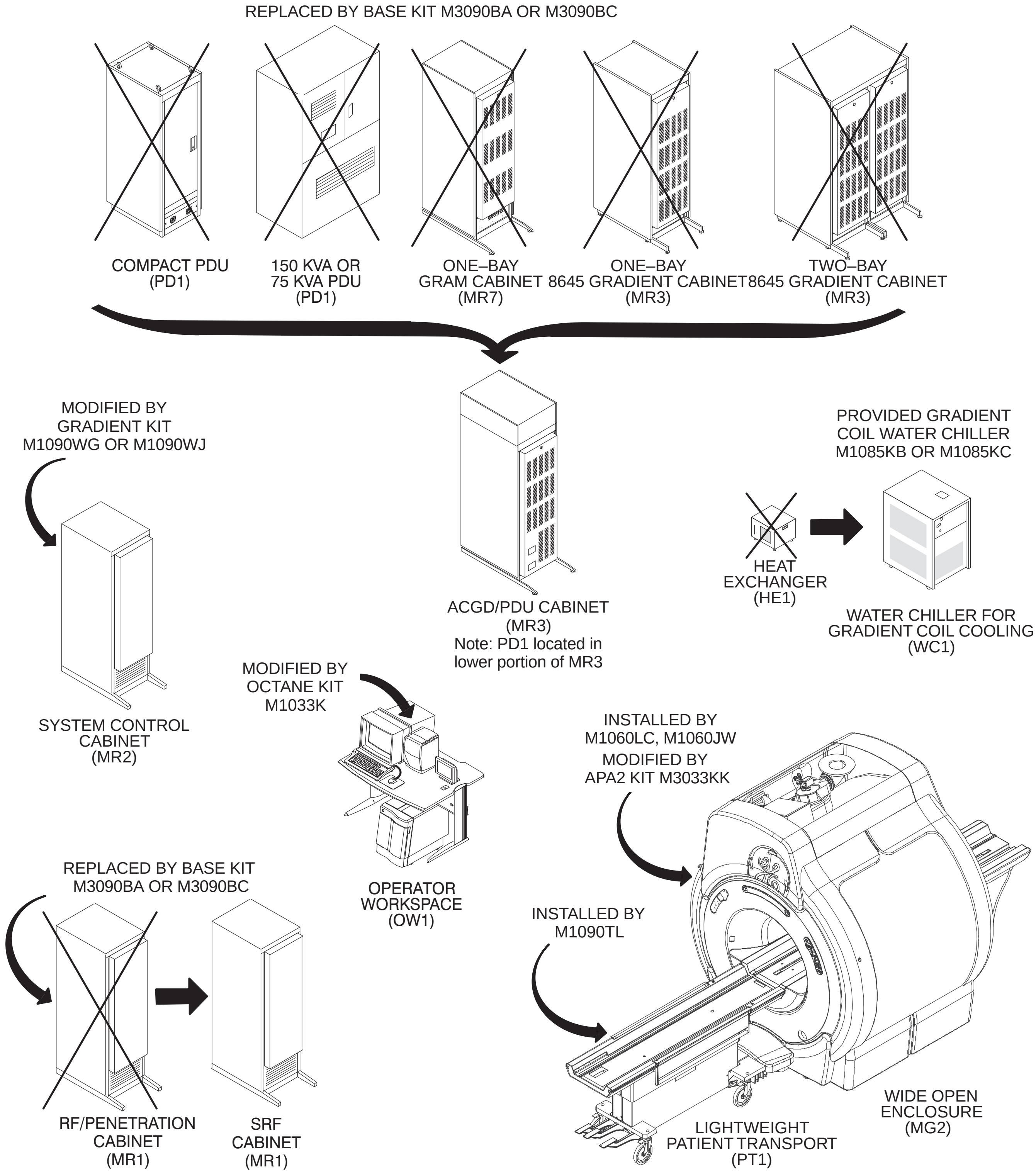
Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1–16

1–10–1 Upgrades from Horizon LX with 8645 Gradients (Continued)



SIGNA HORIZON LX SYSTEM UPGRADE MAJOR EQUIPMENT REPLACED & MODIFIED

ILLUSTRATION 1–17

1–10–2 Upgrades from Horizon 5.x System

Illustrations 1–18 through NO TAG list the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX with ACGD. Illustrations 1–20 and 1–21 show the cabinets replaced or modified and new equipment added by the listed catalogs.

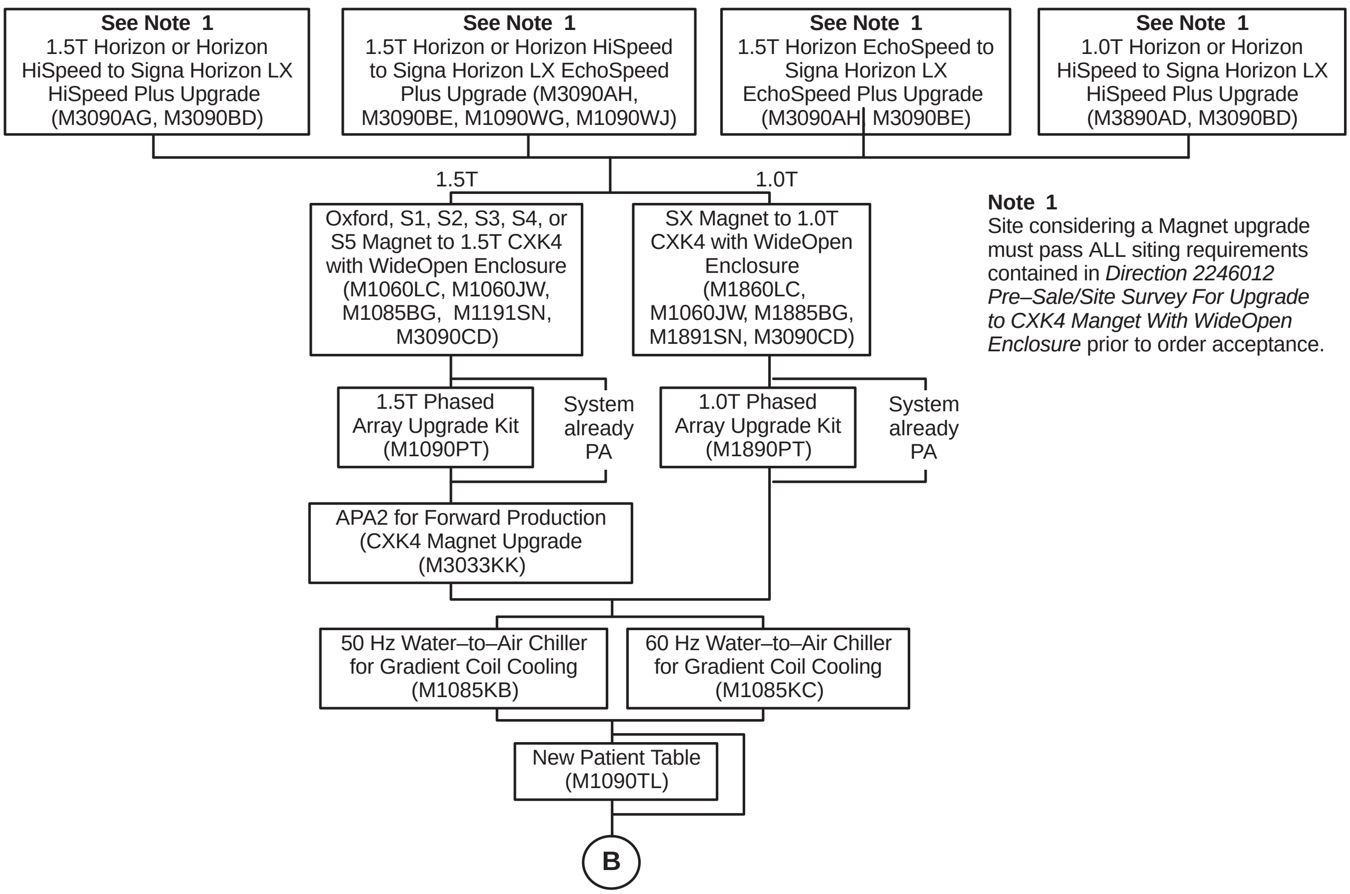
Refer to Section 1–10 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require upgrade catalogs defined in Section NO TAG, SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM UPGRADES.

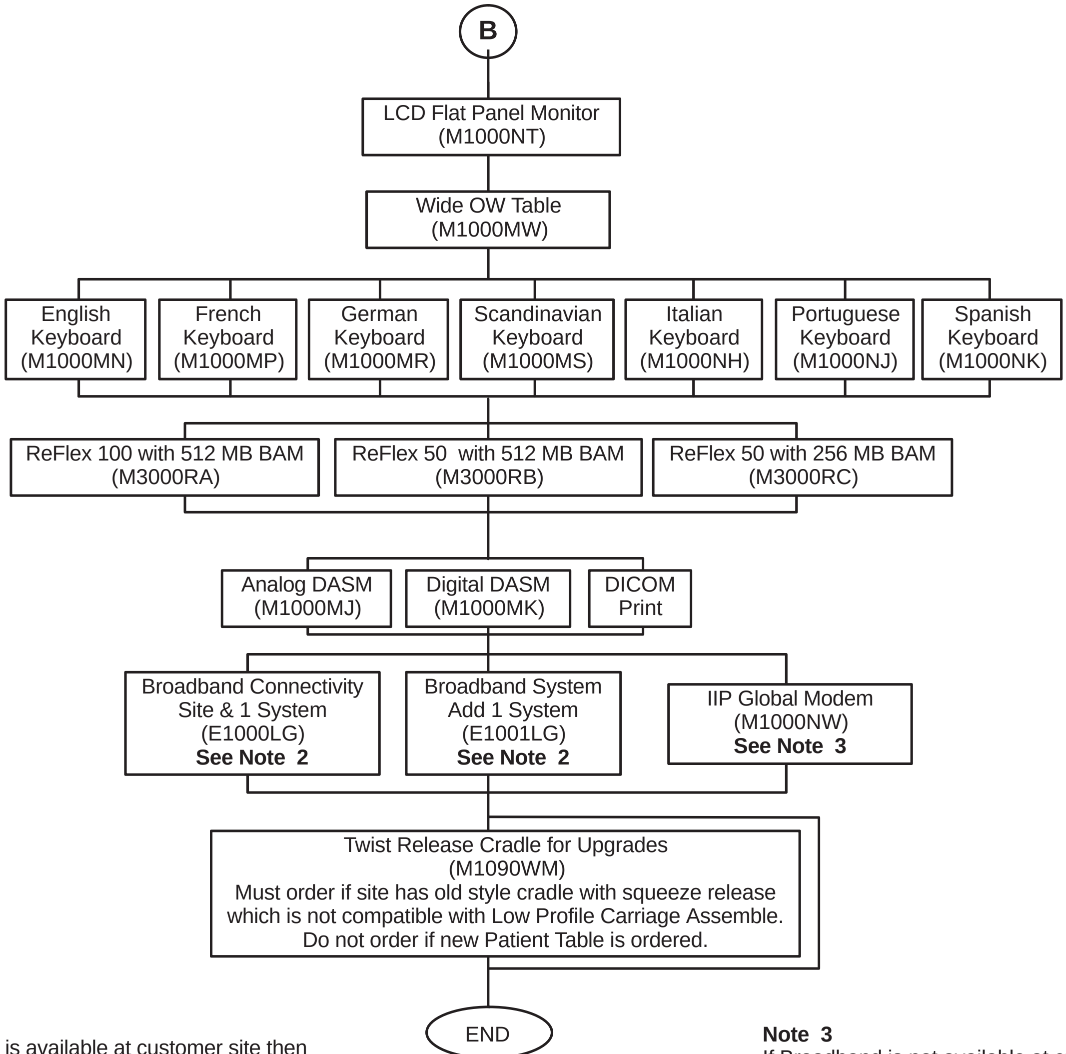
Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1 – 18

1–10–2 Upgrades from Horizon 5.x System (Continued)



Note 2
If Broadband is available at customer site then choose one of the options:

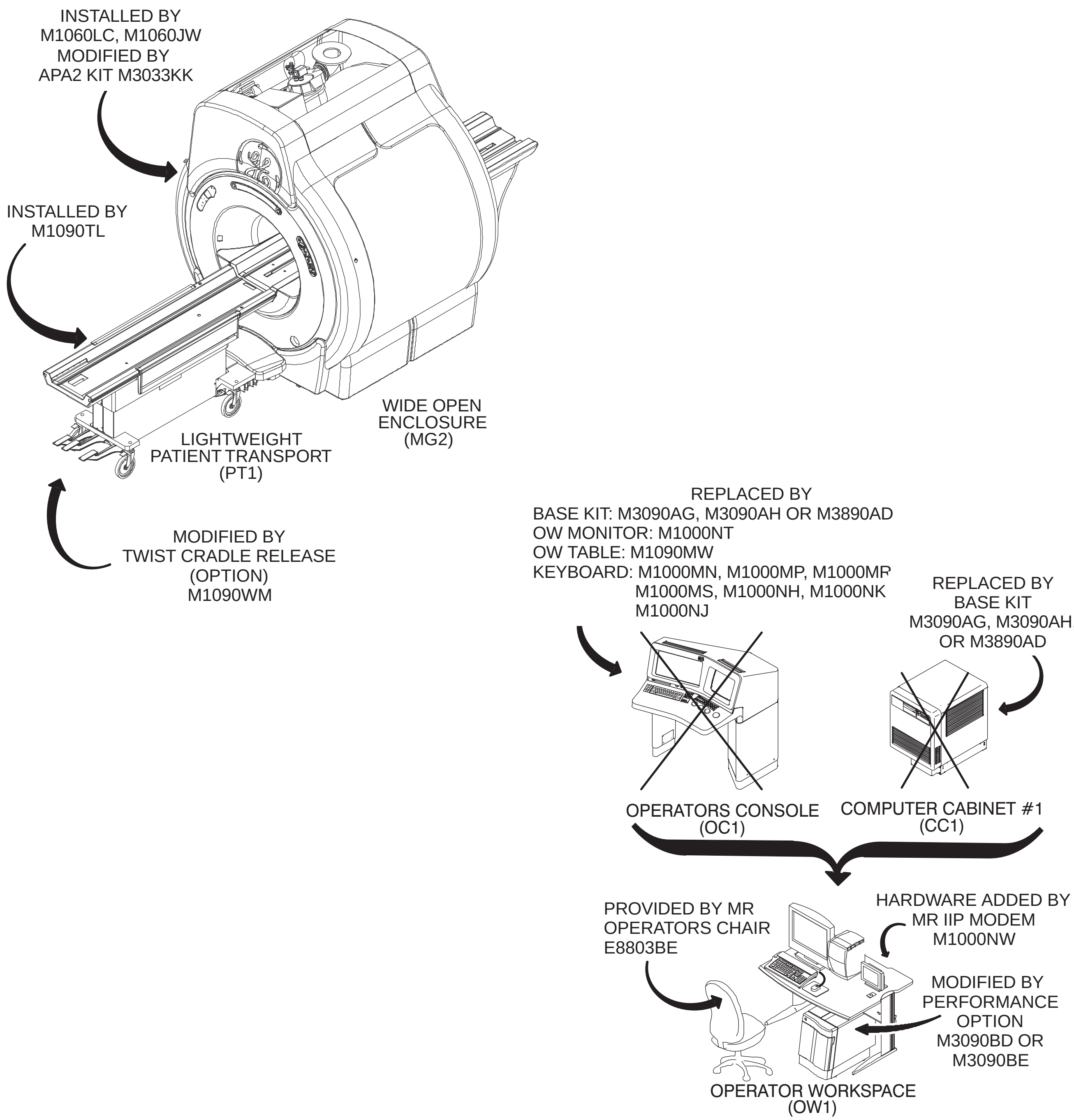
•Order E1000LG if customer site does not have an existing connection.

•Order E1001LG if customer site has previously ordered E1000LG

Note 3
If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1–19

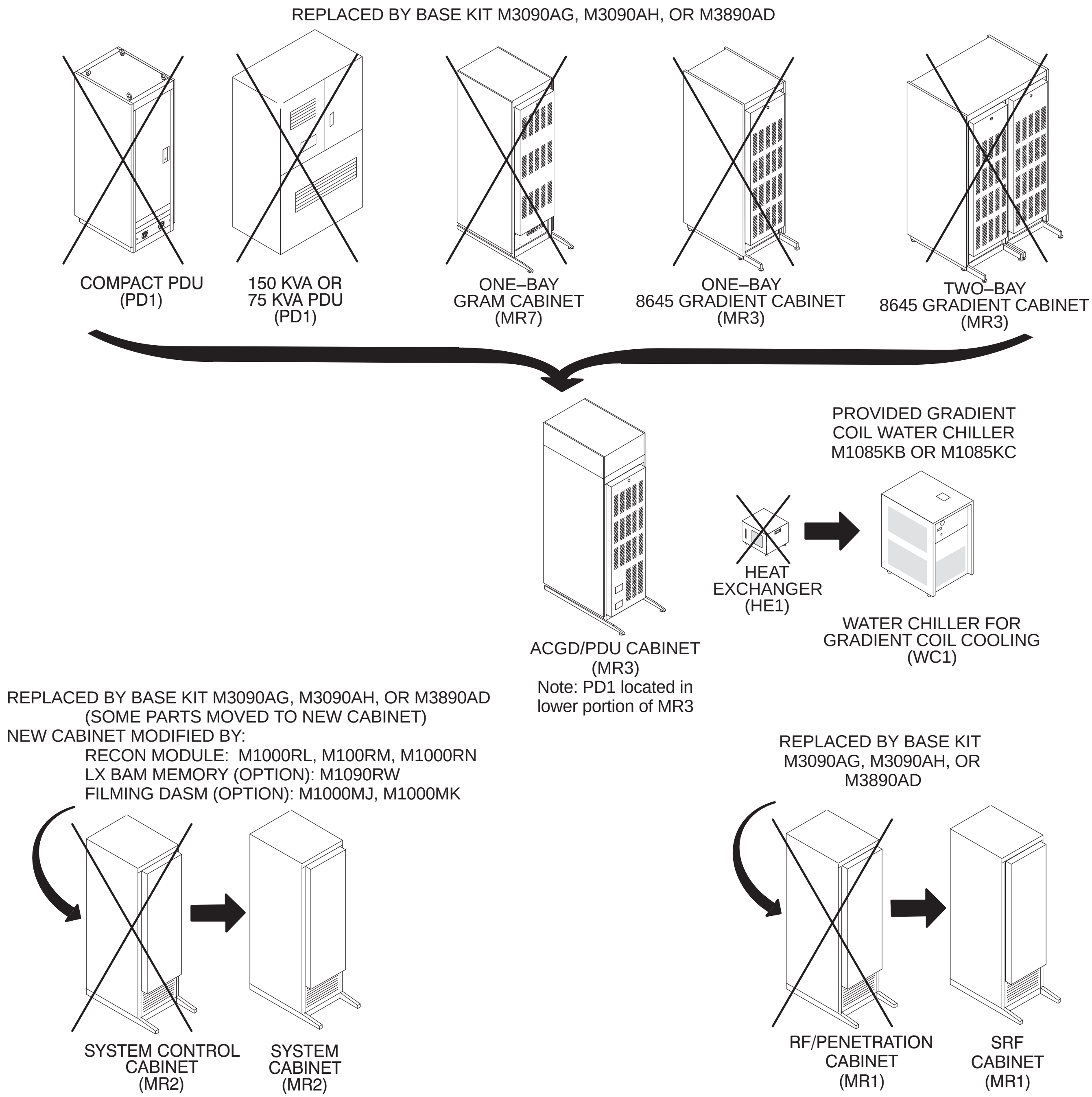
1–10–2 Upgrades from Horizon 5.x System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1–20

1–10–2 Upgrades from Horizon 5.x System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–21

1–10–3 Upgrades from Signa Horizon 5.x Software Only & Signa Advantage 5.x System

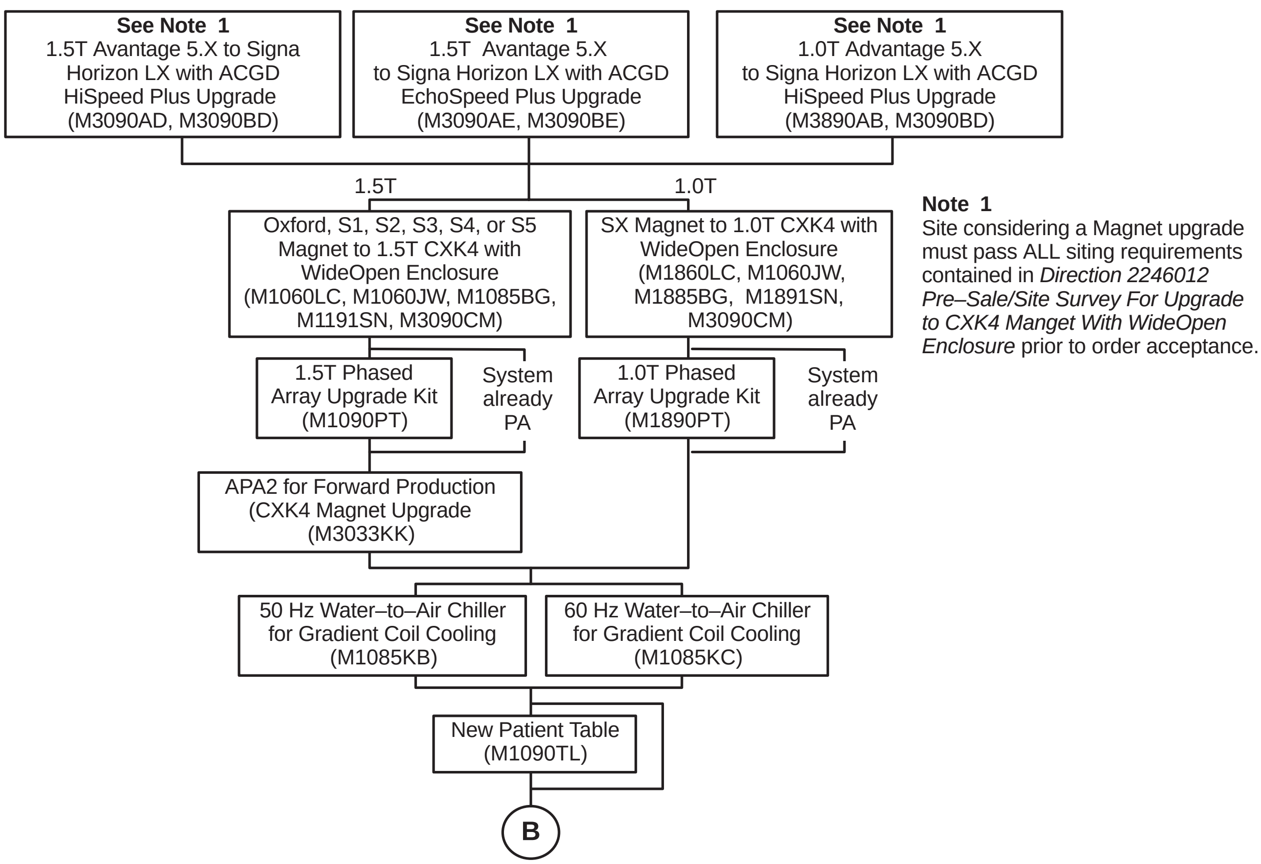
Illustrations 1–22 through 1–23 lists the upgrade catalogs for Signa Advantage 5.X hardware system and Signa Horizon 5.X software only system to Signa LX with ACGD. Illustrations 1–24 and 1–25 show the cabinets replaced or modified and new equipment added by the listed catalogs. Signa 5.1–5.4 hardware systems are the following configurations:

- Signa Horizon 5.X Software only systems which have a 55 cm Body Coil
- Signa Advantage 5.X systems identified by the Genesis Computer, 1–bay System Cabinet, and 8607 Gradient Cabinet.

Refer to Section 1–10 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

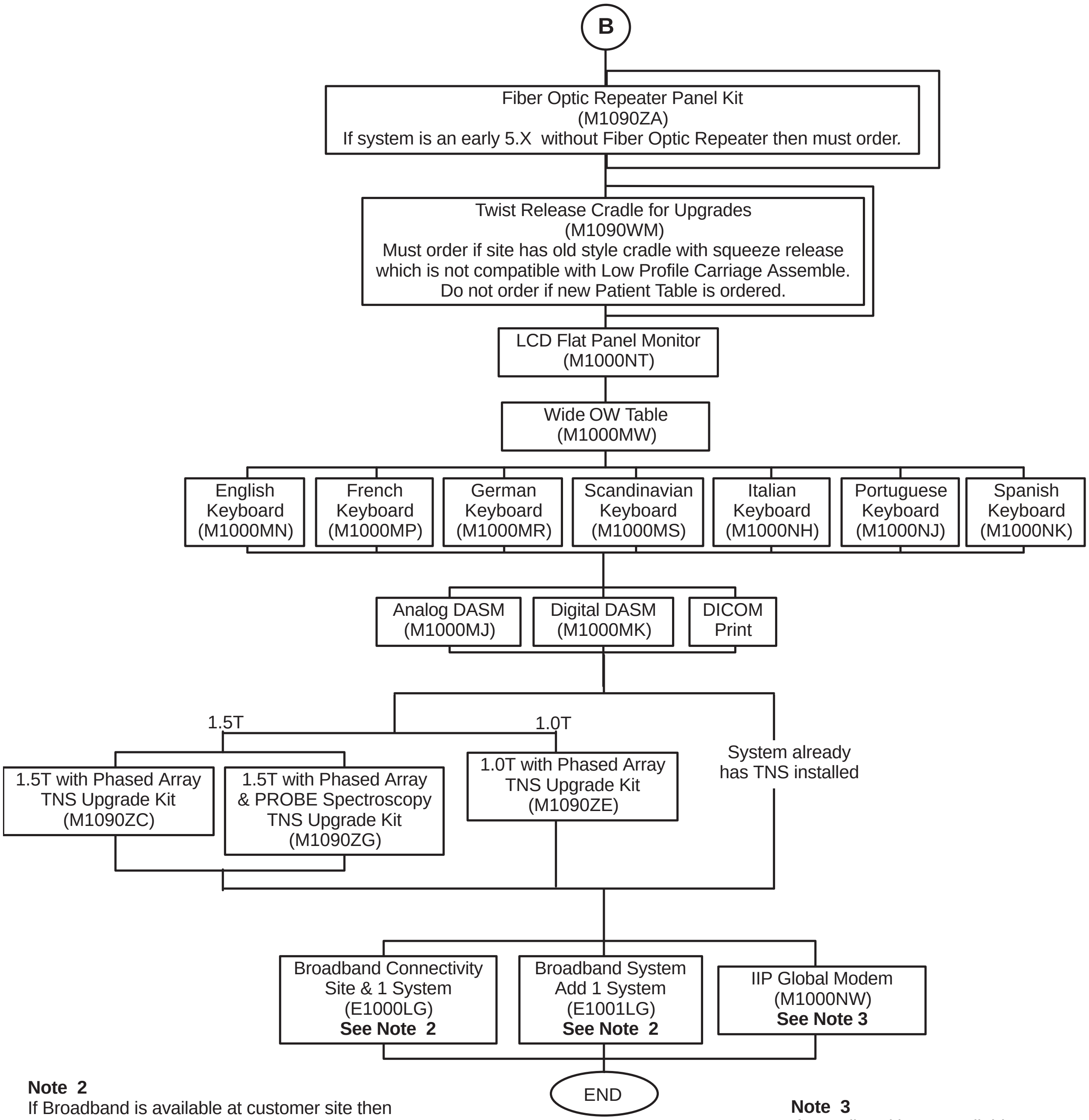
Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1 – 22

1–10–3 Upgrades from Signa Horizon 5.x Software Only & Signa Advantage 5.x System (Continued)



Note 2
If Broadband is available at customer site then choose one of the options:

•Order E1000LG if customer site does not have an existing connection.

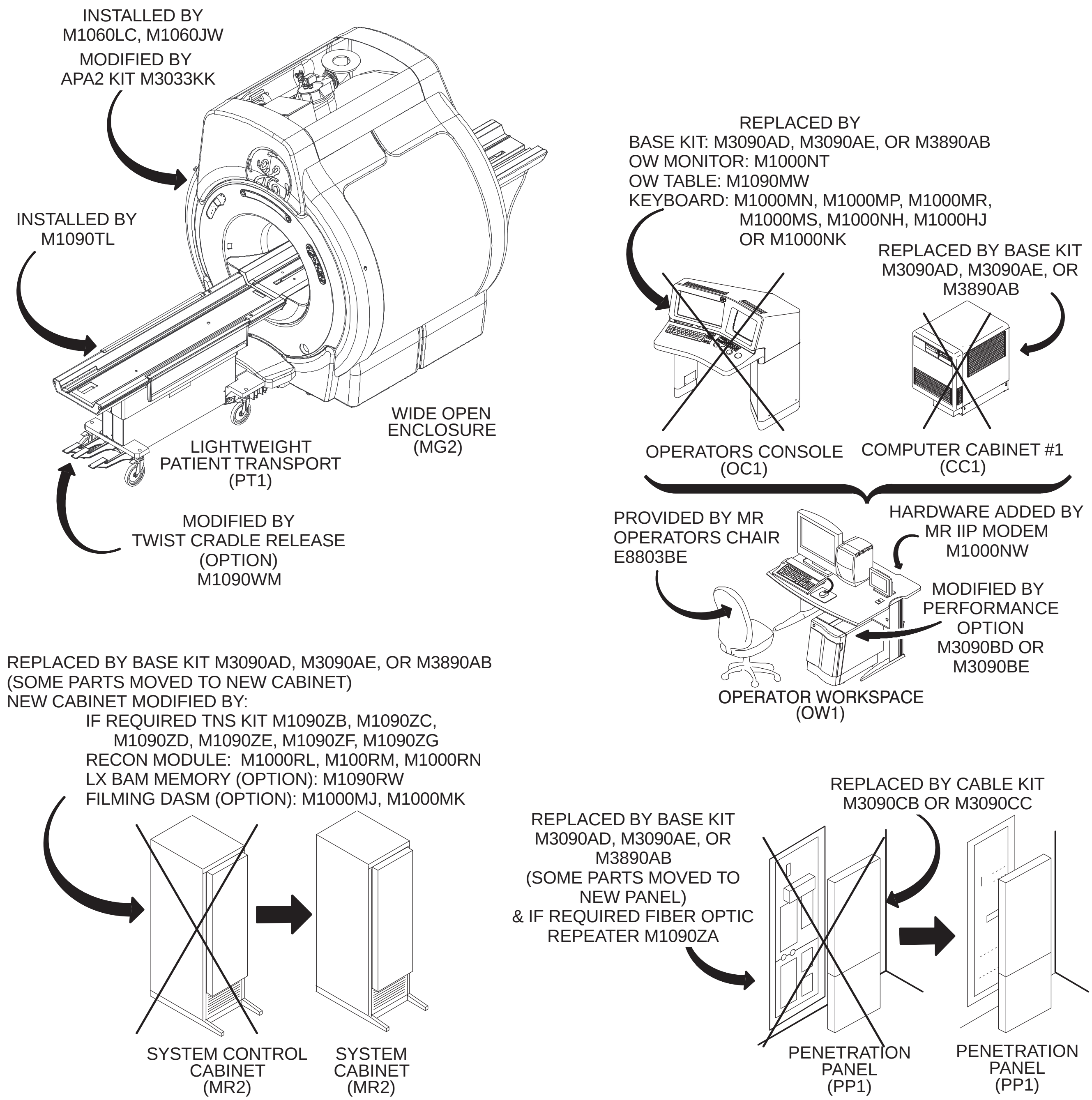
•Order E1001LG if customer site has previously ordered E1000LG

Note 3
If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS

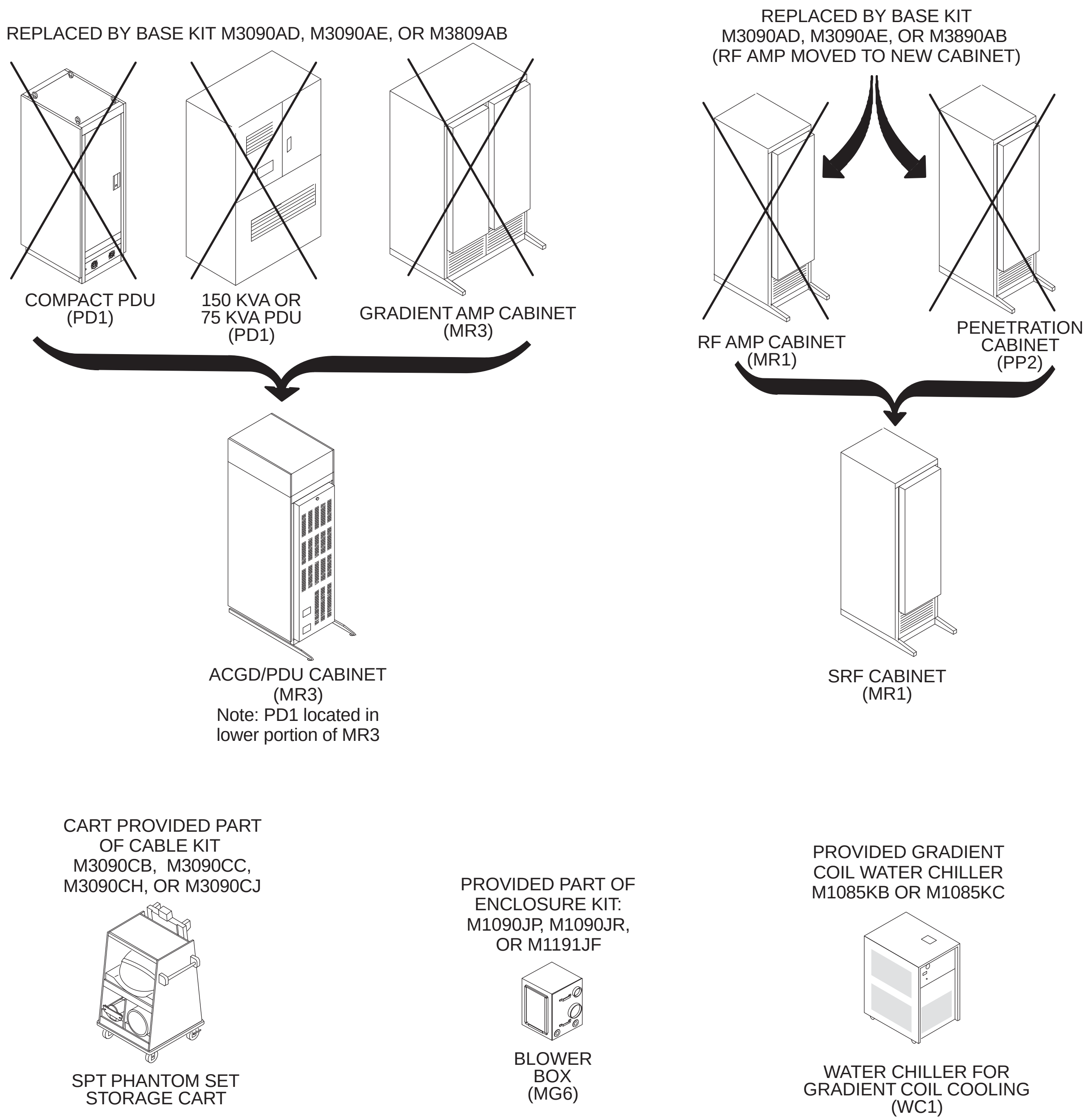
ILLUSTRATION 1–23

1–10–3 Upgrades from Signa Horizon 5.x Software Only & Signa Advantage 5.x System (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–24

1–10–3 Upgrades from Signa Horizon 5.x Software Only & Signa Advantage 5.x System (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–25

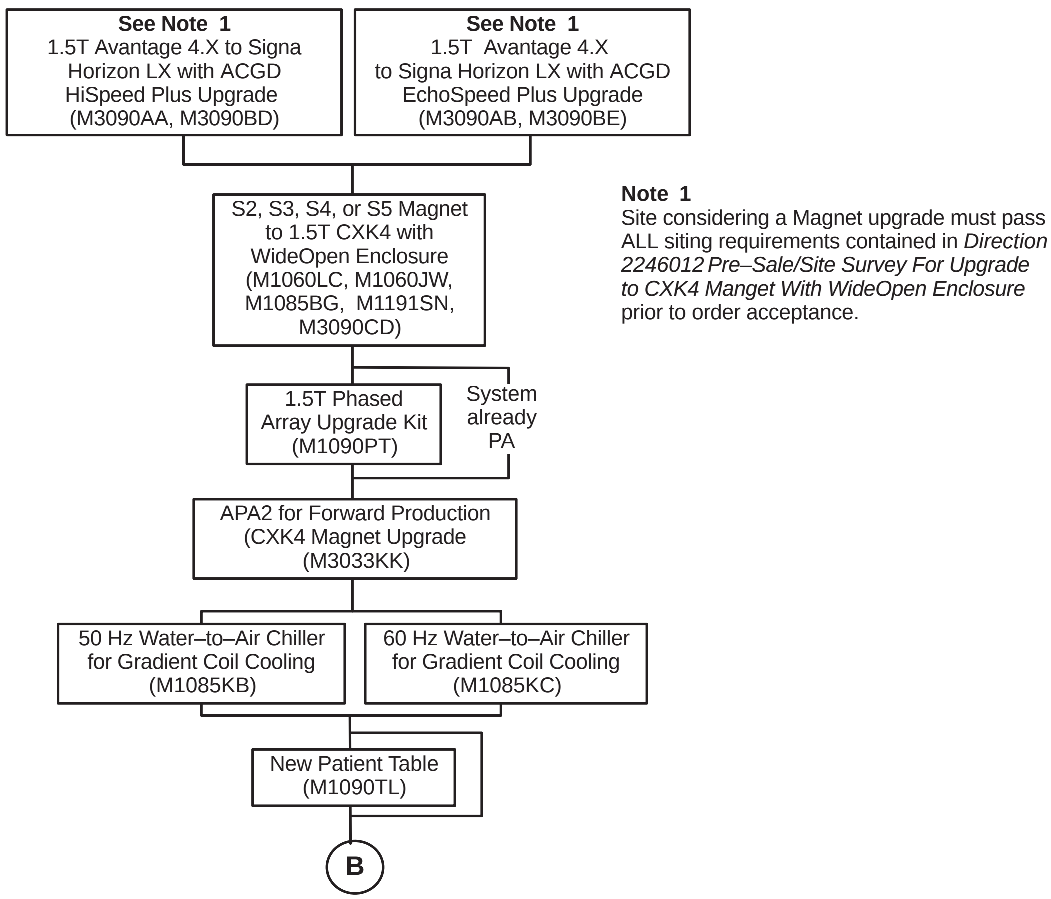
1–10–4 Upgrades from Signa Advantage 4.x System

Illustrations 1–26 through 1–27 lists the upgrade catalogs for Signa Advantage 4.X to Signa Horizon LX with ACGD. Signa Advantage 4.X hardware systems are Signa Advantage 4.5 to 4.8 which can be identified by RF Cabinet model number 46–306900 and Signa Advantage 4.0 to pre–4.5 which can be identified by RF Cabinet model number 46–282900. Illustrations 1–28 and 1–29 show the major equipment replaced or modified and new equipment added by the listed catalogs.

Refer to Section 1–10 SIGNA HORIZON LX SYSTEM OPTIONS for options catalogs.

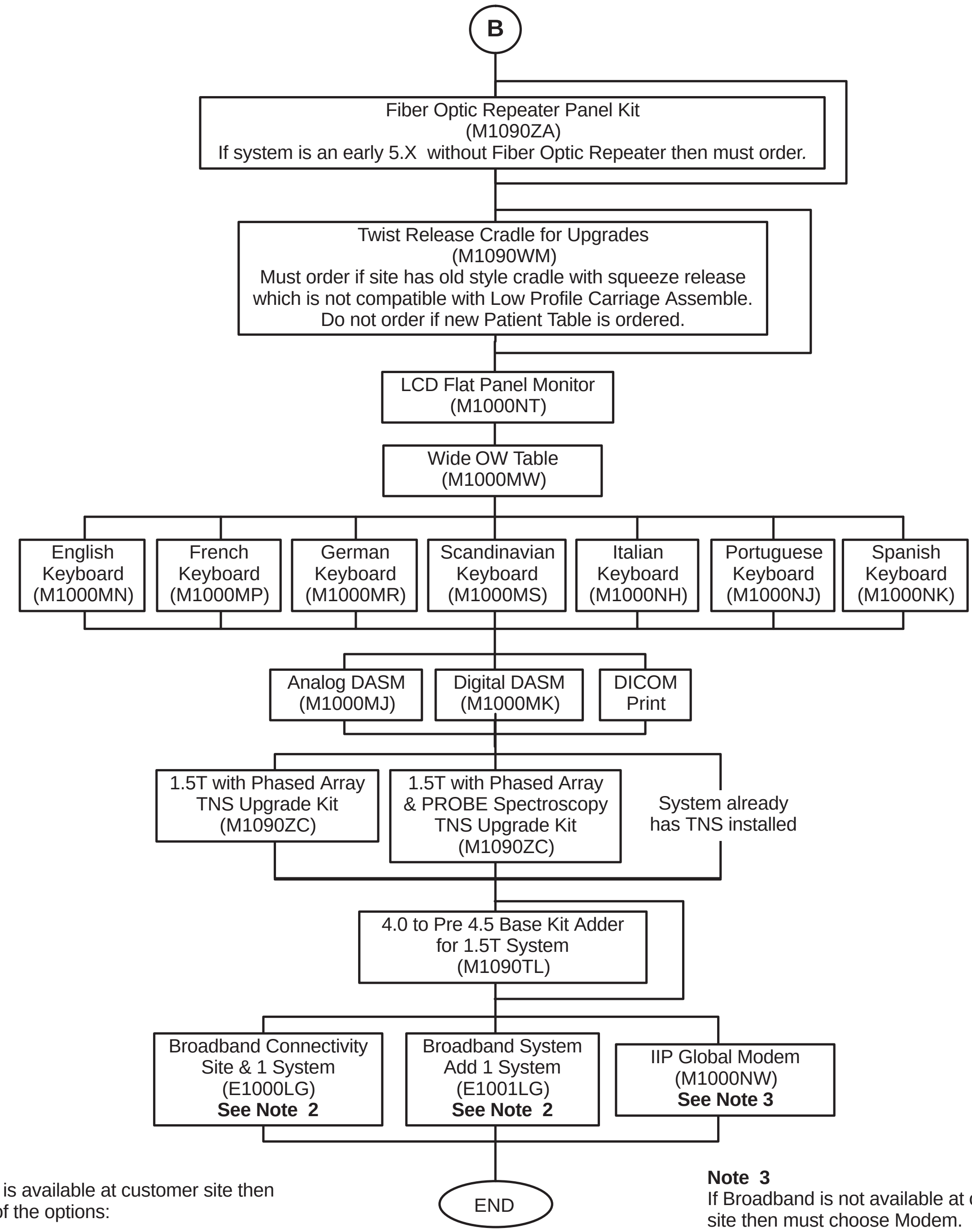
Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1 – 26

1–10–4 Upgrades from Signa Horizon 4.x System (Continued)



Note 2
If Broadband is available at customer site then choose one of the options:

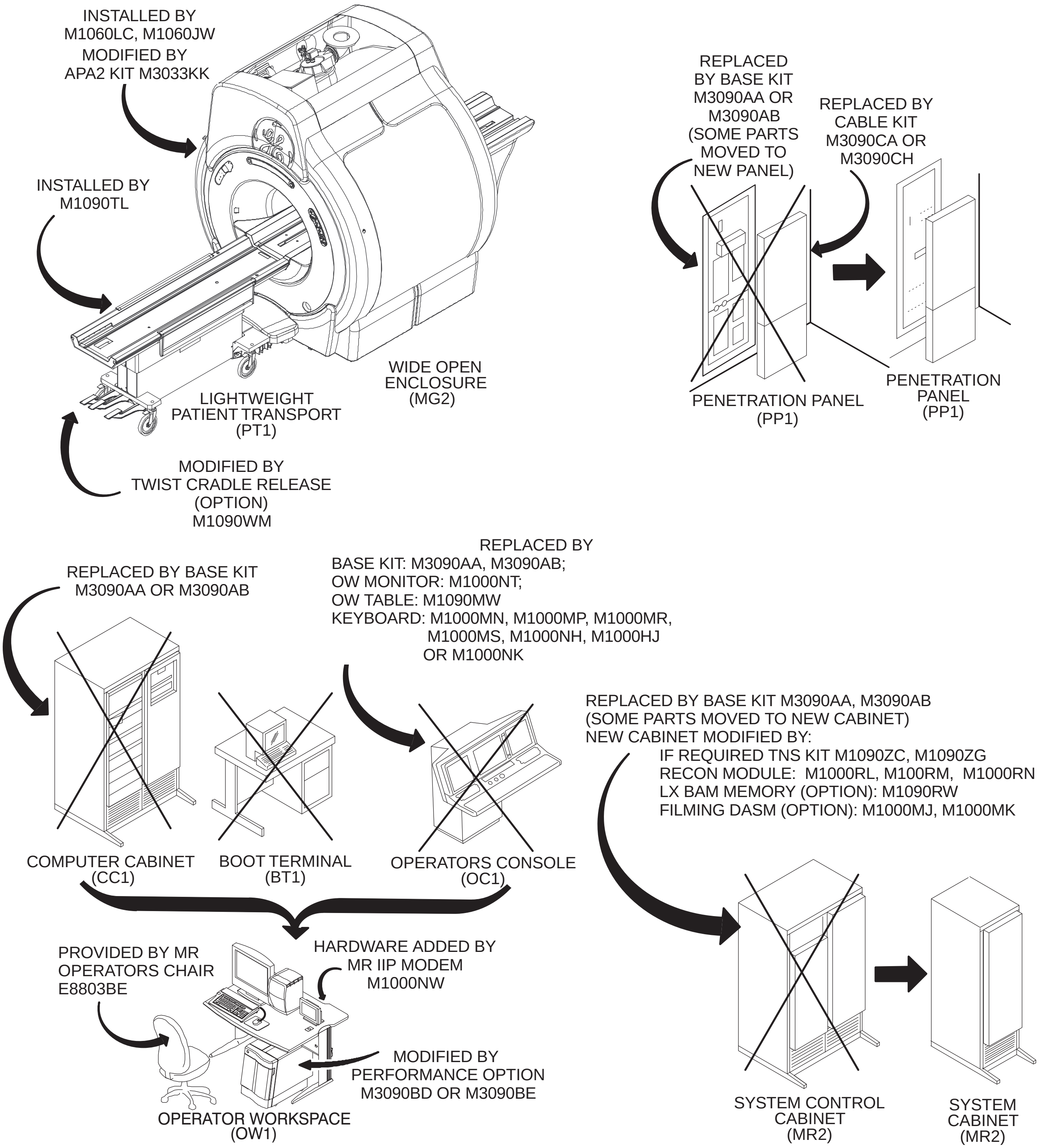
•Order E1000LG if customer site does not have an existing connection.

•Order E1001LG if customer site has previously

Note 3
If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1–27

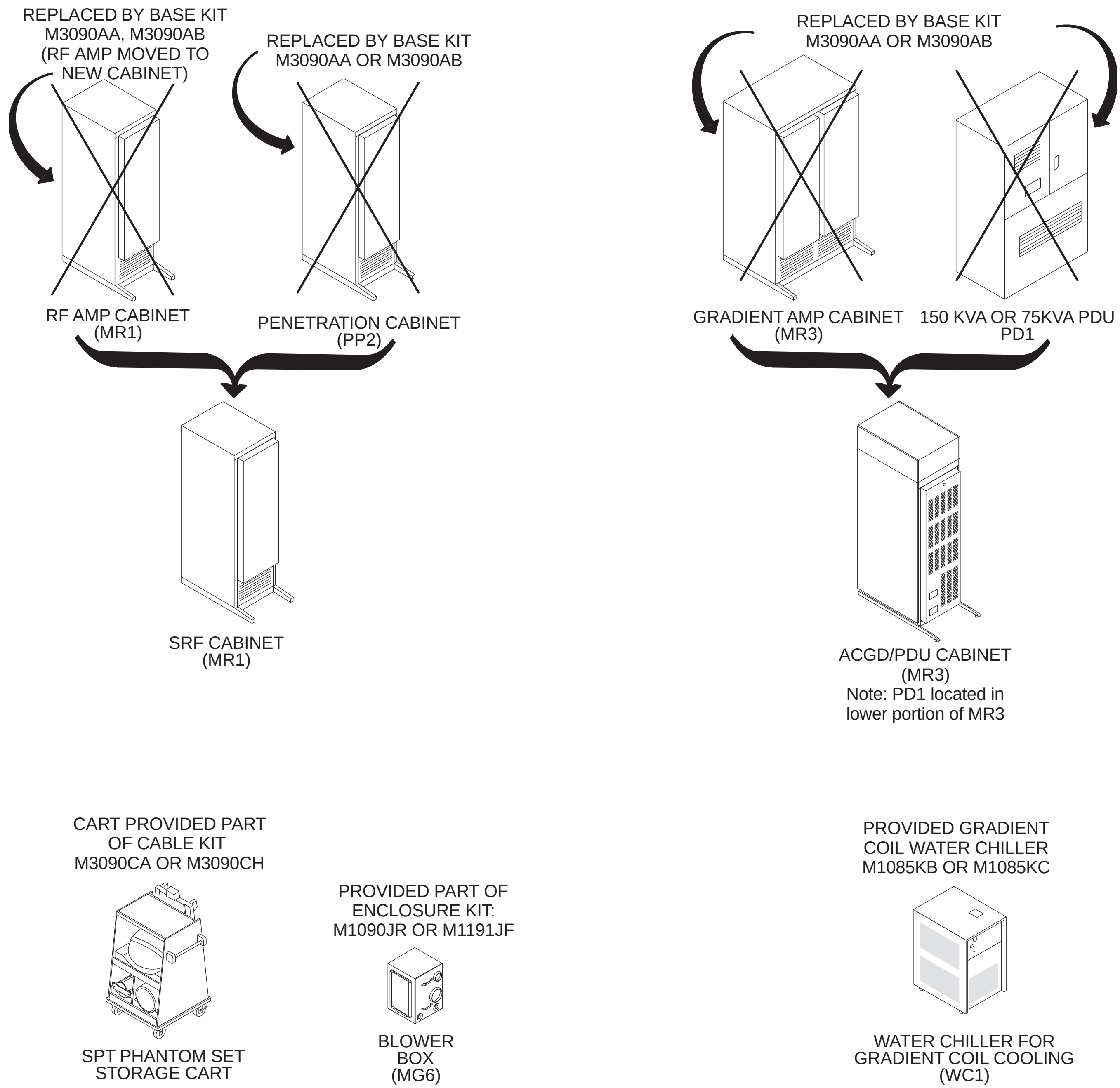
1–10–4 Upgrades from Signa Horizon 4.x System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1–28

1–10–4 Upgrades from Signa Horizon 4.x System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1–29

SECTION 2 – SYSTEM UPGRADE INSTALLATION PROCESS

2–1 INTRODUCTION

This section contains flow charts that will steer you through de–installation of existing systems and installation of material and equipment for the new upgrade.

The installation of the magnet and associated materials are provided in Direction 2226318. This instruction is refered to in the following flow charts.

Refer to the appropriate flow chart listed below for de–installation and installation of equipment room cabinets, operator room equipment, cables, and related magnet enclosure materials for the site being upgraded. They are as follows:

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT, CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

2–2 UPGRADE FLOWCHART EXPLANATION

Shown below are three examples of the flow chart Installation Procedures and an explanation of the areas within the procedures.

Use the CD–ROM that applies to the System being installed to locate the installation procedure listed below

CD–ROM
Operator Workspace:
Set Up/Calibration:
o ***PATIENT INTERCOM BIAS***
o ***INTERCOM ADJUSTMENTS***

Subsystem selection

Main menu selection

Installation Procedure to be performed

Use the correct Magnet Manual that applies to the System being installed to locate the installation procedure listed below

APPLICABLE
* MAGNET MANUAL
Set Up & Calibration Tab
SEC 1–4
o ***MRU INSTALLATION***

Tab within the manual where Installation Procedure is located

Section within Tab where Installation Procedure is located

Tab within this manual where the Direction is located

Installation Procedure to be performed

Specific steps or room of the above Installation Procedure to be performed at this time

TAB 1
DIRECTION 2286976
o ***ACGD GRADIENT OUTPUT CABLE INSTALLATION***
MAGNET ROOM

Direction within the Tab where the Installation Procedure listed below is located

LEGEND FOR FOLLOWING INSTALLATION FLOWCHART

MX — INDICATES MILESTONE COMPLETION GOAL IN DAYS

OR — INDICATES AND OPTIONAL PATH DEPENDING ON SYSTEM CONFIGURATION

INDICATES GE PROPRIETARY PROCEDURE

1 — Indicates an additional proprietary test or alternative proprietary calibration procedure available for GE use or to customers with an advanced service package limited license.

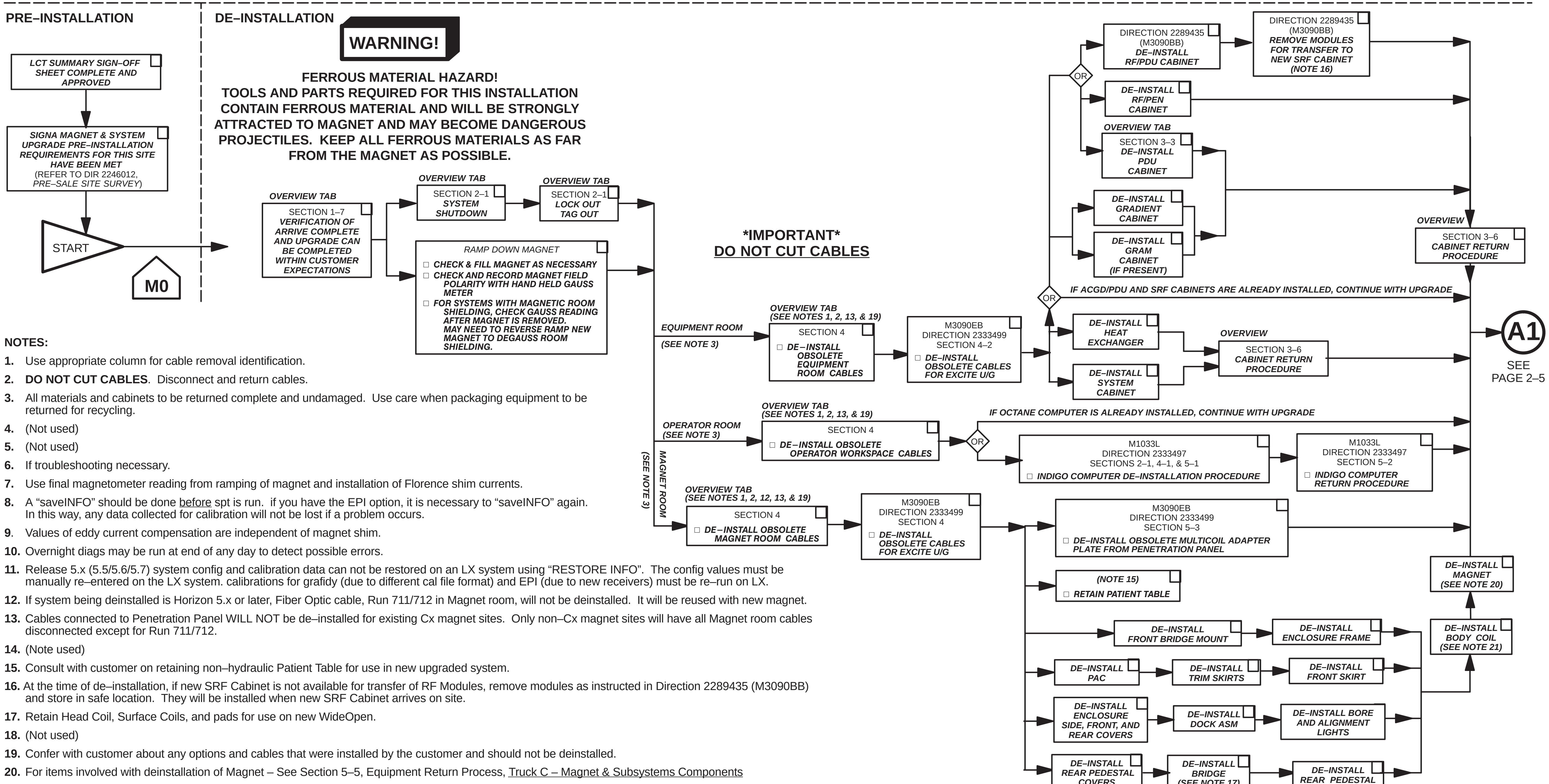
2 — As part of re–check and after quick shim (or LV SHIM). It may be necessary to repark the magnet.

SIGNA WideOpen 10.x (MGD) UPGRADE FROM:

- ***SIGNA HORIZON 8.x***

MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS



NOTES:

1. Use appropriate column for cable removal identification.
2. **DO NOT CUT CABLES**. Disconnect and return cables.
3. All materials and cabinets to be returned complete and undamaged. Use care when packaging equipment to be returned for recycling.
4. (Not used)
5. (Not used)
6. If troubleshooting necessary.
7. Use final magnetometer reading from ramping of magnet and installation of Florence shim currents.
8. A “saveINFO” should be done <u>before</u> spt is run. if you have the EPI option, it is necessary to “saveINFO” again. In this way, any data collected for calibration will not be lost if a problem occurs.
9. Values of eddy current compensation are independent of magnet shim.
10. Overnight diags may be run at end of any day to detect possible errors.
11. Release 5.x (5.5/5.6/5.7) system config and calibration data can not be restored on an LX system using “RESTORE INFO”. The config values must be manually re–entered on the LX system. calibrations for grafidy (due to different cal file format) and EPI (due to new receivers) must be re–run on LX.
12. If system being deinstalled is Horizon 5.x or later, Fiber Optic cable, Run 711/712 in Magnet room, will not be deinstalled. It will be reused with new magnet.
13. Cables connected to Penetration Panel WILL NOT be de–installed for existing Cx magnet sites. Only non–Cx magnet sites will have all Magnet room cables disconnected except for Run 711/712.
14. (Note used)
15. Consult with customer on retaining non–hydraulic Patient Table for use in new upgraded system.
16. At the time of de–installation, if new SRF Cabinet is not available for transfer of RF Modules, remove modules as instructed in Direction 2289435 (M3090BB) and store in safe location. They will be installed when new SRF Cabinet arrives on site.
17. Retain Head Coil, Surface Coils, and pads for use on new WideOpen.
18. (Not used)
19. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
20. For items involved with deinstallation of Magnet – See Section 5–5, Equipment Return Process, <u>Truck C – Magnet & Subsystems Components</u>
21. <u>If Cx magnet, leave Body Coil in bore of the magnet. Do not deinstall.</u>

SIGNA HORIZON 8.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–1 (SHEET 1 OF 3)

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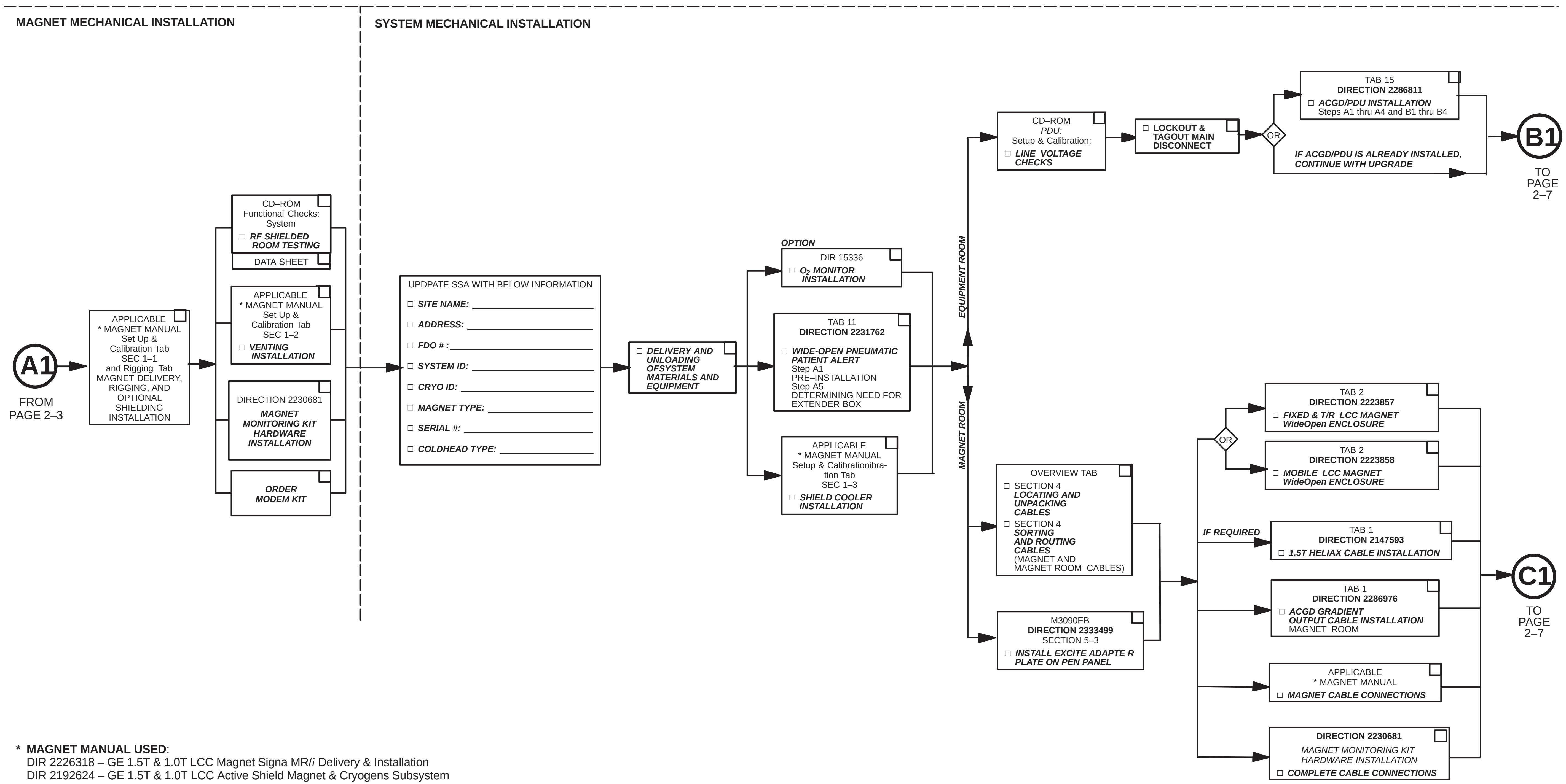
MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS



* **MAGNET MANUAL USED**:
DIR 2226318 – GE 1.5T & 1.0T LCC Magnet Signa MR/*i* Delivery & Installation
DIR 2192624 – GE 1.5T & 1.0T LCC Active Shield Magnet & Cryogens Subsystem

SIGNA HORIZON 8.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–1 (SHEET 2 OF 3)

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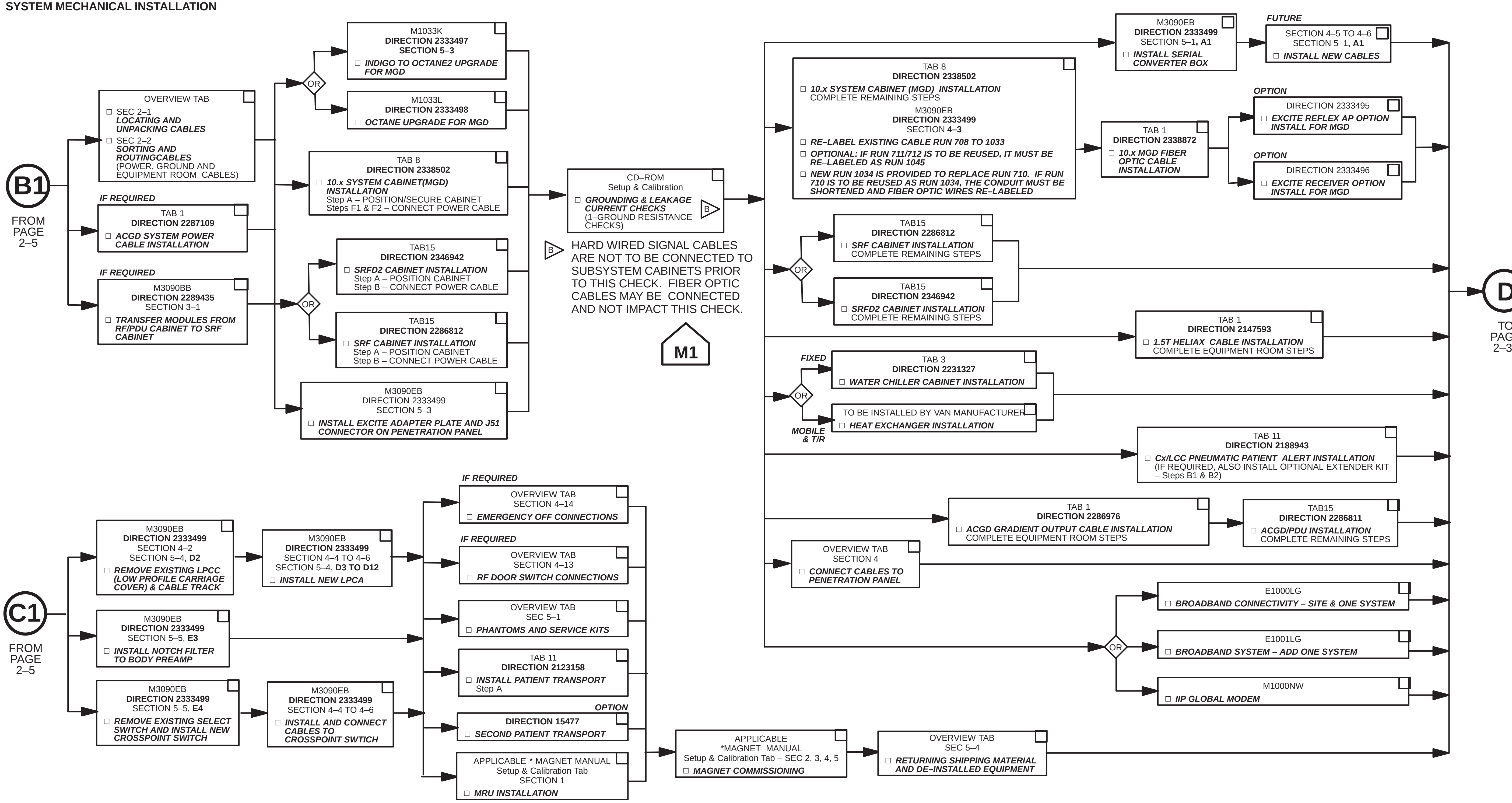
MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM MECHANICAL INSTALLATION



SIGNA HORIZON 8.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–1 (SHEET 3 OF 3)

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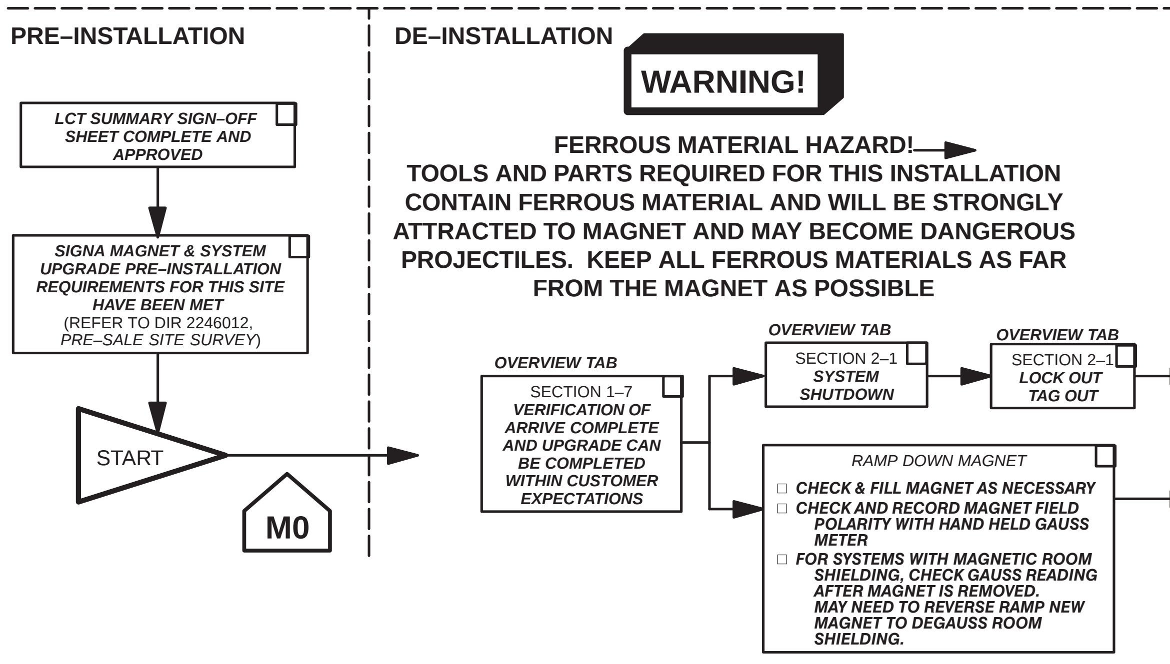
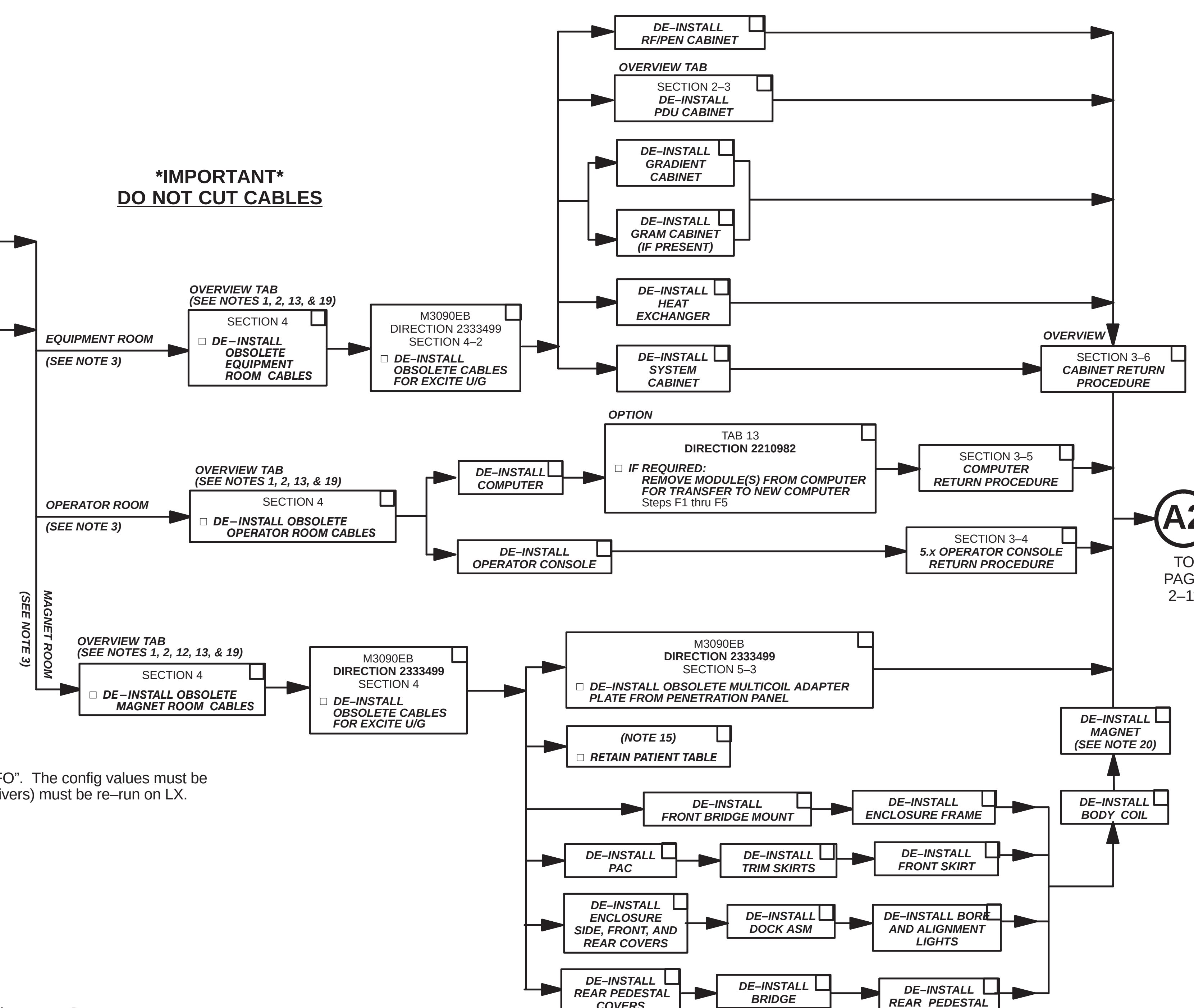
SIGNA WideOpen 10.x (MGD) UPGRADE FROM:
● ***SIGNA HORIZON 5.x***

MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS





NOTES:

1. Use appropriate column for cable removal identification.
2. **DO NOT CUT CABLES**. Disconnect and return cables.
3. All materials and cabinets to be returned complete and undamaged. Use care when packaging equipment to be returned for recycling.
4. (Not used)
5. (Not used)
6. If troubleshooting necessary.
7. Use final magnetometer reading from ramping of magnet and installation of Florence shim currents.
8. A "saveINFO" should be done before spt is run. if you have the EPI option, it is necessary to "saveINFO" again. In this way, any data collected for calibration will not be lost if a problem occurs.
9. Values of eddy current compensation are independent of magnet shim.
10. Overnight diags may be run at end of any day to detect possible errors.
11. Release 5.x (5.5/5.6/5.7) system config and calibration data can not be restored on an LX system using "RESTORE INFO". The config values must be manually re–entered on the LX system. calibrations for grafidy (due to different cal file format) and EPI (due to new receivers) must be re–run on LX.
12. Fiber Optic cable, Run 711/712 in Magnet room, will not be deinstalled. It will be reused with new magnet.
13. (Not used)
14. (Not used)
15. Consult with customer on retaining non–hydraulic Patient Table for use in new upgraded system.
16. (Not used)
17. Retain Head Coil, Surface Coils, and pads for use on new WideOpen.
18. (Not used)
19. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
20. For items involved with deinstallation of Magnet – See Section 5–5, Equipment Return Process, Truck C – Magnet & Subsystems Components

SIGNA HORIZON 5.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–2 (SHEET 1 OF 3)

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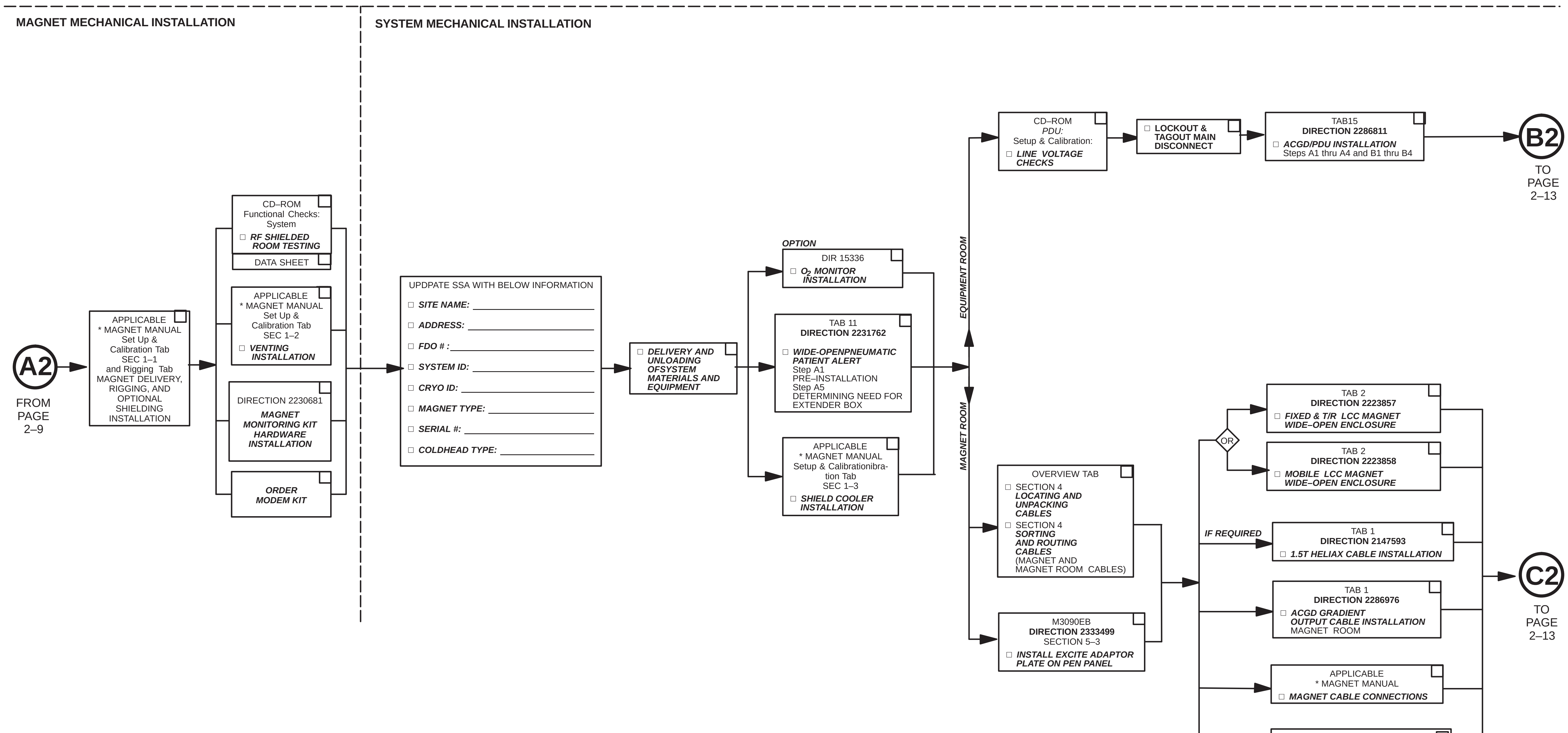
MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS



*** MAGNET MANUAL USED**:
DIR 2226318 – GE 1.5T & 1.0T LCC Magnet Signa MR/*i* Delivery & Installation
DIR 2192624 – GE 1.5T & 1.0T LCC Active Shield Magnet & Cryogens Subsystem

SIGNA HORIZON 5.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–2 (SHEET 2 OF 3)

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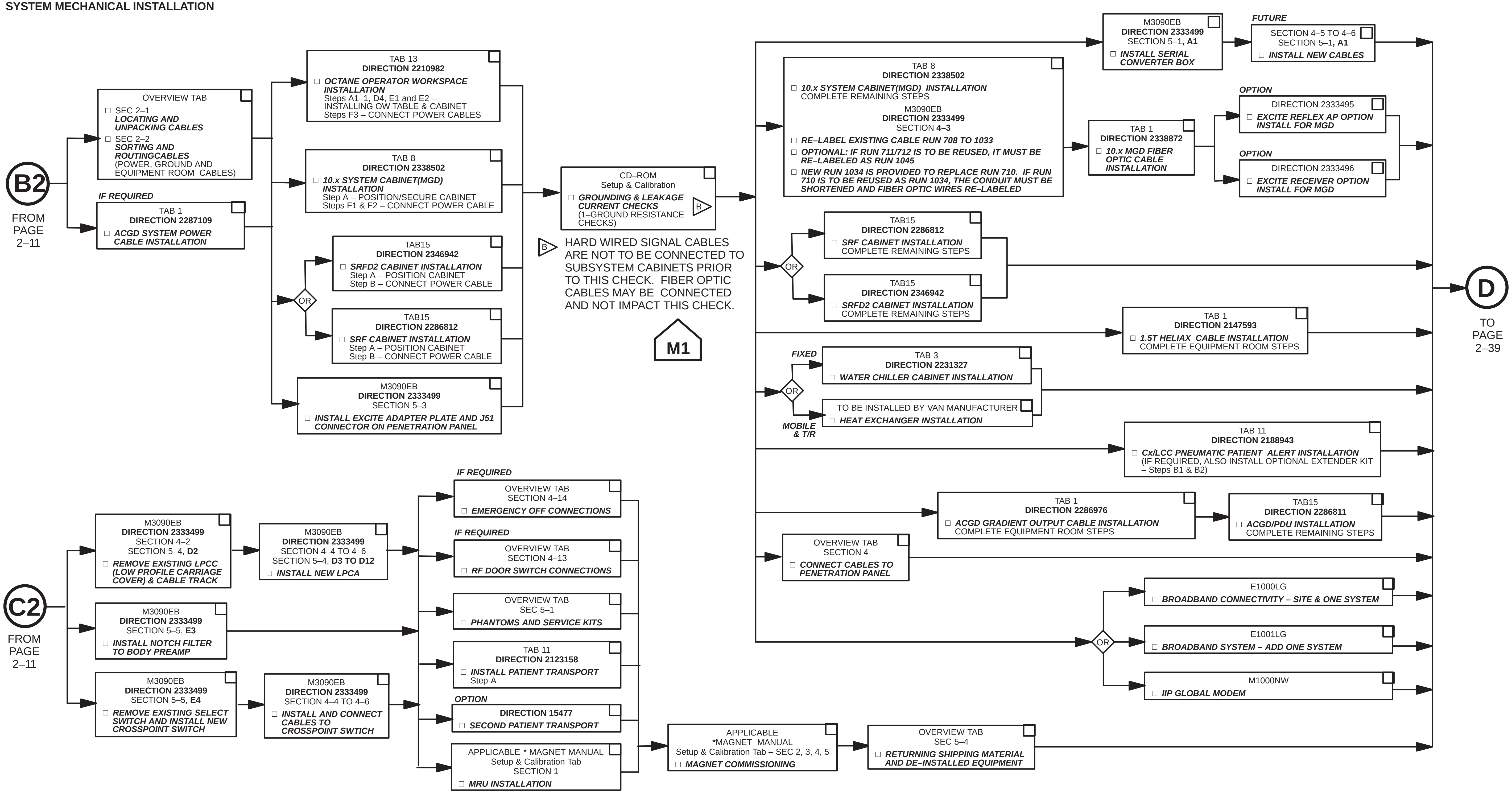
MARK OFF PERFORMED ITEMS

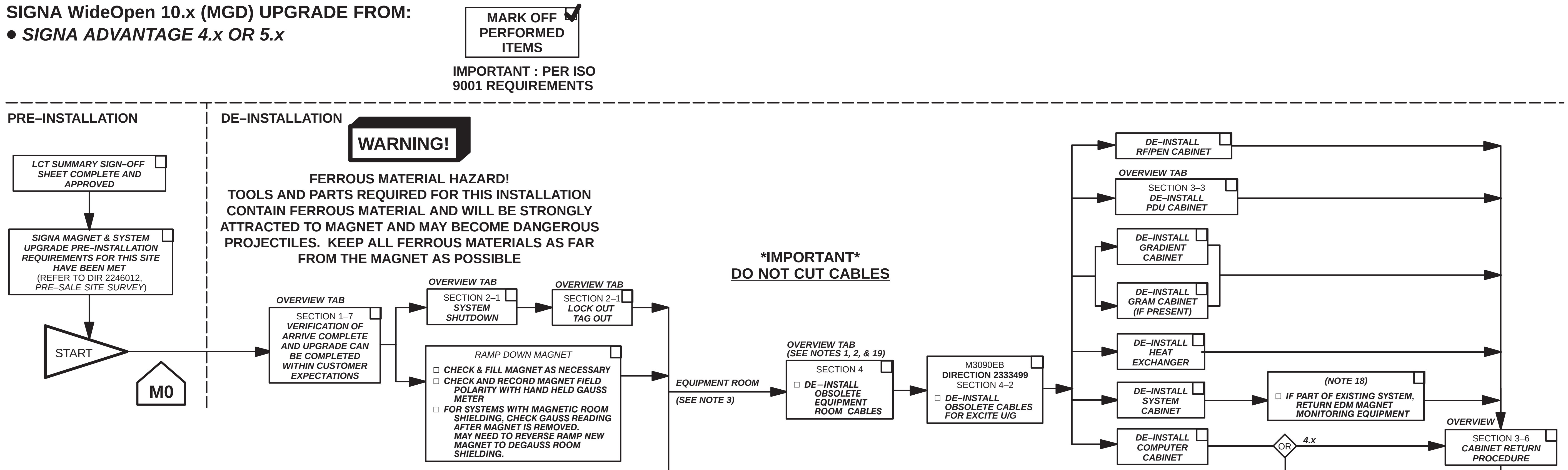
IMPORTANT: PER ISO 9001 REQUIREMENTS

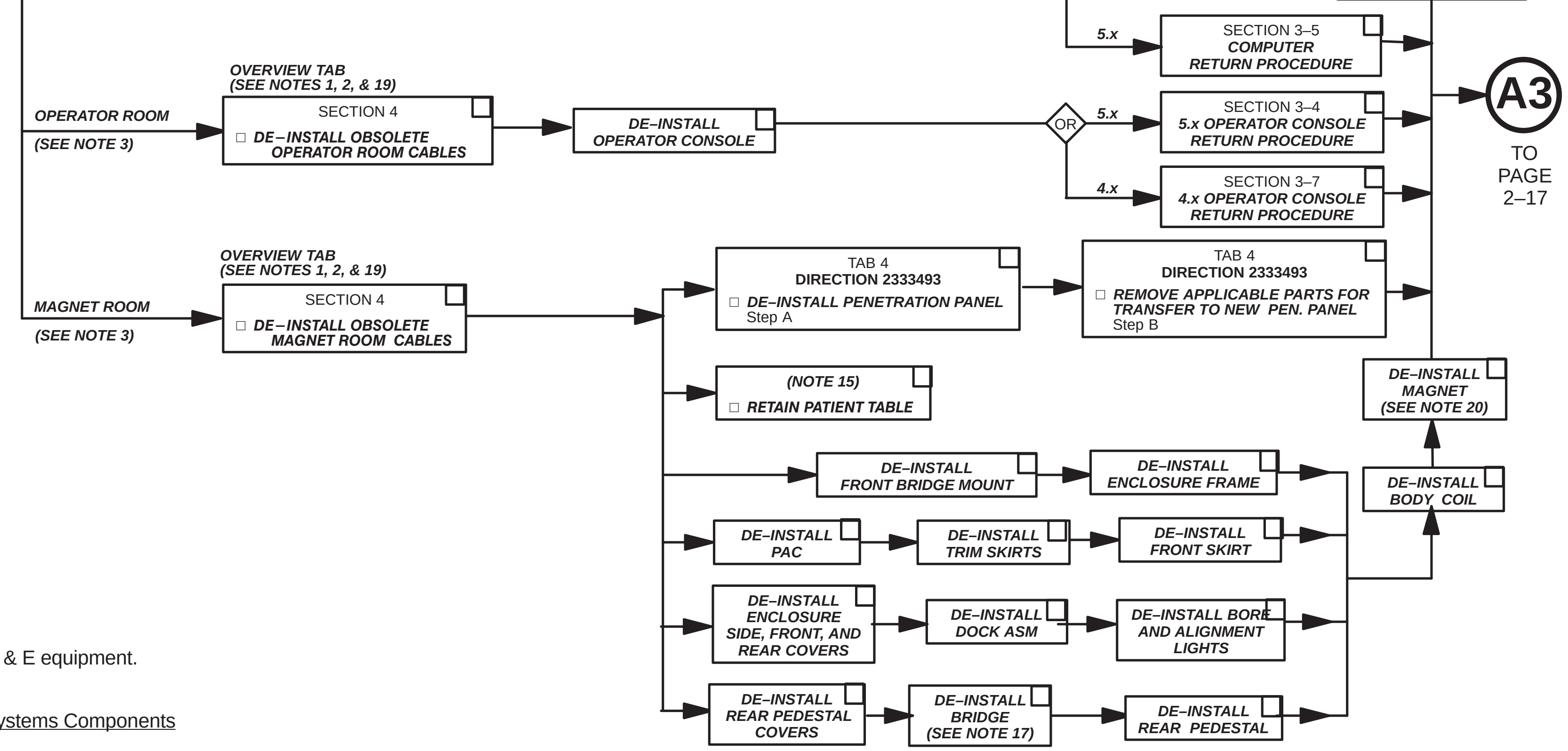
SYSTEM MECHANICAL INSTALLATION



SIGNA HORIZON 5.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–2 (SHEET 3 OF 3)

SIGNA WideOpen 10.x (MGD) UPGRADE FROM:
- ***SIGNA ADVANTAGE 4.x OR 5.x***





NOTES:

1. Use appropriate column for cable removal identification.
2. **DO NOT CUT CABLES**. Disconnect and return cables.
3. All materials and cabinets to be returned complete and undamaged. Use care when packaging equipment to be returned for recycling.
4. (Not used)
5. (Not used)
6. If troubleshooting necessary.
7. Use final magnetometer reading from ramping of magnet and installation of Florence shim currents.
8. A "saveINFO" should be done before spt is run. if you have the epi option, it is necessary to "saveINFO" again. In this way, any data collected for calibration will not be lost if a problem occurs.
9. Values of eddy current compensation are independent of magnet shim.
10. Overnight diags may be run at end of any day to detect possible errors.
11. (Not used)
12. (Not used)
13. (Not used)
14. (Not used)
15. Consult with customer on retaining non–hydraulic Patient Table for use in new upgraded system.
16. (Not used)
17. Retain Head Coil, Surface Coils and pads for use on new WideOpen.
18. If your system has EDM Magnet Monitoring, contact OnLine Magnet Support at 1–800–321–7937 for details on returning P & E equipment.
19. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
20. For items involved with deinstallation of Magnet – See Section 5–5, Equipment Return Process, Truck C – Magnet & Subsystems Components

SIGNA ADVANTAGE 4.x OR 5.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–3 (SHEET 1 OF 3)

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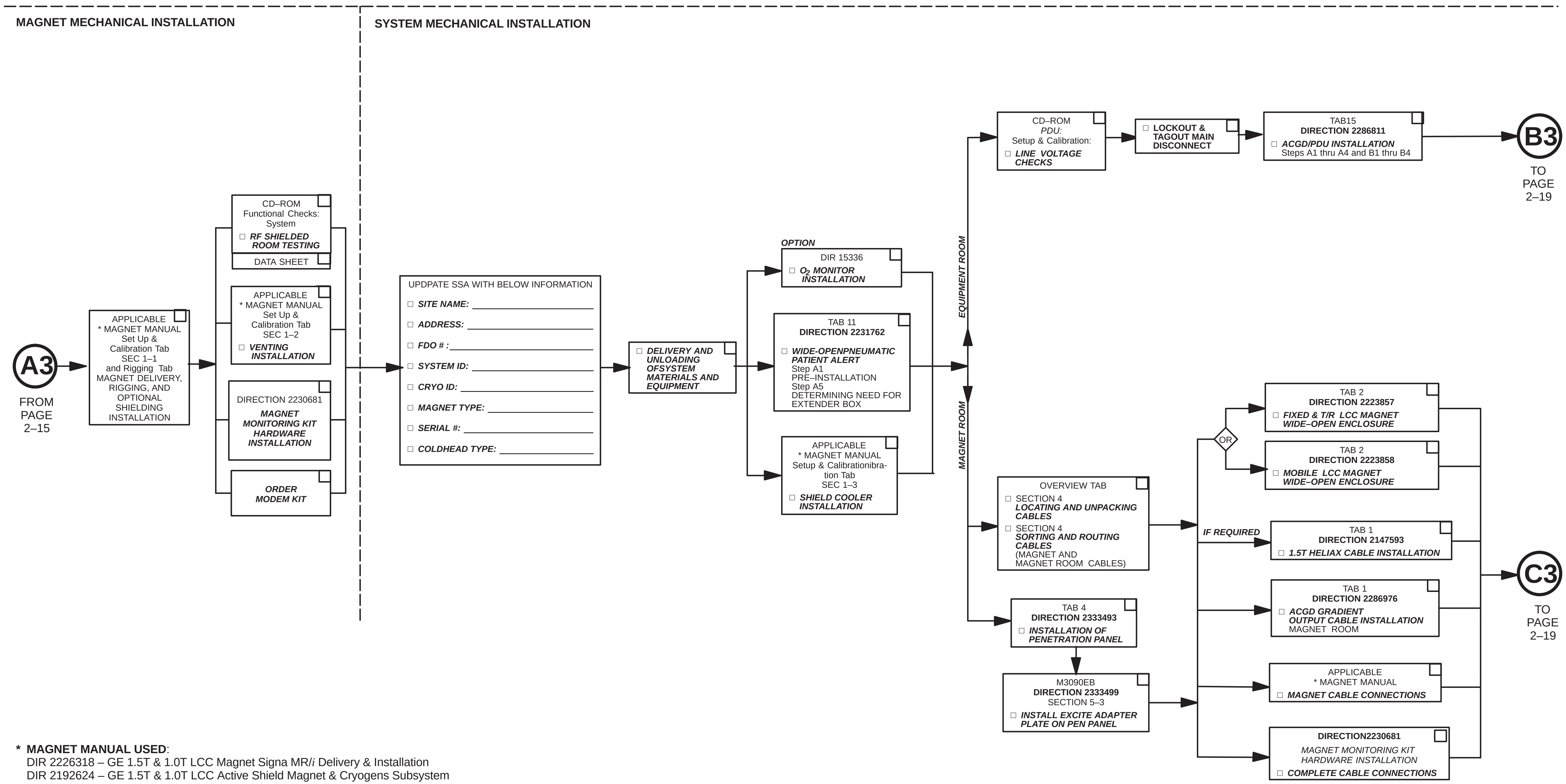
IMPORTANT: PER ISO 9001 REQUIREMENTS



* **MAGNET MANUAL USED**:
DIR 2226318 – GE 1.5T & 1.0T LCC Magnet Signa MR/*i* Delivery & Installation
DIR 2192624 – GE 1.5T & 1.0T LCC Active Shield Magnet & Cryogens Subsystem

SIGNA ADVANTAGE 4.x OR 5.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–3 (SHEET 2 OF 3)

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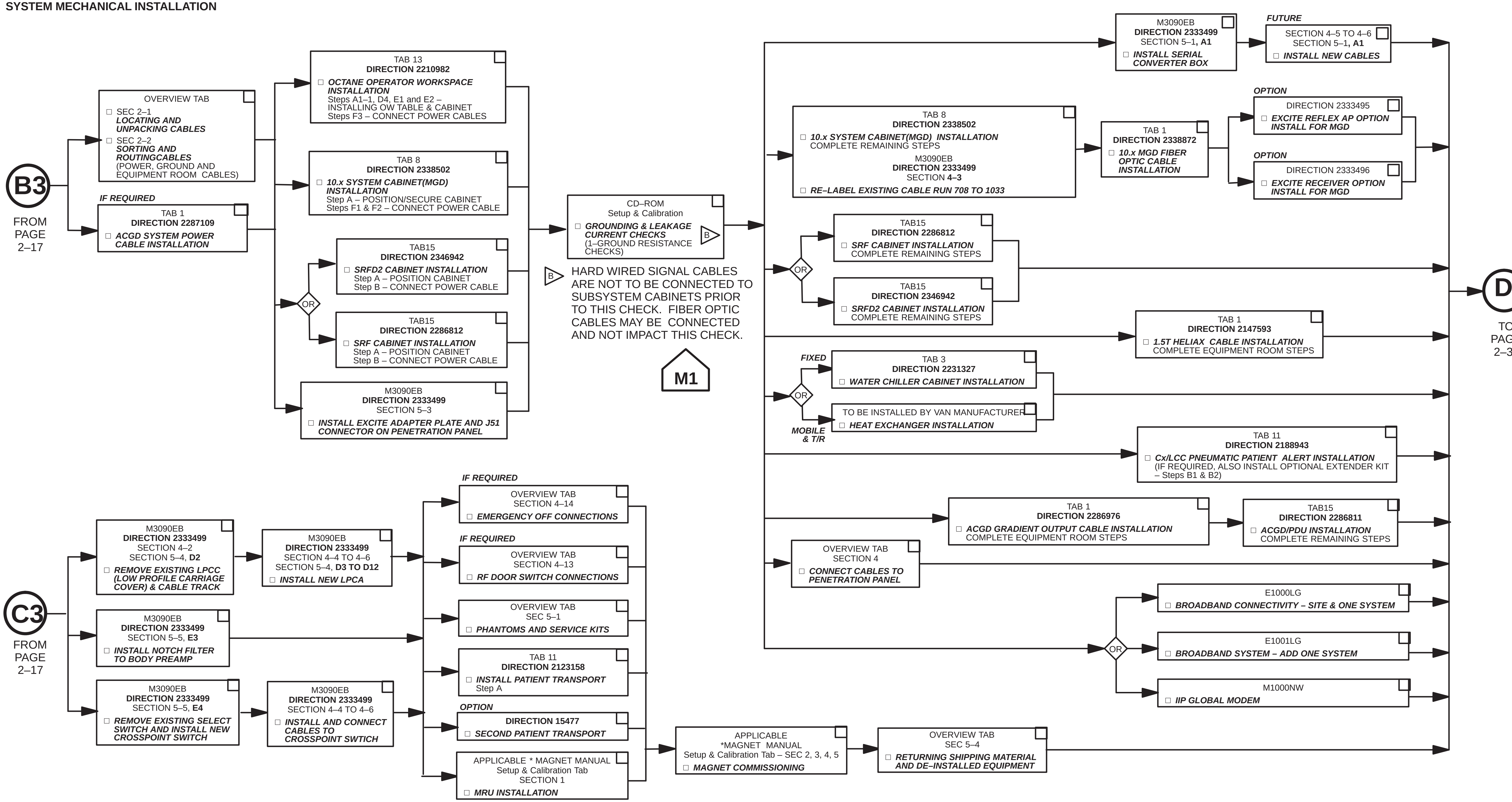
MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM MECHANICAL INSTALLATION



SIGNA ADVANTAGE 4.x OR 5.x to SIGNA WideOpen 10.x (MGD) UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–3 (SHEET 3 OF 3)

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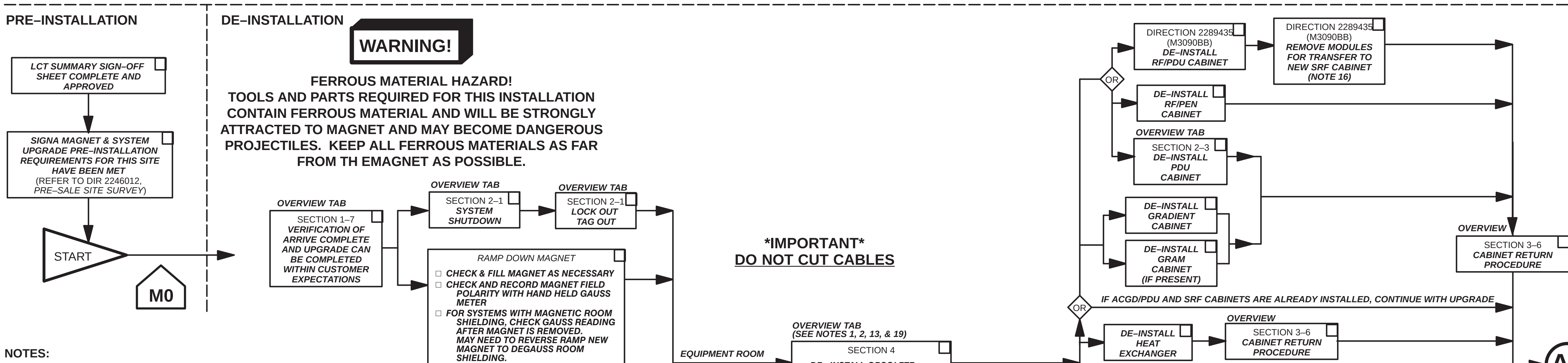
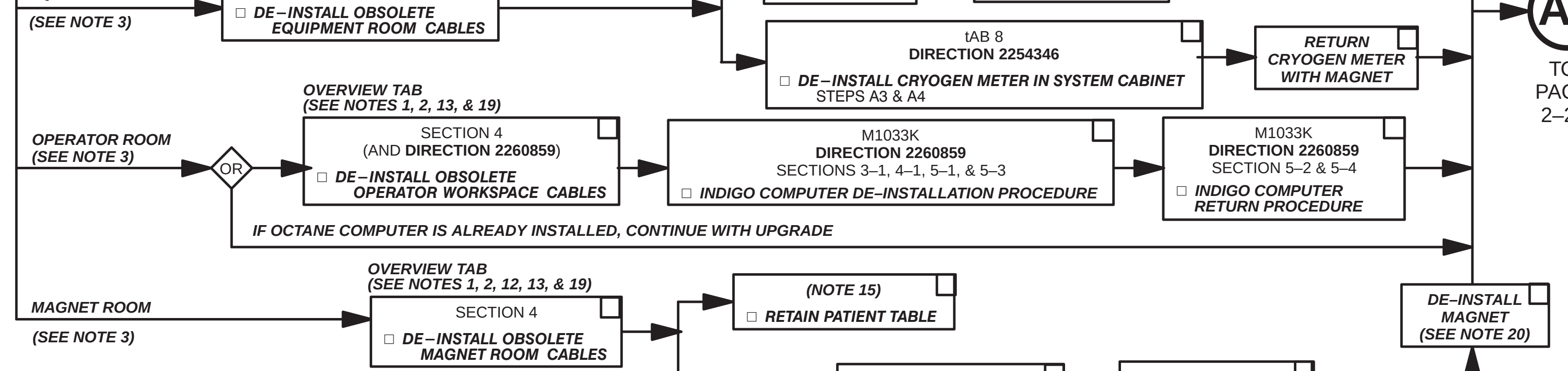
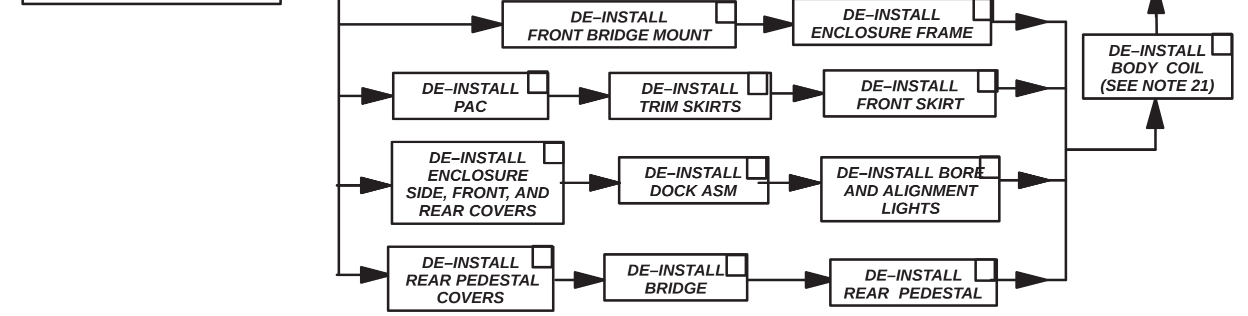
SIGNA WideOpen 9.x UPGRADE FROM:
- ***SIGNA HORIZON 8.x***

MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS







NOTES:

1. Use appropriate column for cable removal identification.
2. **DO NOT CUT CABLES**. Disconnect and return cables.
3. All materials and cabinets to be returned complete and undamaged. Use care when packaging equipment to be returned for recycling.
4. (Not used)
5. (Not used)
6. If troubleshooting necessary.
7. Use final magnetometer reading from ramping of magnet and installation of Florence shim currents.
8. A "saveINFO" should be done before spt is run. if you have the EPI option, it is necessary to "saveINFO" again. In this way, any data collected for calibration will not be lost if a problem occurs.
9. Values of eddy current compensation are independent of magnet shim.
10. Overnight diags may be run at end of any day to detect possible errors.
11. Release 5.x (5.5/5.6/5.7) system config and calibration data can not be restored on an LX system using "RESTORE INFO". The config values must be manually re–entered on the LX system. calibrations for grafidy (due to different cal file format) and EPI (due to new receivers) must be re–run on LX.
12. If system being deinstalled is Horizon 5.x or later, Fiber Optic cable, Run 711/712 in Magnet room, will not be deinstalled. It will be reused with new magnet.
13. Cables connected to Penetration Panel WILL NOT be de–installed for existing Cx magnet sites. Only non–Cx magnet sites will have all Magnet room cables disconnected except for Run 711/712.
14. (Note used)
15. Consult with customer on retaining non–hydraulic Patient Table for use in new upgraded system.
16. At the time of de–installation, if new SRF Cabinet is not available for transfer of RF Modules, remove modules as instructed in Direction 2289435 (M3090BB) and store in safe location. They will be installed when new SRF Cabinet arrives on site.
17. Retain Head Coil, Surface Coils, and pads for use on new WideOpen.
18. (Not used)
19. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
20. For items involved with deinstallation of Magnet – See Section 5–5, Equipment Return Process, Truck C – Magnet & Subsystems Components
21. If Cx magnet, leave Body Coil in bore of the magnet. Do not deinstall.

SIGNA HORIZON 8.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–4 (SHEET 1 OF 3)

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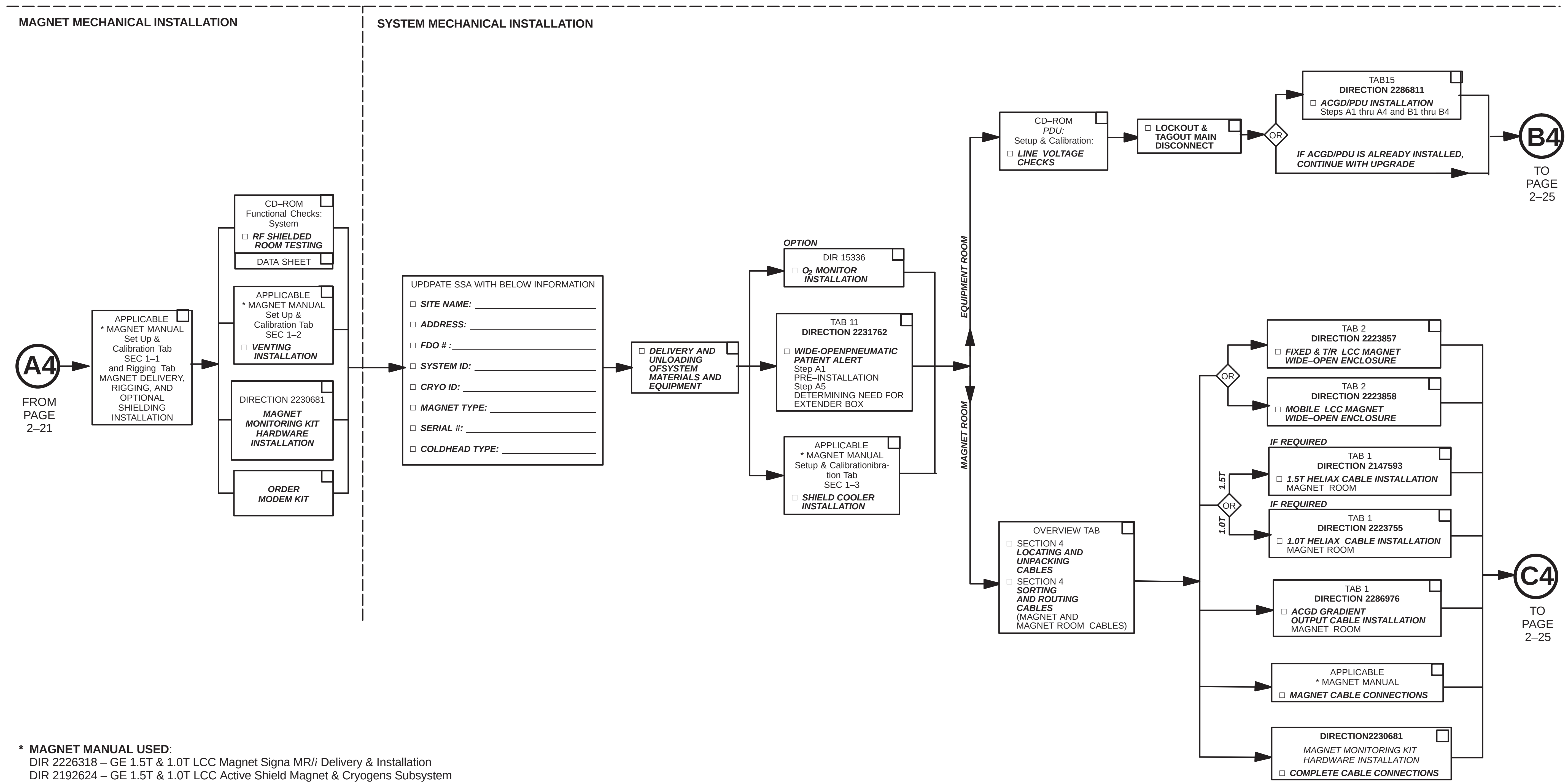
IMPORTANT: PER ISO 9001 REQUIREMENTS



* **MAGNET MANUAL USED**:
DIR 2226318 – GE 1.5T & 1.0T LCC Magnet Signa MR/*i* Delivery & Installation
DIR 2192624 – GE 1.5T & 1.0T LCC Active Shield Magnet & Cryogens Subsystem

SIGNA HORIZON 8.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–4 (SHEET 2 OF 3)

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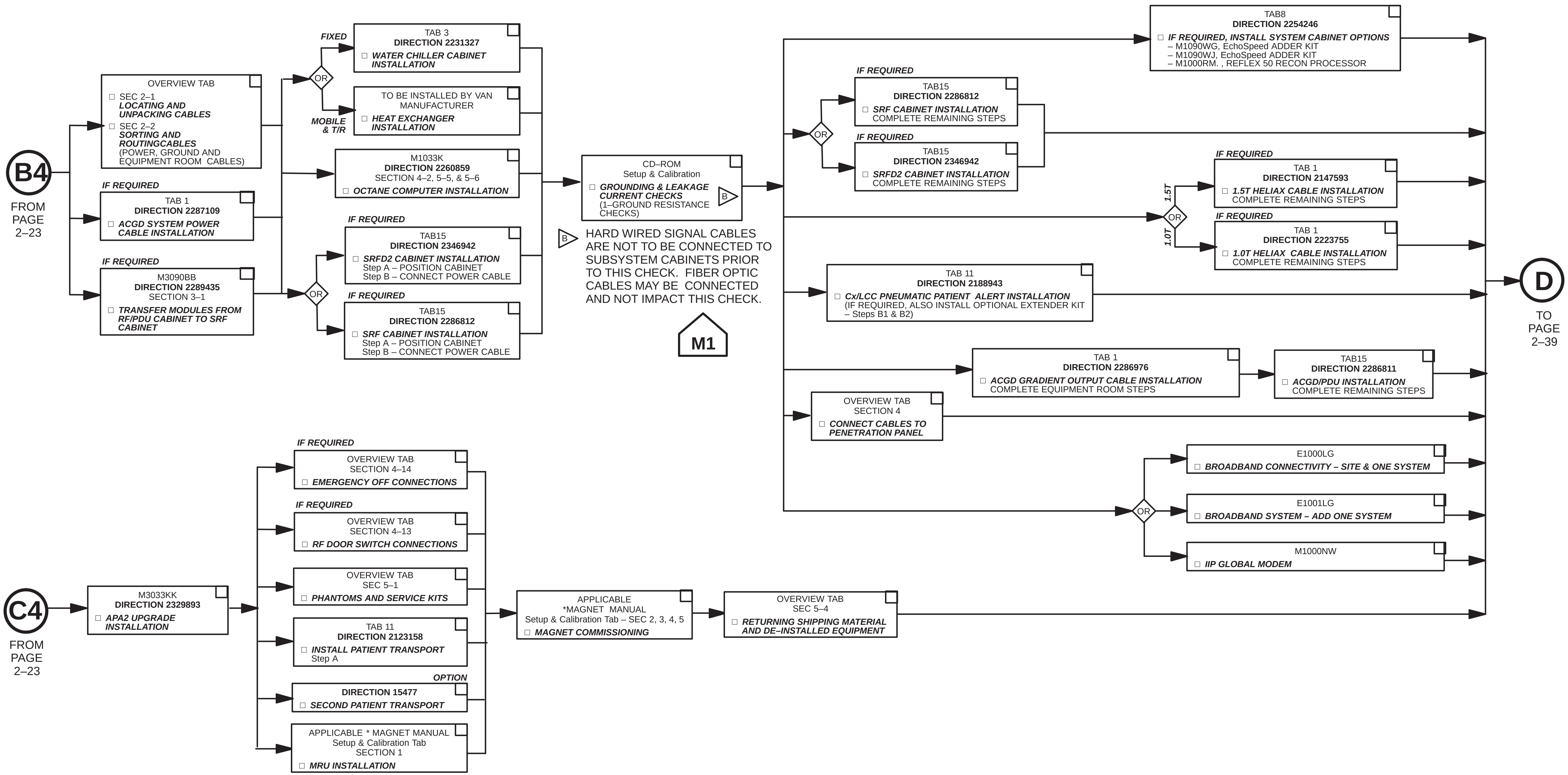
MARK OFF PERFORMED ITEMS

IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM MECHANICAL INSTALLATION



SIGNA HORIZON 8.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–4 (SHEET 3 OF 3)

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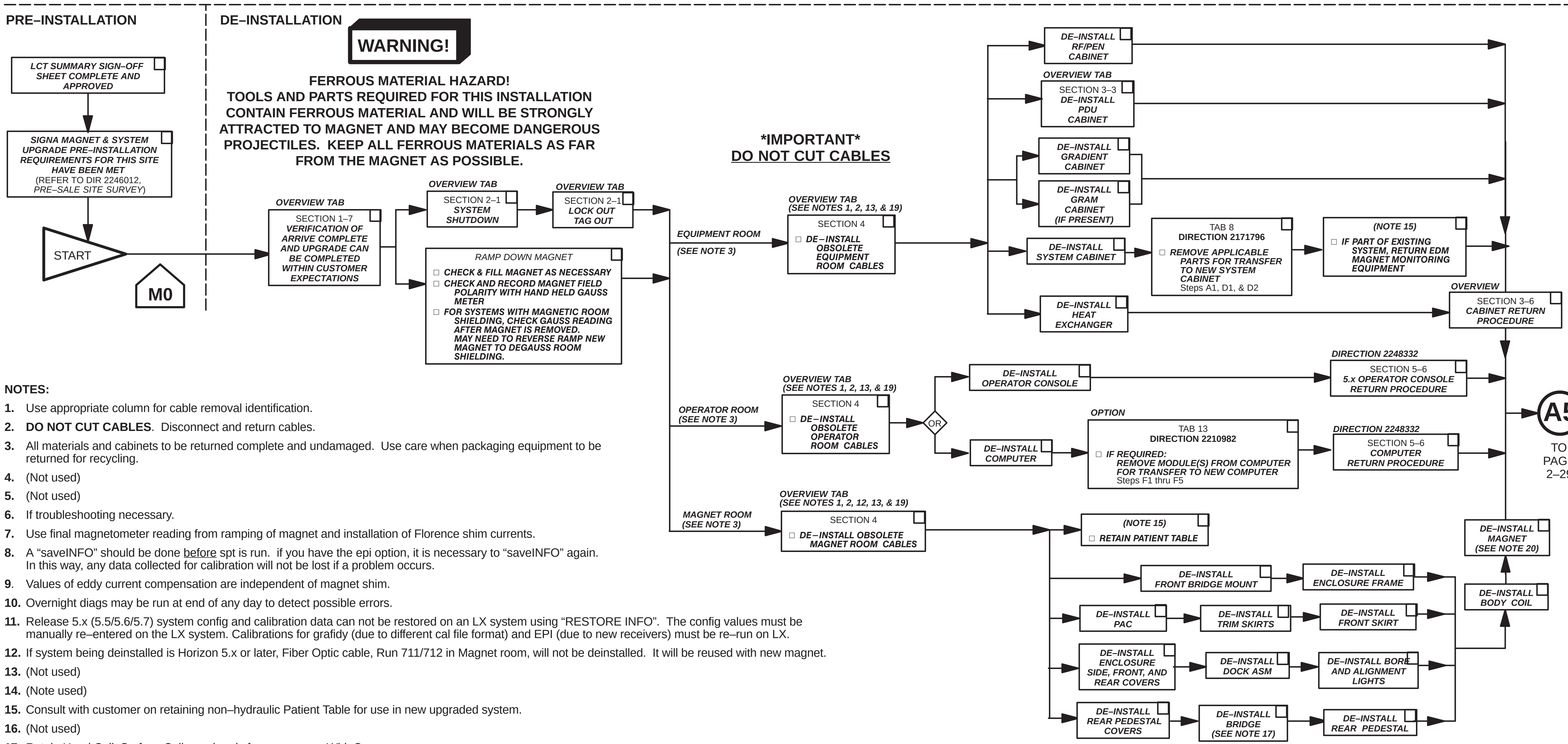
SIGNA WideOpen 9.x UPGRADE FROM:

- ***SIGNA HORIZON 5.x***

MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS



NOTES:

1. Use appropriate column for cable removal identification.
2. **DO NOT CUT CABLES**. Disconnect and return cables.
3. All materials and cabinets to be returned complete and undamaged. Use care when packaging equipment to be returned for recycling.
4. (Not used)
5. (Not used)
6. If troubleshooting necessary.
7. Use final magnetometer reading from ramping of magnet and installation of Florence shim currents.
8. A "saveINFO" should be done before spt is run. if you have the epi option, it is necessary to "saveINFO" again. In this way, any data collected for calibration will not be lost if a problem occurs.
9. Values of eddy current compensation are independent of magnet shim.
10. Overnight diags may be run at end of any day to detect possible errors.
11. Release 5.x (5.5/5.6/5.7) system config and calibration data can not be restored on an LX system using "RESTORE INFO". The config values must be manually re–entered on the LX system. Calibrations for grafidy (due to different cal file format) and EPI (due to new receivers) must be re–run on LX.
12. If system being deinstalled is Horizon 5.x or later, Fiber Optic cable, Run 711/712 in Magnet room, will not be deinstalled. It will be reused with new magnet.
13. (Not used)
14. (Note used)
15. Consult with customer on retaining non–hydraulic Patient Table for use in new upgraded system.
16. (Not used)
17. Retain Head Coil, Surface Coils, and pads for use on new WideOpen.
18. If your system has EDM Magnet Monitoring, contact OnLine Magnet Support at 1–800–321–7937 for details on returning P & E equipment.
19. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
20. For items involved with deinstallation of Magnet – See Section 5–5, Equipment Return Process, Truck C – Magnet & Subsystems Components

SIGNA HORIZON 5.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–5 (SHEET 1 OF 3)

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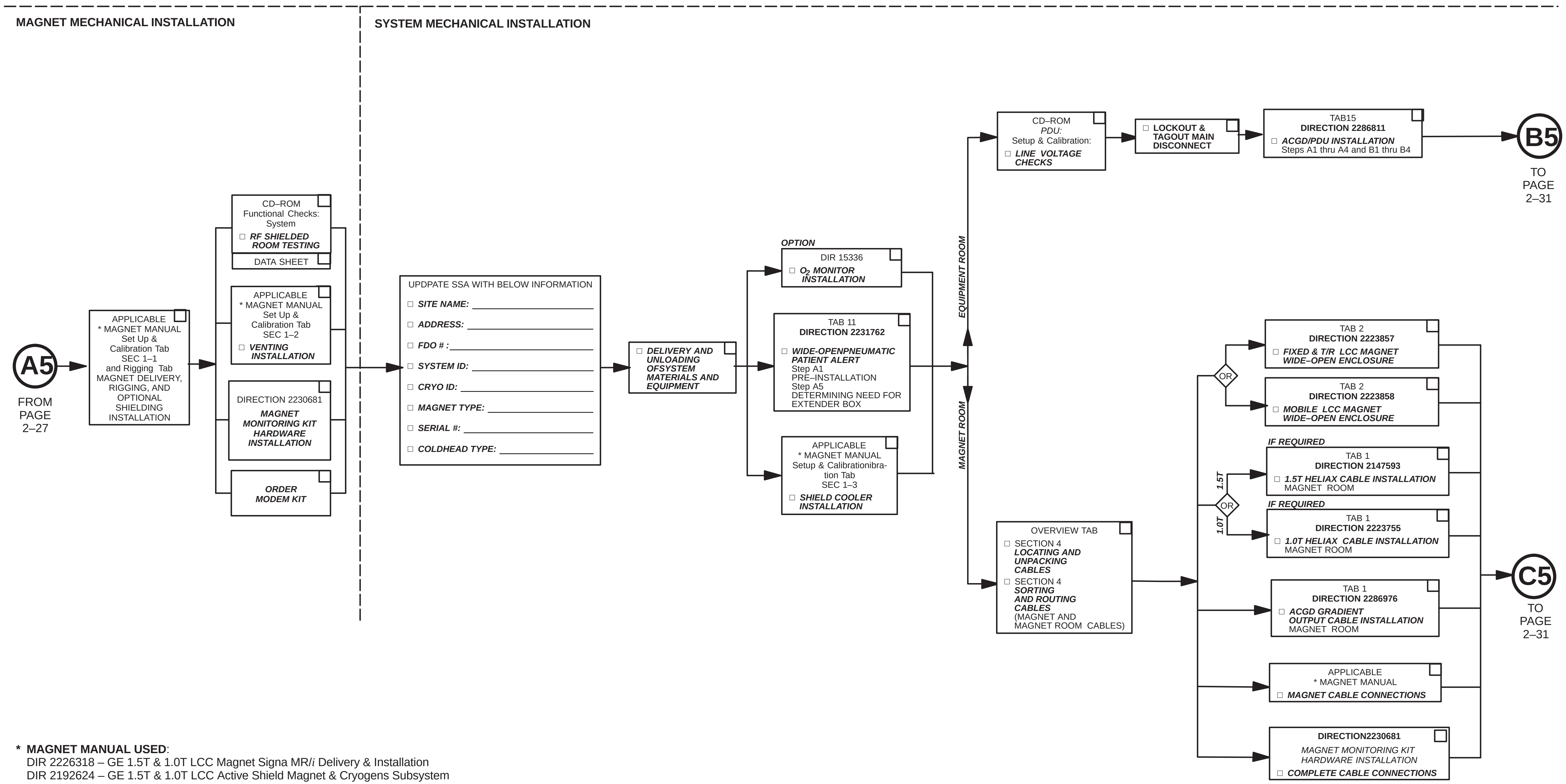
IMPORTANT: PER ISO 9001 REQUIREMENTS



* **MAGNET MANUAL USED**:
DIR 2226318 – GE 1.5T & 1.0T LCC Magnet Signa MR/*i* Delivery & Installation
DIR 2192624 – GE 1.5T & 1.0T LCC Active Shield Magnet & Cryogens Subsystem

SIGNA HORIZON 5.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–5 (SHEET 2 OF 3)

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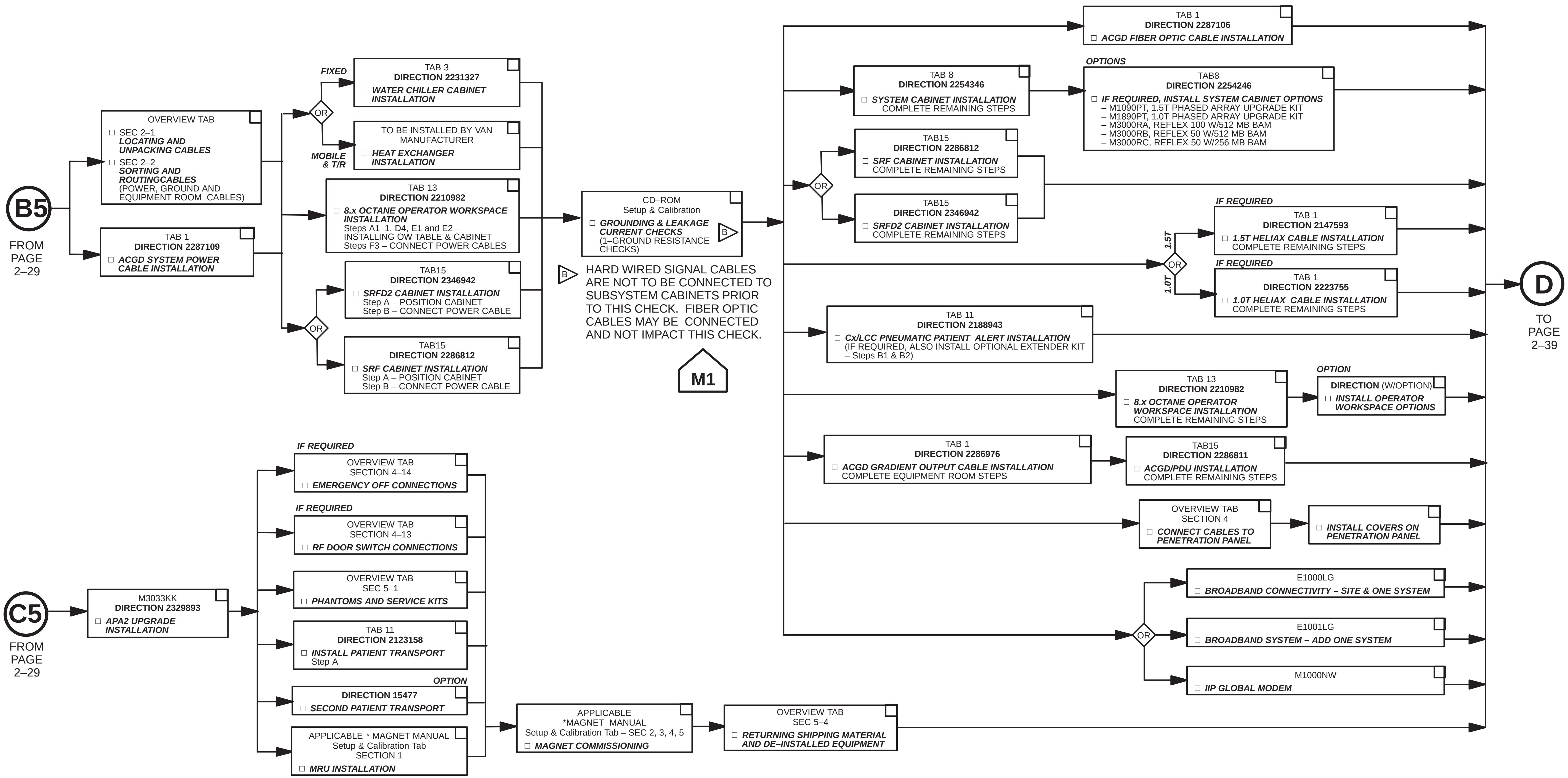
IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM MECHANICAL INSTALLATION



SIGNA HORIZON 5.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–5 (SHEET 3 OF 3)

SIGNA WideOpen 9.x UPGRADE FROM:

● ***SIGNA ADVANTAGE 4.x OR 5.x***

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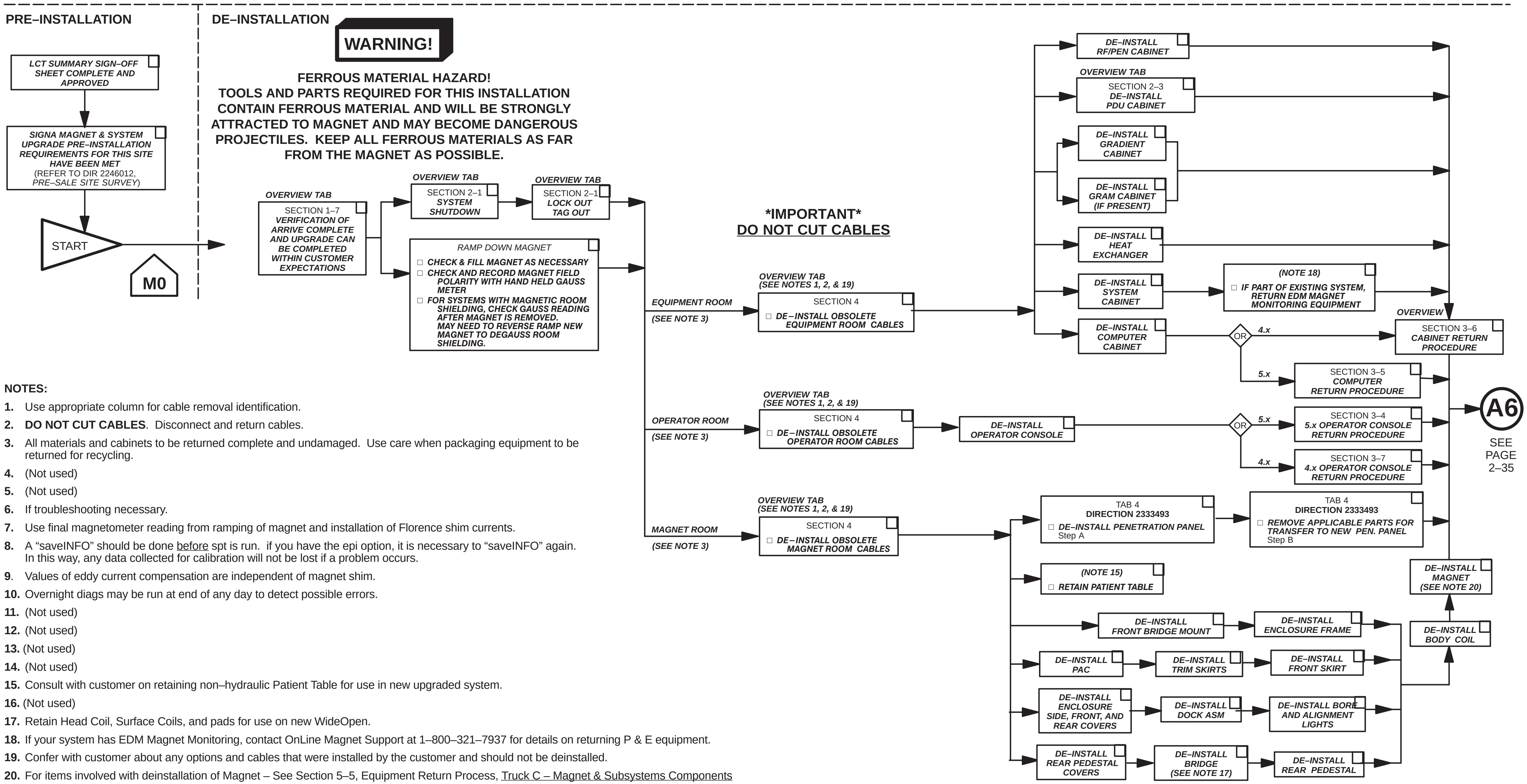
IMPORTANT : PER ISO 9001 REQUIREMENTS



NOTES:

1. Use appropriate column for cable removal identification.
2. **DO NOT CUT CABLES**. Disconnect and return cables.
3. All materials and cabinets to be returned complete and undamaged. Use care when packaging equipment to be returned for recycling.
4. (Not used)
5. (Not used)
6. If troubleshooting necessary.
7. Use final magnetometer reading from ramping of magnet and installation of Florence shim currents.
8. A “saveINFO” should be done before spt is run. if you have the epi option, it is necessary to “saveINFO” again. In this way, any data collected for calibration will not be lost if a problem occurs.
9. Values of eddy current compensation are independent of magnet shim.
10. Overnight diags may be run at end of any day to detect possible errors.
11. (Not used)
12. (Not used)
13. (Not used)
14. (Not used)
15. Consult with customer on retaining non–hydraulic Patient Table for use in new upgraded system.
16. (Not used)
17. Retain Head Coil, Surface Coils, and pads for use on new WideOpen.
18. If your system has EDM Magnet Monitoring, contact OnLine Magnet Support at 1–800–321–7937 for details on returning P & E equipment.
19. Confer with customer about any options and cables that were installed by the customer and should not be deinstalled.
20. For items involved with deinstallation of Magnet – See Section 5–5, Equipment Return Process, Truck C – Magnet & Subsystems Components

SIGNA ADVANTAGE 4.x OR 5.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–6 (SHEET 1 OF 3)

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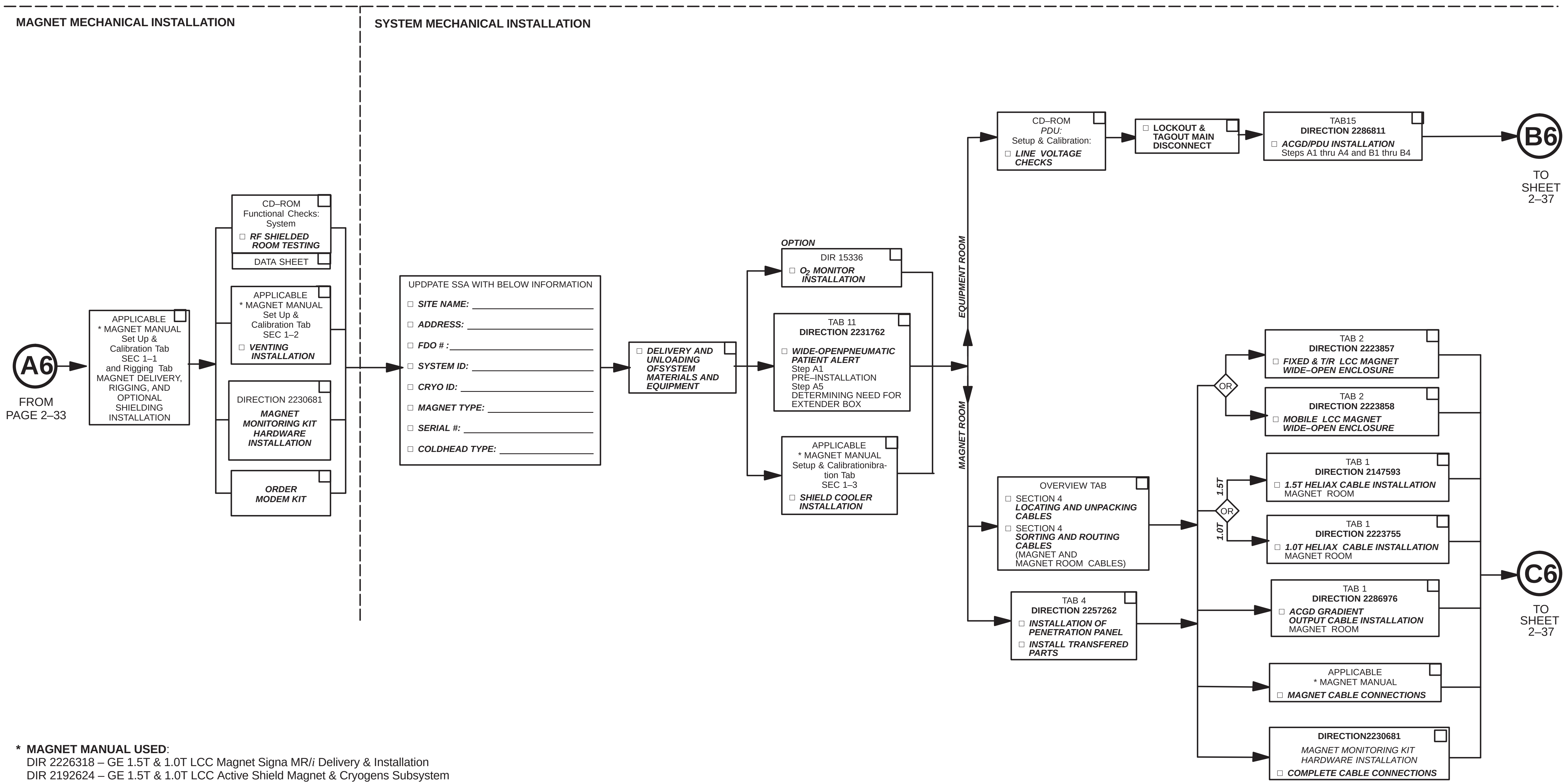
IMPORTANT: PER ISO 9001 REQUIREMENTS



* **MAGNET MANUAL USED**:
DIR 2226318 – GE 1.5T & 1.0T LCC Magnet Signa MR/*i* Delivery & Installation
DIR 2192624 – GE 1.5T & 1.0T LCC Active Shield Magnet & Cryogens Subsystem

SIGNA ADVANTAGE 4.x OR 5.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–6 (SHEET 2 OF 3)

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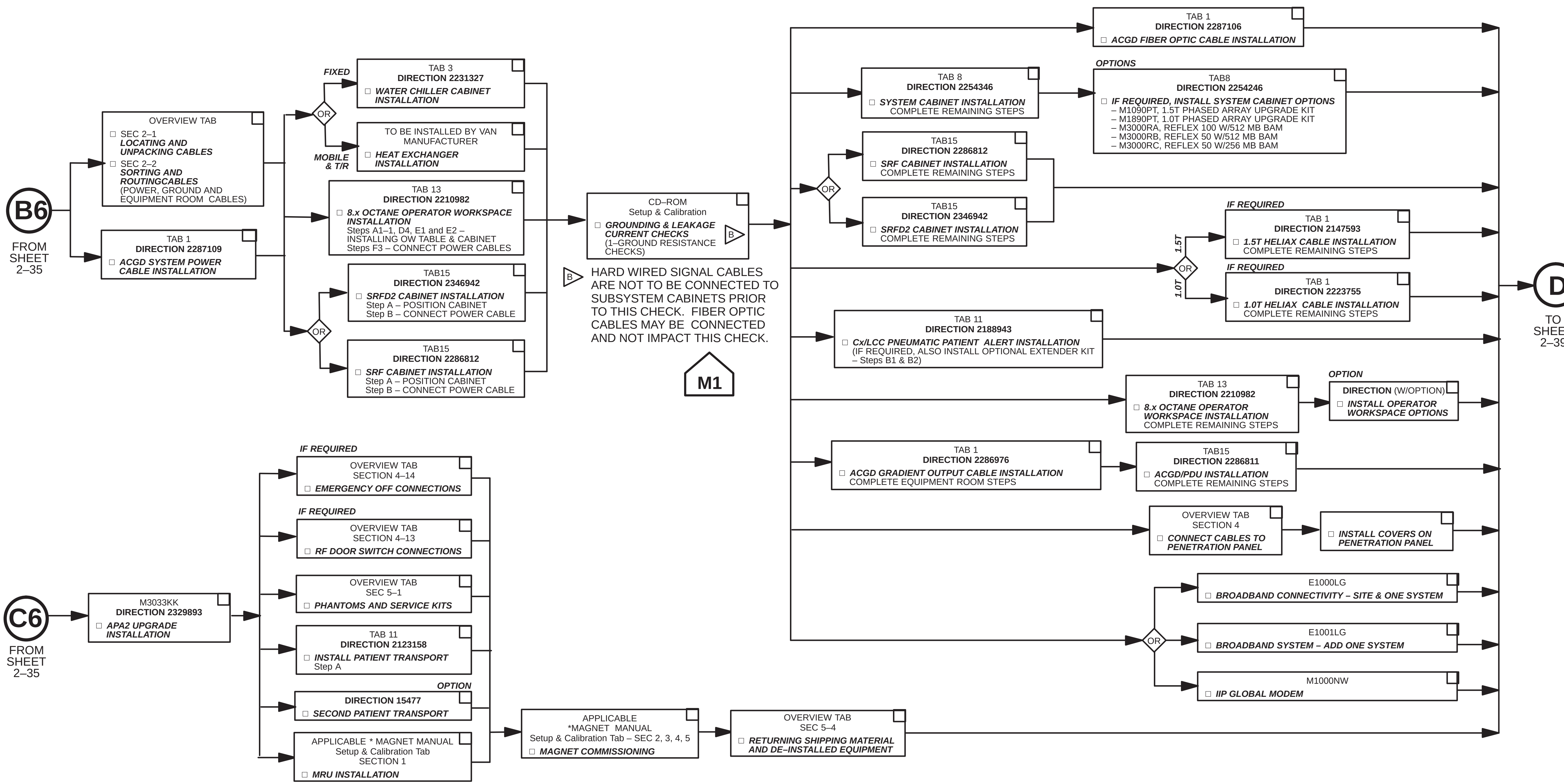
IMPORTANT: PER ISO 9001 REQUIREMENTS

SYSTEM MECHANICAL INSTALLATION



SIGNA ADVANTAGE 4.x OR 5.x to SIGNA WideOpen 9.x UPGRADE FLOW CHART–MECHANICAL
ILLUSTRATION 2–6 (SHEET 3 OF 3)

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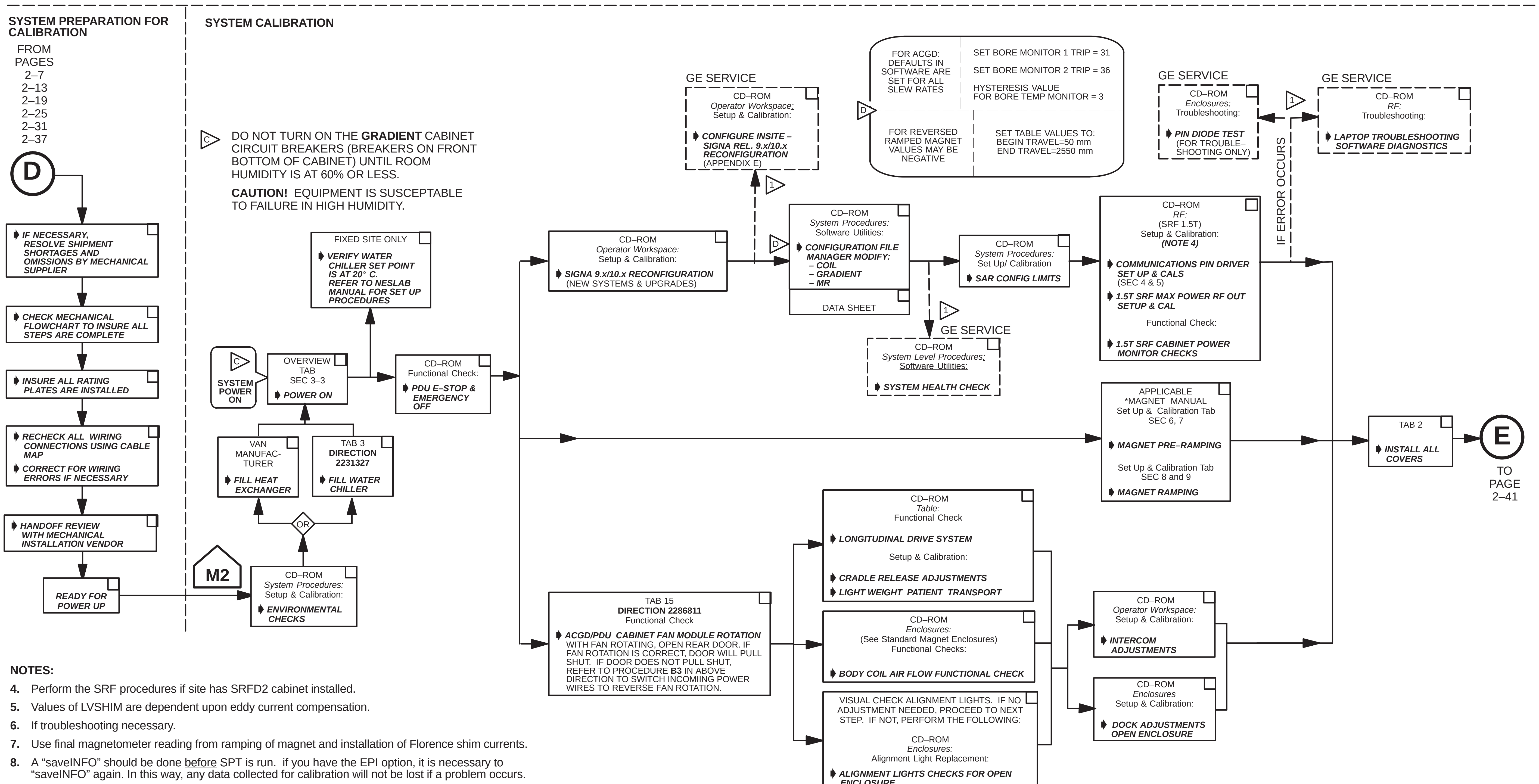
IMPORTANT: PER ISO 9001 REQUIREMENTS



NOTES:

4. Perform the SRF procedures if site has SRFD2 cabinet installed.

5. Values of LVSHIM are dependent upon eddy current compensation.

6. If troubleshooting necessary.

7. Use final magnetometer reading from ramping of magnet and installation of Florence shim currents.

8. A “saveINFO” should be done before SPT is run. if you have the EPI option, it is necessary to “saveINFO” again. In this way, any data collected for calibration will not be lost if a problem occurs.

9. Values of eddy current compensation are independent of magnet shim.

10. Overnight diags may be run at end of any day to detect possible errors.

SIGNA WideOpen 9.x OR 10.x UPGRADE FLOW CHART–CALIBRATION
ILLUSTRATION 2–7 (SHEET 1 OF 3)

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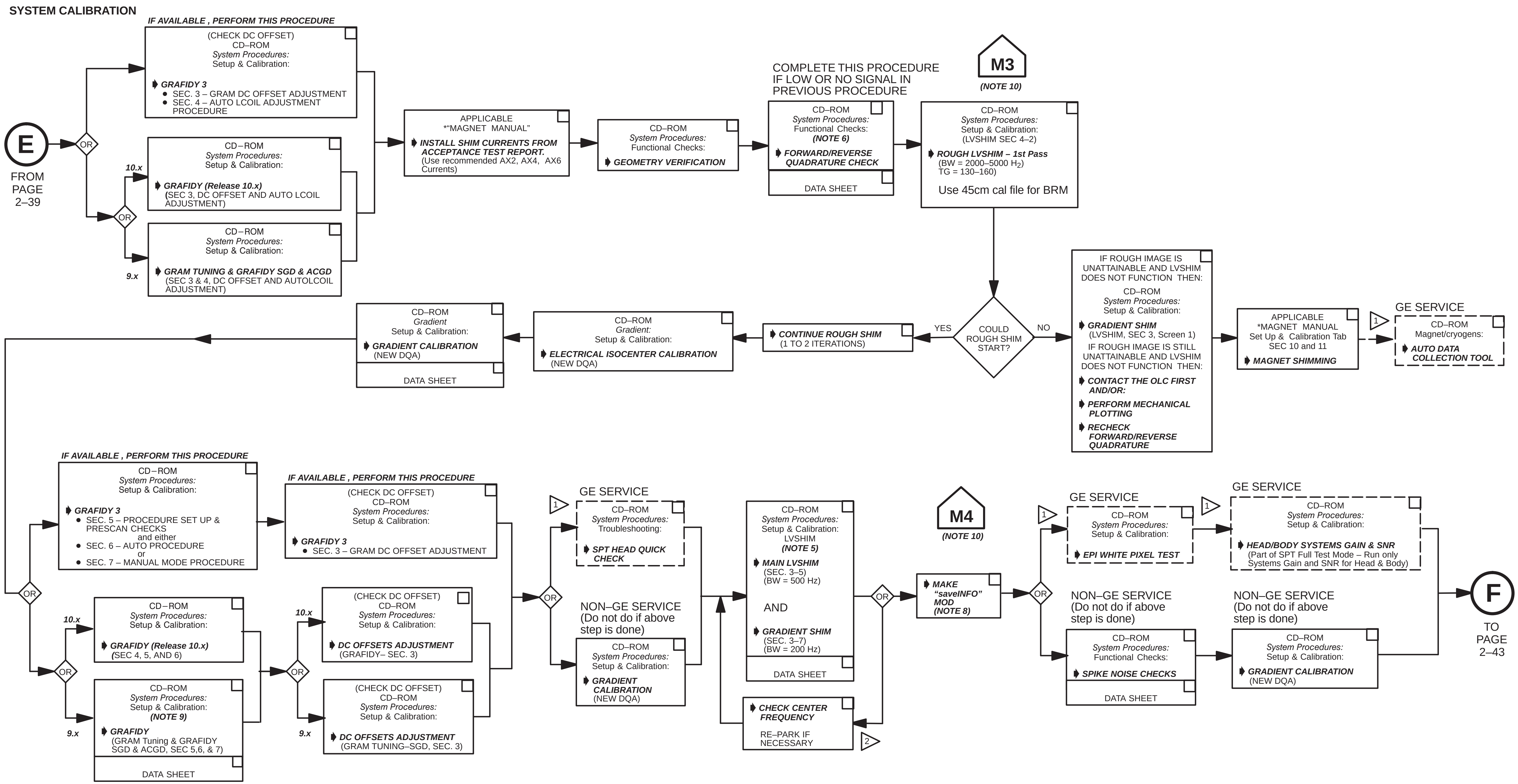
IMPORTANT: PER ISO 9001 REQUIREMENTS



SIGNA WideOpen 9.x OR 10.x UPGRADE FLOW CHART–CALIBRATION
ILLUSTRATION 2–7 (SHEET 2 OF 3)

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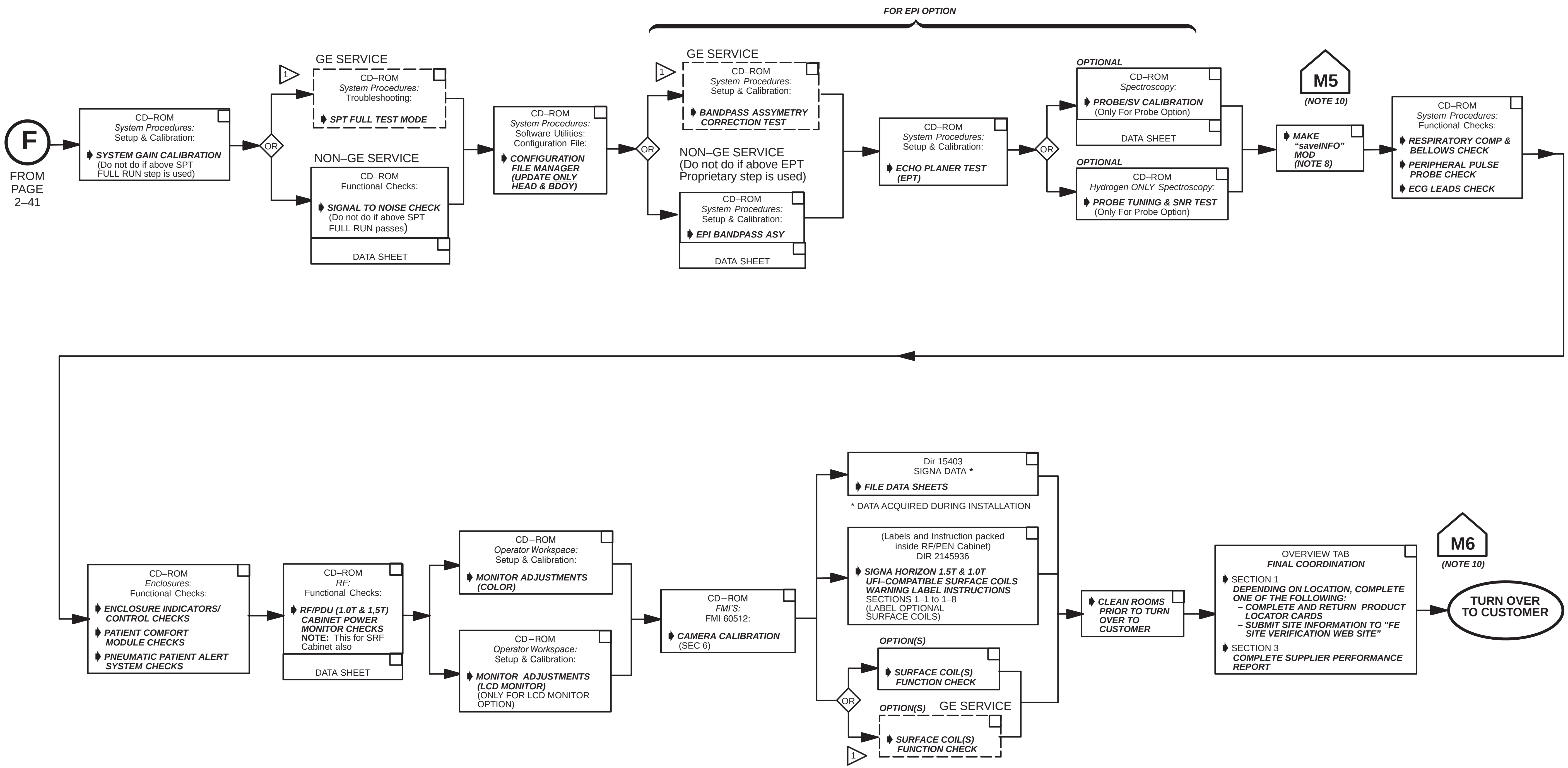
IMPORTANT : PER ISO 9001 REQUIREMENTS

SYSTEM CALIBRATION



SIGNA WideOpen 9.x OR 10.x UPGRADE FLOW CHART–CALIBRATION
ILLUSTRATION 2–7 (SHEET 3 OF 3)

SECTION 3 – DE-INSTALLATION PROCESS

3–1 SYSTEM POWER SHUTDOWN

DANGER!!

FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance baseline measurements have been obtained. and center frequency should be recorded before system is shut down.
2. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
3. Press the FULL OFF button on the PDU front control panel.

DANGER!!

ELECTRIC SHOCK HAZARD!! HIGH VOLTAGES ARE PRESENT ON INTERCONNECTING CABLES USED WITH SPECIAL FIELD INSTALLED ION–PUMPS USED WITH OXFORD MAGNETS. THE ION PUMP POWER SUPPLY IS USUALLY CONNECTED TO A POWER SOURCE THAT IS NOT CONTROLLED BY THE PDU AND HIGH VOLTAGE WILL BE PRESENT EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK, DISCONNECT POWER TO THE ION PUMP POWER SUPPLY, LOCK OUT, AND TAG BEFORE PERFORMING PROCEDURES IN THE FOLLOWING SECTIONS. ALSO CHECK FOR OTHER INTERCONNECTING EQUIPMENT WHICH WAS FIELD INSTALLED AND DISCONNECT POWER, LOCK OUT, AND TAG.

4. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
5. Locate and shut off main disconnect supplying power to the PDU.
6. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
7. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

3–2 RETURN MATERIAL INFORMATION FORMS

Fill in the information on “Attention: GEMS GoldSeal” form for Genesis Console (page 3–5), Genesis Computer (page 3–9), and Cabinets (page 3–13). Attach the appropriate sheet to each item.

3–3 REMOVING FULL–SIZE PDU

Sites that are upgrading to ACGD require that the full–size PDU or Compact PDU be de–installed.

The Compact PDU has built–in casters for ease of movement. The full–size PDU does not have casters and will have to be moved with two shipping dollies placed on each side of the PDU and clamped in place using straps with turnbuckles as shown in illustration below. If dollies are not provided, the Full–size PDU will have to removed by riggers or other locally supplied equipment moving personnel. See Illustration 3–1.

CAUTION

PDU weight is about 1000 pounds (455 kilograms). Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.

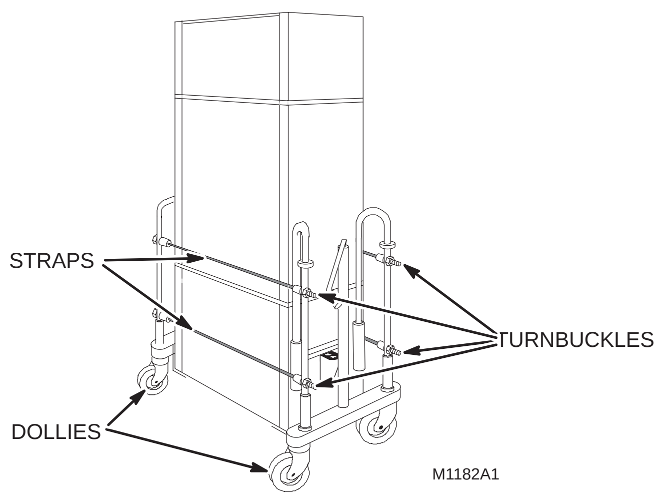
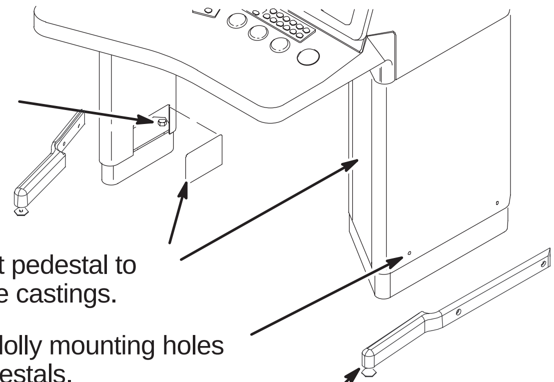


ILLUSTRATION 3–1
FULL–SIZE PDU WITH SHIPPING DOLLIES

3–4 5.x OPERATOR CONSOLE RETURN PROCEDURE

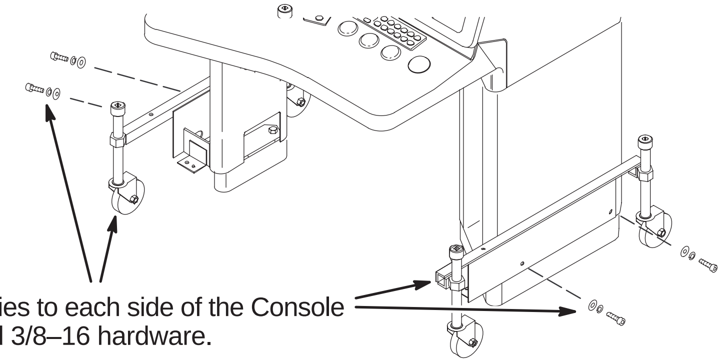
Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 5.x Operator Console.

❑ A1 – PREPARING OPERATOR CONSOLE REMOVAL



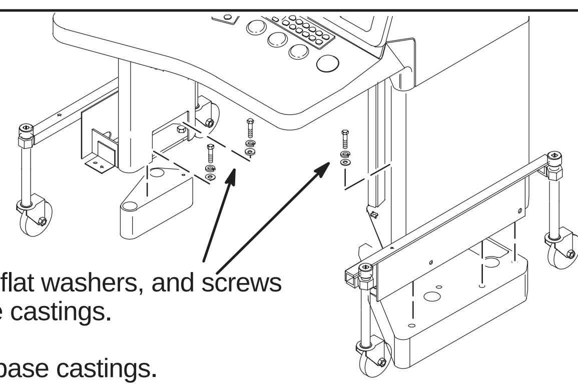
4. Loosen screws holding Console to pedestal risers.
3. Remove covers on sides of support pedestal to access mounting holes on the Base castings.
2. Remove Plug Buttons from dolly mounting holes on each side of Console pedestals.
1. If installed, unfasten and remove legs from console.

❑ A2 – ATTACHING DOLLY TO OPERATOR CONSOLE



1. Attach shipping dollies to each side of the Console legs using furnished 3/8–16 hardware.

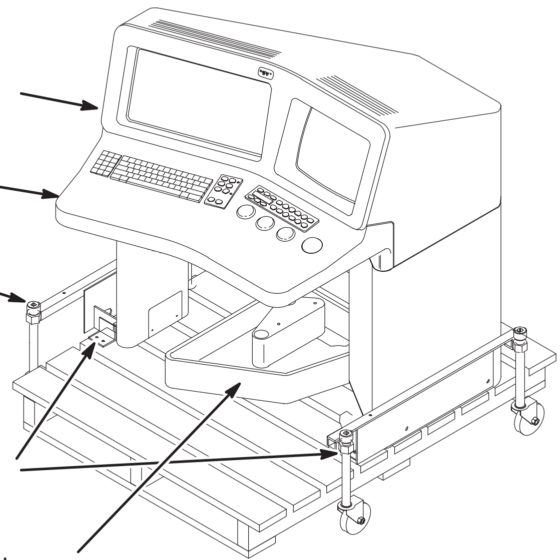
❑ A3 – REMOVING OPERATOR CONSOLE



1. Remove lock washers, flat washers, and screws connecting legs to base castings.
2. Raise console up over base castings.

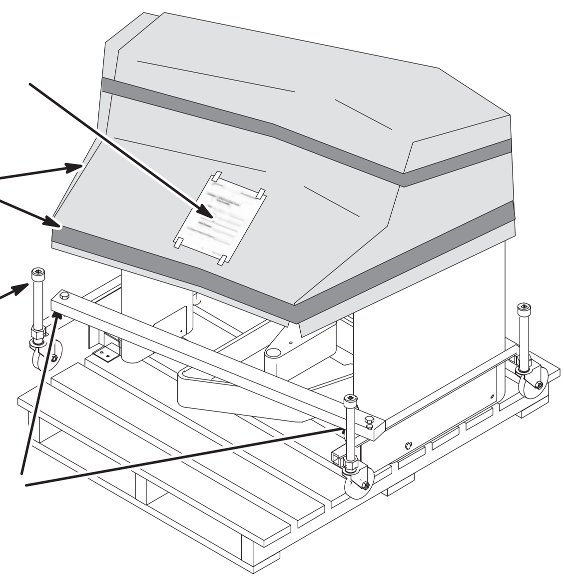
3–4 5.x OPERATOR CONSOLE RETURN PROCEDURE (Cont'd)

❑ A4 – FASTENING OPERATOR CONSOLE TO SHIPPING PALLET



1. Carefully roll Operator Console over an available shipping pallet.
2. Position Console so that Keyboard Housing does not extend beyond shipping pallet.
3. Use a 1/2 inch socket wrench to turn the jack screws and lower the Operator Console onto pallet.
4. Locate (4) four provided 5/16 x 1–1/2 long lag bolts.
5. Install two of the lag bolts thru each shipping dolly bracket to fasten to shipping pallet.
6. Nest together Riser castings on pallet as shown.

❑ A5 – FINAL PREPERATION BEFORE SHIPPING



4. Fill out *GEMS Recycling Form* on page 3–5 and attach to top of cardboard covering console.
3. Using any available cardboard (i.e. empty shipping boxes), cover Operator Console and tape in place. This is to prevent damage when shipping console back to manufacturer.
2. Make sure that all four wheel assemblies are completely raised. If not, use 1/2 inch socket wrench to turn jack screws.
1. Install provided dolly stiffener with furnished 1/4–20 x 1–1/2 long screws and nuts.

Return Console to GE Medical Systems with other system equipment that has been replaced during this upgrade.

Attention: GEMS GoldSeal

FDO: ______________________________

Site Name: ______________________________

Removal Date: ______________________________

Field Engineer: ______________________________

Genesis Console Serial Number: ______________________________

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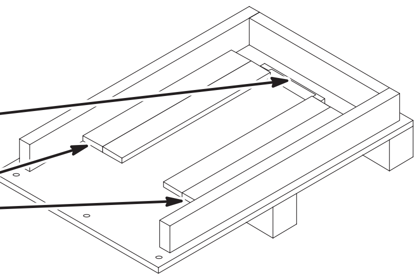
3–5 5.x COMPUTER RETURN PROCEDURE

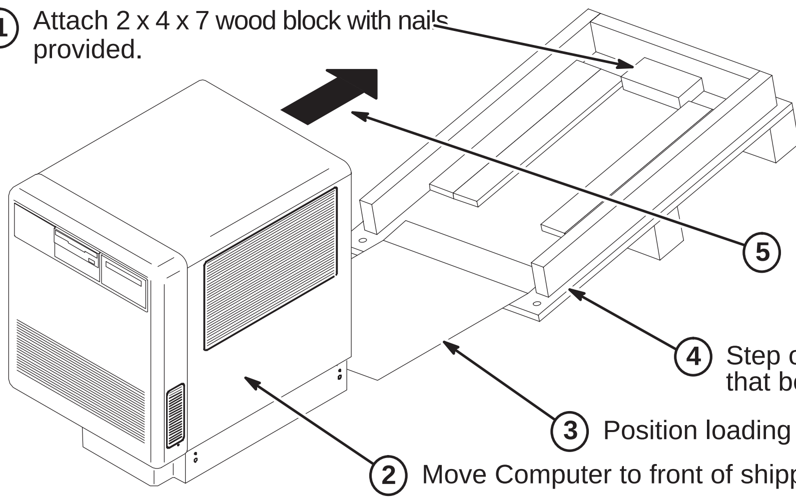
Confirm that all cables are disconnected from Computer. Perform the following procedure for return of 5.x Computer reusing the shipping pallet that the new Operator Workspace Cabinet arrived in.

❑ A1 – PREPARING SHIPPING PALLET



1. Locate the Operator Workspace Cabinet shipping pallet.
2. Remove existing bracket.
3. Re-position the protective foam on each side from a vertical to a horizontal position between the side of the pallet and the existing horizontal foam.

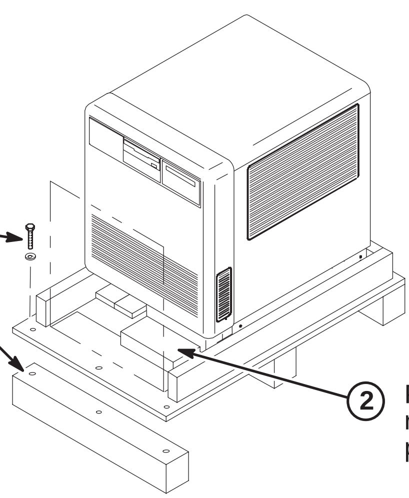
❑ A2 – LOADING COMPUTER ONTO SHIPPING PALLET



1. Attach 2 x 4 x 7 wood block with nails provided.
2. Move Computer to front of shipping pallet.
3. Position loading board at front of pallet.
4. Step on front corner of shipping pallet so that bottom board of pallet touches floor.
5. With another person, carefully roll Computer onto shipping pallet so it will rest on the horizontal foam pieces.

Note:
Pallet will return to horizontal position as Computer is rolled toward back of pallet.

❑ A3 – REASSEMBLE SHIPPING PALLET



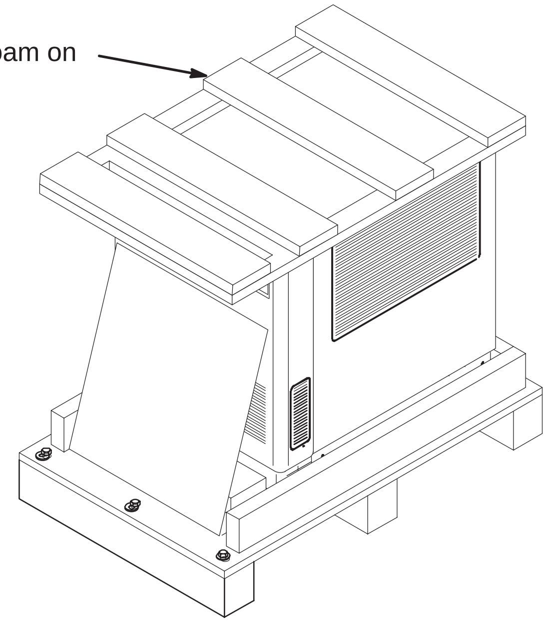
1. Re-attach wood block to pallet using the three previously removed fasteners.
2. Place 2 x 4 wood block against front right side of Computer and attach to pallet with nails provided.

3–5 5.x COMPUTER RETURN PROCEDURE (Cont'd)

❑ A4 – INSTALLING COMPUTER TOP SHIPPING FOAM

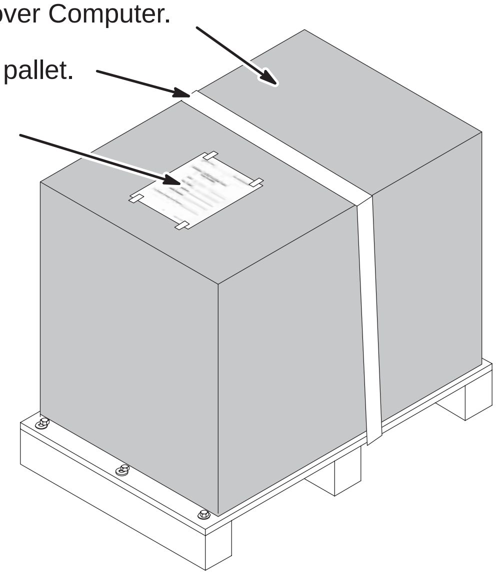
1. Install the Operator Workspace Cabinet shipping foam on top of Computer.

   Note: Shipping foam insert will not fit as well as it did with the new Operator Workspace Cabinet.



❑ A5 – FINAL PREPERATION BEFORE SHIPPING

1. Place Operator Workspace shipping box over Computer.
2. Secure to shipping pallet.
3. Fill out *GEMS Recycling Form* on page 3–9 and attach to top of shipping box.



Return Computer to GE Medical Systems with other system equipment that has been replaced during this upgrade.

Attention: GEMS GoldSeal

FDO: ______________________________

Site Name: __________________________

Removal Date: _______________________

Field Engineer: ______________________

Computer Serial Number: ______________________

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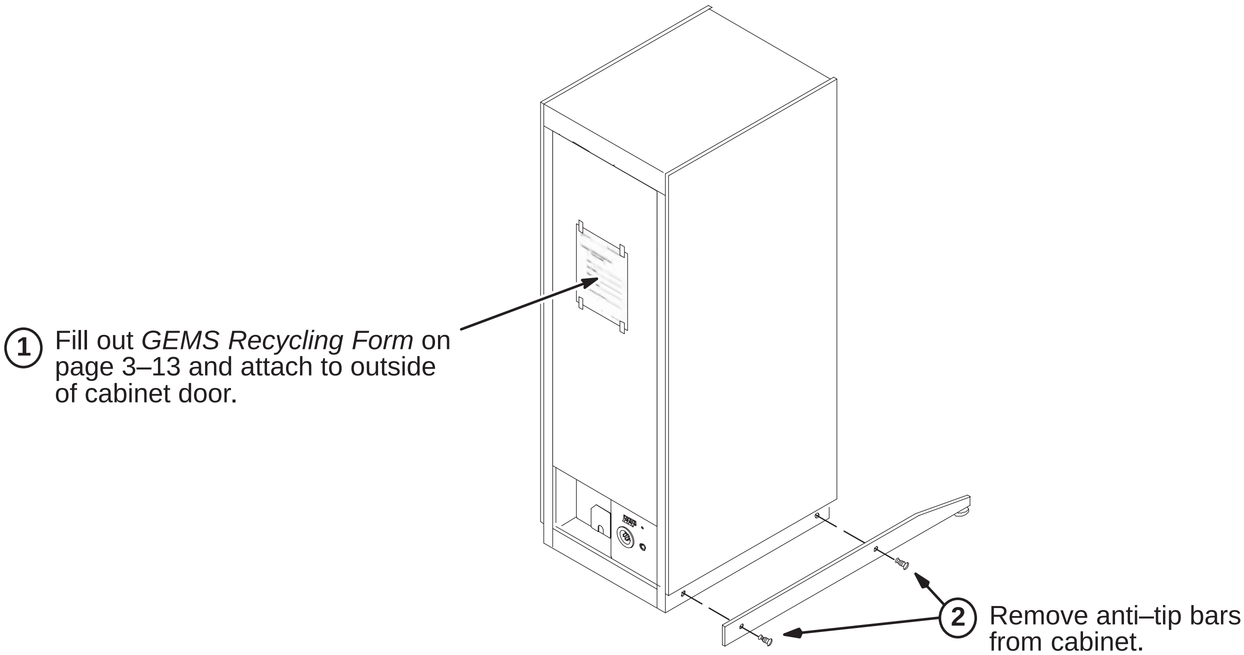
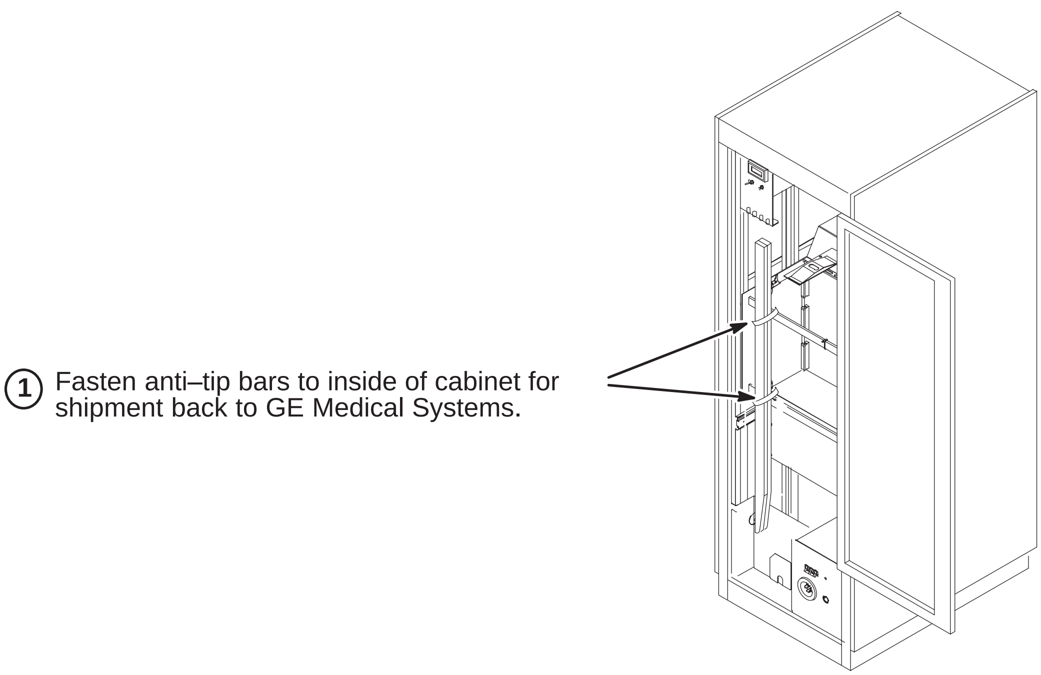
3–6 CABINET RETURN PROCEDURE

Confirm that all cables are disconnected from applicable Cabinet. The System Cabinet is shown below. The following procedure should be used for return of any cabinet with anti–tip bars.

❑ A1 – REMOVING ANTI–TIP BARS



❑ A2 – ATTACHING ANTI–TIP BARS AND FORM FOR RETURN SHIPMENT



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Attention: GEMS GoldSeal

FDO: ______________________________

Site Name: __________________________

Removal Date: _______________________

Field Engineer: ______________________

System Cabinet Serial Number: ___________________

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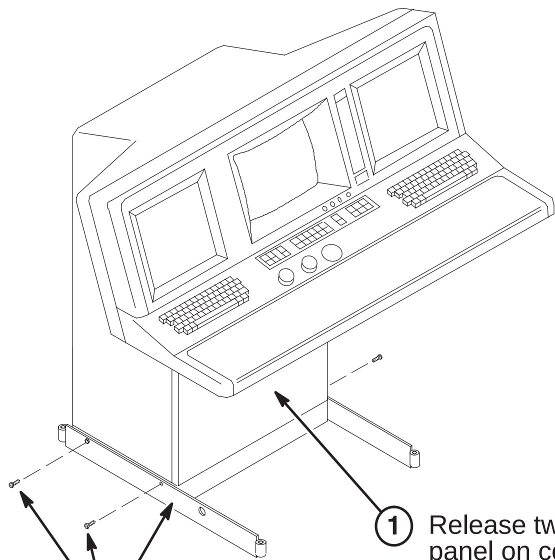
3–7 4.x OPERATOR CONSOLE RETURN PROCEDURE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 3.x/4.x Operator Console.

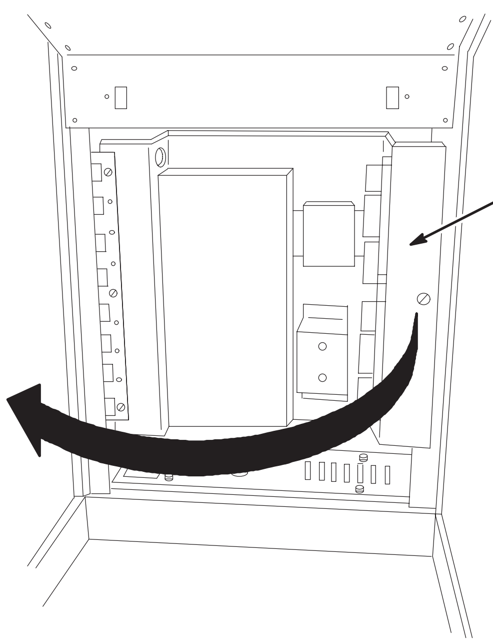
❑ A1 – INSTALL OPERATOR CONSOLE SHIPPING LEGS



(1) Release two captive fasteners near top of lower front panel on cosole and remove panel.

(1) Fasten to each side of Console:
(1) 46–225881G1 Anti–Tip Shipping Leg
(2) 46–208561P19 1/4–20x 3/4 Long Hex Head Screw

A2 – OPERATOR CONSOLE REMOVAL



1. Unlatch and swing open panel assembly.
2. Remove hardware that attaches console to floor.
3. Close and latch panel assembly.
4. Re-attach Front Panel.

SECTION 4 – SYSTEM CABLES

4–1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading a magnet and system to Signa MR/*i* 9.x or 10.x (MGD) WideOpen.

- Cable removal information for upgrade to Signa 10.x (MGD) w/EXCITE Technology begins with Section 4–2.
- Cable removal information for upgrade to Signa 9.x begins with Section 4–7.
- Preparation for new cable installation begins with Section 4–12.
- Cable install information for upgrade to Signa 10.x (MGD) w/EXCITE Technology begins with Section 4–16.
- Cable install information for upgrade to Signa 9.x begins with Section 4–19.
- Cable technical information is located in Section 4–22.

4–2 CABLE REMOVAL FOR UPGRADE TO SIGNA 10.x MR/i w/EXCITE TECHNOLOGY (MGD)

The upgrade to a Signa 10.x (MGD) will have many new cables delivered to the site. Table 4–1 lists all cables to be removed for upgrading to Signa 10.x MR/i WideOpen with EXCITE Technology HiSpeed or EchoSpeed from various early system configurations, including Advantage 5.4, Horizon 5.x & 8.x, and Horizon 8.x with Cx magnet. Refer to Section 4–6 for MGD cable map information.

Note

Magnet room Fiber Optic runs 711/712 will not be removed during Magnet room deinstallation for existing Horizon 5.x and 8.x systems.

The letters on the right side of table 4–5 are defined by the following cable removal explanations:

A – Existing system is Cx Magnet Signa Horizon 8.x with ACGD & Octane upgrading to MGD

B– Existing system is Cx Magnet Signa Horizon 8.x with SGD & Octane upgrading to MGD

C – Existing system is Cx Magnet Signa Horizon 8.x w/SGD or Gradient/GRAM & Indigo upgrading to MGD

D – Existing system is Pre–Cx Signa Horizon 8.x with ACGD & Octane upgrading to MGD

E – Existing system is Pre–Cx Signa Horizon 8.x with SGD & Octane upgrading to MGD

F – Existing system is Pre–Cx Signa Horizon 8.x with SGD or Gradient/GRAM & Indigo upgrading to MGD

G – Existing system is Pre–Cx Signa Horizon 5.x with Gradient/GRAM upgrading to MGD

H – Existing system is Pre–Cx Signa Advantage 5.x upgrading to MGD

I – Existing system is Pre–Cx Signa Advantage 4.5 or later but pre 5.4 Advantage upgrading to MGD

J – Existing system is Pre–Cx Signa Advantage 4.x but pre 4.5 Signa Advantage upgrading to MGD

There are instances where the cable Run to be removed is located in the Equipment and Magnet room. These cables are identified in the table by the letters **E** and **M** and are defined by the following explanations:

E= Equipment Room side of cable Run to be removed.

M = Magnet Room side of cable Run to be removed.

X = Cable to be removed and returned for recycling. **DO NOT CUT CABLES.**

4–2 CABLE REMOVAL FOR UPGRADE TO SIGNA 10.x MR/i w/EXCITE TECHNOLOGY (MGD) (Cont'd)

TABLE 4–1
CABLES TO BE REMOVED AND RETURNED WHEN UPGRADING TO SIGNA 10.x with EXCITE (MGD)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J
002	PD1–A2–TS1–32/NEUT/GND	MR2–A12–SERVICE									X	X
003	PD1–A2–TS1–1,2,3/NEUT/GND	MR3–A10–AC IN								X	X	X
007	PD1–FEEDER 15	PP2–A6–AC IN									X	X
008	PD1–FEEDER 16	PP2–A6–SERVICE									X	X
009	PD1–A2–TS1–10,11,12/NEUT/GND	MR1–J1		X	X		X	X	X	X	X	X
012	PD1–A2–TS1–21/NEUT/GND	OC1–A8–J3 (AC IN)									X	X
013	PD1–A2–TS1–33/NEUT/GND	OC1–A8–J6 (SERVICE)									X	X
014	PD1–A2–TS1–34/NEUT/GND	CC1–A20–SERVICE									X	X
015	PD1–A2–TS1–13,14,15/NEUT/GND	CC1–A20–AC IN									X	X
016	PD1–A2–TS1–16,17,18/NEUT/GND	CC2–A20–AC IN									X	X
017	PD1–A2–TS1–34/NEUT/GND	CC2–A20–SERVICE									X	X
021	PD1–A2–GND	CC1–A20–GND									X	X
022	PD1–A2–GND	CC2–A20–GND									X	X
023	PD1–GND BUS	OC1–GND							X	X	X	X
027	PD1–A2–TS1–32/NEUT/GND	PP2–A15–J1–SERVICE								X	X	X
028	PD1–A2–TS1–31/NEUT/GND	PP2–A15–J2–AC IN								X	X	X
030	PD1–A2–TS1–10	MR2–A11–AC IN	X	X	X	X	X	X	X	X	X	X
035	PD1–A2–TS1–22,23,24/NEUT/GND	CC2–A20–AC IN									X	X
036	PD1–FEEDER 5–TS1–4,5,6 (Spect.)	MR6 (Spectroscopy Only)								X	X	X
037	PD1–GND BUS	MR2–A11 GND								X	X	X
038	PD1–GND BUS	MR1–A7 GND		X	X		X	X	X	X	X	X
040	RF SHIELD ROOM COMMON GND	MS1–GND	X	X	X	X	X	X	X	X	X	X
041	PD1–TS1–1,2,3,GND	MR3–A6		X	X		X		X			
042	PD1–TS1–4,5,6,GND	MR3–A7		X	X		X		X			
043	PD1–TS1–7,8,9,GND,NEU	MR7–A7							X			
045	PP1–GND **(See Note 2)**	FAN MODULE GND **(Note 2)**							X			
046	PD1–TS1–20,GND,NEU	HE1	X	X	X	X	X	X	X			
047	PD1–TS1–XX	OW1–A2–A5–J1	X	X	X	X	X	X	X			
048	PD1–GND	OW1–A2–A5–GND										
049	OW1–A1–A1	OW1–A2–A5–J2										
050	OC1–AC–IN	PD1–TS1–13,14,15,NUE,GND							X	X		
064	MR2–A12–J1–AC OUT	CC1							X	X		
200	MR2–A11–J19	MR3–A9–J1									X	X
201	MR2–A11–J20	MR3–A9–J2									X	X
202	MR2–A11–J21	MR3–A9–J3									X	X
203	MR2–A11–J22	MR3–A9–J4									X	X
207	RF DOOR SWITCH	MR2–A11–J15										
223	OC1–A8–J13	MR2–A1–J36									X	X
224	CC1–A21–J19	MR1–A7–J8									X	X
226	MR1–A7–J5	MR2–A11–J14									X	X
227	PD1–A1–A2–J2	MR2–A11–J16										
228	CC1–A21–J10	MR2–A11–J17									X	X

4–2 CABLE REMOVAL FOR UPGRADE TO SIGNA 10.x MR/i w/EXCITE TECHNOLOGY (MGD) (Cont'd)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J
229	MR1–A7–J3	MR2–A11–J1										
231	MR2–A11–J2	PP1–J8										
233	MR2–A11–J5	PP1–J7										
235	MR1–A7–J13 **(See Note 1)**	PP1–J44 **(See Note 1)**					X	X	X	X		
236	MR1–A7–J12 **(See Note 1)**	PP1–J4 **(See Note 1)**					X	X	X	X		
238	CC1–A21–J14	OC1–A8–J8									X	X
239	CC1–A21–J15	OC1–A8–J10									X	X
244	CC1–A21–J13	BTI–HOST									X	
248	CC2–A21–J1	OC1–A9–J1									X	X
249	CC2–A21–J22	OC1–A8–J12									X	X
256	PP1–J12	OC1–A8–J11									X	X
257	PP1–J44 **(See Note 1)**	MG3–A3–A1–J1 **(See Note 1)**				X	X	X	X	X	X	X
258	PP1–J4 **(See Note 1)**	MG3–A17–J1 **(See Note 1)**				X	X	X	X	X	X	X
260	PP1–J7	MG3–A17–J2									X	X
261	PP1–J6	MG2–A11–A1–J1									X	X
262	PP1–J8	MG3–A3–A5–OUT				X	X	X	X	X	X	X
274	CC1–A21–J20	MR1–A7–J7									X	X
280	PP1–J15	PP2–A1–J2									X	X
281	PP1–J11	PP2–A1–J3									X	X
282	PP1–J12	MG3–A2–J6				X	X	X	X	X	X	X
287	CC2–A21–J15	CC1–A21–J2									X	X
296	FACILITY DISCONNECT	PP1–J18										
297	EO1	PP1–J18										
300	PP1–J15	MG2–A15	X	X	X	X	X	X	X	X	X	X
301	CC2–A21–J20	CC1–A21–J28									X	X
302	CC2–A21–J21	CC1–A21–J29									X	X
318	MG3–A7	PP1–J11				X	X	X	X	X	X	X
322	MR2–A11–J11	PP2–A6–J18									X	X
325	CC1–A21–J27	CC2–A21–J19									X	X
327	CC1–A21–J25	CC2–A21–J17									X	X
346	MR2–A11–J41	CM1–FRONT									X	X
347	MR2–A11–J42	CM1–BACK									X	X
354	CM1	ST1									X	X
370	PP1–A9–J46	MR1–A7–J19										X
384	PP1–J20	PP2–A15–J17								X	X	X
385	PP1–J20	MG2–A30	X	X	X	X	X	X	X	X	X	X
386	PP1–J47	PP2–A15–J21								X	X	X
387	PP1–J47	MG2–A29				X	X	X	X	X	X	X
388	PP1–J11	PP2–A15–J22								X	X	X
389	PP1–J15	PP2–A15–J20								X	X	X
404	MG3–A17–J1	MG2–A16–A5–J6	X	X	X	X	X	X	X	X	X	X
405	MR3–A9–TS1	MG3–TS1–1,2								X	X	X
406	MR3–A9–TS1	MG3–TS1–3,4								X	X	X
407	MR3–A9–TS1	MG3–TS1–5,6								X	X	X
408	PD1–A1–A3–J3	PP1–J16								X	X	X

4–2 CABLE REMOVAL FOR UPGRADE TO SIGNA 10.x MR/i w/EXCITE TECHNOLOGY (MGD) (Cont'd)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J
409	PP1–J16	MG2–A12–A1–J7								X	X	X
410	PP1–J14	PP2–A15–J23										X
411	PP1–J14	MG3–A13–J1										X
412	PP1–J51	MG3–A32–J1								X	X	X
413	PP1–J51	PP2–A15–J28(A16–J9)								X	X	X
414	MG3–A35	MG3–A2–J9	X	X	X	X	X	X	X	X	X	X
415	PP1–A8–J45	MG3–A33–J7										X
416	MR1–A7–J21	PP1–A8–J45										X
417	PP2–A15–J19	MR2–A11–J32									X	X
418	MR2–A11–J39	PP2–A15–J3										X
419	MR2–A11–J26	PP1–A10–J1									X	X
420	MG3–A33–J1	PP1–A10–J2									X	X
421	MG2–A33–J3	MG3–A31–J1	X	X	X	X	X	X	X	X	X	X
422	MG2–A33–J2	MG3–A1–J4	X	X	X	X	X	X	X	X	X	X
424	PP1–A10–J5/J6	MG2–A33–J11/J12									X	X
425	MR2–A11–J12	PP1–J6										X
426	MR1–A7–J9	MR3–A9–J5									X	X
427	MR2–A11–J17	CC1–A21–J13									X	X
428	MR2–A11–J44	CC1–A21–J35									X	X
429	BT1	CC1–A21–J25									X	X
430	MR2–A11–J24	CC1–A21–J26									X	X
431	MR2–A11–J25	CC1–A21–J27									X	X
432	PD1–A1–A3–J2	MR2–A11–J45									X	X
433	OC1–A9–J1	CC1–A21–J10									X	X
434	CC1–A8–J12	CC1–A21–J29									X	X
435	MR2–A11–J13	PP1–A10–J3									X	X
436	MG2–A12–A1–J1	MG3–A33–J3									X	X
437	MG2–A12–A1–J2	MG3–A33–J4									X	X
439	MG3–A3–J4	MG3–A33–J1									X	X
440	MG3–A3–J3	MG3–A34–J2									X	X
441	MG3–A33–J7	MG3–A34–J8									X	X
446	PP1–A9–J46	MG2–A12–A13–J5										X
447	PP1–A9–J46	MG2–A12–A13–J3										X
448	MR2–A11–J44	CC1–A21–J1									X	
449	MR2–A11–J24	CC1–A21–J34									X	
450	MR2–A11–J25	CC1–A21–J35									X	
453	PP1–J50	MG2–A11–A1–J4									X	X
454	MR2–A11–J34	PP1–J50									X	X
470	MR1–A7–J17	PP1–A11–J71								X	X	
474	PP1–A11–J72	MG3–A12–A1–J3								X	X	
475	PP1–A11–J73	MG3–A12–A1–J4								X	X	
476	PP1–A11–J74	MG3–A33–J7								X	X	
477	PP1–A11–J75	MG3–A12–A1–J5								X	X	
478	PP1–A11–J76	MG3–A12–A1–J6								X	X	
485	MG3–A17–J2	MG2–A16–A3–OUT	X	X	X	X	X	X	X	X	X	X

4–2 CABLE REMOVAL FOR UPGRADE TO SIGNA 10.x MR/i w/EXCITE TECHNOLOGY (MGD) (Cont'd)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J
486	PP1–A14–J85	MR1–A7–J22	X	X	X	X	X	X	X	X	X	X
487	PP1–A15–J87	MR1–A7–J20	X	X	X	X	X	X	X	X	X	X
488	PP1–J80	MR2–A11–J8	X	X	X	X	X	X	X	X	X	X
489	PP1–J81	MR2–A11–J9	X	X	X	X	X	X	X	X	X	X
490	PP1–J82	MR2–A11–J10	X	X	X	X	X	X	X	X	X	X
491	PP1–J80	MG3–A36–J7	X	X	X	X	X	X	X	X	X	X
492	PP1–J81	MG3–A36–J8	X	X	X	X	X	X	X	X	X	X
493	PP1–J82	MG3–A36–J9	X	X	X	X	X	X	X	X	X	X
494	PP1–A14–J87	MG3–A36–J6	X	X	X	X	X	X	X	X	X	X
495	PP1–A15–J78	MG3–A36–J10	X	X	X	X	X	X	X	X	X	X
499	PP1–A14–J84	MR2–A11–J5	X	X	X	X	X	X	X	X		
536	HE1 (Coolant Hose)	MG2–A12–A1–IN	X	X	X	X	X	X	X			
537	HE1 (Coolant Hose)	MG2–A12–A1–OUT	X	X	X	X	X	X	X			
601	MS1 (RED)	MS6 (RED)	X	X	X	X	X	X	X	X	X	X
602	MS1 (BLK)	MS6 (BLK)	X	X	X	X	X	X	X	X	X	X
603	MS1–A13–A1	MS7	X	X	X	X	X	X	X	X	X	X
604	MS1–A13–A1	MS6/MS7	X	X	X	X	X	X	X	X	X	X
605	PP1–J10 (Magnet Room)	MS1–A3–A1 (Magnet Room)	M	M	M	M	M	M	M	M	M	M
605	PP1–J10 (Equipment Room)	MR1–J11 (Equipment Room)	E	E	E	E	E	E	E	E	E	E
606	MS4	MS1–A3–A1	X	X	X	X	X	X	X	X	X	X
621	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	X	X	X	X	X
622	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	X	X	X	X	X
623	MS5–A1	PP1–F1,F2,F3,F4	X	X	X	X	X	X	X	X	X	X
624	MS1–A2	PP1–F1,F2,F3,F4	X	X	X	X	X	X	X	X	X	X
626	MS5–A1	BALZER REMOTE	X	X	X	X	X	X	X	X	X	X
700	PP1–A14–J86	MG3–A17–J2	X	X	X	X	X	X	X	X		
701	RF DOOR SWITCH	MR2–A11–J15										
702	MR2–A11–J18	MR1–A7–J5										
703	PD1–A14–A1–J2	MR2–A11–J23										
704	PP1–J12	OC1–A10–J9					X	X	X	X		
705	PP2–A15–J19	MR2–A11–J25										
706	PD1–A14–A1–J3	MR2–A11–J14	X	X	X	X	X	X	X	X		
707	MR2–A11–J26–ESTOP	OC1–J8							X	X		
708	MR2–A11–J16	MR1–A7–J27								X		
709	MR1–A7–J26	MR3–A9–J1								X		
710	MR2–A11–J17	MR3–A9–J2								X		
711	PP1–A17–A1–J91	MG2–A33								M		
712	PP1–A17–A1–J91	MG2–A11–A1								M		
711/ 712	PP1–A17–A1–J91	MR2–A11–J13	E	E	E	E	E	E	E	E		
715	PP1–J14	MG2–A33–J1				X	X	X	X	X		
716	PP1–J89	MG2–A11–A1–J2				X	X	X	X	X		
718	MR2–A11–J29	OC1–A10–J4							X	X		
719	CC1–A1–A6–J1	MR2–A15–A3–J1							X	X		
726	PP1–A16–J77	MR1–A7–J18	X	X	X	X	X	X	X	X	X	X

4–2 CABLE REMOVAL FOR UPGRADE TO SIGNA 10.x MR/i w/EXCITE TECHNOLOGY (MGD) (Cont'd)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J
727	PP1–A16–J77	MG3–A36–J5	X	X	X	X	X	X	X	X	X	X
728	MG3–A36–J11	MG2–A16–OUT–J1	X	X	X	X	X	X	X	X	X	X
729	MG3–A36–J12	MG2–A16–OUT–J2	X	X	X	X	X	X	X	X	X	X
730	MG3–A36–J13	MG2–A16–OUT–J3	X	X	X	X	X	X	X	X	X	X
731	MG3–A36–J14	MG2–A16–OUT–J4	X	X	X	X	X	X	X	X	X	X
732	MG3–A36–J15	MG2–A16–OUT–J5	X	X	X	X	X	X	X	X	X	X
733	MG3–A36–J16	MG2–A16–OUT–J6	X	X	X	X	X	X	X	X	X	X
734	MG3–A36–J17	MG2–A16–OUT–J7	X	X	X	X	X	X	X	X	X	X
735	MG3–A36–J18	MG2–A16–OUT–J8	X	X	X	X	X	X	X	X	X	X
736	MG3–A36–J19	MG2–A16–OUT–J9	X	X	X	X	X	X	X	X	X	X
739	MG3–A3–A1–J4	MG2–A12–A1–J1 OR J2								X		
740	MG3–A3–A1–J3	MG2–A12–A1–J2 OR J1								X		
743	MG3–A3–J3	MG2–A12–J2	X	X	X	X	X	X	X			
744	MG3–A3–J4	MG2–A12–J1	X	X	X	X	X	X	X			
745	PP1–J44 **(See Note 1)**	MR1–A7–J13 **(See Note 1)**										
746	PP1–J44 **(See Note 1)**	MG3–A3–A1–J1 **(See Note 1)**										
747	PP1–J4 **(See Note 1)**	MG3–A17–J1 **(See Note 1)**										
748	PP1–J4 **(See Note 1)**	MR1–A7–J12 **(See Note 1)**										
749	MG2–A36	MG2–A33–J10	X	X	X	X	X	X	X	X		
753	MR7–A4–J1	MR3–A11–J5							X			
754	MR7–A4–J2	MR3–A11–J4							X			
755	MR7–A4–J3	MR3–A11–J3							X			
756	MR7–A4–J4	MR3–A11–J8							X			
757	MR7–A4–J5	MR3–A11–J7							X			
758	MR7–A4–J6	MR3–A11–J6							X			
759	MR3–A11–J12	MR7–A4–J8							X			
760	MR3–A11–J11	MR7–A4–J9							X			
761	MR3–A11–J10	MR7–A4–J10							X			
762	MG3–A32–TS1/2 **(See Note 3)**	PP1–A7–1/2 **(See Note 3)**		M	M	M	M	M	M			
762	MR3–J3 **(See Note 3)**	PP1–A7–1/2 **(See Note 3)**		E	E		E	E	E			
763	MG3–A32–TS3/4 **(See Note 3)**	PP1–A7–3/4 **(See Note 3)**		M	M	M	M	M	M			
763	MR3–J4 **(See Note 3)**	PP1–A7–3/4 **(See Note 3)**		E	E		E	E	E			
764	MG3–A32–TS5/6 **(See Note 3)**	PP1–A7–5/6 **(See Note 3)**		M	M	M	M	M	M			
764	MR3–J5 **(See Note 3)**	PP1–A7–5/6 **(See Note 3)**		E	E		E	E	E			
767	MR1–A7–J37 **(See Note 4)**	PP1–J51 **(See Note 4)**							X	X		
771	PP1–J47	MR1–A7–J36										
772	PP1–J14	MR1–A7–J40										
773	PP1–A17–J1	MR1–A7–J35										
774	MR2–A11–J25	MR1–A7–J32										
778	MG2–A12–A13–J3	PP1–A11–J72				X	X	X	X			
779	MG2–A12–A13–J4	PP1–A11–J73				X	X	X	X			
780	MG3–A3–J6	PP1–A11–J74				X	X	X	X			
781	MG2–A12–A12–J5	PP1–A11–J75				X	X	X	X			
782	MG2–A12–A12–J6	PP1–A11–J76				X	X	X	X			
784	PP1–F5/F6	MG6–J1							X			

4–2 CABLE REMOVAL FOR UPGRADE TO SIGNA 10.x MR/i w/EXCITE TECHNOLOGY (MGD) (Cont'd)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J
786	PP1–F7/F8	MG6–J2							X			
787	MR2–A11–J30	CC1–A1–A1–SERIAL C							X			
788	OW1–A1–A4–J7	PP1–J12										
789	OW1–A1–A4–J18	MR2–A11–J26										
790	OW1–A2–A6–BIT3	MR2–A11–J21	X	X	X	X	X	X				
791	OW1–A16–PORT1	MR2–A11–J30			X			X				
792	OW1–A1–A4–J11–J13	CUSTOMER MODEM			X			X				
793	OW1–A2—A1–COM1	OW1–A1–A4–J9			X			X				
794	OW1–A2–A3–MIC,AUDIO	OW1–A1–A4–J12,J14			X			X				
795	OW1–A2–A3–PWR I/O	OW1–A1–A4–J17			X			X				
796	OW1–A2–A3–KBD/MS	OW1–A1–A4–J8			X			X				
797	OW1–A2–A1–PC	OW1–A3–LCD										
798	OW1–A2–A3–MONITOR	OW1–A6–R,B,G										
799	OW1–A2–A3–MONITOR	OW1–A4–B/W										
800	OC1–J1	CC1–A2–J1 (MUX)							X	X		
801	OC1–J10	CC1–A2–J2 (SECUR. KEY)							X	X		
802	OC1–J12	CC1–A1–J9 (PLASMA)							X	X		
803	OC1–J13	CC1–A1–A10 (VIDEO)							X	X		
804	OC1–J3	CC1–A1–A1 (BOOT)							X	X		
805	OW1–A1–A4–J6	OW1–A7										
806	OW1–A2–A3–SER 1	OW1–A13			X			X				
807	OW1–A2–A3–HOST	OW1–A5										
816	OW1–A2–A6–J1	OW1–A15–BIT3	X	X		X	X					
820	OW1–A2–A3–MONITOR	OW1–A4–B/W										
821	OW1–A2–A3–MONITOR	OW1–A6–R,B,G										
836	MR2–A11–J21	OW1–A15–BIT3	X	X		X	X					
971	MR2–A11–J23	MR3–PD1–J2										
972	MR2–A11–J14	MR3–PD1–J3	X	X	X	X	X	X				

Notes:

1. 200 series cables are 1.5T specific. 700 series cables are 1.0T specific. These cable will have labels added after they are routed and terminated.
2. Only present if system is at Horizon
3. The cable is divided between the magnet room and equipment room.
4. Run 767 is only found on a few early Horizon 5.x (5.5) systems.

4–3 RE–LABELING AND RE–ROUTING OF EXISTING PRE–HORIZON CABLES TO 10.x WideOpen

Several existing cables need to be re–labeled and then re–routed per the new labels. Refer to Table 4–2.

Runs 772, 773, and 774 are only re–labled for Advantage 5.4 systems upgraded to 10.x WideOpen. All pre–5.4 system upgrades are provided new cables for Runs 772, 773, and 774.

TABLE 4–2
PRE–HORIZON CABLES RE–LABELED AND RE–LOCATED FOR UPGRADE TO 10.x WideOpen

OLD RUN	SYSTEM TYPE	NEW RUN	ONE END		OTHER END	
			OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
207	3.x 4.x	701	RUN #207 MR2–A11–J6 46–243775G411	RUN #701 MR2–A11–J15 46–243775G807	RUN #207 DOOR SWITCH 46–243775G411	RUN #701 RF DOOR SWITCH 46–243775G807
227	3.x 4.x	971	RUN #227 MR2–A11–J16 46–243775G464	RUN #703/971 MR2–A11–J23 46–243775G809	RUN #227 PD1–A1–A2–J2 46–243775G464	RUN #703/971 PD1–A1–A2–J2 46–243775G809
387	3.x	387	RUN #387 IS NOT LABELED USE LABELS PROVIDED	RUN #387 PP1–J47 46–243775G531	RUN #387 IS NOT LABELED USE LABELS PROVIDED	RUN #387 MG2–A29–P1 46–243775G531
705	5.x	774	RUN #705 MR2–A11–J25 46–243775G663	RUN #774 MR2–A11–J25 46–243775G811	RUN #705 PP2–A15–J19 46–243775G663	RUN #774 MR1–A7–A32 46–243775G811
713	5.x	772	RUN #713 PP2–A15–J23 46–243775G676	RUN #772 MR1–A7–J40 46–243775G808	RUN #713 PP1–J14 46–243775G676	RUN #772 PP1–J14 46–243775G808
714	5.x	773	RUN #714 PP2–A5–J27 46–243775G676	RUN #773 MR1–A7–J35 46–234775G808	RUN #714 PP1–J89 46–243775G676	RUN #773 PP1–J89 46–234775G808

4–4 RE–LABELING & RE–ROUTING HORIZON 5.x OR LX CABLES TO 10.x WideOpen

Run 709 is re–labeled only for Horizon 5.x systems upgraded to 10.x. All pre–5.x system upgrades are provided a new cable for Run 709. Refer to Table 4–3.

Run 708 is re–labeled for all pre–MGD systems to Run 1033. Refer to Table 4–3. Refer to 10.x Fiber Optic inistallation in Tab1 for marking indivdual fiber optic wires for connection to modules.

TABLE 4–3
HORIZON 5.x CABLES RE–LABELED AND RE–LOCATED FOR UPGRADE 8.x WideOpen

OLD RUN	ONE END		OTHER END	
	OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
708	RUN #708 MR2–A11–J16 46–317077G802	RUN #1033 MR2–A11–J16 2330480	RUN #708 MR1–A7–TX 46–317077G802	RUN #1033 MR1–A7–TX 2330480
709	RUN #709TR–1 MR3–A11–J1 46–317077G802	RUN #709TR–1 MR3CABI/FJ2 46–317077G803	RUN #709TR–1 MR1–A7–RX 46–317077G802	RUN #709TR–1 MR1–A7–RX 46–317077G803

4–5 LABELS ADDED TO NEW MAGNET ROOM CABLES (Non–Cx Magnet Only)

The following table defines the additional cables which need to be relabeled when upgrading to a LCC WideOpen magnet, unless the magnet being replaced is a Cx magnet. **In the case of a Cx magnet being replaced by a LCC magnet, the cables in the following table do not need to be relabeled.**

TABLE 4–4
MAGNET ROOM LABELS TO BE RE–LABELED

OLD RUN	ONE END		OTHER END	
	OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
257 (1.5T)	NONE	RUN #257 MG3 A3 J1 46–	NONE	RUN #257 PP1–J44 46–
258 (1.5T)	NONE	RUN #258 MG3 A17 J1 46–	NONE	RUN #257 PP1–J4 46–
711/712	NONE	RUN #711/712 MG2 A33 46–328079G6	NONE	RUN #711/712 PP1–J91 46–328079G6
045	NONE	RUN #045 PP1 GND 46–	NONE	RUN #045 MG6 GND 46–

4–6 CABLE MAPS OF AFFECTED CABLES FOR UPGRADING TO SIGNA 10.x MR/i with EXCITE (MGD)

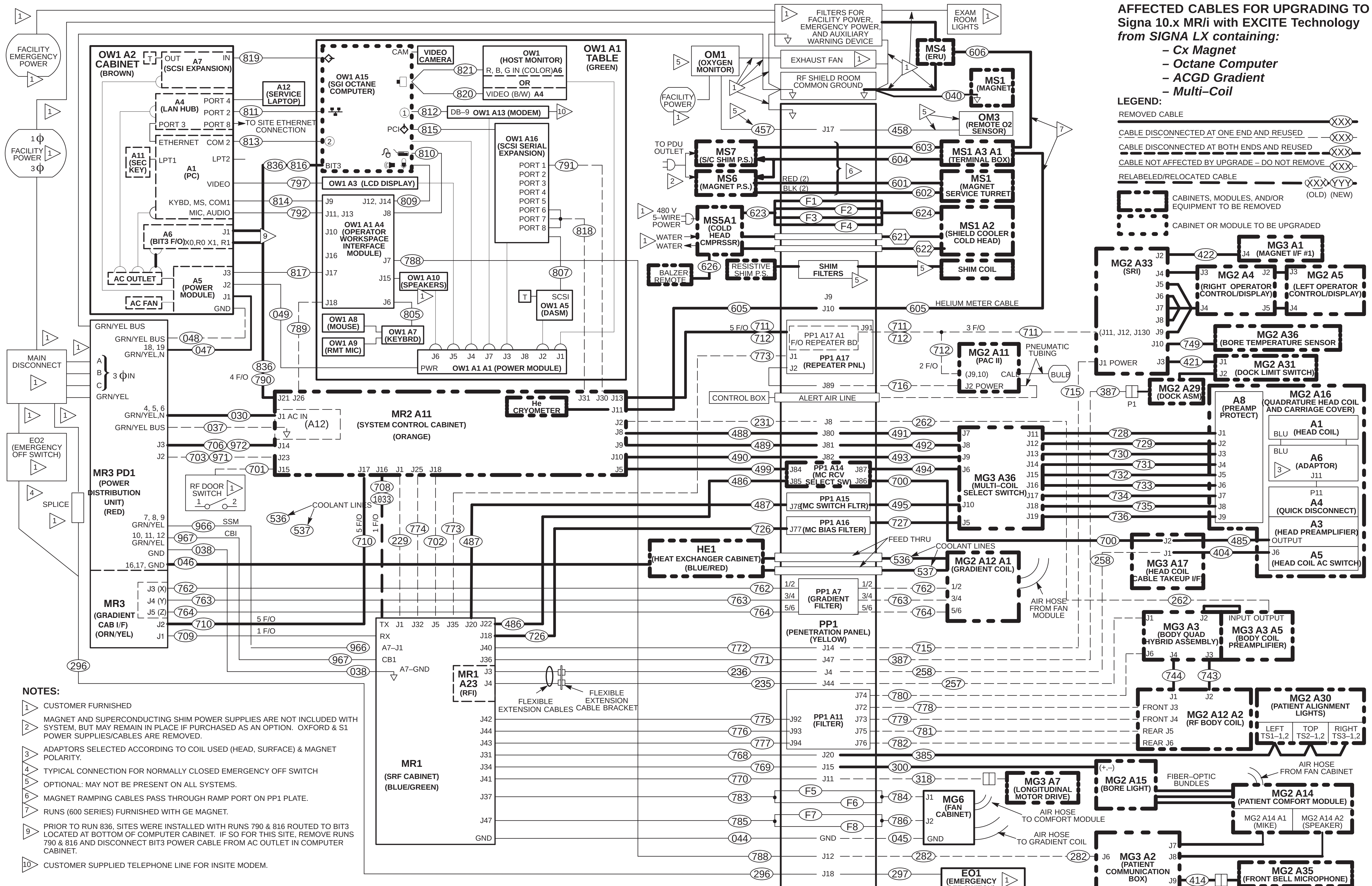
The following “Cable Removal” interconnect maps refer only to cables removed, relabeled, and/or relocated when upgrading:

AFFECTED CABLES FOR UPGRADING TO Signa 10.x MR/i with EXCITE Technology *from SIGNA LX containing:*

- *Cx Magnet*
- *Octane Computer*
- *ACGD Gradient*
- *Multi–Coil*

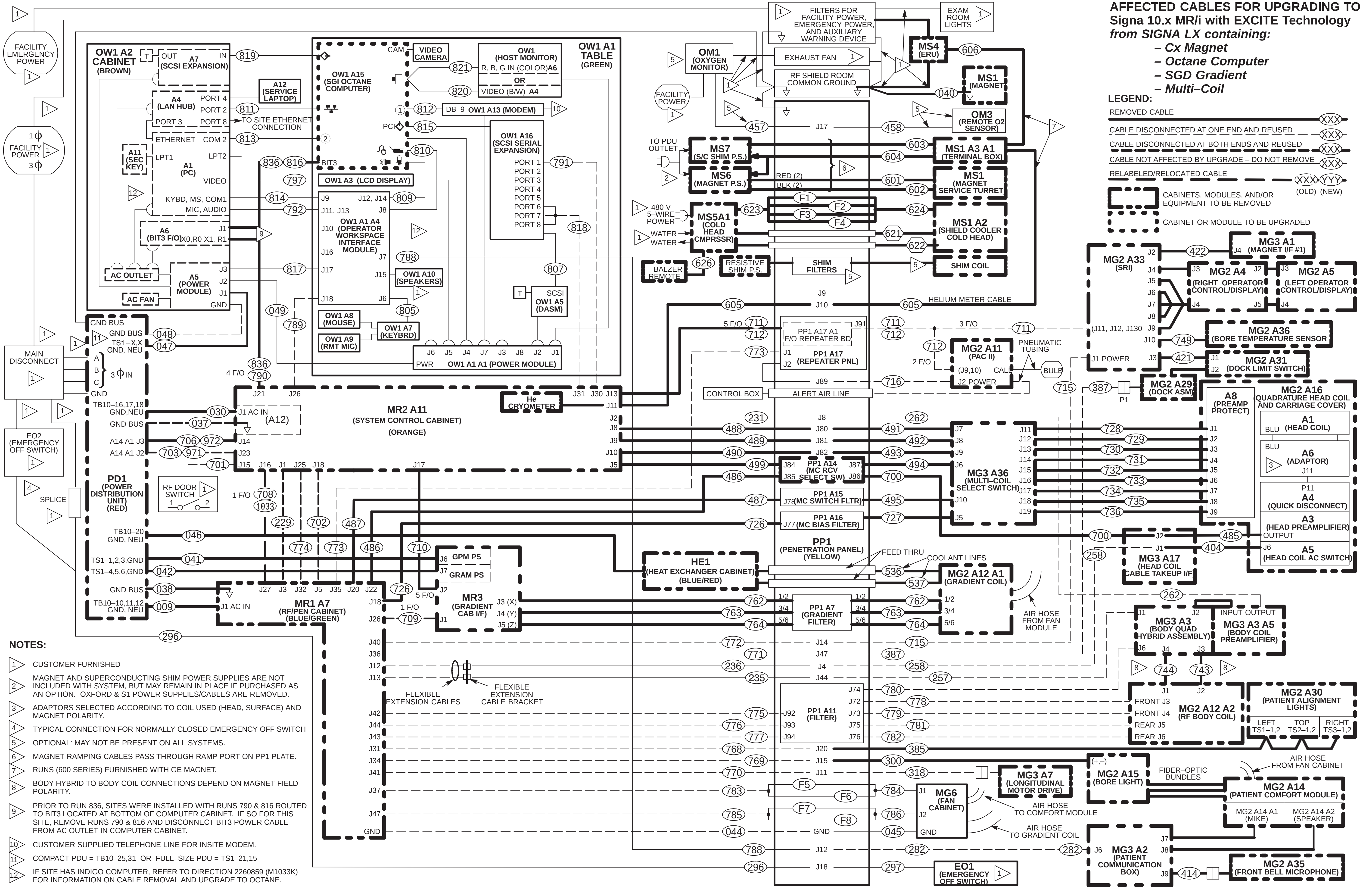
LEGEND:

- REMOVED CABLE (XXX)
- CABLE DISCONNECTED AT ONE END AND REUSED (XXX)
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE (XXX)
- RELABELED/RELOCATED CABLE (XXX) (YYY) (OLD) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED
- CABINET OR MODULE TO BE UPGRADED



NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) & MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
9. PRIOR TO RUN 836, SITES WERE INSTALLED WITH RUNS 790 & 816 ROUTED TO BIT3 LOCATED AT BOTTOM OF COMPUTER CABINET. IF SO FOR THIS SITE, REMOVE RUNS 790 & 816 AND DISCONNECT BIT3 POWER CABLE FROM AC OUTLET IN COMPUTER CABINET.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.



AFFECTED CABLES FOR UPGRADING TO Signa 10.x MR/i with EXCITE Technology *from SIGNA LX containing:*

- *Cx Magnet*
- *Octane Computer*
- *SGD Gradient*
- *Multi–Coil*

LEGEND:

- REMOVED CABLE
- CABLE DISCONNECTED AT ONE END AND REUSED
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE
- RELABELED/RELOCATED CABLE (OLD) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED
- CABINET OR MODULE TO BE UPGRADED

NOTES:

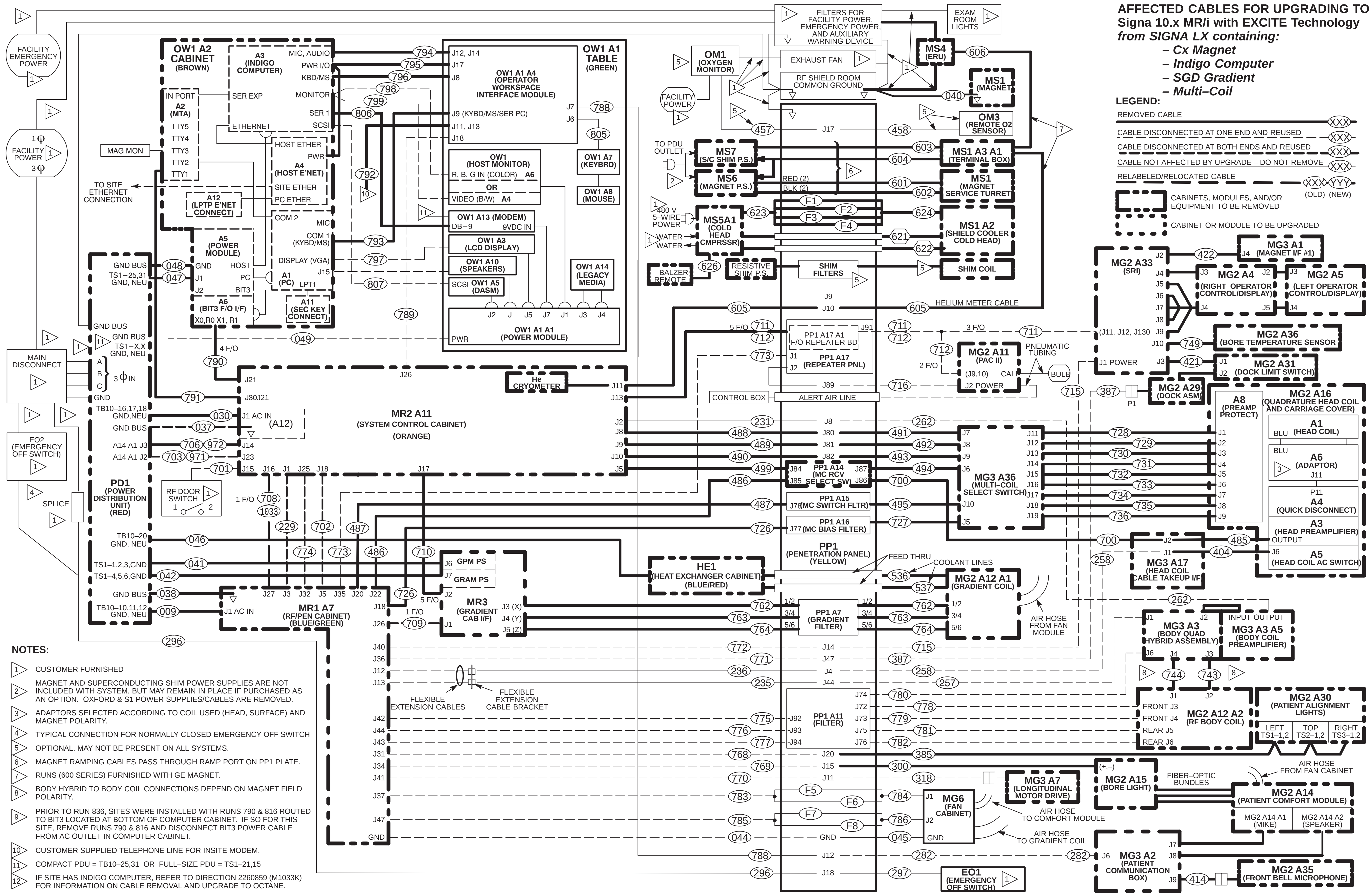
1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. PRIOR TO RUN 836, SITES WERE INSTALLED WITH RUNS 790 & 816 ROUTED TO BIT3 LOCATED AT BOTTOM OF COMPUTER CABINET. IF SO FOR THIS SITE, REMOVE RUNS 790 & 816 AND DISCONNECT BIT3 POWER CABLE FROM AC OUTLET IN COMPUTER CABINET.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.
11. COMPACT PDU = TB10–25,31 OR FULL–SIZE PDU = TS1–21,15
12. IF SITE HAS INDIGO COMPUTER, REFER TO DIRECTION 2260859 (M1033K) FOR INFORMATION ON CABLE REMOVAL AND UPGRADE TO OCTANE.

AFFECTED CABLES FOR UPGRADING TO Signa 10.x MR/i with EXCITE Technology *from SIGNA LX containing:*

- *Cx Magnet*
- *Indigo Computer*
- *SGD Gradient*
- *Multi–Coil*

LEGEND:

- REMOVED CABLE (XXX)
- CABLE DISCONNECTED AT ONE END AND REUSED (XXX)
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE (XXX)
- RELABELED/RELOCATED CABLE (XXX) (YYY) (OLD) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED
- CABINET OR MODULE TO BE UPGRADED

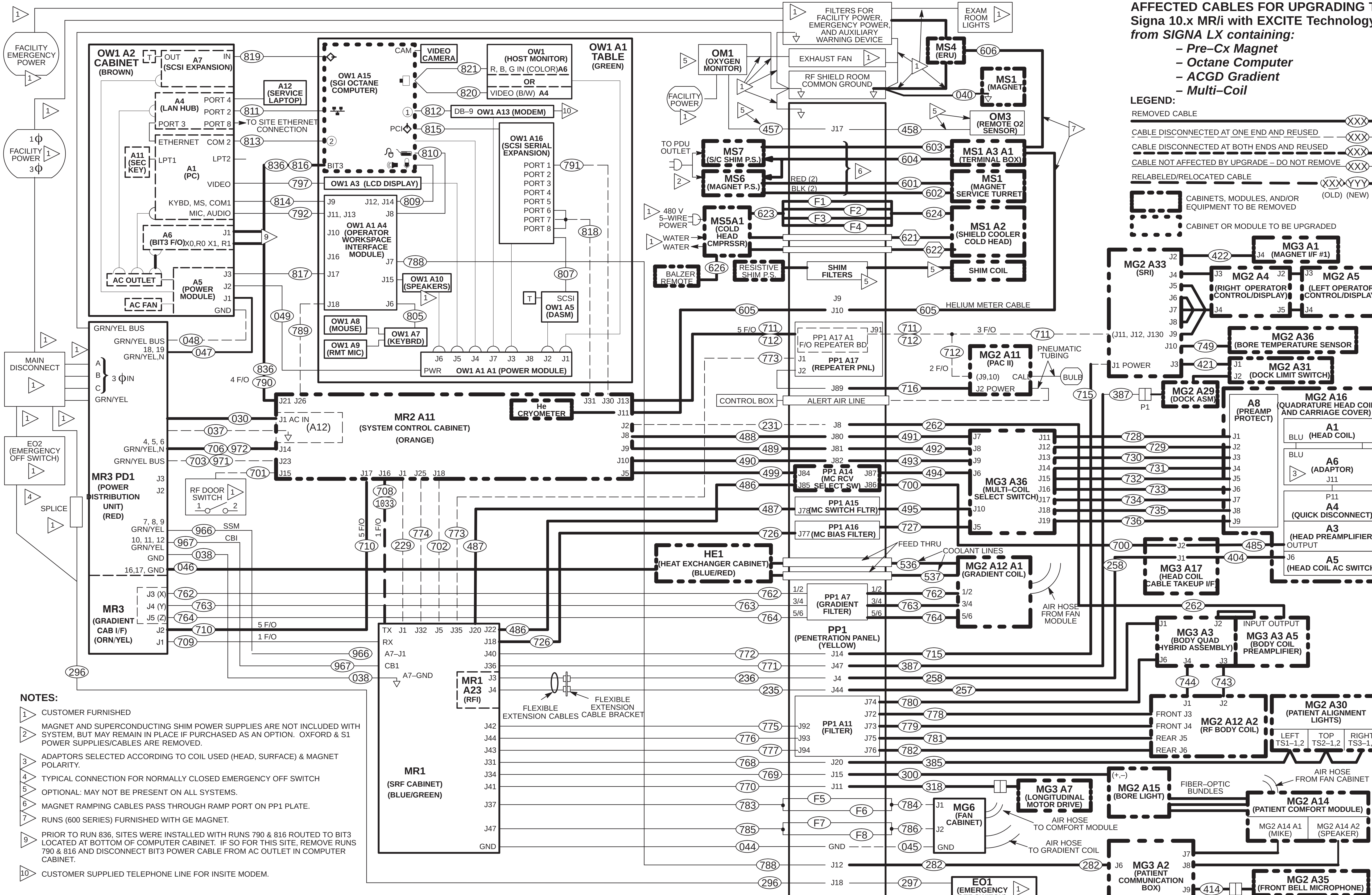


NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. PRIOR TO RUN 836, SITES WERE INSTALLED WITH RUNS 790 & 816 ROUTED TO BIT3 LOCATED AT BOTTOM OF COMPUTER CABINET. IF SO FOR THIS SITE, REMOVE RUNS 790 & 816 AND DISCONNECT BIT3 POWER CABLE FROM AC OUTLET IN COMPUTER CABINET.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.
11. COMPACT PDU = TB10–25,31 OR FULL–SIZE PDU = TS1–21,15
12. IF SITE HAS INDIGO COMPUTER, REFER TO DIRECTION 2260859 (M1033K) FOR INFORMATION ON CABLE REMOVAL AND UPGRADE TO OCTANE.

AFFECTED CABLES FOR UPGRADING TO Signa 10.x MR/i with EXCITE Technology *from SIGNA LX containing:*

- *Pre–Cx Magnet*
- *Octane Computer*
- *ACGD Gradient*
- *Multi–Coil*



NOTES:

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5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
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7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
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10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.

AFFECTED CABLES FOR UPGRADING TO Signa 10.x MR/i with EXCITE Technology *from SIGNA LX containing:*

- *Pre–Cx Magnet*
- *Octane Computer*
- *SGD Gradient*
- *Multi–Coil*

LEGEND:

- REMOVED CABLE (XXX)
- CABLE DISCONNECTED AT ONE END AND REUSED (XXX)
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE (XXX)
- RELABELED/RELOCATED CABLE (XXX) (YYY) (OLD) (NEW)
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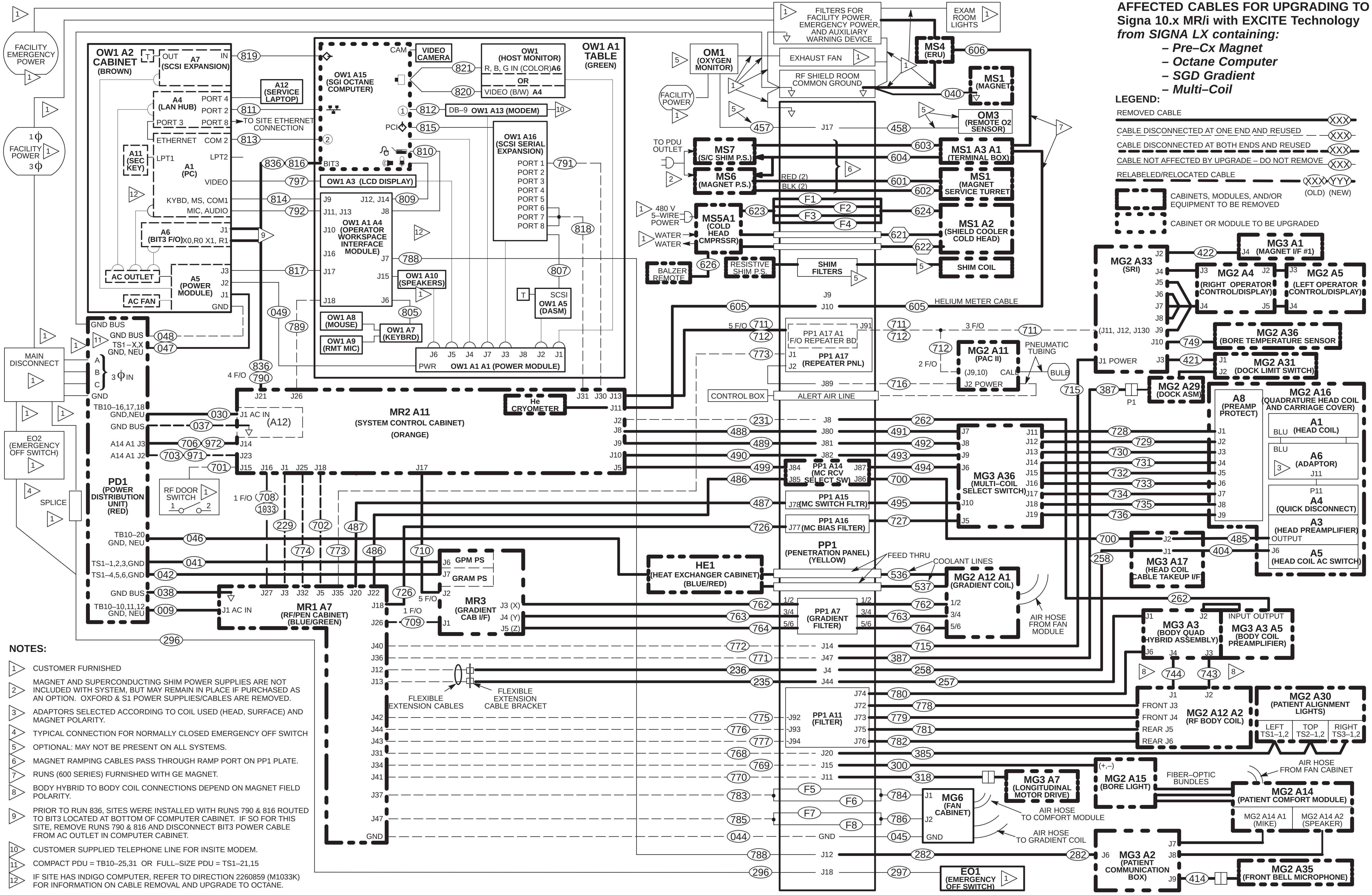

NOTES:

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2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
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4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. PRIOR TO RUN 836, SITES WERE INSTALLED WITH RUNS 790 & 816 ROUTED TO BIT3 LOCATED AT BOTTOM OF COMPUTER CABINET. IF SO FOR THIS SITE, REMOVE RUNS 790 & 816 AND DISCONNECT BIT3 POWER CABLE FROM AC OUTLET IN COMPUTER CABINET.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.
11. COMPACT PDU = TB10–25,31 OR FULL–SIZE PDU = TS1–21,15
12. IF SITE HAS INDIGO COMPUTER, REFER TO DIRECTION 2260859 (M1033K) FOR INFORMATION ON CABLE REMOVAL AND UPGRADE TO OCTANE.

AFFECTED CABLES FOR UPGRADING TO Signa 10.x MR/i with EXCITE Technology *from SIGNA LX containing:*

- *Non–Cx Magnet*
- *Indigo Computer*
- *SGD Gradient*
- *Multi–Coil*

LEGEND:

- REMOVED CABLE (XXX)
- CABLE DISCONNECTED AT ONE END AND REUSED (XXX)
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE (XXX)
- RELABELED/RELOCATED CABLE (XXX) (YYY) (OLD) (NEW)
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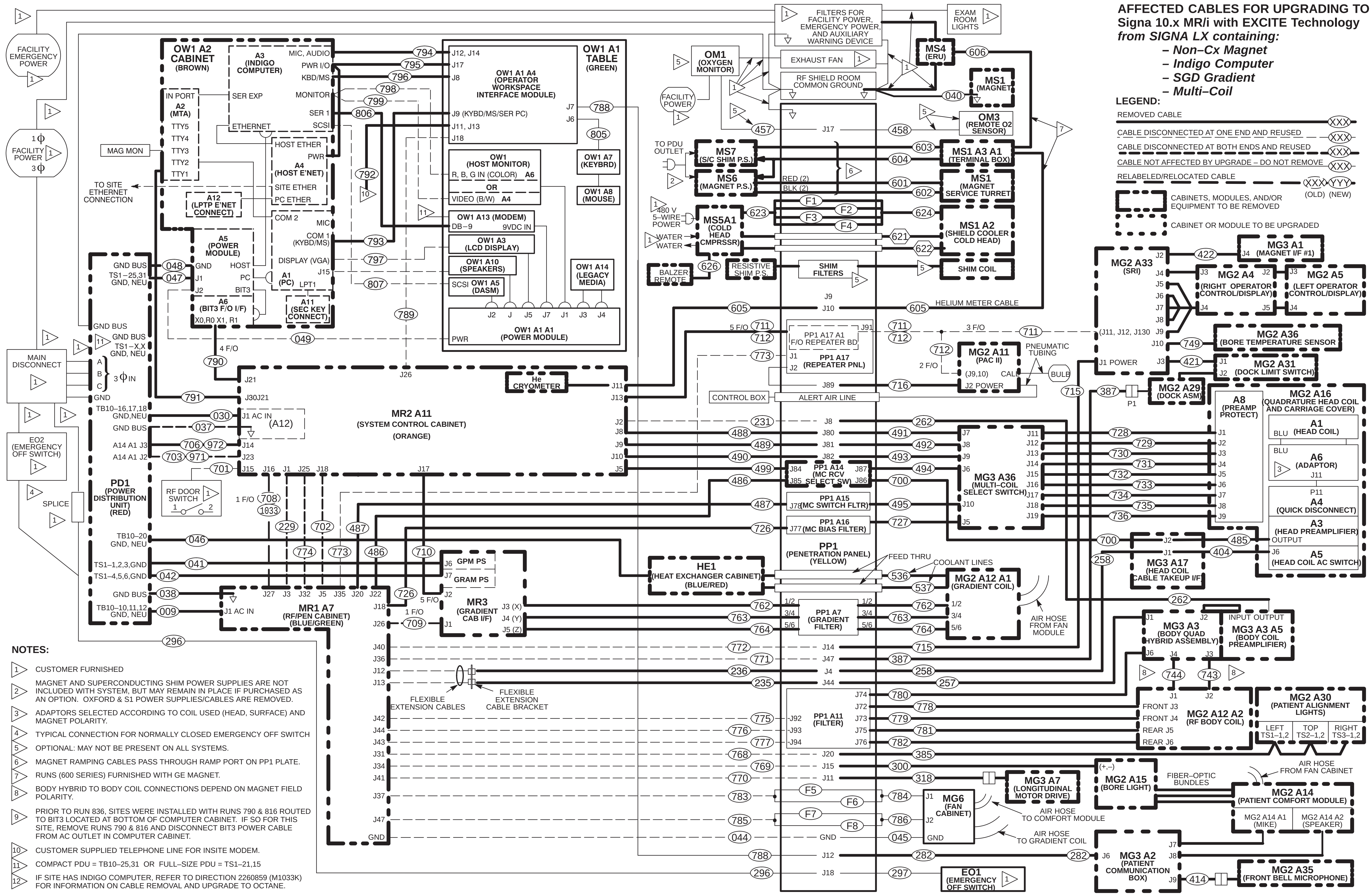


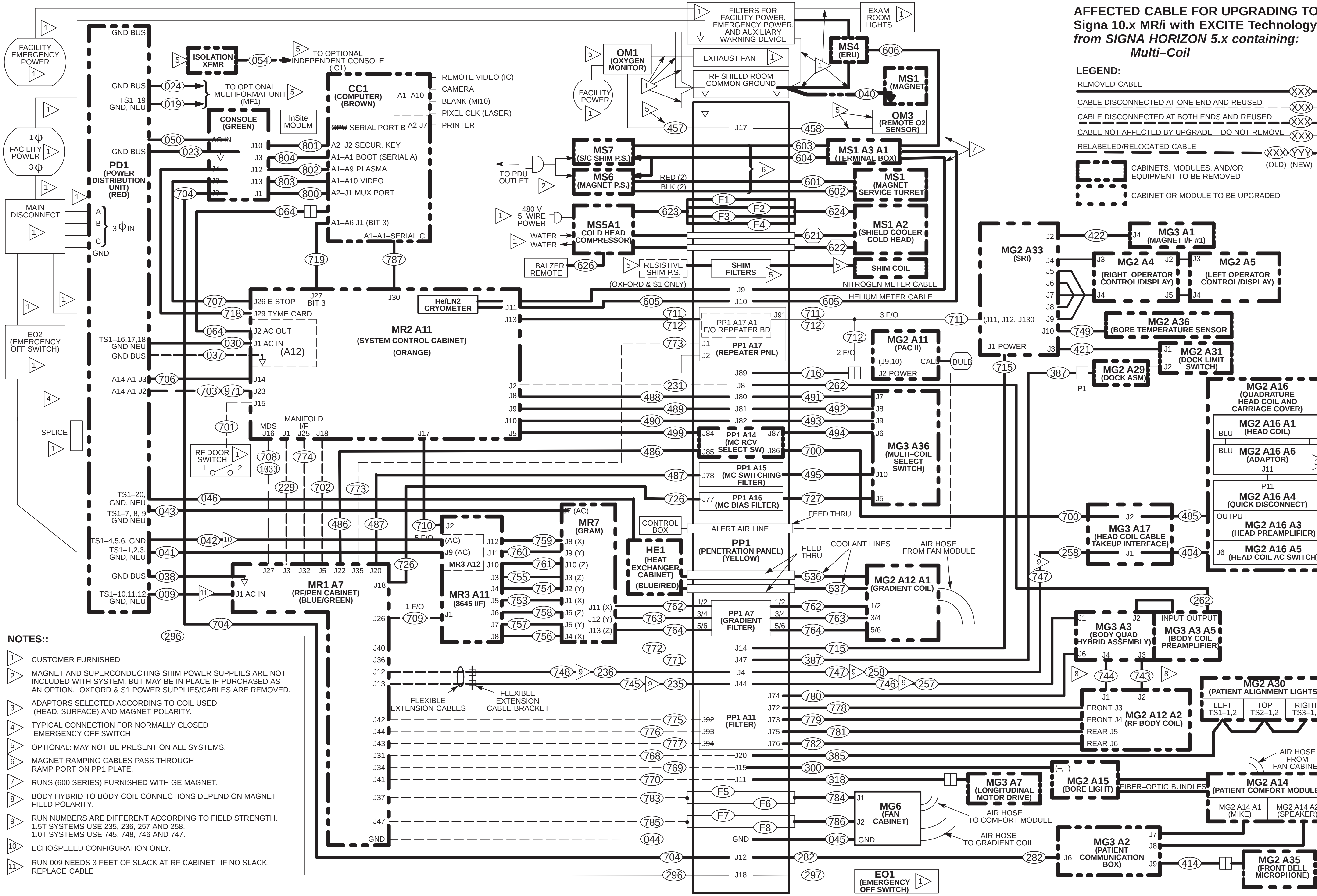
NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. PRIOR TO RUN 836, SITES WERE INSTALLED WITH RUNS 790 & 816 ROUTED TO BIT3 LOCATED AT BOTTOM OF COMPUTER CABINET. IF SO FOR THIS SITE, REMOVE RUNS 790 & 816 AND DISCONNECT BIT3 POWER CABLE FROM AC OUTLET IN COMPUTER CABINET.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.
11. COMPACT PDU = TB10–25,31 OR FULL–SIZE PDU = TS1–21,15
12. IF SITE HAS INDIGO COMPUTER, REFER TO DIRECTION 2260859 (M1033K) FOR INFORMATION ON CABLE REMOVAL AND UPGRADE TO OCTANE.

AFFECTED CABLE FOR UPGRADING TO Signa 10.x MR/i with EXCITE Technology *from SIGNA HORIZON 5.x containing: Multi–Coil*

LEGEND:

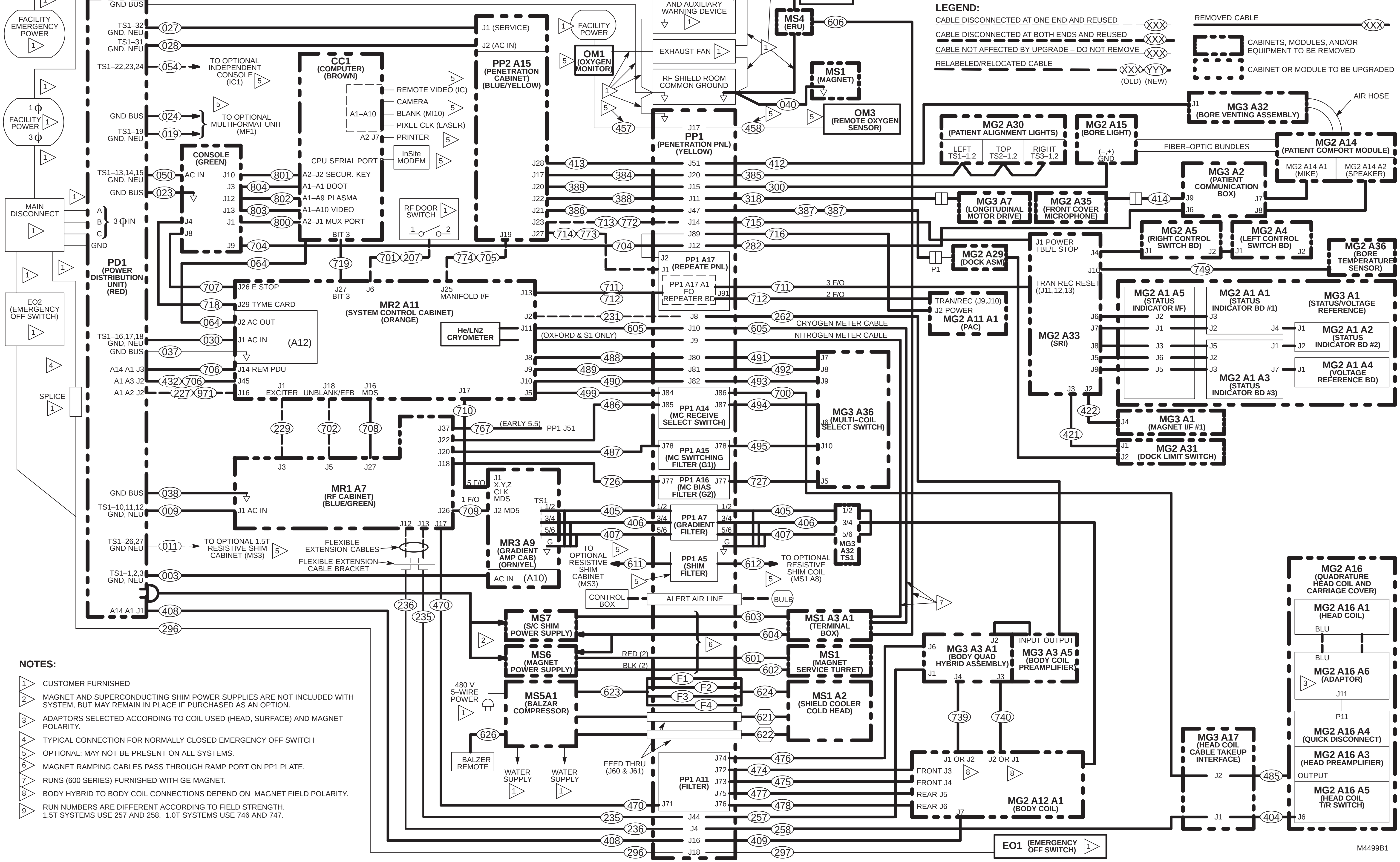
- REMOVED CABLE
- CABLE DISCONNECTED AT ONE END AND REUSED
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE
- RELABELED/RELOCATED CABLE (OLD) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED
- CABINET OR MODULE TO BE UPGRADED



NOTES::

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY BE IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. ECHOSPEEED CONFIGURATION ONLY.
11. RUN 009 NEEDS 3 FEET OF SLACK AT RF CABINET. IF NO SLACK, REPLACE CABLE

AFFECTED CABLE FOR UPGRADING TO Signa 10.x MR/i w/EXCITE Technology *from SIGNA ADVANTAGE 5.x*



NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 257 AND 258. 1.0T SYSTEMS USE 746 AND 747.

AFFECTED CABLE FOR UPGRADING TO SIGNA Signa 10.x MR/i with EXCITE Technology *from SIGNA ADVANTAGE 4.x*

The upgrade of a Signa Advantage 4.x system to current non–MGD system requires the de–installation and return of all cabinets, equipment, cables, magnet, and magnet enclosure parts except for the following:

- Emergency Off Switch and Runs 296 and 297
- Oxygen Monitor and Runs 457 and 458 (if installed)
- RF Door Switch and Run 205 (to be relabeled as Run 701)
- Run 229
- Run 231
- Heliax connector on Penetration Panel
- Consult with customer on any other options that the customer may have installed and could be used with this current magnet and system upgrade .

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4–7 CABLE REMOVAL FOR UPGRADING TO SIGNA 9.x MR/i WideOpen

The upgrade to a Signa 9.x (LX) will have many new cables delivered to the site. Table 4–5 lists all cables to be removed for upgrading to Signa 9.x MR/i WideOpen HiSpeed or EchoSpeed from various early system configurations, including Signa Advantage 4.x, Advantage 5.x, Horizon 5.x & 8.x, and Horizon 8.x with Cx magnet.

Note

Magnet room Fiber Optic runs 711/712 will not be removed during Magnet room deinstallation for existing Horizon 5.x and 8.x systems.

The letters on the right side of table 4–5 are defined by the following cable removal explanations:

A – Existing system is Cx Magnet Signa Horizon 8.x with SGD Cabinet upgrading to 9.x MR/*i*

B – Existing system is Cx Magnet Signa Horizon 8.x with Gradient/GRAM Cabinets upgrading to 9.x MR/*i*

C – Existing system is Signa Horizon 8.x with SGD Cabinet upgrading to MR/*i*

D – Existing system is Signa Horizon 8.x with Gradient/GRAM Cabinets upgrading to MR/*i*

E – Existing system is Signa Horizon 5.x with SGD Cabinet upgrading to MR/*i*

F – Existing system is Signa Horizon 5.x with Gradient/GRAM cabinets upgrading to MR/*i*

G – Existing system is 5.4 Signa Advantage upgrading to MR/*i*

H – Existing system is 4.5 or later but pre 5.4 Signa Advantage upgrading to MR/*i*

I – Existing system is 4.x but pre 4.5 Signa Advantage upgrading to MR/*i*

There are instances where the cable Run to be removed is located in the Equipment and Magnet room. These cables are identified in the table by the letters **E** and **M** and are defined by the following explanations:

E= Equipment Room side of cable Run to be removed.

M = Magnet Room side of cable Run to be removed.

X = Cable to be removed and returned for recycling. **DO NOT CUT CABLES.**

Note

Discard means to remove and return to GE for recycling

Refer to Section 4–11 for cable map information.

4–7 CABLE REMOVAL FOR UPGRADING TO SIGNA 9.x MR/i WideOpen (Continued)

TABLE 4–5
CABLES TO BE REMOVED AND RETURNED WHEN UPGRADING TO SIGNA 9.x MR/i WideOpen

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
002	PD1–A2–TS1–32/NEUT/GND	MR2–A12–SERVICE								X	X
003	PD1–A2–TS1–1,2,3/NEUT/GND	MR3–A10–AC IN							X	X	X
007	PD1–FEEDER 15	PP2–A6–AC IN								X	X
008	PD1–FEEDER 16	PP2–A6–SERVICE								X	X
009	PD1–A2–TS1–10,11,12/NEUT/GND	MR1–J1	X	X	X	X	X	X	X	X	X
012	PD1–A2–TS1–21/NEUT/GND	OC1–A8–J3 (AC IN)								X	X
013	PD1–A2–TS1–33/NEUT/GND	OC1–A8–J6 (SERVICE)								X	X
014	PD1–A2–TS1–34/NEUT/GND	CC1–A20–SERVICE								X	X
015	PD1–A2–TS1–13,14,15/NEUT/GND	CC1–A20–AC IN								X	X
016	PD1–A2–TS1–16,17,18/NEUT/GND	CC2–A20–AC IN								X	X
017	PD1–A2–TS1–34/NEUT/GND	CC2–A20–SERVICE								X	X
021	PD1–A2–GND	CC1–A20–GND								X	X
022	PD1–A2–GND	CC2–A20–GND								X	X
023	PD1–GND BUS	OC1–GND					X	X	X	X	X
027	PD1–A2–TS1–32/NEUT/GND	PP2–A15–J1–SERVICE							X	X	X
028	PD1–A2–TS1–31/NEUT/GND	PP2–A15–J2–AC IN							X	X	X
030	PD1–A2–TS1–10	MR2–A11–AC IN	X	X	X	X	X	X	X	X	X
035	PD1–A2–TS1–22,23,24/NEUT/GND	CC2–A20–AC IN								X	X
036	PD1–FEEDER 5–TS1–4,5,6 (Spect.)	MR6 (Spectroscopy Only)							X	X	X
037	PD1–GND BUS	MR2–A11 GND							X	X	X
038	PD1–GND BUS	MR1–A7 GND	X	X	X	X	X	X	X	X	X
040	RF SHIELD ROOM COMMON GND	MS1–GND	X	X	X	X	X	X	X	X	X
041	PD1–TS1–1,2,3,GND	MR3–A6	X	X	X	X	X				
042	PD1–TS1–4,5,6,GND	MR3–A7	X	X	X	X	X				
043	PD1–TS1–7,8,9,GND,NEU	MR7–A7			X	X	X				
045	PP1–GND **(See Note 2)**	FAN MODULE GND **(Note 2)**			X	X	X	X			
046	PD1–TS1–20,GND,NEU	HE1	X	X	X	X	X				
047	PD1–TS1–XX	OW1–A2–A5–J1	X	X	X	X					
048	PD1–GND	OW1–A2–A5–GND									
049	OW1–A1–A1	OW1–A2–A5–J2									
050	OC1–AC–IN	PD1–TS1–13,14,15,NUE,GND					X	X	X		
064	MR2–A12–J1–AC OUT	CC1					X	X	X		
200	MR2–A11–J19	MR3–A9–J1								X	X
201	MR2–A11–J20	MR3–A9–J2								X	X
202	MR2–A11–J21	MR3–A9–J3								X	X
203	MR2–A11–J22	MR3–A9–J4								X	X
207	RF DOOR SWITCH	MR2–A11–J15									
223	OC1–A8–J13	MR2–A1–J36								X	X
224	CC1–A21–J19	MR1–A7–J8								X	X
226	MR1–A7–J5	MR2–A11–J14								X	X
227	PD1–A1–A2–J2	MR2–A11–J16									
228	CC1–A21–J10	MR2–A11–J17								X	X

4–7 CABLE REMOVAL FOR UPGRADING TO SIGNA 9.x MR/i WideOpen (Continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
229	MR1–A7–J3	MR2–A11–J1									
231	MR2–A11–J2	PP1–J8									
233	MR2–A11–J5	PP1–J7									
235	MR1–A7–J13 **(See Note 1)**	PP1–J44 **(See Note 1)**			X	X	X	X	X		
236	MR1–A7–J12 **(See Note 1)**	PP1–J4 **(See Note 1)**			X	X	X	X	X		
238	CC1–A21–J14	OC1–A8–J8								X	X
239	CC1–A21–J15	OC1–A8–J10								X	X
244	CC1–A21–J13	BTI–HOST								X	
248	CC2–A21–J1	OC1–A9–J1								X	X
249	CC2–A21–J22	OC1–A8–J12								X	X
256	PP1–J12	OC1–A8–J11								X	X
257	PP1–J44 **(See Note 1)**	MG3–A3–A1–J1 **(See Note 1)**			X	X	X	X	X	X	X
258	PP1–J4 **(See Note 1)**	MG3–A17–J1 **(See Note 1)**			X	X	X	X	X	X	X
260	PP1–J7	MG3–A17–J2								X	X
261	PP1–J6	MG2–A11–A1–J1								X	X
262	PP1–J8	MG3–A3–A5–OUT	X	X	X	X	X	X	X	X	X
274	CC1–A21–J20	MR1–A7–J7								X	X
280	PP1–J15	PP2–A1–J2								X	X
281	PP1–J11	PP2–A1–J3								X	X
282	PP1–J12	MG3–A2–J6			X	X	X	X	X	X	X
287	CC2–A21–J15	CC1–A21–J2								X	X
296	FACILITY DISCONNECT	PP1–J18									
297	EO1	PP1–J18									
300	PP1–J15	MG2–A15	X	X	X	X	X	X	X	X	X
301	CC2–A21–J20	CC1–A21–J28								X	X
302	CC2–A21–J21	CC1–A21–J29								X	X
318	MG3–A7	PP1–J11			X	X	X	X	X	X	X
322	MR2–A11–J11	PP2–A6–J18								X	X
325	CC1–A21–J27	CC2–A21–J19								X	X
327	CC1–A21–J25	CC2–A21–J17								X	X
346	MR2–A11–J41	CM1–FRONT								X	X
347	MR2–A11–J42	CM1–BACK								X	X
354	CM1	ST1								X	X
370	PP1–A9–J46	MR1–A7–J19									X
384	PP1–J20	PP2–A15–J17							X	X	X
385	PP1–J20	MG2–A30	X	X	X	X	X	X	X	X	X
386	PP1–J47	PP2–A15–J21							X	X	X
387	PP1–J47	MG2–A29			X	X	X	X	X	X	X
388	PP1–J11	PP2–A15–J22							X	X	X
389	PP1–J15	PP2–A15–J20							X	X	X
404	MG3–A17–J1	MG2–A16–A5–J6	X	X	X	X	X	X	X	X	X
405	MR3–A9–TS1	MG3–TS1–1,2							X	X	X
406	MR3–A9–TS1	MG3–TS1–3,4							X	X	X
407	MR3–A9–TS1	MG3–TS1–5,6							X	X	X
408	PD1–A1–A3–J3	PP1–J16							X	X	X

4–7 CABLE REMOVAL FOR UPGRADING TO SIGNA 9.x MR/i WideOpen (Continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
409	PP1–J16	MG2–A12–A1–J7							X	X	X
410	PP1–J14	PP2–A15–J23									X
411	PP1–J14	MG3–A13–J1									X
412	PP1–J51	MG3–A32–J1							X	X	X
413	PP1–J51	PP2–A15–J28(A16–J9)							X	X	X
414	MG3–A35	MG3–A2–J9	X	X	X	X	X	X	X	X	X
415	PP1–A8–J45	MG3–A33–J7									X
416	MR1–A7–J21	PP1–A8–J45									X
417	PP2–A15–J19	MR2–A11–J32								X	X
418	MR2–A11–J39	PP2–A15–J3									X
419	MR2–A11–J26	PP1–A10–J1								X	X
420	MG3–A33–J1	PP1–A10–J2								X	X
421	MG2–A33–J3	MG3–A31–J1	X	X	X	X	X	X	X	X	X
422	MG2–A33–J2	MG3–A1–J4	X	X	X	X	X	X	X	X	X
424	PP1–A10–J5/J6	MG2–A33–J11/J12								X	X
425	MR2–A11–J12	PP1–J6									X
426	MR1–A7–J9	MR3–A9–J5								X	X
427	MR2–A11–J17	CC1–A21–J13								X	X
428	MR2–A11–J44	CC1–A21–J35								X	X
429	BT1	CC1–A21–J25								X	X
430	MR2–A11–J24	CC1–A21–J26								X	X
431	MR2–A11–J25	CC1–A21–J27								X	X
432	PD1–A1–A3–J2	MR2–A11–J45								X	X
433	OC1–A9–J1	CC1–A21–J10								X	X
434	CC1–A8–J12	CC1–A21–J29								X	X
435	MR2–A11–J13	PP1–A10–J3								X	X
436	MG2–A12–A1–J1	MG3–A33–J3								X	X
437	MG2–A12–A1–J2	MG3–A33–J4								X	X
439	MG3–A3–J4	MG3–A33–J1								X	X
440	MG3–A3–J3	MG3–A34–J2								X	X
441	MG3–A33–J7	MG3–A34–J8								X	X
446	PP1–A9–J46	MG2–A12–A13–J5									X
447	PP1–A9–J46	MG2–A12–A13–J3									X
448	MR2–A11–J44	CC1–A21–J1								X	
449	MR2–A11–J24	CC1–A21–J34								X	
450	MR2–A11–J25	CC1–A21–J35								X	
453	PP1–J50	MG2–A11–A1–J4								X	X
454	MR2–A11–J34	PP1–J50								X	X
470	MR1–A7–J17	PP1–A11–J71							X	X	
474	PP1–A11–J72	MG3–A12–A1–J3							X	X	
475	PP1–A11–J73	MG3–A12–A1–J4							X	X	
476	PP1–A11–J74	MG3–A33–J7							X	X	
477	PP1–A11–J75	MG3–A12–A1–J5							X	X	
478	PP1–A11–J76	MG3–A12–A1–J6							X	X	
485	MG3–A17–J2	MG2–A16–A3–OUT			X	X	X	X	X	X	X

4–7 CABLE REMOVAL FOR UPGRADING TO SIGNA 9.x MR/i WideOpen (Continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
486	PP1–A14–J85	MR1–A7–J22								X	X
487	PP1–A15–J87	MR1–A7–J20								X	X
488	PP1–J80	MR2–A11–J8								X	X
489	PP1–J81	MR2–A11–J9								X	X
490	PP1–J82	MR2–A11–J10								X	X
491	PP1–J80	MG3–A36–J7			X	X	X	X	X	X	X
492	PP1–J81	MG3–A36–J8			X	X	X	X	X	X	X
493	PP1–J82	MG3–A36–J9			X	X	X	X	X	X	X
494	PP1–A14–J87	MG3–A36–J6			X	X	X	X	X	X	X
495	PP1–A15–J78	MG3–A36–J10			X	X	X	X	X	X	X
536	HE1 (Coolant Hose)	MG2–A12–A1–IN									
537	HE1 (Coolant Hose)	MG2–A12–A1–OUT									
601	MS1 (RED)	MS6 (RED)	X	X	X	X	X	X	X	X	X
602	MS1 (BLK)	MS6 (BLK)	X	X	X	X	X	X	X	X	X
603	MS1–A13–A1	MS7	X	X	X	X	X	X	X	X	X
604	MS1–A13–A1	MS6/MS7	X	X	X	X	X	X	X	X	X
605	PP1–J10 (Magnet Room)	MS1–A3–A1 (Magnet Room)	M	M	M	M	M	M	M	M	M
605	PP1–J10 (Equipment Room)	MR1–J11 (Equipment Room)	E	E	E	E	E	E	E	E	E
606	MS4	MS1–A3–A1	X	X	X	X	X	X	X	X	X
621	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	X	X	X	X
622	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	X	X	X	X
623	MS5–A1	PP1–F1,F2,F3,F4	X	X	X	X	X	X	X	X	X
624	MS1–A2	PP1–F1,F2,F3,F4	X	X	X	X	X	X	X	X	X
626	MS5–A1	BALZER REMOTE			X	X	X	X	X	X	X
700	PP1–A14–J86	MG3–A17–J2			X	X	X	X	X		
701	RF DOOR SWITCH	MR2–A11–J15									
702	MR2–A11–J18	MR1–A7–J5									
703	PD1–A14–A1–J2	MR2–A11–J23									
704	PP1–J12	OC1–A10–J9					X	X	X		
705	PP2–A15–J19	MR2–A11–J25									
706	PD1–A14–A1–J3	MR2–A11–J14									
707	MR2–A11–J26–ESTOP	OC1–J8					X	X	X		
708	MR2–A11–J16	MR1–A7–J27							X		
709	MR1–A7–J26	MR3–A9–J1							X		
710	MR2–A11–J17	MR3–A9–J2							X		
711	PP1–A17–A1–J91	MG2–A33							M		
712	PP1–A17–A1–J91	MG2–A11–A1							M		
715	PP1–J14	MG2–A33–J1			X	X	X	X	X		
716	PP1–J89	MG2–A11–A1–J2			X	X	X	X	X		
718	MR2–A11–J29	OC1–A10–J4					X	X	X		
719	CC1–A1–A6–J1	MR2–A15–A3–J1					X	X	X		
726	PP1–A16–J77	MR1–A7–J18								X	X
727	PP1–A16–J77	MG3–A36–J5			X	X	X	X	X	X	X
728	MG3–A36–J11	MG2–A16–OUT–J1	X	X	X	X	X	X	X	X	X
729	MG3–A36–J12	MG2–A16–OUT–J2	X	X	X	X	X	X	X	X	X

4–7 CABLE REMOVAL FOR UPGRADING TO SIGNA 9.x MR/i WideOpen (Continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
730	MG3–A36–J13	MG2–A16–OUT–J3	X	X	X	X	X	X	X	X	X
731	MG3–A36–J14	MG2–A16–OUT–J4	X	X	X	X	X	X	X	X	X
732	MG3–A36–J15	MG2–A16–OUT–J5	X	X	X	X	X	X	X	X	X
733	MG3–A36–J16	MG2–A16–OUT–J6	X	X	X	X	X	X	X	X	X
734	MG3–A36–J17	MG2–A16–OUT–J7	X	X	X	X	X	X	X	X	X
735	MG3–A36–J18	MG2–A16–OUT–J8	X	X	X	X	X	X	X	X	X
736	MG3–A36–J19	MG2–A16–OUT–J9	X	X	X	X	X	X	X	X	X
739	MG3–A3–A1–J4	MG2–A12–A1–J1 OR J2							X		
740	MG3–A3–A1–J3	MG2–A12–A1–J2 OR J1							X		
743	MG3–A3–J3	MG2–A12–J2	X	X	X	X	X	X			
744	MG3–A3–J4	MG2–A12–J1	X	X	X	X	X	X			
745	PP1–J44 **(See Note 1)**	MR1–A7–J13 **(See Note 1)**			X	X	X	X	X		
746	PP1–J44 **(See Note 1)**	MG3–A3–A1–J1 **(See Note 1)**			X	X	X	X	X		
747	PP1–J4 **(See Note 1)**	MG3–A17–J1 **(See Note 1)**			X	X	X	X	X		
748	PP1–J4 **(See Note 1)**	MR1–A7–J12 **(See Note 1)**			X	X	X	X	X		
749	MG2–A36	MG2–A33–J10	X	X	X	X	X	X	X		
753	MR7–A4–J1	MR3–A11–J5		X		X		X			
754	MR7–A4–J2	MR3–A11–J4		X		X		X			
755	MR7–A4–J3	MR3–A11–J3		X		X		X			
756	MR7–A4–J4	MR3–A11–J8		X		X		X			
757	MR7–A4–J5	MR3–A11–J7		X		X		X			
758	MR7–A4–J6	MR3–A11–J6		X		X		X			
759	MR3–A11–J12	MR7–A4–J8		X		X		X			
760	MR3–A11–J11	MR7–A4–J9		X		X		X			
761	MR3–A11–J10	MR7–A4–J10		X		X		X			
762	MG3–A32–TS1/2 **(See Note 3)**	PP1–A7–1/2 **(See Note 3)**			X	X	X	X			
762	MR3–J3 **(See Note 3)**	PP1–A7–1/2 **(See Note 3)**	X	X	X	X	X	X			
763	MG3–A32–TS3/4 **(See Note 3)**	PP1–A7–3/4 **(See Note 3)**			X	X	X	X			
763	MR3–J4 **(See Note 3)**	PP1–A7–3/4 **(See Note 3)**	X	X	X	X	X	X			
764	MG3–A32–TS5/6 **(See Note 3)**	PP1–A7–5/6 **(See Note 3)**			X	X	X	X			
764	MR3–J5 **(See Note 3)**	PP1–A7–5/6 **(See Note 3)**	X	X	X	X	X	X			
767	MR1–A7–J37 **(See Note 4)**	PP1–J51 **(See Note 4)**						X	X		
771	PP1–J47	MR1–A7–J36									
772	PP1–J14	MR1–A7–J40									
773	PP1–A17–J1	MR1–A7–J35									
774	MR2–A11–J25	MR1–A7–J32									
778	MG2–A12–A13–J3	PP1–A11–J72			X	X	X	X			
779	MG2–A12–A13–J4	PP1–A11–J73			X	X	X	X			
780	MG3–A3–J6	PP1–A11–J74			X	X	X	X			
781	MG2–A12–A12–J5	PP1–A11–J75			X	X	X	X			
782	MG2–A12–A12–J6	PP1–A11–J76			X	X	X	X			
784	PP1–F5/F6	MG6–J1			X	X	X	X			
786	PP1–F7/F8	MG6–J2			X	X	X	X			
787	MR2–A11–J30	CC1–A1–A1–SERIAL C					X	X			
788	OW1–A1–A4–J7	PP1–J12									

4–7 CABLE REMOVAL FOR UPGRADING TO SIGNA 9.x MR/i WideOpen (Continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
789	OW1–A1–A4–J18	MR2–A11–J26									
790	OW1–A2–A6–BIT3	MR2–A11–J21									
797	OW1–A2–A1–PC	OW1–A3–LCD									
798	OW1–A2–A3–MONITOR	OW1–A6–R,B,G									
799	OW1–A2–A3–MONITOR	OW1–A4–B/W									
800	OC1–J1	CC1–A2–J1 (MUX)					X	X	X		
801	OC1–J10	CC1–A2–J2 (SECUR. KEY)					X	X	X		
802	OC1–J12	CC1–A1–J9 (PLASMA)					X	X	X		
803	OC1–J13	CC1–A1–A10 (VIDEO)					X	X	X		
804	OC1–J3	CC1–A1–A1 (BOOT)					X	X	X		
805	OW1–A1–A4–J6	OW1–A7									
807	OW1–A2–A3–HOST	OW1–A5									
816	OW1–A2–A6–J1	OW1–A15–BIT3									
820	OW1–A2–A3–MONITOR	OW1–A4–B/W									
821	OW1–A2–A3–MONITOR	OW1–A6–R,B,G									
836	MR2–A11–J21	OW1–A15–BIT3									
971	MR2–A11–J23	MR3–PD1–J2									
972	MR2–A11–J14	MR3–PD1–J3									

Notes:

1. 200 series cables are 1.5T specific. 700 series cables are 1.0T specific. These cable will have labels added after they are routed and terminated.
2. Only present if system is at Horizon
3. The cable is divided between the magnet room and equipment room.
4. Run 767 is only found on a few early Horizon 5.x (5.5) systems.

4–8 RE–LABELING AND RE–ROUTING OF EXISTING PRE–HORIZON CABLES TO 9.x WideOpen

Several existing cables need to be re–labeled and then re–routed per the new labels. Refer to Table 4–6.

Runs 706, 772, 773, and 774 are only re–labled for Advantage 5.4 systems upgraded to 9.x WideOpen. All pre–5.4 system upgrades are provided new cables for Runs 706, 772, 773, and 774.

TABLE 4–6
PRE–HORIZON CABLES RE–LABELED AND RE–LOCATED FOR UPGRADE TO 9.x

OLD RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
207	4.x	RUN #207 MR2–A11–J6 46–243775G411	RUN #701 MR2–A11–J15 46–243775G807	RUN #207 DOOR SWITCH 46–243775G411	RUN #701 RF DOOR SWITCH 46–243775G807
227	4.x	RUN #227 MR2–A11–J16 46–243775G464	RUN #703/971 PD1–A1–A2–J2 PD1–A14–A1–J2 MR3–PD1–J2	RUN #227 PD1–A1–A2–J2 46–243775G464	RUN #703/971 MR2–A11–J23 46–258431G941
432	4.x	RUN #432 PD1–A1–A3–J2 46–243775G676	RUN #706/972 PD1–A1–A3–J2 PD1–A14–A1–J3 MR3–PD1–J3	RUN #432 MR2–A11–J45 46–243775G676	RUN #706/972 MR2–A11–J14 46–258431G940
705	5.x	RUN #705 MR2–A11–J25 46–243775G663	RUN #774 MR2–A11–J25 46–243775G811	RUN #705 PP2–A15–J19 46–243775G663	RUN #774 MR1–A7–A32 46–243775G811
713	5.x	RUN #713 PP2–A15–J23 46–243775G676	RUN #772 MR1–A7–J40 46–243775G808	RUN #713 PP1–J14 46–243775G676	RUN #772 PP1–J14 46–243775G808
714	5.x	RUN #714 PP2–A5–J27 46–243775G676	RUN #773 MR1–A7–J35 46–234775G808	RUN #714 PP1–J89 46–243775G676	RUN #773 PP1–J89 46–234775G808

4–9 RE–LABELING & RE–ROUTING HORIZON 5.x CABLES TO 9.x WideOpen WITH ACGD

Several existing Horizon 5.x cables need to be re–labeled and then re–routed per the new labels. Refer to Table 4–7.

Runs 041, 042, 709, and 710 are re–labled only for Horizon 5.x systems upgraded to 9.x with ACGD. All pre–5.x system upgrades are provided new cables for Runs 041, 042, 709, and 710

TABLE 4–7
HORIZON 5.x & 8.x CABLES RE–LABELED AND RE–LOCATED FOR UPGRADE TO ACGD

RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
703	5.x 8.x	RUN #703 PD1–A1–A2–J2 PD1–A14–A1–J2 46–258431G809	RUN #703/971 PD1–A1–A2–J2 PD1–A14–A1–J2 MR3–PD1–J2	RUN #703 MR2–A11–J23 46–228431G809	RUN #703/971 MR2–A11–J23 46–258431G941
706	5.x 8.x	RUN #706 MR1–PD1–J3 PD1–A14–A1–J3 46–258431G808	RUN #706/972 PD1–A1–A3–J2 PD1–A14–A1–J3 MR3–PD1–J3	RUN #706 MR2–A11–J14 46–258431G808	RUN #706/972 MR2–A11–J14 46–258431G940
709	5.x	RUN #709TR–1 MR3–A11–J1 46–317077G802	RUN #709TR–1 MR3CABI/FJ2 46–317077G803	RUN #709TR–1 MR1–A7–RX 46–317077G802	RUN #709TR–1 MR1–A7–RX 46–317077G803
710	5.x	RUN #710TR–1 MR3–A11–J2 46–317085G602	RUN #710TR–1 MR3CABI/FJ1 46–317085G603	RUN #710TR–1 MR2–A11–J17 46–317085G602	RUN #710TR–1 MR2–A11–J17 46–317085G603
710–1	5.x	RUN #710–1	RUN #710–1 MR3GIPJ2	NO CHANGE	NO CHANGE
710–2	5.x	RUN #710–2	RUN #710–2 MR3GIPJ3	NO CHANGE	NO CHANGE
710–3	5.x	RUN #710–3	RUN #710–3 MR3GIPJ4	NO CHANGE	NO CHANGE
710–4	5.x	RUN #710–4	RUN #710–4 MR3GIPJ1	NO CHANGE	NO CHANGE
710–5	5.x	RUN #710–5	RUN #710–5 MR3GIPJ5	NO CHANGE	NO CHANGE

4–10 LABELS ADDED TO NEW MAGNET ROOM CABLES (Non–Cx Magnet Only)

The following table defines the additional cables which need to be relabeled when upgrading to a LCC WideOpen magnet, unless the magnet being replaced is a Cx magnet. **In the case of a Cx magnet being replaced by a LCC magnet, the cables in the following table do not need to be relabeled.**

TABLE 4–8
MAGNET ROOM LABELS TO BE RE–LABELED

OLD RUN	ONE END OLD LABEL	ONE END NEW LABEL	OTHER END OLD LABEL	OTHER END NEW LABEL
257 (1.5T)	NONE	RUN #257 MG3 A3 J1 46–	NONE	RUN #257 PP1–J44 46–
258 (1.5T)	NONE	RUN #258 MG3 A17 J1 46–	NONE	RUN #257 PP1–J4 46–
746 (1.0T)	NONE	RUN #746 MG3 A3 J1 46–	NONE	RUN #746 PP1–J44 46–
747 (1.0T)	NONE	RUN #747 MG3 A17 J1 46–	NONE	RUN #747 PP1–J4 46–
711/712	NONE	RUN #711/712 MG2 A33 46–328079G6	NONE	RUN #711/712 PP1–J91 46–328079G6
045	NONE	RUN #045 PP1 GND 46–	NONE	RUN #045 MG6 GND 46–

4–11 CABLE MAPS OF AFFECTED CABLES FOR UPGRADING TO SIGNA 9.x MR/i WideOpen

The following “Affected Cables” interconnect maps refer only to cables removed, relabeled, and/or relocated when upgrading from:

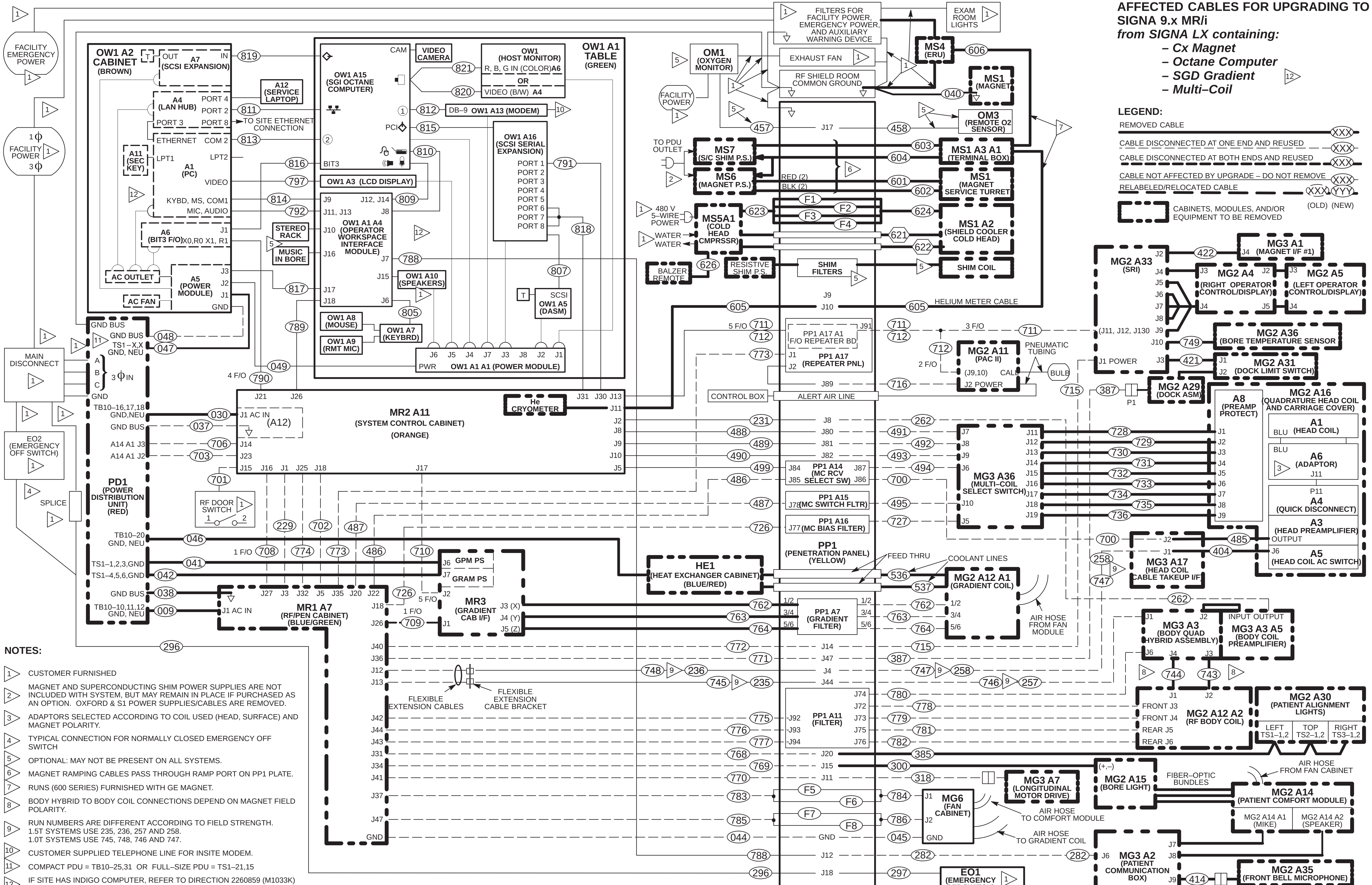
AFFECTED CABLES FOR UPGRADING TO SIGNA 9.x MR/i

from SIGNA LX containing:

- *Cx Magnet*
- *Octane Computer*
- *SGD Gradient* 12
- *Multi–Coil*

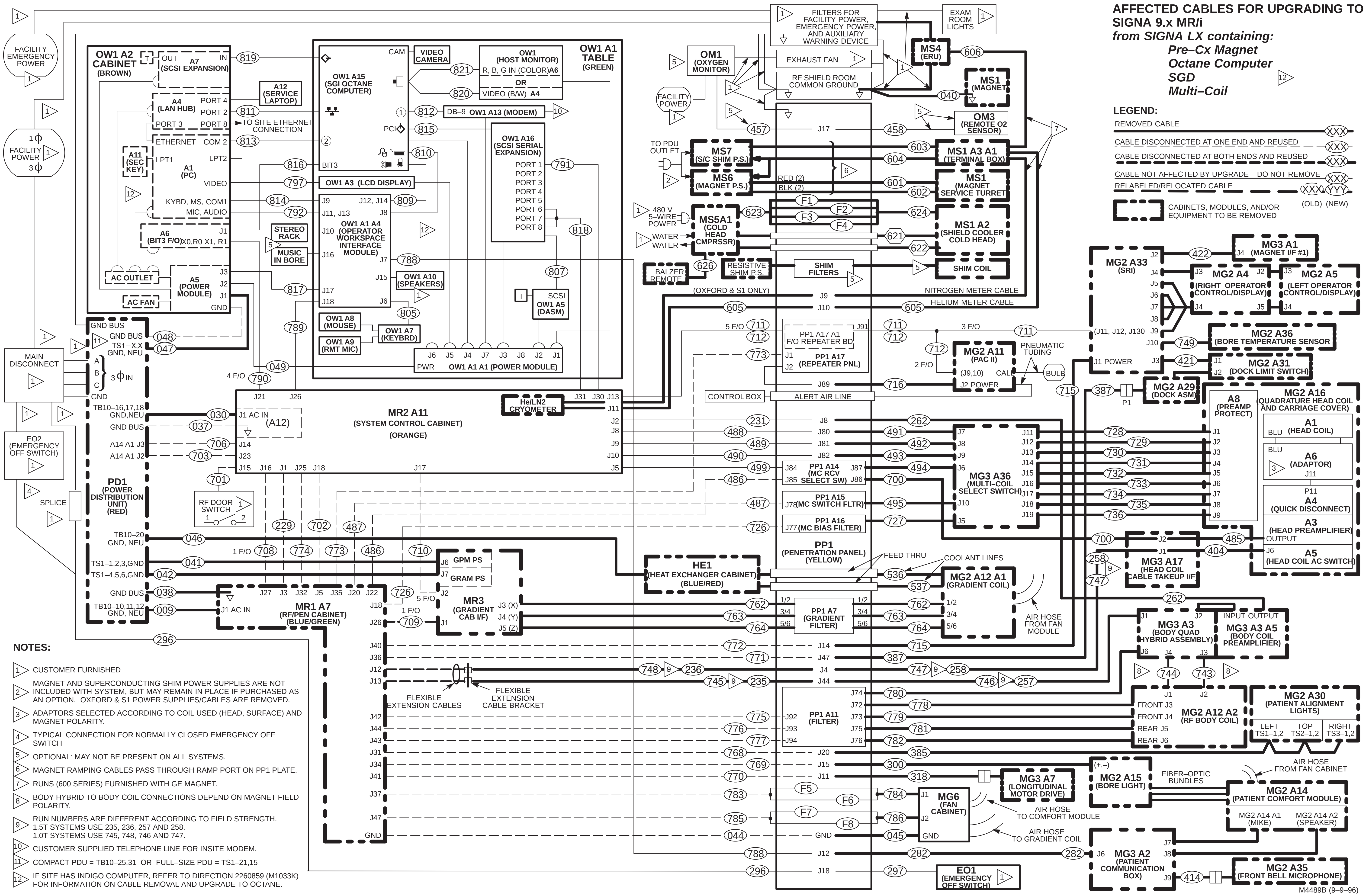
LEGEND:

- REMOVED CABLE (XXX)
- CABLE DISCONNECTED AT ONE END AND REUSED (XXX)
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE (XXX)
- RELABELED/RELOCATED CABLE (XXX) (YYY) (OLD) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED



NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.
11. COMPACT PDU = TB10–25,31 OR FULL–SIZE PDU = TS1–21,15
12. IF SITE HAS INDIGO COMPUTER, REFER TO DIRECTION 2260859 (M1033K) FOR INFORMATION ON CABLE REMOVAL AND UPGRADE TO OCTANE.



AFFECTED CABLES FOR UPGRADING TO SIGNA 9.x MR/i

from SIGNA LX containing:

Pre–Cx Magnet
Octane Computer
SGD
Multi–Coil

LEGEND:

- REMOVED CABLE
- CABLE DISCONNECTED AT ONE END AND REUSED
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE
- RELABELED/RELOCATED CABLE (XXX) (OLD) (YYY) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED

NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.
11. COMPACT PDU = TB10–25,31 OR FULL–SIZE PDU = TS1–21,15
12. IF SITE HAS INDIGO COMPUTER, REFER TO DIRECTION 2260859 (M1033K) FOR INFORMATION ON CABLE REMOVAL AND UPGRADE TO OCTANE.

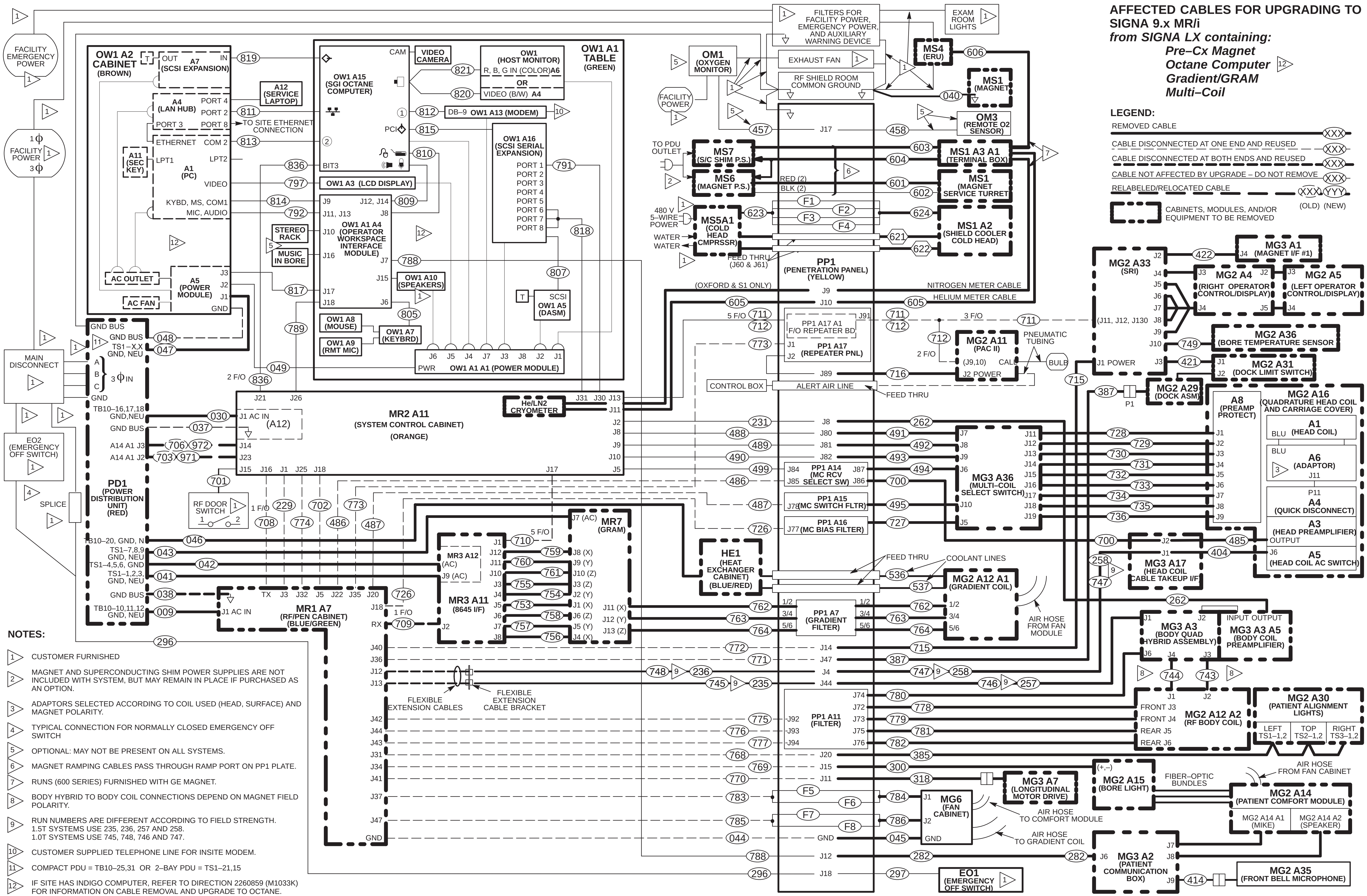
AFFECTED CABLES FOR UPGRADING TO SIGNA 9.x MR/i

from SIGNA LX containing:

Pre–Cx Magnet
Octane Computer 12
Gradient/GRAM
Multi–Coil

LEGEND:

- REMOVED CABLE (XXX)
- CABLE DISCONNECTED AT ONE END AND REUSED (XXX)
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE (XXX)
- RELABELED/RELOCATED CABLE (XXX) (YYY) (OLD) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED



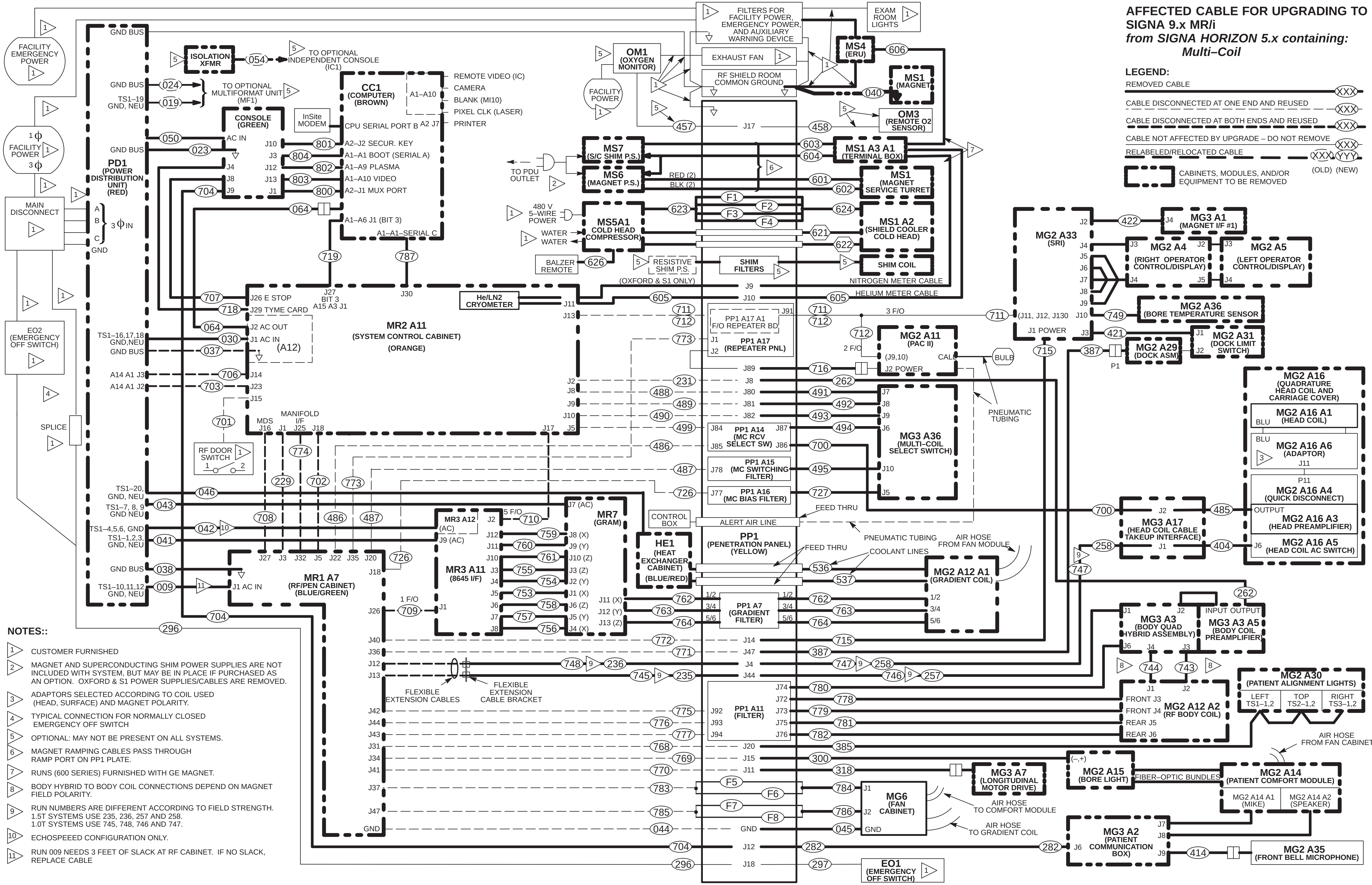
NOTES:

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. CUSTOMER SUPPLIED TELEPHONE LINE FOR INSITE MODEM.
11. COMPACT PDU = TB10–25,31 OR 2–BAY PDU = TS1–21,15
12. IF SITE HAS INDIGO COMPUTER, REFER TO DIRECTION 2260859 (M1033K) FOR INFORMATION ON CABLE REMOVAL AND UPGRADE TO OCTANE.

AFFECTED CABLE FOR UPGRADING TO SIGNA 9.x MR/i
from SIGNA HORIZON 5.x containing: Multi–Coil

LEGEND:

- REMOVED CABLE (XXX)
- CABLE DISCONNECTED AT ONE END AND REUSED (XXX)
- CABLE DISCONNECTED AT BOTH ENDS AND REUSED (XXX)
- CABLE NOT AFFECTED BY UPGRADE – DO NOT REMOVE (XXX)
- RELABELED/RELOCATED CABLE (XXX) (YYY) (OLD) (NEW)
- CABINETS, MODULES, AND/OR EQUIPMENT TO BE REMOVED

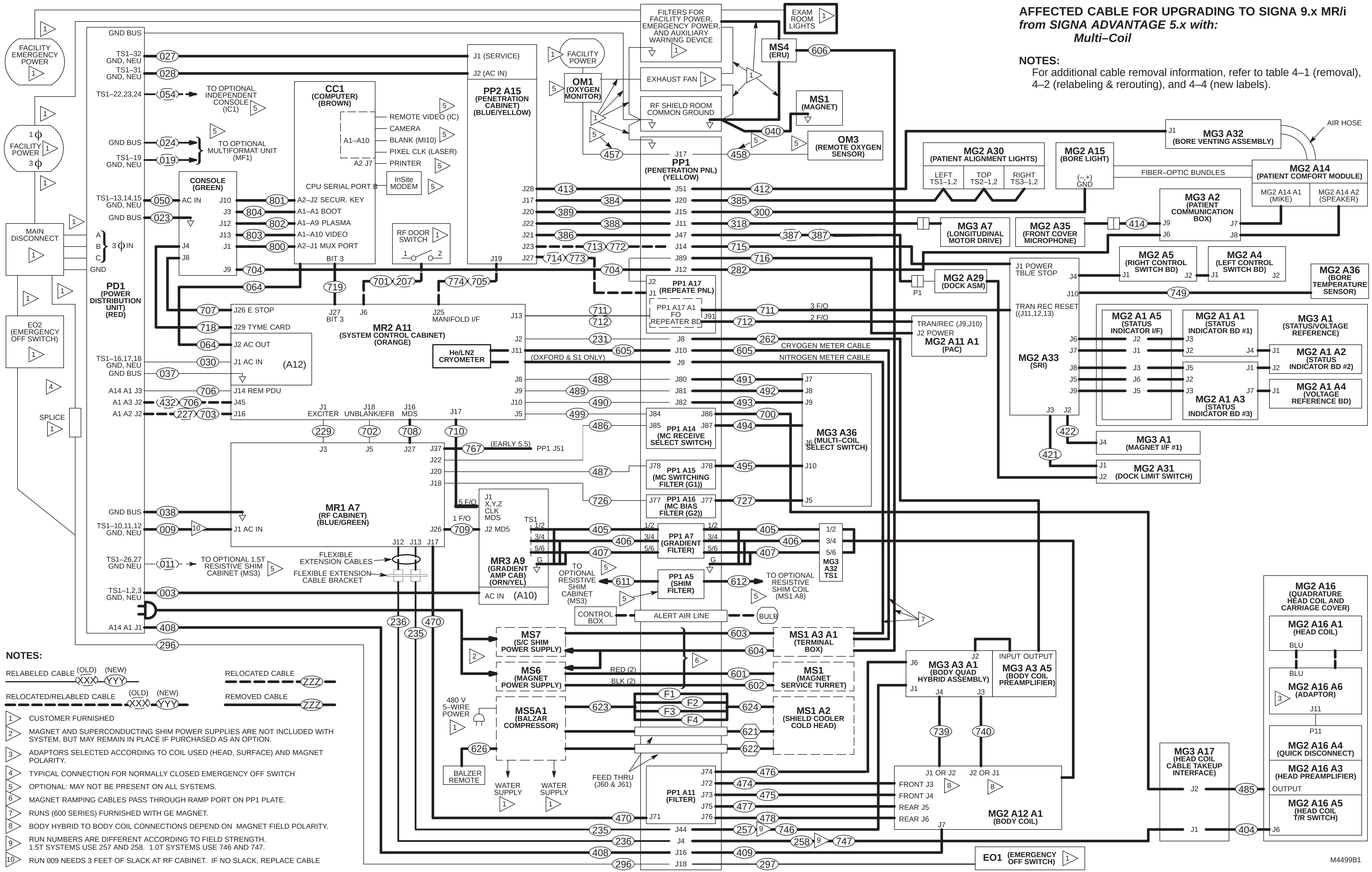


NOTES::

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY BE IN PLACE IF PURCHASED AS AN OPTION. OXFORD & S1 POWER SUPPLIES/CABLES ARE REMOVED.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 235, 236, 257 AND 258. 1.0T SYSTEMS USE 745, 748, 746 AND 747.
10. ECHOSPEEED CONFIGURATION ONLY.
11. RUN 009 NEEDS 3 FEET OF SLACK AT RF CABINET. IF NO SLACK, REPLACE CABLE

AFFECTED CABLE FOR UPGRADING TO SIGNA 9.x MR/i
from SIGNA ADVANTAGE 5.x with: Multi–Coil

NOTES:
For additional cable removal information, refer to table 4–1 (removal), 4–2 (relabeling & rerouting), and 4–4 (new labels).



NOTES:

RELABELED CABLE (OLD) XXX (NEW) YYY

RELOCATED CABLE ZZZ

RELOCATED/RELABLED CABLE (OLD) XXX (NEW) YYY

REMOVED CABLE ZZZ

1. CUSTOMER FURNISHED
2. MAGNET AND SUPERCONDUCTING SHIM POWER SUPPLIES ARE NOT INCLUDED WITH SYSTEM, BUT MAY REMAIN IN PLACE IF PURCHASED AS AN OPTION.
3. ADAPTORS SELECTED ACCORDING TO COIL USED (HEAD, SURFACE) AND MAGNET POLARITY.
4. TYPICAL CONNECTION FOR NORMALLY CLOSED EMERGENCY OFF SWITCH
5. OPTIONAL: MAY NOT BE PRESENT ON ALL SYSTEMS.
6. MAGNET RAMPING CABLES PASS THROUGH RAMP PORT ON PP1 PLATE.
7. RUNS (600 SERIES) FURNISHED WITH GE MAGNET.
8. BODY HYBRID TO BODY COIL CONNECTIONS DEPEND ON MAGNET FIELD POLARITY.
9. RUN NUMBERS ARE DIFFERENT ACCORDING TO FIELD STRENGTH. 1.5T SYSTEMS USE 257 AND 258. 1.0T SYSTEMS USE 746 AND 747.
10. RUN 009 NEEDS 3 FEET OF SLACK AT RF CABINET. IF NO SLACK, REPLACE CABLE

M4499B1

AFFECTED CABLE FOR UPGRADING TO SIGNA 9.x MR/i
from SIGNA ADVANTAGE 4.x with:
Multi–Coil

The upgrade of a Signa Advantage 4.x system to current non–MGD system requires the de–installation and return of all cabinets, equipment, cables, magnet, and magnet enclosure parts except for the following:

- Emergency Off Switch and Runs 296 and 297
- Oxygen Monitor and Runs 457 and 458 (if installed)
- RF Door Switch and Run 205 (to be relabeled as Run 701)
- Run 229
- Run 231
- Heliax connector on Penetration Panel
- Consult with customer on any other options that the customer may have installed and could be used with this current magnet and system upgrade .

4–12 LOCATING AND UNPACKING CABLES

1. Unpack and move power and ground cable boxes to the PDU area where they will be routed. Set Run 044 and 045 aside.
2. Unpack and move fixed site magnet room interconnect cables to magnet room.
   a. Move Run 045 Ground Cable into magnet room .
   b. Move applicable magnet room box of Fixed Site Cables into magnet room.
   c. Locate Body Coil RF input Cables (ship with Body Coil) and move cables to magnet room.
   d. Move gradient cables into magnet room. Gradient cables will be cut, re–terminated and routed according to instructions in Tab 1.
3. Unpack and move equipment room system cables into routing location (cabinets, Operator Workspace, PDU). Note that for specific system upgrades, labels are supplied to re–number cables 387, 701, 703, 706, 772, 773, and 774.
4. Move remainder of gradient cables to equipment room. Gradient cables will be cut, re–terminated and routed according to instructions in Tab 1.
5. Unpack and move fiber optic cables into routing location.

4–13 SORTING AND ROUTING CABLES

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

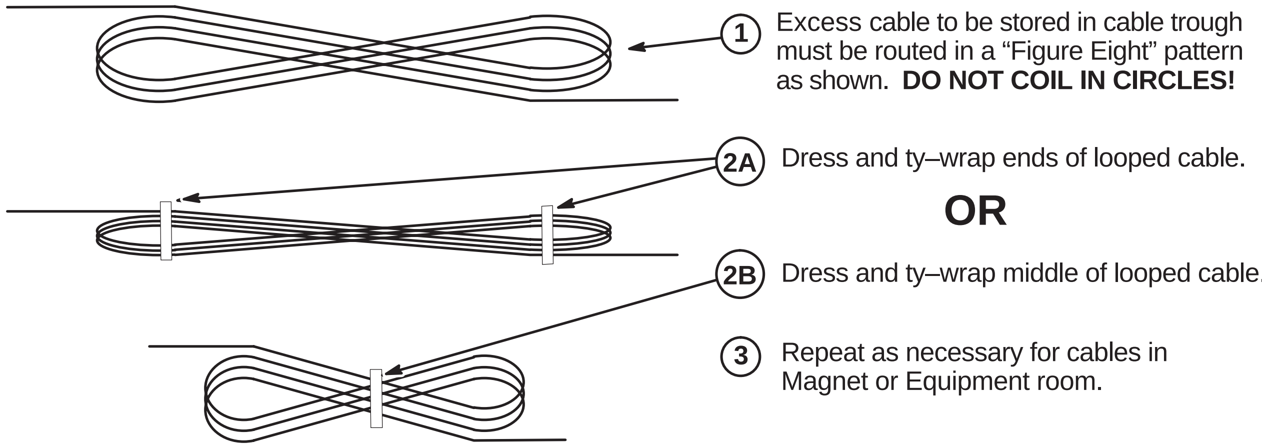
1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

   Note

   Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4–1. Some cables are usually cut to length such as gradient cables, power cables, Heliax, and Fiber–optic cables.

4–13 SORTING AND ROUTING CABLES (continued)



PROPER STORAGE OF EXCESS CABLES
ILLUSTRATION 4 – 1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

4–14 RUN 701 (RF DOOR SWITCH) CONNECTIONS

For most upgrades, Run 701 is not removed since it is routed within the magnet room wall or other conduit and can be reused for this upgrade. If Run 701 needs to be installed, complete remainder of this section.

1. Route Run #701 (from System Cabinet) to RF Door Switch. This 100 ft cable has a 9–pin subminature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminature D plug.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

4–15 EMERGENCY OFF CONNECTIONS

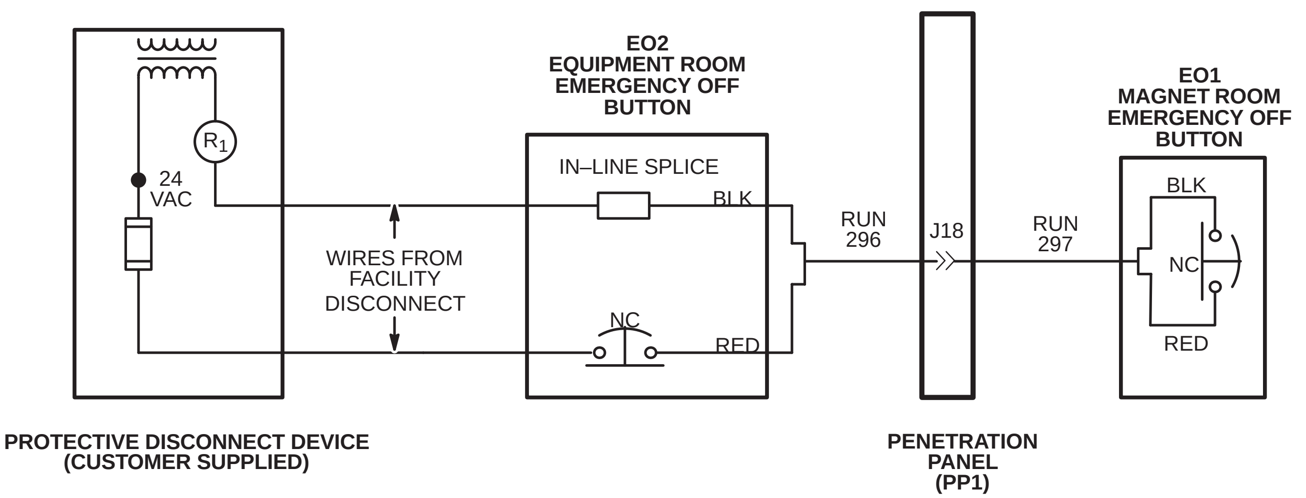
For most upgrades, Runs 296 and 297 are not removed since they are routed within the magnet room walls or other conduit and can be reused for this upgrade. If Runs 296 and 297 need to be installed, complete remainder of this section.

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to *Direction 2223170, Signa MR/i Pre–Installation*, Section 5–3–1, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

   Aside: In Illustration 4–1, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

2. At equipment room room “Emergency Off” (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2). See Illustration 4–1.
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2). See Illustration 4–1.
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push–on terminal.
5. Terminate black wire at end of Run 296 with local supplied “push–on” terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in–line splice. See Illustration 4–1.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1). See Illustration 4–1.



EMERGENCY OFF WIRING
ILLUSTRATION 4–1

4–16 IDENTIFYING NEW CABLES FOR SIGNA 10.x WITH EXCITE TECHNOLOGY (MGD)

The information in this section assume Signa 10.x WideOpen MR/*i* with EXCITE Technolgy upgrade components, appropriate Site Kit, and options are delivered complete with all cables present.

Cable shipment containers have color coded labels which identify the cable routing areas:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room

Table 4–9 lists all cables added for upgrading to Signa 10.x WideOpen MR/i from various early configurations. Select the column(s) that pertains to the upgrade being performed. The letters on the right side of the table are defined as follows:

A– Cables added if upgrading from LX system with Cx magnet, ACGD, and Octane

B– Cables added if upgrading from LX system with Cx magnet, SGD, and Octane

C– Cables added if upgrading from LX system with Cx magnet, SGD, and Indigo

D– Cables added if upgrading from LX system with Pre–Cx magnet, ACGD, and Octane

E– Cables added if upgrading from LX system with Pre–Cx magnet, SGD, and Octane

F– Cables added if upgrading from LX system with Pre–Cx magnet, SGD, and Indigo

G – Cables added if upgrading from Horizon 5.x system with pre–Cx magnet and Gradient/GRAM

H – Cables added if upgrading from Advantage 5.x or Horizon 5.x software only system with pre–Cx magnet

I – Cables added if upgrading from Advantage 4.x system with pre–Cx magnet

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

There are instances where the cable Run to be installed is located in the Equipment and Magnet room. These cables are identified in the table by the letters **E** and **M** and are defined by the following explanations:

E= Equipment Room side of cable Run to be isntalled.

M = Magnet Room side of cable Run to be installed.

X = Complete cable Run to be installed.

4–16 IDENTIFYING NEW CABLES FOR SIGNA 10.x WITH EXCITE TECHNOLOGY (MGD) (continued)

TABLE 4–9
CABLES ADDED FOR UPGRADING TO 9.x WideOpen FROM VARIOUS PRE–WideOpen CONFIGURATIONS

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
038	MR1–A7–GND	PD1–GND BUS		X	X	X	X	X	X	X	X
040	MS1	RF COMMON GND	X	X	X	X	X	X	X	X	X
044	MR1–A7–GND	PP1–GND							X	X	X
045	PP1–GND **(See Note 1)**	MG6 GND (FAN) **(See Note 1)**				X	X	X	X	X	X
048	OW1–A2–A5–GND	PD1–GND							X	X	X
049	OW1–A1–A1–PWR	OW1–A2–A5–J2							X	X	X
229	MR2–A11–J1	MR1–A23–J1									
231	MR2–A11–J2	PP1–J8									
235	PP1–J44	MR1–A7–J13				X	X	X	X	X	X
236	PP1–J4	MR1–A7–J12				X	X	X	X	X	X
257	PP1–J44	MG3–A3–A1–J1				X	X	X	X	X	X
258	PP1–J4	MG3–A17–J1				X	X	X	X	X	X
262	MG3–A3–A5–OUT	PP1–J8				X	X	X	X	X	X
282	MG3–A2–J6	PP1–J12				X	X	X	X	X	X
296	EO2/MAIN DISCONNECT	PP1–J18									
297	EO1	PP1–J18									
300	MG2–A15	PP1–J15	X	X	X	X	X	X	X	X	X
318	MG3–A7	PP1–J11				X	X	X	X	X	X
385	MG2–A30	PP1–J20	X	X	X	X	X	X	X	X	X
387	MG2–A29	PP1–J47				X	X	X	X	X	X
404	MG3–A17–J1	MG2–A16–A5–J6	X	X	X	X	X	X	X	X	X
414	MG3–A35	MG3–A2–J9	X	X	X	X	X	X	X	X	X
421	MG2–A33–J3	MG2–A31–J1	X	X	X	X	X	X	X	X	X
422	MG3–A1–J4	MG2–A33–J2	X	X	X	X	X	X	X	X	X
457	OM1 (Option)	PP1–J17 (Option)									
458	OM3 (Option)	PP1–J17 (Option)									
488	PP1–J80	MR2–A11–J8						X	X	X	X
489	PP1–J81	MR2–A11–J9						X	X	X	X
490	PP1–J82	MR2–A11–J10						X	X	X	X
491	MG3–A36–J7	PP1–J80		X	X	X	X	X	X	X	X
492	MG3–A36–J8	PP1–J81		X	X	X	X	X	X	X	X
493	MG3–A36–J9	PP1–J82		X	X	X	X	X	X	X	X
499	MR2–A11–J5	PP1–J84	X	X	X	X	X	X	X	X	X
536	HE1 (Coolant Hose)	MG2–A12–A1–IN	X	X	X	X	X	X	X	X	X
537	HE1 (Coolant Hose)	MG2–A12–A1–OUT	X	X	X	X	X	X	X	X	X
601	MS1 (RED)	MS6 (RED)	X	X	X	X	X	X	X	X	X
602	MS1 (BLK)	MS6 (BLK)	X	X	X	X	X	X	X	X	X
603	MS1–A13–A1	MS7	X	X	X	X	X	X	X	X	X
604	MS1–A13–A1	MS6/MS7	X	X	X	X	X	X	X	X	X
606	MS4	MS1–A3–A1	X	X	X	X	X	X	X	X	X
621	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	X	X	X	X
622	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	X	X	X	X

4–16 IDENTIFYING NEW CABLES FOR SIGNA 10.x WITH EXCITE TECHNOLOGY (MGD) (continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
623	MS5–A1	PP1–F1,F2,F3,F4	X	X	X	X	X	X	X	X	X
624	MS1–A2	PP1–F1,F2,F3,F4	X	X	X	X	X	X	X	X	X
701	MR2–A11–J15	RF DOOR SWITCH									
702	MR2–A11–J18	MR1–A7–J5									X
708	MR2–A11–J18 MDS **(See Note 4)**	MR1–A7–J27 **(See Note 4)**								X	X
709	MR1–A7–J26	MR3–A11–J2 CLK								X	X
711/ 712	PP1–A17–A1–J91 (F/O REPEATER BD)	MG2–A33–J11,12,13 MG2–A11–A1–J9,10							X	X	X
715	MG2–A33–J1	PP1–J14				X	X	X	X	X	X
716	MG2–A11–A1–J2	PP1–J89				X	X	X	X	X	X
728	MG3–A36–J11	MG2–A16–OUT–J1	X	X	X	X	X	X	X	X	X
729	MG3–A36–J12	MG2–A16–OUT–J2	X	X	X	X	X	X	X	X	X
730	MG3–A36–J13	MG2–A16–OUT–J3	X	X	X	X	X	X	X	X	X
731	MG3–A36–J14	MG2–A16–OUT–J4	X	X	X	X	X	X	X	X	X
732	MG3–A36–J15	MG2–A16–OUT–J5	X	X	X	X	X	X	X	X	X
733	MG3–A36–J16	MG2–A16–OUT–J6	X	X	X	X	X	X	X	X	X
734	MG3–A36–J17	MG2–A16–OUT–J7	X	X	X	X	X	X	X	X	X
735	MG3–A36–J18	MG2–A16–OUT–J8	X	X	X	X	X	X	X	X	X
743	MG2–A12–A1–J1	MG3–A3–J4	X	X	X	X	X	X	X	X	X
744	MG2–A12–A1–J2	MG3–A3–J3	X	X	X	X	X	X	X	X	X
749	MG3–A36	MG2–A33–J10	X	X	X	X	X	X	X	X	X
762	MR3CABI/FJ3	PP1–A7–1/2		E	E		E	E	E	E	E
762	PP1–A7–1/2	MG3–A32–TS1/2		M	M	M	M	M	M	M	M
763	MR3CABI/FJ4	PP1–A7–1–3/4		E	E		E	E	E	E	E
763	PP1–A7–1–3/4	MG3–A32–TS3/4		M	M	M	M	M	M	M	M
764	MR3CABI/FJ5	PP1–A7–1–5/6		E	E		E	E	E	E	E
764	PP1–A7–1–5/6	MG3–A32–TS5/6		M	M	M	M	M	M	M	M
768	PP1–J20	MR1–A7–J31							X	X	X
769	PP1–J15	MR1–A7–J34							X	X	X
770	PP1–J11	MR1–A7–J41							X	X	X
771	PP1–J47	MR1–A7–J36							X	X	X
772	PP1–J14	MR1–A7–J40							X	X	X
773	PP1–A17–J1	MR1–A7–J35							X	X	X
774	MR2–A11–J25	MR1–A7–J32							X	X	X
775	MR1–A7–J42	PP1–A11–J92							X	X	X
776	MR1–A7–J44	PP1–A11–J93							X	X	X
777	MR1–A7–J43	PP1–A11–J94							X	X	X
778	MG2–A12–A1–J3	PP1–A11–J72				X	X	X	X	X	X
779	MG2–A12–A1–J4	PP1–A11–J73				X	X	X	X	X	X
780	MG3–A3–J6	PP1–A11–J74				X	X	X	X	X	X
781	MG2–A12–A1–J5	PP1–A11–J75				X	X	X	X	X	X
782	MG2–A12–A1–J6	PP1–A11–J76				X	X	X	X	X	X
783	MR1–A7–J37	PP1–F5/F6							X	X	X
784	PP1–F5/F6 **(See Note 1)**	MG6–J1 (FAN) **(See Note 1)**				X	X	X	X	X	X
785	MR1–A7–J18	PP1–F7/F8							X	X	X
786	PP1–F7/F8 **(See Note 1)**	MG6–J2 (FAN) **(See Note 1)**				X	X	X	X	X	X

4–16 IDENTIFYING NEW CABLES FOR SIGNA 10.x WITH EXCITE TECHNOLOGY (MGD) (continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
788	OW1–A1–A2–J7	PP1–J12							X	X	X
789	OW1–A1–12–J18	MR2–A11–J26							X	X	X
791	OW1–A2–A5–J2	OW1–A11–J30							X	X	X
797	OW1 – A3 (LCD DISPLAY)	OW1 – A2 – A1 (VIDEO)							X	X	X
805	OW1–A1–A4–J6	OW1–A7–KYBRD							X	X	X
807	OW1–A16	OW1–A5 (DASM)							X	X	X
810	OW1–A1–A2–J8	OW1–A15–KBD,MOUSE			X			X	X	X	X
811	OW1–A2–A4–PORT2	OW1–A15–ETHERNET			X			X	X	X	X
812	OW1–A13 (MODEM)	OW1–A15–COM1			X			X	X	X	X
813	OW1–A2–A1–COM2	OW1–A15–COM2			X			X	X	X	X
814	OW1–A1–A4–J9	OW1–A2–A1–KBD,MOUSE			X			X	X	X	X
815	OW1–A16–SCSI	OW1–A15–SCSI (PCI)			X			X	X	X	X
817	OW1–A1–A2–J17	OW1–A2–A5–J3			X			X	X	X	X
818	OW1–A16–PORT6,7,8	MR2–A11–J31							X	X	X
819	OW1–A2–A7–SCSI	OW1–A15–SCSI			X			X	X	X	X
821	OW1–A15–MONITOR **(See Note 2)**	OW1–A6–R,B,G (COLOR) **(See Note 2)**							X	X	X
823	MSM1–A1–J10	MR2–A11–J24	X	X	X	X	X	X	X	X	X
824	PP1–J10	FJ1	X	X	X	X	X	X	X	X	X
825	PP1–J48	MSM1–A1–J8	X	X	X	X	X	X	X	X	X
826	MSM1–A1–J9	FJ3	X	X	X	X	X	X	X	X	X
827	FJ3	MS5–A6–A1–J1/FJ4	X	X	X	X	X	X	X	X	X
828	PP1–J10	MA1–A3–A1–P403	X	X	X	X	X	X	X	X	X
829	PP1–J48	MS1–FJ2	X	X	X	X	X	X	X	X	X
830	MS1–FJ2	MS1–A2–J2/MS1–A5–A2–J1/MG3–TB1	X	X	X	X	X	X	X	X	X
831	FJ1	MSM1–A1–J7	X	X	X	X	X	X	X	X	X
832	MS1–A2–J1	MS1–SLEEVE/MS1–COLDHEAD	X	X	X	X	X	X	X	X	X
833	FJ4	MS5–A1–A6–JR	X	X	X	X	X	X	X	X	X
838	PP1–J51 (Future) **(See Note 3)**	OW1–A17–J3 (Future) **(See Note 3)**									
848	MG2–A11–J4	MG2–A4–J5	X	X	X	X	X	X	X	X	X
861	MG2–A4–J4	MG2–A33–J5,6,7,8,9	X	X	X	X	X	X	X	X	X
862	MG2–A4–J1	MG2–A6–J1	X	X	X	X	X	X	X	X	X
863	MG2–A4–J2	MG2–A6–J2	X	X	X	X	X	X	X	X	X
869	MG3–A2–J8	MG2–A14–A2–SPEAKER	X	X	X	X	X	X	X	X	X
893	MG3–A3–J2	MG3–A3–A5–INPUT	X	X	X	X	X	X	X	X	X
897	MG2–A4–J3	MG2–A33–J4	X	X	X	X	X	X	X	X	X
911	OW1–A17–J1 (Future) **(See Note 3)**	OW1–A16–PORT 5 (Future) **(See Note 3)**									
940	MSM3 RS232	MSM1 J1		X	X	X	X	X	X	X	X
966	MR3–PD1–7,8,9,GND	MR1–A7–J1		X	X	X	X	X	X	X	X
967	MR3–PD1–10,11,12,GND	MR1–CB1		X	X	X	X	X	X	X	X
968	MR3–PD1–4,5,6,GND	MR2–A12–J1		X	X	X	X	X	X	X	X
969	MR3–PD1–18,19,GND,N	OW1–A2–A5–J1		X	X	X	X	X	X	X	X
970	MR3–PD1–16,17,GND	WC1		X	X	X	X	X	X	X	X
1025	MR2–A11–J40	PP1–J6	X	X	X	X	X	X	X	X	X
1026	MR2–A11–J41	PP1–J7	X	X	X	X	X	X	X	X	X
1027	MR2–A11–J42	PP1–J58	X	X	X	X	X	X	X	X	X
1028	MR2–A11–J43	PP1–J59	X	X	X	X	X	X	X	X	X

4–16 IDENTIFYING NEW CABLES FOR SIGNA 10.x WITH EXCITE TECHNOLOGY (MGD) (continued)

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I
1029	MR2–A11–J62	PP1–A16–J77	X	X	X	X	X	X	X	X	X
1030	MR2–A11–J61	PP1–A16–J78	X	X	X	X	X	X	X	X	X
1031	MR2–A11–J63	OW1–A15–SLOT1	X	X	X	X	X	X	X	X	X
1032	MR2–A11–J31	OW1–A16–FJ100			X			X			
1033	MR2–A11–J16	MR1–A7–TX									
1034	MR2–A11–J17	MR3–A11–J1	X	X	X	X	X	X	X	X	X
1035	MG2–A16–A3/A6–J4	MG3–A36–J21	X	X	X	X	X	X	X	X	X
1036	PP1–J84	MG3–A36–J6	X	X	X	X	X	X	X	X	X
1037	PP1–J6	MG3–A36–J1	X	X	X	X	X	X	X	X	X
1038	PP1–J7	MG3–A36–J2	X	X	X	X	X	X	X	X	X
1039	PP1–J58	MG3–A36–J3	X	X	X	X	X	X	X	X	X
1040	PP1–J59	MG3–A36–J4	X	X	X	X	X	X	X	X	X
1041	PP1–A15–J78	MG3–A36–J10	X	X	X	X	X	X	X	X	X
1042	PP1–A16–J77	MG3–A36–J5	X	X	X	X	X	X	X	X	X
1043	MG2–A16–A8–J11	MG2–A16–A3/A5–J1	X	X	X	X	X	X	X	X	X
1044	MR2–A11–J22	OW1–A15–SLOT2	X	X	X	X	X	X	X	X	X
1045	MR2–A11–J13	PP1–A17–A1	X	X	X	X	X	X	X	X	X
1046	FJ100	OW1–A16–PORT 1,6,7,8			X			X			
1047	PP1–J51 (Future) **(See Note 3)**	MG3–A36–J (Future) **(See Note 3)**									
1048	MG2–A33–J (Future) **(See Note 3)**	MG3–A36–J (Future) **(See Note 3)**									
1049	MG3–A36–J19	MG2–A16–A8–J9	X	X	X	X	X	X	X	X	X
MIC	MG3–A2–J7	MG2–A14–A1–MICROPHONE	X	X	X	X	X	X	X	X	X

Notes:

1. Runs 045, 784, and 786 may not have to be installed if existing Blower Box has not been deinstalled and existing runs are still in place. If new Runs 045, 784, 786, and new Blower Box are not used, return for recycling.
2. Run 798 is equivalent to Run 821 and Run 799 is equivalent to Run 820.
3. Cable Runs 838, 911, 1047, and 1048 are to be used with Coil ID when available.
4. Run 708 should be relabeled as Run 1033. Refer to Table 4–10.

4–17 FIBER OPTIC CABLE LABEL INFORMATION

Runs 708 needs to be re–labeled as indicated in Table 4–10 below.

TABLE 4–10
SYSTEM CABINET FIBER OPTIC CONNECTIONS

RUN	SYSTEM CABINET				SRF CABINET	
	CABLE LABEL AT ONE END AT INTERFACE PANEL		FIBER OPTIC WIRE LABEL AT STIF BD WITHIN CABINET		CABLE LABEL AT OTHER END AT INTERFACE PANEL	
	OLD LABEL	*NEW LABEL*	*OLD LABEL*	*NEW LABEL*	*OLD LABEL*	*NEW LABEL*
708	RUN #708 MR2–A11–J16	RUN #1033 MR2–A11–J16 2330480	MR2–A23–A2–J5 OR A30–A23–J19–RX	MGD SLOT 13R–J21	RUN #708 MR1–A7–TX	RUN #1033 MR1–A7–TX 2330480

4–18 SIGNA 10.x (MGD) INTERCONNECT MAPS FOR UPGRADES

For Signa 10.x cable installation, fold–out Interconnect Maps are provided on pages 4–57 thru 4–65. These maps show bolded and dashed lines that represent new, re–used, and/or re–labeled cables for existing systems being upgraded to Signa MR/*i* with EXCITE Technology (MGD).

Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.

Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation instruction (PIP Direction) within this manual.

From any early configuration, the cable maps listed below illustrate the final configuration of various magnet systems after the completion of the Signa 10.x upgrade. The cable map illustrates the Signa 10.x EXCITE System interconnect for all supplied cables and interface with customer supplied wiring.

Use only the applicable Upgrade Interconnect Cable Map that applies to the Signa 10.x system being upgraded. Tape map in a convenient location to check off routed cable. The fold–out map should be returned later to the binder for future reference.

The Signa 10.x Upgrade Interconnect Cable Maps can be found on the following pages:

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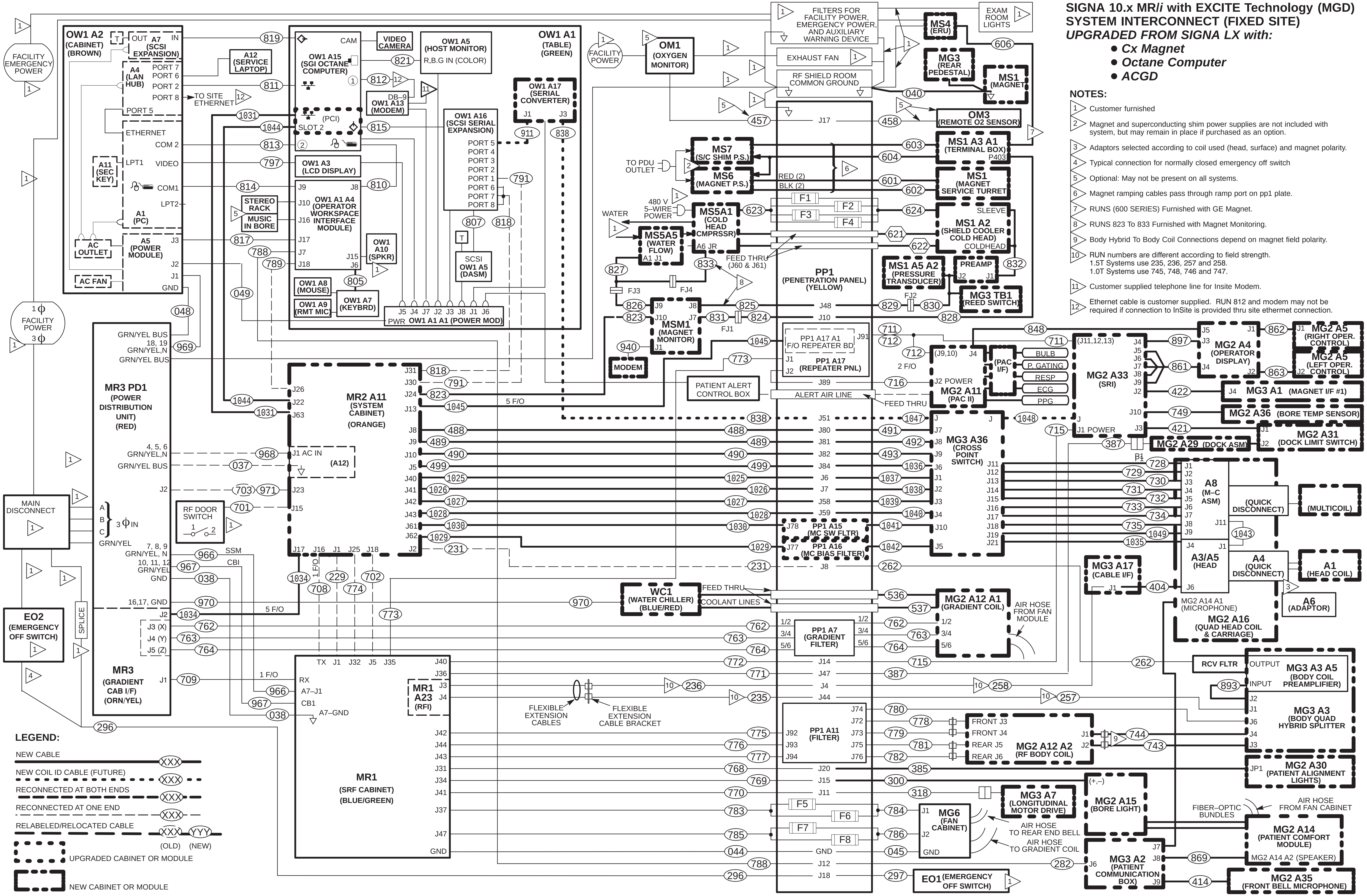
SIGNA 10.x MR/*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE)

UPGRADED FROM SIGNA LX with:

- ***Cx Magnet***
- ***Octane Computer***
- ***ACGD***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



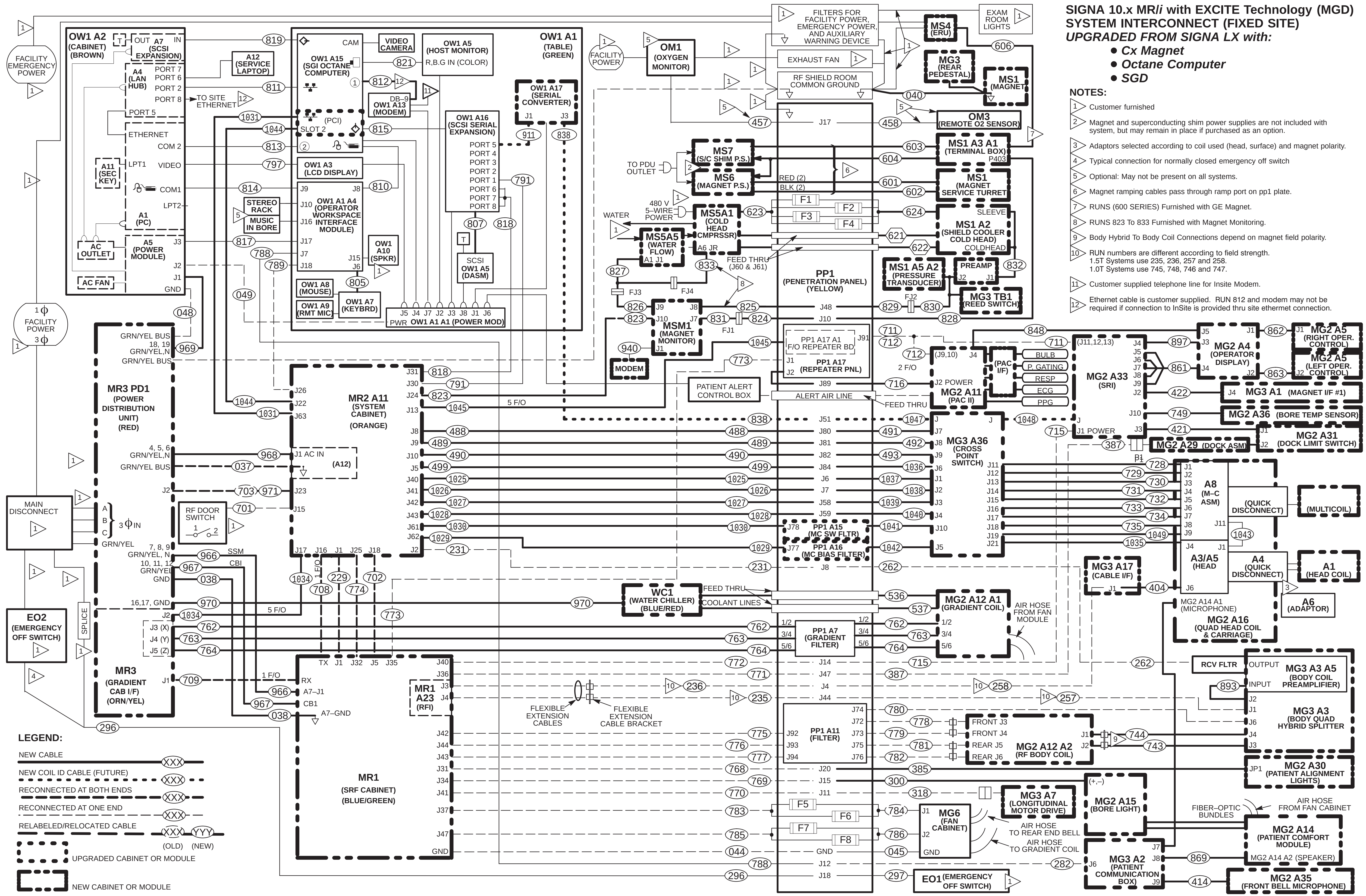
SIGNA 10.x MR/*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE)

UPGRADED FROM SIGNA LX with:

- ***Cx Magnet***
- ***Octane Computer***
- ***SGD***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



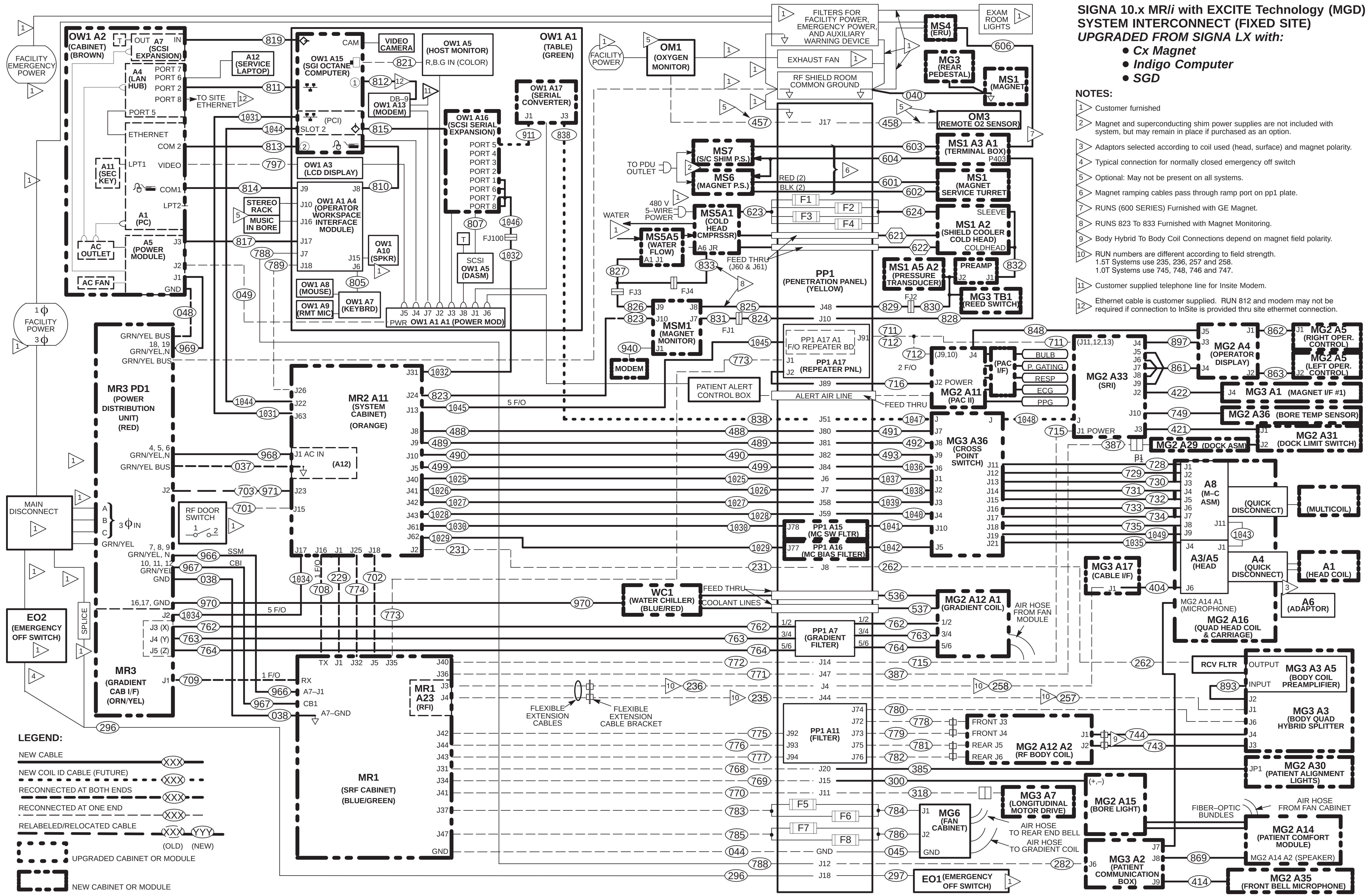
SIGNA 10.x MR/*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE)

UPGRADED FROM SIGNA LX with:

- ***Cx Magnet***
- ***Indigo Computer***
- ***SGD***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



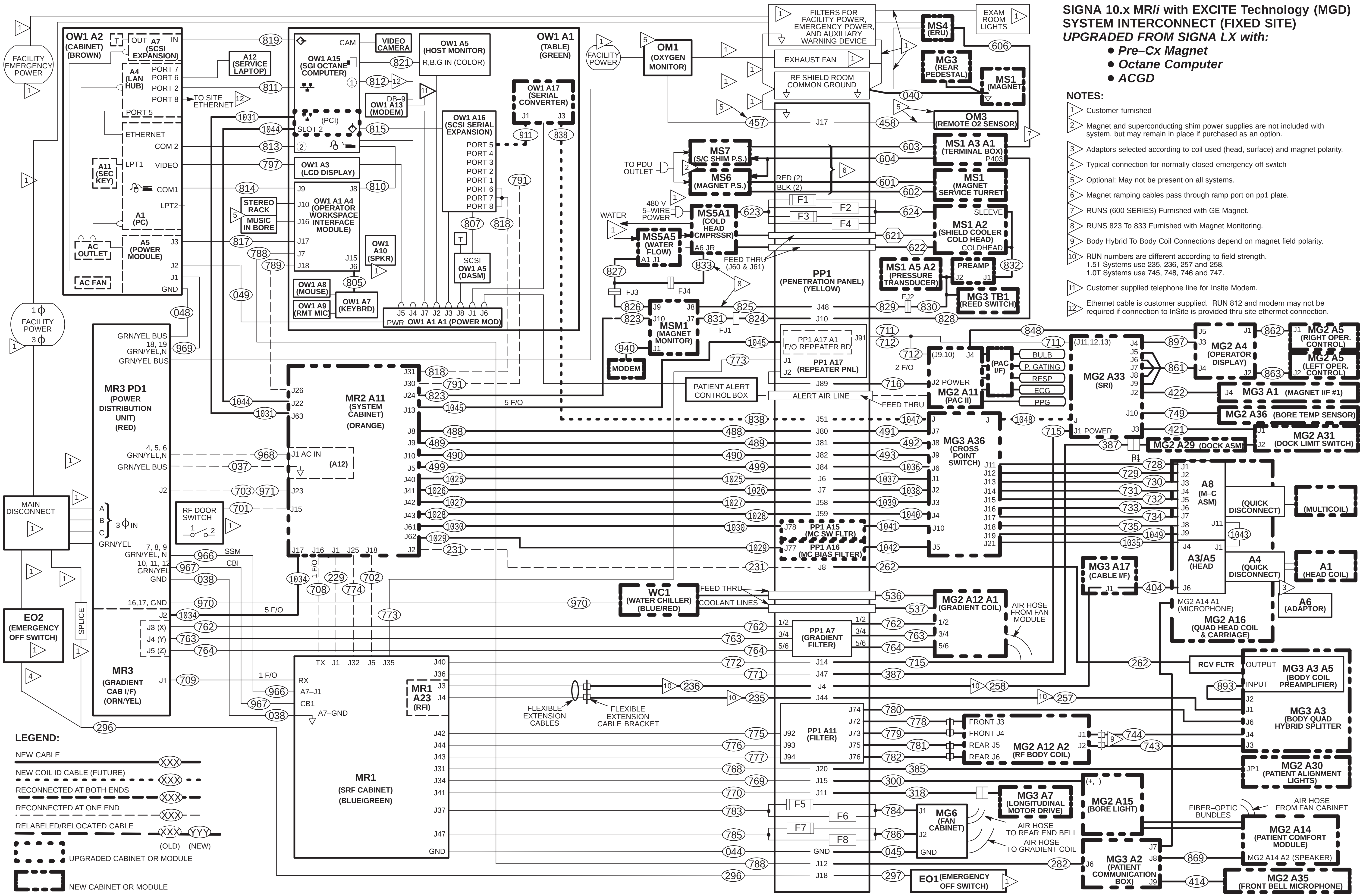
SIGNA 10.x MR*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE)

UPGRADED FROM SIGNA LX with:

- ***Pre–Cx Magnet***
- ***Octane Computer***
- ***ACGD***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



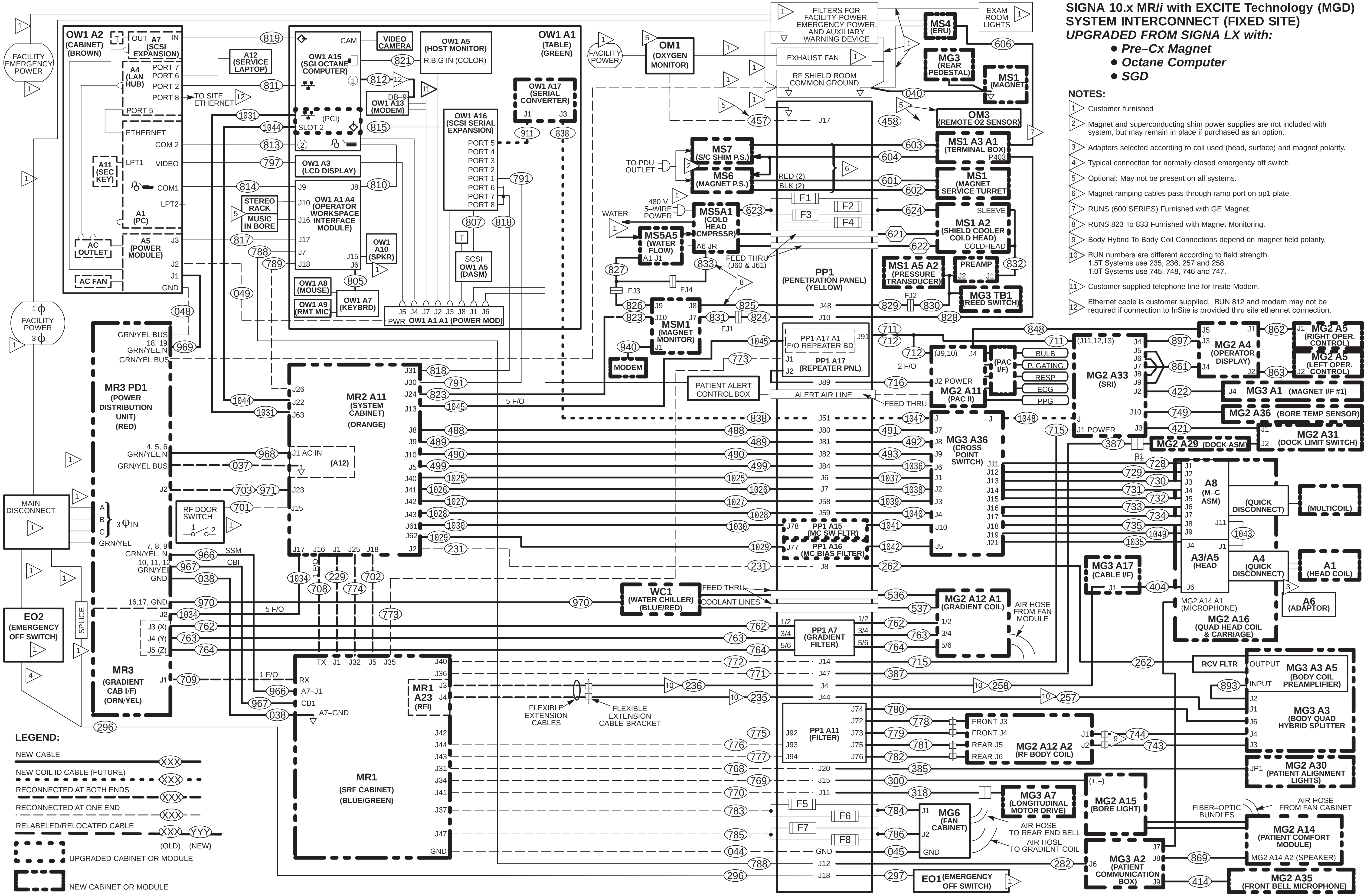
SIGNA 10.x MR*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE)

UPGRADED FROM SIGNA LX with:

- ***Pre–Cx Magnet***
- ***Octane Computer***
- ***SGD***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



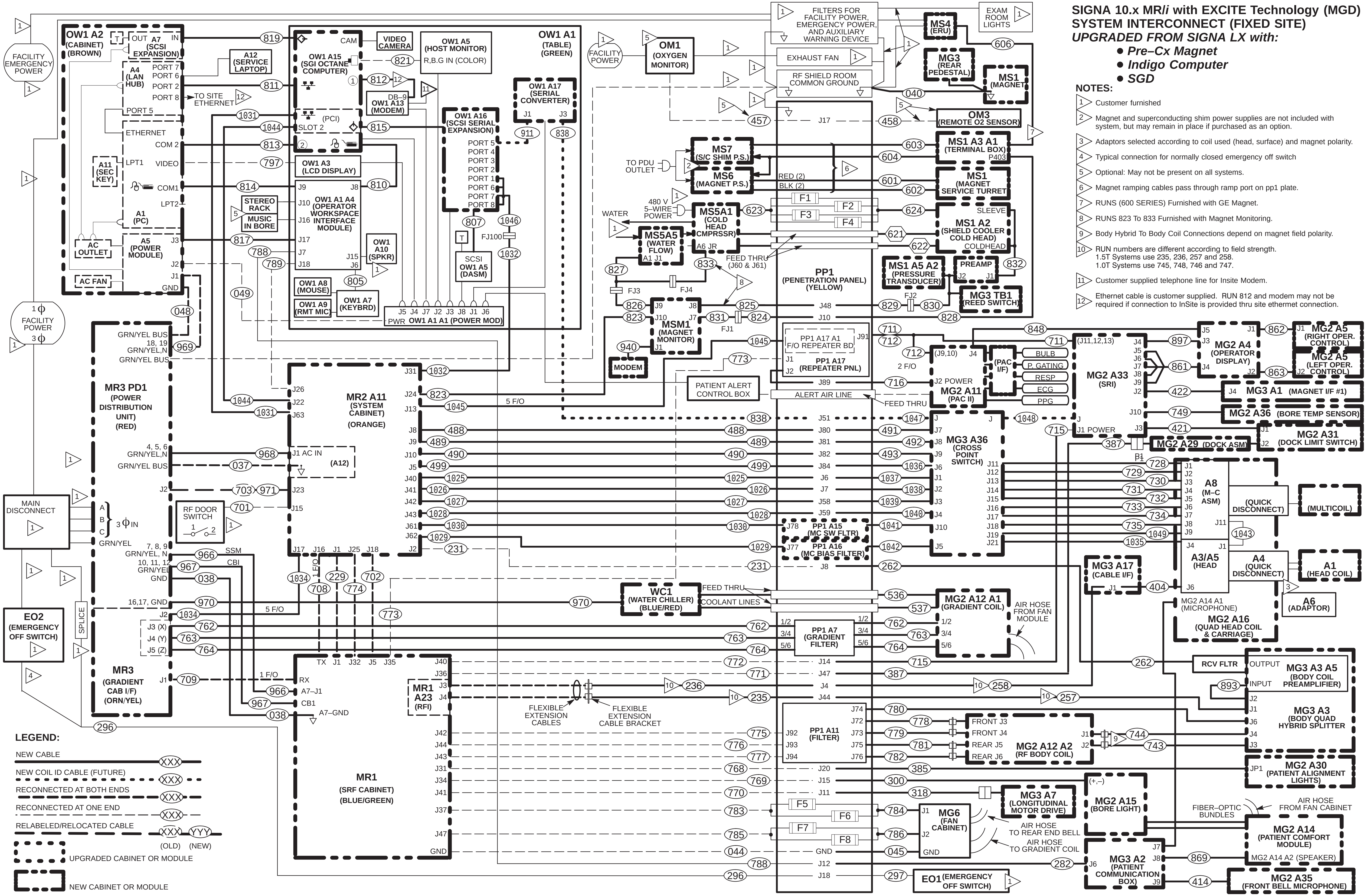
SIGNA 10.x MR/*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE)

UPGRADED FROM SIGNA LX with:

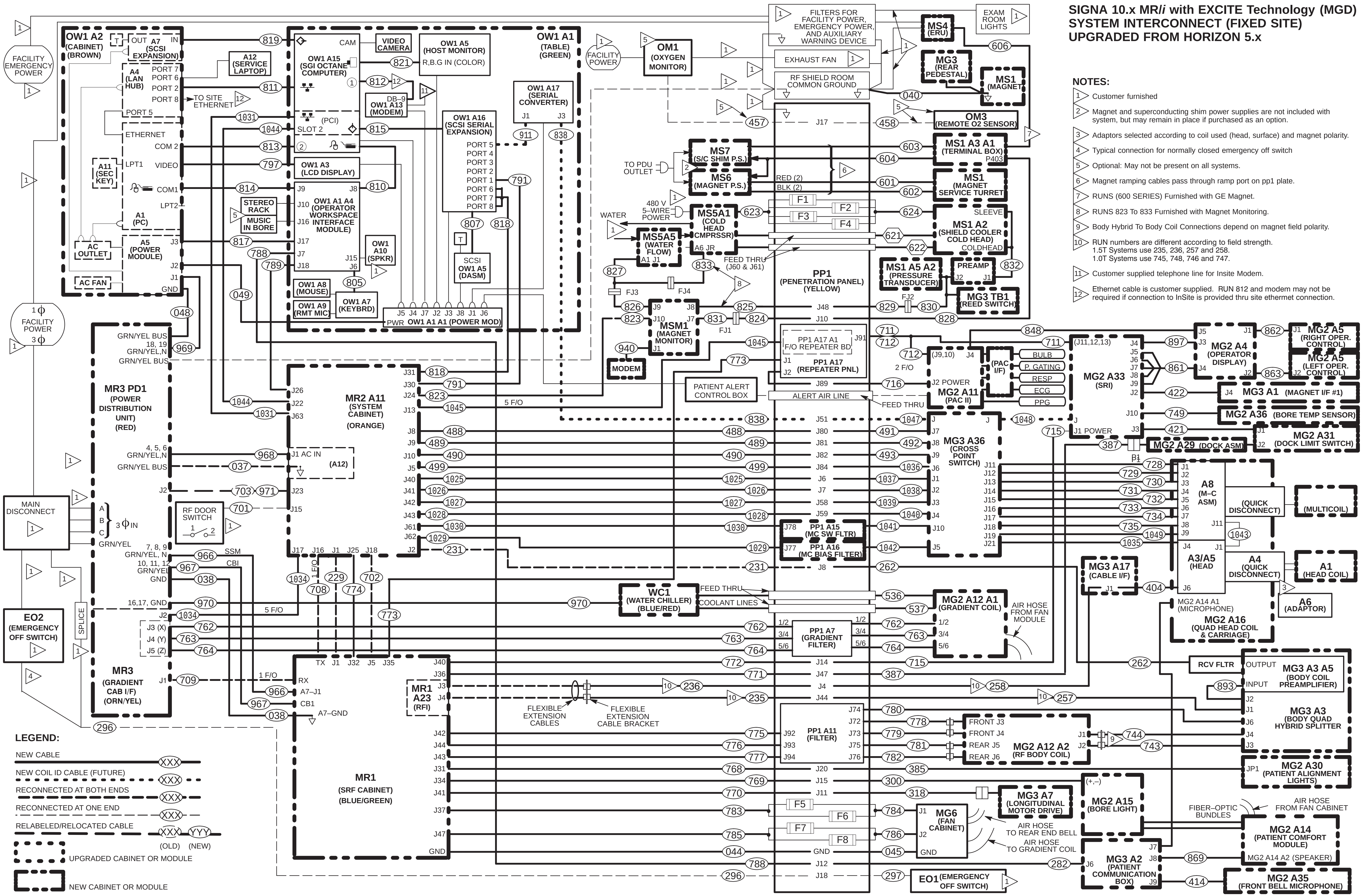
- ***Pre–Cx Magnet***
- ***Indigo Computer***
- ***SGD***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



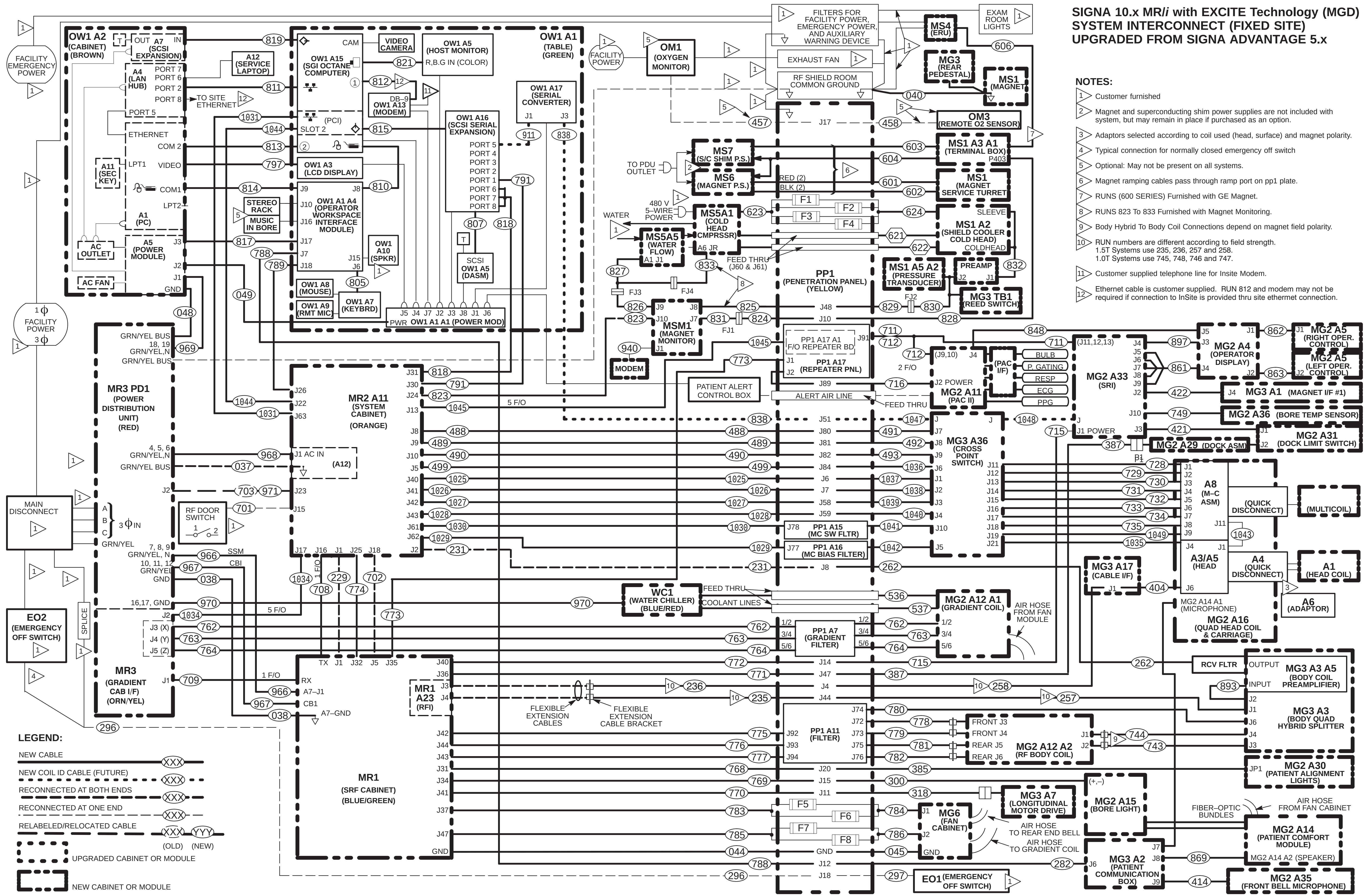
SIGNA 10.x MR/*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE) UPGRADED FROM HORIZON 5.x



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

SIGNA 10.x MR/*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE) UPGRADED FROM SIGNA ADVANTAGE 5.x



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

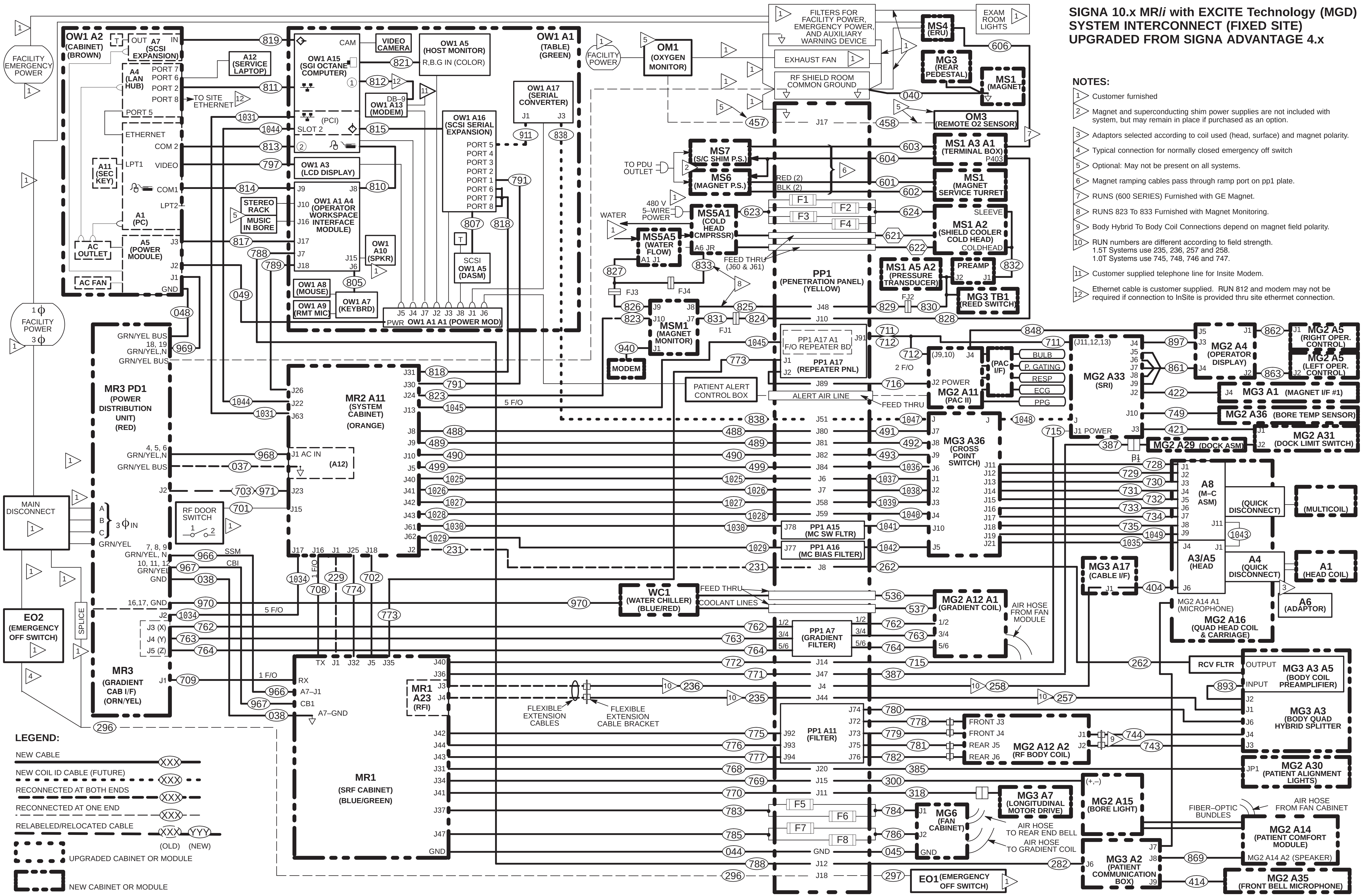
LEGEND:

- NEW CABLE
- NEW COIL ID CABLE (FUTURE)
- RECONNECTED AT BOTH ENDS
- RECONNECTED AT ONE END
- RELABELED/RELOCATED CABLE (XXX) (OLD) (YYY) (NEW)
- UPGRADED CABINET OR MODULE
- NEW CABINET OR MODULE

SIGNA 10.x MR/*i* with EXCITE Technology (MGD) SYSTEM INTERCONNECT (FIXED SITE) UPGRADED FROM SIGNA ADVANTAGE 4.x

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) Furnished with GE Magnet.
8. RUNS 823 To 833 Furnished with Magnet Monitoring.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



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4–19 IDENTIFYING NEW CABLES FOR SIGNA 9.x WITH ACGD

The information in this section assume Signa 9.x WideOpen MR/*i* HiSpeed or EchoSpeed upgrade componenets, appropriate Site Kit, and options are delivered complete with all cables present.

Cable shipment containers have color coded labels which identify the cable routing areas:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room

Table 4–11 lists all cables added for upgrading to Signa WideOpen 9.x MR/i HiSpeed or EchoSpeed from various early configurations. Select the column(s) that pertains to the upgrade being performed. The letters on the right side of the table are defined as follows:

A– Cables added if upgrading from 8.x system with Cx magnet, SGD, and Octane

B– Cables added if upgrading from 8.x system with pre–Cx magnet, SGD (or Gradient/GRAM) and Octane

C – Cables added if upgrading from Horizon 5.x system with pre–Cx magnet and Gradient/GRAM

D – Cables added if upgrading from Advantage 5.x or Horizon 5.x software only system with pre–Cx magnet

E – Cables added if upgrading from Advantage 4.x system with pre–Cx magnet

Instructions for the following upgrade are delivered with the option.

F – Cables added if upgrading from Indigo to Octane computer, M1033K.

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

TABLE 4–11
CABLES ADDED FOR UPGRADING TO 9.x WideOpen FROM VARIOUS PRE–WideOpen CONFIGURATIONS

RUN#	FROM	TO	A	B	C	D	E	F
038	MR1–A7–GND	PD1–GND BUS	X	X	X	X	X	
040	MS1	RF COMMON GND	X	X	X	X	X	
044	MR1–A7–GND	PP1–GND				X	X	
045	PP1–GND **(See Note 5)**	MG6 GND (FAN) **(See Note 5)**		X	X	X	X	
048	OW1–A2–A5–GND	PD1–GND			X	X	X	
049	OW1–A1–A1–PWR	OW1–A2–A5–J2			X	X	X	
235	PP1–J44 **(See Note 2)**	MR1–A7–J13 **(See Note 2)**		X	X	X	X	
236	PP1–J4 **(See Note 2)**	MR1–A7–J12 **(See Note 2)**		X	X	X	X	
257	PP1–J44 **(See Note 2)**	MG3–A3–A1–J1 **(See Note 2)**		X	X	X	X	
258	PP1–J4 **(See Note 2)**	MG3–A17–J1 **(See Note 2)**		X	X	X	X	
262	MG3–A3–A5–OUT	PP1–J8		X	X	X	X	
282	MG3–A2–J6	PP1–J12		X	X	X	X	
296	EO2/MAIN DISCONNECT	PP1–J18						
297	EO1	PP1–J18						
300	MG2–A15	PP1–J15	X	X	X	X	X	
318	MG3–A7	PP1–J11		X	X	X	X	
385	MG2–A30	PP1–J20	X	X	X	X	X	

4–19 IDENTIFYING NEW CABLES FOR SIGNA 9.x WITH ACGD (continued)

RUN#	FROM	TO	A	B	C	D	E	F
387	MG2–A29	PP1–J47		X	X	X	X	
404	MG3–A17–J1	MG2–A16–A5–J6	X	X	X	X	X	
414	MG3–A35	MG3–A2–J9	X	X	X	X	X	
421	MG2–A33–J3	MG2–A31–J1	X	X	X	X	X	
422	MG3–A1–J4	MG2–A33–J2	X	X	X	X	X	
485	MG3–A17–J2	MG2–A16–A3–J4	X	X	X	X	X	
486	PP1–A14–J85 **(See Note 1)**	MR1–A23–J22 **(See Note 1)**						
487	PP1–A15–J78 **(See Note 1)**	MR1–A23–J20 **(See Note 1)**						
488	PP1–J80 **(See Note 1)**	MR2–A11–J8 **(See Note 1)**						
489	PP1–J81 **(See Note 1)**	MR2–A11–J9 **(See Note 1)**						
490	PP1–J82 **(See Note 1)**	MR2–A11–J10 **(See Note 1)**						
491	MG3–A36–J7	PP1–J80		X	X	X	X	
492	MG3–A36–J8	PP1–J81		X	X	X	X	
493	MG3–A36–J9	PP1–J82		X	X	X	X	
494	MG3–A36–J6	PP1–A14–J87		X	X	X	X	
495	MG3–A36–J10	PP1–A15–J78		X	X	X	X	
499	PP1–A14–J84 **(See Note 1)**	MR2–A11–J5 **(See Note 1)**						
536	HE1 (Coolant Hose)	MG2–A12–A1–IN	X	X	X	X	X	
537	HE1 (Coolant Hose)	MG2–A12–A1–OUT	X	X	X	X	X	
601	MS1 (RED)	MS6 (RED)	X	X	X	X	X	
602	MS1 (BLK)	MS6 (BLK)	X	X	X	X	X	
603	MS1–A13–A1	MS7	X	X	X	X	X	
604	MS1–A13–A1	MS6/MS7	X	X	X	X	X	
606	MS4	MS1–A3–A1	X	X	X	X	X	
621	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	
622	MS5–A1 (Gas Line)	MS1–A2 (Gas Line)	X	X	X	X	X	
623	MS5–A1	PP1–F1,F2,F3,F4	X	X	X	X	X	
624	MS1–A2	PP1–F1,F2,F3,F4	X	X	X	X	X	
700	MG3–A17–J2	PP1–A14–J86		X	X	X	X	
702	MR2–A11–J18	MR1–A7–J5					X	
706	MR2–A11–J14	PD1–A14–A1–J3						
708*	MR2–A11–J18 MDS	MR1–A7–J27				X	X	
709	MR1–A7–J26	MR3–A11–J2 CLK				X	X	
710*	MR2–A11–J17	MR3–A11–J1 MDS				X	X	
711/712	PP1–A17–A1–J91 (F/O REPEATER BD)	MG2–A33–J11,12,13 MG2–A11–A1–J9,10				X	X	
711/712*	MR2–A11–J13	PP1–A17–A1					X	
715	MG2–A33–J1	PP1–J14		X	X	X	X	
716	MG2–A11–A1–J2	PP1–J89		X	X	X	X	
726	PP1–A16–J77 **(See Note 1)**	MR1–A23–J18 **(See Note 1)**						
727	MG3–A36–J5	PP1–A16–J77		X	X	X	X	
728	MG3–A36–J11	MG2–A16–OUT–J1	X	X	X	X	X	
729	MG3–A36–J12	MG2–A16–OUT–J2	X	X	X	X	X	
730	MG3–A36–J13	MG2–A16–OUT–J3	X	X	X	X	X	
731	MG3–A36–J14	MG2–A16–OUT–J4	X	X	X	X	X	

4–19 IDENTIFYING NEW CABLES FOR SIGNA 9.x WITH ACGD (continued)

RUN#	FROM	TO	A	B	C	D	E	F
732	MG3–A36–J15	MG2–A16–OUT–J5	X	X	X	X	X	
733	MG3–A36–J16	MG2–A16–OUT–J6	X	X	X	X	X	
734	MG3–A36–J17	MG2–A16–OUT–J7	X	X	X	X	X	
735	MG3–A36–J18	MG2–A16–OUT–J8	X	X	X	X	X	
736	MG3–A36–J19	MG2–A16–OUT–J9	X	X	X	X	X	
743	MG2–A12–A1–J1	MG3–A3–J4	X	X	X	X	X	
744	MG2–A12–A1–J2	MG3–A3–J3	X	X	X	X	X	
745	PP1–J44 **(See Note 2)**	MR1–A7–J13 **(See Note 2)**		X	X	X		
746	PP1–J44 **(See Note 2)**	MG3–A3–A1–J1 **(See Note 2)**		X	X	X		
747	PP1–J4 **(See Note 2)**	MG3–A17–J1 **(See Note 2)**		X	X	X		
748	PP1–J4 **(See Note 2)**	MR1–A7–J12 **(See Note 2)**		X	X	X		
749	MG3–A36	MG2–A33–J10	X	X	X	X	X	
762	MR3CABI/FJ3	MG3–A32–TS1/2		X	X	X	X	
763	MR3CABI/FJ4	MG3–A32–TS3/4		X	X	X	X	
764	MR3CABI/FJ5	MG3–A32–TS5/6		X	X	X	X	
768	PP1–J20	MR1–A7–J31				X	X	
769	PP1–J15	MR1–A7–J34				X	X	
770	PP1–J11	MR1–A7–J41				X	X	
771	PP1–J47	MR1–A7–J36				X	X	
772	PP1–J14	MR1–A7–J40					X	
773	PP1–A17–J1	MR1–A7–J35					X	
774	MR2–A11–J25	MR1–A7–J32					X	
775	MR1–A7–J42	PP1–A11–J92				X	X	
776	MR1–A7–J44	PP1–A11–J93				X	X	
777	MR1–A7–J43	PP1–A11–J94				X	X	
778	MG2–A12–A1–J3	PP1–A11–J72		X	X	X	X	
779	MG2–A12–A1–J4	PP1–A11–J73		X	X	X	X	
780	MG3–A3–J6	PP1–A11–J74		X	X	X	X	
781	MG2–A12–A1–J5	PP1–A11–J75		X	X	X	X	
782	MG2–A12–A1–J6	PP1–A11–J76		X	X	X	X	
783	MR1–A7–J37	PP1–F5/F6				X	X	
784	PP1–F5/F6 **(See Note 5)**	MG6–J1 (FAN) **(See Note 5)**		X	X	X	X	
785	MR1–A7–J18	PP1–F7/F8				X	X	
786	PP1–F7/F8 **(See Note 5)**	MG6–J2 (FAN) **(See Note 5)**		X	X	X	X	
788	OW1–A1–A2–J7	PP1–J12			X	X	X	
789	OW1–A1–12–J18	MR2–A11–J26			X	X	X	
791	OW1–A2–A5–J2	OW1–A11–J30			X	X	X	
797	OW1 – A3 (LCD DISPLAY)	OW1 – A2 – A1 (VIDEO)			X	X	X	X
805	OW1–A1–A4–J6	OW1–A7–KYBRD			X	X	X	X
807	OW1–A16	OW1–A5 (DASM)			X	X	X	X
810	OW1–A1–A2–J8	OW1–A15–KBD,MOUSE			X	X	X	X
811	OW1–A2–A4–PORT2	OW1–A15–ETHERNET			X	X	X	X
812	OW1–A13 (MODEM)	OW1–A15–COM1			X	X	X	X
813	OW1–A2–A1–COM2	OW1–A15–COM2			X	X	X	X
814	OW1–A1–A4–J9	OW1–A2–A1–KBD,MOUSE			X	X	X	X
815	OW1–A16–SCSI	OW1–A15–SCSI (PCI)			X	X	X	X

4–19 IDENTIFYING NEW CABLES FOR SIGNA 9.x WITH ACGD (continued)

RUN#	FROM	TO	A	B	C	D	E	F
817	OW1–A1–A2–J17	OW1–A2–A5–J3			X	X	X	X
818	OW1–A16–PORT6,7,8	MR2–A11–J31			X	X	X	X
819	OW1–A2–A7–SCSI	OW1–A15–SCSI			X	X	X	X
821	OW1–A15–MONITOR **(See Note 4)**	OW1–A6–R,B,G (COLOR) **(See Note 4)**			X	X	X	X
823	MSM1–A1–J10	MR2–A11–J24	X	X	X	X	X	
824	PP1–J10	FJ1	X	X	X	X	X	
825	PP1–J48	MSM1–A1–J8	X	X	X	X	X	
826	MSM1–A1–J9	FJ3	X	X	X	X	X	
827	FJ3	MS5–A6–A1–J1/FJ4	X	X	X	X	X	
828	PP1–J10	MA1–A3–A1–P403	X	X	X	X	X	
829	PP1–J48	MS1–FJ2	X	X	X	X	X	
830	MS1–FJ2	MS1–A2–J2/MS1–A5–A2–J1/MG3–TB1	X	X	X	X	X	
831	FJ1	MSM1–A1–J7	X	X	X	X	X	
832	MS1–A2–J1	MS1–SLEEVE/MS1–COLDHEAD	X	X	X	X	X	
833	FJ4	MS5–A1–A6–JR	X	X	X	X	X	
836	OW1–A15–BIT3–XMT1/RCV1	MR2–A11–A30–RCV1/XMT1			X	X	X	
848	MG2–A11–J4	MG2–A4–J5	X	X	X	X	X	
861	MG2–A4–J4	MG2–A33–J5,6,7,8,9	X	X	X	X	X	
862	MG2–A4–J1	MG2–A6–J1	X	X	X	X	X	
863	MG2–A4–J2	MG2–A6–J2	X	X	X	X	X	
869	MG3–A2–J8	MG2–A14–A2–SPEAKER	X	X	X	X	X	
893	MG3–A3–J2	MG3–A3–A5–INPUT	X	X	X	X	X	
897	MG2–A4–J3	MG2–A33–J4	X	X	X	X	X	
940	MSM3 RS232	MSM1 J1	X	X	X	X	X	
966	MR3–PD1–7,8,9,GND	MR1–A7–J1	X	X	X	X	X	
967	MR3–PD1–10,11,12,GND	MR1–CB1	X	X	X	X	X	
968	MR3–PD1–4,5,6,GND	MR2–A12–J1	X	X	X	X	X	
969	MR3–PD1–18,19,GND,N	OW1–A2–A5–J1	X	X	X	X	X	
970	MR3–PD1–16,17,GND	WC1	X	X	X	X	X	
1050	MG3–A37–J1 **(See Note 3)**	MG3–A36–J19 **(See Note 3)**	X	X	X	X	X	
1051	MG3–A37–J2 **(See Note 3)**	MG2–A33–J7, RUN 861 **(See Note 3)**	X	X	X	X	X	
1052	MG3–A37–J3 (Future) **(See Note 3)**	MG2–A33–J12 (Future) **(See Note 3)**						
MIC	MG3–A2–J7	MG2–A14–A1–MICROPHONE	X	X	X	X	X	

* See Section 4–17 table for connection to TYME boards within 9.x System Cabinet.

Notes:

1. If not part of current system, runs 486, 487, 488, 489, 490, 499, and 726 will be installed as part of Multi–coil Option.
2. 200 series cables are 1.5T specific. 700 series cables are 1.0T specific. These cable will have labels added after they are routed and terminated.
3. Installed on 1.5T systems only.
4. Run 798 is equivalent to Run 821 and Run 799 is equivalent to Run 820.
5. Runs 045, 784, and 786 may not have to be installed if existing Blower Box has not been deinstalled and existing runs are still in place. If new Runs 045, 784, 786, and new Blower Box are not used, return for recycling.

4–20 TYME BOARD FIBER OPTIC CABLE LABEL CONNECTION INFORMATION

Runs 708, 710, 711, and 712 are used for both Horizon 5.x and 9.x. The fiber optic connections to the TYME board are different for 5.x and 9.x. Table 4–12 below indicates the connection labels as they should be marked on the TYME board end of the fiber optic cables.

TABLE 4–12
SYSTEM CABINET FIBER OPTIC CONNECTIONS

RUN	SYSTEM CABINET ONE END		OTHER END	
	5.x ONE END	9.x ONE END	5.x OTHER END	9.x OTHER END
708–1	MR2–A15–A23–J5	MR2–A30–A23–J19–RXMDS	NO CHANGE	NO CHANGE
710–1	MR2–A15–A23–J13	MR2–A30–A23–J12–X	NO CHANGE	NO CHANGE
710–2	MR2–A15–A23–J15	MR2–A30–A23–J13–Y	NO CHANGE	NO CHANGE
710–3	MR2–A15–A23–J17	MR2–A30–A23–J15–Z	NO CHANGE	NO CHANGE
710–4	MR2–A15–A23–J14	MR2–A30–A23–J17–CLOCK	NO CHANGE	NO CHANGE
710–5	MR2–A15–A23–J4	MR2–A30–A23–J18–TXMDS	NO CHANGE	NO CHANGE
711–1	MR2–A15–A23–J6,J7	MR2–A30–A23–J25,J27	NO CHANGE	NO CHANGE
711–2	MR2–A15–A23–J8	MR2–A30–A23–J23–RSTSRI	NO CHANGE	NO CHANGE
712–1	MR2–A15–A23–J9,J10	MR2–A30–A23–J20,J21	NO CHANGE	NO CHANGE

4–21 SIGNA 9.x INTERCONNECT MAPS FOR UPGRADES

For Signa 9.x cable installation, fold–out Interconnect Maps are provided on pages 4–45 thru 4–49. These maps show bolded and dashed lines that represent new, re–used, and/or re–labeled cables for existing systems being upgraded to Signa 9.x.

Interconnects for additional cables supplied with other options (Spectroscopy, etc.) are illustrated in the installation manual shipped with the option.

Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation instruction (PIP Direction) within this manual.

From any early configuration, the cable maps listed below illustrate the final configuration of various magnet systems after the completion of the Signa 9.x upgrade. The cable map illustrates the Signa 9.x HiSpeed or EchoSpeed System interconnect for all supplied cables and interface with customer supplied wiring.

<u>Use only the applicable Upgrade Interconnect Cable Map that applies to the 9.x HiSpeed or EchoSpeed system being upgraded.</u> Tape map in a convenient location to check off routed cable. The fold–out map should be returned later to the binder for future reference.

The Signa 9.x Upgrade Interconnect Cable Maps can be found on the following pages:

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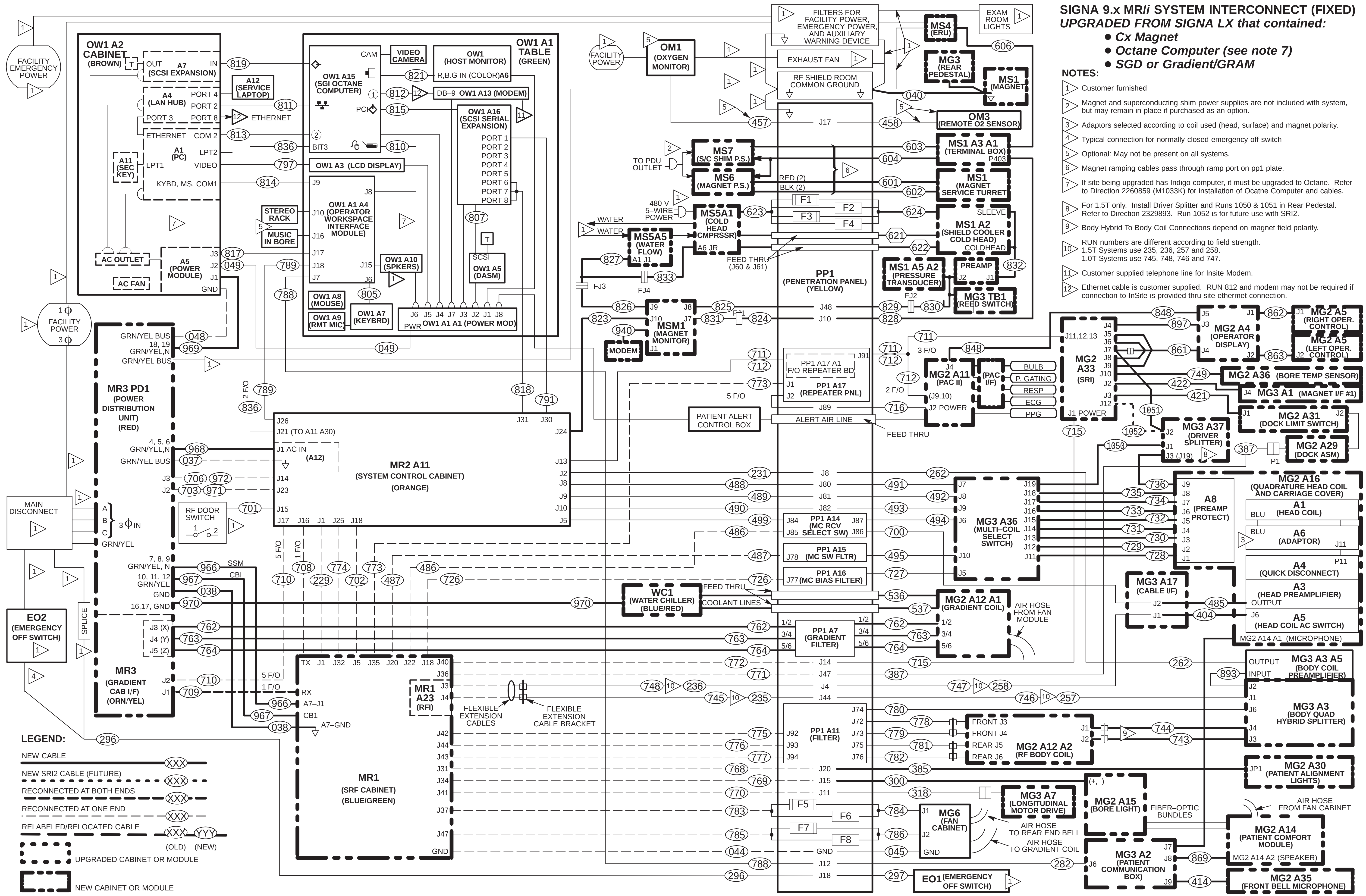
SIGNA 9.x MR/*i* SYSTEM INTERCONNECT (FIXED)

UPGRADED FROM SIGNA LX that contained:

- ***Cx Magnet***
- ***Octane Computer (see note 7)***
- ***SGD or Gradient/GRAM***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. If site being upgraded has Indigo computer, it must be upgraded to Octane. Refer to Direction 2260859 (M1033K) for installation of Ocatne Computer and cables.
8. For 1.5T only. Install Driver Splitter and Runs 1050 & 1051 in Rear Pedestal. Refer to Direction 2329893. Run 1052 is for future use with SRI2.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



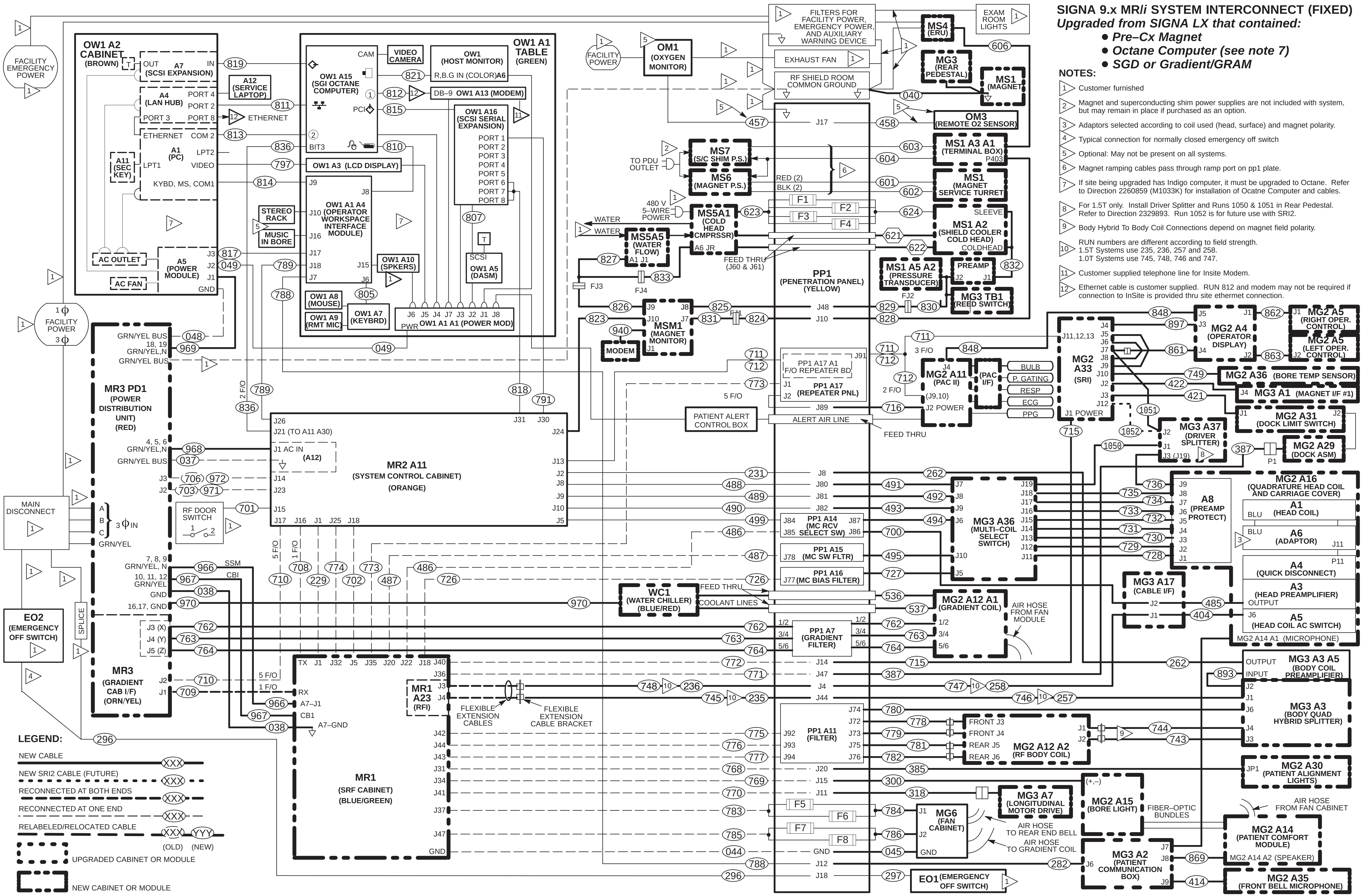
SIGNA 9.x MR/*i* SYSTEM INTERCONNECT (FIXED)

Upgraded from SIGNA LX that contained:

- ***Pre–Cx Magnet***
- ***Octane Computer (see note 7)***
- ***SGD or Gradient/GRAM***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. If site being upgraded has Indigo computer, it must be upgraded to Octane. Refer to Direction 2260859 (M1033K) for installation of Ocatne Computer and cables.
8. For 1.5T only. Install Driver Splitter and Runs 1050 & 1051 in Rear Pedestal. Refer to Direction 2329893. Run 1052 is for future use with SRI2.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



LEGEND:

- NEW CABLE
- NEW SRI2 CABLE (FUTURE)
- RECONNECTED AT BOTH ENDS
- RECONNECTED AT ONE END
- RELABELED/RELOCATED CABLE (OLD) (NEW)
- UPGRADED CABINET OR MODULE
- NEW CABINET OR MODULE

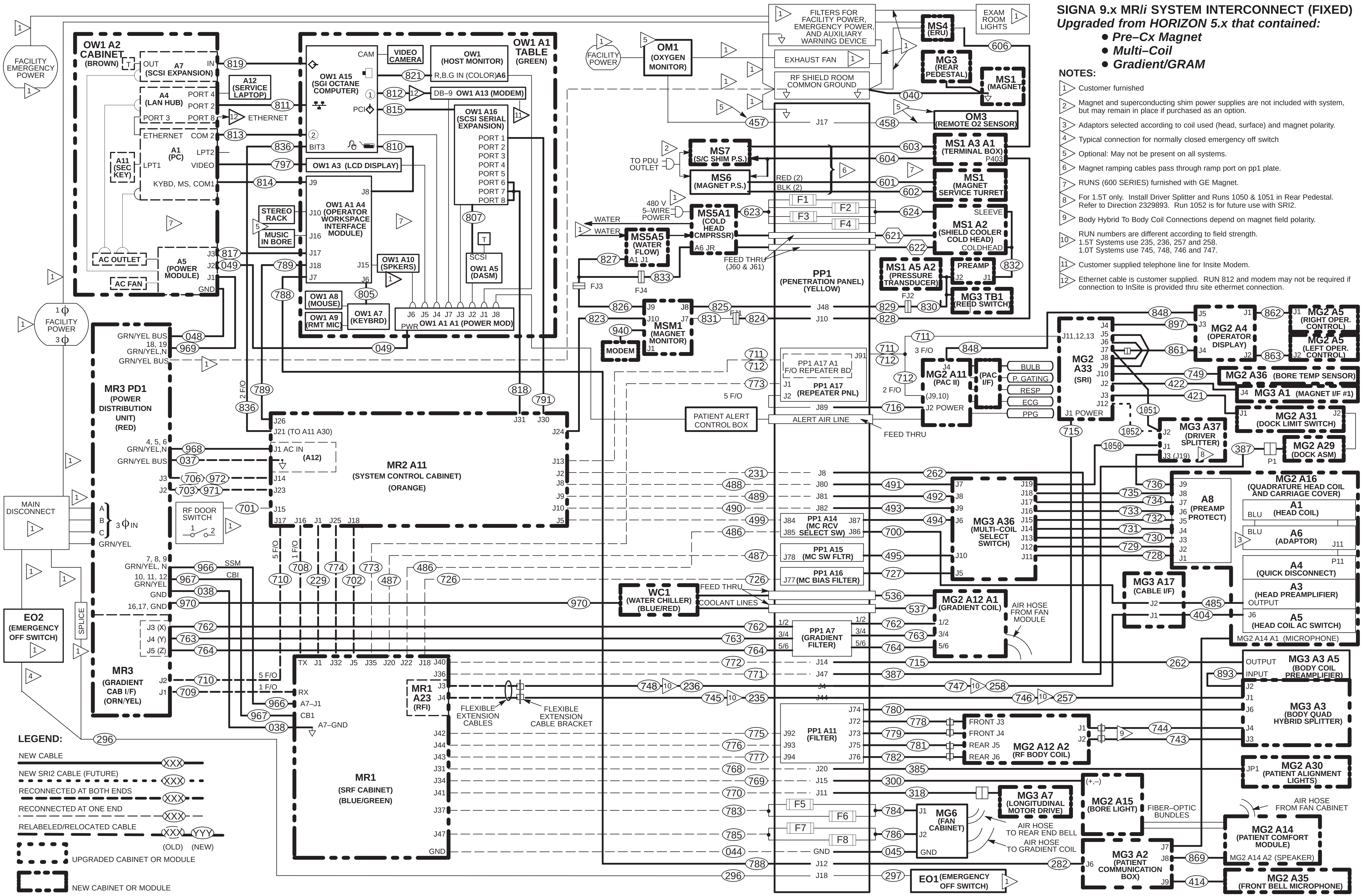
SIGNA 9.x MR/*i* SYSTEM INTERCONNECT (FIXED)

Upgraded from HORIZON 5.x that contained:

- ***Pre–Cx Magnet***
- ***Multi–Coil***
- ***Gradient/GRAM***

NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) furnished with GE Magnet.
8. For 1.5T only. Install Driver Splitter and Runs 1050 & 1051 in Rear Pedestal. Refer to Direction 2329893. Run 1052 is for future use with SRI2.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



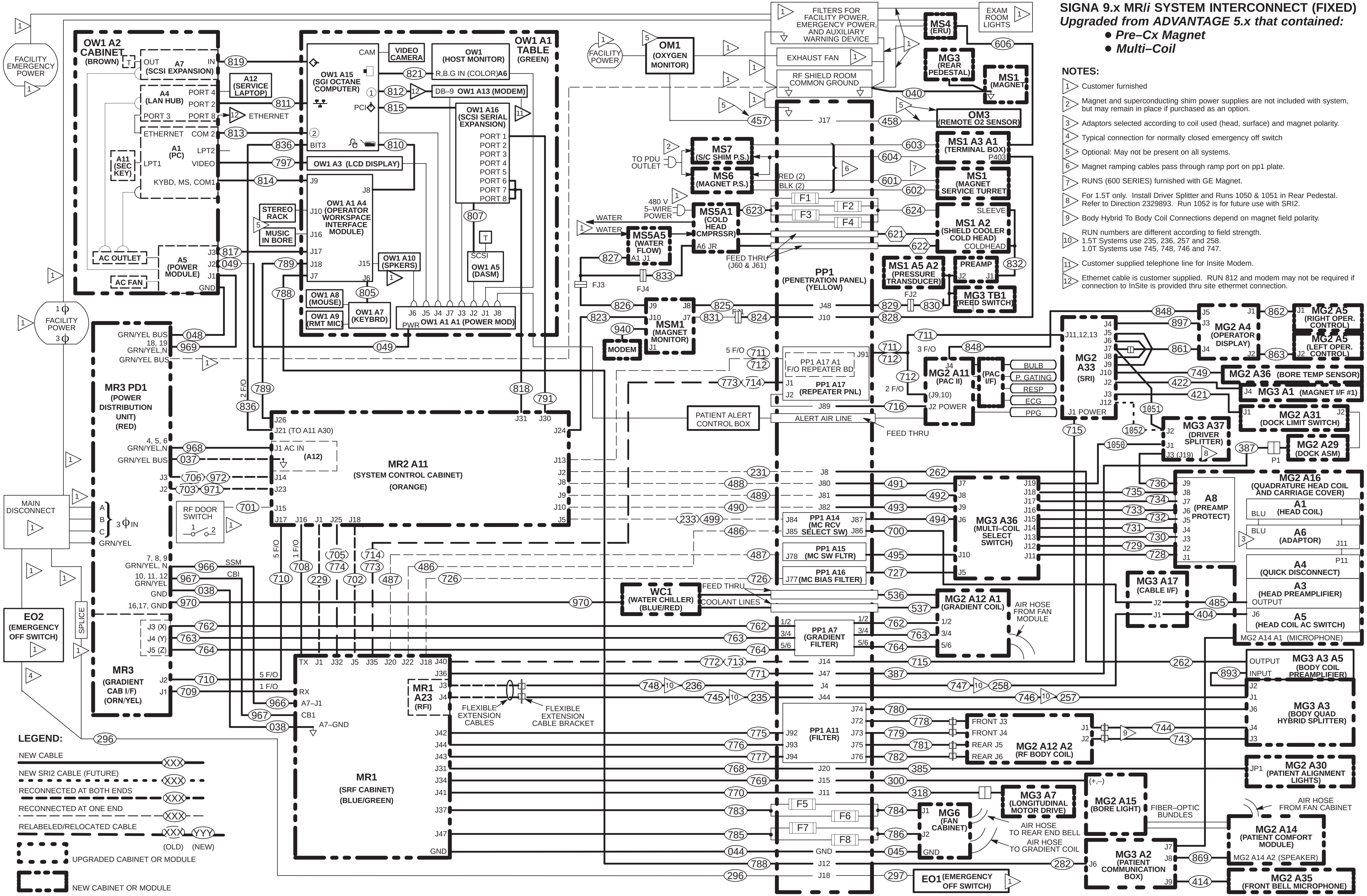
SIGNA 9.x MR/*i* SYSTEM INTERCONNECT (FIXED)

Upgraded from ADVANTAGE 5.x that contained:

- ***Pre–Cx Magnet***
- ***Multi–Coil***

NOTES:

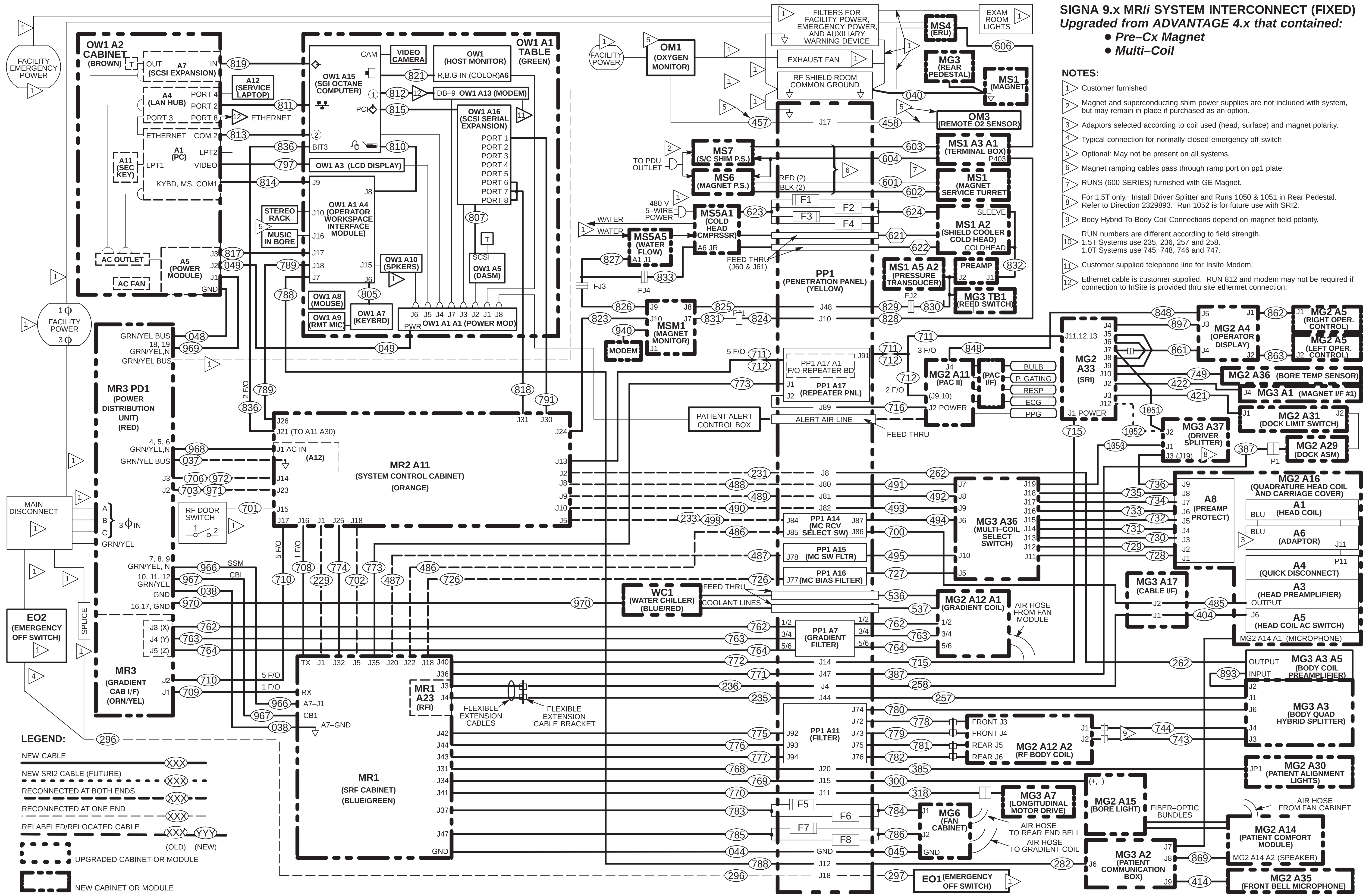
1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) furnished with GE Magnet.
8. For 1.5T only. Install Driver Splitter and Runs 1050 & 1051 in Rear Pedestal. Refer to Direction 2329893. Run 1052 is for future use with SRI2.
9. Body Hybrid To Body Coil Connections depend on magnet field polarity.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



SIGNA 9.x MR/*i* SYSTEM INTERCONNECT (FIXED)

Upgraded from ADVANTAGE 4.x that contained:

- ***Pre–Cx Magnet***
- ***Multi–Coil***



NOTES:

1. Customer furnished
2. Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
3. Adaptors selected according to coil used (head, surface) and magnet polarity.
4. Typical connection for normally closed emergency off switch
5. Optional: May not be present on all systems.
6. Magnet ramping cables pass through ramp port on pp1 plate.
7. RUNS (600 SERIES) furnished with GE Magnet.
8. For 1.5T only. Install Driver Splitter and Runs 1050 & 1051 in Rear Pedestal. Refer to Direction 2329893. Run 1052 is for future use with SRI2.
9. Body Hybrid To Body Coil Connections depend on magnet field strength.
10. RUN numbers are different according to field strength. 1.5T Systems use 235, 236, 257 and 258. 1.0T Systems use 745, 748, 746 and 747.
11. Customer supplied telephone line for Insite Modem.
12. Ethernet cable is customer supplied. RUN 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.

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4–22 SIGNA CABLE INFORMATION

Many of the cables are common to both the 9.x and 10.x System. The cables that are specific to 9.x or 10.x installation are designated in the applicable cable maps and in the first column of the following tables as:

- * Cable used on 9.x installation only
- ** Cable used on 10.x (MGD) installation only

CAUTION

Ty–wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
“FROM” & “TO”	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material my be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material’s ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable’s use in the MR system	

4–22 SIGNA CABLE INFORMATION (continued)

1. Unpack magnet room cables from the GE Magnet accessory box(s) and move them into the magnet or equipment room for routing. The shield cooler runs should have been installed previously when magnet was delivered.

   a. For magnet and equipment room standard runs delivered with GE Magnet, refer to Table 4–16.

2. Unpack and move power and ground cable boxes to the PDU area where they will be routed. Set Run 040, 044, and 045 aside. Power and ground cables are listed in Table 4–13.

TABLE 4–13
PDU POWER AND GROUND CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW–1	MR2 "permanent" (UL) ground
038	PD1 A2 A2	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW–1	SRF Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	208Y	105	VW–1	Magnet Ground Cable
044	PP1 GND	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW–1	RF/Pen Cabinet Separate ground
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW–1	Blower Cabinet Separate ground
048	PD1 GND	OW1 A2 A5 GND	51 ft Ground Wire (#10/1)	SO	600	0	105	VW–1	Operator Workspace Cabinet ground
049	OW1 A1 A1	OW1 A2 A5 J2	Power Cord	STO	600	120	90	VW–1	Operator Workspace Table to OW Cabinet
966	MR1 A7 J1	MR3 PD1	35 ft Power (5–#8)	SO	600	208Y	75	FT–4	ACGD SSM Module Power
967	MR1 CB1	MR3 PD1	35.5 ft Power (4–#8)	SO	600	208	90	FT–4	ACGD SRF Cabinet Power
968	MR1 A12 J1	MR3 PD1	35 ft Power (5–#8)	SO	600	208Y	75	FT–4	ACGD System Cabinet Power
969	OW1 A2 A5 J1	MR3 PD1	65.5 ft Power (4–#10)	SO	600	120	75	FT–4	ACGD Operator Workspace Cabinet Power
970	WC1	MR3 PD1	82 ft Power (3–#10)	STO	600	208	90	FT–4	ACGD Water Chiller Power
973	HE1	MR3 PD1	82 ft Power (3–#12)	SO	600	208	75	FT–4	ACGD Heat Exchanger Power
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5–#8) With 30A Plug and Receptacle						Cable for Compact PDU to GE Magnet Power Supply or SC Shim Supply Power

4–22 SIGNA CABLE INFORMATION (continued)

3. Unpack and move fixed site magnet room interconnect cables to magnet room. Refer to Table 4–14.
   a. Locate Body Coil RF input Cables (ship with Body Coil) and move cables to magnet room.

TABLE 4–14
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
262	PP1 J8	MG3 A3 A5 OUT	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Body Coil receive
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Table Elevation motor power
404	MG3 A17 J1	MG2 A16 A5 J6	13.67 ft. RG58/U Coax	CI2	1900 VRMS	<30	80	VW–1	Head transmit cable take up
414	MG3 A35	MG3 A2 J9	30 ft. 9–pin Sub–D	CI3	300	<30	80	FT–4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	Long Drive Encoder/Limit Sw
458	PP1 J17	OM3	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional Remote O2 Sensor
485*	MG3 A17 J2	MG2 A16 A3 OUT	13.67 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW–1	Head receive cable take up
491	PP1–J80	MG3–A36–J7	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
492	PP1–J81	MG3–A36–J8	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC3 Receive
493	PP1–J82	MG3–A36–J9	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
494*	PP1–A14–J87	MG3–A36–J6	80 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC1 Receive
495*	PP1–A15–J78	MG3–A36–J10	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Switching
700*	PP1 A14 J86	MG2/MG3 A17 J2	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Head or Surface Coil receive
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	PAC Power
727*	PP1–A16–J77	MG3–A36–J5	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Bias
728	MG3 A36 J11	MG2 A16 OUT J1	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
729	MG3 A36 J12	MG2 A16 OUT J2	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
730	MG3 A36 J13	MG2 A16 OUT J3	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
731	MG3 A36 J14	MG2 A16 OUT J4	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
732	MG3 A36 J15	MG2 A16 OUT J5	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
733	MG3 A36 J16	MG2 A16 OUT J6	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable

4–22 SIGNA CABLE INFORMATION (continued)

TABLE 4–14 (CONTINUED)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
734	MG3 A36 J17	MG2 A16 OUT J7	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
735	MG3 A36 J18	MG2 A16 OUT J8	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW–1	RF Cable
736*	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	RF Cable
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F “N”	STO	2500 VDC	950	85	VW–1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F “N”	STO	2500 VDC	950	85	VW–1	Body Coil Drive
749	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9–Sub D	CI3	300	<30	105	FT–4	Bore Temperature Sensor
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power	SO	600	120	80	FT–4	Patient cooling blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub–D/rt.	CI3	300	<30	80	FT–4	Operator Display to PAC II
861	MG2 A4 J4	MG2 A33J5,6,7,8,9	20 ft 5/37 pin Sub–D/80 pin	CI2	300	<30	80	VW–1	Operator Display to SRI
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display – Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW–1	Operator Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW–1	Spliiter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub–D/26 pin	CI3	300	<30	80	FT–4	Operator Display to SRI
1035**	MG2 A16 A3/A5 J4	MG3 A36 J21	40 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW–1	Head or Surface Coil Receive
1036**	PP1 J84	MG3 A36 J6	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1037**	PP1 J6	MG3 A36 J1	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1038**	PP1 J7	MG3 A36 J2	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1039**	PP1 J58	MG3 A36 J3	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1040**	PP1 J59	MG3 A36 J4	BNC	CMX	1900 VRMS	<30	80	VW–1	MC4 Receive
1041**	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	SRI Power
1042**	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Monitor to Compressor I/F
1049**	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Multi–Coil Select Cable
1050*	MG3 A37 J1	MG3 A36 J19							APA2 Cable
1051*	MG3 A37 J2	MG2 A33 J7							APA2 Cable
1052*	MG3 A37 J2	MG2 A33 J12							APA2 Cable (Future)

4–22 SIGNA CABLE INFORMATION (continued)

4. Unpack and move fiber optic cables into routing location. Refer to Table 4–15.

TABLE 4–15
FIBER OPTIC CABLES DELIVERED WITH FIXED–SITE KIT

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
708*	MR2 A11 J16	MR1 A7 TX	20 ft. 1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
709	MR3 A11 J2	MR1 A7 RX	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
710*	MR2 A11 J17	MR3 A11 J1	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
711/ 712*	MR2 A11 J13	PP1–J91	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
711/ 712	PP1–J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC–II
836*	MR2 A11 J30 BIT3	OW1 A15 BIT3	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables wiith two spares
1033**	MR2 A11 J16	MR1 A7 TX	20 ft. 1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
1034**	MR2 A11 J17	MR3 A11 J1	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
1044**	MR2 A11 J22	OW1 A15 SLOT2	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1045**	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables

TABLE 4–16
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW–1	Ramp and Shim Cable Assem-bly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12–#22)	CI3	150	60 VDC	80	FT–4	Switch Heaters
605	MS1	MR2 A8	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Instrumentation Cable, 80 feet
606	MS4	MS1 A3 A1	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Emergency Run Down Unit
608	MS1	MR2 A8	Data Cable (4–#22)	CI3	300	<30	105	FT–4	Instrumentation Cable, 15 feet
609	MS7 OR MS8	PDU	Power Cable (5–#16)	SO	300	208Y	105	FT–4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW–1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4–#18)	SO	600	200	105	VW–1	Cold Head Control Cable

4–22 SIGNA CABLE INFORMATION (continued)

5. Unpack and move interconnect cables which are cut to length to the applicable room. Refer to Table 4–17.

   a. Move gradient cable Runs 982 thru 987 into magnet room. Move gradient cable Runs 976 thru 981 into equipment room. Gradient cables will be cut, re–terminated and routed according to appropriate Direction in Tab 1.

TABLE 4–17
EQUIPMENT ROOM AND MAGNET ROOM CABLES CUT TO LENGTH

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
235	MR1 A7 J13	PP1 J44	7/8 in. HELIAX® Coax	STO	6000 VDC	<30	120	VW–1	1.5T Body Coil Run created from kit
236	MR1 A7 J12	PP1 J4	1/2 in. HELIAX® Coax	STO	4000 VDC	<30	120	VW–1	1.5T Head Coil Run created from kit
257	PP1 J44	MG3 A3 A1 J1	7/8 in. HELIAX® Coax	STO	6000 VDC	<30	120	VW–1	1.5T Body Coil Transmit
258	PP1 J4	MG3 A17 J1	1/2 in. HELIAX® Coax	STO	4000 VDC	<30	120	VW–1	1.5T Head Coil Transmit
745	MG3 A7 J13	PP1 J44	50 ft 1/2” Heliax (HN)	STO	4000 VDC	950	120	VW–1	1.0T Body RF Transmit
746	PP1 J44	MG3 A3 J1	55 ft 1/2” Heliax (HN)	STO	4000 VDC	950	120	VW–1	1.0T Body Coil Transmit
747	PP1 J4	MG3 A17 J1	50 ft 3/8” Heliax (N)	STO	2500 VDC	950	85	VW–1	1.0T Head Coil Transmit
748	MG3 A7 J12	PP1 J4	50 ft 3/8” Heliax (N)	STO	2500 VDC	950	85	VW–1	1.0T Head RF Transmit
762	MR3 CABI/F J3	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
762	MG3 A32 1/2	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT–4	X Gradient output
763	MR3 CABI/F J4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
763	MG3 A32 3/4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT–4	Y Gradient output
764	MR3 CABI/F J5	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output
764	MG3 A32 5/6	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT–4	Z Gradient output

4–22 SIGNA CABLE INFORMATION (continued)

6. Verify magnet monitoring cables for GE Magnet are delivered (or if previously installed, remain in place). Refer to Table 4–18.

TABLE 4–18
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15–#22)	CI3	300	<30	80	FT–4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8–#22) & Data Cable (4–#22)	CI3	300	<30	80	FT–4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25–#22)	CI3	300	<30	80	FT–4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT–4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8–#24)	CI3	300	<30	80	FT–4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8–#22)	CM	300	<30	80	FT–4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6–#24)	CM	300	<30	80	FT–4	Compressor to Compressor I/F
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9–#22)	CI3	300	<30	80	FT–4	Monitor to Modem
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

4–22 SIGNA CABLE INFORMATION (continued)

7. Unpack and move equipment room system cables into routing location (cabinets, Operator Console, Operator Workstation, PDU). Refer to Table 4–19.

TABLE 4–19
EQUIPMENT ROOM RUNS

RUN	“FROM”	“TO”	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
229	MR2 A11 J1	MR1 A7 J3	20 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Exciter RF output
231	MR2 A11 J2	PP1 J8	40 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW–1	Body Coil receive
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub–D	CI3	300	<30	80	FT–4	Optional O2 Monitor cable
486*	PP1–A14–J85	MR1–A7–J22	30 ft. 9 Sub–D	CI3	300	<30	80	FT–4	Head/MC1 Receive Select Switch
487*	PP1–A15–J78	MR1–A7–J20	40 ft.15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Switching
488	PP1–J80	MR2–A11–J8	40 ft. BNC	CMX	300	<30	80	VW–1	MC2 Receive
489	PP1–J81	MR2–A11–J9	40 ft. BNC	CMX	300	<30	80	VW–1	MC3 Receive
490	PP1–J82	MR2–A11–J10	40 ft. BNC	CMX	300	<30	80	VW–1	MC4 Receive
499/ 233	MR2 A11 J5	PP1 J7	40 ft RG223/U Coax	CMX	300	<30	80	VW–1	Head and Surface Coil receive
611	TAC J22	PP1 SHIM							Resiisteive Shim PS Option
612	MG2 A12 A1	PP1 SHIM							Resiisteive Shim PS Option
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	RF Door switch
702	MR2 A11 J18	MR1 A7 J5	20 ft 9 pin Sub–D	CI3	300	<30	80	FT–4	Unblank/EFB
726*	PP1–A16–J77	MR1–A7–J18	30 ft. 15 Sub–D	CI3	300	<30	80	FT–4	Multi–coil Bias
768	MR1 A7 J31	PP1 J20	30 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	Patient Alignment Light power
769	MR1 A7 J34	PP1 J15	30 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	Bore Light power
770	MR1 A7 J41	PP1 J11	30 ft 9 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Long Drive Motor power
771	MR1 A7 J36	PP1 J47	30 ft 25 pin Sub–D	CI3	300	38.5 VDC	80	FT–4	Dock Power
772	MR1 A7 J40	PP1 J14	30 ft 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
773	MR1 A7 J35	PP1 A17 J1	30 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	PAC & Fiber Optic I/F Power
774	MR2 A11 J25	MR1 A7 J32	20 ft 25 pin Sub–D	CI3	300	<30	80	FT–4	Hydraulic Manifold Interface
775	MR1 A7 J42	PP1 A11 J92	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	100V Dynamic Disable
776	MR1 A7 J44	PP1 A11 J93	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	100V Dynamic Disable
777	MR1 A7 J43	PP1 A11 J94	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW–1	100V Dynamic Disable
783	MR1 A7 J37	PP1 F5/F6	30 ft 2–#8 AWG	SO	600	120	80	FT–4	Blower Cabinet Power
788	OW1 A1 A4 J7	PP1 J12	40 ft Data (25–#22)	CI3	300	<30	80	FT–4	Intercom
789	OW1 A1 A4 J18	MR2 A11 J26	51 ft Data (9–#22)	CI3	300	<30	80	FT–4	E–stop
791	OW1 A16 PORT1	MR2 A11 J30	51 ft Data (25–#22)	CI3	300	<30	80	FT–4	TNF monitoring (Octane)
797	OW1 A2 A1 PC	OW1 A3 LCD	2 line; 3.5 mm jacks	CI3	300	<30	80	FT–4	LCD Display

4–22 SIGNA CABLE INFORMATION (continued)

TABLE 4–19 (CONT'D)
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
798	OW1 A2 A3 HOST	OW1 A6	10.5 ft. Data (3–#24)	CI3	300	<30	80	FT–4	Color Monitor cable
805	OW1 A1 A4 J6	OW1 A7	3.3 ft. Data	CI3	300	<30	80	FT–4	Keyboard cable
807	OW1 A2 A3 HOST	OW1 A5	9.8 ft. Data	CI3	300	<30	80	FT–4	DASM SCSI cable
810	OW1 A1 A2 J8	OW1 A15 KBD,MS	9 ft. 25 pin Sub–D	CI3	300	<30	80	UL–1581	Keyboard/Mouse cable (Octane)
811	OW1 A2 A4	OW1 A15	12 ft. Ethernet	CI3	300	<30	80	FT–4	Communication
812	OW1 A15	OW1 A13	12 ft. 25/9 pn Sub–D	CM	300	<30	80	UL–1581	Modem
813	OW1 A2 A1 COM2	OW1 A415 PORT7	12 ft. 9 pin Sub–D	CI3	600	<30	105	VW–1	Communication
814	OW1 A2 A1 KBD,MS	OW1 A1 A4 J9	9 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Keyboard/Mouse cable (PC)
815	OW1 A16	OW1 A15 SCSI	6 ft. 68/50 pin	CI3	300	<30	80	FT–4	SCSI Cable (Octane)
817	OW1 A2 A5 J3	OW1 A1 A2 J17	8 ft. 9 pin Sub –D	CI3	300	<30	80	FT–4	Workspace I/F Module
818	OW1 A16 J6,7,8	MR2 A11 J31	70 ft. 25/3–25 pin	CI3	300	<30	80	FT–4	TNF monitoring
819	OW1 A2 A7 SCSI	OW1 A15	10 ft. 68/50 pin	CI3	300	<30	80	FT–4	SCSI Cable
821	OW1 A15 MON-ITOR	OW1 A6 R,B,G	10.5 ft. 3 BNC/SubD	CI3	300	<30	80	FT–4	Color Monitor cable
838**	OW1 A17 J3	PP1 J51	65 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Serial Converter to Pen Panel
911**	OW1 A17 J1	OW1 A16 PRT 5	6 ft. 25 pin Sub–D						
971	MR2 A11 J23	MR3 PD1 J2	40 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	PDU Status (ACGD)
972*	MR2 A11 J14	MR3 PD1 J3	40 ft. 9 pin Sub–D	CI3	300	<30	80	FT–4	Remote PDU (ACGD)
1025**	MR2 A11 J40	PP1 J6	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1026**	MR2 A11 J41	PP1 J7	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1027**	MR2 A11 J42	PP1 J58	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1028**	MR2 A11 J43	PP1 J59	40 ft. BNC	CMX	1900 VRMS	<30	80	VW–1	MC2 Receive
1029**	MR2 A11 J62	PP1 A16 J77	60 ft. 15 pin Sub–D	CI3	300	<30	80	FT–4	Monitor to Compresssor I/F
1030**	MR2 A11 J61	PP1 A16 J78	30 ft. 37 pin Sub–D	CI3	300	<30	80	FT–4	SRI Power
1031**	MR2 A11 J63	OW1 A15 SLOT1	65 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Communication
1032	MR2–A11–J31	OW1–A16–FJ100	65 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Serial Communication
1046	FJ100	OW1–A16–PORT 1,6,7,8	4.5 ft. 25 pin Sub–D	CI3	300	<30	80	FT–4	Serial Communication

8. Unpack additional equipment room cables for Signa Options:
   a. Refer to the service/installation manual shipped with options.

SECTION 5 – COMPLETION

5–1 PHANTOMS AND SERVICE KITS

1. Locate SPT Phantom Cart Kit.
2. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
3. Place service phantoms into SPT Phantom Cart or in a service storage cabinet to be organized and ready for be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.

5–2 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD–ROM, Set–Up & Calibration, and perform the following:
   - PDU Voltage Checks Compact
   - Startup Checks Compact
   - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the Gradient Cabinet 8645 power modules, switch all cabinet circuit breakers to ON. The 8645 power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.

5–3 CALIBRATION AND SYSTEM CHECKS

Refer to Installation Flow Chart, Section 2, to perform:

- Software load and Configuration file edit.
- Functional checks
- Set–up and calibration
- System performance checks

5–4 RETURNING SHIPPING MATERIAL

The shipping material must be returned **as soon as possible.** Prompt return is required so that production shipments can continue with these labor saving dollies. These should have been included with the “EQUIPMENT RETURN PROCESS” as part of the deinstallation.

5–5 EQUIPMENT RETURN PROCESS

Getting the de–installed equipment sent back to the correct location is another essential step in the Magnet Upgrade process. The de–installed equipment may be sent back to a variety of places depending upon the type of upgrade being performed. Advance planning of equipment return will aid in the upgrade process. The three categories of equipment to be returned are as follows:

1. P & E hardware - i.e. InSite hardware, EDM Magnet Monitor Hardware.
2. System cables, cabinets, magnet enclosure, and miscellaneous hardware.
3. Magnet and associated magnet subsystem components.

For Magnet Monitor return (P&E Hardware) Refer to Direction 2108906, Magnet Monitoring Subsystem (Restricted), SECTION 7, DE–INSTALLATION PROCEDURES, for process and contacts for returning the equipment.

For all other equipment returns use the following process:

1. Contact the MR Product Specialist at GoldSeal Logistics phone number 877–251–4468.
2. Provide the MR Product Specialist with the Magnet type and system hardware configuration being de–installed. If GoldSeal is accepting the returned equipment then all transportation arrangements, empty equipment dollies, coil carts, skids, and packaging will be processed through them. If GoldSeal is not accepting the return then they will advise you to contact GEMS Renewable Resources at phone number 414–747–6997. In this case, GEMS Renewable Resources will be responsible for coordinating the transportation arrangements, empty equipment dollies, coil carts, skids, and packaging.

Note

All equipment being returned to GoldSeal will be sold as used equipment. Special care <u>must</u> be taken by the mechanical de–installers not to cut any cables or damage any of the hardware to be returned.

The equipment will generally be returned on different trucks. The following lists may be useful in the collection and coordination of the equipment to be returned.

<u>Truck A - P & E Hardware</u>

☐ EDM and mounting hardware
☐ InSite Modem and external Power Supply
☐ Magnet Monitor Modem and external Power Supply
☐ Modem Cables
☐ Magnet Monitor Sensor Kit - Water Flow/Temperature Sensor, Pressure Transducer, Reed Switch and Key Switch
☐ Magnet Monitor Cables (equipment room and magnet room)
☐ Magnet Monitor Interface Box, external AC/DC Power Supply and ribbon cables

5–5 EQUIPMENT RETURN PROCESS (continued)

Truck B - System Hardware

Note

The list will change depending on the type of electronics upgrade performed.

☐ System Cabinet
☐ RF Cabinet
☐ Gradient Cabinet
☐ Penetration Cabinet
☐ Penetration Panel
☐ Lytron - Heat Exchanger (air to water)
☐ Heat Exchanger Fluids (empty container and ethylene glycol label provided with upgrade hardware)
☐ Operator Console
☐ System Computer
☐ Magnet Enclosure Frame & mounting hardware
☐ Magnet Enclosure - All fiberglass parts and trim skirts
☐ Front & Rear Endbells
☐ Bridge & Carriage Cover
☐ Rear Pedestal & Electronic Components mounted within
☐ Rear Pedestal Covers
☐ Dock Assembly
☐ PAC Module
☐ SRI Module
☐ Patient Comfort Module
☐ RF/Gradient Body Coil
☐ System Cables removed from the Equipment Room
☐ System Cables removed from the Magnet Room
☐ System Cables removed from Operator Room
☐ Phantoms
☐ Non–conforming surface coils and extremity coils.
☐ Patient Table (optional - only if customer ordered new one and does not want the old one)

5–5 EQUIPMENT RETURN PROCESS (continued)

Truck C - Magnet & Subsystem Components

Note

All Oxford and S–I magnets will be returned per instructions supplied by GEMS Renewable Resources for disposal.

☐ Magnet with cryogens
☐ Cryogen Meters, instrumentation cables and power cable(s).
☐ ERU/MRU, instrumentation cable and power cable
☐ Magnet vent pipe and fittings
☐ Magnet Ramp Supply, Ramp Cables and Ramp Lead Extensions or Current Probes (Used on Oxford & S–I Magnets)
☐ Magnet Shim (Resistive) Power Supply and Shim Cables (Used on Oxford and optional for S–I, S–II, S–III and S–IV Magnets)
☐ Magnet Shim (Superconducting) Power Supply and Shim Cables (Used on S–I Magnets)
☐ Magnet Shim (Superconducting) Power Supply and Shim Cables (Used on S–I Magnets)
☐ Shield Cooler Compressor, Compressor Power Cable, Coldhead Power Cable, Supply & Return Gas Lines, and Remote Control Box (Balzers only).
☐ Magnet spares kit consisting of burst disc, baffles, vacuum release tool, o–rings, miscellaneous fittings etc.
☐ Magnet service ladder
☐ Magnet service platform
☐ Resistive Shim FIlter Box on Penetration Panel
☐ Cable support bracket
☐ Field mapping fixture

When calling, the following information is needed:

- FDO (Future Delivery Order) number under which the upgrade was shipped
- Site name, address, and phone number
- Type of magnet being returned
- Date of completed Horizon upgrade.
- Name and Beeper Number of Field Engineer that could be available when the equipment is picked up.

1. Prepare the return items for shipment as follows:
   a. Package other removed modules and boards in a cardboard box and secure inside System Cabinet (or any other cabinet where they fit).
   b. Use ramp to move cabinets on to shipping pallets. Three installers are required for this heavy lifting task. Fasten with shipping brackets that were removed previously from new cabinets.
2. Process product locater cards for all serialized components removed from the system according to policy.
3. Regulations governing the disposal of this material being deinstalled vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all deinstalled materials per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Renewable Resources for assistance.

5–6 RETURN ITEMS CHECKLIST

The following items are designated to be returned after the upgrade. All materials should be able to be packaged for return on the two provided cover dollies and pallets. Refer to Section 5–5, EQUIPMENT RETURN PROCESS.

Magnet Room

☐ 1. Magnet
☐ 2. Magnet Enclosure Front and Rear Cover
☐ 3. Magnet Enclosure Side Covers (4)
☐ 4. Magnet Enclosure Frame
☐ 5. Front End Bell
☐ 6. Rear End Bell
☐ 7. Left and Right Side Slide Support assemblies
☐ 8. Service Platform
☐ 9. Rear Pedestal Frame
☐ 10. Rear Pedestal Covers
☐ 11. Gradient/RF Coil in shipping container
☐ 12. Bore Lamp Asm with Fiber Optic bundle
☐ 13. Fiberglass Bridge
☐ 14. Obsolete Cables
☐ 15. Heat Exchanger (later systems only)
☐ 16. Coolant removed from Heat Exchanger and Body Coil (later systems only)
☐ 17. EDM Magnet Monitor Sensor Cables
☐ 18. ERU/MRU Module

Depending on the type of system being upgraded, some of the items listed below may not be returned. Refer to the appropriate Upgrade Flowchart in Section 3 for details.

Equipment Room

☐ 1. Obsolete Cables
☐ 2. Penetration Panel
☐ 3. System Cabinet
☐ 4. Gradient Cabinet
☐ 5. RF Cabinet
☐ 6. Penetration Cabinet
☐ 7. Computer Cabinet
☐ 8. Balzer/Leybold Shield Cooler Compressor, Gase Lines, and Cold Head Power Cables
☐ 9. EDM Magnet Monitor Computer, Water Flow Switch, and Cables

Operator Room

☐ 1. Console
☐ 2. Obsolete Cables
☐ 3. Balzer Remote Box

5–7 PHANTOM REPLACEMENT FOR SIGNA ADVANTAGE 4.x

A new TLT Head Phantom and a new TLT Body Phantom are delivered with this upgrade. These new phantoms contain Nickle Chloride solution which has been found to be more temperature stable than the Copper Sulfate solution in the existing phantoms.

WARNING!

NEW TLT PHANTOMS CONTAIN NICKEL – A SUSPECTED CARCINOGEN. DO NOT INGEST. DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO REGULATIONS.

CAUTION

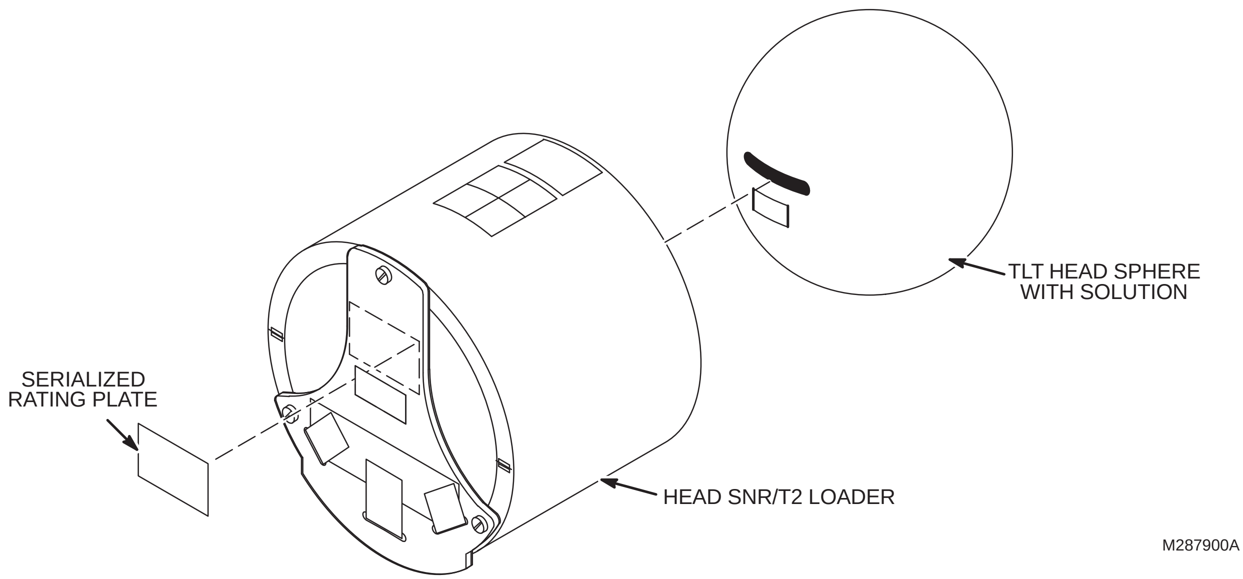
Old phantoms are filled with a solution of Copper Sulfate. Copper Sulfate is considered an environmentally hazardous material and must be disposed of properly. Regulations governing the disposal of this material vary from location to location and from country to country. In order to insure compliance with all local, state or national regulations, these old phantoms must be returned to one of the locations listed at the end of this section. Failure to comply will put you at personal risk of violating environmental regulations.

1. Locate model 46–265826G**6** Head TLT Sphere; Return Label, Rating plate, and Product locator card set.
2. Locate model 46–265635G**6** Body TLT Sphere; Return Label, Rating plate, and Product locator card set.
3. Unpack the phantoms carefully to enable re–use of packing material to return old phantoms.
4. Locate Material Safety Data Sheets (MSDS) packed with new phantoms, they must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
5. Remove old 46–265826G**5** Head TLT Sphere(Copper Sulfate Solution) from service storage cabinet. Pack it in carton that new Nickel Chloride TLT Head Sphere was shipped in.
6. Remove old 46–265635G**5** Body TLT Sphere(Copper Sulfate Solution) from service storage cabinet. Pack it in carton that new Nickel Chloride TLT Body Sphere was shipped in.
7. Seal each phantom box and place boxes on hazardous material return pile.

5–8 PHANTOM REMANUFACTURE FOR SIGNA ADVANTAGE 4.x

Because these phantoms are being replaced, it is necessary to change the rating plates on the Head Loader Assembly and the Body Loader Assembly and process Product Locator Remanufacture Cards.

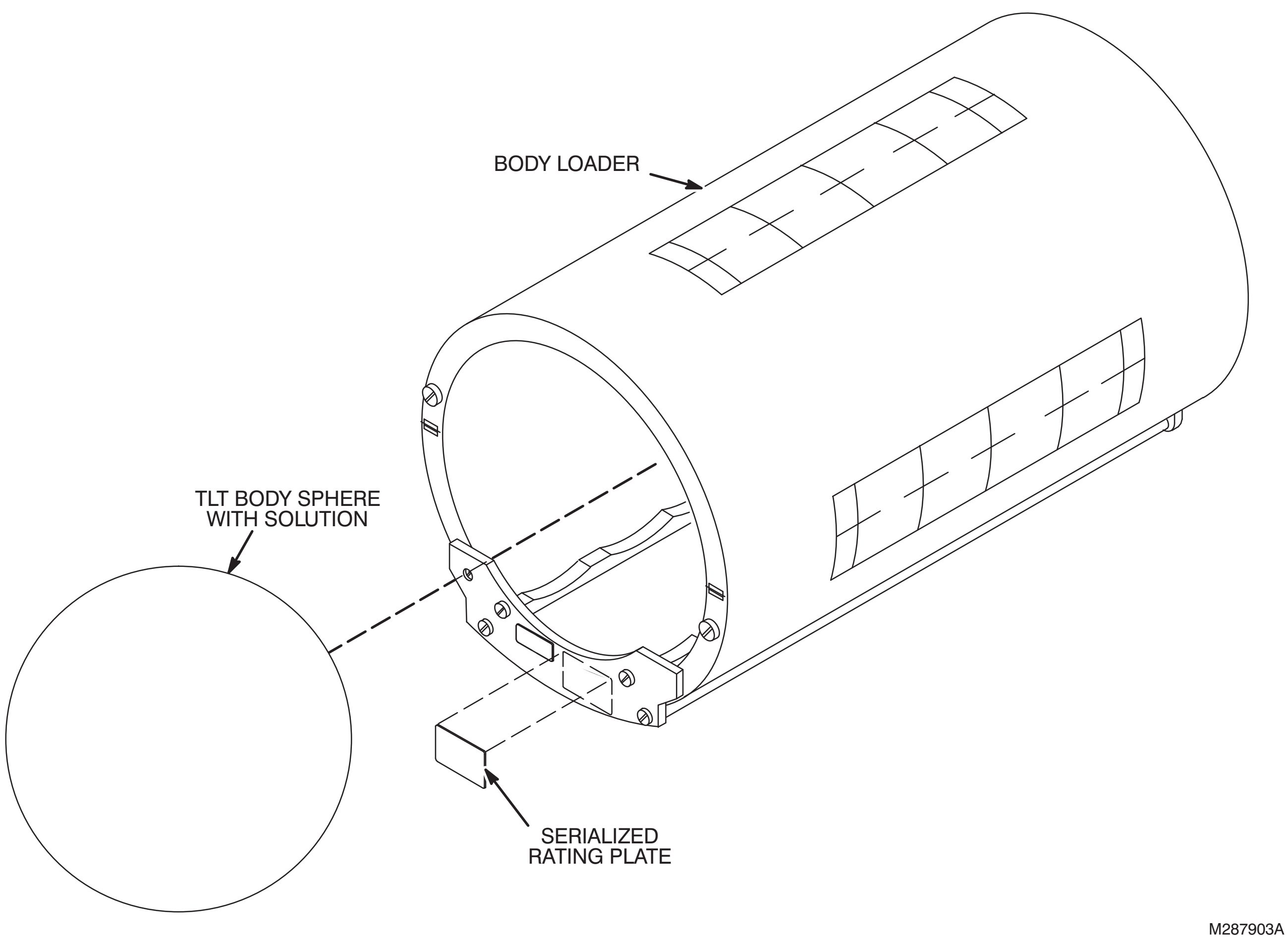
1. Remove Head SNR/T2 Loader from service storage cabinet. Record the serial number from the Head Loader Assembly rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5–3).
2. Remove old Model 46–287900G1 rating plate from end. See Illustration 5–1. Attach new Model 46–287900G2 rating plate in same location (above "Property of General Electric" label. Seal the new rating plate with clear acrylic aerosol spray to help insure that label remains securely attached.
3. Fill out furnished Product Locater Remanufacture Card for Head Loader (Illustration 5–3). Attach old 46–287900G1 Head Loader rating plate to card.
4. Check serial number on Installation Card and rating plate. Fill out Product Locator Installation Card for Head Loader.
5. Record the serial number from the new rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5–3).
6. Sign and date the Product Locator Remanufacture Card. Return both cards to Product Locator.



HEAD TLT PHANTOM
ILLUSTRATION 5–5

5–8 PHANTOM REMANUFACTURE FOR SIGNA ADVANTAGE 4.x (continued)

7. Remove Body SNR/T2 Loader from service storage cabinet. Record the serial number from the Head Loader Assembly rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5–2).
8. Remove old 46–287903G1 rating plate from Body Loader. See Illustration 5–2. Attach new 46–287903G2 rating plate. Seal the new rating plate with clear acrylic aerosol spray to help insure that label remains securely attached.
9. Fill out furnished Product Locater Remanufacture Card for Body Loader (Illustration 5–4). Attach old 46–287903G1 Head Loader rating plate to card.
10. Check serial number on Installation Card and rating plate. Fill out Product Locator Installation Card for Body Loader.
11. Record the serial number from the new rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5–2).
12. Sign and date the Product Locator Remanufacture Card. Return both cards to Product Locator.
13. Place Head and Body Loaders, and new Head and Body TLT spheres back into service storage cabinet.



BODY TLT PHANTOM
ILLUSTRATION 5–6

5–8 PHANTOM REMANUFACTURE FOR SIGNA ADVANTAGE 4.x (continued)

14. Fill in the four shaded boxes on each of the two Product Locator Remanufacture Cards and return to:

GE Medical Systems
Product Locator File
P.O. Box 414, W–523
Milwaukee, Wi 53201–0414

NEW PRODUCT IDENTIFICATION	LOT	**HEAD LOADER** DESCRIPTION	**46–287900G2** MODEL	SERIAL
OLD PRODUCT IDENTIFICATION	LOT	**HEAD LOADER** DESCRIPTION	**46–287900G1** MODEL	SERIAL
			SIGNATURE	DATE

TC 180 REMANUFACTURE

GE Medical Systems
Product Locator File

P.O. Box 414, W–523,
Milwaukee, WI 53201–0414

Product Locator
Remanufacture Card

46–265826G5, Head TLT Phantom was removed and replaced by 46–265826G6, Head TLT Phantom.

The above product was remanufactured per the above description.

HEAD TLT PHANTOM REMANUFACTURE CARD
ILLUSTRATION 5–7

NEW PRODUCT IDENTIFICATION	LOT	**BODY LOADER** DESCRIPTION	**46–287903G2** MODEL	SERIAL
OLD PRODUCT IDENTIFICATION	LOT	**BODY LOADER** DESCRIPTION	**46–287903G1** MODEL	SERIAL
			SIGNATURE	DATE

TC 180 REMANUFACTURE

GE Medical Systems
Product Locator File

P.O. Box 414, W–523,
Milwaukee, WI 53201–0414

Product Locator
Remanufacture Card

46–265635G5, Body TLT Phantom was removed and replaced by 46–265635G6, Body TLT Phantom.

The above product was remanufactured per the above description.

BODY TLT PHANTOM REMANUFACTURE CARD
ILLUSTRATION 5–8

5–9 PHANTOM RETURN GUIDANCE

The “hazardous material return pile” consists of phantoms that contain Copper Sulfate. Regulations governing the disposal of this material vary from location to location and from country to country. In order to insure compliance with all local, state or national regulations, these old phantoms must be returned to one of the locations listed below:

If you are performing this upgrade on a system **within the Americas**, return the old Copper Sulfate filled phantoms to the following address:

G.E. Medical Systems W–853
3200 N. Grandview Blvd.
Waukesha, Wisconsin 53188
Attn: CHEM LAB

NOTE

Unfortunately, the U.S. Post Office does NOT deliver to the above address. Return phantoms via a commercial carrier such as RPS, Federal Express etc.

If you are performing this upgrade on a system within the area supported by **GE Medical Systems – Europe**, return the old Copper Sulfate filled phantoms to the following address:

GE Medical Systems – Europe
198 Avenue du President Wilson
93210 La Plaine Saint Denis, FRANCE

5–9 PHANTOM RETURN GUIDANCE (continued)

If you are performing this upgrade on a system within the area supported by **YMS/GEMSA** (Asia and Pacific), return the old Copper Sulfate filled phantoms to one of the the following:

Hong Kong: **G.E. Medical Systems**
Attn: Service Manager
37F Shun Tak Centre
West Tower
200 Connaufgt Road
Central Hong Kong

Singapore: **G.E. Medical Systems**
Attn: Regional Tech Support Manager
16th Floor, The Gateway, West Tower
150 Beach Road, Singapore

India: **Vice President, Customer Service**
Wipro GE
A–1 Block, 1st Floor
Golden Enclave, Corporate Powers
Airport Road, Bangalore 560017
India

Tiawan: **Service Manager**
GETSCO – Medical Systems
7th Floor, Formosia Plastics Building
201 Tun–Hau North Road
Taipei
Tiawan, Republic of China

Korea: **Samsung Medical Systems**
Attn: Service Manager
15th Floor, Seo Woo Building
837–12 Yeoksan–dong Kangnam–gu
Seoul, Korea

If none of the above work, Field Engineers in Asia should contact YMS Legal Console, at 0425–85–5055 for assistance.

5–10 FINAL COORDINATION

1. Confirm that the blower box has been anchored properly and installed outside the minimum service area.
2. Record and enter applicable data into applicable site configuration files and records.
3. Process product locater installation information. Refer to Section 1–6. If method 2 is used for submitting information, return to:

Product Locator File
P.O. Box 414, W–523
Milwaukee, WI 53201–0414

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

4. Process the **Supplier Performance Report** located in Common Forms. A reference copy of the form is provided at the end of this section.
5. Leave the site's set of Manuals and CD–ROM at site. Organize and set up reference cabinet for the Manuals. Since considerable hardware remains from initial installation, (electronics cabinets, operator workspace, etc.) retain *Direction 2138247 Signa Renewal Parts Release 5.x & 8.x*
6. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
7. Verify that coordination of application support for instruction of site users on Software Release 8.x has been made before returning system to the users. Turn site over to applications who will instruct users.

GE Medical Systems Field Service

SUPPLIER PERFORMANCE REPORT

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: ____ Date: ____________________

Work Performed / Evaluated: ____

Commodity / Item Purchased: ____

GE Customer Name: ____

FDO Number: ____________________________ Test Equipment Bar–code No; _______________________

GEMS Evaluator Name: ____

Please rate suppliers as 1 to 5 on each of the 15 questions below:
1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

	Supplier schedules well and:					
1	starts on time?	1	2	3	4	5
2	finishes on time?	1	2	3	4	5
3	avoids disruptions?	1	2	3	4	5
	Supplier is flexible and:					
4	responds to GEMS/customer requests?	1	2	3	4	5
5	cooperates with changes in schedule?	1	2	3	4	5
6	deals with unexpected problems?	1	2	3	4	5
	Supplier tools and equipment are:					
7	appropriate for contracted work?	1	2	3	4	5
8	adequate for contracted work?	1	2	3	4	5
9	calibrated and in good working order?	1	2	3	4	5
	Supplier meets GEMS quality standards with employees who:					
10	are trained, skilled and technically competent?	1	2	3	4	5
11	always keep job sites clean and free of debris?	1	2	3	4	5
12	ensure both function and appearance are as expected?	1	2	3	4	5
	Supplier employees enhance GEMS image by their:					
13	appearance?	1	2	3	4	5
14	behavior?	1	2	3	4	5
15	attitude?	1	2	3	4	5

Examples / Details: ____

____

____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

5. Senior Operations Specialist for your LCT
6. The contractor who performed the work
7. ISS (already in default distribution)
8. Sourcing Supplier Quality (already in default distribution)

SYSTEM INSTALLATION INSTRUCTIONS

Tab 1, System, contains installation instructions for the following:

Use only the applicable System Installation Instruction Directions

- *DIRECTION 2147593* – 1.5T Heliax Cable Installation Instructions
- *DIRECTION 2223755* – 1.0T Heliax Cable Installation Instructions
- *DIRECTION 2286976* – Signa ACGD Gradient Cable Installation Instructions
- *DIRECTION 2287106* – Signa ACGD Fiber Optic Cable Installation Instructions
- *DIRECTION 2338872* – Signa 10.x MGD Fiber Optic Cable Installation Instructions
- *DIRECTION 2287109* – Signa ACGD System Power Cable Installation Instructions

1.5T HELIAX CABLE INSTALLATION

INSTALLATION STEPS SUMMARY

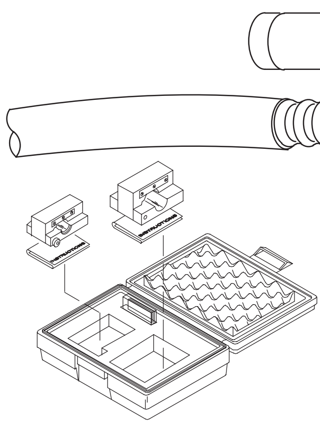
- ❑ **A – Routing, Cutting, Terminating Heliax Cables**
- ❑ **B – Connecting Runs 235, 236, 257, and 258**

CAUTION

Do not subject 1/2 inch diameter heliax cable to a bend radius less than 5 inches (127mm). Do not subject 7/8 inch diameter heliax cable to a bend radius less than 10 inches (254mm). This cable will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

❑ A – ROUTING, CUTTING, TERMINATING HELIAX CABLES

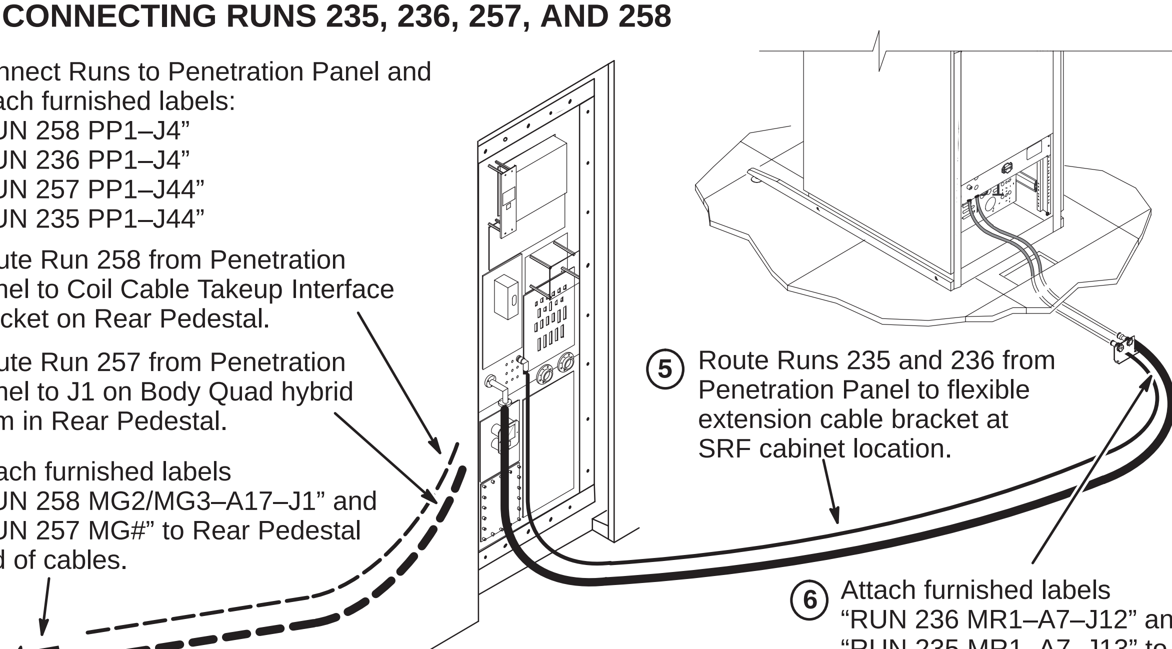
Runs 236 and 258 are created during installation from 50 foot and/or 55 foot cables along with four pre–marked labels. The installer may use both lengths of cables as provided, or if storage of excess length is a problem, cut one length (If runs are short, it may be feasible to cut one cable into two lengths) or both lengths and re–terminate each cut end. Either way, labels are attached after length is decided and cables are installed. Runs 235 and 257 are created during installation from a 100 foot cable along with four pre–marked labels.



1. Decide if cables are to be cut and re–terminated. If yes, continue to Step 2. If no, proceed to Step B.
2. Procure a new connector (46–230011P1) for each cable that needs to be shortened and re–terminated.
3. Route cables and mark at length for cut–off.
4. Cut cables to length with a fine–tooth hack saw. Remove about one inch of jacket to expose outer conductor.
5. Obtain the Heliax Termination Kit (46–320362G1) field service tool and follow instructions supplied with the kit. This will enable precise cut–off of cable.
6. Attach new connector(s) to cable(s) according to instruction sheets supplied with connector.

❑ B – CONNECTING RUNS 235, 236, 257, AND 258

1. Connect Runs to Penetration Panel and attach furnished labels:
   "RUN 258 PP1–J4"
   "RUN 236 PP1–J4"
   "RUN 257 PP1–J44"
   "RUN 235 PP1–J44"
2. Route Run 258 from Penetration Panel to Coil Cable Takeup Interface bracket on Rear Pedestal.
3. Route Run 257 from Penetration Panel to J1 on Body Quad hybrid Asm in Rear Pedestal.
4. Attach furnished labels "RUN 258 MG2/MG3–A17–J1" and "RUN 257 MG#" to Rear Pedestal end of cables.



5. Route Runs 235 and 236 from Penetration Panel to flexible extension cable bracket at SRF cabinet location.
6. Attach furnished labels "RUN 236 MR1–A7–J12" and "RUN 235 MR1–A7–J13" to RF Cabinet end of cables.

1.0T HELIAX CABLE INSTALLATION

INSTALLATION STEPS SUMMARY

- ❑ **A – Routing, Cutting, Terminating Heliax Cables**
- ❑ **B – Connecting Runs 745, 746, 747, and 748**

CAUTION

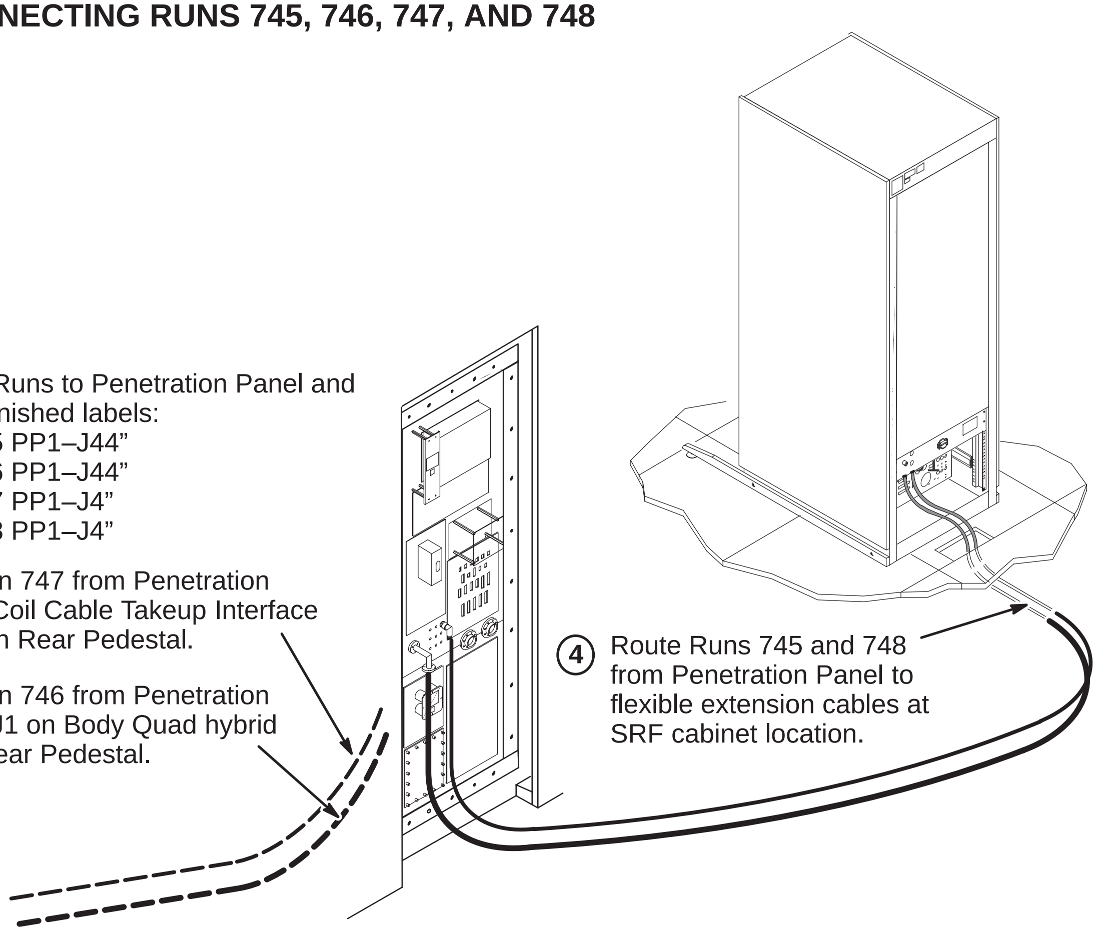
Do not subject 0.6 inch diameter heliax cable to a bend radius less than 1.5 inches (138mm). Do not subject 7/8 inch diameter heliax cable to a bend radius less than 3 inches (75mm). This cable will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

❑ A – ROUTING, CUTTING, TERMINATING HELIAX CABLES

Runs 747 and 748 are created during installation from 100 foot cables along with two labels. Labels are attached after length is decided and cables are installed. Runs 745 and 746 are created during installation from a 100 foot cable along with two labels. Kit (2207145) includes cable with terminal ends and 4 connectors.

1. Route cables and mark at length for cut–off.
2. Cut cables to length with a fine–tooth hack saw.
3. Follow instructions supplied with the kit. This will enable precise cut–off of cable.
4. Attach new connector(s) to cable(s) according to instruction sheets supplied with connector.

❑ B – CONNECTING RUNS 745, 746, 747, AND 748



1. Connect Runs to Penetration Panel and attach furnished labels:
   “RUN 745 PP1–J44”
   “RUN 746 PP1–J44”
   “RUN 747 PP1–J4”
   “RUN 748 PP1–J4”
2. Route Run 747 from Penetration Panel to Coil Cable Takeup Interface bracket on Rear Pedestal.
3. Route Run 746 from Penetration Panel to J1 on Body Quad hybrid Asm in Rear Pedestal.
4. Route Runs 745 and 748 from Penetration Panel to flexible extension cables at SRF cabinet location.

SIGNA ACGD GRADIENT OUTPUT CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

- ❑ **A – Cut cables to length and route in Equipment and Magnet Rooms**
- ❑ **B – Terminate Runs 762, 763, and 764 for ACGD**
- ❑ **C – Terminate Runs 762, 763, and 764 in Magnet Room**
- ❑ **D – Connect Cables at Penetration Panel**
- ❑ **E – Connection of Body Coil Gradient Cables to Runs 762, 763 & 764**

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

❑ A – CUT CABLES TO LENGTH AND ROUTE IN EQUIPMENT AND MAGNET ROOMS

1. Verify that 100 ft. (30.5 M) length of supplied Run 762, 763, and 764 cable is sufficient to route from ACGD/PDU Cabinet to Penetration Panel and Penetration Panel to Body Coil Gradient Cables at rear of magnet.
   Each cable has identical terminations at each end for connection to <u>both sides of the Penetration Panel Gradient Filter</u> after completing Steps 2 and 3 below.
2. Unroll each cable. Route cables from equipment room side of Penetration Panel to ACGD/PDU Cabinet. The end of cable with terminals will be connected to Penetration Panel. Make sure enough slack is allowed for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
3. Move cable to magnet room. Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
4. Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. **Do not cut Body Coil gradient cables.**
5. Locate furnished bag of parts and attach terminal and labels to each cable which was cut to length. See Procedures **B** and **C.**
6. Route Run 762, 763, and 764 from equipment room side of Penetration Panel to ACGD/PDU cabinet.
7. See Procedures **D** for connection to Penetration Panel.
8. Route Run 762, 763, and 764 from magnet room side of Penetration Panel to Body Coil gradient cables.
9. See Procedure **D** for connection to Penetration Panel.
10. See Procedure **E** for connection of Runs 762, 763, and 764 to Body Coil gradient cables.
11. See ACGD/PDU Cabinet installation instructions **(DIRECTION 2286811 IN TAB 15)** for connection of Runs 762, 763, and 764 to ACGD/PDU cabinet.

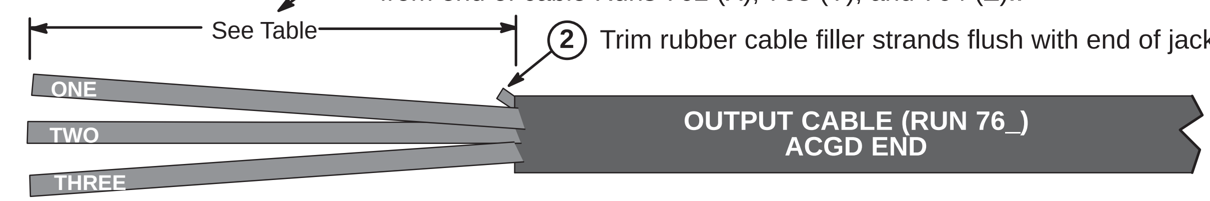
TOOLS REQUIRED: 2135776 – Cable Crimper/Stripper Kit

❑ B – TERMINATE RUNS 762, 763, AND 764 FOR ACGD

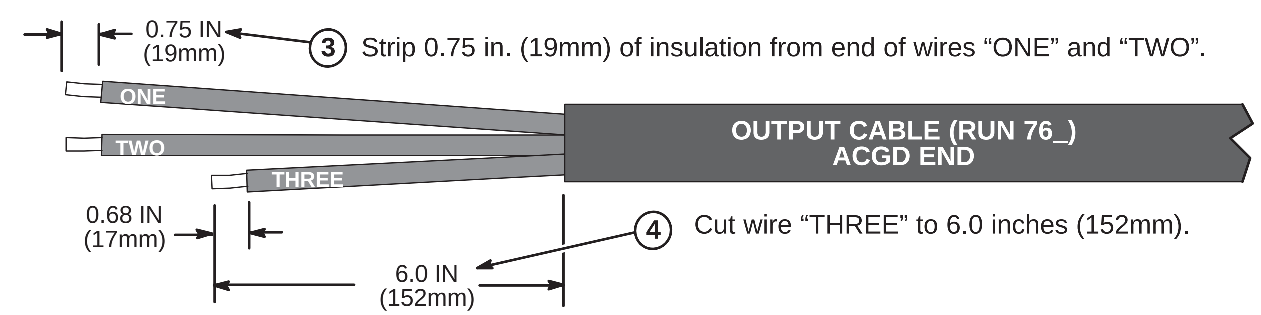
Note: The following steps are required to terminate the end of the cable that was cut in Step **A3**. The other end of the cable is connected to the Penetration Panel.

	Run 762	**Run 763**	**Run 764**
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 762 (X), 763 (Y), and 764 (Z)..
2. Trim rubber cable filler strands flush with end of jacket.



3. Strip 0.75 in. (19mm) of insulation from end of wires "ONE" and "TWO".
4. Cut wire "THREE" to 6.0 inches (152mm).



Note:
Steps in illustration below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y–) and 764 (Z+, Z–).

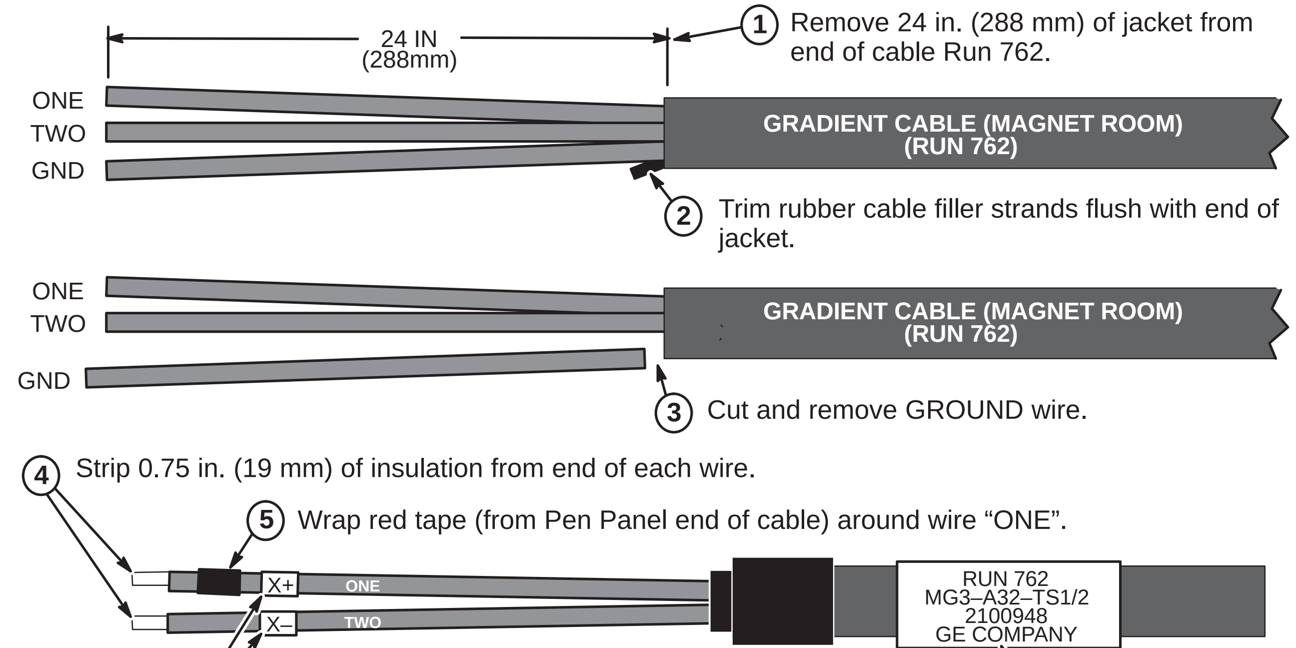
Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire "ONE".**

5. Place furnished "X+" label on wire "ONE".
   Place furnished "X–" label on wire "TWO".
   Place furnished "GND" label on wire "THREE".
6. Wrap red tape around wire "ONE".

7. Crimp a furnished T&B #G2–14 terminal on GND wire end using a Ratcheting Crimper with #2 die.
8. Place furnished "RUN 762" label on cable.

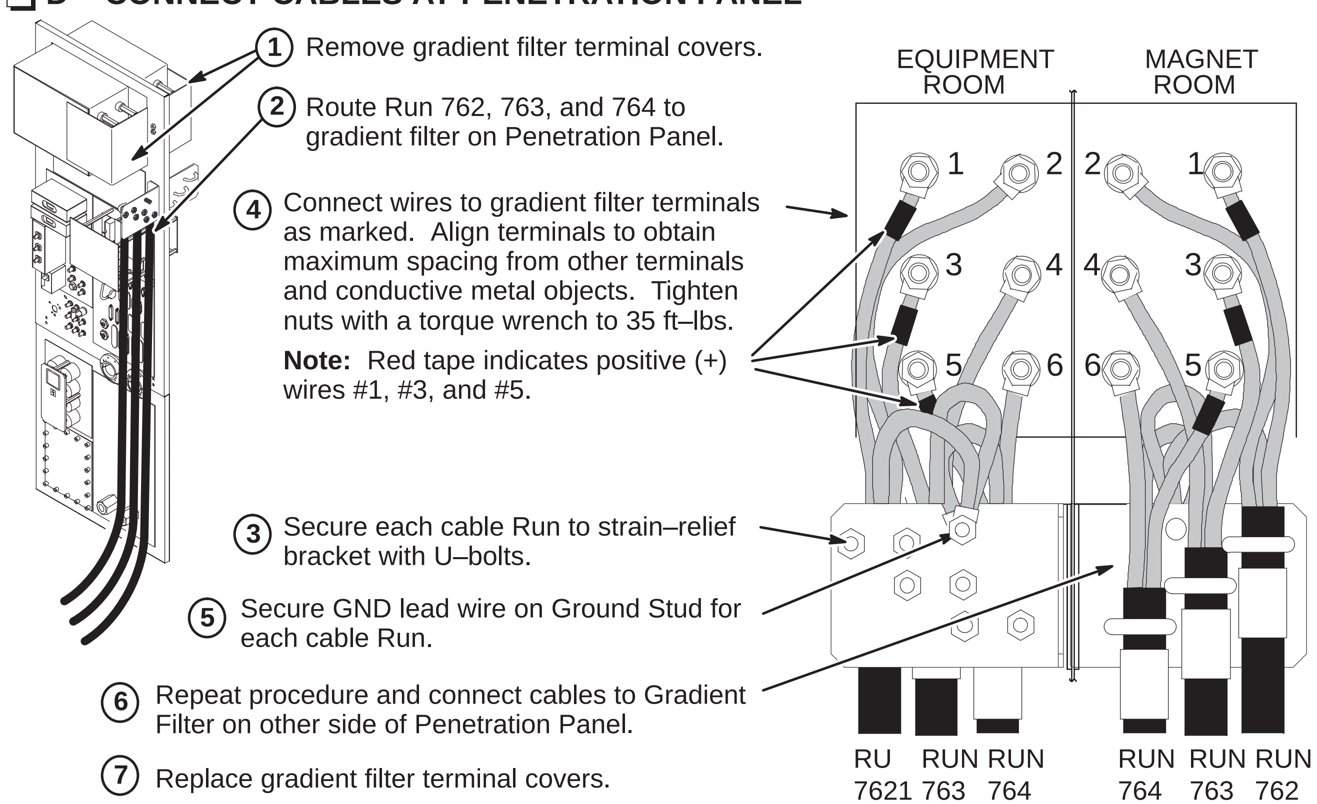
❑ C – TERMINATE RUNS 762, 763, AND 764 FOR MAGNET ROOM

Note: The following steps are required to terminate the end of the cable that was cut in Step **A3**. The other end of the cable is connected to the Penetration Panel.



1. Remove 24 in. (288 mm) of jacket from end of cable Run 762.
2. Trim rubber cable filler strands flush with end of jacket.
3. Cut and remove GROUND wire.
4. Strip 0.75 in. (19 mm) of insulation from end of each wire.
5. Wrap red tape (from Pen Panel end of cable) around wire "ONE".
6. Place furnished "X+" label on wire "ONE".
   Place furnished "X–" label on wire "TWO".
7. Place furnished "Run 762" label on cable.
8. Repeat above process for Runs 763 and 764.
   Note: Red tape, Y+ (RUN 763) **and Z+** (RUN 764) **labels are attached to wire "ONE".**

❑ D – CONNECT CABLES AT PENETRATION PANEL

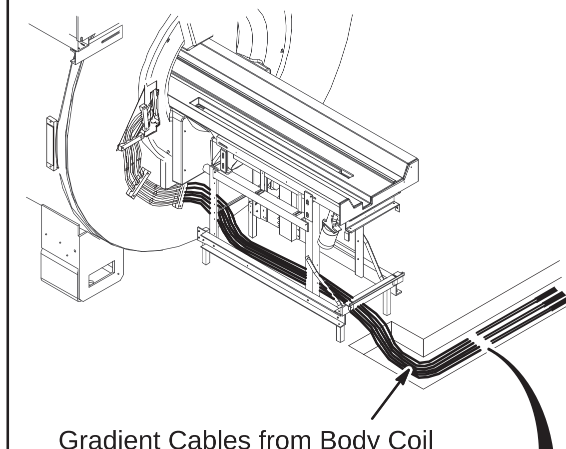


1. Remove gradient filter terminal covers.
2. Route Run 762, 763, and 764 to gradient filter on Penetration Panel.
4. Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft–lbs.
   Note: Red tape indicates positive (+) wires #1, #3, and #5.
3. Secure each cable Run to strain–relief bracket with U–bolts.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Repeat procedure and connect cables to Gradient Filter on other side of Penetration Panel.
7. Replace gradient filter terminal covers.

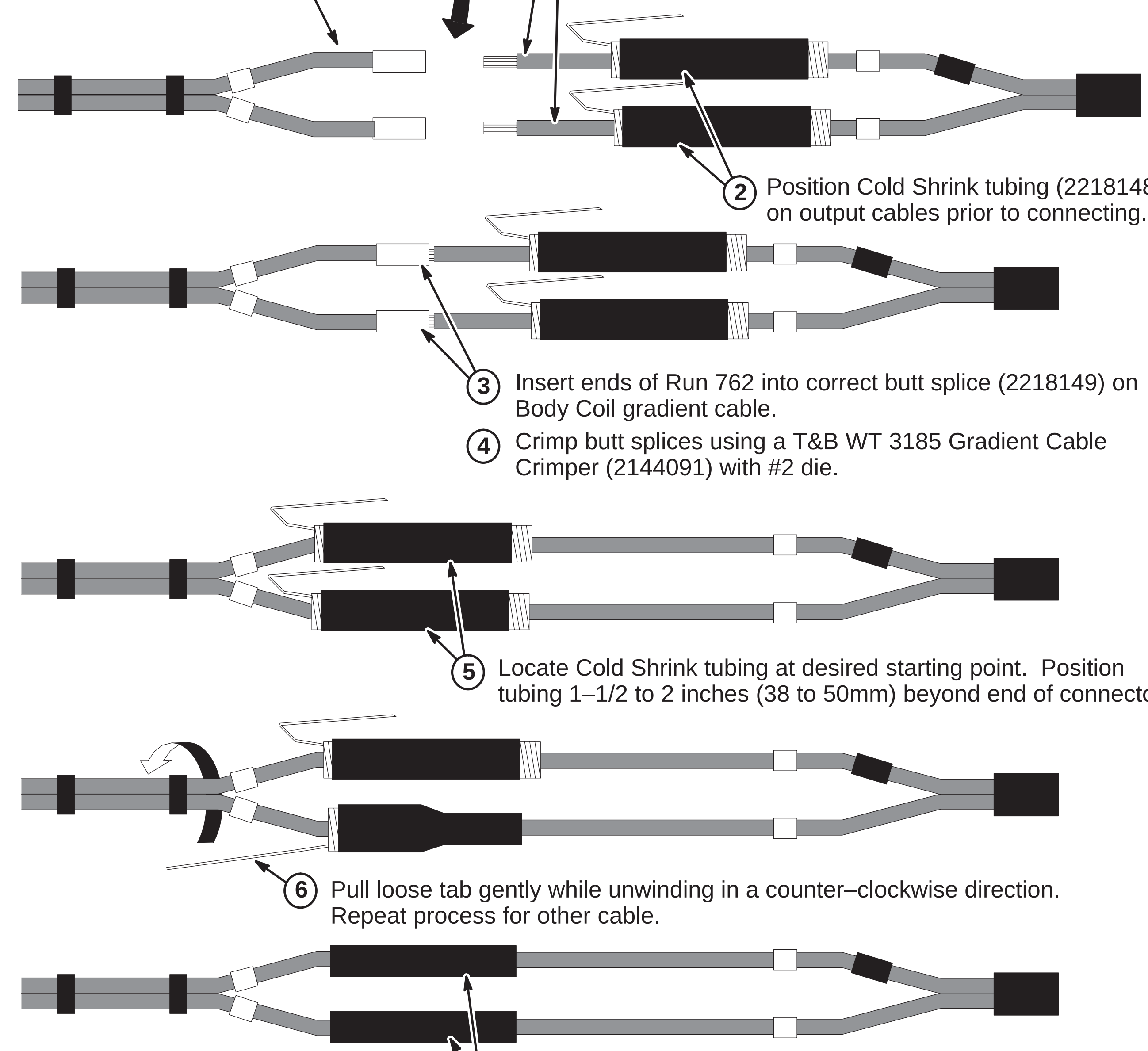
❑ E – CONNECTION OF BODY COIL GRADIENT CABLES TO RUNS 762, 763 & 764

Note:
Illustration below shows a typical cable routing for connection of gradient output cables within an enclosed cable trough.

Note:
Cold Shrink Tubing (2218148) can be found in the Magnet Field Spares Kit delivered with the Magnet



1. Cable connection process shown below is for Run 762. Repeat process for Runs 763 and 764.
   Note: Red tape, X+ (RUN 762), **Y+** (RUN 763) **and Z+** (RUN 764) **labels are attached to wire "ONE".**



2. Position Cold Shrink tubing (2218148) on output cables prior to connecting.
3. Insert ends of Run 762 into correct butt splice (2218149) on Body Coil gradient cable.
4. Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die.
5. Locate Cold Shrink tubing at desired starting point. Position tubing 1–1/2 to 2 inches (38 to 50mm) beyond end of connector.
6. Pull loose tab gently while unwinding in a counter–clockwise direction. Repeat process for other cable.
7. Connection of cables and installation of Cold Shrink tubing is now complete.

SIGNA ACGD FIBER OPTIC CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

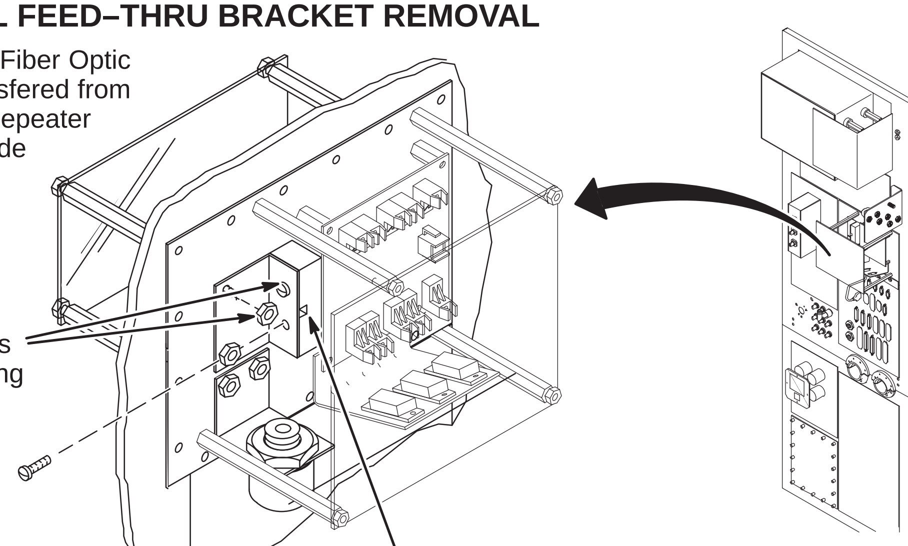
- ❑ **A1 – Repeater Panel Feed–thru Bracket Removal**
- ❑ **A2 – Routing Run 711/712 in Magnet Room**
- ❑ **A3 – Connecting Magnet Room Run 711/712 to Repeater Panel**
- ❑ **A4 – Route/Connect Equipment Room Run 711/712 to Repeater Panel**
- ❑ **A5 – Route/Connect Fiber Optic Equipment Room Runs 708, 709, 710, 711/712, and 836**

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

❑ A1 – REPEATER PANEL FEED–THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transfered from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



1. Unfasten two #10–32 screws and two #10–32 nuts securing feed–thru access bracket to J91 feed–thru block.
2. Remove feed–thru access bracket and fastening hardware from panel.

❑ A2 – ROUTING RUN 711/712 IN MAGNET ROOM

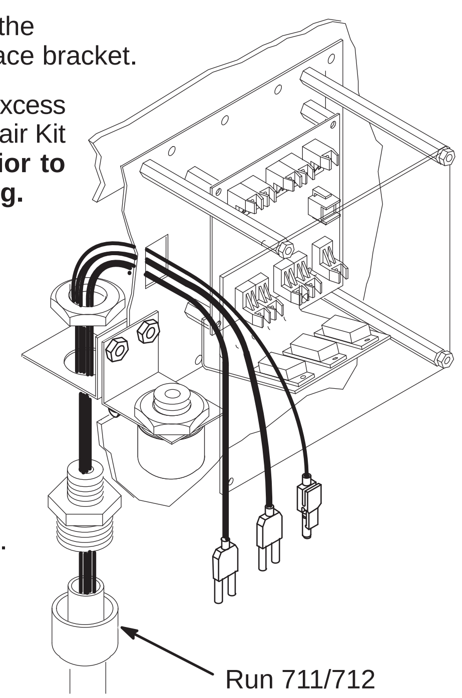
1. Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The "PP1–J91" end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

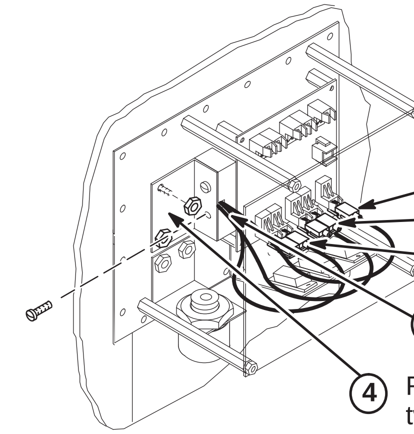
2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
3. Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
4. Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC–II.

5. Route Run 711/712 to the front right side of the magnet (SRI location). .
6. Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC–II.



❑ A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

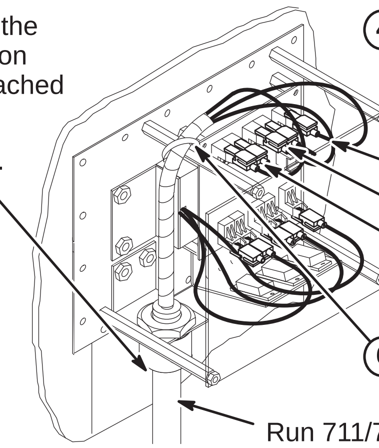


Note: Take care to maintain the minimum 2 inch bend radius.

1. Adjust route of individual fiber optic runs to maintain large radius.
2. Connect Runs 712–1, 711–1, and 711–2 as shown.
   - Run 711–2 (J5)
   - Run 711–1 (J3, J4)
   - Run 712–1 (J1, J2)
3. Gather fiber optic runs and route through wave guide notch in feed–thru block.
4. Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

❑ A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 711/712 TO REPEATER PANEL

1. Move Run 711/712 (46–328078G5) to the Equipment Room side of the Penetration Panel. The "PP1–A17" end will be attached to interface bracket.
2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
3. Route other end of Run 711/712 to System Cabinet. Refer to Direction 2170004 (Tab 8) for connecting fiber optics in System Cabinet.

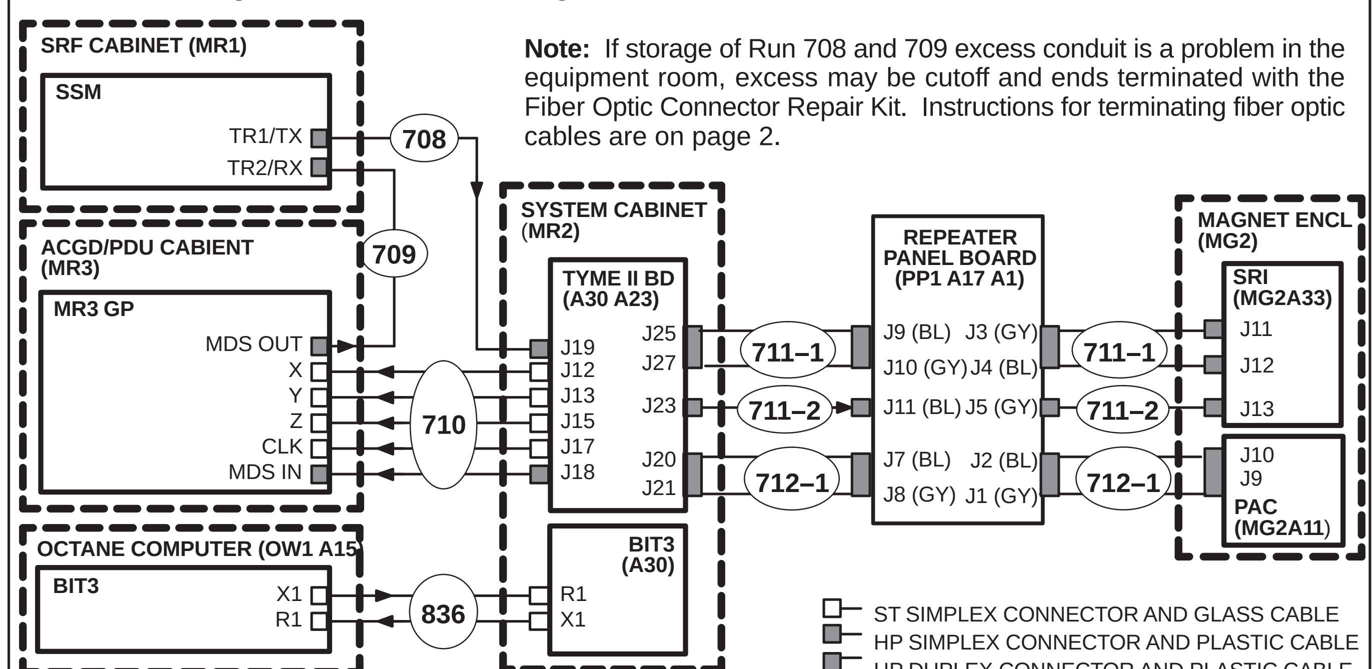


4. Adjust route of individual fiber optic runs to maintain large radius.
5. Connect Runs 712–1, 711–1, and 711–2 as shown.
   - Run 711–2 (J11)
   - Run 711–1 (J9, J10)
   - Run 712–1 (J7, J8)
6. Adjust position of protective spiral wrap and secure with tie–wrap.

❑ A5 – ROUTE/CONNECT F/O EQUIPMENT ROOM RUNS 708, 709, 710, 711/712 & 836

1. Unroll coiled fiber optic conduits for Runs 708, 709, and 710 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 710 conduit of as it cannot be reterminated.

Note: If storage of Run 708 and 709 excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are on page 2.



INSTALLATION STEPS SUMMARY

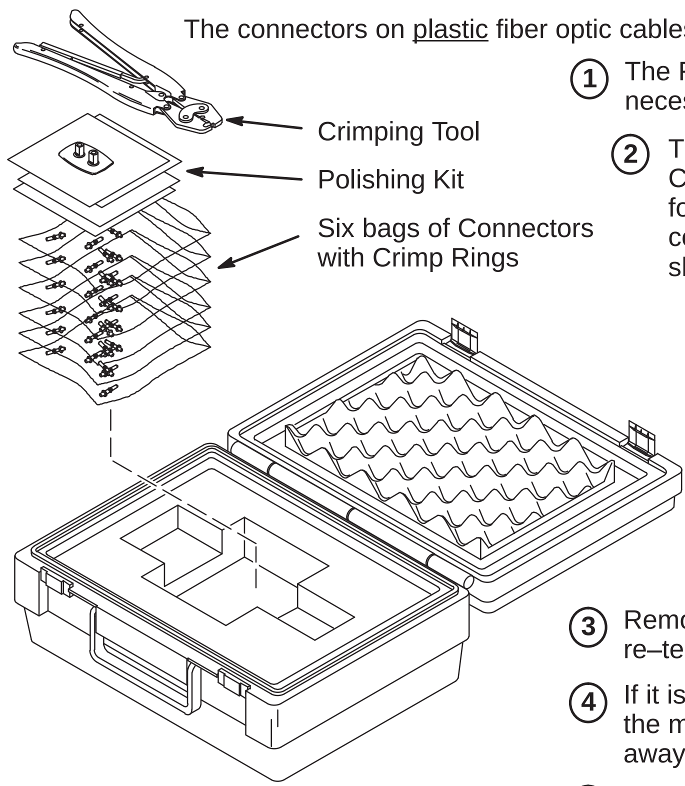
- ❑ **B1 – Fiber Optic Cable Termination Kit**
- ❑ **B2 – Stripping Fiber Optic Outer Jacket**
- ❑ **B3 – Crimping Connector to Fiber Optic Cable**
- ❑ **B4 – Trimming Excess Fiber Optic from Connector**
- ❑ **B5 – Polishing Fiber Optic End**

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

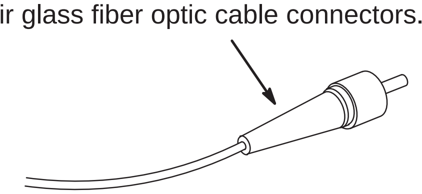
❑ B1 – FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.



1. The Fiber Optic Repair Kit, 46–301450G1, contains all the necessary materials to replace a damaged connector.
2. The "Spares Kit", delivered with the Shipping Collector, contains a 46–320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



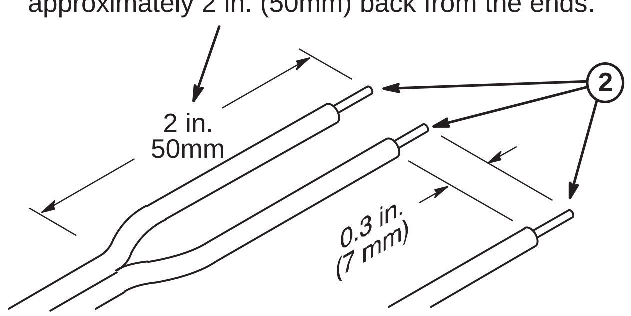
3. Remove the end of fiber optic cable to be re–terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

1. For Duplex Cables, seperate the two fibers approximately 2 in. (50mm) back from the ends.

Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

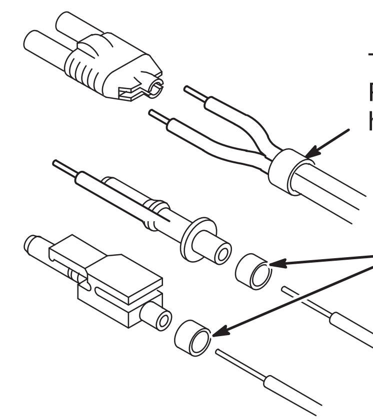
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

1. Place the Crimp Ring and Connector over the end of the cable.

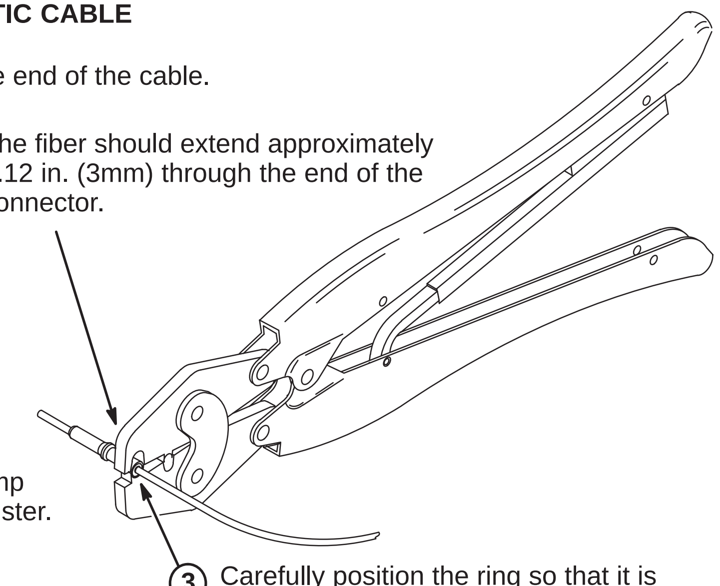
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.



The Duplex Crimp Ring is shiny and has a thinner wall

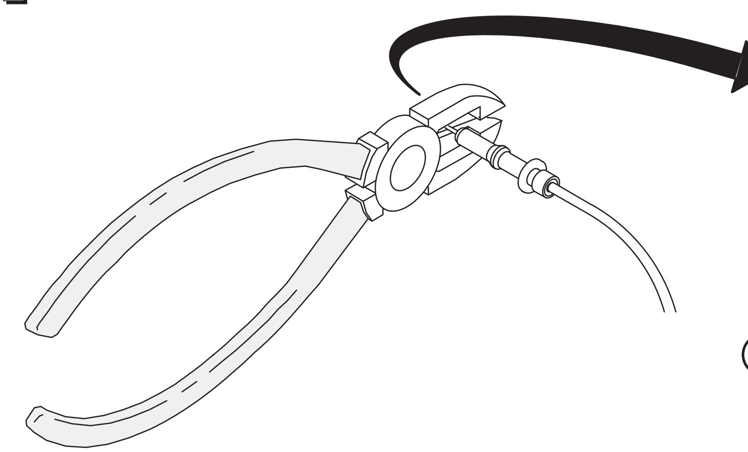
The Simplex Crimp Ring has a dull luster.

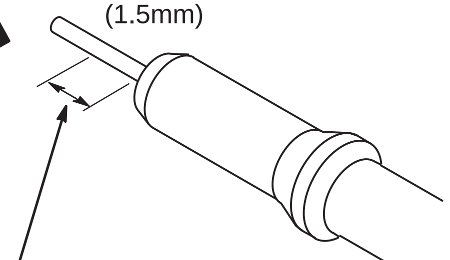
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

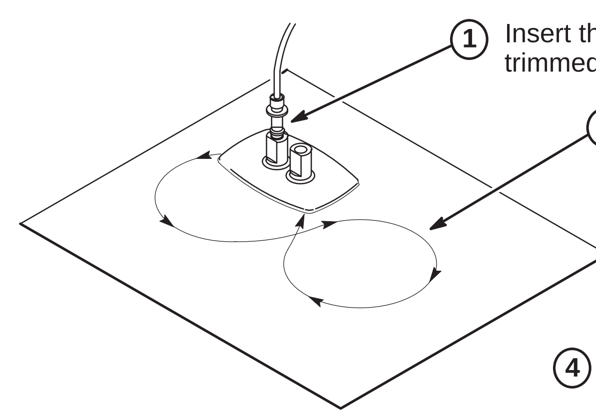
❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR





1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1–1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

SIGNA 10.x MGD FIBER OPTIC CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

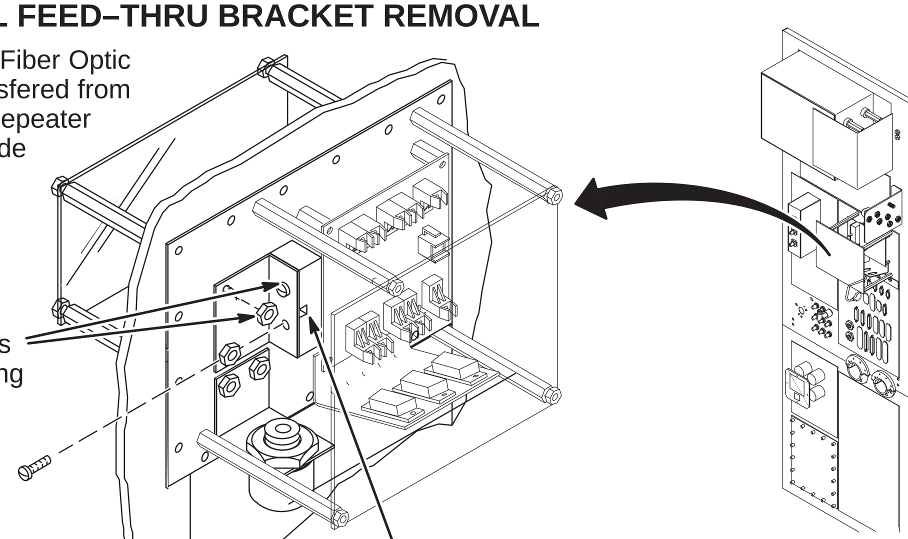
- ❑ **A1 – Repeater Panel Feed–thru Bracket Removal**
- ❑ **A2 – Routing Run 711/712 in Magnet Room**
- ❑ **A3 – Connecting Magnet Room Run 711/712 to Repeater Panel**
- ❑ **A4 – Route/Connect Equipment Room Run 1045 to Repeater Panel**
- ❑ **A5 – Cable Map of Fiber Optic Runs 709, 711/712, 1032, 1033, 1034, 1044, and 1045**

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

❑ A1 – REPEATER PANEL FEED–THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transfered from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



1. Unfasten two #10–32 screws and two #10–32 nuts securing feed–thru access bracket to J91 feed–thru block.
2. Remove feed–thru access bracket and fastening hardware from panel.

❑ A2 – ROUTING RUN 711/712 IN MAGNET ROOM

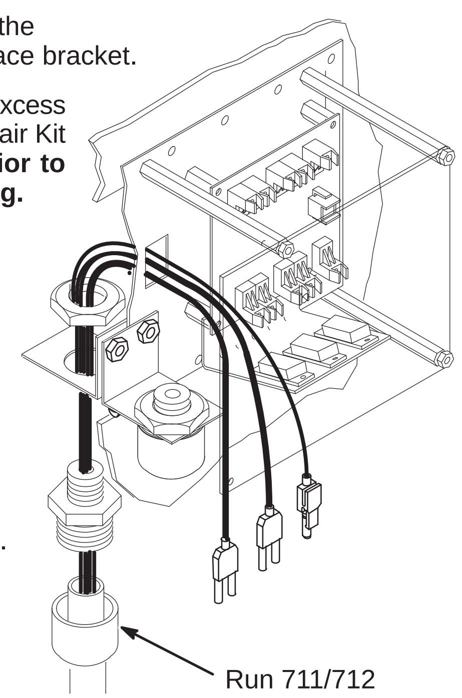
1. Move Run 711/712 (46–328079G9) to the Magnet Room side of the Penetration Panel. The “PP1–J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

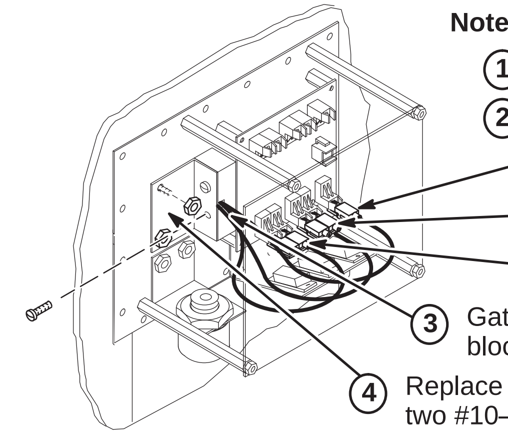
2. Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
3. Route ends of individual fiber–optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
4. Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC–II.

5. Route Run 711/712 to the front right side of the magnet (SRI location). .
6. Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC–II.



❑ A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

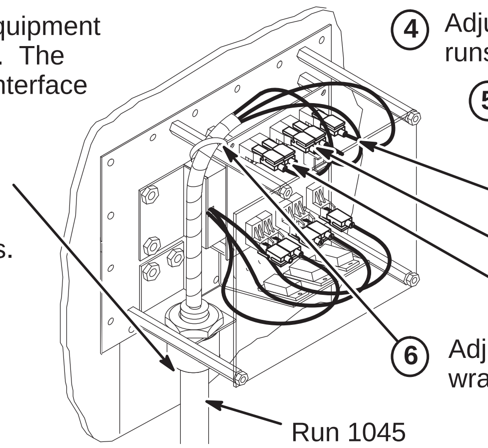


Note: Take care to maintain the minimum 2 inch bend radius.

1. Adjust route of individual fiber optic runs to maintain large radius.
2. Connect Runs 712–1, 711–1, and 711–2 as shown.
   - Run 711–2 (J5)
   - Run 711–1 (J3, J4)
   - Run 712–1 (J1, J2)
3. Gather fiber optic runs and route through wave guide notch in feed–thru block.
4. Replace feed–thru access bracket and fasten with two #10–32 screws and two #10–32 nuts removed in step A1.

❑ A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 1045 TO REPEATER PANEL

1. Move Run 1045 (2331846) to the Equipment Room side of the Penetration Panel. The “PP1–A17” end will be attached to interface bracket.
2. Attach Run 1045 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
3. Route other end of Run 1045 to System Cabinet. Refer to Direction 2338502 (Tab 8) for connecting fiber optics in System Cabinet.

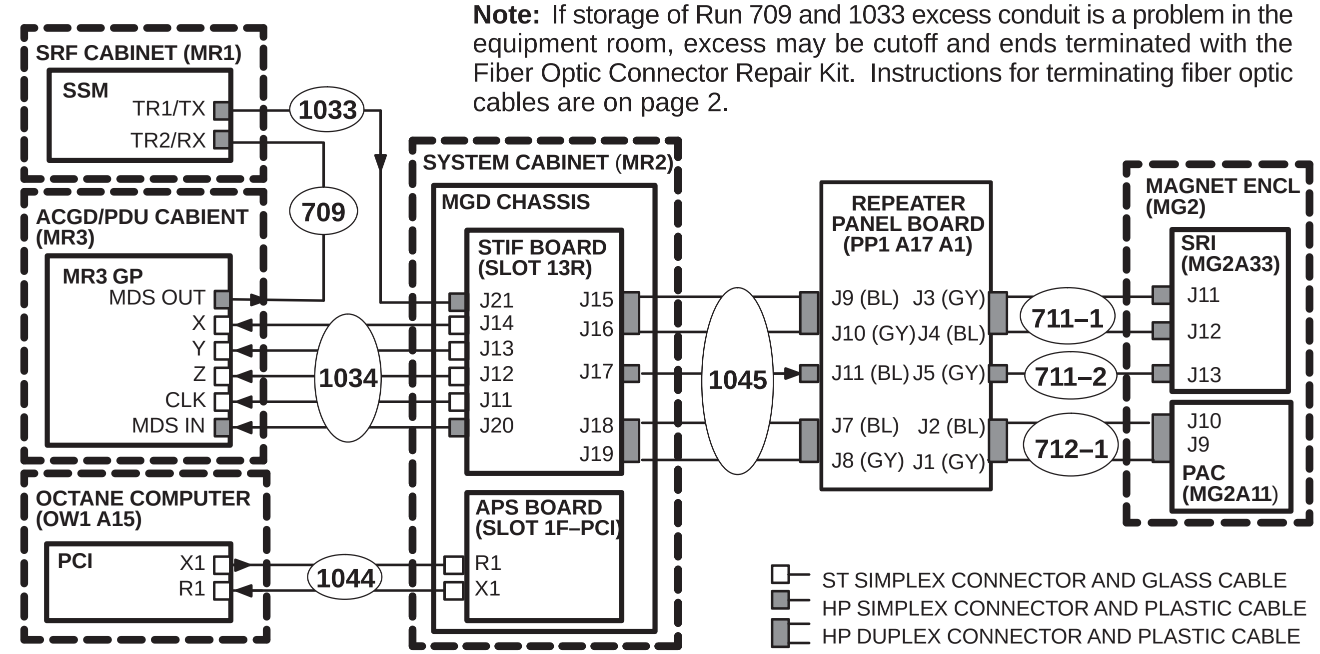


4. Adjust route of individual fiber optic runs to maintain large radius.
5. Connect Run 1045 as shown.
   - 2338146 (J11)
   - 2338146–3 (J9, J10)
   - 2338146–2 (J7, J8)
6. Adjust position of protective spiral wrap and secure with tie–wrap.

❑ A5 – CABLE MAP OF F/O RUNS 709, 711/712, 1032, 1033, 1034, 1044, AND 1045

1. Unroll coiled fiber optic conduits for Runs 709, 1033, and 1034 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.

Note: If storage of Run 709 and 1033 excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are on page 2.



SIGNA 10.x MGD FIBER OPTIC CABLE INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

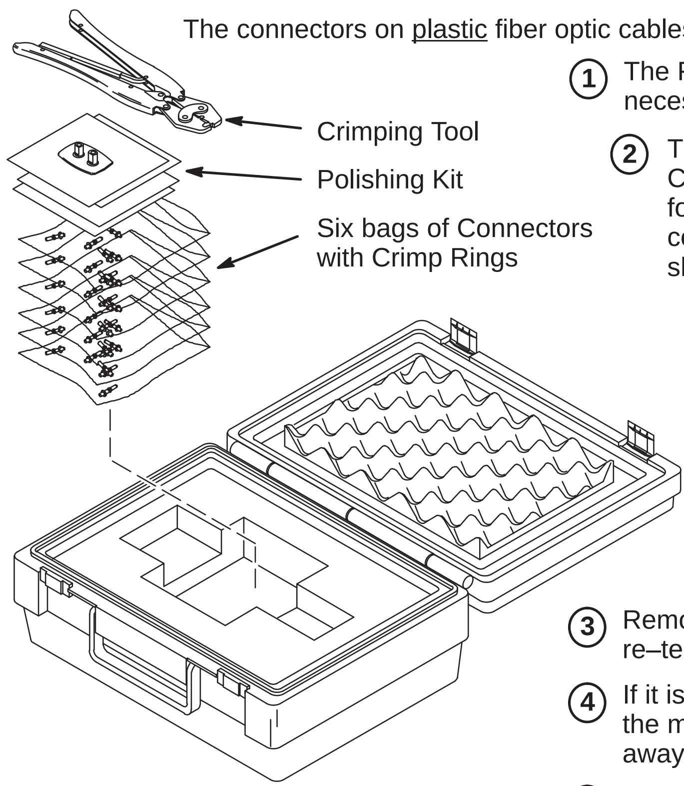
- ❑ **B1 – Fiber Optic Cable Termination Kit**
- ❑ **B2 – Stripping Fiber Optic Outer Jacket**
- ❑ **B3 – Crimping Connector to Fiber Optic Cable**
- ❑ **B4 – Trimming Excess Fiber Optic from Connector**
- ❑ **B5 – Polishing Fiber Optic End**

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

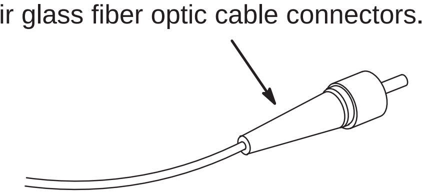
❑ B1 – FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.



1. The Fiber Optic Repair Kit, 46–301450G1, contains all the necessary materials to replace a damaged connector.
2. The “Spares Kit”, delivered with the Shipping Collector, contains a 46–320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

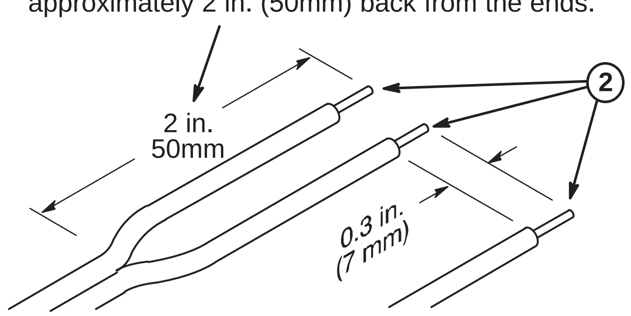
Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



3. Remove the end of fiber optic cable to be re–terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

1. For Duplex Cables, seperate the two fibers approximately 2 in. (50mm) back from the ends.



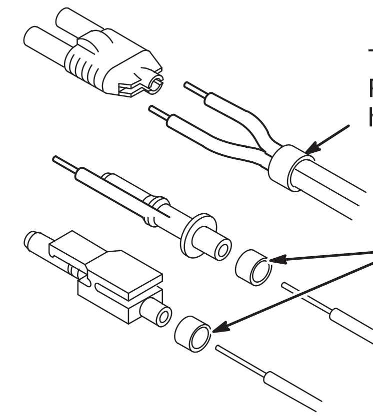
Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

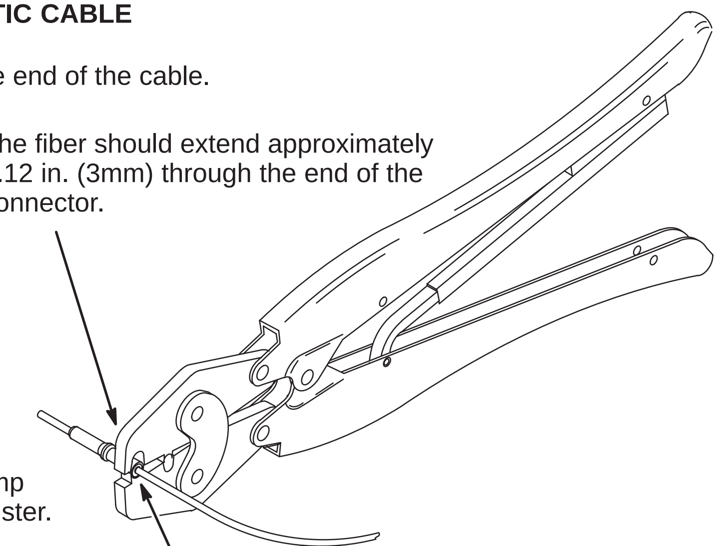
❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

1. Place the Crimp Ring and Connector over the end of the cable.

Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

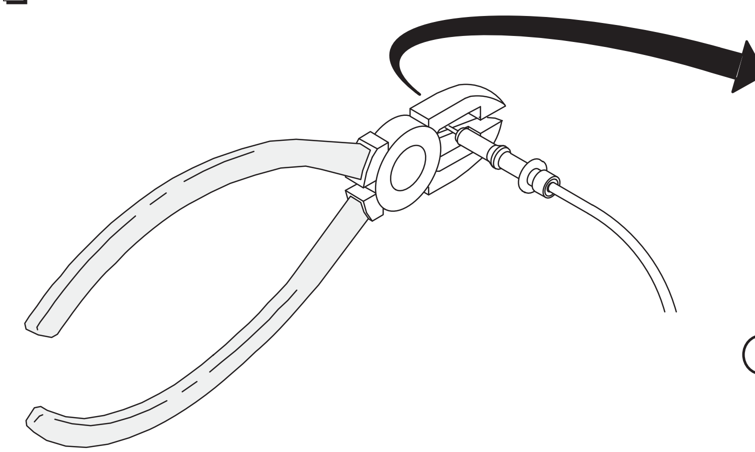


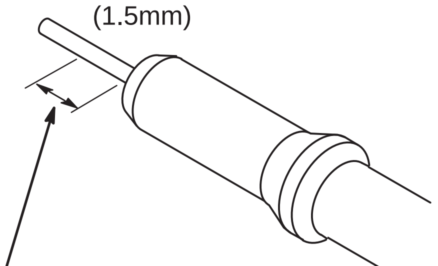
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

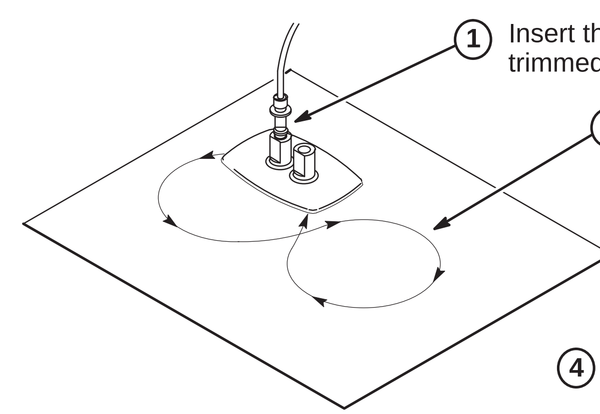
❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR





1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1–1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

INSTALLATION STEPS SUMMARY

- ❑ **A – System Power Cable Installation**
- ❑ **B – Power Cable Re–termination**
- ❑ **C – Power Cable Pre–connect checks**
- ❑ **D – Power Cable Connections Details**

❑ A – SYSTEM POWER CABLE INSTALLATION

1. Verify that all power and ground cables for the Signa System with ACGD are present. Refer to Section 2–1, LOCATING AND UNPACKING CABLES.
2. **Lengths of PDU Power cables for:**

	FIXED SITE	MOBILE	T/R
Run 966, SSM	35.0 ft. (10.7M)	35.0 ft. (10.7M)	48.0 ft. (14.6M)
Run 967, SRF Cabinet	35.5 ft. (11.1M)	29.5 ft. (9.0M)	49.0 ft. (15.0M)
Run 968, System Cabinet	35.0 ft. (10.7M)	25.0 ft. (7.6M)	45.0 ft. (13.7M)
Run 969, Operator Workspace Computer	65.5 ft. (20.0M)	33.0 ft. (10.0M)	33.0 ft. (10.0M)
Run 970, Water Chiller (Fixed–Site) **(Note 1)**	82.0 ft. (25.0M)		
Run 973, Heat Exchanger (Mob & T/R) **(Note 1)**		82.0 ft. (25.0M)	82.0 ft. (25.0M)
Run 989, Twin Accessory Cabinet **(Note 1)**	25.0 ft. (7.62M)	25.0 ft. (7.62M)	25.0 ft. (7.62M)

Note 1: Runs 970 and 973 are for Non–Twin*Speed* systems, Run 989 is only used on Twin*Speed.*

If insufficient room exists to store excess cables:
Plan for storage location if cables are not being cut to length. Proceed to Step **C**.

If insufficient room exists to store excess cables:
They can be cut shorter and re–terminated at the PDU. Complete Step **B**. Then proceed to Step **C**.

3. Route all power cables per applicable System Interconnect Diagram. Unroll cables to lay without twists or kinks from cabinet to PDU through provided troughs, conduit, or ducts.

❑ B – POWER CABLE RE–TERMINATION

1. Cut power and ground cables to length as necessary. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans.
2. Cut and trim insulation for connection of power and ground cable conductors to connectors at PDU.
3. Transfer color code tape, run–number labels, and applicable wire marking labels as each cable is cut to size.

❑ C – POWER CABLE PRE–CONNECT CHECKS

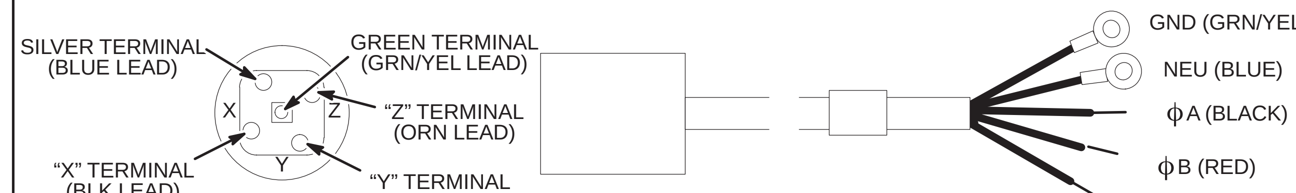
Checks for opens, shorts, and crossed connections are made for field re–terminated power cables prior to connection to PDU and cabinets. This does not provide ground resistance checks for cabinets. Ground resistance checks are in MR CD–ROM, Set Up and Calibration: System, GROUNDING AND LEAKAGE CURRENT CHECKS.

1. Use the ohmmeter and the cable plug connector Illustration to check each power cable for the following:
   - Open circuit between cable neutral and cable ground.
   - Open circuit between cable neutral and each line on the cable.
   - Open circuit between cable ground and each line on the cable.
   - Less than 1 ohm between cable neutral and the light blue (or white) conductor.
   - Less than 1 ohm between cable ground and the green/yellow conductor.
2. Any out–of–spec conditions must be immediately corrected. See Procedure **D** for plug connection details. After all power cables have been satisfactorily checked out, continue installation.

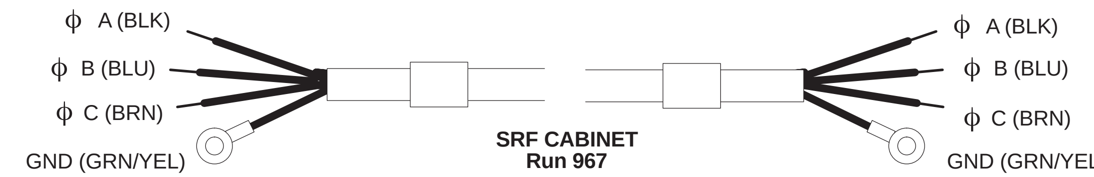
❑ D – POWER CABLE CONNECTIONS DETAILS

Note

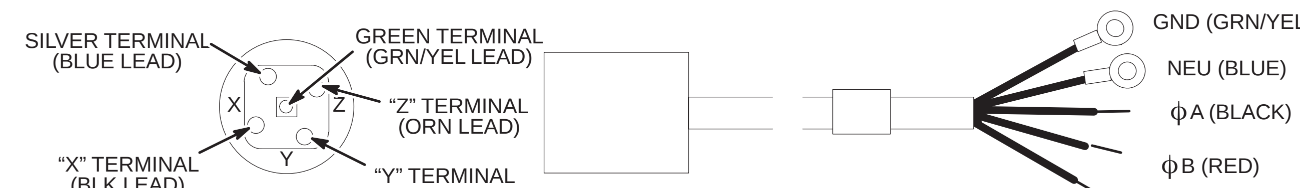
After performing cable pre–connect checks, **avoid potential problems by taping up exposed ACGD Gradient Cabinet cables** or immediately install ACGD/PDU Gradient Cabinet and connect terminals to power modules before PDU power is turned on.



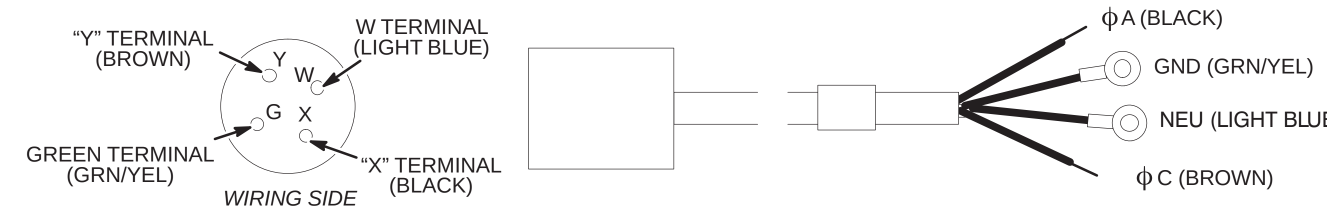
SSM
Run 966



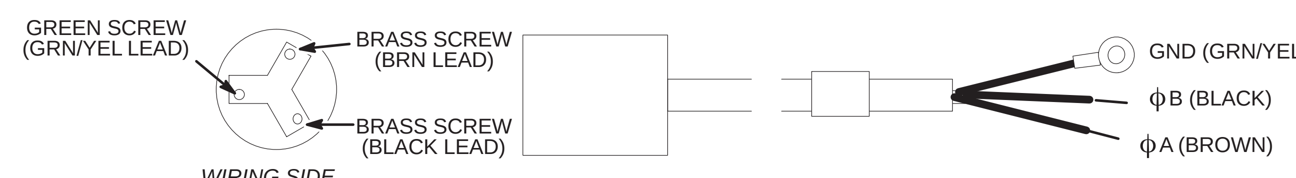
SRF CABINET
Run 967



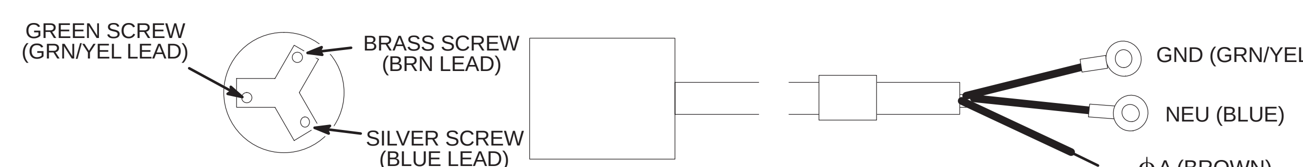
SYSTEM CABINET
Run 968



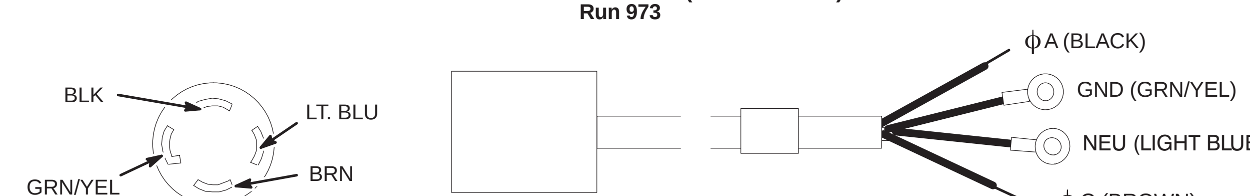
OPERATOR WORKSPACE COMPUTER CABINET
Run 969



WATER CHILLER CABINET (Non–Twin Site)
Run 970



HEAT EXCHANGER CABINET (Non–Twin Site)
Run 973



TWIN ACCESSORY CABINET
Run 989

MAGNET ENCLOSURE INSTALLATION INSTRUCTIONS

Tab 2, Magnet Enclosure, contains installation instructions for the following:

Use only the applicable Magnet Enclosure Installation Instruction Directions

9.x or **10.x**

- *DIRECTION 2223857* – Fixed & T/R LCC Magnet WideOpen Enclosure Installation Instructions
- *DIRECTION 2223858* – Mobile LCC Magnet WideOpen Enclosure Installation Instructions

FIXED & T/R LCC MAGNET WideOpen ENCLOSURE INSTALLATION INSTRUCTIONS

GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.
- A 1–ounce tube of Permatex "Anti–Seize" lubricant (2119594) is provided. Follow instructions on back of tube. "Anti–Seize" lubricant is to be applied to threads of stainless–steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

LIST OF EFFECTIVE PAGES

BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.
2. If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.

WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

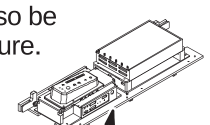
IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

INSTALLATION PROCEDURES SUMMARY

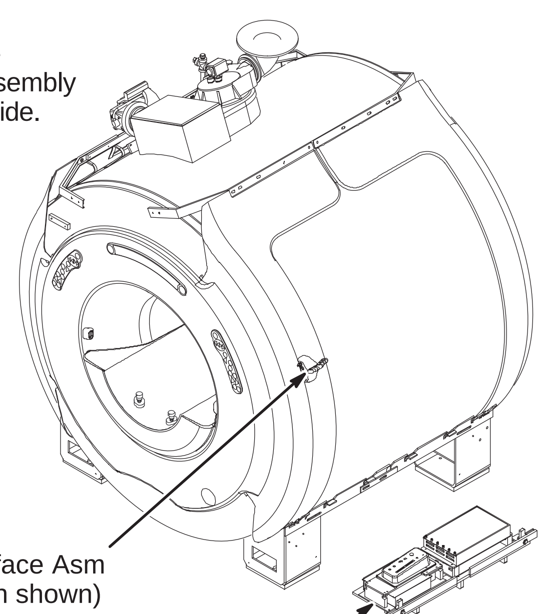
- ❑ **A1 – Determine PAC/SRI Platform and PAC Remote Interface Location**
- ❑ **A2 – PAC/SRI Platform Left Hand Configuration (If Required)**
- ❑ **A3 – Location of Magnet Ground Cable**

❑ A1 – DETERMINE PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

1. Consult with customer to determine which side of magnet the PAC/SRI Platform (2223440) and PAC Remote Interface Assembly (2227142) should be located. They have to be on the same side.
2. The PAC/SRI Platform arrives at the site in a right hand configuration.See Procedure **A2** for left hand configuration.
3. The PAC Remote Interface Assembly is shown below installed in the right side location. It can also be mounted on the left side of magnet enclosure.



PAC/SRI Platform
(Left hand configuration)

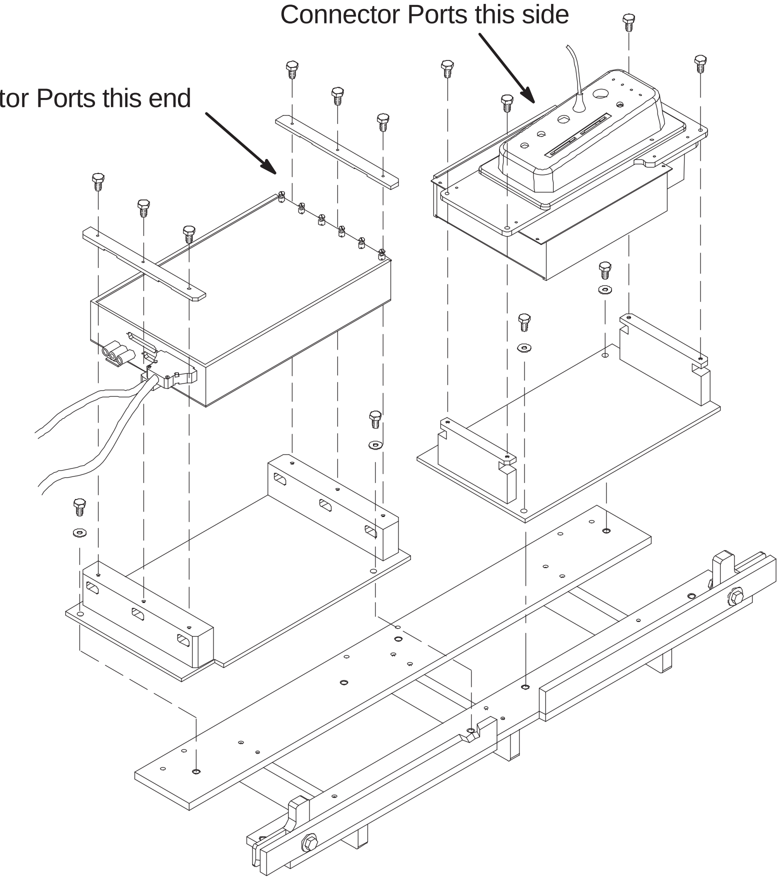


PAC Remote Interface Asm
(Right side location shown)

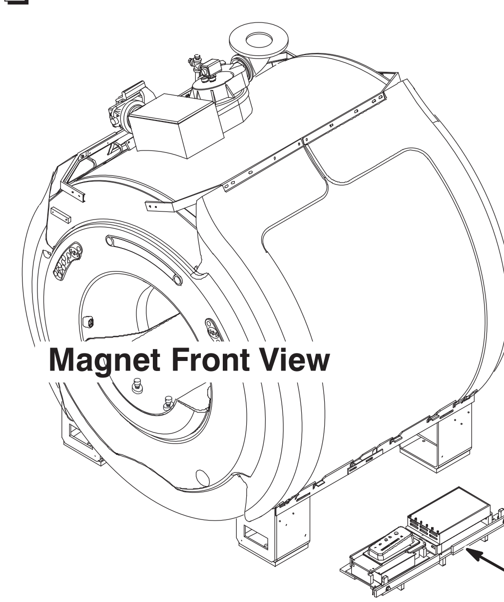
PAC/SRI Platform
(Right hand configuration)

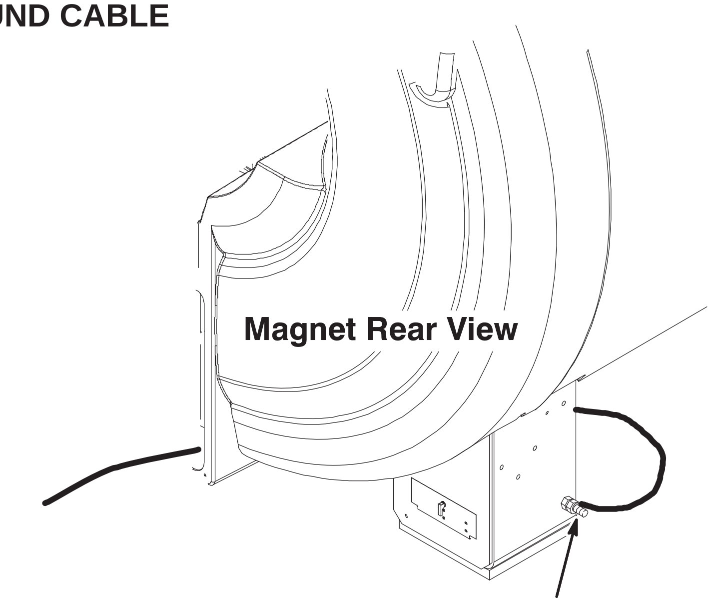
❑ A2 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)

1. For left hand configuration, reassemble PAC/SRI as shown.



❑ A3 – LOCATION OF MAGNET GROUND CABLE





1. If PAC/SRI Platform is located on right side of magnet, then the Magnet ground cable , RUN 040, has to be connected to the opposite side rear left leg as shown above. The other end of Run 040 is connected to the RF Screen Room common ground.
2. If PAC/SRI Platform is located on left side of magnet, then the Magnet ground cable has to be connected to the opposite side rear right leg.

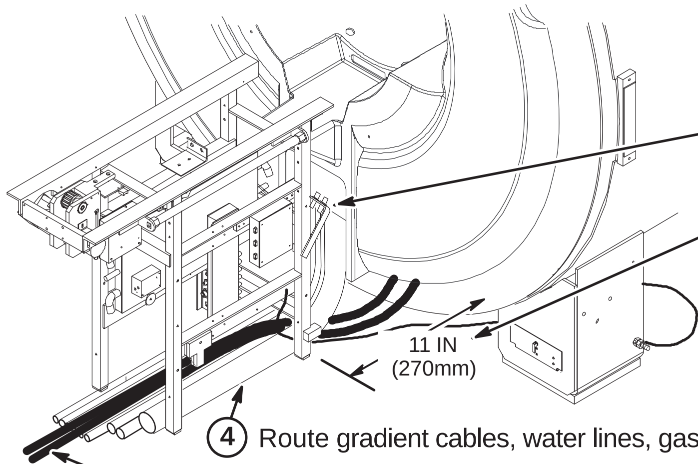
INSTALLATION PROCEDURES SUMMARY

- ❑ **B1 – Location of Rear Pedestal**
- ❑ **B2 – Routing and Connecting Gradient Output Cables to Runs 762, 763, and 764**
- ❑ **B3 – Route/Connect Runs 257/746, 262, 743, 744, and 780**
- ❑ **B4 – Route/Connect Runs 258/747, 318, 700, and Drive Cables**

❑ B1 – LOCATION OF REAR PEDESTAL

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

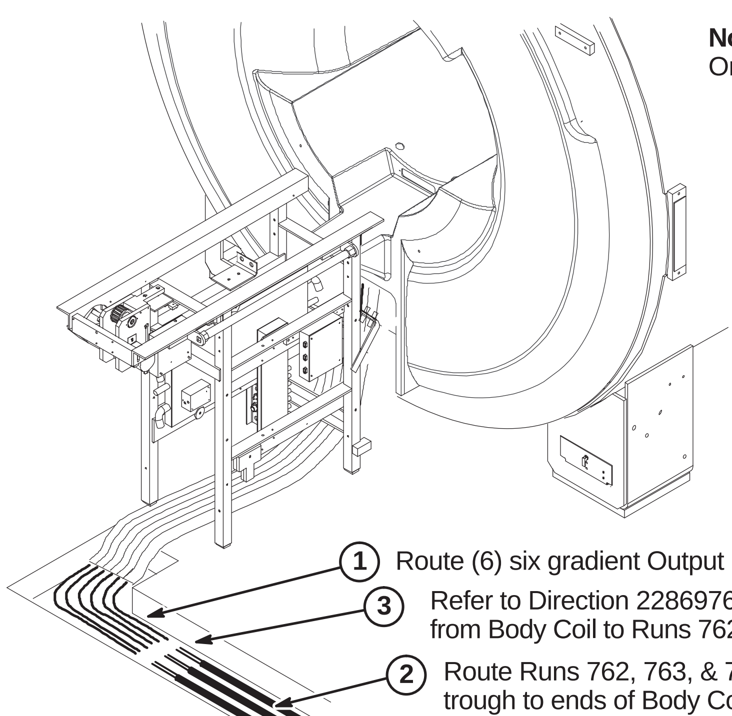


Note:
Cables and hoses can be routed to exit out of rear or side of Rear Pedestal.

1. For **CV/*i*** install bracket on other side.
2. Center Rear Pedestal on Rear Endbell and locate approximately 11 inches (270mm) from face of magnet.
3. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.
4. Route gradient cables, water lines, gas lines, and 2" air hose under Rear Pedestal as shown.
5. Route and connect coolant hosees from Water Chiller to 10 ft. (300cm) coolant hoses attached to Body Coil.
6. Route all other cables on top of the above cables and hoses.

❑ B2 – ROUTING AND CONNECTING GRADIENT OUTPUT CABLES

Note:
Only Gradient Cables are shown for clarity.



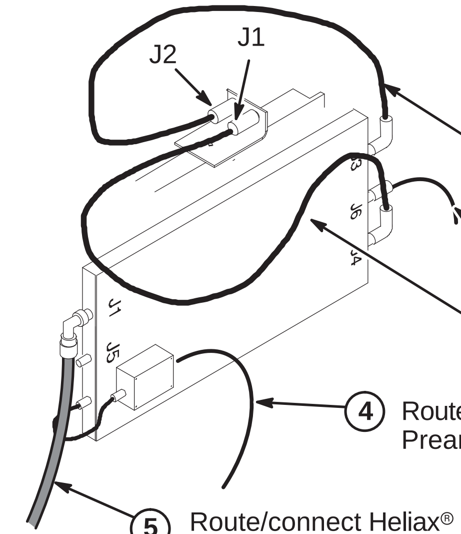
1. Route (6) six gradient Output Cables from Body Coil to cable trough.
3. Refer to Direction 2286976 in Tab 1 for connecting Output cables from Body Coil to Runs 762, 763, & 764.
2. Route Runs 762, 763, & 764 from Penetration Panel through cable trough to ends of Body Coil gradient cables. Make sure that there is at least two feet (610mm) of slack before cutting to length.

❑ B3 – ROUTE/CONNECT RUNS 257/746, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

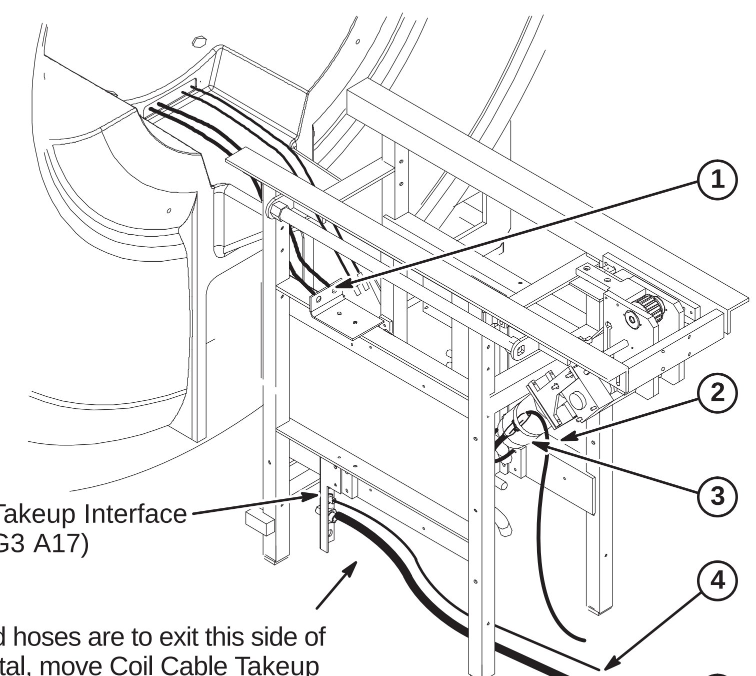
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

1. Route and connect Run 744 from J2 on bracket to J4 on the RF Splitter.
2. Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
3. Route and connect Run 743 from J1 on the bracket to J3 on the RF Splitter.
4. Route/connect Run 262 from J8 on Pen Panel to Body Coil Preamplifier. (For MGD, connect to Receive Filter on Preamp)
5. Route/connect Heliax® to J1 on RF Splitter:
   Run 257 (1.5T, 7/8") or
   Run 746 (1.0T, 1/2")

❑ B4 – ROUTE/CONNECT RUNS 258/747, 318, 700, AND DRIVE CABLES



1. Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
2. Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
3. Ty–wrap cable to motor to prevent movement.
4. Route/connect Run 700 to J2 on Coil Cable Takeup Interface (NON–MGD ONLY)
5. Route/connect to J1 on Coil Cable Takeup Interface:
   Run 258 (1.5T, 1/2") or
   Run 747 (1.0T, 3/8")

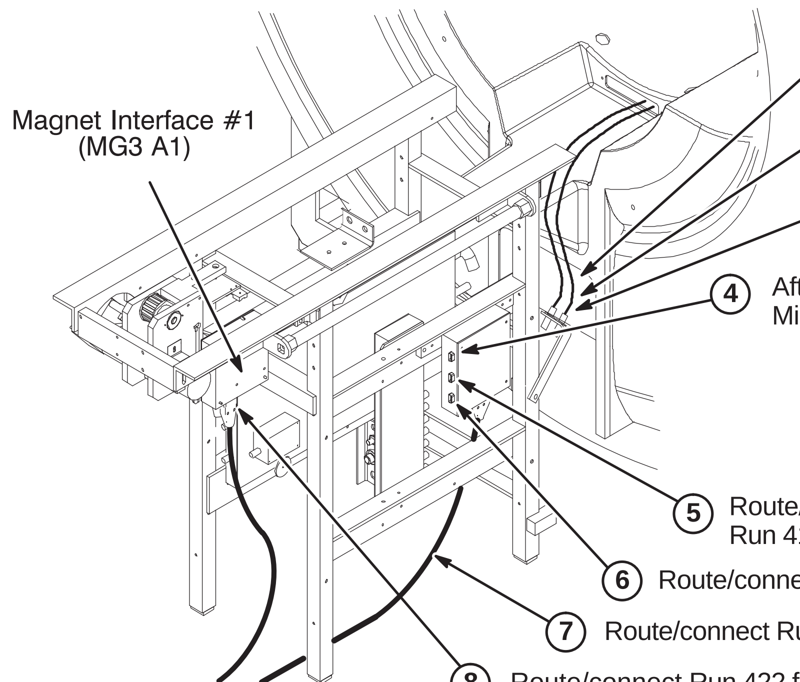
Note:
If cables and hoses are to exit this side of Rear Pedestal, move Coil Cable Takeup Interface to opposite side of Rear Pedestal.

If Interface bracket is moved to other side, connectors on bracket need to be removed, turned end for end, and reinstalled.

INSTALLATION PROCEDURES SUMMARY

- ❑ **C1 – Route/Connect Runs 282, 414, 422, 781, 782, and Front End Bell Speaker Cable**
- ❑ **C2 – Install Bore Lamp Assembly**
- ❑ **C3 – Route/Connect Runs 257/746, 262, 743, 744, and 780**
- ❑ **C4 – Install Blower Box and Route/Connect Air Hoses**

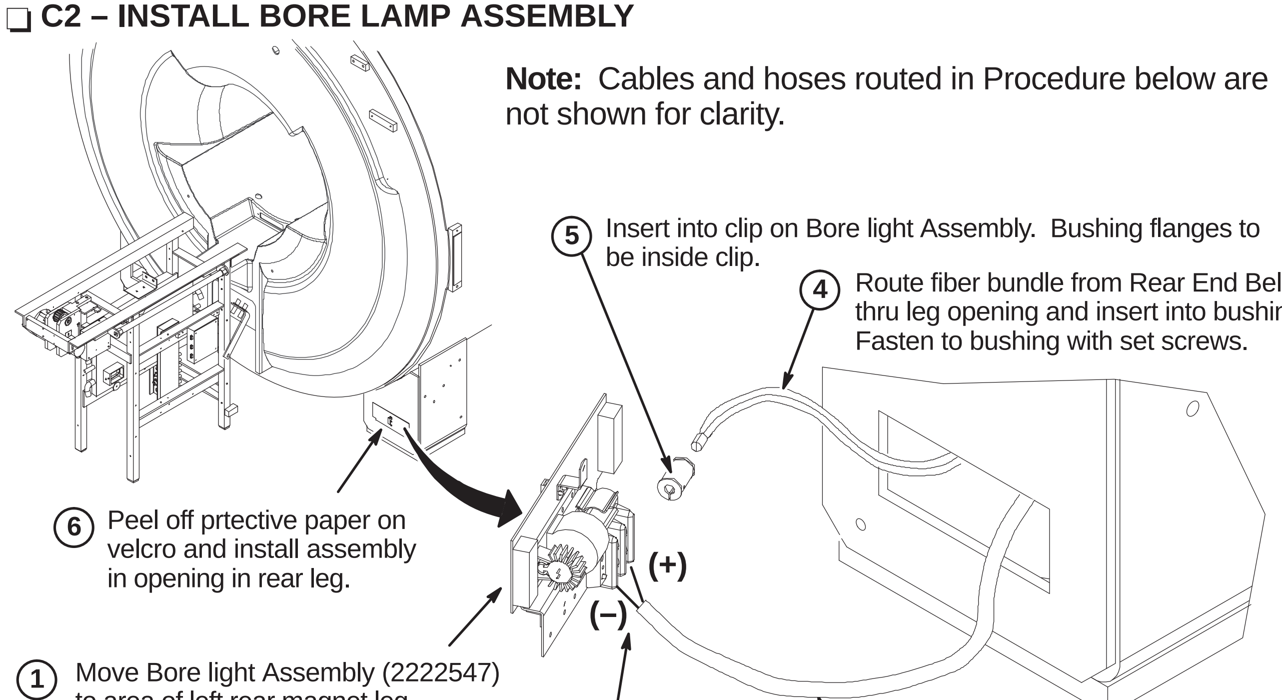
❑ C1 – ROUTE/CONNECT RUNS 282, 414, 422, 781, 782, & FRONT END BELL CABLE



1. Route and connect RF Bias cables from RF Coil to bracket as marked.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. After Carriage cover is installed, connect Microphone cable from Carriage Cover to J7.
5. Route/connect Front End Bell microphone cable Run 414 to J9.
6. Route/connect speaker cable from Front End Bell to J8.
7. Route/connect Run 282 from J12 on Pen Panel to J6.
8. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

❑ C2 – INSTALL BORE LAMP ASSEMBLY

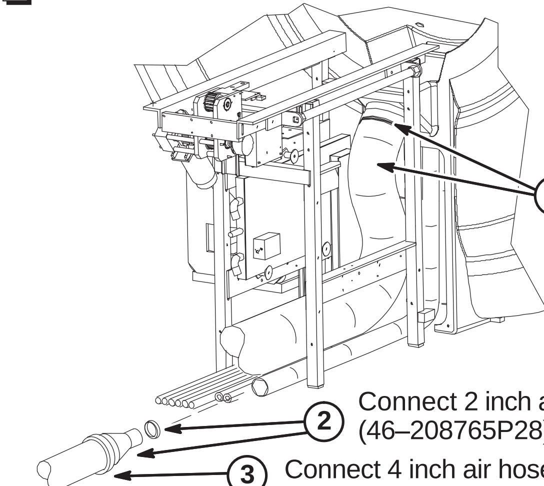
Note: Cables and hoses routed in Procedure below are not shown for clarity.



1. Move Bore light Assembly (2222547) to area of left rear magnet leg.
2. Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
3. Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
4. Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
5. Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
6. Peel off prtective paper on velcro and install assembly in opening in rear leg.

❑ C3 – CONNECT AIR HOSES TO REAR OF MAGNET

Note:
The following items are part of Blower Hose Kit, 2129412–3 or 2129412–4.



1. Connect 6 inch air hose (46–282666P12) to Rear End Bell with 6 inch clamp (46–208675P29).
2. Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). **Do not cut or reduce the length of the 2 inch hose.**
3. Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). **If necessary, the 4 inch hose may need to be cut to reduce length.**

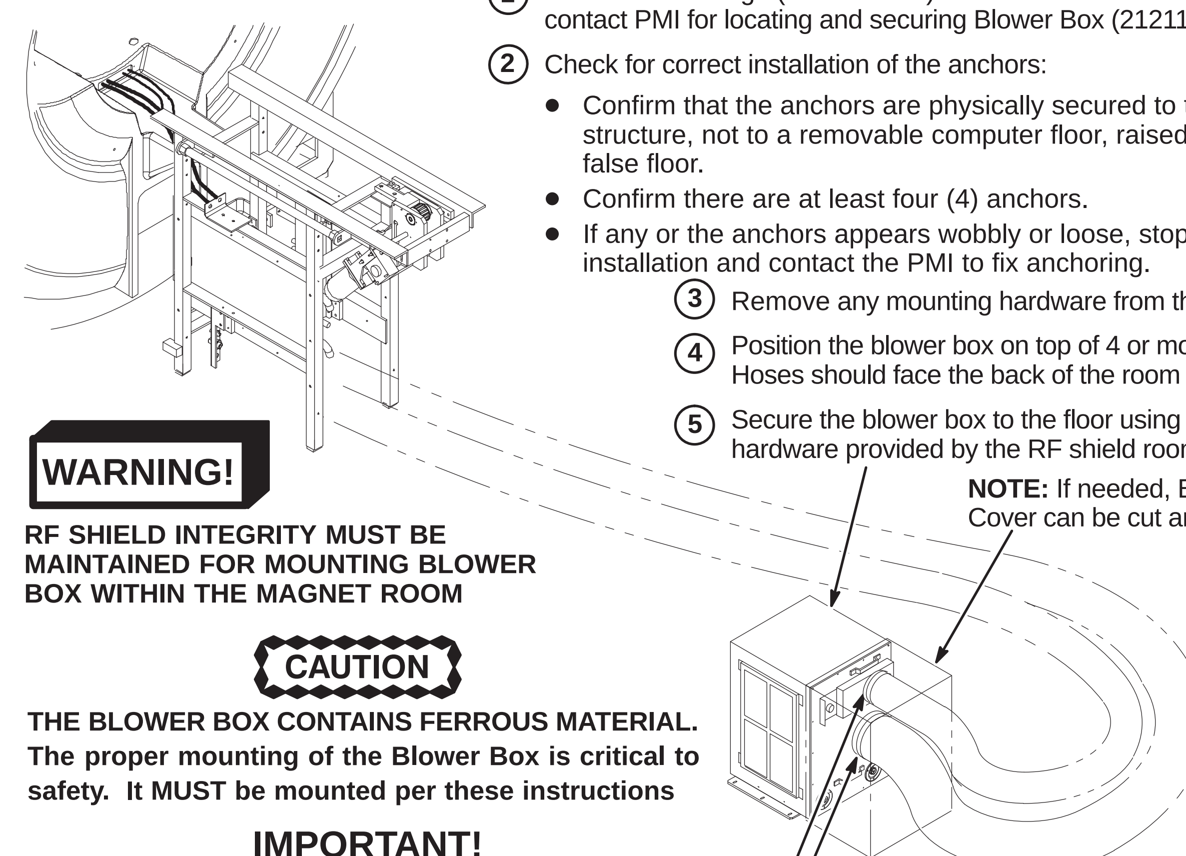
WARNING!

THE BLOWER BOX IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE. TO LOCATE THE 200 GAUSS ZONE, REFER TO THE TABLE IN SECTION 1–1 INTRODUCTION.

❑ C4 – INSTALL BLOWER BOX AND ROUTE/CONNECT AIR HOSES

1. Refer to site design (architectural)/customer construction drawings or contact PMI for locating and securing Blower Box (2121199).
2. Check for correct installation of the anchors:
   - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
   - Confirm there are at least four (4) anchors.
   - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
3. Remove any mounting hardware from the anchors.
4. Position the blower box on top of 4 or more anchors. Hoses should face the back of the room
5. Secure the blower box to the floor using the fastening hardware provided by the RF shield room vendor.

NOTE: If needed, Blower Hose Cover can be cut and modified.



WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

6. Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
   (1) 46–208765P34 4 inch Hose Clamp
   (1) 46–208765P29 6 inch Hose Clamp

INSTALLATION PROCEDURES SUMMARY

- ❑ **D1 – Prepare Cables and Tubes for Routing**
- ❑ **D2 – Position Remote PAC Interface Mounting Clip**
- ❑ **D3 – Route Remote PAC Interface Cables and Tubes**
- ❑ **D4 – Installing Remote PAC Interface to Arc Cover**
- ❑ **D5 – Install PAC Hanger Bar**

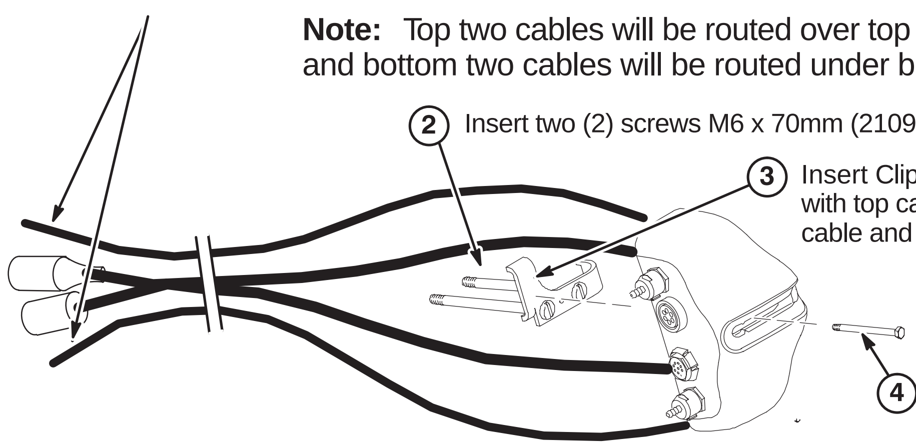
❑ D1 – PREPARE CABLES AND TUBES FOR ROUTING

Note:
Consult with customer for correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform.

1. Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

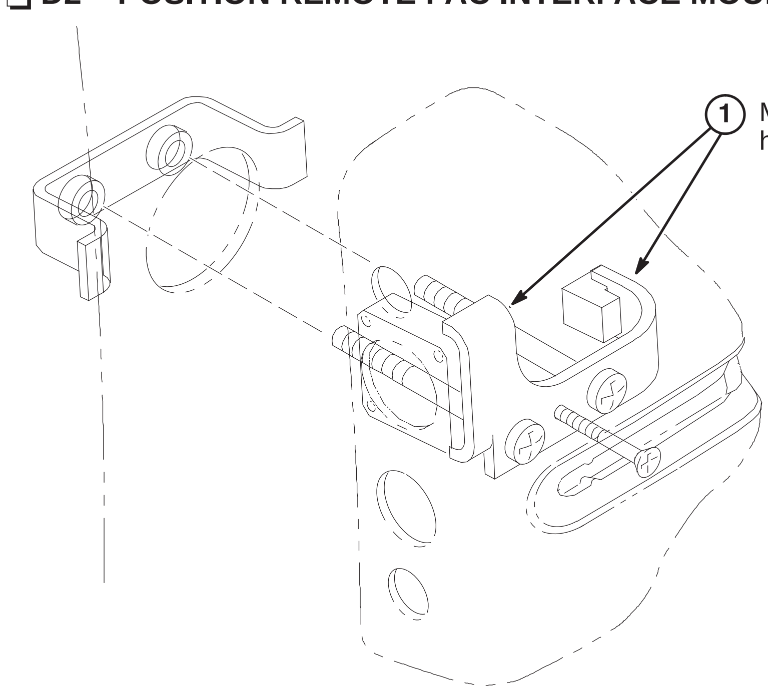
Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

2. Insert two (2) screws M6 x 70mm (2109873–33) screws into Clip (2221023).
3. Insert Clip with screws into shell as shown with top cable and tube above Clip and bottom cable and tube below.
4. Hold in place by inserting provided M4 screw thru slot and turning about 3 or 4 turns into Clip.

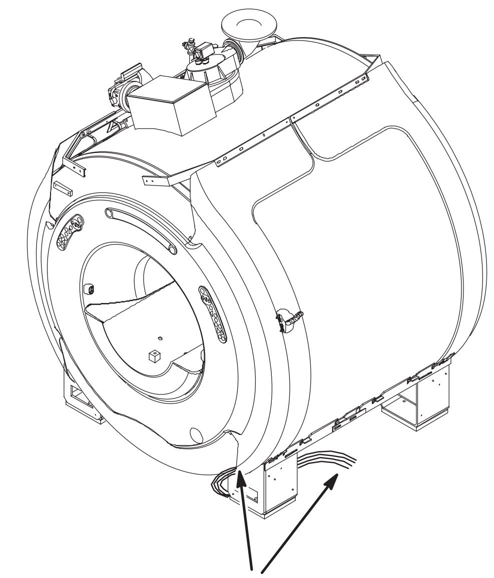


❑ D2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP

1. Make sure each side of clip is mounted behind housing blocks.



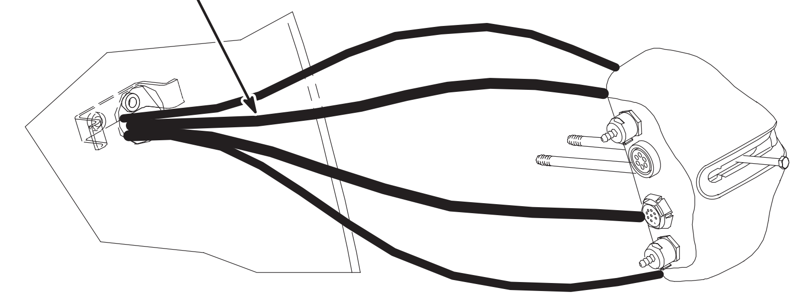
❑ D3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES



Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

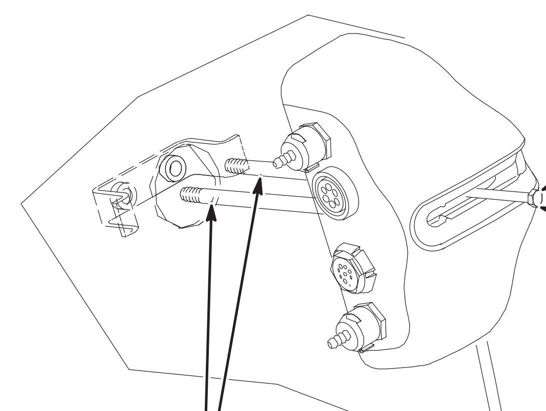
Note: Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.

1. **Insert ECG Cable first.** Route this thru pre–mounted foam tube (if available) located on inside of Arc cover.
2. Route Peripheral Pulse cable and tubes next.



3. Route around foot of magnet to PAC/SRI Platform location.

❑ D4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER

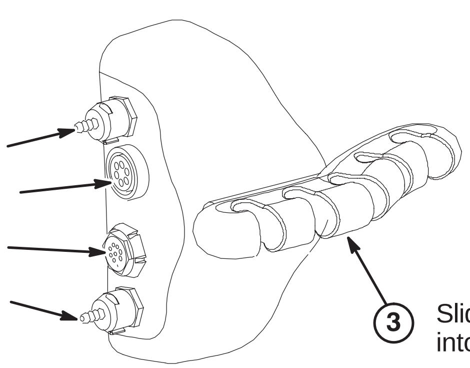


Note:
Routed cables and tubes not shown for clarity.

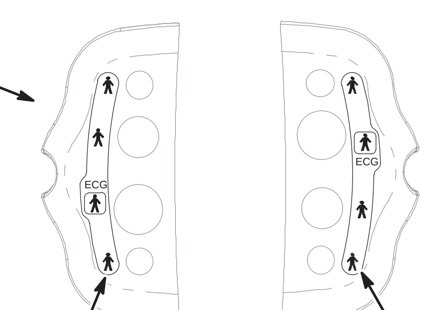
1. Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.
3. Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.
OR
Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in **T5** into place.
2. Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place

❑ D5 – INSTALL PAC HANGER BAR

1. Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.
2. Attach cables as indicated below.



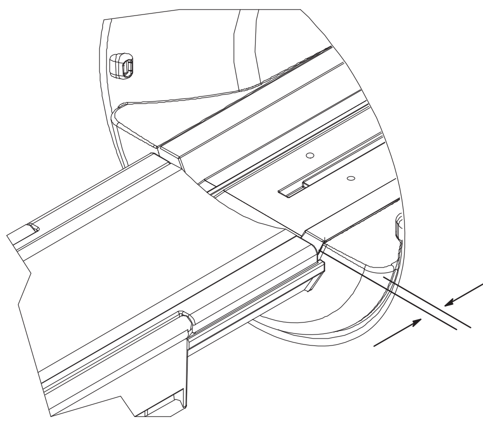
3. Slide PAC Hanger Bar (2221023) into position.



INSTALLATION PROCEDURES SUMMARY

- ❑ E1 – Install Dock Mounting Bracket and Dock
- ❑ E2 – Route and Connect Magnet Enclosure Front Area Cables
- ❑ E3 – Cable Connection to PAC Module

❑ E1 –INSTALL DOCK MOUNTING BRACKET AND DOCK



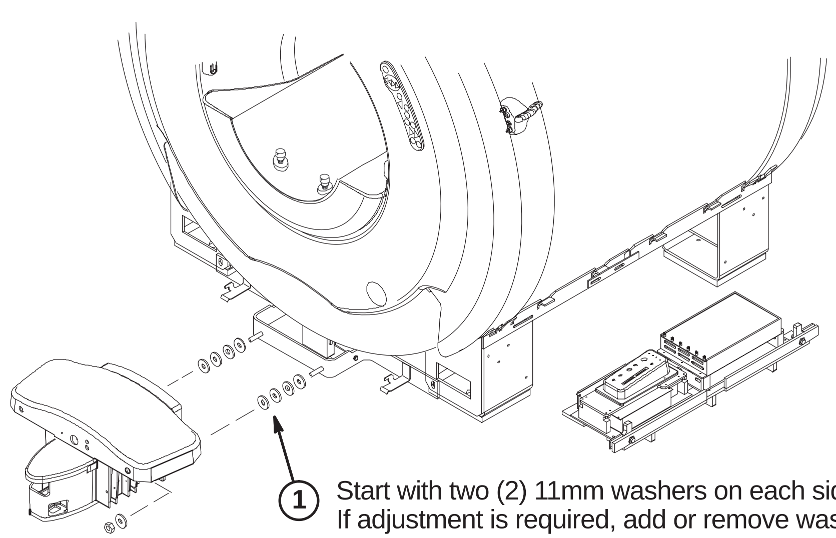
Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches ± 0.25 (19.0±6.4mm). See Step 2 below.

WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY.

WARNING!

DO NOT LIFT THE DOCK MORE THAN 2 INCHES (5 CM) OFF THE FLOOR. If the height (to clear the dock anchor) exceeds two inches (5 cm), the magnet must be ramped down prior to removing or installling the dock as there is significant risk of the magnet attracting the dock if it needs to be raised more that 2 inches (5 cm).



1. Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is acheived.

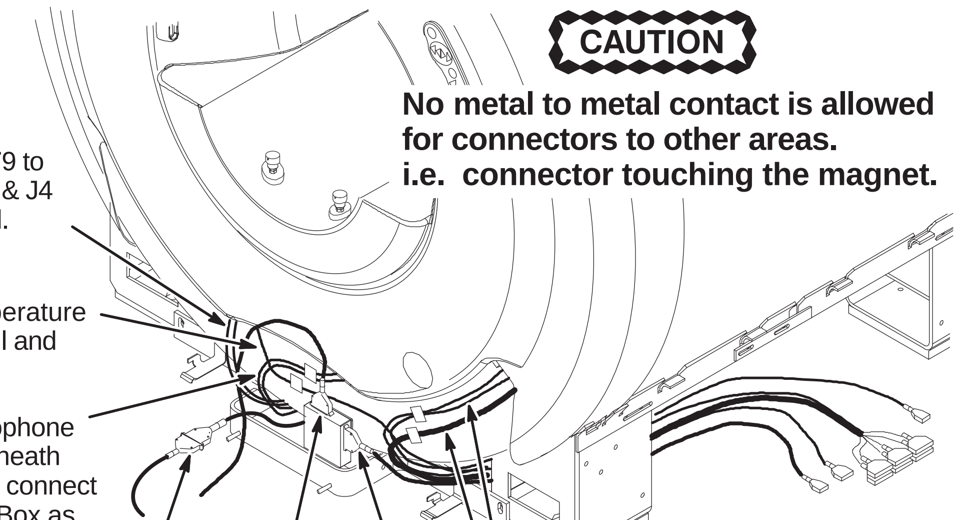
Note:
Floor stud installed by RF room vendor/mechanical contractor.

❑ E2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Dock Assembly not shown for clarity.

CAUTION

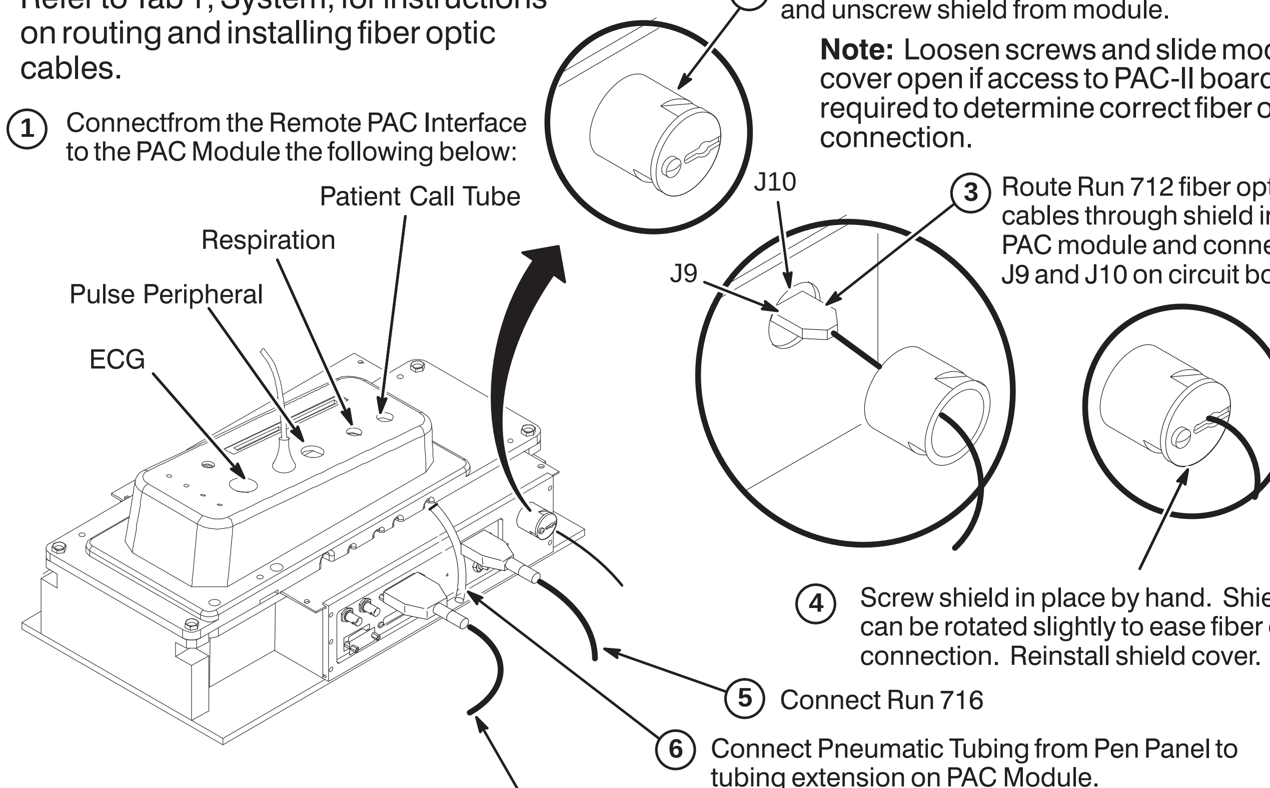
No metal to metal contact is allowed for connectors to other areas. i.e. connector touching the magnet.



1. Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.
2. Route Run 749, Bore Temperature Sensor, from Front End Bell and connect ot J10 on SRI.
3. Route Front End Bell Microphone and Speaker cables underneath manet to ReaPedestal and connect to Patient Communication Box as marked.
4. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.
5. Connect Dock Cable to J2 on Dock Limit Box
6. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.
7. Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

❑ E3 – CABLE CONNECTION TO PAC MODULE

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

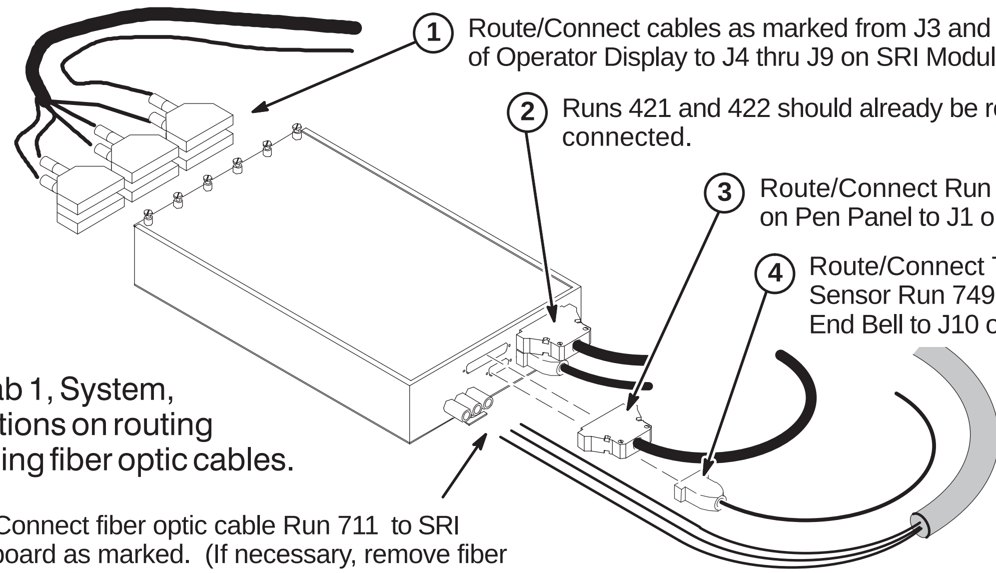


1. Connectfrom the Remote PAC Interface to the PAC Module the following below:
2. Remove shield cover by removing one screw and unscrew shield from module.
   Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.
3. Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.
4. Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.
5. Connect Run 716
6. Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.
7. Connect cable from J5 on Operator Display to J4 on PAC Module.

INSTALLATION PROCEDURES SUMMARY

- ❑ **F1 – Connect Cables to SRI Module**
- ❑ **F2 – Routing of Photo–Plethysmograph Cable**
- ❑ **F3 – Route and Tie–wrap Cables to PAC/SRI Platform**
- ❑ **F4 – Install PAC/SRI Platform under Magnet**

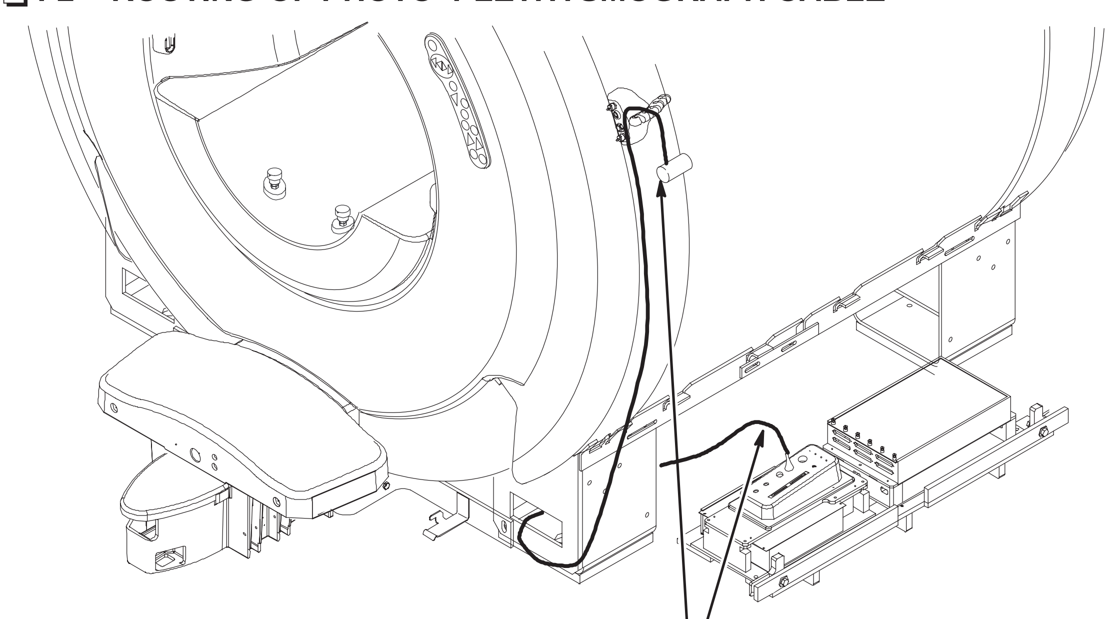
❑ F1 – CONNECT CABLES TO SRI MODULE



1. Route/Connect cables as marked from J3 and J4 of Operator Display to J4 thru J9 on SRI Module.
2. Runs 421 and 422 should already be routed and connected.
3. Route/Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
4. Route/Connect Temperture Sensor Run 749 from Front End Bell to J10 on SRI Module
5. Route/Connect fiber optic cable Run 711 to SRI circuit board as marked. (If necessary, remove fiber optic cable guides and pass fiber optic cables thru) Close SRI cover and tighten captive screws.

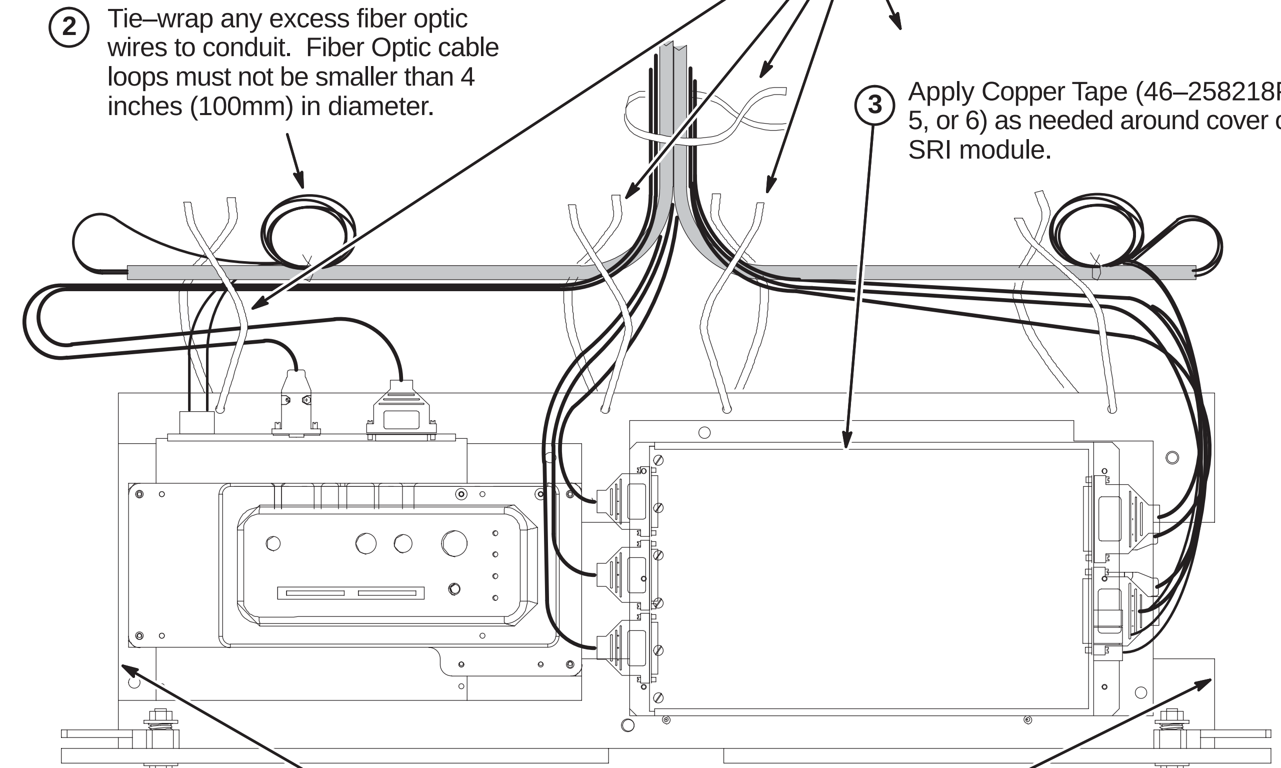
Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

❑ F2 – ROUTING OF PHOTO–PLETHYSMOGRAPH CABLE



1. Route photo–plethysmograph cable from PAC module thru opening in front leg and up to PAC Hanger Bar.
2. Cable must fit between front Foot Cover and side Trim Cover (which are installed later).

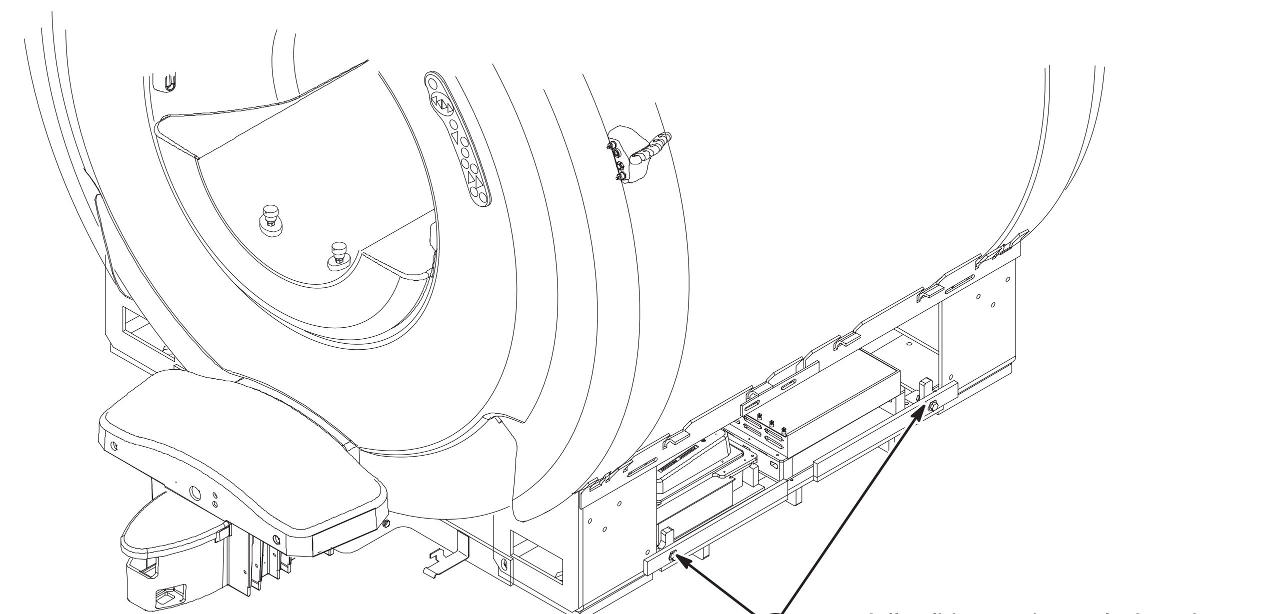
❑ F3 – ROUTE AND TIE–WRAP CABLES TO PAC/SRI PLATFORM



1. Dress all cables and fiber optic conduits as shown and tie–wrap together and to PAC/SRI Platform.
2. Tie–wrap any excess fiber optic wires to conduit. Fiber Optic cable loops must not be smaller than 4 inches (100mm) in diameter.
3. Apply Copper Tape (46–258218P4, 5, or 6) as needed around cover of SRI module.

Note: Keep cables dressed inside edge so unit slides between legs easily for service.

❑ F4 – INSTALL PAC/SRI PLATFORM UNDER MAGNET



1. Carefully slide PAC/SRI Platform between legs of magnet and lock in place at each end.

BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. **If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.**

2. **If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.**

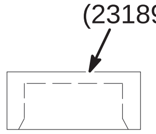
INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)
- ❑ **G1 – Front Bridge Support Adjustments**
- ❑ **G2 – Bridge Support Adjustment Tolerances**
- ❑ **G3 – Bridge Assembly Preparation Before Installing in Magnet**
- ❑ **G4 – Install Bridge**

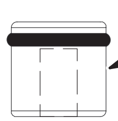
NOTE:
Measure to confirm that the distance between the top of the Support Cap Cup and the flat surface of the End Bell is110mm as shown in Procedure **G1**. Also measure to confirm that the 10mm and 62mm M10 Jam Nut distance locations on the Stud are also in place.

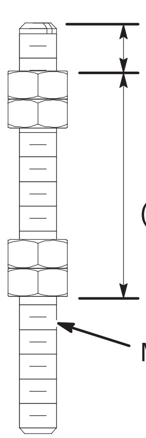
- If the dimensions are found to be in place for this site, proceed to Procedure **G2**.
- If these dimensions are not in place, complete Steps 1 thru 6 in Procedure **G1** and then proceed to **G2**.

❑ G1 – FRONT BRIDGE SUPPORT ADJUSTMENTS

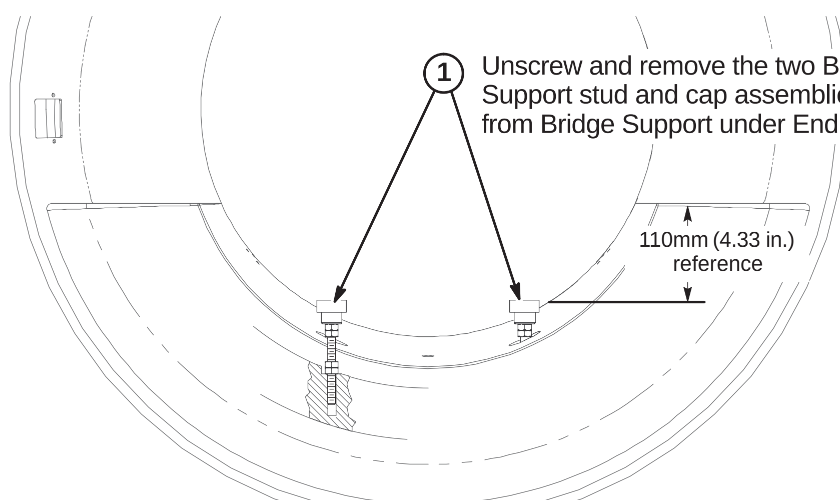
2 Remove Support Cap Cup and unscrew Support Cap


Support Cap Cup (2318948)


Support Cap (2318950) & O–Ring (46–136374P37)



M10 x 110mm Stud (2116325–5) & M10 Nuts (2109875–4))



1 Unscrew and remove the two Bridge Support stud and cap assemblies from Bridge Support under End Bell.

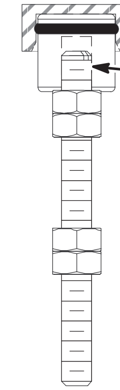
3 Set top M10 Nut at a distance of 10mm from top of stud. Lock in position with second M10 Nut.

4 Set bottom M10 Nut at a distance of 62mm from top of first M10 Nut. Lock in position with fourth M10 Nut.

5 Fasten Support Cap to top of Stud and tighten to top M10 Nut. Install Support Cap Cup to Support Cap

6 Re–install the two Bridge Support stud and cap assemblies into Bridge Support.

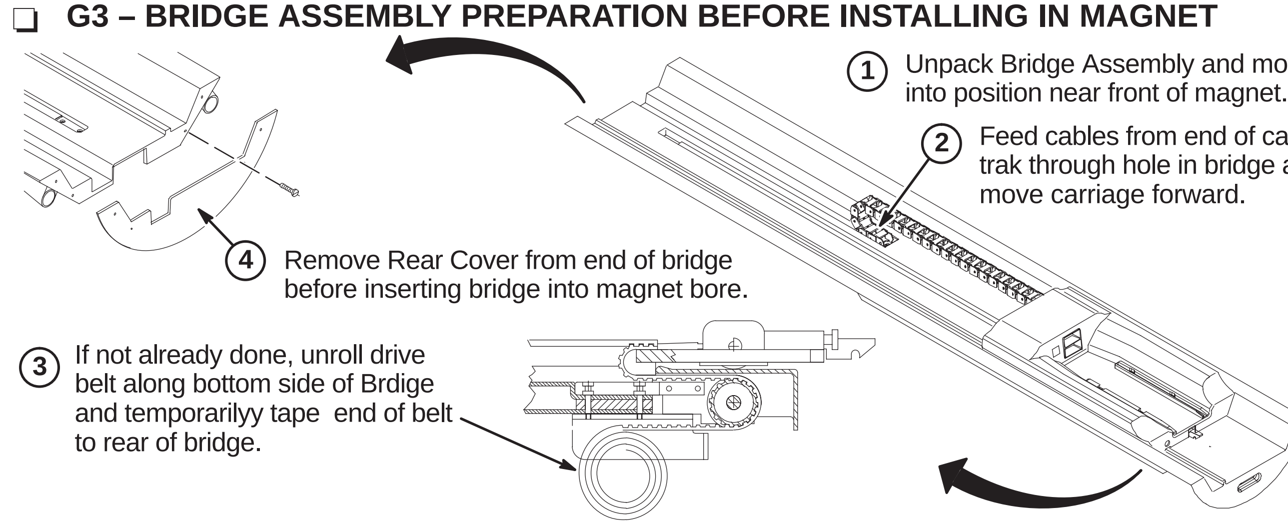
❑ G2 – BRIDGE SUPPORT ADJUSTMENT TOLERANCES



1 The 10mm location distance noted in Procedure **G1** is a starting point for installation and leveling of Bridge.

This starting point gives the installer an adjustment of +7.5mm and –5mm. The amount of adjustment will be determined during bridge leveling.

❑ G3 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET



1 Unpack Bridge Assembly and move into position near front of magnet.

2 Feed cables from end of cable trak through hole in bridge and move carriage forward.

3 If not already done, unroll drive belt along bottom side of Brdige and temporarilyy tape end of belt to rear of bridge.

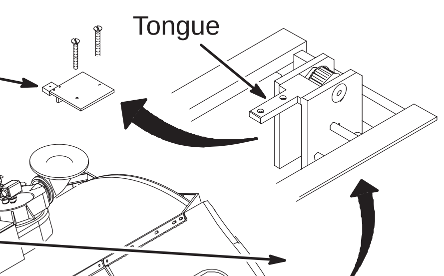
4 Remove Rear Cover from end of bridge before inserting bridge into magnet bore.

❑ G4 – INSTALL BRIDGE

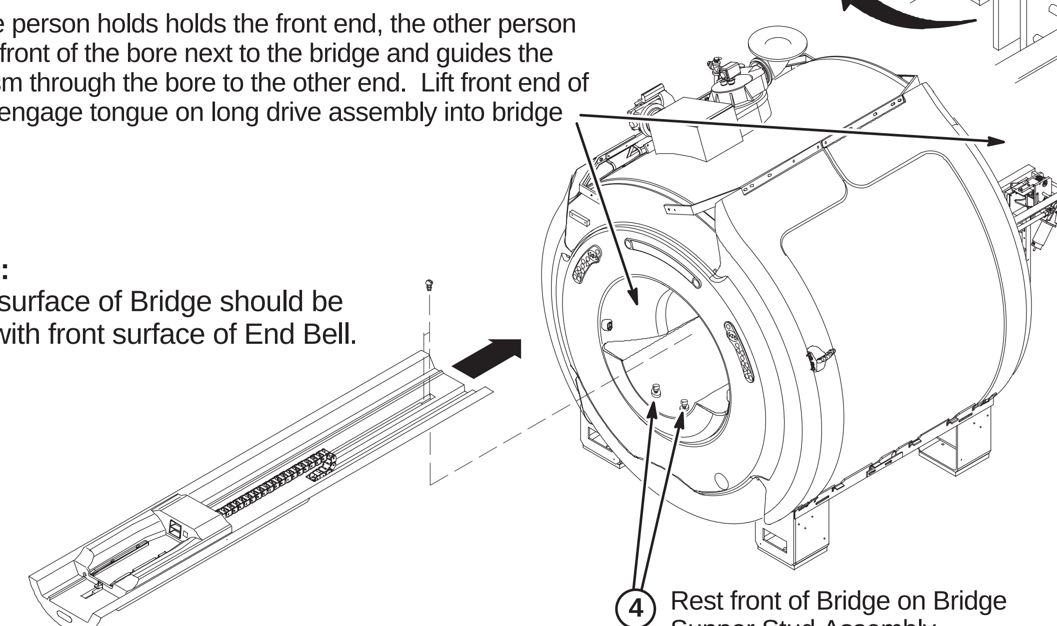
1 Unfasten Rear Bridge Mount Asm. It will be reinstalled later.

2 Position rear end of Bridge Asm at front of Body Coil.

3 While one person holds holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.



NOTE:
Front surface of Bridge should be flush with front surface of End Bell.



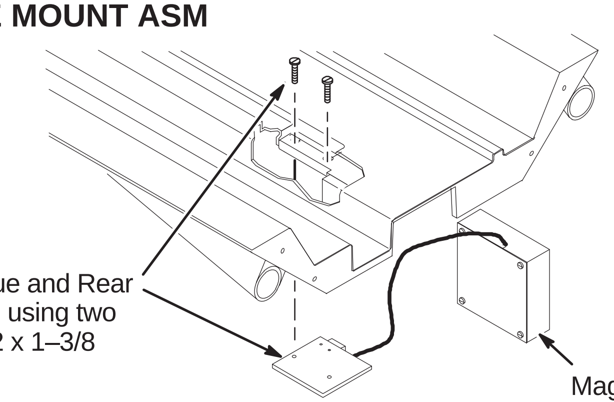
4 Rest front of Bridge on Bridge Suppor Stud Assembly.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

- ❑ **H1 – Reattach Bridge Mount Assembly**
- ❑ **H2 – Align Front of Bridge to End Bell**
- ❑ **H3 – Applying Adhesive to Bridge Support Caps**
- ❑ **H4 – Leveling Bridge**

❑ H1 – REATTACH BRIDGE MOUNT ASM

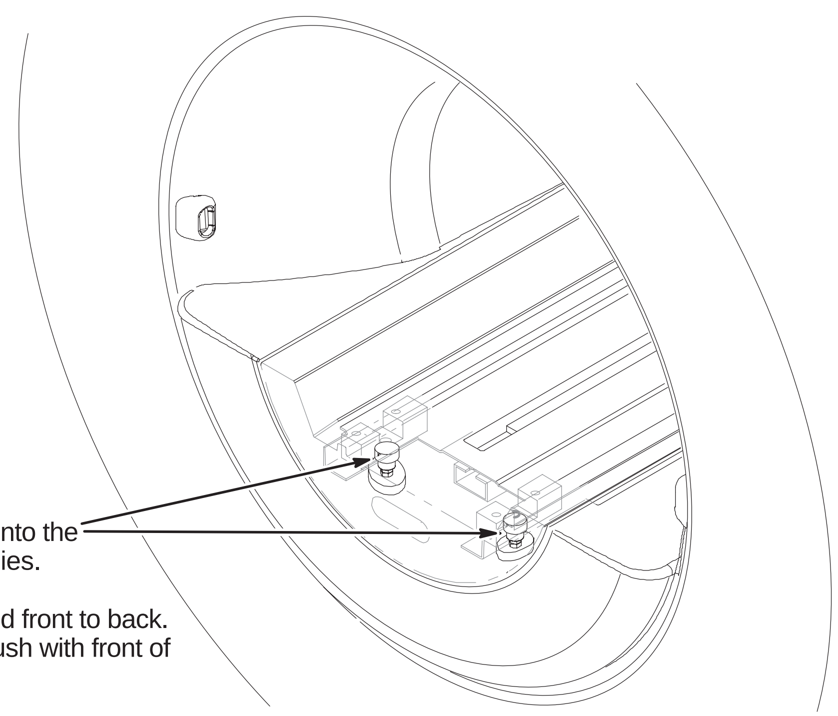
NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.



1. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) previously removed 10–32 x 1–3/8 Screws (46–220184P47)

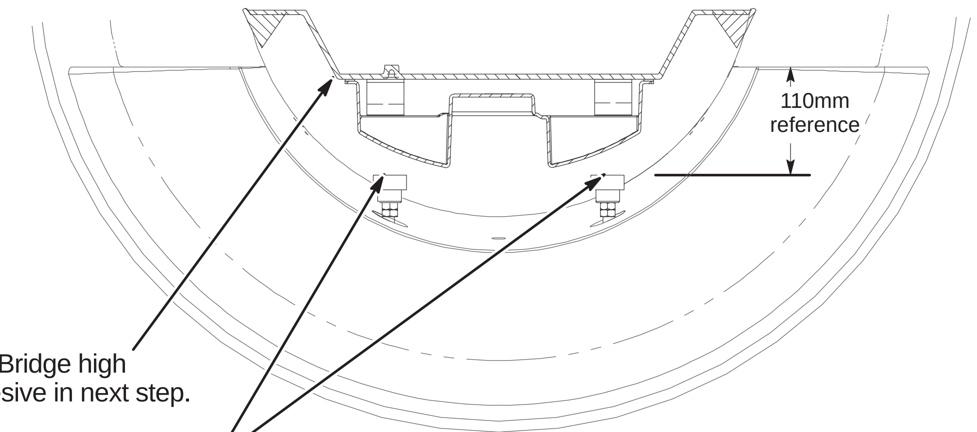
Magnet I/F #1
(Mounted on Rear Pedestal)

❑ H2 – ALIGN FRONT OF BRIDGE TO END BELL



1. Lower the front of the Bridge onto the Bridge Support Stud Assemblies.
2. Align the Bridge left to right and front to back. Make sure front of Bridge is flush with front of end bell.

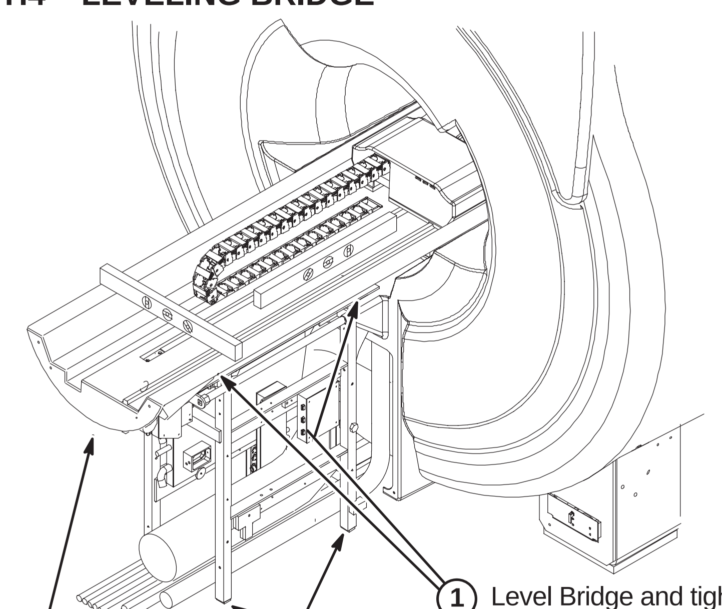
❑ H3 – APPLYING ADHESIVE TO BRIDGE SUPPORT CAPS



1. Raise the front of the Bridge high enough to apply adhesive in next step.
2. Apply epoxy adhesive (46–282264P1) to top of each Bridge Support Cap.
3. Lower the Bridge to rest firmly to the Bridge Support Caps.
4. Allow adhesive to cure.
5. After adhesive has cured, adjust the Bridge to the final vertical position. Make sure bridge is level.

   Note: The final position will leave the top surface of the Bridge 1mm (or more) above the top surface of the end bell.
6. Repeat the vertical adjustment as needed. Make sure the two M10 jam nuts are tight against each Support Cap. This will prevent change of position.

❑ H4 – LEVELING BRIDGE



NOTE:
PVC Spacers positioned thru End Bells are to support Bridge.

NOTE:
Rear Pedestal to be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

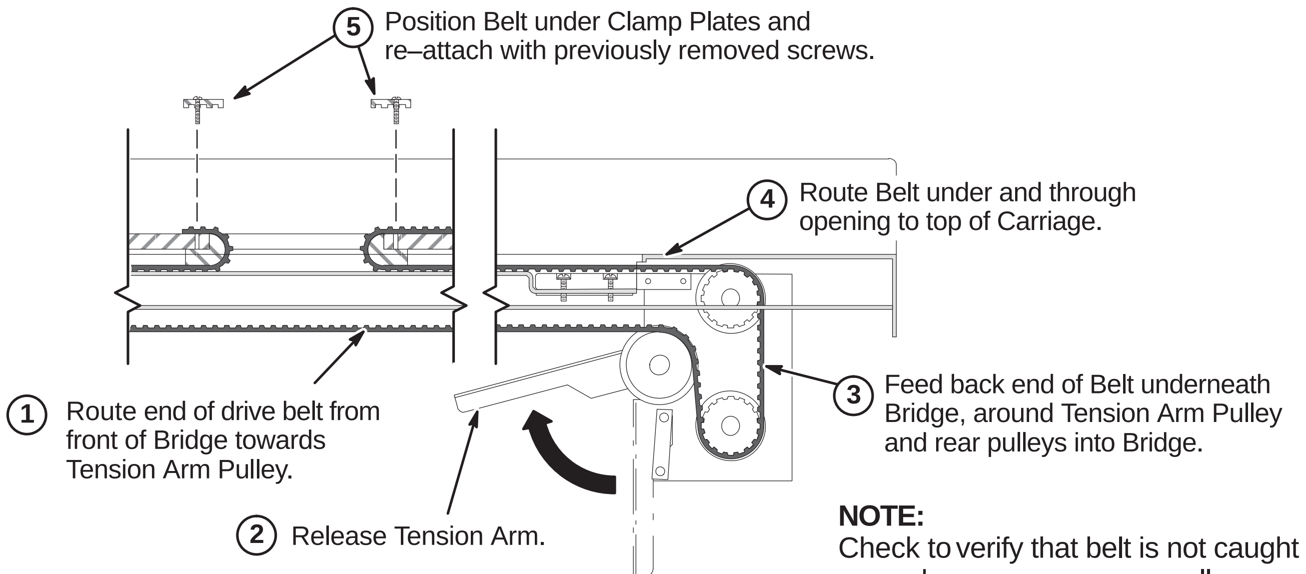
NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

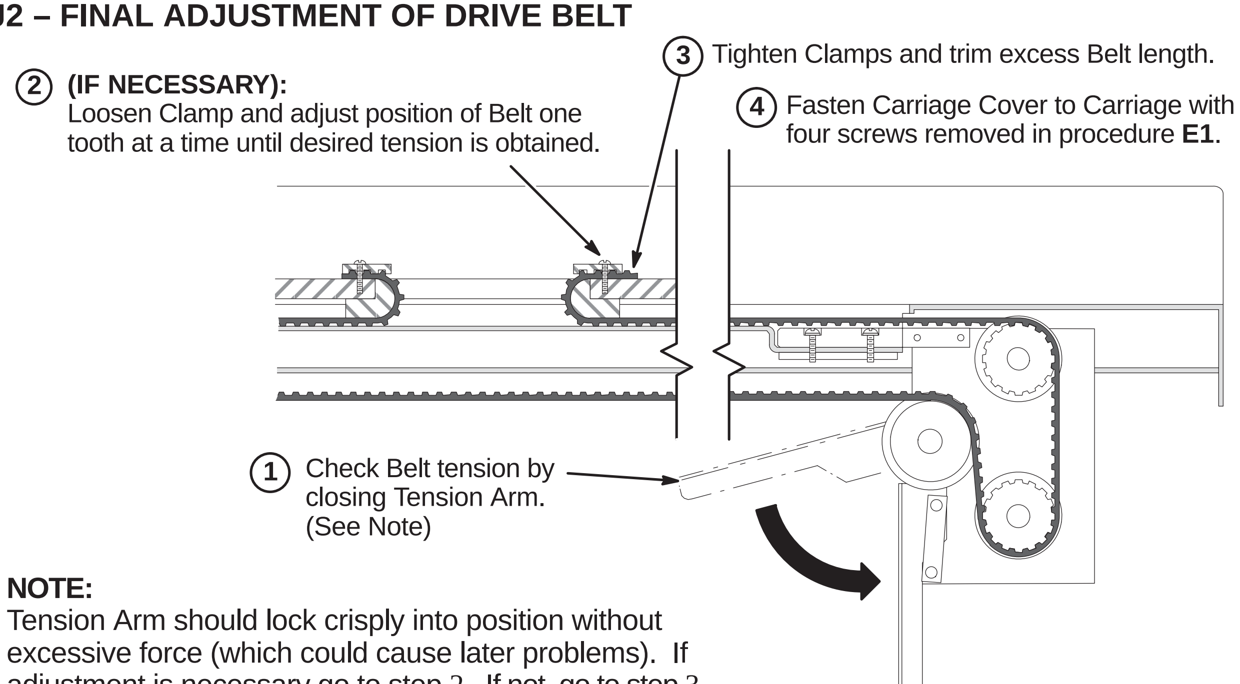
- ❑ J1 – Routing Drive Belt
- ❑ J2 – Final Adjustment of Drive Belt
- ❑ J3 – Installation of Sensor Flag
- ❑ J4 – Attach Bridge Cover
- ❑ J5 – Route/Connect Runs 404 and 485 (and Multi–coil Cables if Required)

❑ J1 – ROUTING OF DRIVE BELT



NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

❑ J2 – FINAL ADJUSTMENT OF DRIVE BELT

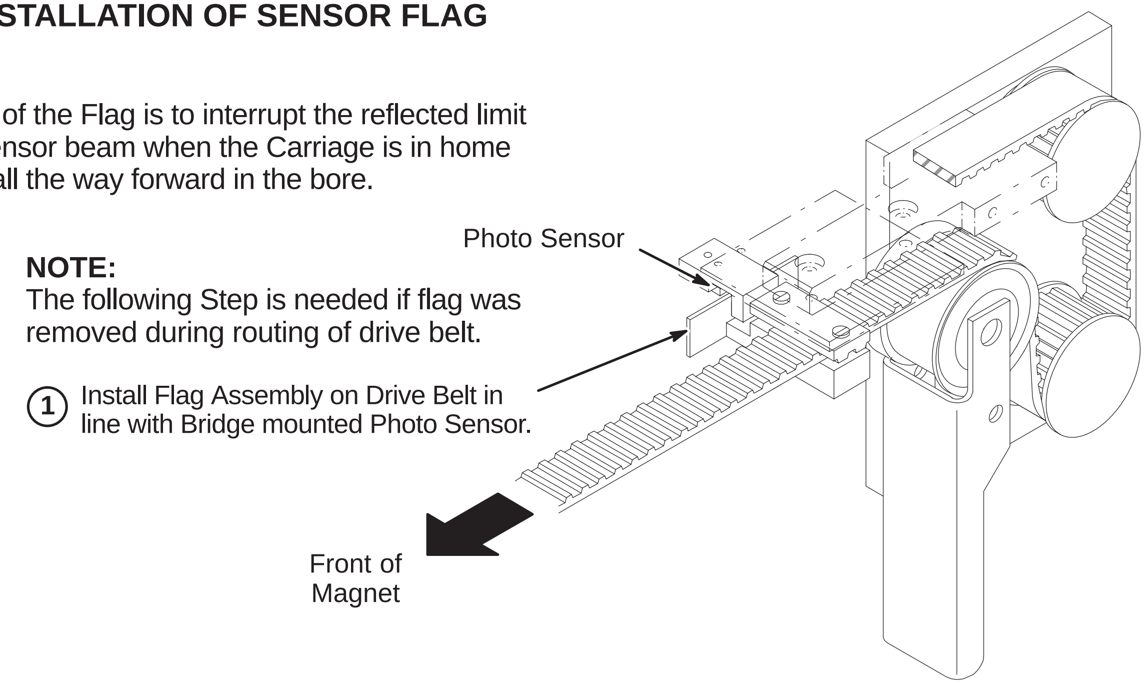


NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

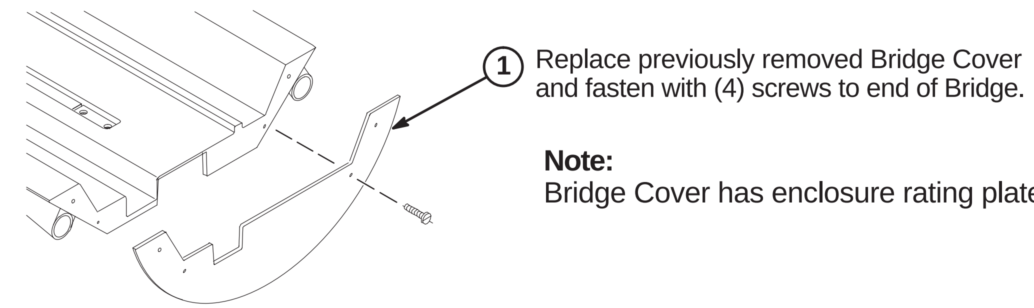
❑ J3 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:
The following Step is needed if flag was removed during routing of drive belt.



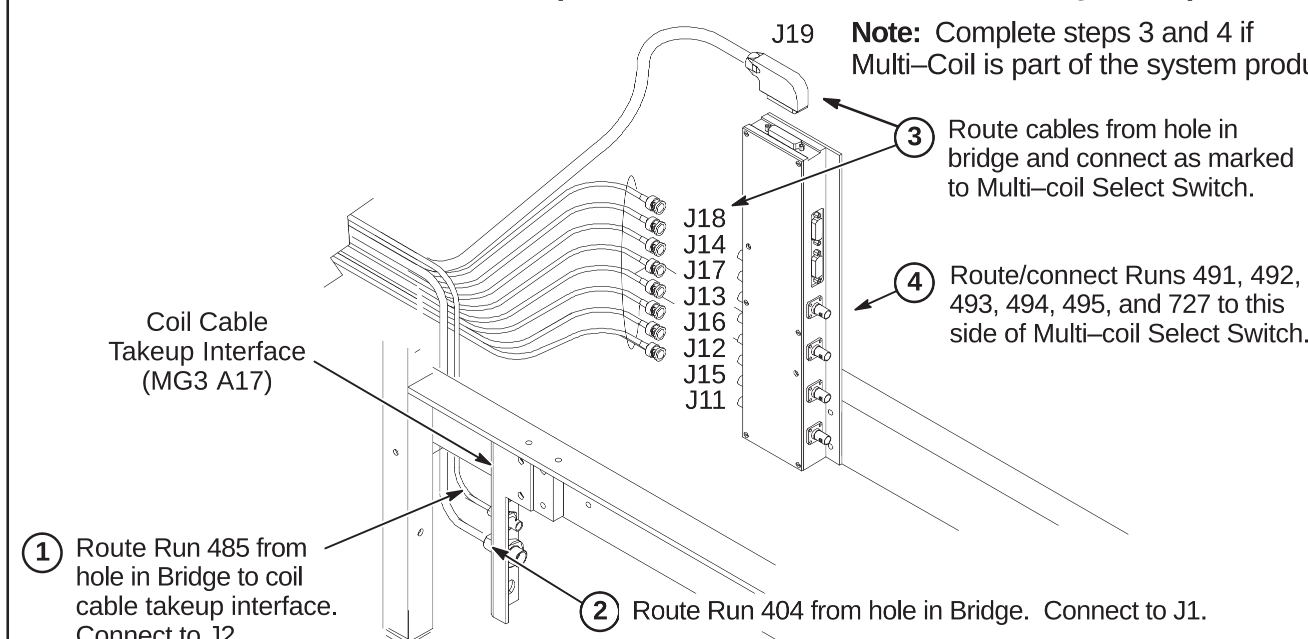
❑ J4 – ATTACH BRIDGE COVER



Note:
Bridge Cover has enclosure rating plates on it.

❑ J5 – ROUTE RUNS 404 AND 485 (AND MULTI–COIL CABLES IF REQUIRED)

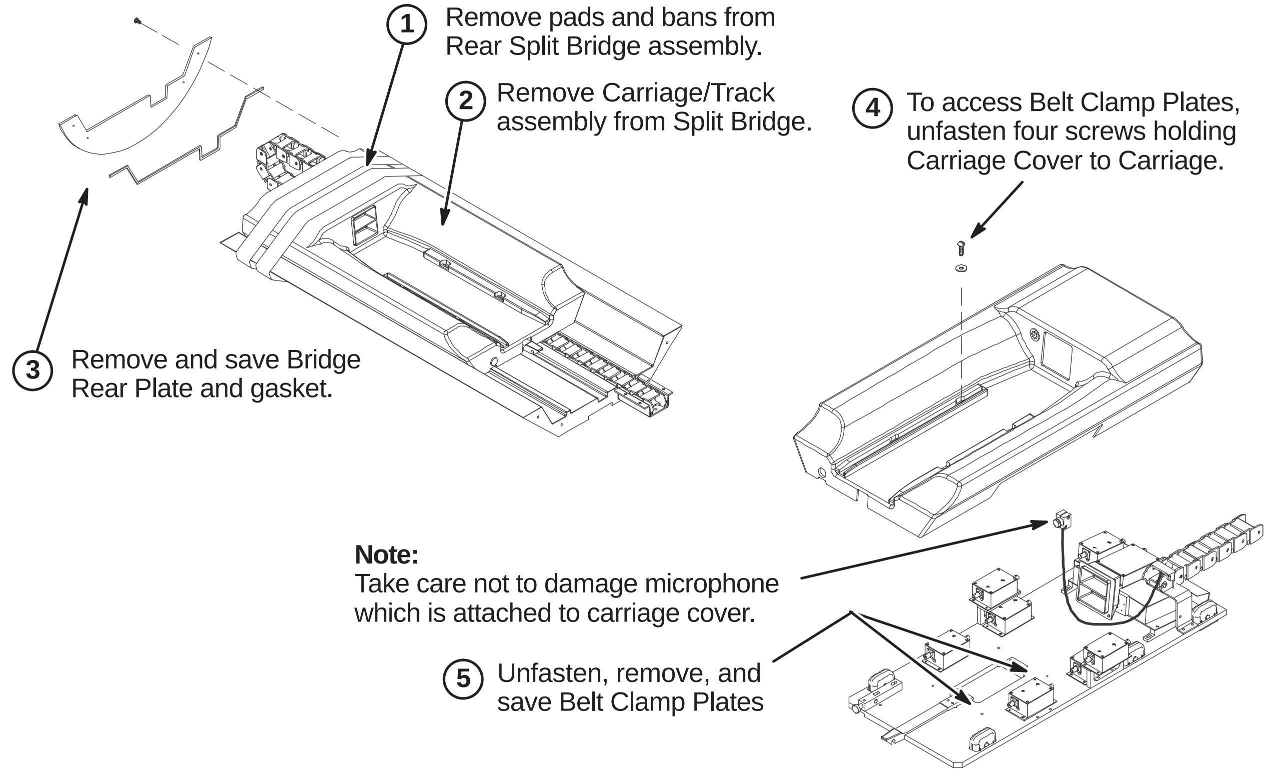
Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ K1 – Preparing Rear Split Bridge for Installation
- ❑ K2 – Attaching Rear Split Bridge to Rear Pedestal
- ❑ K3 – Attaching Rear Split Bridge to Front Split Bridge
- ❑ K4 – Aligning Bridge on Rear Pedestal
- ❑ K5 – Installing Bridge Locators

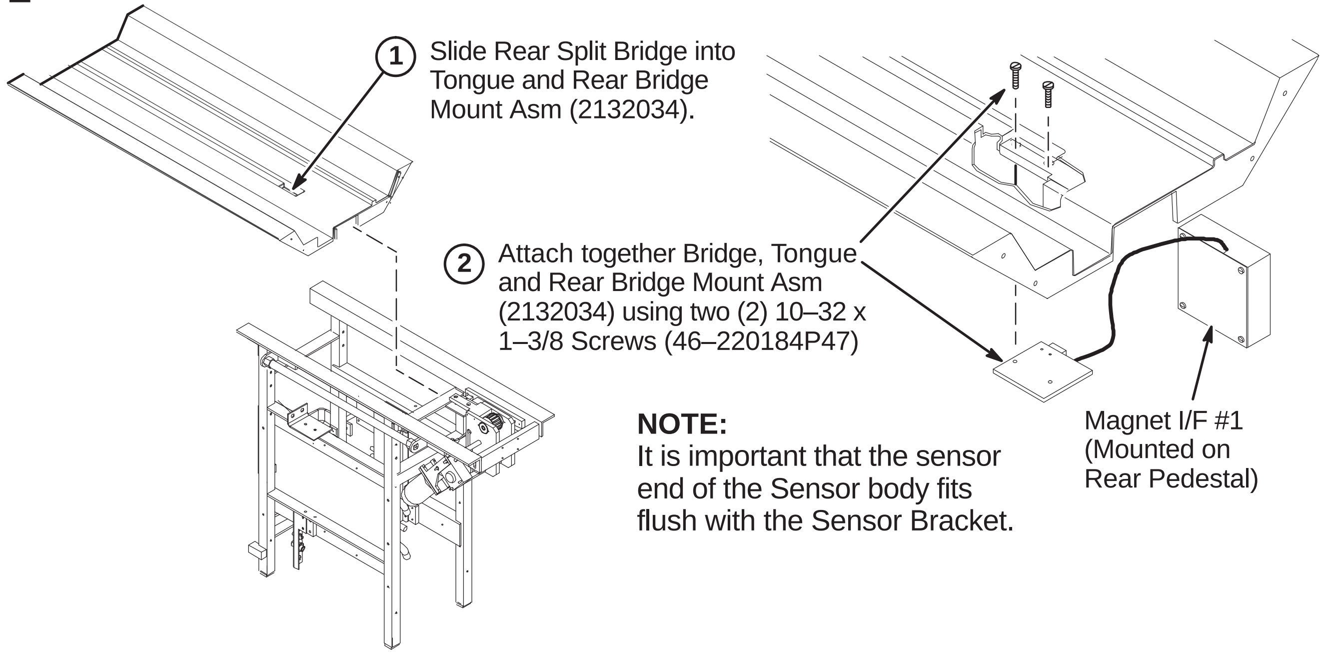
❑ K1 – PREPARING REAR SPLIT BRIDGE FOR INSTALLATION



1. Remove pads and bans from Rear Split Bridge assembly.
2. Remove Carriage/Track assembly from Split Bridge.
3. Remove and save Bridge Rear Plate and gasket.
4. To access Belt Clamp Plates, unfasten four screws holding Carriage Cover to Carriage.
5. Unfasten, remove, and save Belt Clamp Plates

Note:
Take care not to damage microphone which is attached to carriage cover.

❑ K2 – ATTACHING REAR SPLIT BRIDGE TO REAR PEDESTAL



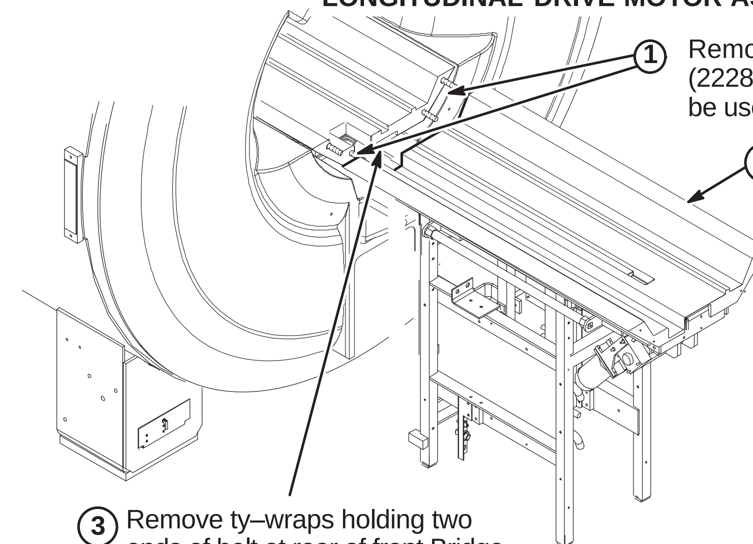
1. Slide Rear Split Bridge into Tongue and Rear Bridge Mount Asm (2132034).
2. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) 10–32 x 1–3/8 Screws (46–220184P47)

Magnet I/F #1 (Mounted on Rear Pedestal)

NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.

❑ K3 – ATTACHING REAR SPLIT BRIDGE TO FRONT SPLIT BRIDGE

WARNING! **THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.**

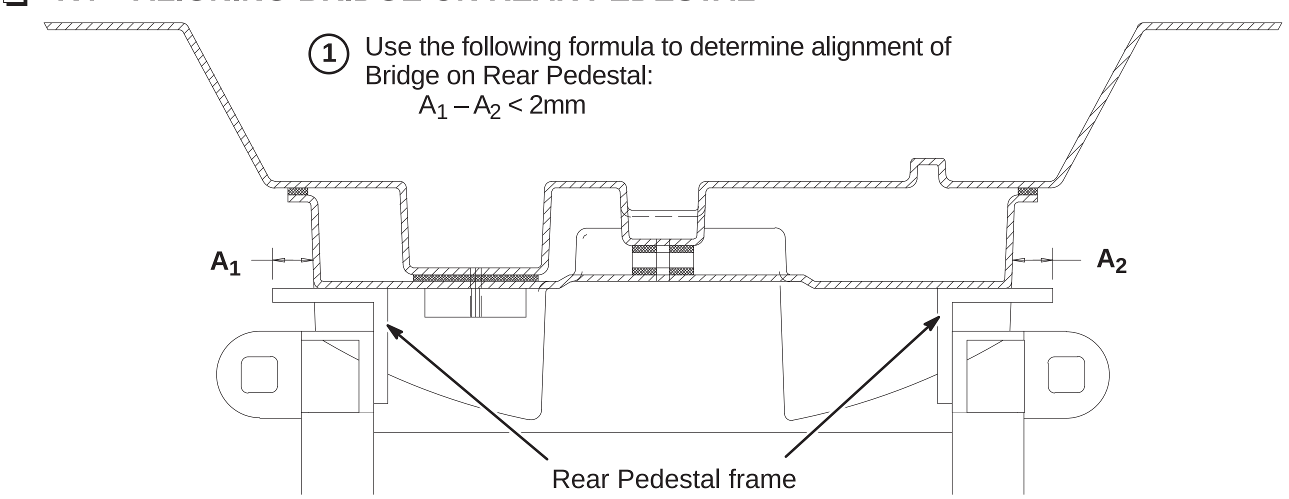


1. Remove four 10M hex nuts (2109877–3) and washers (2228717–2) from end of Bridge in magnet. These will be used for attaching the Rear Pedestal.
2. Move Rear Pedestal/Bridge into position and fasten with four nuts and washers that were removed form end of Front Split Bridge.
3. Remove ty–wraps holding two ends of belt at rear of front Bridge.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

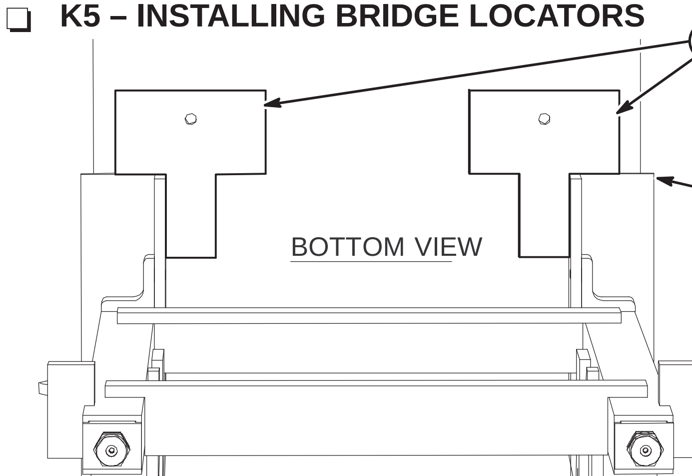
❑ K4 – ALIGNING BRIDGE ON REAR PEDESTAL

1. Use the following formula to determine alignment of Bridge on Rear Pedestal:
   $A_1 – A_2 < 2mm$



❑ K5 – INSTALLING BRIDGE LOCATORS

1. After Bridge and Rear Pedestal are aligned, peel off protective paper on each Bridge Locator and attach to bottom side of Bridge as shown.

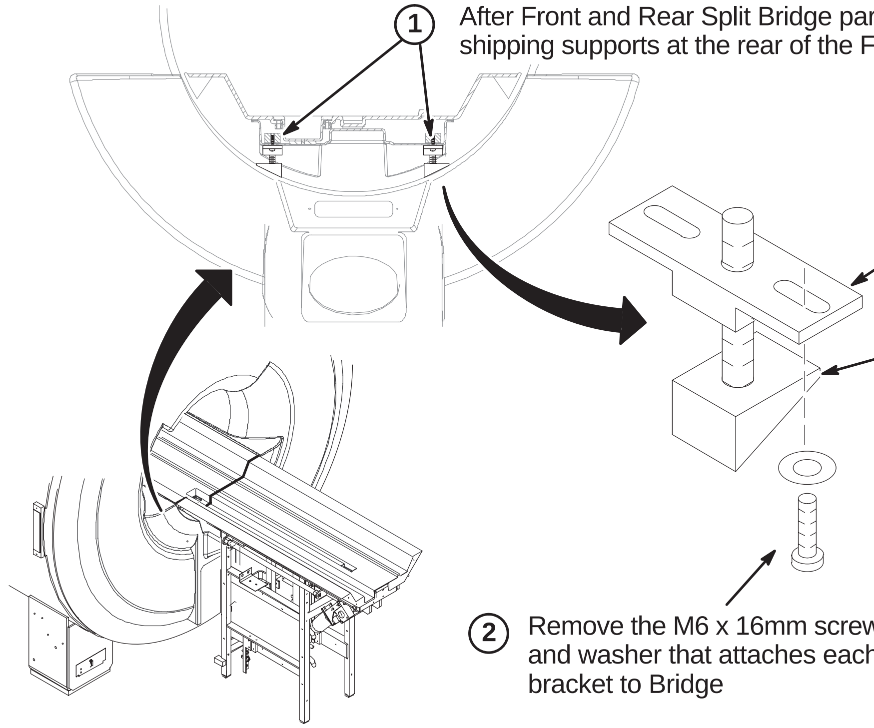


End of Rear Pedestal frame closest to Magnet.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

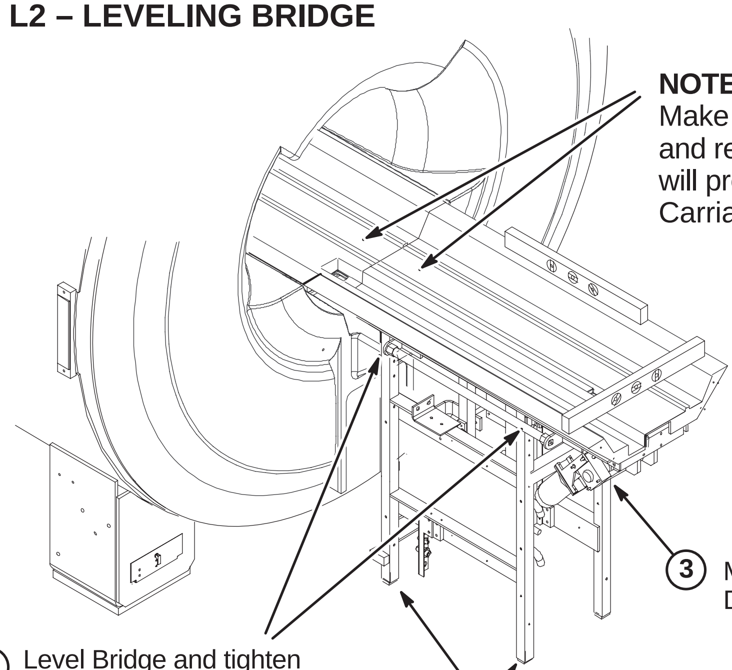
- ❑ L1 – Removing Front Split Bridge Shipping Support Brackets
- ❑ L2 – Leveling Bridge
- ❑ L3 – Height Adjustment at Front of Bridge (if needed)
- ❑ L4 – Installing Carriage on Bridge
- ❑ L5 – Routing of Drive Belt

❑ L1 – REMOVING FRONT SPLIT BRIDGE SHIPPING SUPPORT BRACKETS



1. After Front and Rear Split Bridge parts are fastened together, the two shipping supports at the rear of the Front Split Bridge need to be removed.
2. Remove the M6 x 16mm screw and washer that attaches each bracket to Bridge
3. Unscrew downword the Bridge Mounting Flange.
4. Support Base is attached to End Bell with adhesive tape. To remove, carefully pry base away from end bell. Remove base, stud, and mounting flange

❑ L2 – LEVELING BRIDGE

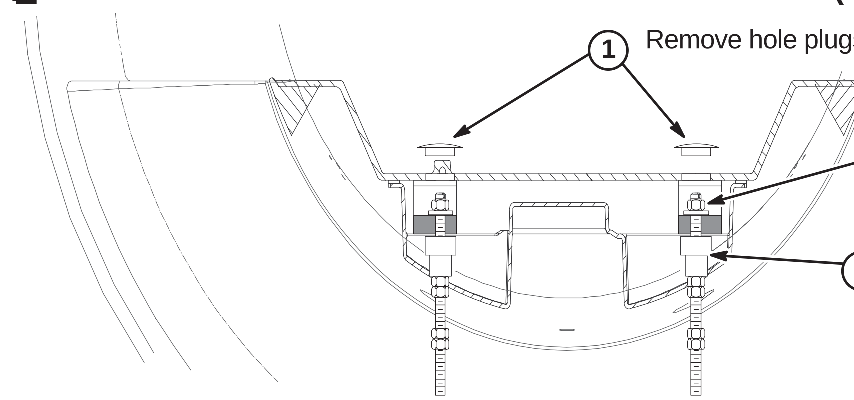


NOTE:
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:
Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

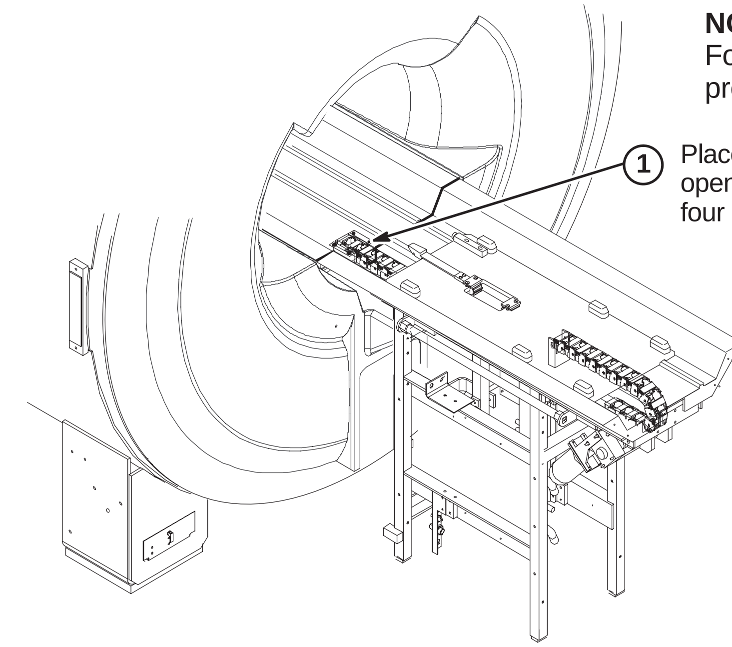
1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

❑ L3 – HEIGHT ADJUSTMENT AT FRONT OF BRIDGE (IF NEEDED)



1. Remove hole plugs from each side of Bridge.
2. Remove M10 nut and washer from each stud.
3. Adjust Support Nut (2293381) as needed to achieve correct height.

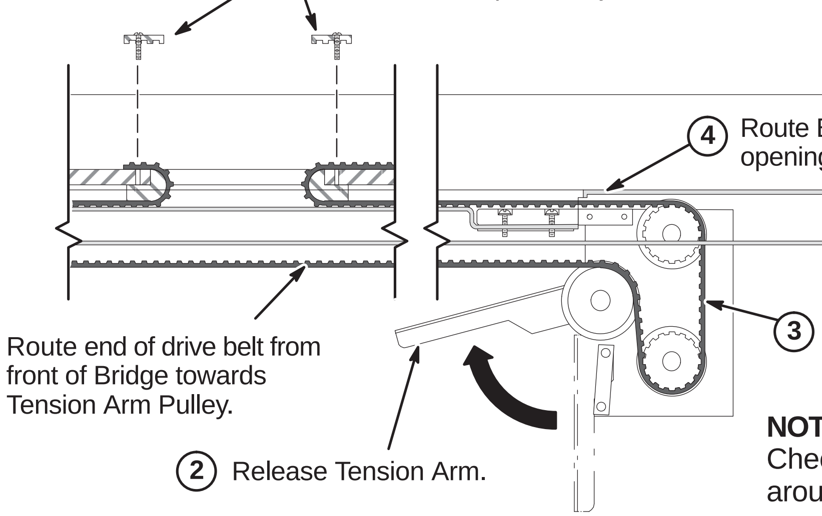
❑ L4 – INSTALLING CARRIAGE ON BRIDGE



NOTE:
For clarity, the carriage is shown without preamps, preamp protect module, and cables.

1. Place Carriage on Bridge and route all cables thru opening in Bridge and fasten plate to Front Bridge with four M6 x 12mm nylon screws (2296958) provided.

❑ L5 – ROUTING OF DRIVE BELT



1. Route end of drive belt from front of Bridge towards Tension Arm Pulley.
2. Release Tension Arm.
3. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
4. Route Belt under and through opening to top of Carriage.
5. Position Belt under Clamp Plates and re–attach with previously removed screws.

NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

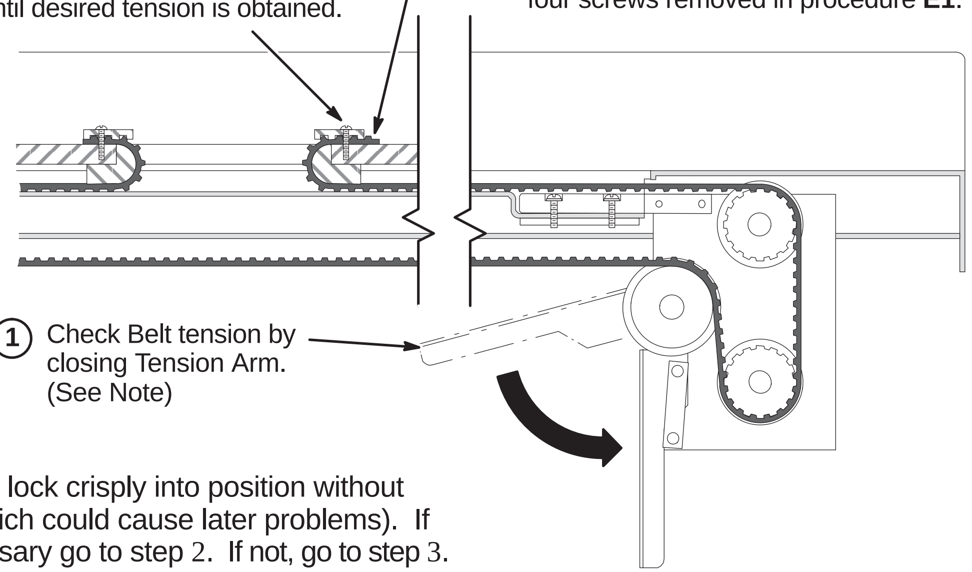
- ❑ **M1 – Final Adjustment of Drive Belt**
- ❑ **M2 – Installation of Sensor Flag**
- ❑ **M3A – Route Non–MGD Runs 404 and 485 (and Multi–coil Cables if Required)**
  OR
- ❑ **M3B – Route/Connect MGD Cables from LPCA to Rear Pedestal**
- ❑ **M4 – Installing Carriage Cover to Trolley and Bridge Cover**

❑ M1 – FINAL ADJUSTMENT OF DRIVE BELT

(2) **(IF NECESSARY):** Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.

(3) Tighten Clamps and trim excess Belt length.

(4) Fasten Carriage Cover to Carriage with four screws removed in procedure **E1**.



(1) Check Belt tension by closing Tension Arm. (See Note)

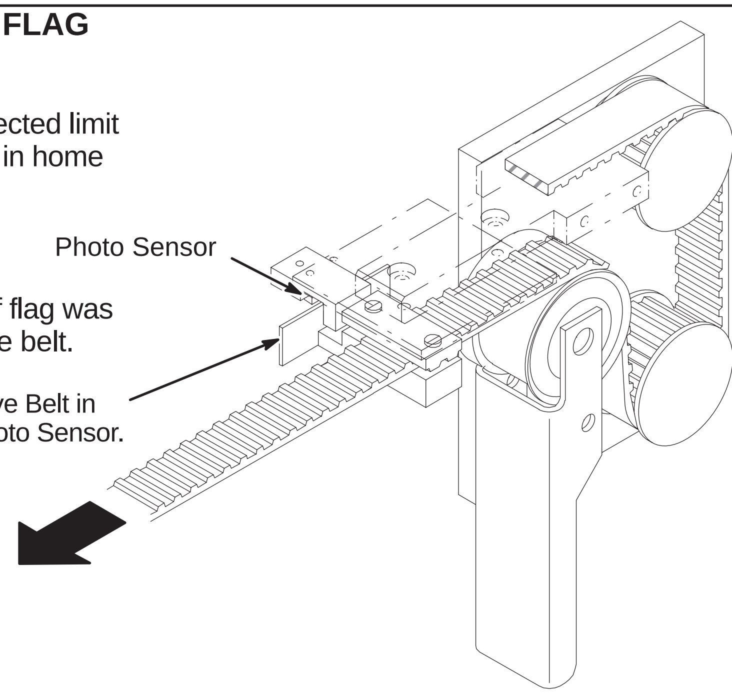
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

❑ M2 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

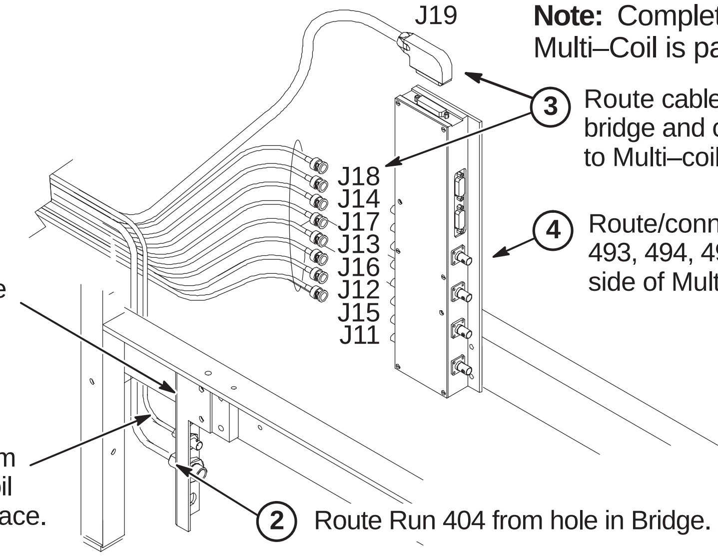
NOTE:
The following Step is needed if flag was removed during routing of drive belt.

(1) Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



❑ M3A – ROUTE NON–MGD RUNS 404 AND 485 (& Multi–Coil Cables if Required)

Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



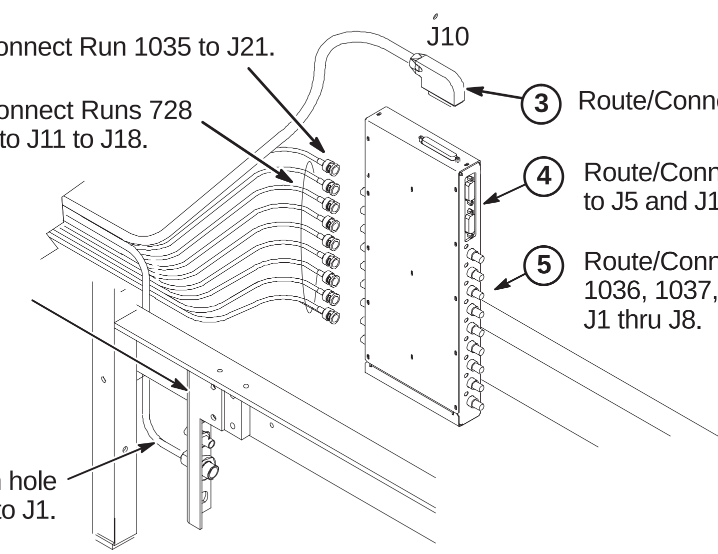
(3) Route cables from hole in bridge and connect as marked to Multi–coil Select Switch.

(4) Route/connect Runs 491, 492, 493, 494, 495, and 727 to this side of Multi–coil Select Switch.

(1) Route Run 485 from hole in Bridge to coil cable takeup interface. Connect to J2.

(2) Route Run 404 from hole in Bridge. Connect to J1.

❑ M3B – ROUTE/CONNECT MGD CABLES FROM LPCA TO REAR PEDESTAL



(1) Route/Connect Run 1035 to J21.

(2) Route/Connect Runs 728 thru 735 to J11 to J18.

(3) Route/Connect Run 1041 to J19.

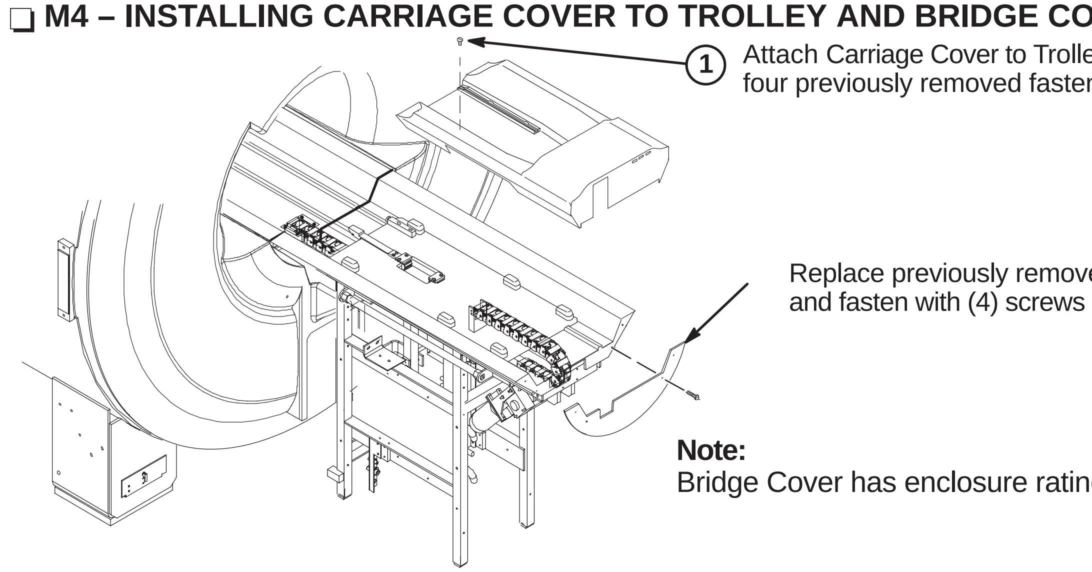
(4) Route/Connect Runs 1042 and 1049 to J5 and J19.

(5) Route/Connect Runs 491, 492, 493, 1036, 1037, 1038, 1039, and 1040 to J1 thru J8.

(6) Route Run 404 from hole in Bridge. Connect to J1.

❑ M4 – INSTALLING CARRIAGE COVER TO TROLLEY AND BRIDGE COVER

(1) Attach Carriage Cover to Trolley using four previously removed fasteners.



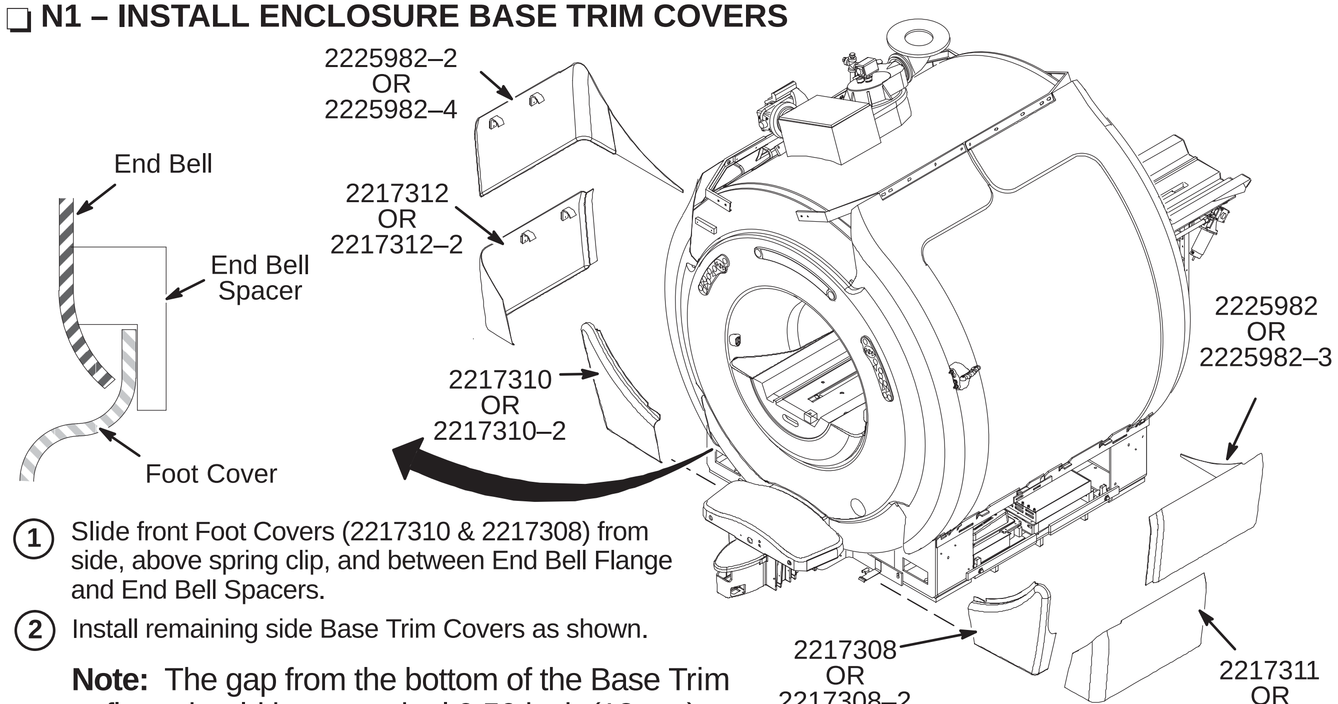
Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

Note:
Bridge Cover has enclosure rating plates on it.

INSTALLATION PROCEDURES SUMMARY

- ❑ **N1 – Install Enclosure Base Trim Covers**
- ❑ **N2 – Install Enclosure Top Covers**
- ❑ **N3 – Install Rear Pedestal Covers**

❑ N1 – INSTALL ENCLOSURE BASE TRIM COVERS



1. Slide front Foot Covers (2217310 & 2217308) from side, above spring clip, and between End Bell Flange and End Bell Spacers.
2. Install remaining side Base Trim Covers as shown.

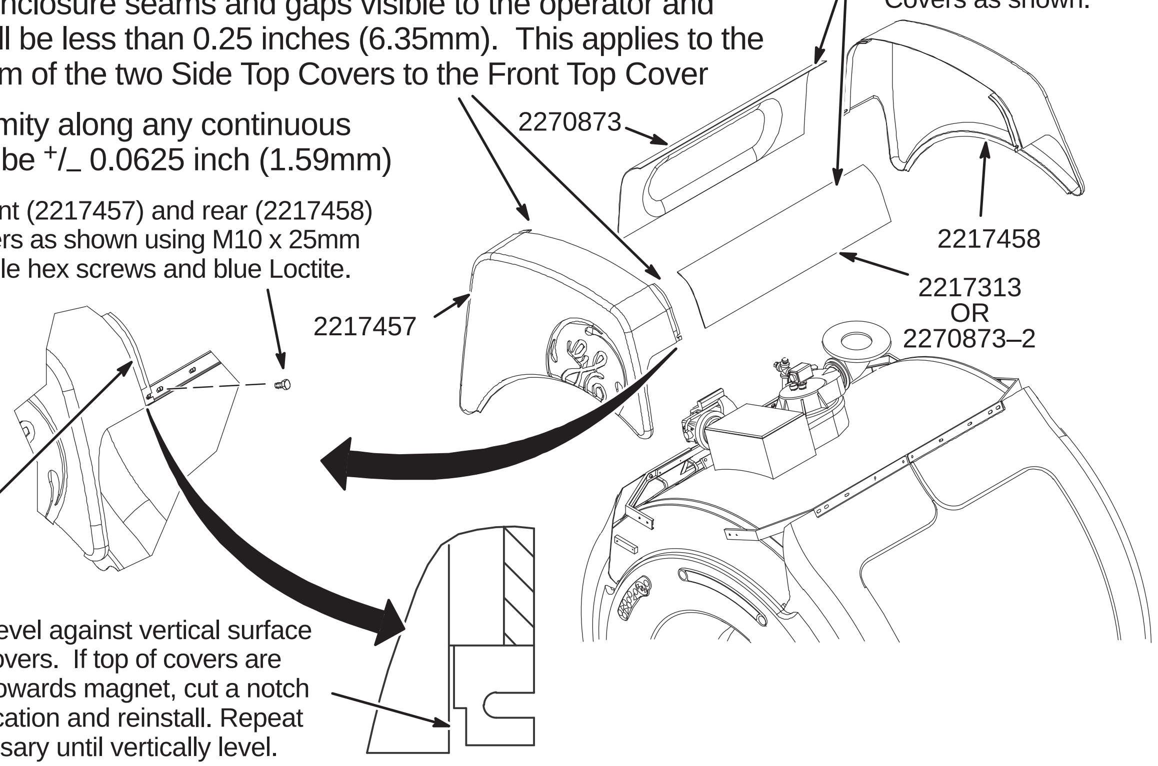
Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

❑ N2 – INSTALL ENCLOSURE TOP COVERS

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be $^{+}/_{-}$ 0.0625 inch (1.59mm)

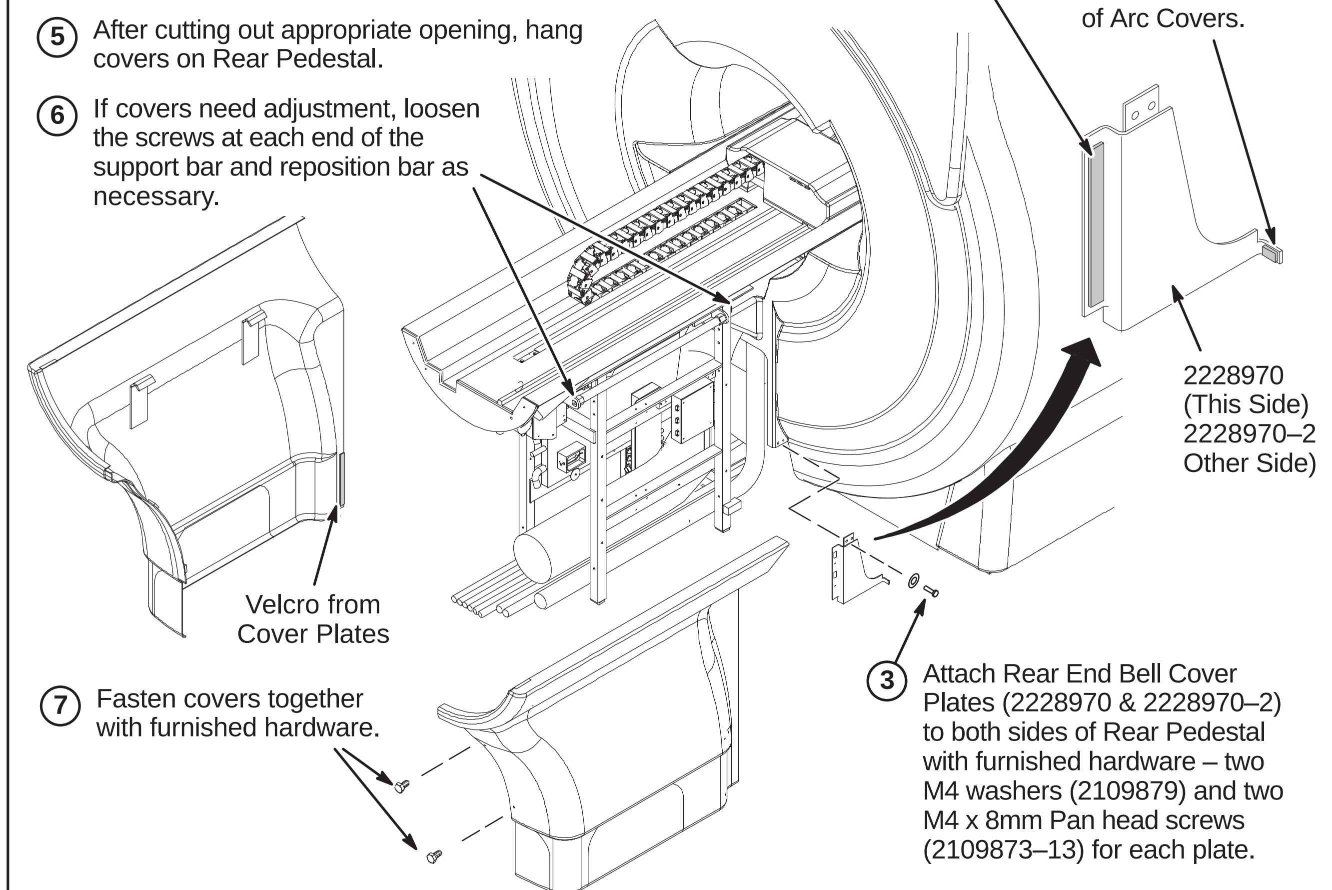
1. Install front (2217457) and rear (2217458) Top Covers as shown using M10 x 25mm Low Profile hex screws and blue Loctite.
2. Install side Top Covers as shown.



2. Place a level against vertical surface of Top Covers. If top of covers are leaning towards magnet, cut a notch at this location and reinstall. Repeat as necessary until vertically level.

❑ N3 – INSTALL REAR PEDESTAL COVERS

1. Remove mating piece of velcro from Rear End Bell Cover Plates (2228970 & 2228970–2) and attach to Rear Pedestal Covers
2. Remove mating velcro and attach to back side of Arc Covers.
3. Attach Rear End Bell Cover Plates (2228970 & 2228970–2) to both sides of Rear Pedestal with furnished hardware – two M4 washers (2109879) and two M4 x 8mm Pan head screws (2109873–13) for each plate.
4. Pedestal Covers (2217704 & 2217705) require an opening to be cut out depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting.
5. After cutting out appropriate opening, hang covers on Rear Pedestal.
6. If covers need adjustment, loosen the screws at each end of the support bar and reposition bar as necessary.
7. Fasten covers together with furnished hardware.



GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.
- A 1–ounce tube of Permatex "Anti–Seize" lubricant (2119594) is provided. Follow instructions on back of tube. <u>"Anti–Seize" lubricant is to be applied to threads of stainless–steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.</u>

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV
1	5	8	2
2	5	9	2
3	5	10	5
4	2	11	2
5	5	12	3
6	2	13	4
7	2		

BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.
2. If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.

WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.

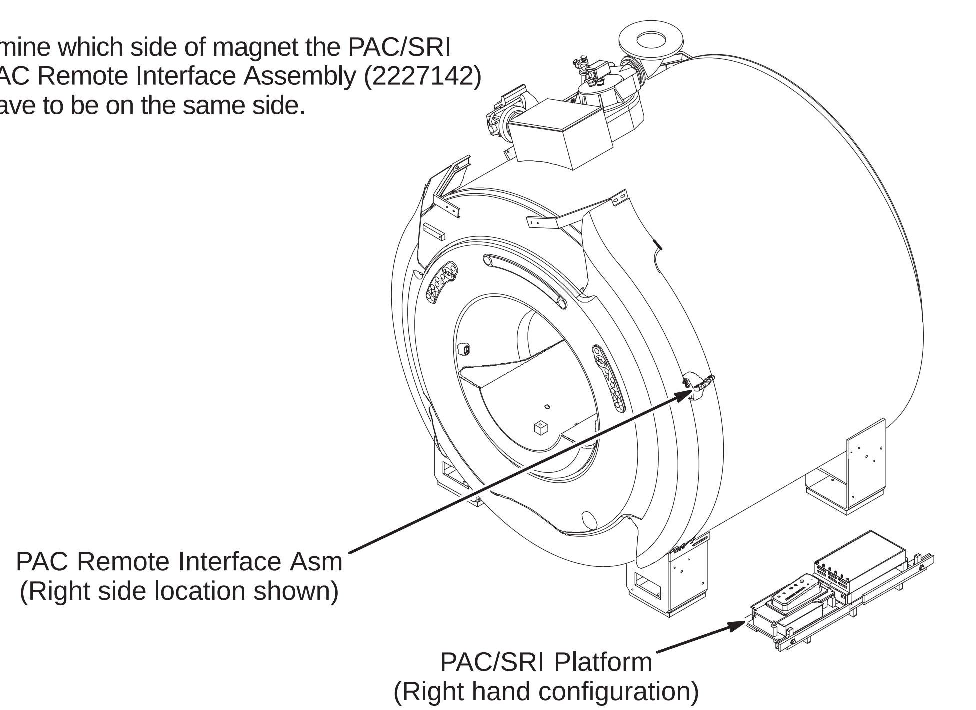
IF MAGNET IS AT FULL FIELD – KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

INSTALLATION PROCEDURES SUMMARY

- ❑ **A1 – PAC/SRI Platform and PAC Remote Interface Location**
- ❑ **A2 – PAC/SRI Platform Left Hand Configuration (If Required)**
- ❑ **A3 – Location of Magnet Ground Cable**

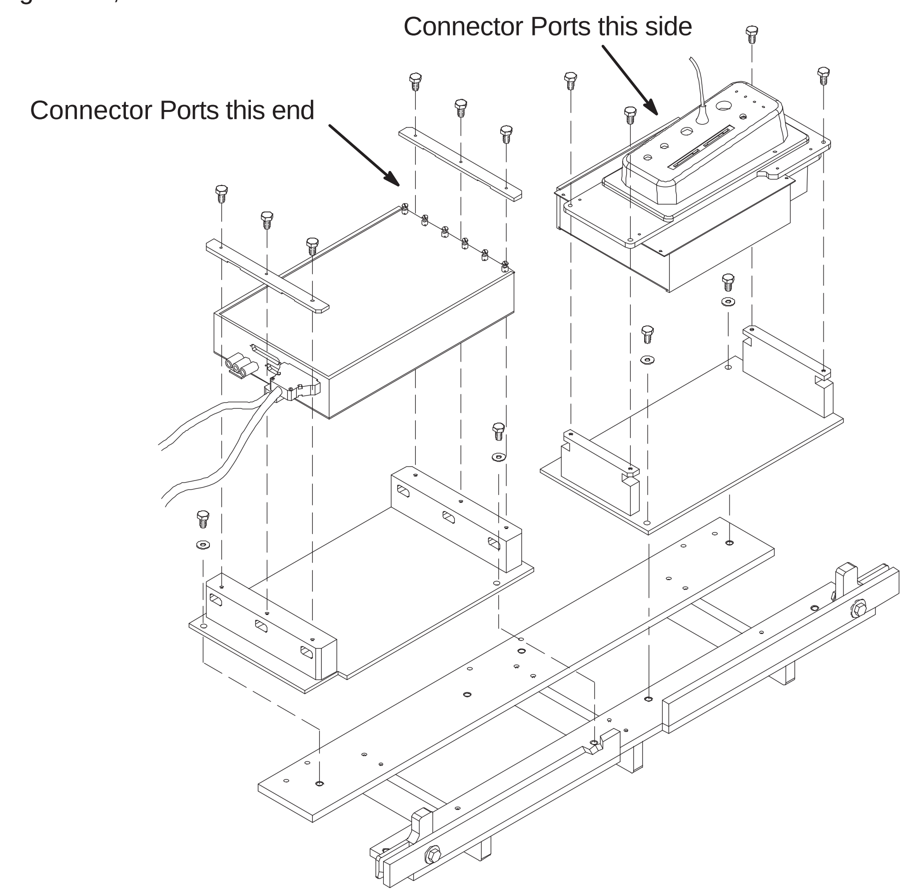
❑ A1 – PAC/SRI PLATFORM AND PAC REMOTE INTERFACE LOCATION

(1) Van manufacturer to determine which side of magnet the PAC/SRI Platform (2223440) and PAC Remote Interface Assembly (2227142) should be located. They have to be on the same side.

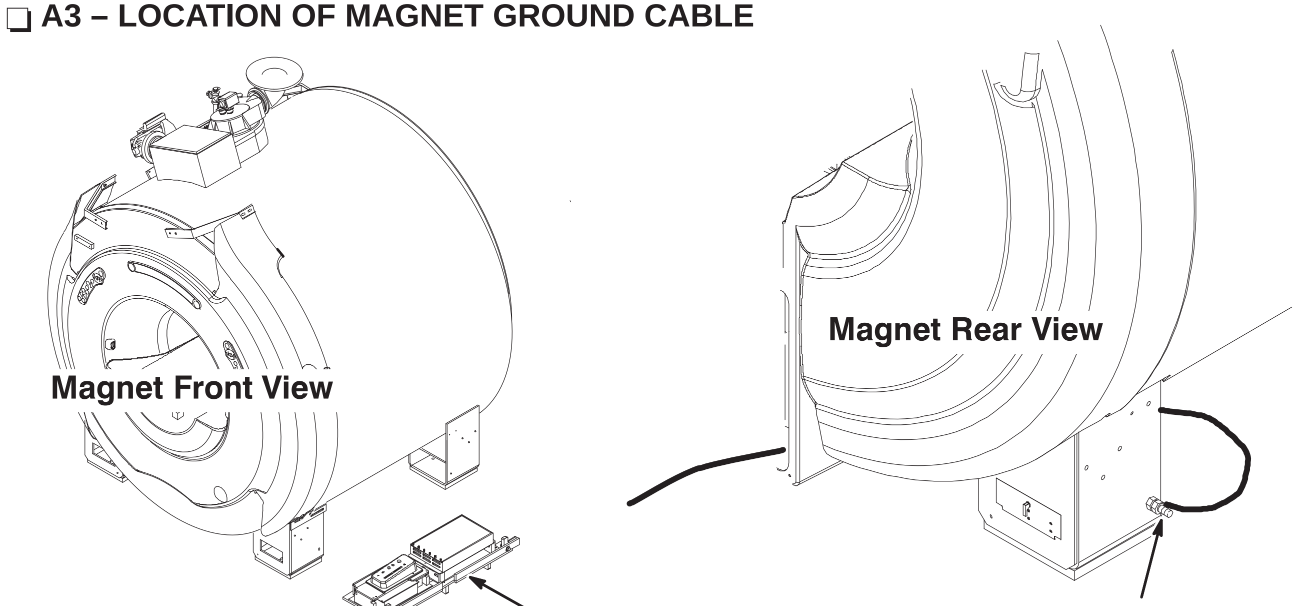


❑ A2 – PAC/SRI PLATFORM LEFT HAND CONFIGURATION (IF REQUIRED)

(1) For left hand configuration, reassemble PAC/SRI as shown.



❑ A3 – LOCATION OF MAGNET GROUND CABLE

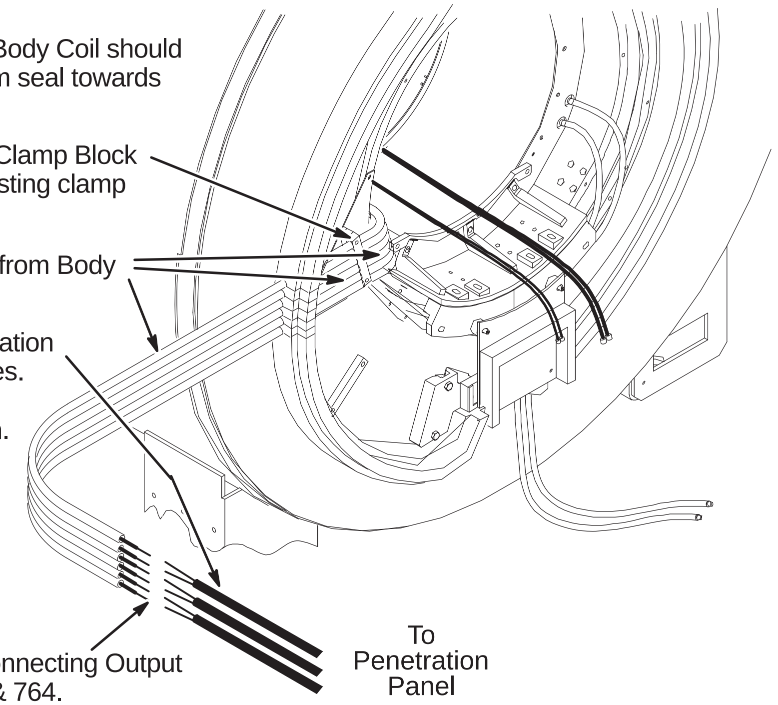


(1) If PAC/SRI Platform is located on right side of magnet, then the Magnet ground cable , RUN 040, has to be connected to the opposite side rear left leg as shown above. The other end of Run 040 is connected to the RF Screen Room common ground.

INSTALLATION PROCEDURES SUMMARY

- ❑ **B1 – Routing and Connecting Gradient Output Cables to Runs 762, 763, and 764 (If Needed)**
- ❑ **B2 – Locating Rear Pedestal and Connecting Run 318 & I and Q Drive Cables**
- ❑ **B3 – Route Cables & Hoses, Connect Ground Cable, and Install Coil Cable Takeup Interface**
- ❑ **B4 – Route/Connect Runs 257/746, 262, 743, 744, and 780**

❑ B1 – ROUTING AND CONNECTING GRADIENT OUTPUT CABLES (If Needed)



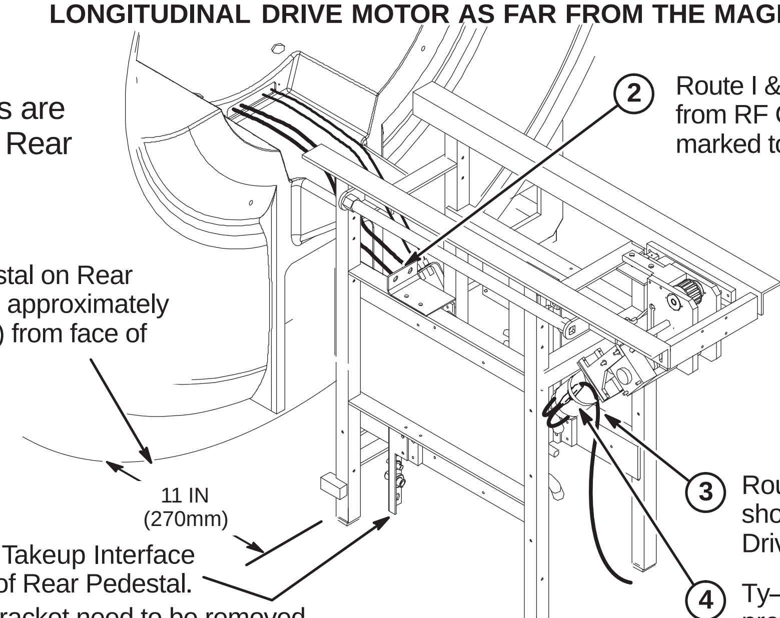
1. The (6) six gradient Output Cables from Body Coil should have been routed through slits in the foam seal towards wall and secured to wall.
2. Hold cables in place with 6 slot Gradient Clamp Block (2214162). This is mounted on top of existing clamp block using longer screws.
3. Around each cable install pipe insulation from Body Coil to Penetration Panel.
4. Route Runs 762, 763, & 764 from Penetration Panel to ends of Body Coil gradient cables. Make sure that there is at least two feet (610mm) of slack before cutting to length.
5. Refer to Direction 2286976 in Tab 1 for connecting Output cables from Body Coil to Runs 762, 763, & 764.

❑ B2 – LOCATING REAR PEDESTAL AND CONNECTING RUN 318 & DRIVE CABLES

WARNING!

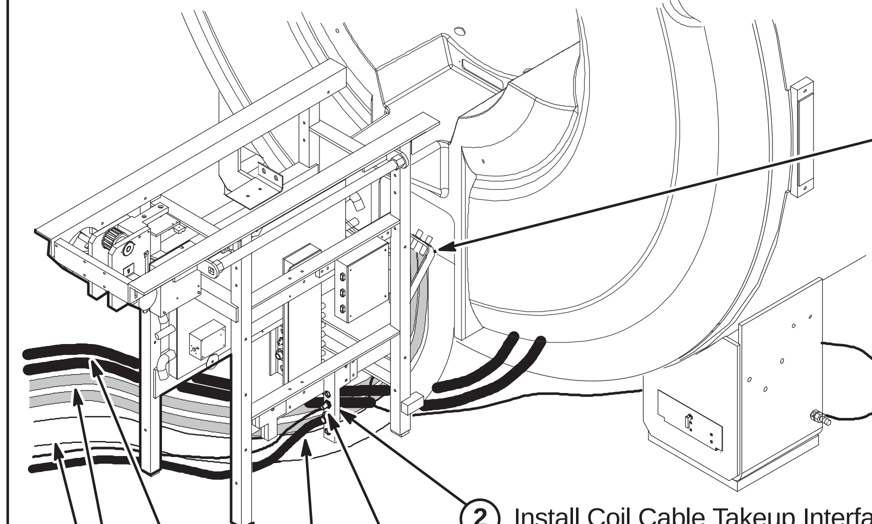
THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

Note:
Cables and hoses are to exit this side of Rear Pedestal



1. Center Rear Pedestal on Rear Endbell and locate approximately 11 inches (270mm) from face of magnet.
2. Route I & Q RF Drive Input cables from RF Coil and connect as marked to J1 and J2 on bracket.
3. Route/connect Run 318 to short cable on Longitudinal Drive Motor (MG3 A7).
4. Ty–wrap cable to motor to prevent movement.
5. Move Coil Cable Takeup Interface to opposite side of Rear Pedestal.
6. Connectors on bracket need to be removed, turned end for end, and reinstalled.

❑ B3 – ROUTE CABLES & HOSES, CONNECT GROUND CABLE, AND INSTALL TAKEUP



Note: For CV/i, install bracket on other side of Reat Pedestal.

1. Route Ground cable from Rear Pedestal under magnet and connect to same Ground Stud as Magnet ground.
2. Install Coil Cable Takeup Interface with mounting block. Make sure connectors have been turned end for end.
3. Route/connect Run 700 to J2 on Coil Cable Takeup Interface.
4. Route/connect to J1 on Coil Cable Takeup Interface: Run 258 (1.5T, 1/2") or Run 747 (1.0T, 3/8")
5. Route and connect coolant hoses from Heat Exchanger to 10 ft. (300cm) coolant hoses attached to Body Coil.
6. Route water lines, gas lines, and 2" air hose under Rear Pedestal as shown.
7. Route all other cables on top of the above cables and hoses.

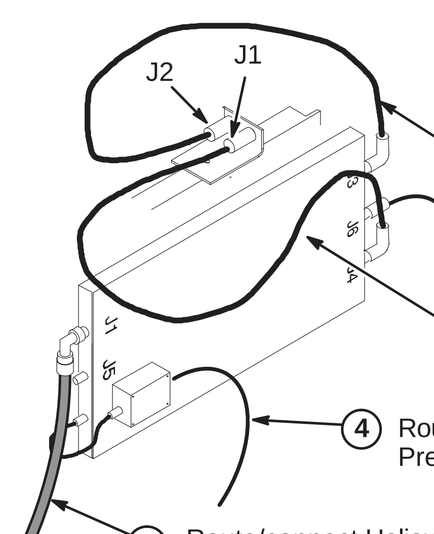
❑ B4 – ROUTE/CONNECT RUNS 257/746, 262, 743, 744, AND 780

CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corregated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

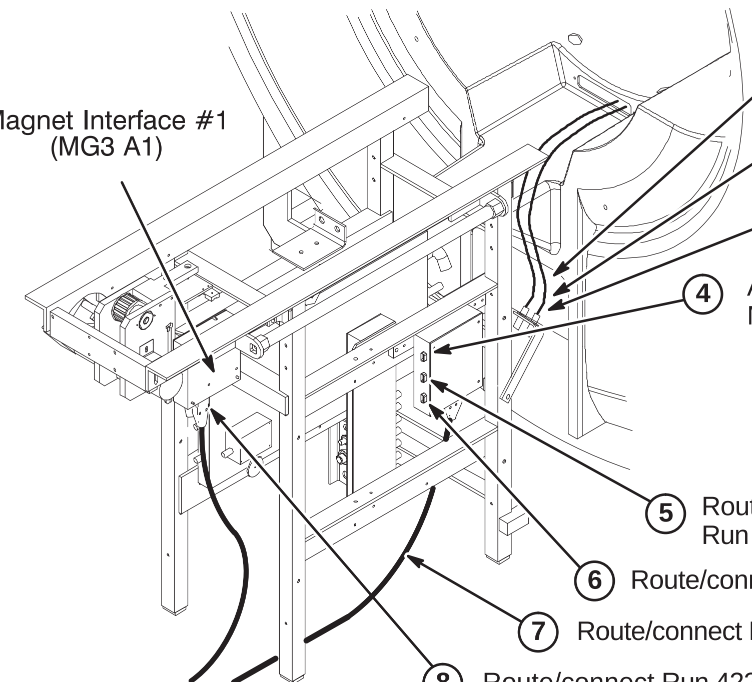


1. Route and connect Run 744 from J2 on bracket to J4 on the RF Splitter.
2. Route/connect Run 780 from J74 on Pen Panel to J6 on RF Splitter.
3. Route and connect Run 743 from J1 on the bracket to J3 on the RF Splitter.
4. Route/connect Run 262 from J8 on Pen Panel to Body Coil Preamplifier. (For MGD, connect to Receive Filter on Preamp)
5. Route/connect Heliax® to J1 on RF Splitter: Run 257 (1.5T, 7/8") or Run 746 (1.0T, 1/2")

INSTALLATION PROCEDURES SUMMARY

- ❑ **C1 – Route/Connect Runs 282, 414, 422, 781, 782, and Front End Bell Speaker Cable**
- ❑ **C2 – Install Bore Lamp Assembly**
- ❑ **C3 – Route/Connect Runs 257/746, 262, 743, 744, and 780**
- ❑ **C4 – Install Blower Box and Route/Connect Air Hoses**

❑ C1 – ROUTE/CONNECT RUNS 282, 414, 422, 781, 782, & FRONT END BELL CABLE

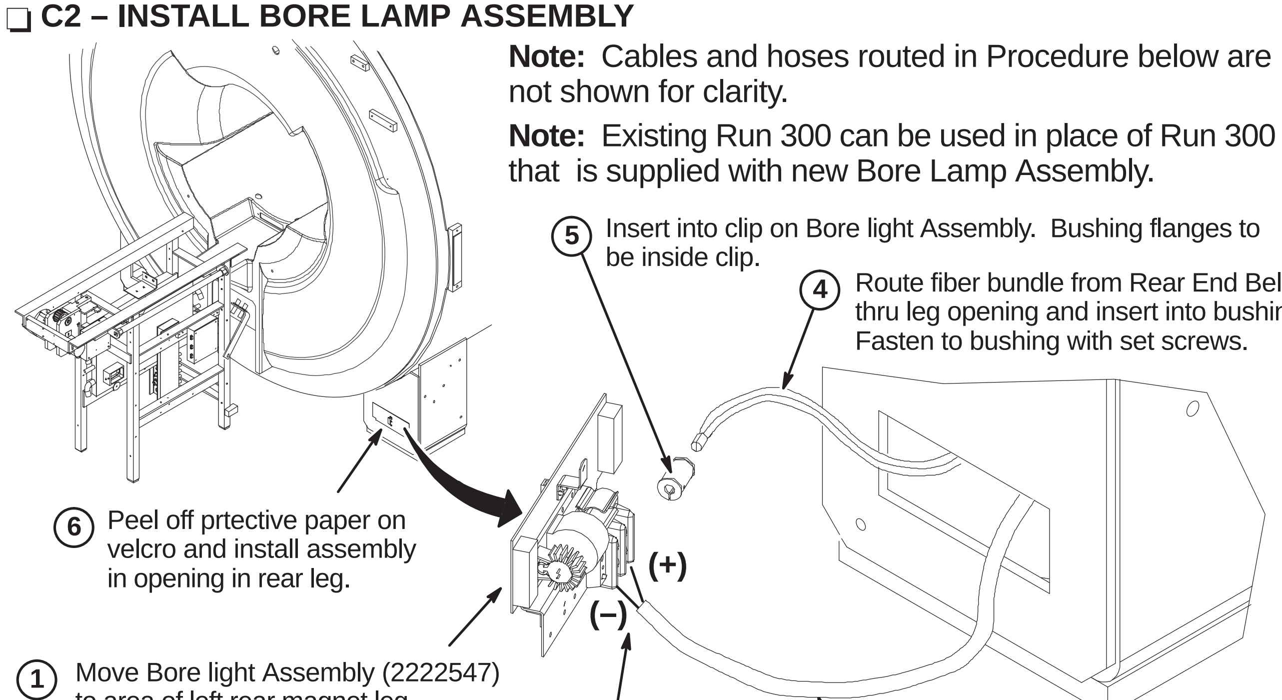


1. Route and connect RF Bias cables from RF Coil to bracket as marked.
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. After Carriage cover is installed, connect Microphone cable from Carriage Cover to J7.
5. Route/connect Front End Bell microphone cable Run 414 to J9.
6. Route/connect speaker cable from Front End Bell to J8.
7. Route/connect Run 282 from J12 on Pen Panel to J6.
8. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.

❑ C2 – INSTALL BORE LAMP ASSEMBLY

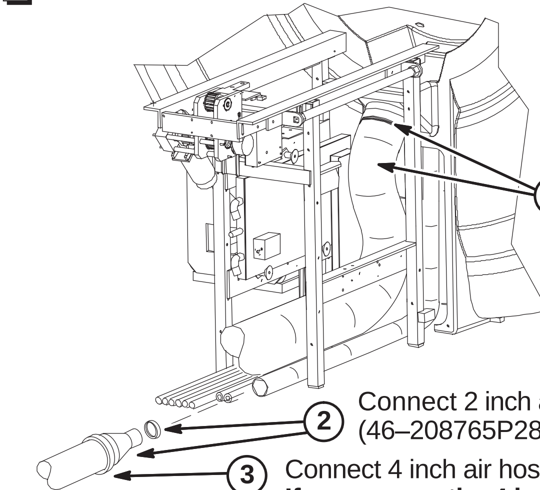
Note: Cables and hoses routed in Procedure below are not shown for clarity.

Note: Existing Run 300 can be used in place of Run 300 that is supplied with new Bore Lamp Assembly.



1. Move Bore light Assembly (2222547) to area of left rear magnet leg.
2. Connect the (+) and (–) wires of Run 300 to the terminal strip as shown.
3. Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
4. Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
5. Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
6. Peel off prtective paper on velcro and install assembly in opening in rear leg.

❑ C3 – CONNECT AIR HOSES TO REAR OF MAGNET



Note:
The following items are part of Blower Hose Kit, 2129412–3 or 2129412–4.

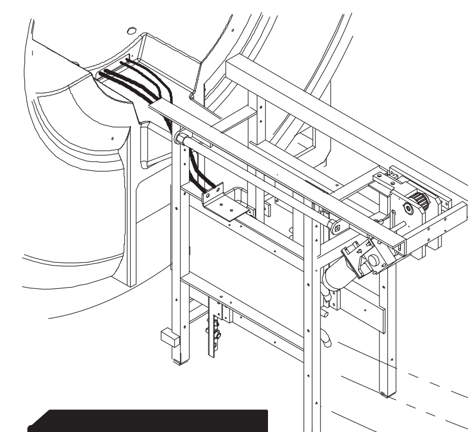
1. Connect 6 inch air hose (46–282666P12) to Rear End Bell with 6 inch clamp (46–208675P29).
2. Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46–208765P28). **Do not cut or reduce the length of the 2 inch hose.**
3. Connect 4 inch air hose (46–282666P11) to reducer with 4 inch clamp (46–208765P34). **If necessary, the 4 inch hose may need to be cut to reduce length.**

WARNING!

THE BLOWER BOX IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE. TO LOCATE THE 200 GAUSS ZONE, REFER TO THE TABLE IN SECTION 1–1 INTRODUCTION.

❑ C4 – INSTALL BLOWER BOX AND ROUTE/CONNECT AIR HOSES

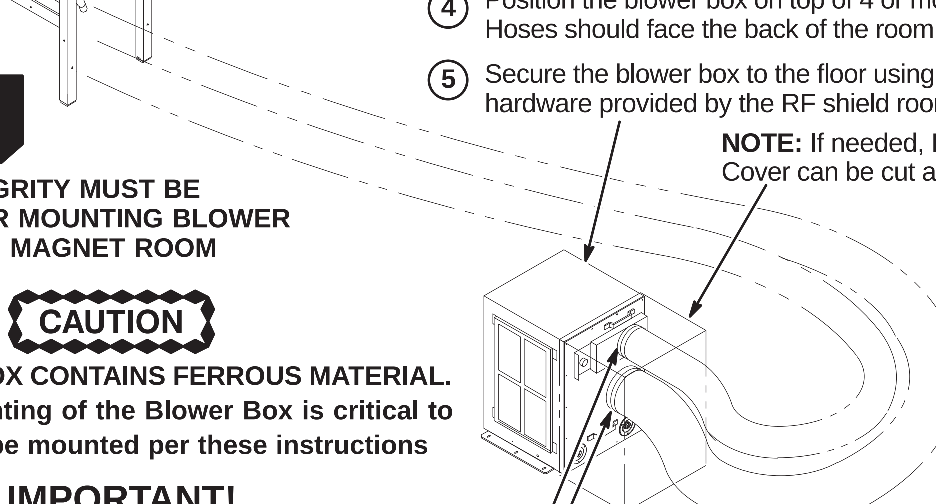
1. Refer to site design (architectural)/customer construction drawings or contact PMI for locating and securing Blower Box (2121199).
2. Check for correct installation of the anchors:
   - Confirm that the anchors are physically secured to the building structure, not to a removable computer floor, raised floor, or false floor.
   - Confirm there are at least four (4) anchors.
   - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.
3. Remove any mounting hardware from the anchors.
4. Position the blower box on top of 4 or more anchors. Hoses should face the back of the room
5. Secure the blower box to the floor using the fastening hardware provided by the RF shield room vendor.



WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

NOTE: If needed, Blower Hose Cover can be cut and modified.



CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions

IMPORTANT!

Location of cabinet and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural/Construction) drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

6. Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
   (1) 46–208765P34 4 inch Hose Clamp
   (1) 46–208765P29 6 inch Hose Clamp

INSTALLATION PROCEDURES SUMMARY

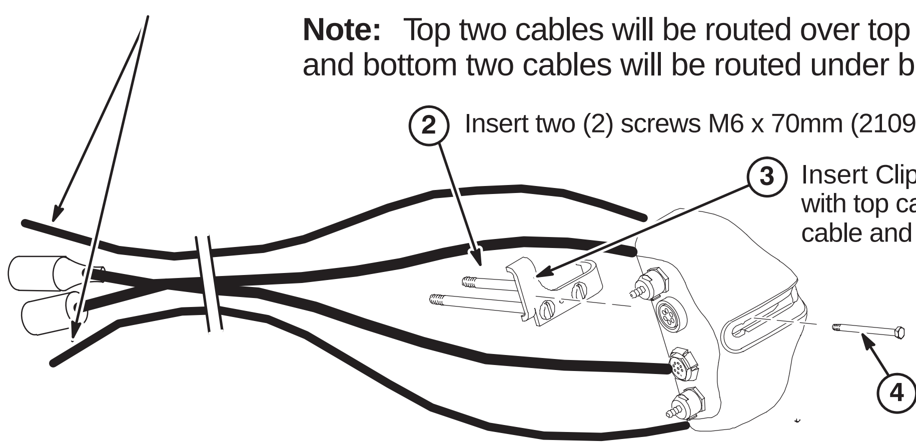
- ❑ **D1 – Prepare Cables and Tubes for Routing**
- ❑ **D2 – Position Remote PAC Interface Mounting Clip**
- ❑ **D3 – Route Remote PAC Interface Cables and Tubes**
- ❑ **D4 – Installing Remote PAC Interface to Arc Cover**
- ❑ **D5 – Install PAC Hanger Bar**

❑ D1 – PREPARE CABLES AND TUBES FOR ROUTING

Note:
Van manufacturer to determine correct location of Remote PAC Interface assembly. It must be on the same side as the PAC/SRI Platform.

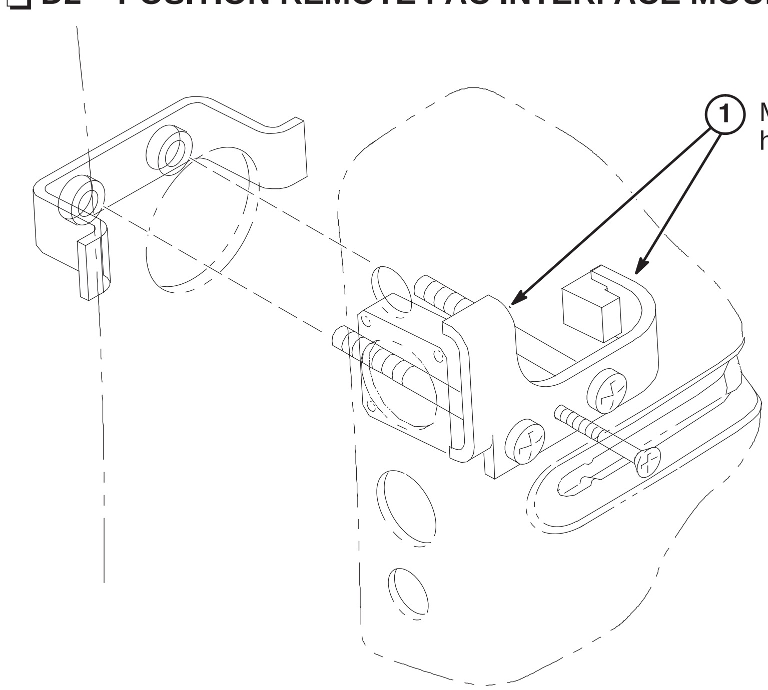
1. Apply tape around end of pneumatic tubes and mark "TOP" to the opposite end of the tube connected to the top connector in the Remote PAC Interface Assembly and "BOTTOM" to the tube connected to the lower connector.

Note: Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.



2. Insert two (2) screws M6 x 70mm (2109873–33) screws into Clip (2221023).
3. Insert Clip with screws into shell as shown with top cable and tube above Clip and bottom cable and tube below.
4. Hold in place by inserting provided M4 screw thru slot and turning about 3 or 4 turns into Clip.

❑ D2 – POSITION REMOTE PAC INTERFACE MOUNTING CLIP



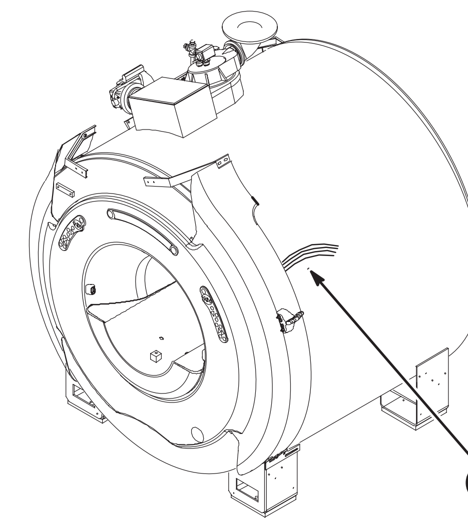
1. Make sure each side of clip is mounted behind housing blocks.

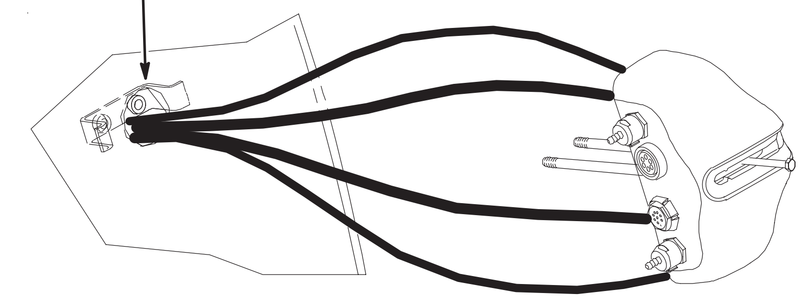
❑ D3 – ROUTE REMOTE PAC INTERFACE CABLES AND TUBES

1. Insert cables/tubes separately thru hole and make sure they come out at bottom of arc cover.

Note:
Top two cables will be routed over top of bracket and bottom two cables will be routed under bracket.

2. Insert ECG Cable first, followed by Peripheral Pulse cable and tubes.

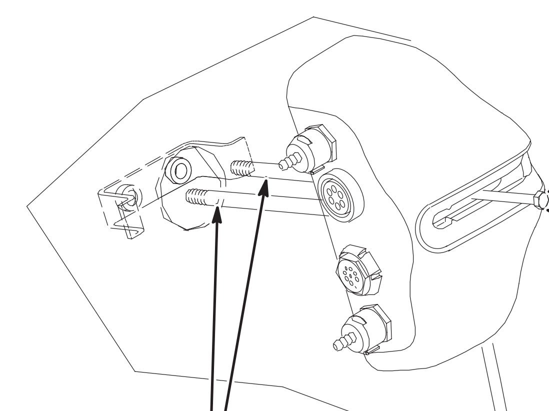




3. Route cable to PAC/SRI Platform location.

❑ D4 – INSTALLING REMOTE PAC INTERFACE TO ARC COVER

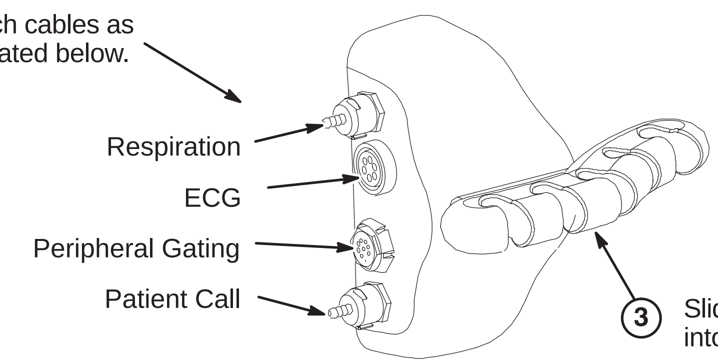
Note:
Routed cables and tubes not shown for clarity.

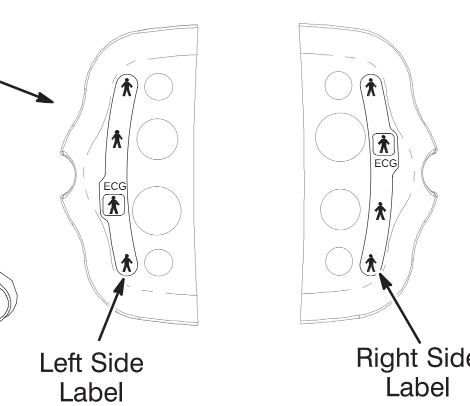


1. Use M4 screw thru slot to lift and guide small end of Clamp over block inside housing.
3. Remove M4 screw and save for use at later time since it would be required if Remote PAC Interface Assembly would need to be removed at any time in the future.
   OR
   Turn the M4 screw inward until there is clearance for sliding the PAC Hanger Bar in **T5** into place.
2. Using screwdriver thru housing slot, tighten down the M6 screws to lock housing in place

❑ D5 – INSTALL PAC HANGER BAR

1. Depending on left or right side installation of the PAC Remote Interface Asm on the magnet enclosure, attach the appropriate label that is provided with the kit.
2. Attach cables as indicated below.



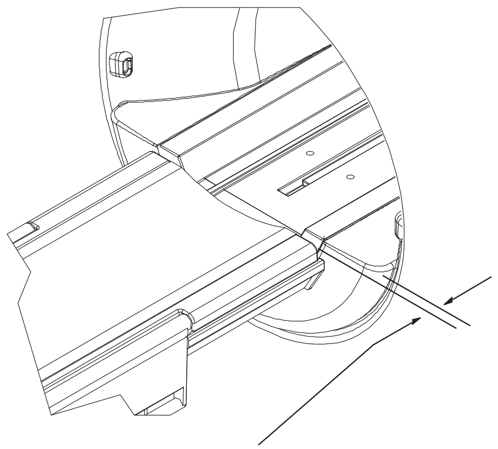


3. Slide PAC Hanger Bar (2221023) into position.

INSTALLATION PROCEDURES SUMMARY

- ❑ **E1 – Install Dock Mounting Bracket and Dock**
- ❑ **E2 – Route and Connect Magnet Enclosure Front Area Cables**

❑ E1 –INSTALL DOCK MOUNTING BRACKET AND DOCK



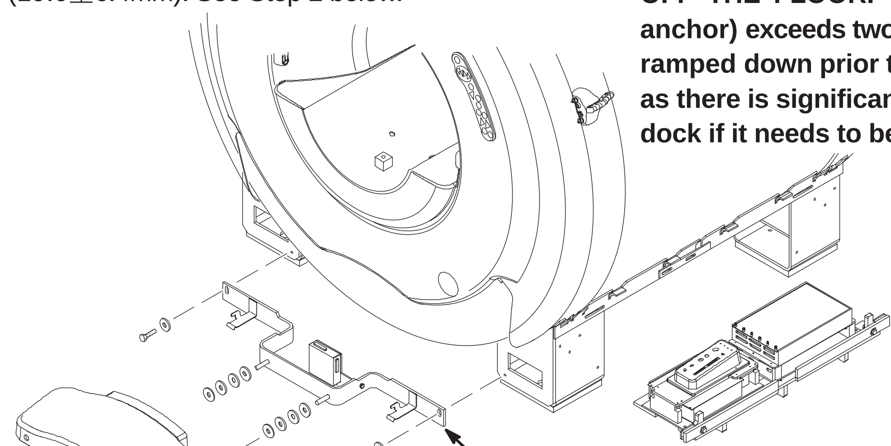
WARNING!

DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION. IF MAGNET IS RAMPED, WHENEVER MOVING DOCK, DOWNWARD PRESSURE MUST BE CONSTANTLY APPLIED TO THE TOP OF THE DOCK ASSEMBLY BY TWO PEOPLE. DOCK MUST BE MOVED SLOWLY ALONG THE GROUND AND NOT LIFTED VERTICALLY

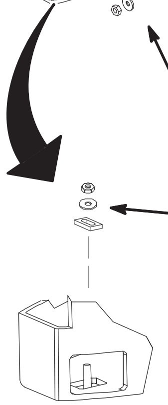
Note:
Gap between Patient Transport front rubber inlay and front of End Bell must be 0.75 Inches ± 0.25 (19.0±6.4mm). See Step 2 below.

WARNING!

DO NOT LIFT THE DOCK MORE THAN 2 INCHES (5 CM) OFF THE FLOOR. If the height (to clear the dock anchor) exceeds two inches (5 cm), the magnet must be ramped down prior to removing or installling the dock as there is significant risk of the magnet attracting the dock if it needs to be raised more that 2 inches (5 cm).



1. Fasten Dock Mounting Support (2226264) to magnet using 11mm inside diameter washers and Hex Screws supplied with support. Apply Loctite 242 as required.
2. Start with two (2) 11mm washers on each side between Dock and Dock Support. If adjustment is required, add or remove washers until proper gap is acheived.
3. Mount Dock Asm (46–271260G3) and pad using (2) each of the M10 Hex Nuts and 11mm washers that are provided with the Dock Mounting Support. Apply Loctite 242 as necessary.



4. Fasten Dock Asm to floor mounted stud with:
   (1) 2178874 Dock Clamp Plate
   (1) 2109878–4 10.5mm ID Washer
   (1) 2109875–4 10mm Hex Nut

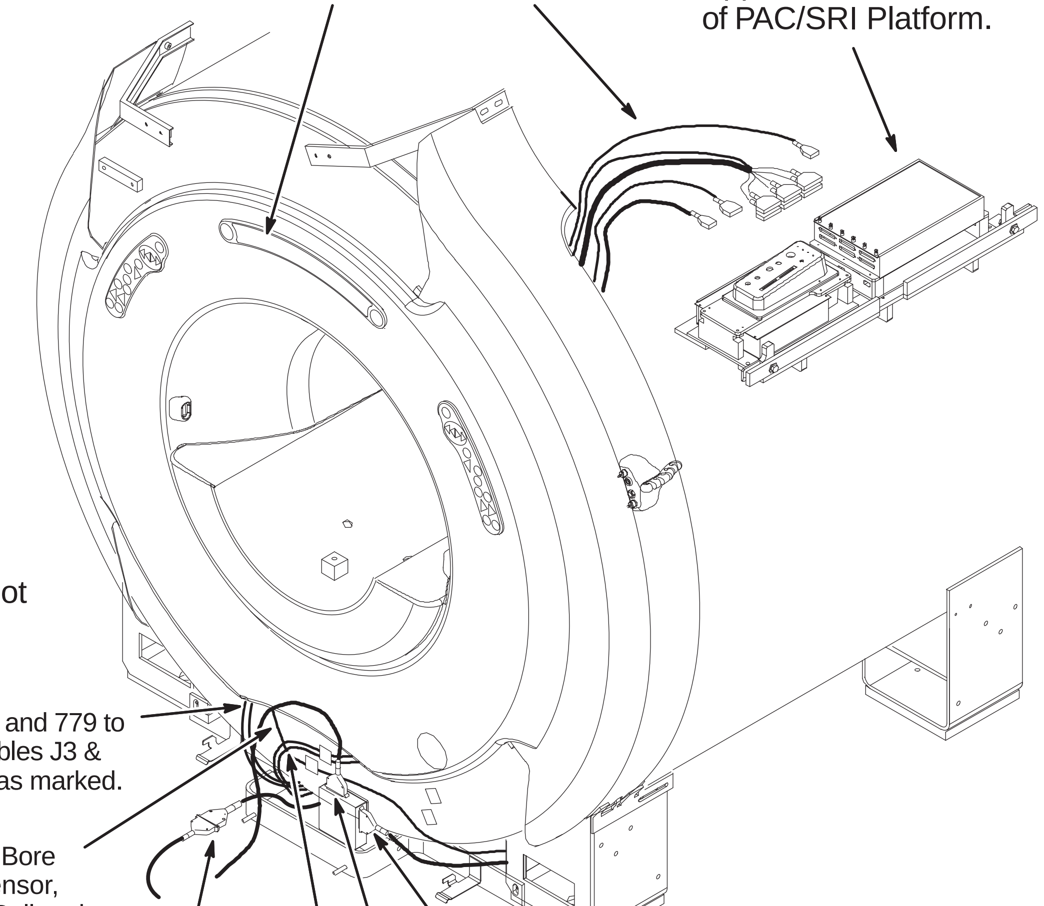
   Apply Loctite 242 as required.

❑ E2 – ROUTE AND CONNECT MAGNET ENCLOSURE FRONT AREA CABLES

Note:
Insulating and securing of cables routed to PAC/SRI Platform assembly to be the responsibility of van manufacturer.

1. Route cables from Operator Display to area of PAC/SRI Platform and connect as marked to SRI and PAC.

Note:
Approximate location of PAC/SRI Platform.



Note:
Dock Assembly not shown for clarity.

2. Route Runs 778 and 779 to front RF Bias cables J3 & J4 and connect as marked.
3. Route Run 749, Bore Temperature Sensor, from Front End Bell and connect ot J10 on SRI.
4. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with tape or with provided velcro jacket.
5. Route Front End Bell Microphone and Speaker cables underneath magnet to Rear Pedestal and connect to Patient Communication Box as marked.
6. Connect Dock Cable to J2 on Dock Limit Box.
7. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.

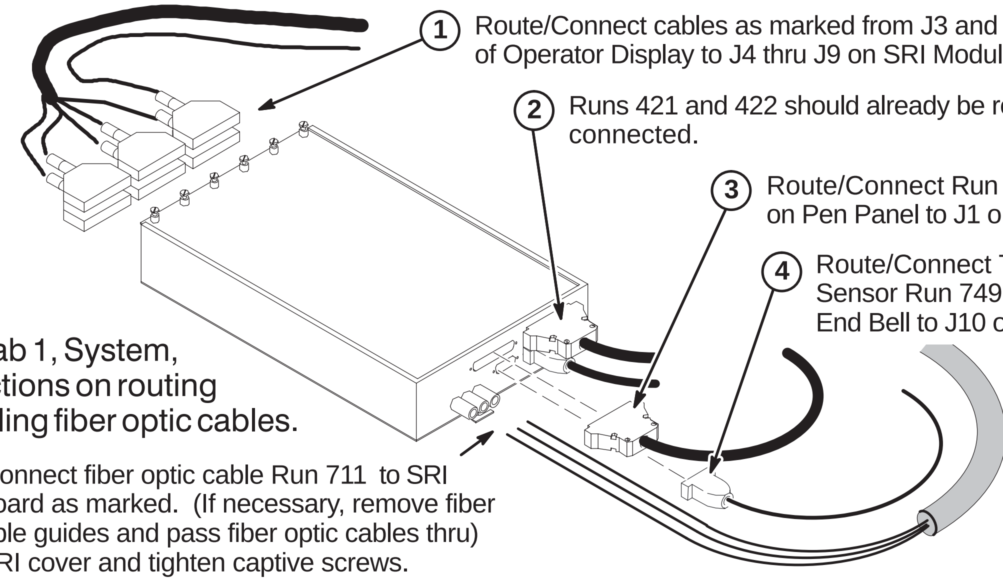
CAUTION

No metal to metal contact is allowed for connectors to other areas. i.e. connector touching the magnet.

INSTALLATION PROCEDURES SUMMARY

- ❑ **F1 – Connect Cables to SRI Module**
- ❑ **F2 – Cable Connection to PAC Module**
- ❑ **F3 – Route and Tie–wrap Cables to PAC/SRI Platform**
- ❑ **F4 – Routing of Photo–Plethysmograph Cable**

❑ F1 – CONNECT CABLES TO SRI MODULE

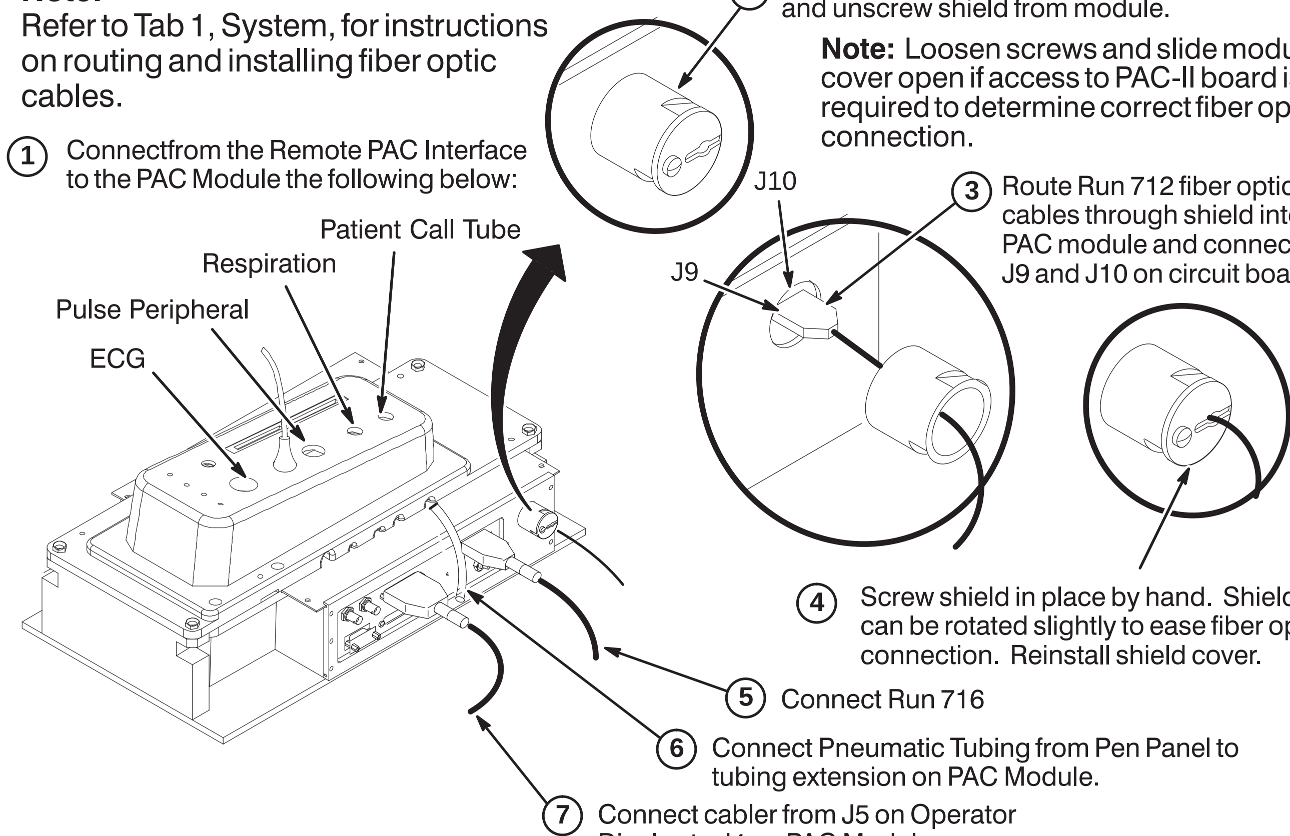


1. Route/Connect cables as marked from J3 and J4 of Operator Display to J4 thru J9 on SRI Module.
2. Runs 421 and 422 should already be routed and connected.
3. Route/Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
4. Route/Connect Temperture Sensor Run 749 from Front End Bell to J10 on SRI Module
5. Route/Connect fiber optic cable Run 711 to SRI circuit board as marked. (If necessary, remove fiber optic cable guides and pass fiber optic cables thru) Close SRI cover and tighten captive screws.

Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.

❑ F2 – CABLE CONNECTION TO PAC MODULE

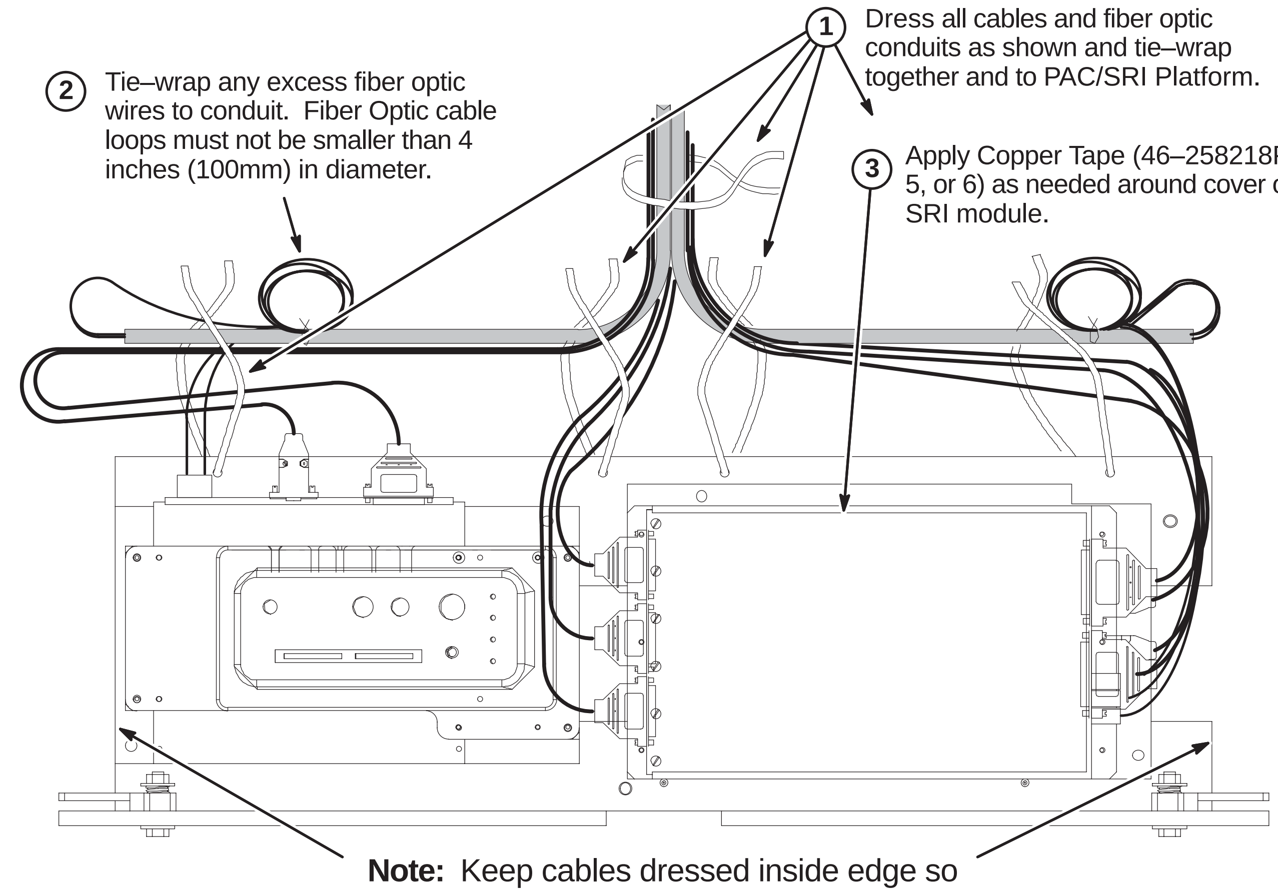
Note:
Refer to Tab 1, System, for instructions on routing and installing fiber optic cables.



1. Connectfrom the Remote PAC Interface to the PAC Module the following below:
2. Remove shield cover by removing one screw and unscrew shield from module.

   Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.
3. Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.
4. Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.
5. Connect Run 716
6. Connect Pneumatic Tubing from Pen Panel to tubing extension on PAC Module.
7. Connect cabler from J5 on Operator Display to J4 on PAC Module.

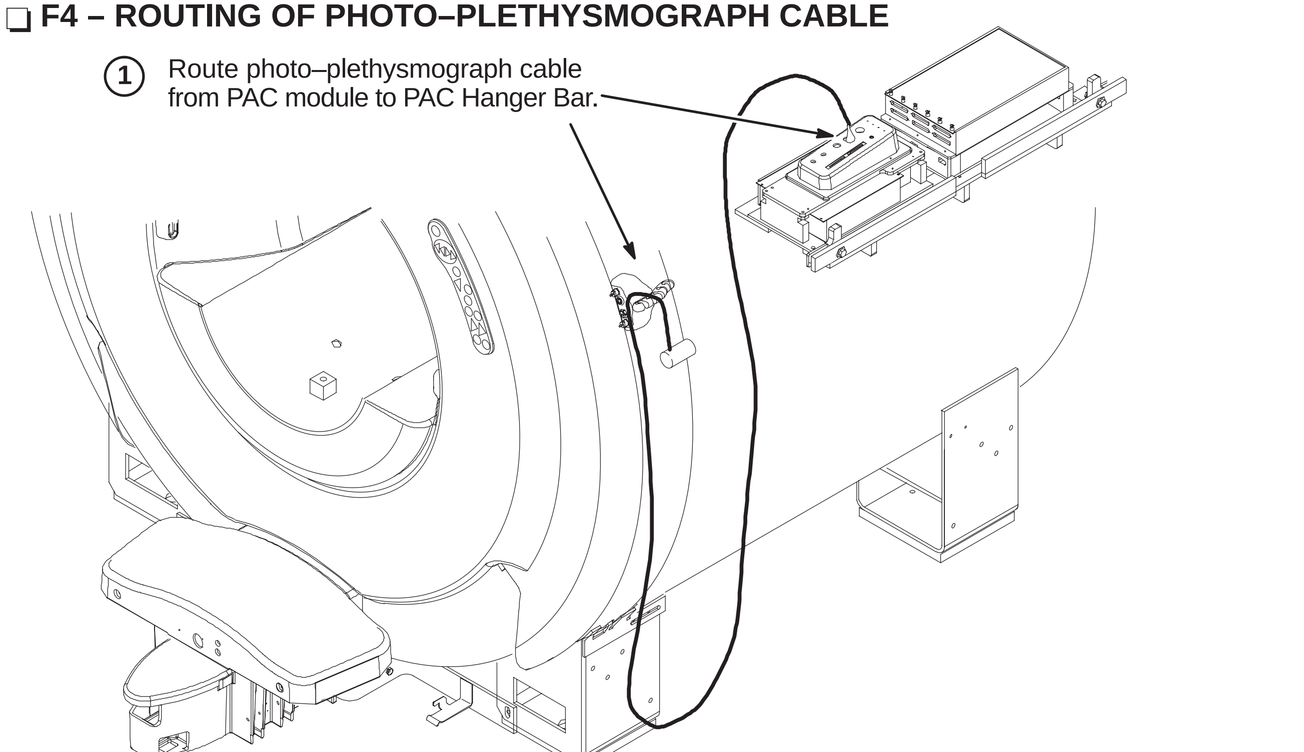
❑ F3 – ROUTE AND TIE–WRAP CABLES TO PAC/SRI PLATFORM



1. Dress all cables and fiber optic conduits as shown and tie–wrap together and to PAC/SRI Platform.
2. Tie–wrap any excess fiber optic wires to conduit. Fiber Optic cable loops must not be smaller than 4 inches (100mm) in diameter.
3. Apply Copper Tape (46–258218P4, 5, or 6) as needed around cover of SRI module.

Note: Keep cables dressed inside edge so unit slides between legs easily for service.

❑ F4 – ROUTING OF PHOTO–PLETHYSMOGRAPH CABLE



1. Route photo–plethysmograph cable from PAC module to PAC Hanger Bar.

BRIDGE INSTALLATION INFORMATION

TWO TYPES OF BRIDGES ARE AVAILABLE FOR INSTALLATION: The original style Long Bridge and the new style Split Bridge. See items 1 and 2 below for explanation of which installation procedures to use.

1. **If original style Long Bridge is delivered to site, complete procedures on pages 7, 8, and 9. This bridge would be used for pre–Split Bridge MR/i and all CV/i installations.**
2. **If new style Split Bridge is delivered to site, complete procedures on pages 10, 11, and 12. This bridge would be used for MR/i installations only, not CV/i.**

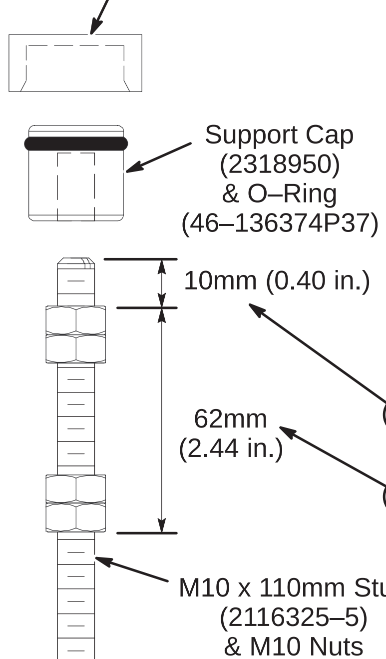
INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

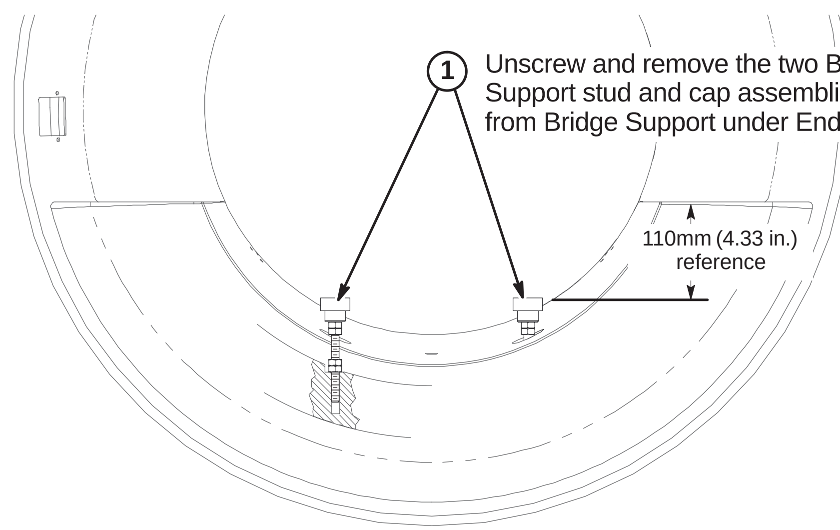
- ❑ **G1 – Front Bridge Support Adjustments**
- ❑ **G2 – Bridge Support Adjustment Tolerances**
- ❑ **G3 – Bridge Assembly Preparation Before Installing in Magnet**
- ❑ **G4 – Install Bridge**

NOTE:
Measure to confirm that the distance between the top of the Support Cap Cup and the flat surface of the End Bell is110mm as shown in Procedure **G1**. Also measure to confirm that the 10mm and 62mm M10 Jam Nut distance locations on the Stud are also in place.

- If the dimensions are found to be in place for this site, proceed to Procedure **G2**.
- If these dimensions are not in place, complete Steps 1 thru 6 in Procedure **G1** and then proceed to **G2**.

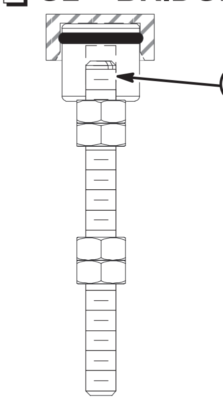
❑ G1 – FRONT BRIDGE SUPPORT ADJUSTMENTS





1. Unscrew and remove the two Bridge Support stud and cap assemblies from Bridge Support under End Bell.
2. Remove Support Cap Cup and unscrew Support Cap
3. Set top M10 Nut at a distance of 10mm from top of stud. Lock in position with second M10 Nut.
4. Set bottom M10 Nut at a distance of 62mm from top of first M10 Nut. Lock in position with fourth M10 Nut.
5. Fasten Support Cap to top of Stud and tighten to top M10 Nut. Install Support Cap Cup to Support Cap
6. Re–install the two Bridge Support stud and cap assemblies into Bridge Support.

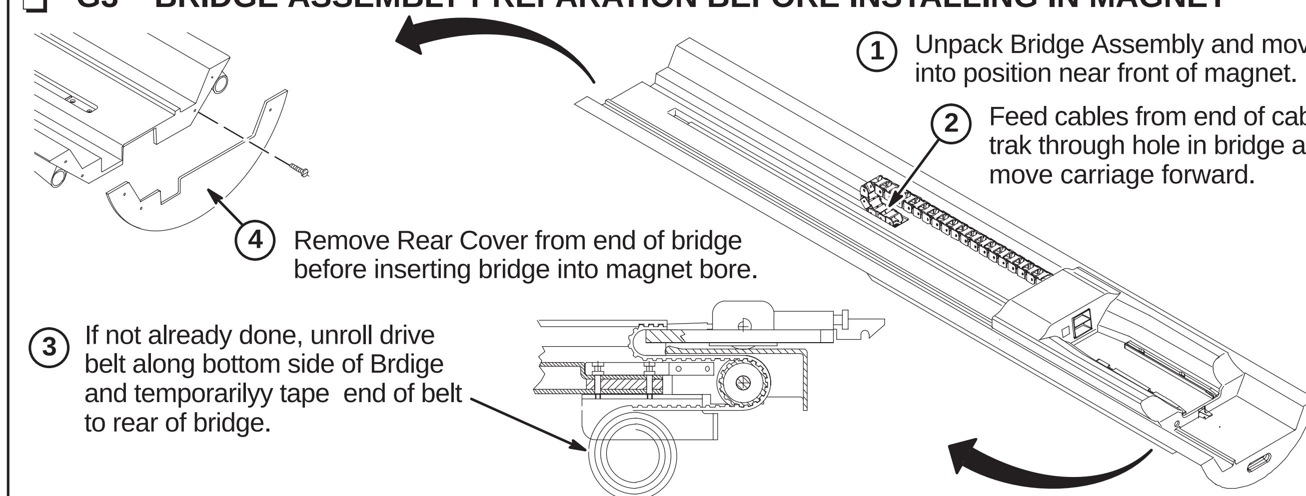
❑ G2 – BRIDGE SUPPORT ADJUSTMENT TOLERANCES



1. The 10mm location distance noted in Procedure **G1** is a starting point for installation and leveling of Bridge.

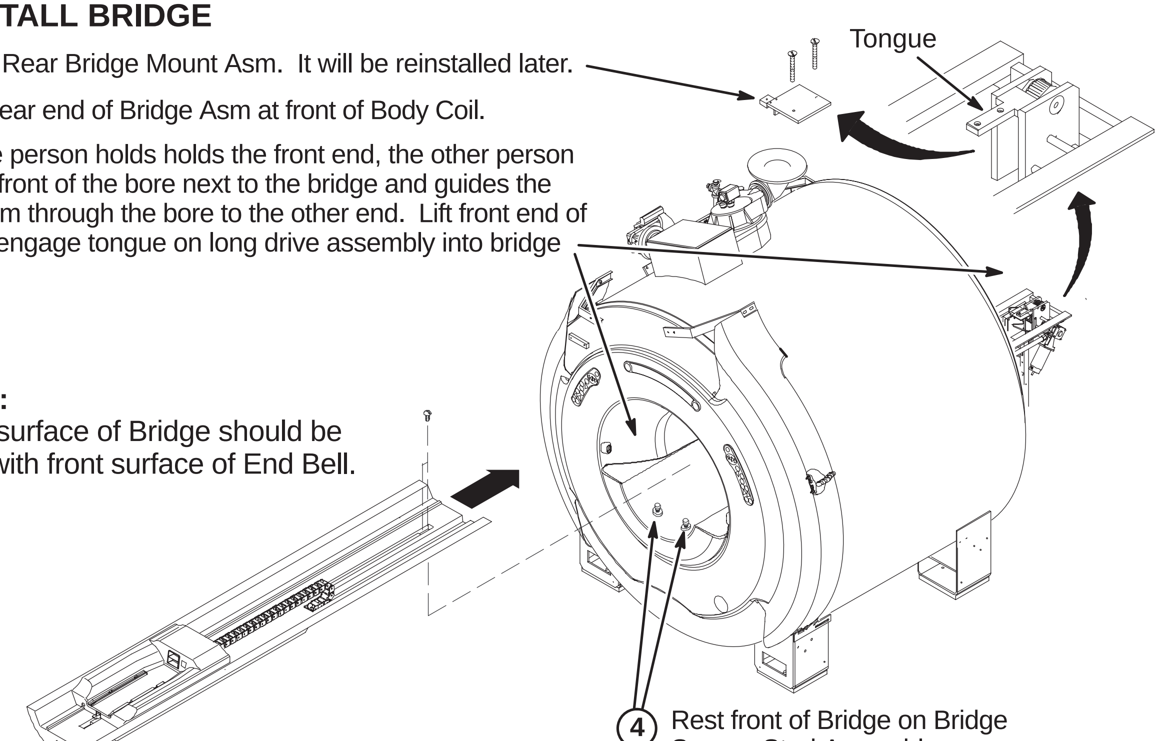
   This starting point gives the installer an adjustment of +7.5mm and –5mm. The amount of adjustment will be determined during bridge leveling.

❑ G3 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET



1. Unpack Bridge Assembly and move into position near front of magnet.
2. Feed cables from end of cable trak through hole in bridge and move carriage forward.
3. If not already done, unroll drive belt along bottom side of Brdige and temporarilyy tape end of belt to rear of bridge.
4. Remove Rear Cover from end of bridge before inserting bridge into magnet bore.

❑ G4 – INSTALL BRIDGE



1. Unfasten Rear Bridge Mount Asm. It will be reinstalled later.
2. Position rear end of Bridge Asm at front of Body Coil.
3. While one person holds holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.
4. Rest front of Bridge on Bridge Suppor Stud Assembly.

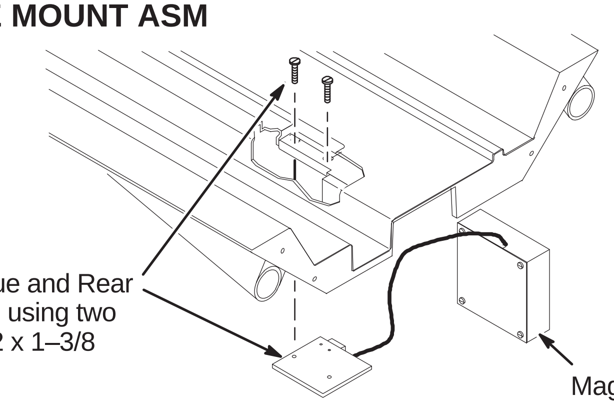
NOTE:
Front surface of Bridge should be flush with front surface of End Bell.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

- ❑ **H1 – Reattach Bridge Mount Assembly**
- ❑ **H2 – Align Front of Bridge to End Bell**
- ❑ **H3 – Applying Adhesive to Bridge Support Caps**
- ❑ **H4 – Leveling Bridge**

❑ H1 – REATTACH BRIDGE MOUNT ASM

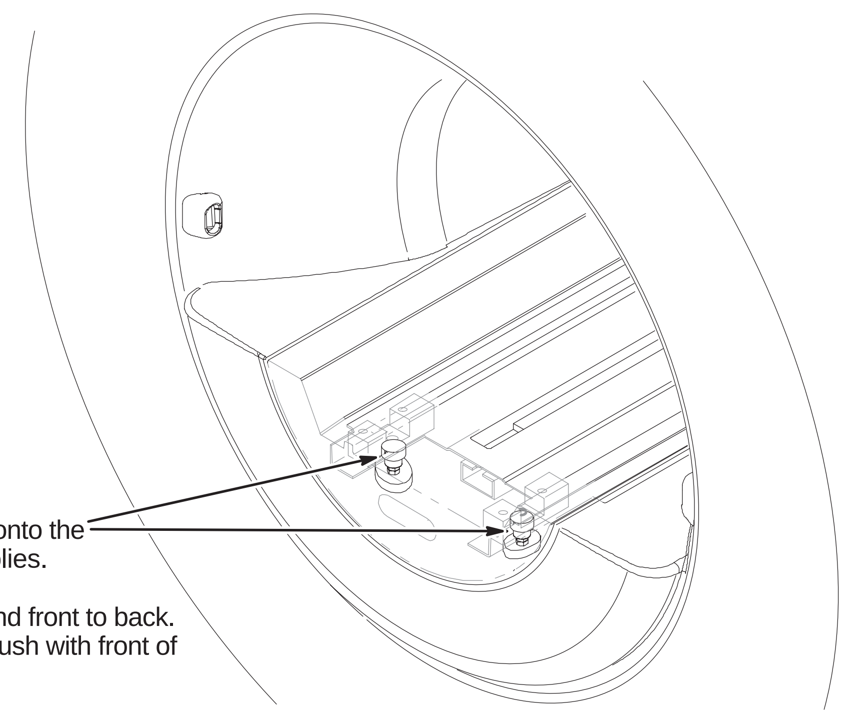
NOTE:
It is important that the sensor end of the Sensor body fits flush with the Sensor Bracket.



1. Attach together Bridge, Tongue and Rear Bridge Mount Asm (2132034) using two (2) previously removed 10–32 x 1–3/8 Screws (46–220184P47)

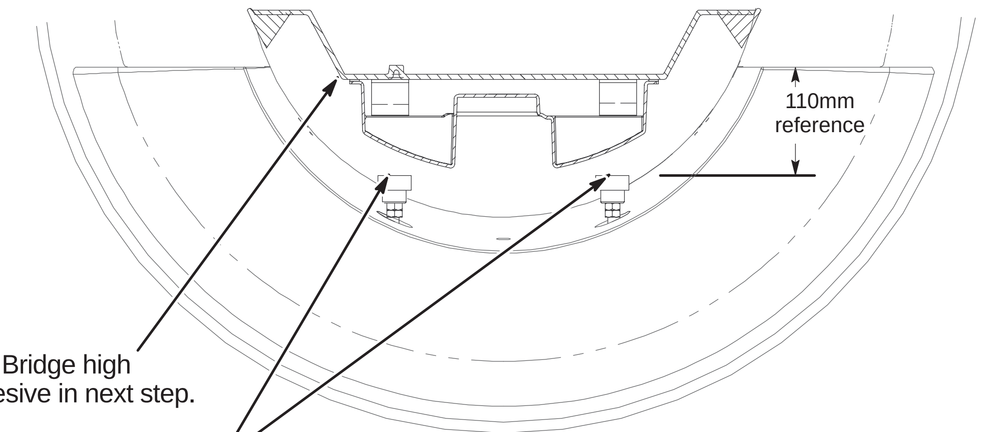
Magnet I/F #1
(Mounted on Rear Pedestal)

❑ H2 – ALIGN FRONT OF BRIDGE TO END BELL



1. Lower the front of the Bridge onto the Bridge Support Stud Assemblies.
2. Align the Bridge left to right and front to back. Make sure front of Bridge is flush with front of end bell.

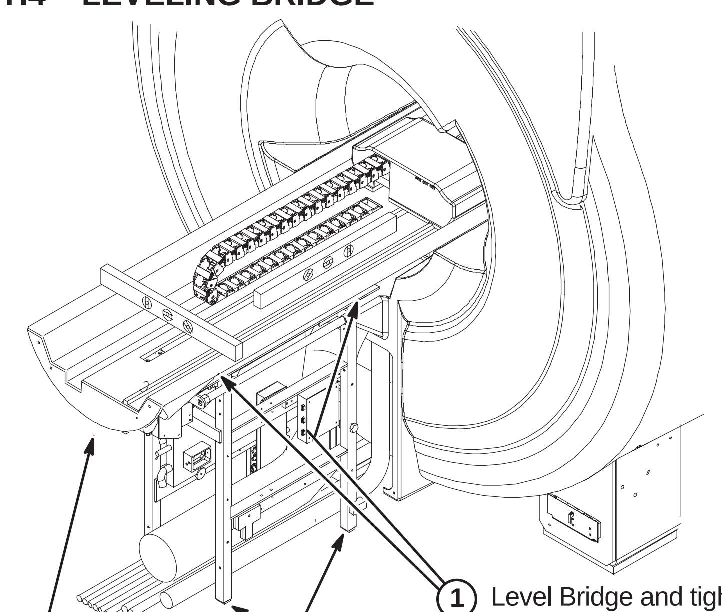
❑ H3 – APPLYING ADHESIVE TO BRIDGE SUPPORT CAPS



1. Raise the front of the Bridge high enough to apply adhesive in next step.
2. Apply epoxy adhesive (46–282264P1) to top of each Bridge Support Cap.
3. Lower the Bridge to rest firmly to the Bridge Support Caps.
4. Allow adhesive to cure.
5. After adhesive has cured, adjust the Bridge to the final vertical position. Make sure bridge is level.

   Note: The final position will leave the top surface of the Bridge 1mm (or more) above the top surface of the end bell.
6. Repeat the vertical adjustment as needed. Make sure the two M10 jam nuts are tight against each Support Cap. This will prevent change of position.

❑ H4 – LEVELING BRIDGE



NOTE:
PVC Spacers positioned thru End Bells are to support Bridge.

NOTE:
Rear Pedestal to be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

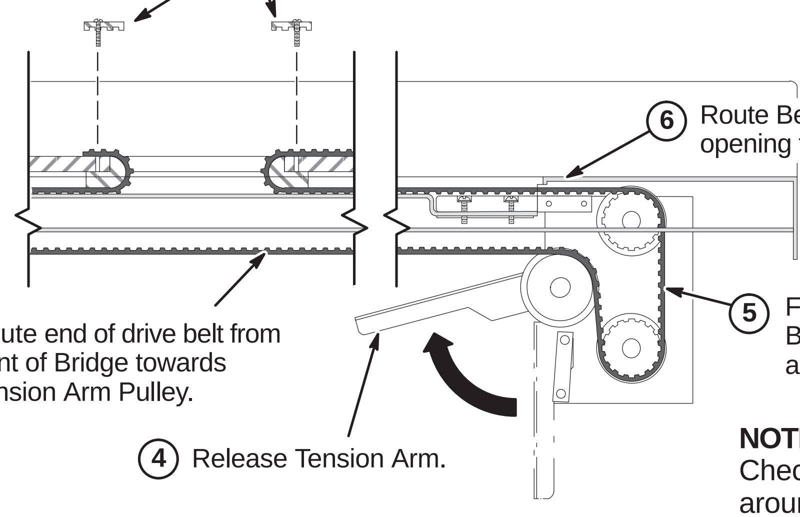
NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

INSTALLATION PROCEDURES SUMMARY (for original Long Bridge)

- ❑ **J1 – Routing Drive Belt**
- ❑ **J2 – Final Adjustment of Drive Belt**
- ❑ **J3 – Installation of Sensor Flag**
- ❑ **J4 – Attach Bridge Cover**
- ❑ **J5 – Route/Connect Runs 404 and 485 (and Multi–coil Cables if Required)**

❑ J1 – ROUTING OF DRIVE BELT



7 Position Belt under Clamp Plates and re–attach with previously removed screws.

6 Route Belt under and through opening to top of Carriage.

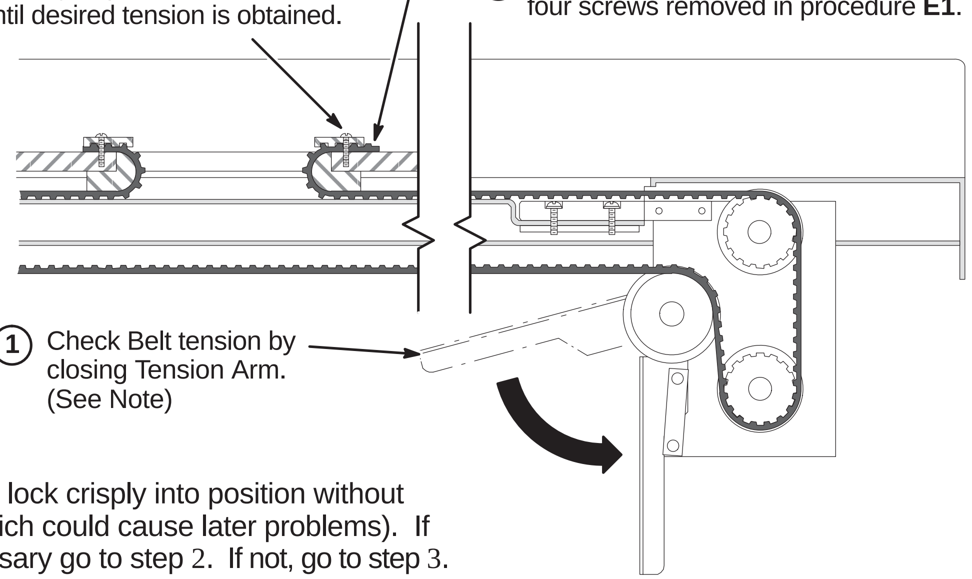
5 Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.

3 Route end of drive belt from front of Bridge towards Tension Arm Pulley.

4 Release Tension Arm.

NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

❑ J2 – FINAL ADJUSTMENT OF DRIVE BELT



2 **(IF NECESSARY):**
Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.

3 Tighten Clamps and trim excess Belt length.

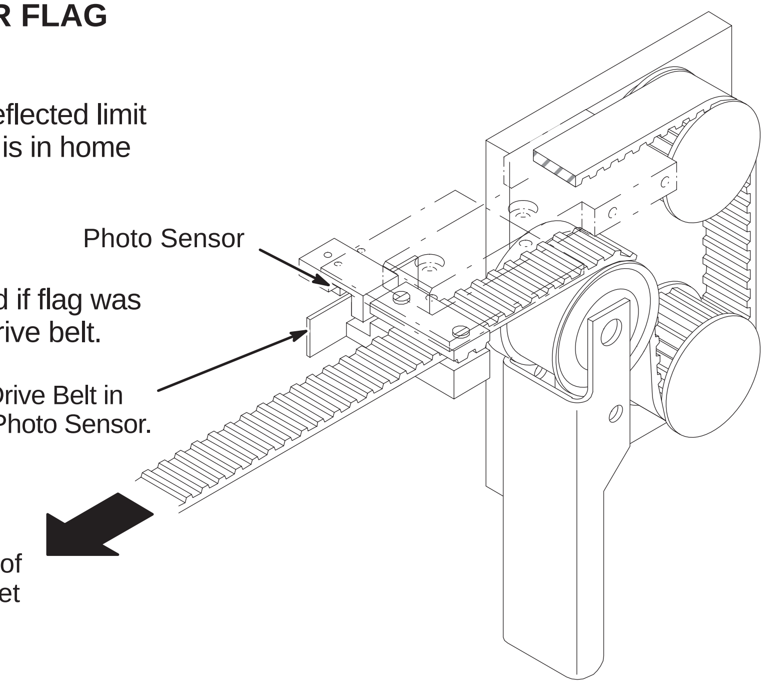
4 Fasten Carriage Cover to Carriage with four screws removed in procedure **E1**.

1 Check Belt tension by closing Tension Arm. (See Note)

NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

❑ J3 – INSTALLATION OF SENSOR FLAG

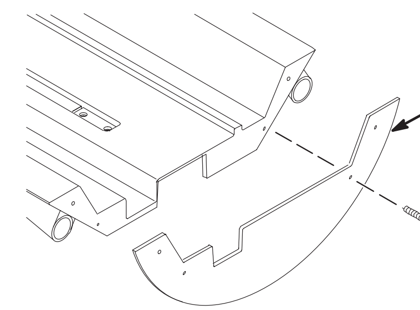
NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.



NOTE:
The following Step is needed if flag was removed during routing of drive belt.

1 Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

❑ J4 – ATTACH BRIDGE COVER

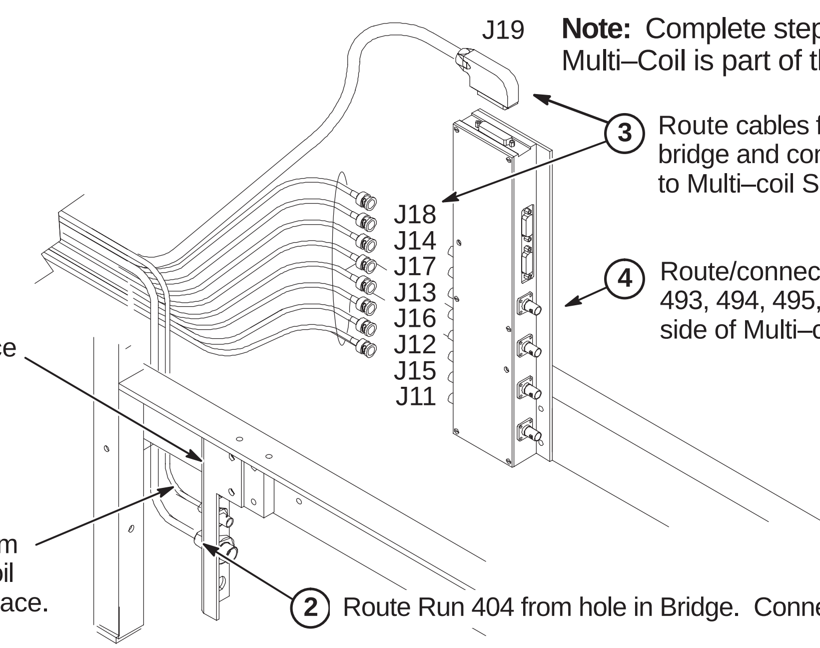


1 Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

Note:
Bridge Cover has enclosure rating plates on it.

❑ J5 – ROUTE RUNS 404 AND 485 (AND MULTI–COIL CABLES IF REQUIRED)

Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



3 Route cables from hole in bridge and connect as marked to Multi–coil Select Switch.

4 Route/connect Runs 491, 492, 493, 494, 495, and 727 to this side of Multi–coil Select Switch.

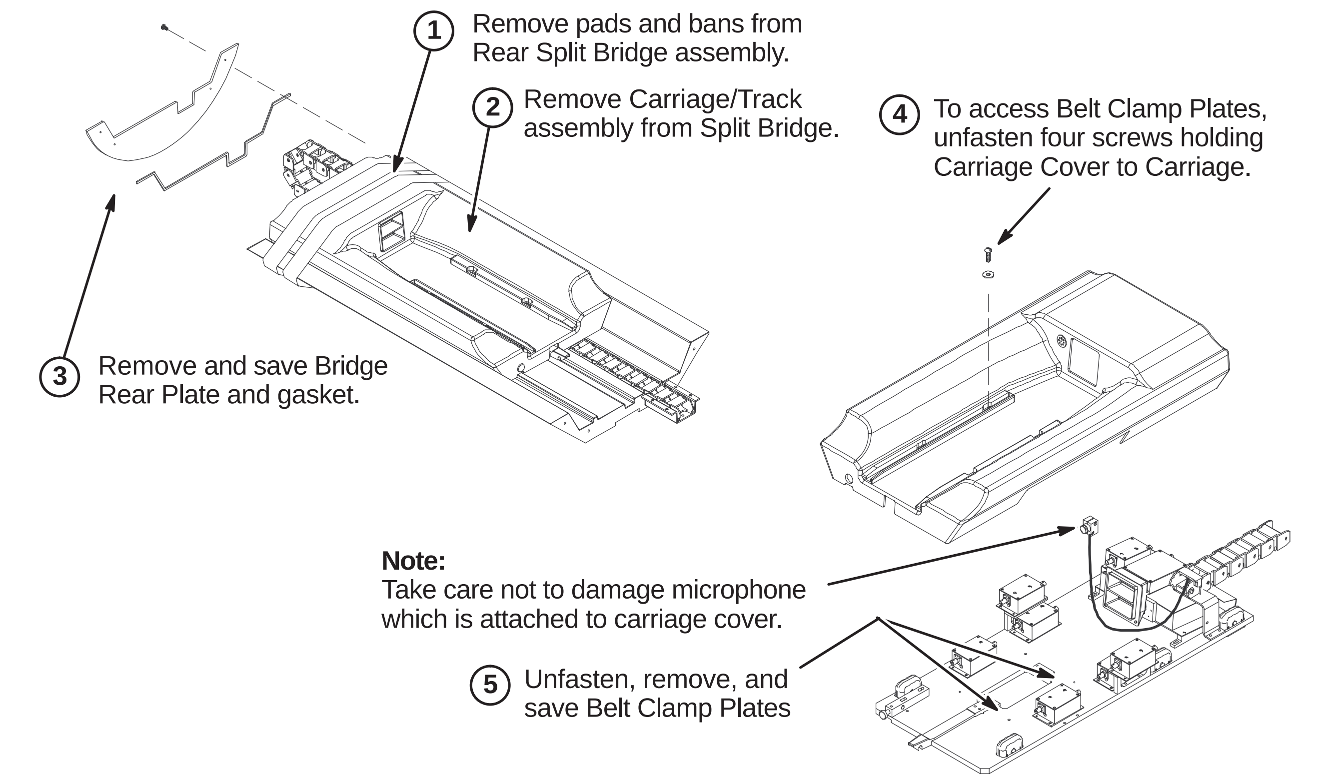
1 Route Run 485 from hole in Bridge to coil cable takeup interface. Connect to J2.

2 Route Run 404 from hole in Bridge. Connect to J1.

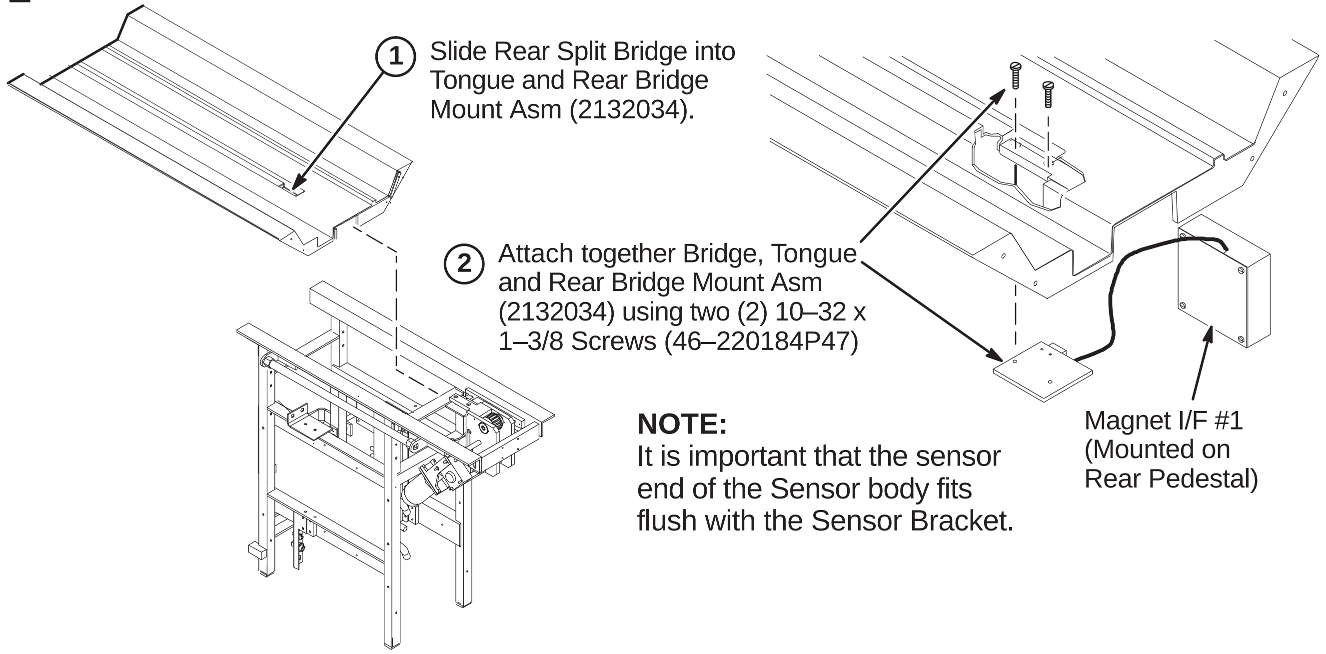
INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

- ❑ K1 – Preparing Rear Split Bridge for Installation
- ❑ K2 – Attaching Rear Split Bridge to Rear Pedestal
- ❑ K3 – Attaching Rear Split Bridge to Front Split Bridge
- ❑ K4 – Aligning Bridge on Rear Pedestal
- ❑ K5 – Installing Bridge Locators

❑ K1 – PREPARING REAR SPLIT BRIDGE FOR INSTALLATION



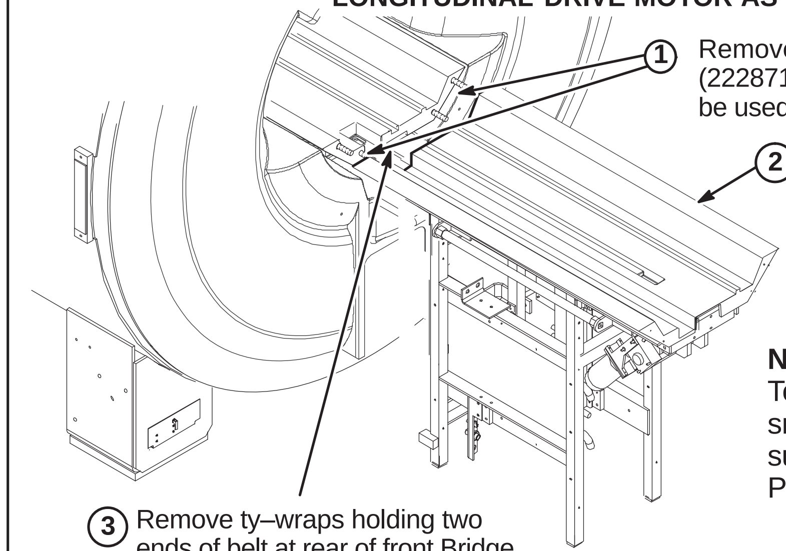
❑ K2 – ATTACHING REAR SPLIT BRIDGE TO REAR PEDESTAL



❑ K3 – ATTACHING REAR SPLIT BRIDGE TO FRONT SPLIT BRIDGE

WARNING!

THE REAR PEDESTAL HOUSES THE LONGITUDINAL DRIVE MOTOR WHICH IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE REAR PEDESTAL REQUIRES THREE PEOPLE AND MUST MAINTAIN THE LONGITUDINAL DRIVE MOTOR AS FAR FROM THE MAGNET AS POSSIBLE.

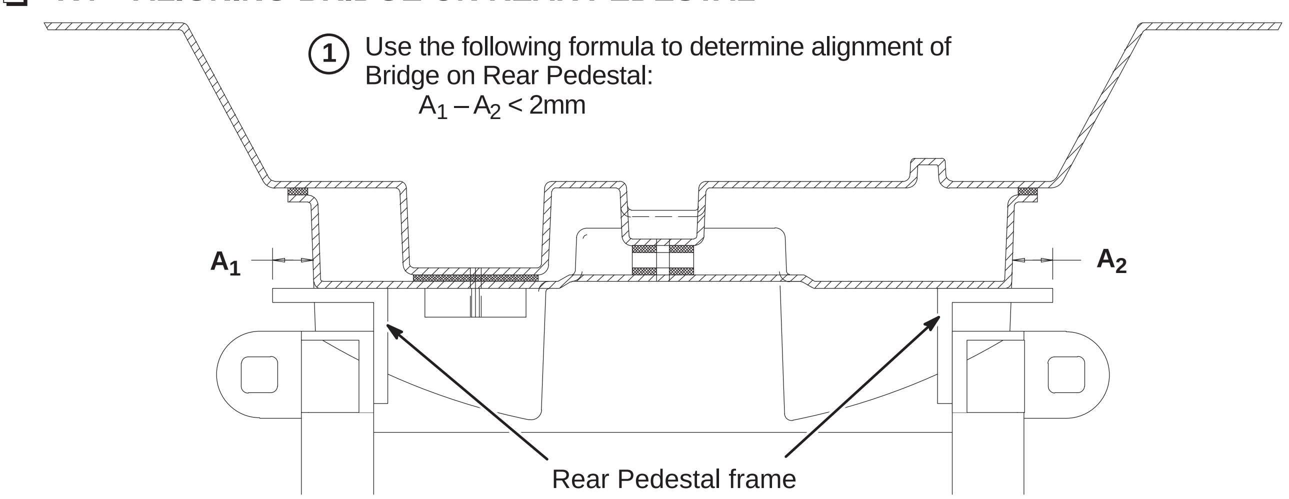


1. Remove four 10M hex nuts (2109877–3) and washers (2228717–2) from end of Bridge in magnet. These will be used for attaching the Rear Pedestal.
2. Move Rear Pedestal/Bridge into position and fasten with four nuts and washers that were removed form end of Front Split Bridge.
3. Remove ty–wraps holding two ends of belt at rear of front Bridge.

NOTE:
Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

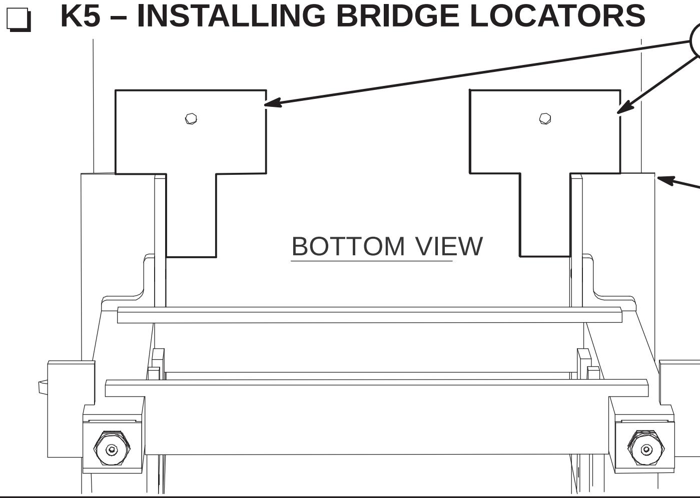
❑ K4 – ALIGNING BRIDGE ON REAR PEDESTAL

1. Use the following formula to determine alignment of Bridge on Rear Pedestal:
   $A_1 - A_2 < 2mm$



❑ K5 – INSTALLING BRIDGE LOCATORS

1. After Bridge and Rear Pedestal are aligned, peel off protective paper on each Bridge Locator and attach to bottom side of Bridge as shown.

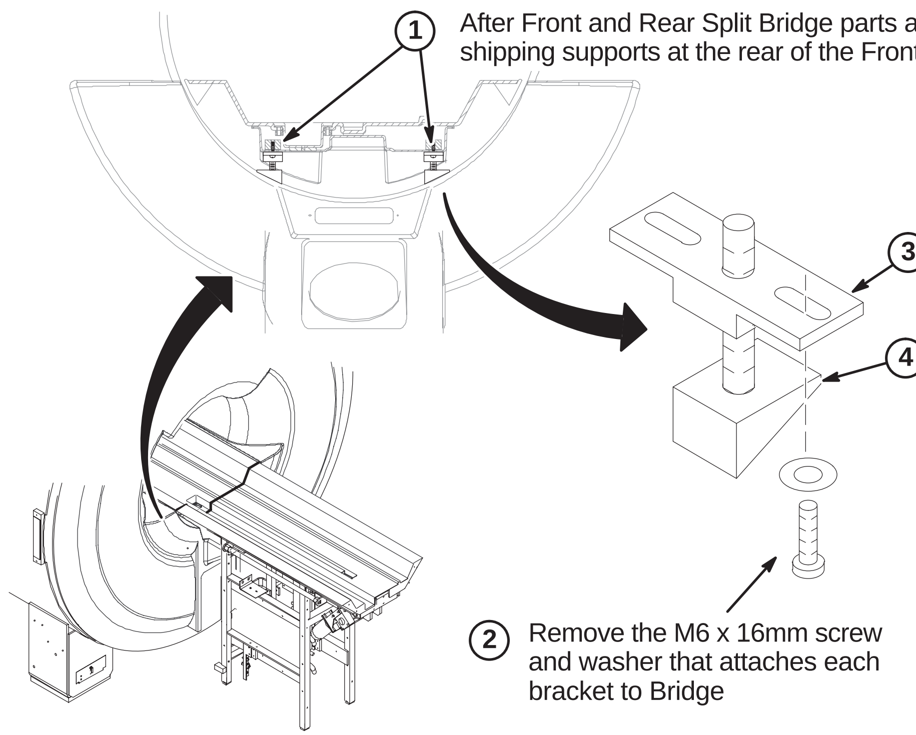


End of Rear Pedestal frame closest to Magnet.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

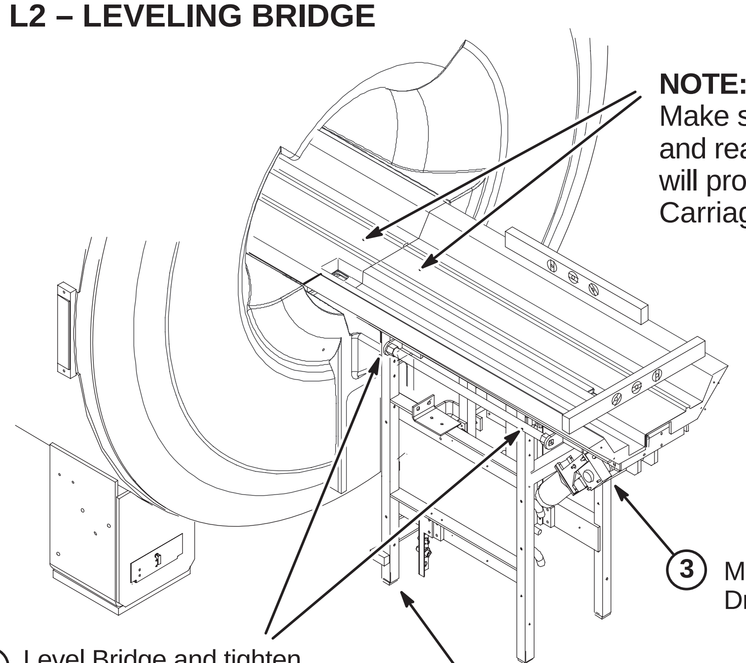
- ❑ **L1 – Removing Front Split Bridge Shipping Support Brackets**
- ❑ **L2 – Leveling Bridge**
- ❑ **L3 – Height Adjustment at Front of Bridge (if needed)**
- ❑ **L4 – Installing Carriage on Bridge**
- ❑ **L5 – Routing of Drive Belt**

❑ L1 – REMOVING FRONT SPLIT BRIDGE SHIPPING SUPPORT BRACKETS



1. After Front and Rear Split Bridge parts are fastened together, the two shipping supports at the rear of the Front Split Bridge need to be removed.
2. Remove the M6 x 16mm screw and washer that attaches each bracket to Bridge
3. Unscrew downword the Bridge Mounting Flange.
4. Support Base is attached to End Bell with adhesive tape. To remove, carefully pry base away from end bell. Remove base, stud, and mounting flange

❑ L2 – LEVELING BRIDGE

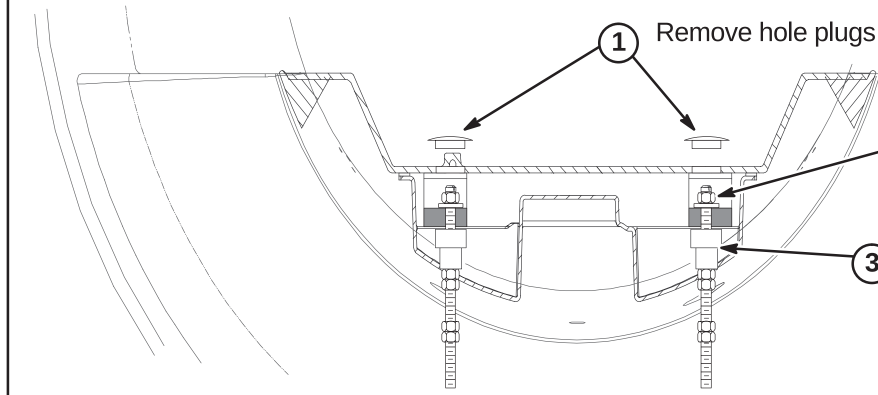


NOTE:
Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:
Rear Pedestal to be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.

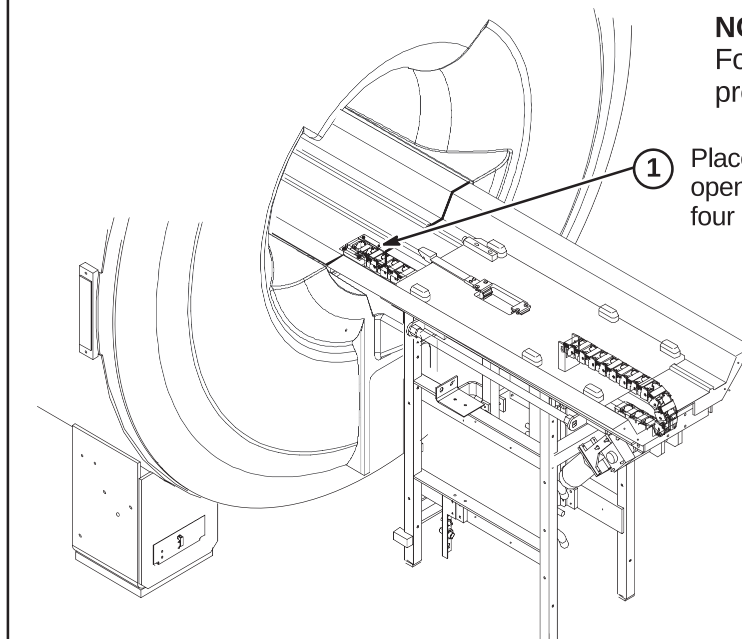
1. Level Bridge and tighten Vertical Adjustment Fasteners.
2. Adjust leg screws on both sides as needed to level Rear Pedetal.
3. Make sure that Longitudinal Drive Asm is level.

❑ L3 – HEIGHT ADJUSTMENT AT FRONT OF BRIDGE (IF NEEDED)



1. Remove hole plugs from each side of Bridge.
2. Remove M10 nut and washer from each stud.
3. Adjust Support Nut (2293381) as needed to achieve correct height.

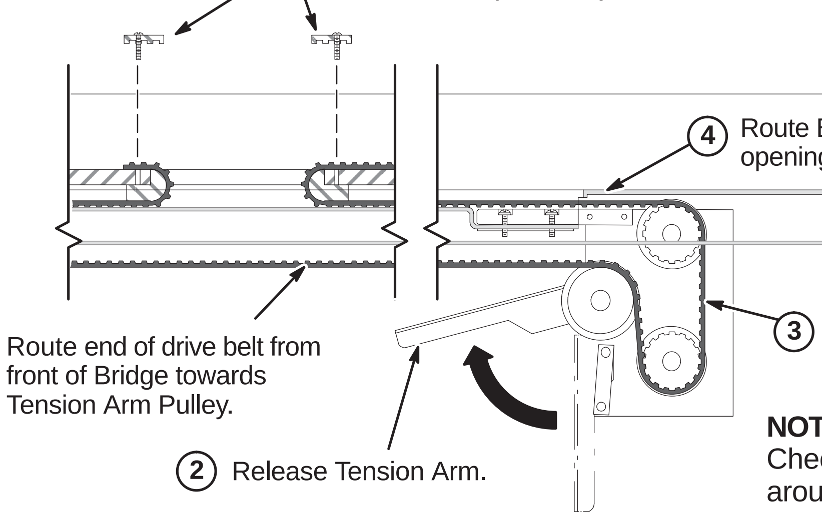
❑ L4 – INSTALLING CARRIAGE ON BRIDGE



NOTE:
For clarity, the carriage is shown without preamps, preamp protect module, and cables.

1. Place Carriage on Bridge and route all cables thru opening in Bridge and fasten plate to Front Bridge with four M6 x 12mm nylon screws (2296958) provided.

❑ L5 – ROUTING OF DRIVE BELT



1. Route end of drive belt from front of Bridge towards Tension Arm Pulley.
2. Release Tension Arm.
3. Feed back end of Belt underneath Bridge, around Tension Arm Pulley and rear pulleys into Bridge.
4. Route Belt under and through opening to top of Carriage.
5. Position Belt under Clamp Plates and re–attach with previously removed screws.

NOTE:
Check to verify that belt is not caught around spacers near rear pulley.

INSTALLATION PROCEDURES SUMMARY (for new style Split Bridge)

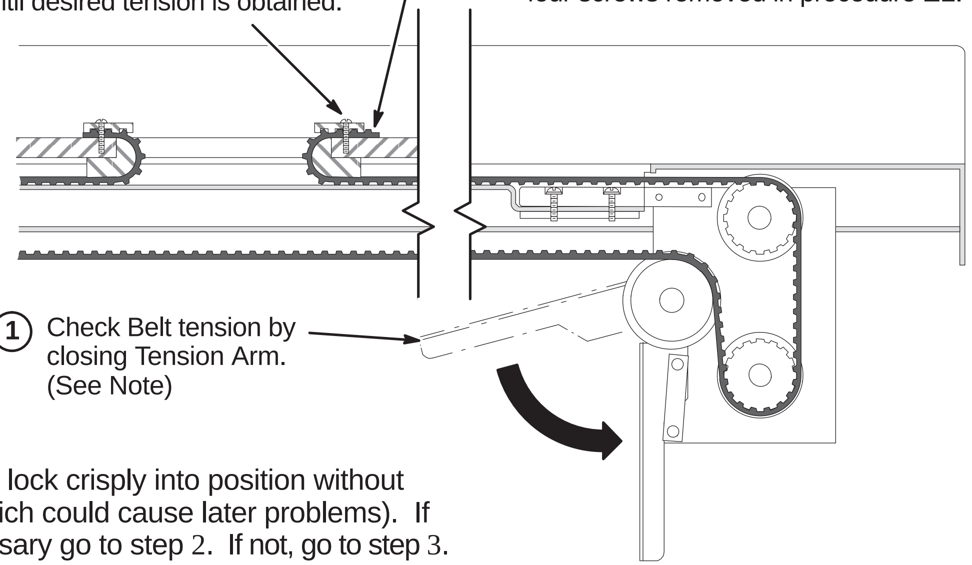
- ❑ **M1 – Final Adjustment of Drive Belt**
- ❑ **M2 – Installation of Sensor Flag**
- ❑ **M3A – Route Non–MGD Runs 404 and 485 (and Multi–coil Cables if Required)**
  OR
- ❑ **M3B – Route/Connect MGD Cables from LPCA to Rear Pedestal**
- ❑ **M4 – Installing Carriage Cover to Trolley and Bridge Cover**

❑ M1 – FINAL ADJUSTMENT OF DRIVE BELT

(3) Tighten Clamps and trim excess Belt length.

(2) **(IF NECESSARY):**
Loosen Clamp and adjust position of Belt one tooth at a time until desired tension is obtained.

(4) Fasten Carriage Cover to Carriage with four screws removed in procedure **E1**.



(1) Check Belt tension by closing Tension Arm. (See Note)

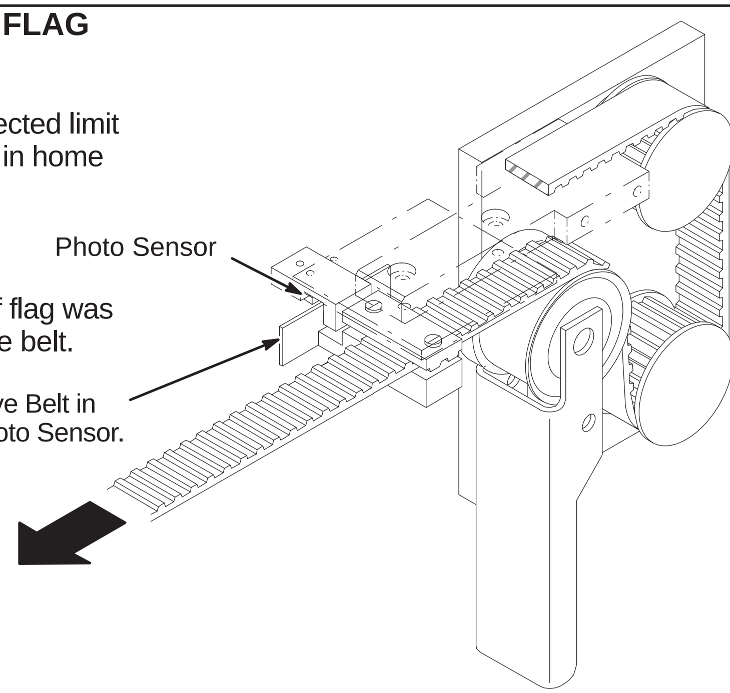
NOTE:
Tension Arm should lock crisply into position without excessive force (which could cause later problems). If adjustment is necessary go to step 2. If not, go to step 3.

❑ M2 – INSTALLATION OF SENSOR FLAG

NOTE:
Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

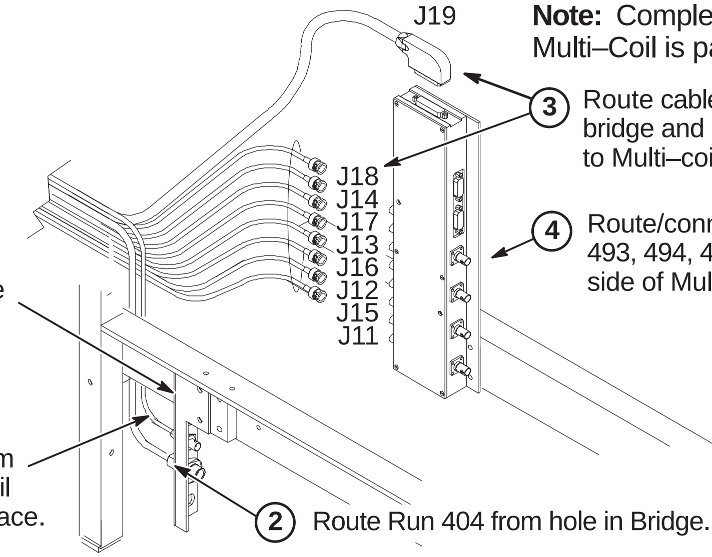
NOTE:
The following Step is needed if flag was removed during routing of drive belt.

(1) Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



❑ M3A – ROUTE NON–MGD RUNS 404 AND 485 (& Multi–Coil Cables if Required)

Note: Complete steps 3 and 4 if Multi–Coil is part of the system product.



(3) Route cables from hole in bridge and connect as marked to Multi–coil Select Switch.

(4) Route/connect Runs 491, 492, 493, 494, 495, and 727 to this side of Multi–coil Select Switch.

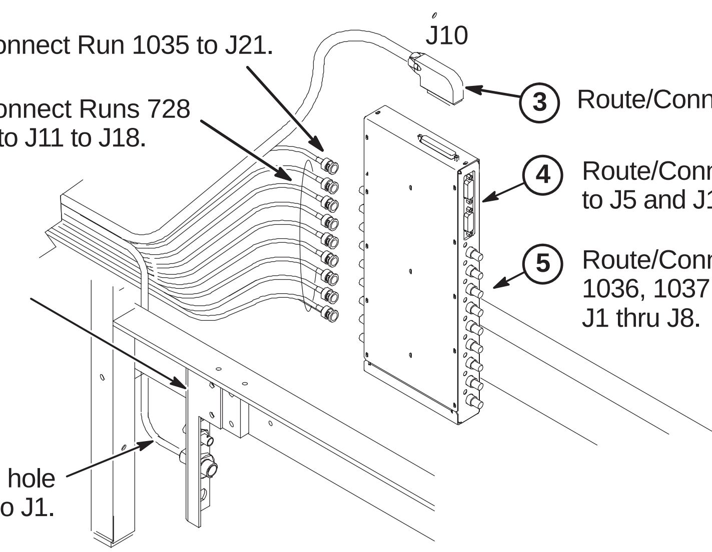
(1) Route Run 485 from hole in Bridge to coil cable takeup interface. Connect to J2.

(2) Route Run 404 from hole in Bridge. Connect to J1.

❑ M3B – ROUTE/CONNECT MGD CABLES FROM LPCA TO REAR PEDESTAL

(1) Route/Connect Run 1035 to J21.

(2) Route/Connect Runs 728 thru 735 to J11 to J18.



(3) Route/Connect Run 1041 to J19.

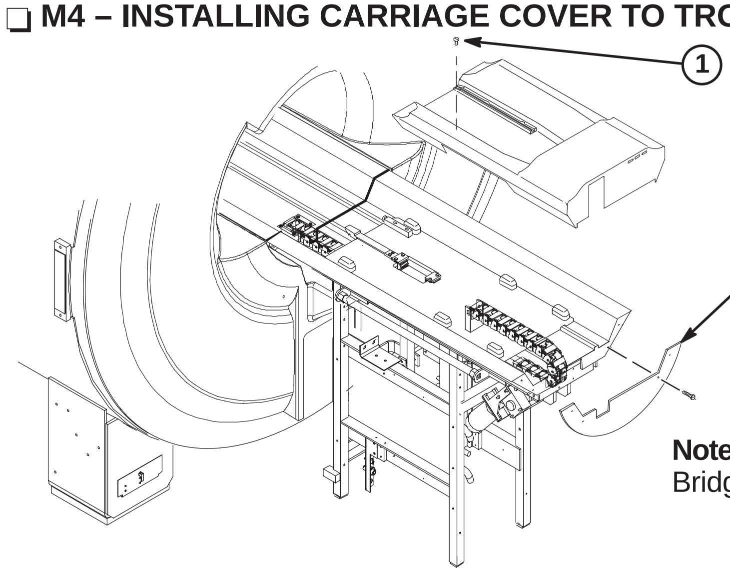
(4) Route/Connect Runs 1042 and 1049 to J5 and J19.

(5) Route/Connect Runs 491, 492, 493, 1036, 1037, 1038, 1039, and 1040 to J1 thru J8.

(6) Route Run 404 from hole in Bridge. Connect to J1.

❑ M4 – INSTALLING CARRIAGE COVER TO TROLLEY AND BRIDGE COVER

(1) Attach Carriage Cover to Trolley using four previously removed fasteners.



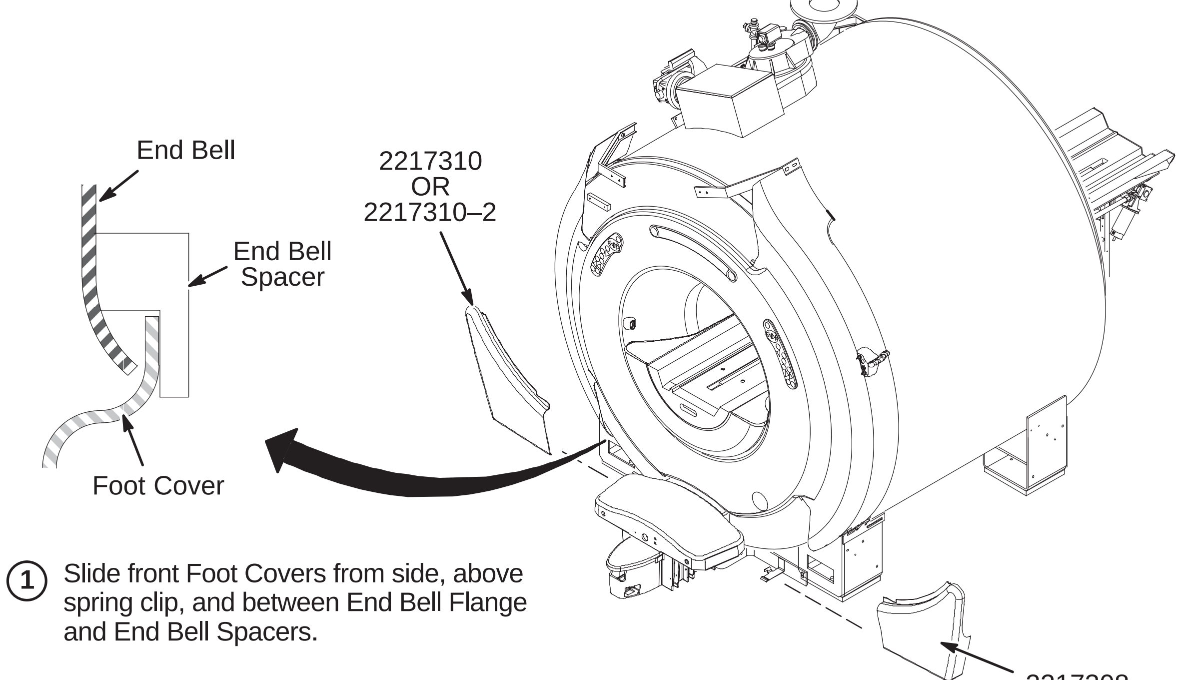
Replace previously removed Bridge Cover and fasten with (4) screws to end of Bridge.

Note:
Bridge Cover has enclosure rating plates on it.

INSTALLATION PROCEDURES SUMMARY

- ❑ **N1 – Install Enclosure Base Front Foot Covers**
- ❑ **N2 – Install Enclosure Top Covers**
- ❑ **N3 – Install Rear Pedestal Covers and Rating Plates**

❑ N1 – INSTALL ENCLOSURE BASE FRONT FOOT COVERS



1. Slide front Foot Covers from side, above spring clip, and between End Bell Flange and End Bell Spacers.

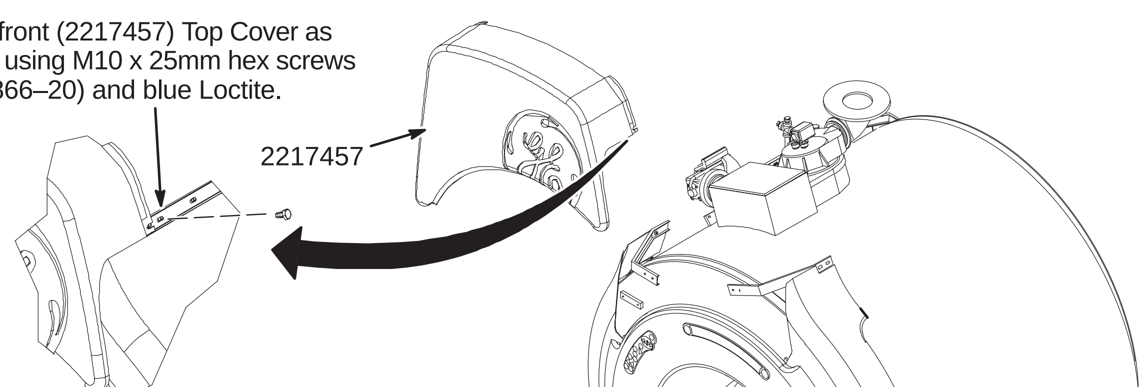
Note: The gap from the bottom of the Base Trim to floor should be a nominal 0.50 inch (13mm).

❑ N2 – INSTALL ENCLOSURE TOP COVERS

Note: All enclosure seams and gaps visible to the operator and patient shall be less than 0.25 inches (6.35mm). This applies to the mating seam of the two Side Top Covers to the Front Top Cover

Gap uniformity along any continuous seam shall be ± 0.0625 inch (1.59mm)

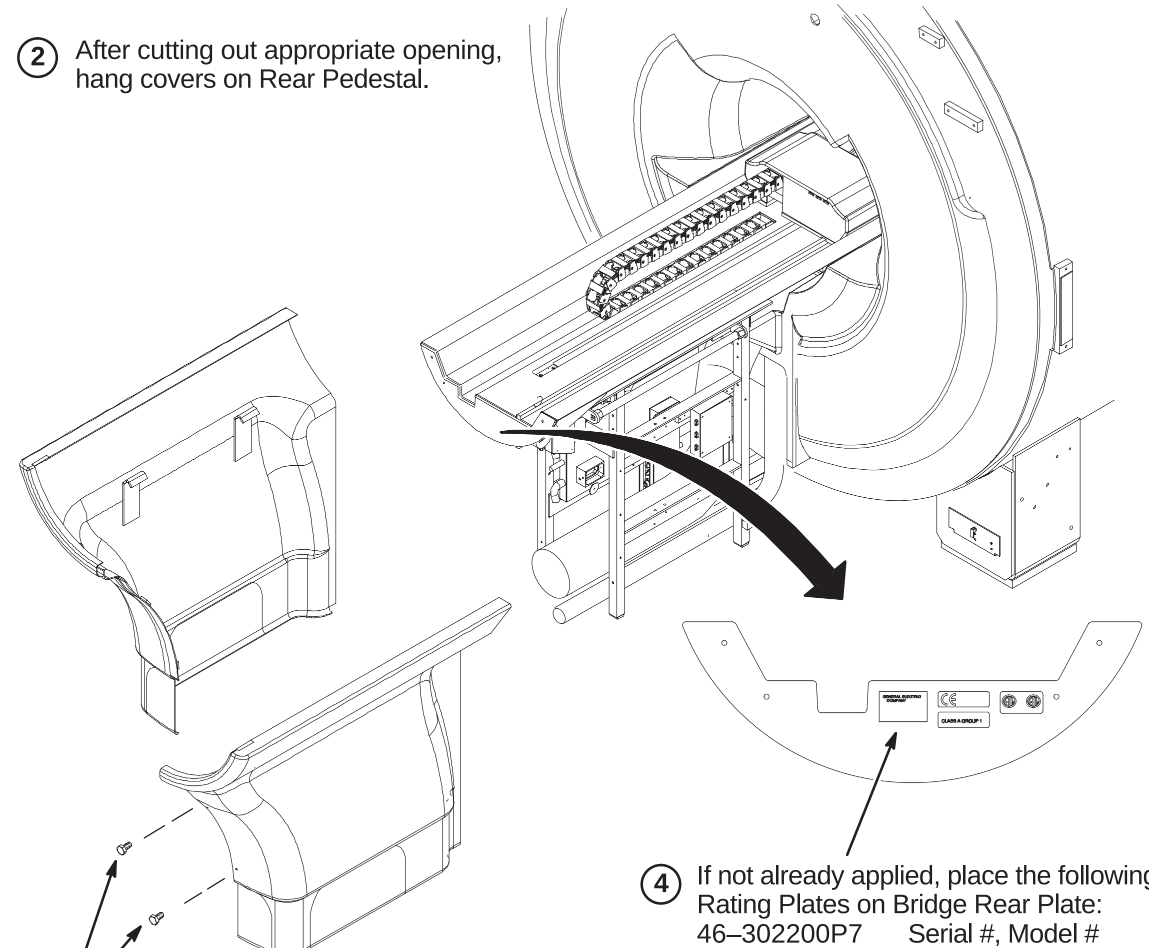
1. Install front (2217457) Top Cover as shown using M10 x 25mm hex screws (2109866–20) and blue Loctite.



❑ N3 – INSTALL REAR PEDESTAL COVERS AND RATING PLATES

1. Pedestal Covers (2217704 & 2217705) **require an opening to be cut out** depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting. Choose the opening that is required for the site.

2. After cutting out appropriate opening, hang covers on Rear Pedestal.



3. Fasten covers together with furnished hardware.

4. If not already applied, place the following Rating Plates on Bridge Rear Plate:

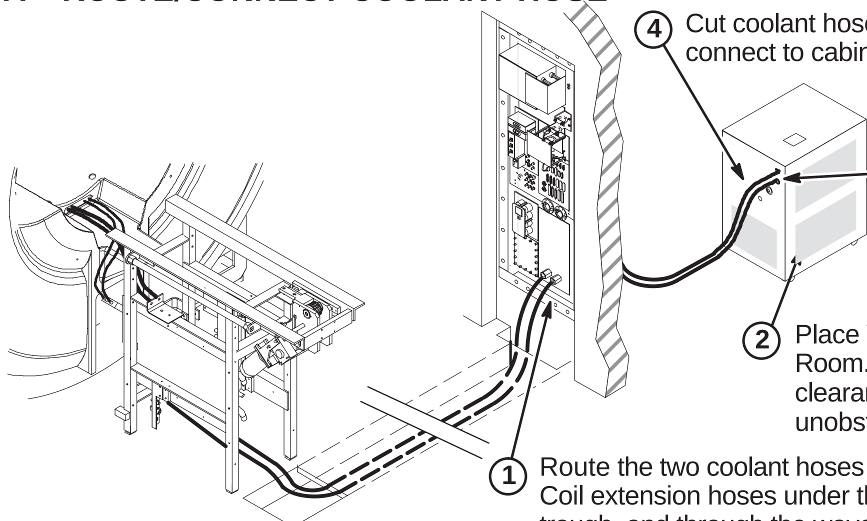
46–302200P7	Serial #, Model #
2142857	ETL List for Global Certification
2117390	CE International Approval
2132620	CE Marking Label Class A Group 1

WATER CHILLER CABINET INSTALLATION

INSTALLATION STEPS SUMMARY

- ❑ **A – Route/Connect Coolant Hose**
- ❑ **B – Install Power Cord**
- ❑ **C – Filling the System with Coolant**

❑ A – ROUTE/CONNECT COOLANT HOSE



4 Cut coolant hoses to appropriate length and connect to cabinet with Hose Clamps (2128542–2).

3 Remove two protective plugs. Wrap threads with Teflon tape and install provided $^{1}/_{2}$ inch hose barb fittings in these connections.

2 Place Water Chiller Cabinet in Equipment Room. Twenty–four inch (610mm) clearance on all sides is required for unobstructed flow of cooling air.

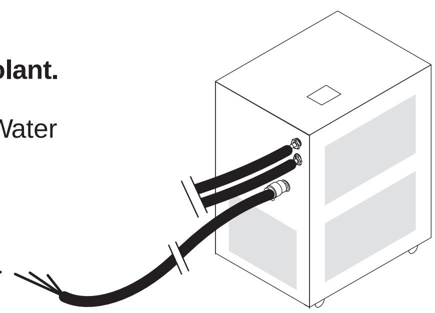
1 Route the two coolant hoses that were connected to the Body Coil extension hoses under the Rear Pedestal, through the cable trough, and through the wave guides on the Penetration Panel.

❑ B – INSTALL POWER CORD

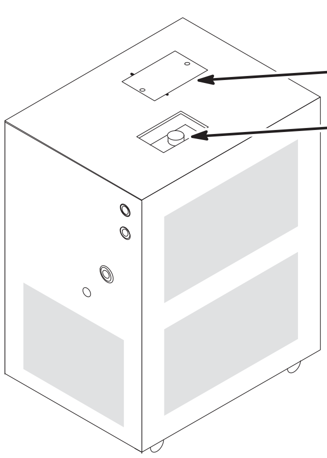
CAUTION:
Do not start Water Chiller pump before filling reservoir with coolant.
Consult vendor Water Chiller manual –
Installation, Operation, and Service Manual that is shipped with the Water Chiller.

1 **FOR RF/PDU:** Connect power cable Run 065 (2223355) to PDU as marked (TS1–26, 27, GND).

FOR ACGD/PDU: Connect power cable Run 970 (2280289) to PDU as markec (TS1–16,17,GND).



❑ C– FILLING THE SYSTEM WITH COOLANT AND START UP



1 Locate the twelve (12) gallons of Body Coil Coolant (2138791) that was shipped with System.

2 Remove cover.

3 Consult vendor Water Chiller manual – *Installation: Filling Requirements*. Initial Filling of Reservoir, and the following steps for the proper procedure on priming and filling the system with coolant.

4 Consult vendor Water Chiller manual – *Operation: Start Up.* Follow instructions in manual for starting water chiller operation.

5 **Verify unit is set to control at 20° C.**

6 If supplied, attach plastic sleeve to side of Water Chiller and insert vendor manual. This will allow easy access to any future service/ maintenance procedures. If plastic sleeve is not present, attach vendor manual to Water Chiller.

PENETRATION PANEL UPGRADE INSTRUCTIONS

Tab 4, Penetration Panel, contains upgrade instructions for the following configurations:

Signa LX

- *DIRECTION 2257262* – Signa LX WideOpen Penetration Panel Upgrade Instructions

Signa 10.x (MGD)

- *DIRECTION 2333493* – Signa 10.x (MGD) Penetration Panel Upgrade Instructions

Use only the applicable Penetration Panel Instruction Direction for the site being upgraded

SIGNA LX WideOpen PENETRATION PANEL UPGRADE INSTRUCTIONS

INSTALLATION STEPS SUMMARY

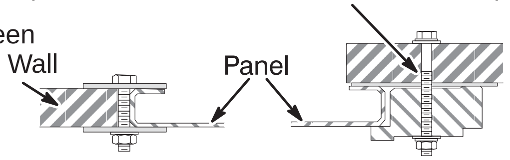
- ❑ **A – Remove/Install Penetration Panel**
- ❑ **B – Transfer of Penetration Panel Components**
- ❑ **C – Install Penetration Panel Floor Plate**
- ❑ **D – Install Penetration Panel Stand–Off Hardware**
- ❑ **E – Install Penetration Panel Covers**

❑ A – REMOVE/INSTALL PENETRATION PANEL

① If still in place, remove existing Penetration Panel. Install new Penetration Panel using the same mounting hardware of either method shown below. Apply copper tape (46–258218P4, 5, or 6) to seam between Panel and wall.

③ To insure good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**

② Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb–ft (1.5 to 2 turns after lock washer flattens).

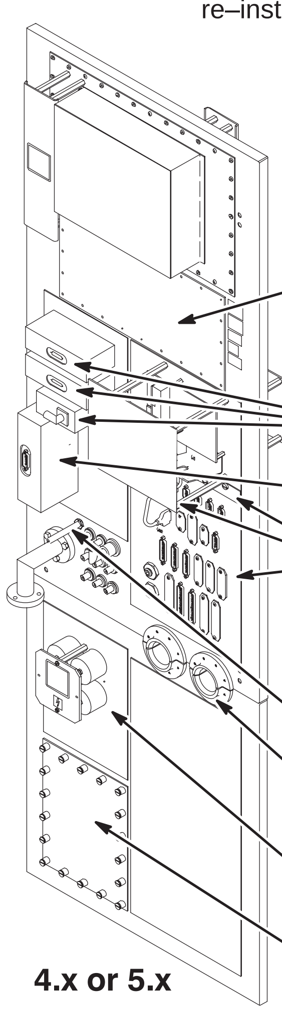


Brass Sealing Plate Method **Hold Down Method**

❑ B – TRANSFER OF PENETRATION PANEL COMPONENTS

CAUTION: All cable connections MUST BE LABELLED before disconnect.

- Customer may have added connections for options to Penetration Panel which are not labelled but must be reconnected on new panel.
- If the site has a Patient Alert Extender Box mounted on or near the Panel, re–install on new Penetration Panel.



4.x or 5.x

NOTE: **If old system is Signa Advantage 4.x**
Transfer the heliax connector, Step 4, to new Penetration Panel. Mount new Fiber Optic Repeater, Bar, and Filter Panel. Discard the rest.

If old system is a Signa Advantage 5.x
Perform the following steps.

① Transfer blank panel to new Penetration Panel re–using mounting hardware. If existing Pen Panel has Shim Filter, do not remove. A new blank panel (46–282638G1) and 24 mounting nuts (46–208937P5) should have been ordered for this upgrade.

② If part of existing system, transfer (3) Multi–coil boxes.

③ If present, do not transfer 20dB Gain Block (Spectroscopy).

④ Transfer Fiber Optic Repeater, Bar, and Filter Panel to new Penetration Panel re–using mounting hardware.

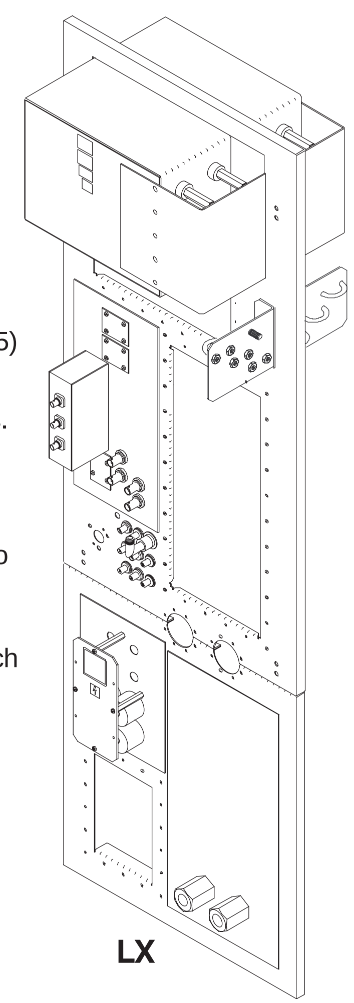
Note: If this is an early 5.X system without Repeater, Catalog M1090ZA should be part of this upgrade which includes this component.

⑤ Transfer Patient Alert Wave Guide.

⑥ **CAUTION! –** Older systems used berylium paste on wave guides and gas lines. WASH HANDS IF YOU COME IN CONTACT WITH THIS MATERIAL.

⑦ Transfer Filters to holes on new Penetration Panel re–using mounting hardware.

⑧ Transfer Panel to new Penetration Panel re–using mounting hardware.

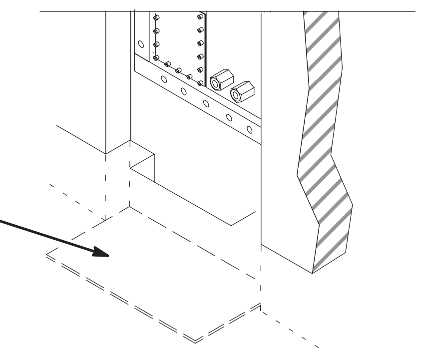


LX

❑ C – INSTALL PENETRATION PANEL FLOOR PLATE

Note: The Penetration Panel Floor plate (2120824) that is supplied with the cover kit is to be installed prior to routing and connecting cables to Penetration Panel.

① Remove backing from tape and stick Floor Plate to floor beneath Penetration Panel.



❑ D – INSTALL PENETRATION PANEL STAND–OFF HARDWARE

Note: The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

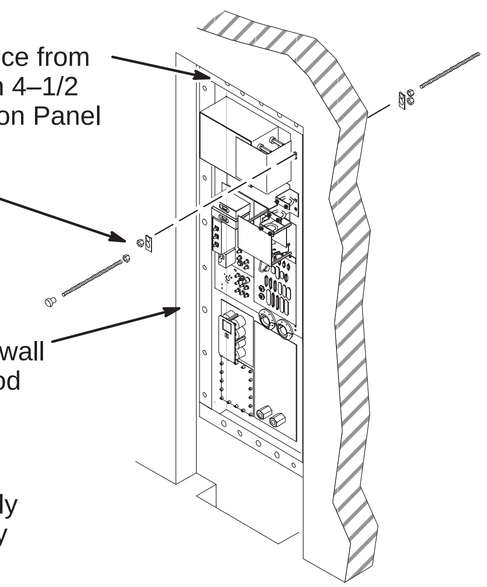
① Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4–1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.

② Place (8) Reinforcing Plates (46–301211P1) over each hole pair on both sides of Penetration Panel and attach using:
(16) 5/16–18 x 9/16 Brass Nut (46–208935P9)
(8) 5/16–18 x 18 inch Threaded Rod (46–301213P2)
(8) 5/16–18 x 1 inch Barrel Nut (46–301212P1)

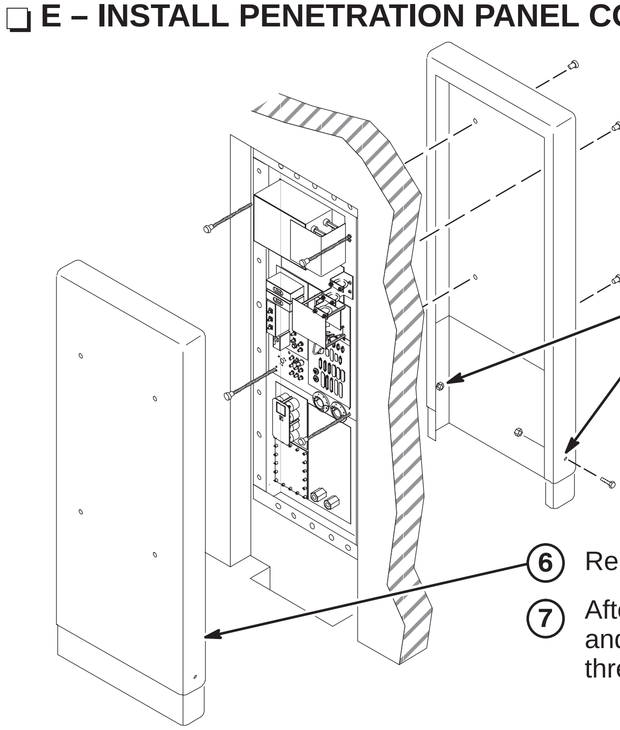
③ Measure distance from face of Pen Panel to equipment room wall surface. Add 7.5 inches (190mm) to obtain length threaded rod should extend from panel.

④ Repeat step 3 procedure for exam room cover.

Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



❑ E – INSTALL PENETRATION PANEL COVERS



① Remove all barrel nuts.

② In the Magnet room, temporarily install Top (46–306936P1) and Bottom (46–306964P1) Cover assembly on threaded rods and secure with four barrel nuts.

③ Adjust position of bottom cover section and mark two lower fastener holes.

④ Remove covers and drill hole in lower cover at each mark.

⑤ Attach top and bottom covers with furnished 10–32 nuts and screws.

⑥ Repeat procedure for Equipment room cover.

⑦ After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

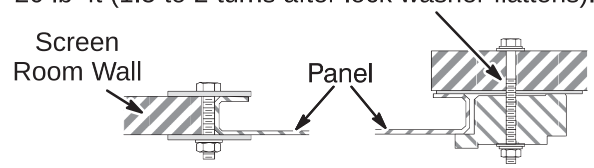
INSTALLATION STEPS SUMMARY

- ❑ **A – Remove/Install Penetration Panel**
- ❑ **B – Transfer of Penetration Panel Components**
- ❑ **C – Install Penetration Panel Floor Plate**
- ❑ **D – Install Penetration Panel Stand–Off Hardware**
- ❑ **E – Install Penetration Panel Covers**

❑ A – REMOVE/INSTALL PENETRATION PANEL

WARNING:
Sites with Oxford Magnets may have an Ion Pump installed on the magnet. The Ion Pump Power supply provides 4,000 volts to the Ion Pump via a cable connected through the penetration panel. To prevent electrical shock during servicing of the Penetration Panel, remove AC power to the Ion Pump Power Supply, located in the Equipment Room.

1. If still in place, remove existing Penetration Panel. Install new Penetration Panel using the same mounting hardware of either method shown below. Apply copper tape (46–258218P4, 5, or 6) to seam between Panel and wall.
2. Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb–ft (1.5 to 2 turns after lock washer flattens).
3. To insure good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**

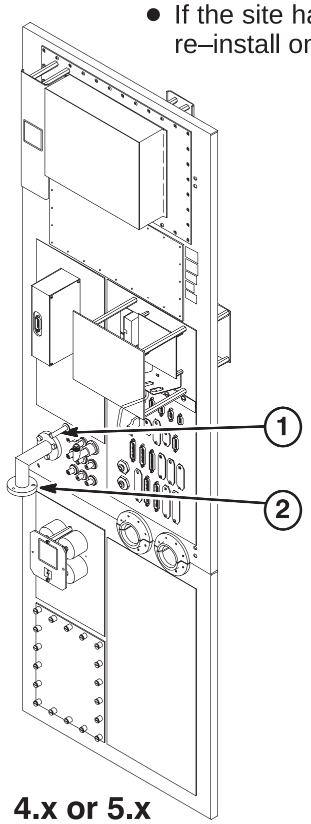


Brass Sealing Plate Method **Hold Down Method**

❑ B – TRANSFER OF PENETRATION PANEL COMPONENTS

CAUTION: All cable connections **MUST BE LABELLED** before disconnect.

- Customer may have added connections for options to Penetration Panel which are not labelled but must be reconnected on new panel.
- If the site has a Patient Alert Extender Box mounted on or near the Panel, re–install on new Penetration Panel.



4.x or 5.x

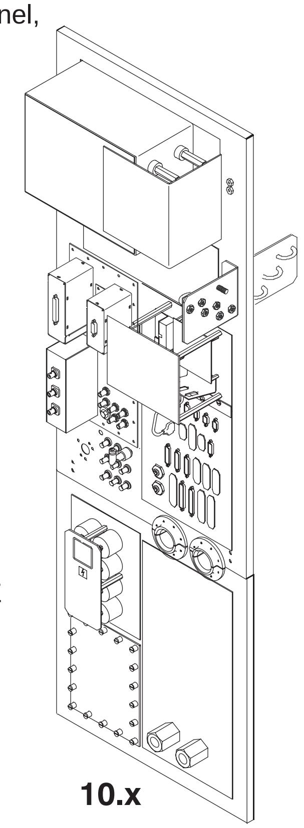
NOTE:

If old system is Advantage 4.x or 5.x
Transfer the heliax connector and patient alert wave guide to new Penetration Panel. Recycle the rest of old penetration panel.

If old system is a Horizon 5.x or LX
Refer to Direction 2333499 for installation of new EXCITE Adapter plate and connector at J51.

1. Transfer Patient Alert Wave Guide to new Penetration Panel re–using mounting hardware.
2. Transfer Heliax Connector to new Penetration Panel re–using mounting hardware.

CAUTION! – OLDER SYSTEMS USED BERYLIUM PASTE ON WAVE GUIDES AND GAS LINES. WASH HANDS IF YOU COME IN CONTACT WITH THIS MATERIAL.

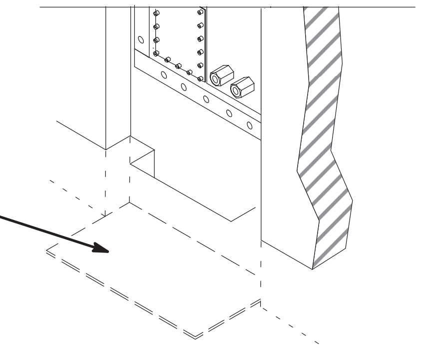


10.x

❑ C – INSTALL PENETRATION PANEL FLOOR PLATE

Note: The Penetration Panel Floor plate (2120824) that is supplied with the cover kit is to be installed prior to routing and connecting cables to Penetration Panel.

1. Remove backing from tape and stick Floor Plate to floor beneath Penetration Panel.

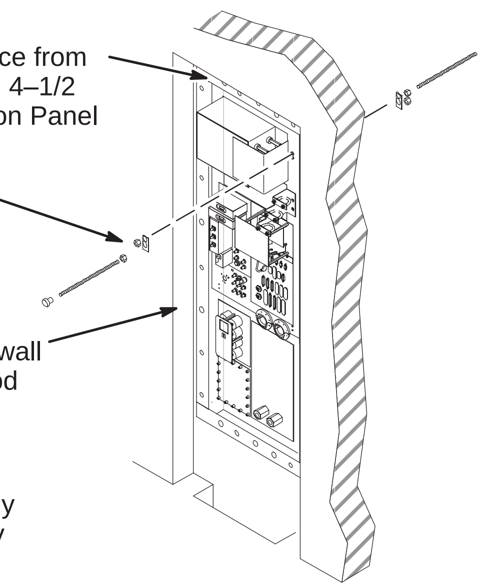


❑ D – INSTALL PENETRATION PANEL STAND–OFF HARDWARE

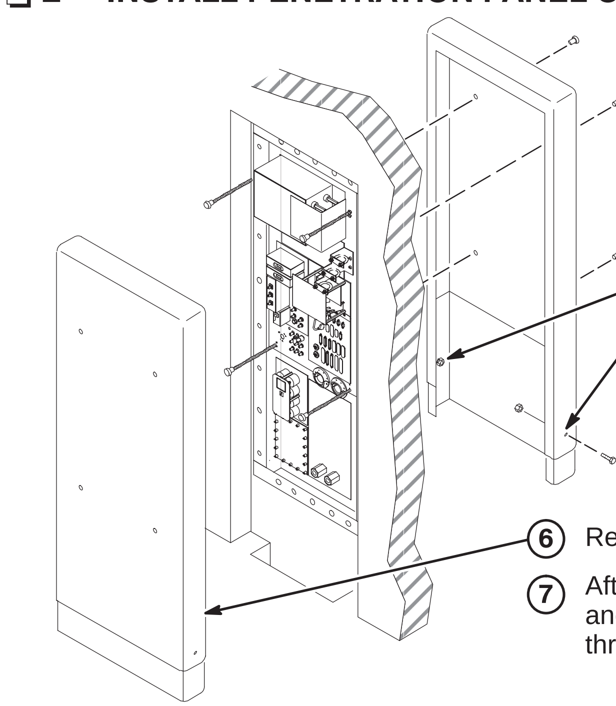
Note: The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

1. Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4–1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
2. Place (8) Reinforcing Plates (46–301211P1) over each hole pair on both sides of Penetration Panel and attach using:
   (16) 5/16–18 x 9/16 Brass Nut (46–208935P9)
   (8) 5/16–18 x 18 inch Threaded Rod (46–301213P2)
   (8) 5/16–18 x 1 inch Barrel Nut (46–301212P1)
3. Measure distance from face of Pen Panel to equipment room wall surface. Add 7.5 inches (190mm) to obtain length threaded rod should extend from panel.
4. Repeat step 3 procedure for exam room cover.

   Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



❑ E – INSTALL PENETRATION PANEL COVERS



1. Remove all barrel nuts.
2. In the Magnet room, temporarily install Top (46–306936P1) and Bottom (46–306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
3. Adjust position of bottom cover section and mark two lower fastener holes.
4. Remove covers and drill hole in lower cover at each mark.
5. Attach top and bottom covers with furnished 10–32 nuts and screws.
6. Repeat procedure for Equipment room cover.
7. After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

SYSTEM CABINET UPGRADE INSTRUCTIONS

Tab 8, System Cabinet, contains upgrade instructions for the following configurations:

Signa LX

- *DIRECTION 2170004* – Signa 8.x/9.x System Cabinet Installation Instructions
- *DIRECTION 2254246* – Signa WideOpen LX System Cabinet Upgrade Instructions

Signa 10.x (MGD)

- *DIRECTION 2338502* – Signa 10.x System Cabinet (MGD) Installation Instructions

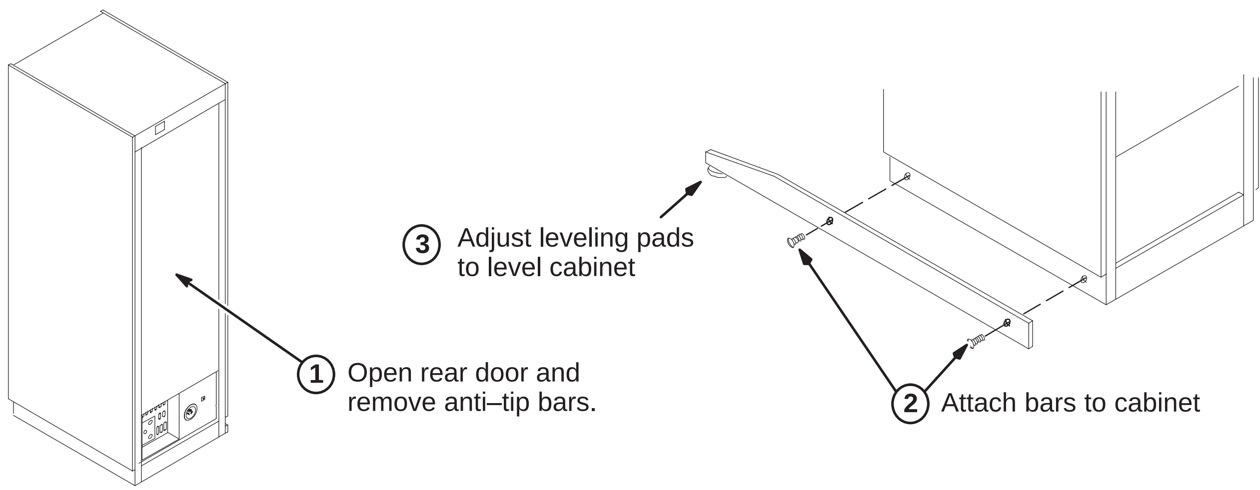
Use only the applicable System Cabinet Upgrade Instruction Direction for the site being upgraded

INSTALLATION STEPS SUMMARY

- ❑ **A – Position/Secure System Cabinet**
- ❑ **B – Install Magnet Monitoring Cables**
- ❑ **C – Connect Fiber Optic Runs 708, 710, 711/712, and 836 to I/F Panel**
- ❑ **D – Route/Connect Fiber Optic Runs 836 to BIT3 Module**
- ❑ **E – Route/Connect Fiber Optic Runs 708, 710, and 711/712 to Tyme II**
- ❑ **F – Connect Power, Ground, and Data Cables**
- ❑ **G – Install InSite "SYSTEMS" Cabinet Magnet**

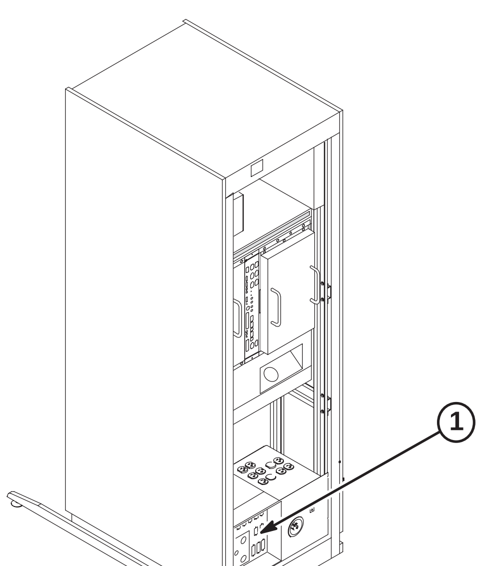
❑ A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements by either installing anti tip legs (see illustration below) or securing the cabinet to the floor with local supplied brackets and floor anchors.



❑ B – INSTALL MAGNET MONITORING CABLES

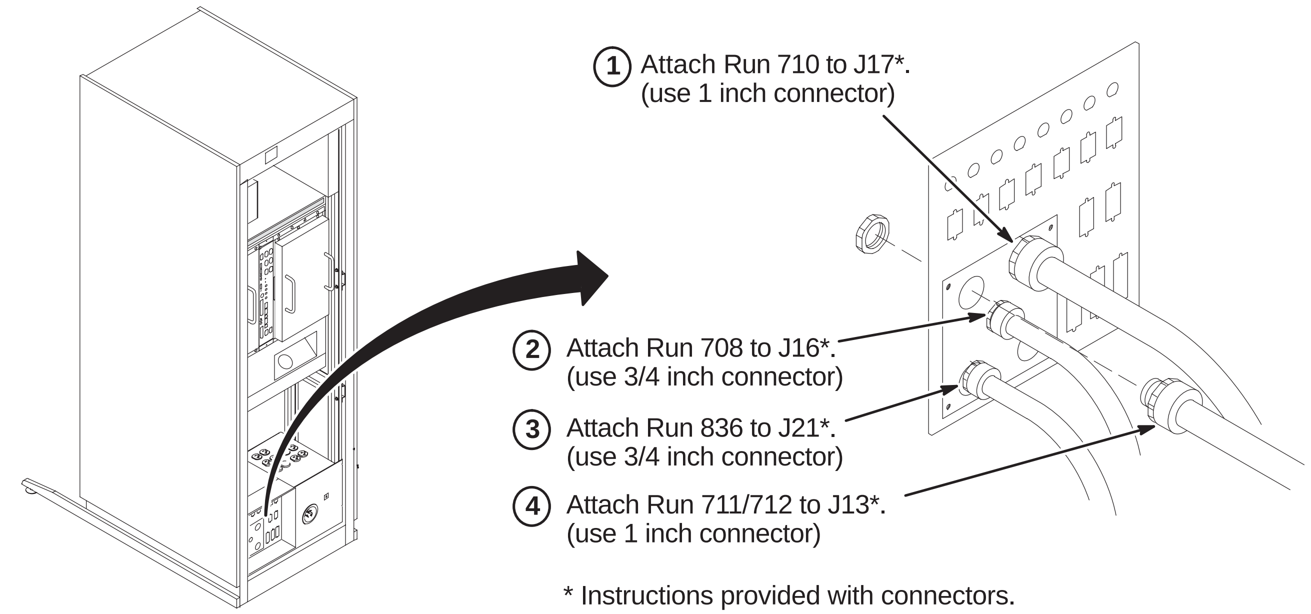
Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual,* for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



❑ C – CONNECT FIBER OPTIC RUNS 708, 710, 711/712, AND 836 TO I/F PANEL

CAUTION

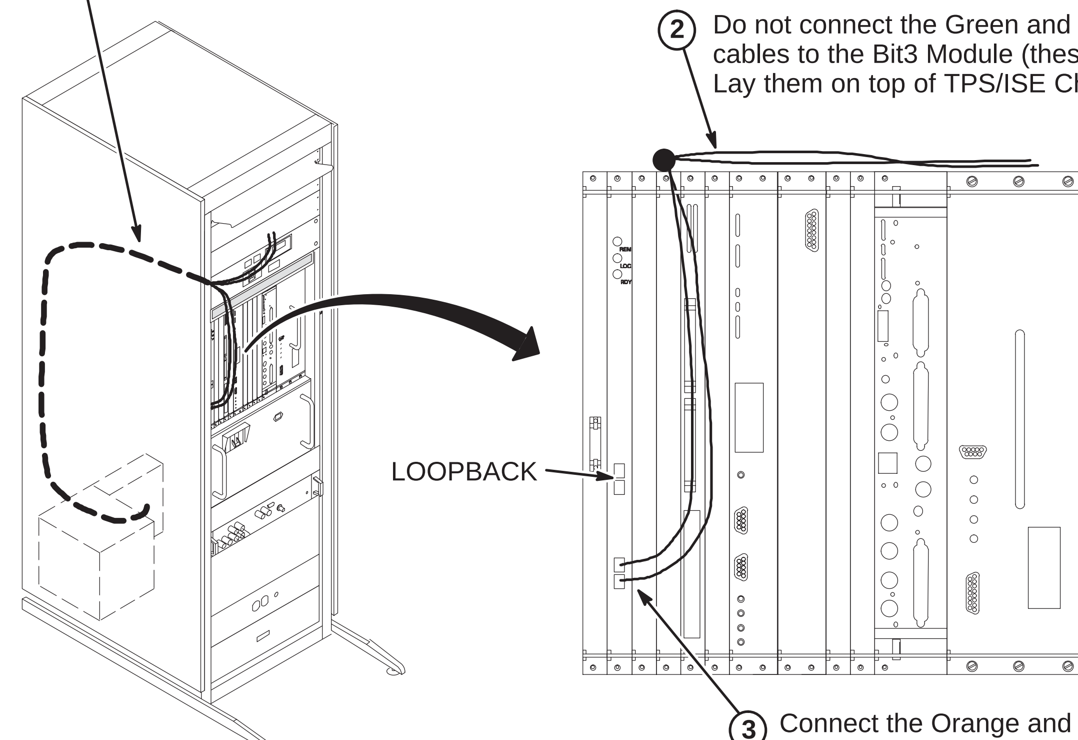
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



* Instructions provided with connectors.

❑ D – ROUTE/CONNECT FIBER OPTIC RUN 836 TO BIT3 MODULE

1. Route Run 836 fiber optic cables from cabinet I/F J21 to front of Bit3 module (MR2 A11 A30). Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**



❑ E – ROUTE/CONNECT FIBER OPTIC RUNS 708, 710, AND 711/712 TO TYME II

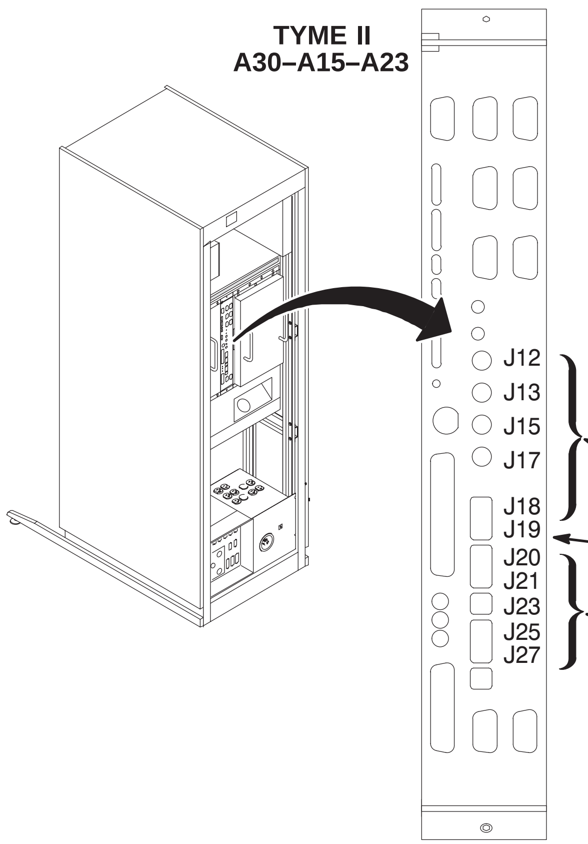
CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.

Note:
Runs 708, 710, 711, and 712 are used for both Horizon 5.x and 8.x. The fiber optic connections to the TYME board are different for 5.x and 8.x. The labels on the cables include both the 5.x and 8.x connections.

The table below indicates the connection labels as they should be marked on this end of the fiber optic cables. **Use only the 8.x connections.**

TYME II
A30–A15–A23



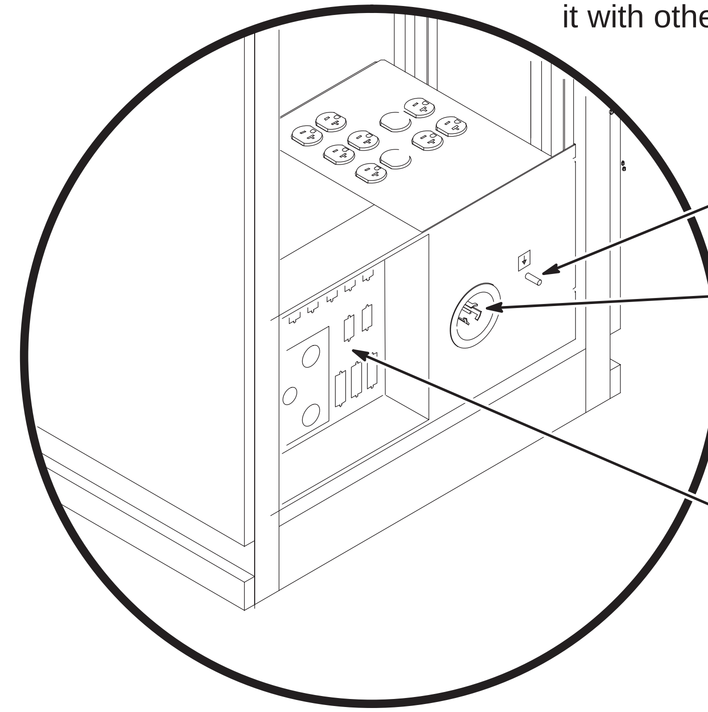
RUN #	5.x	8.x
708–1	J5	J19
710–1	J13	J12
710–2	J15	J13
710–3	J17	J15
710–4	J14	J17
710–5	J4	J18
711–1	J6,J7	J25,J27
711–2	J8	J23
712–1	J9,J10	J20,J21

1. Route/Connect as marked Run 710 fiber optic cables to J12, J13, J15, J17 and J18 on the TYME II Board.
2. Route/Connect as marked Run 708 fiber optic cable to J19 on the TYME II Board.
3. Route/Connect as marked Run 711/712 fiber optic cables to J20, J21, J23, J25, and J27 on the TYME II Board.

Note:
Use tie-wraps to secure fiber optic cables to ISE Chassis cables.

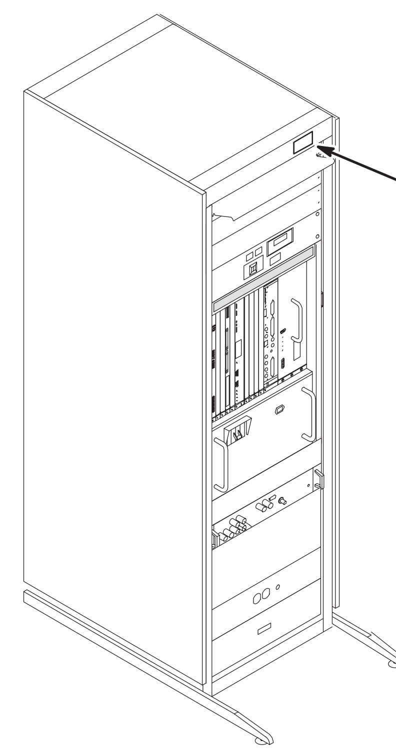
❑ F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.



1. Connect Run 037 GND cable to ground stud.
2. Connect Run 030 or 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.
4. Connect remaining cabinet cables as marked.

❑ G – INSTALL INSITE "SYSTEMS" CABINET MAGNET



1. From the 8.0 LX Insite Kit (46–301708G14), locate the "Systems" cabinet magnet (46–320095P5) and attach to front of cabinet as shown.

INSTALLATION STEPS SUMMARY

- ❑ A1 – System Cabinet Upgrade Guidelines
- ❑ A2 – Position/Secure System Cabinet
- ❑ A3 – Cryogen Meter Removal for Oxford or S2 to SX System Cabinets
- ❑ A4 – Cryogen Meter Removal for S1 System Cabinets

❑ A1 – SYSTEM CABINET UPGRADE GUIDELINES

The Signa LX System Cabinet Upgrade includes a new cabinet which may require items to be transferred from the old cabinet, such as the TNF module.

This Upgrade also provides instructions for other components and options being transferred to the new System Cabinet. Note that various parts will be left over according to the type of system being upgraded. Return these parts for recycling.

Deinstalled System cabients with cryogen meters must be returned for recycling.

For any of the upgrades below, make sure to attach Rating Plate as shown in procedure **D4.**

UPGRADE FROM SIGNA ADVANTAGE 4.x SYSTEM CABINET
Any Signa Advantage 4.x System being upgraded to Signa WideOpen LX HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- To position cabinet and connect system cables, see prodedures **A2, C1, C2, C3,** and **C4.**
- The new Horizon LX System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old two–bay System Cabinet, it will need to be transferred. Refer to procedure **B1, B2,** and **B3 or B4.** If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade. If multi–coil is being installed, Catalogs M1090ZC or M1090ZG should have been ordered prior to upgrade.

UPGRADE FROM SIGNA ADVANTAGE 5.x SYSTEM CABINET
Any Signa Advantage 5.x System being upgraded to Signa WideOpen LX HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

- To position cabinet and connect system cables, see prodedures **A2, C1, C2, C3,** and **C4.**
- The new Horizon LX System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Advantage 5.x System Cabinet, it will need to be transferred. Refer to procedure **B1, B2,** and **B3 or B4.** If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade. If multi–coil is being installed, Catalogs M1090ZC or M1090ZG should have been ordered prior to upgrade.

UPGRADE FROM SIGNA HORIZON 5.X SYSTEM CABINET
Any Signa Horizon 5.x System being upgraded to Signa WideOpen LX HiSpeed or EchoSpeed will receive a new System Cabinet without the TNF Module.

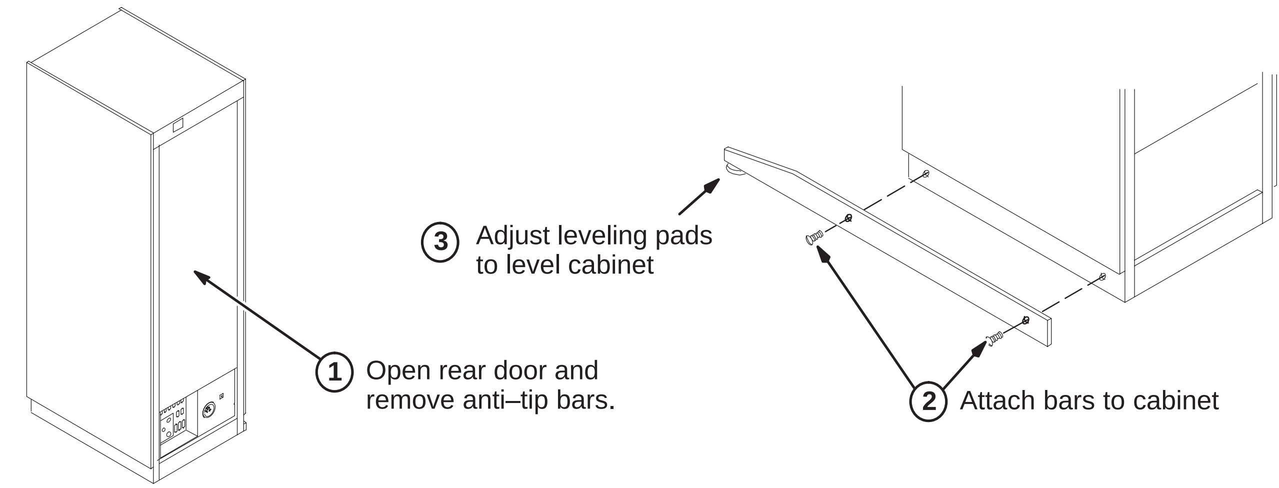
- To position cabinet and connect system cables, see prodedures **A2, C1, C2, C3,** and **C4.**
- The new Horizon LX System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Horizon 5.x System Cabinet, it will need to be transferred. Refer to procedure **B1, B2,** and **B3 or B4.** If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade. If multi–coil is being installed, Catalogs M1090ZC or M1090ZG should have been ordered prior to upgrade.
- To position cabinet and connect system cables, see prodedures **A2, C1, C2, C3,** and **C4.**
- For upgrading from a Horizon 5.x EchoSpeed to a Horizon LX Echospeed, the Fast Receiver and cable (2137306) will need to be transferred, refer to procedures **D1, D2,** and **D3.**

UPGRADE FROM SIGNA HORIZON 8.X SYSTEM CABINET
Any Signa Horizon 8.x System being upgraded to Signa WideOpen LX HiSpeed or EchoSpeed will have Magnet Monitoring installed.

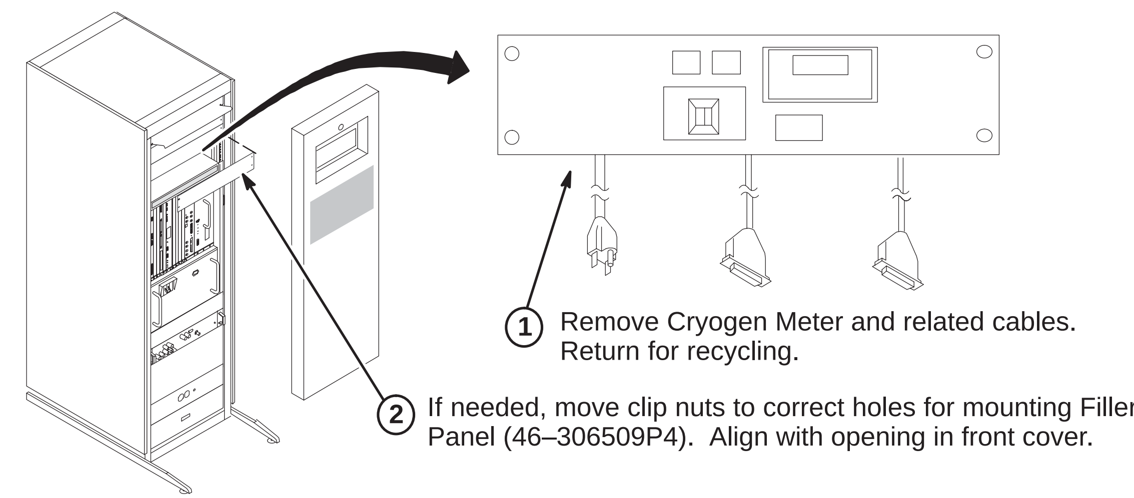
- See procedures **A3** or **A4** for removal of cryogen meters and installation of filler panel(s).

❑ A2 – POSITION/SECURE SYSTEM CABINET

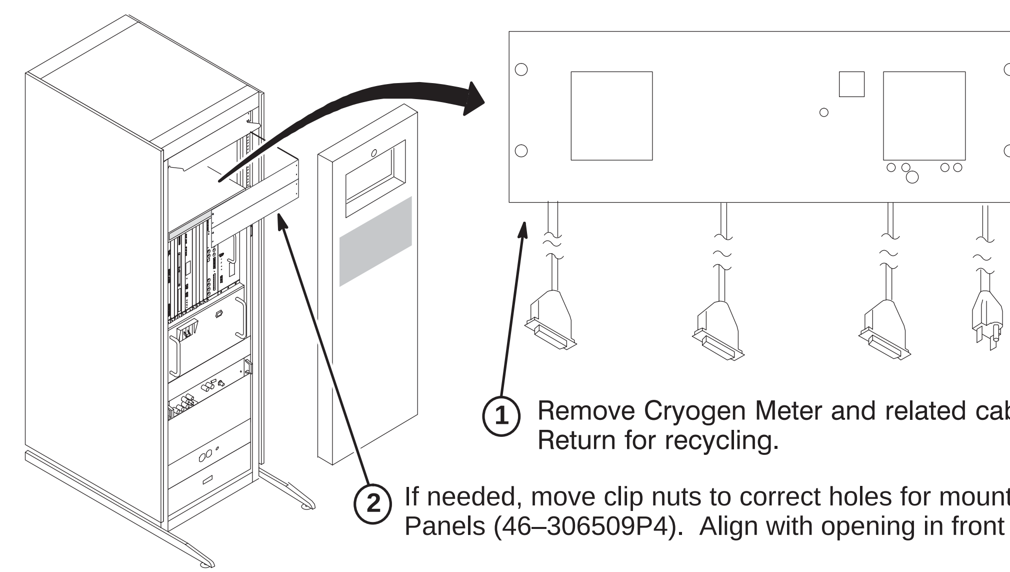
Comply with UL requirements by either installing anti tip legs (see illustration below) or securing the cabinet to the floor with local supplied brackets and floor anchors.



❑ A3 – CRYOGEN METER REMOVAL FOR OXFORD OR GE S2–SX SYSTEM CABINET



❑ A4 – CRYOGEN METER REMOVAL FOR GE S1 SYSTEM CABINET



INSTALLATION STEPS SUMMARY

- ❑ **B1 – Remove TNF Module from Existing Cabinet**
- ❑ **B2 – Install TNF Module into New 8.x Cabinet**
- ❑ **B3 – TNF Module Wiring Diagram**
- ❑ **B4 – TNF Module with Expansion Assembly Wiring Diagram**

❑ B1 – REMOVE TNF MODULE FROM EXISTING CABINET

For Upgrading from Horizon 5.x: TNF Assembly is present in existing cabinet for transfer. The TNF Expansion Asm (2118107) should also have been installed if Multi–coil is being transferred.

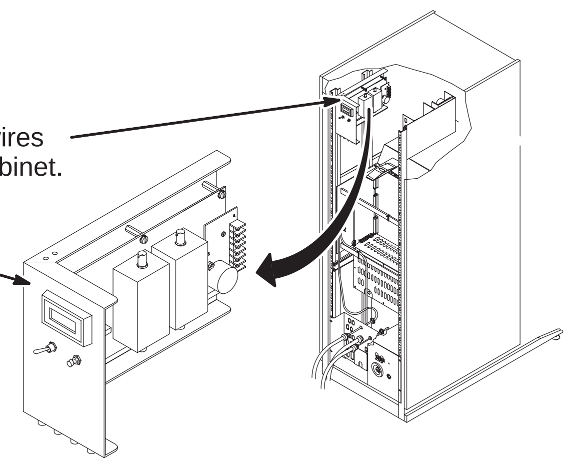
For Upgrading from Advantage 5.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi–coil system or 2118106 and 2118107 for Multi–coil.

For Upgrading from Advantage 4.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi–coil system or 2118106 and 2118107 for Multi–coil.

For Upgrading from Advantage 3.x: TNF Assembly should be present in new 8.x cabinet. The TNF Expansion Asm should also have been installed if Multi–coil is part of new 8.x system. If not present, order 2118107.

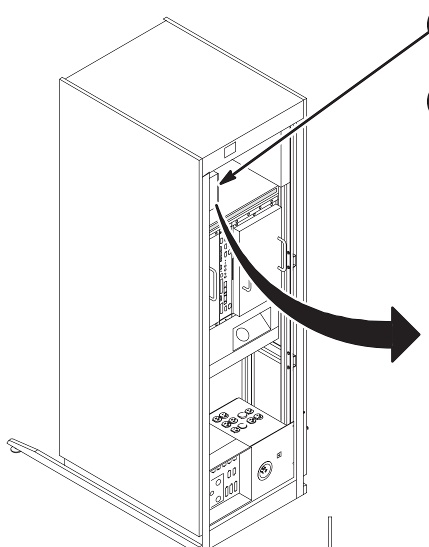
Note: TNF Module is a static sensitive component and needs to be handled as such (wrist strap).

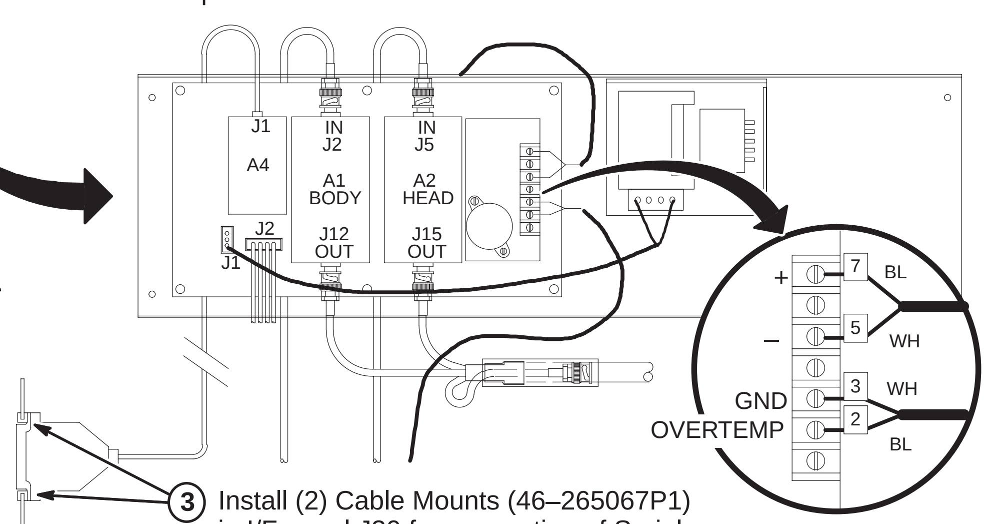
1. Disconnect and discard all cables and wires connected to TNF Module in Horizon cabinet.
2. Unfasten and remove TNF Assembly Module (with Expansion Asm if present). Save mounting hardware.



❑ B2 – INSTALL TNF MODULE INTO NEW 8.x CABINET

1. Install TNF Module into new Horizon 8.x Cabinet with hardware remove in Procedure **B1.**
2. Refer to Procedures **B3** and **B4** for wiring diagrams with and without Expansion Module**.**

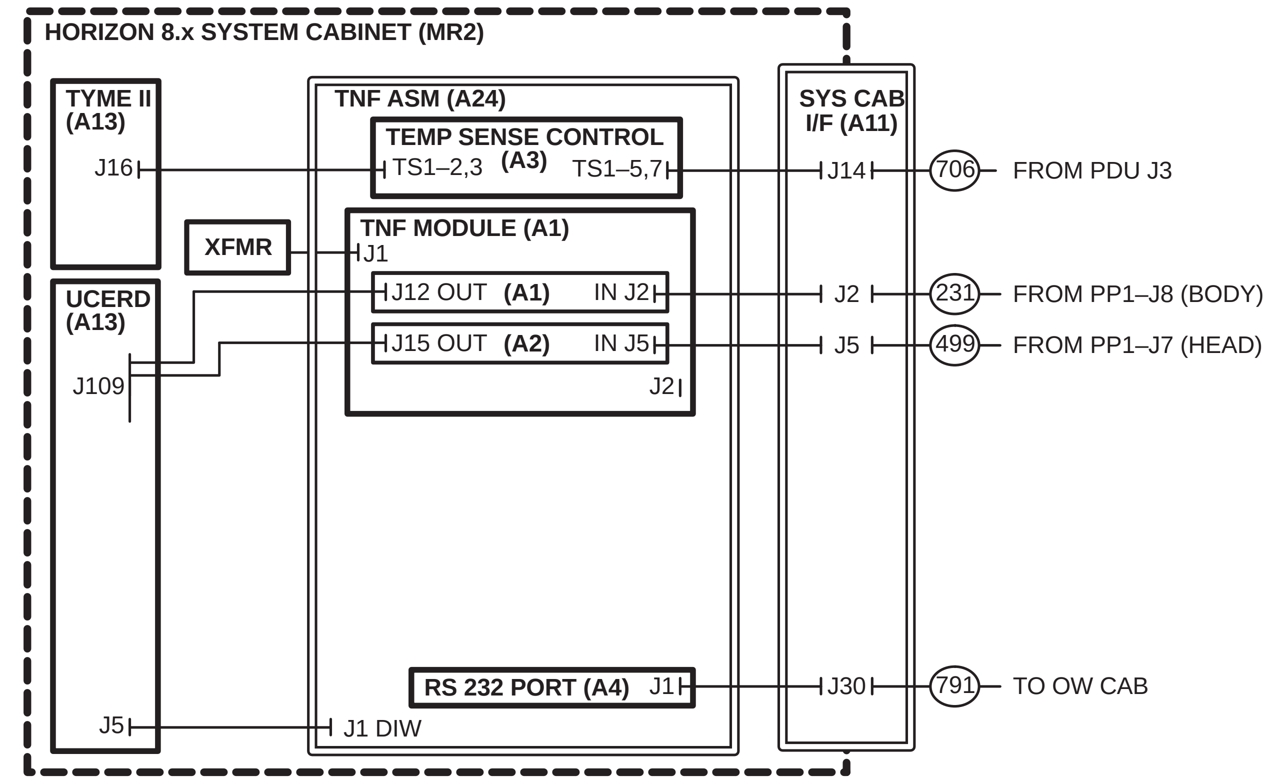




3. Install (2) Cable Mounts (46–265067P1) in I/F panel J30 for connection of Serial Port Cable to Run 787.

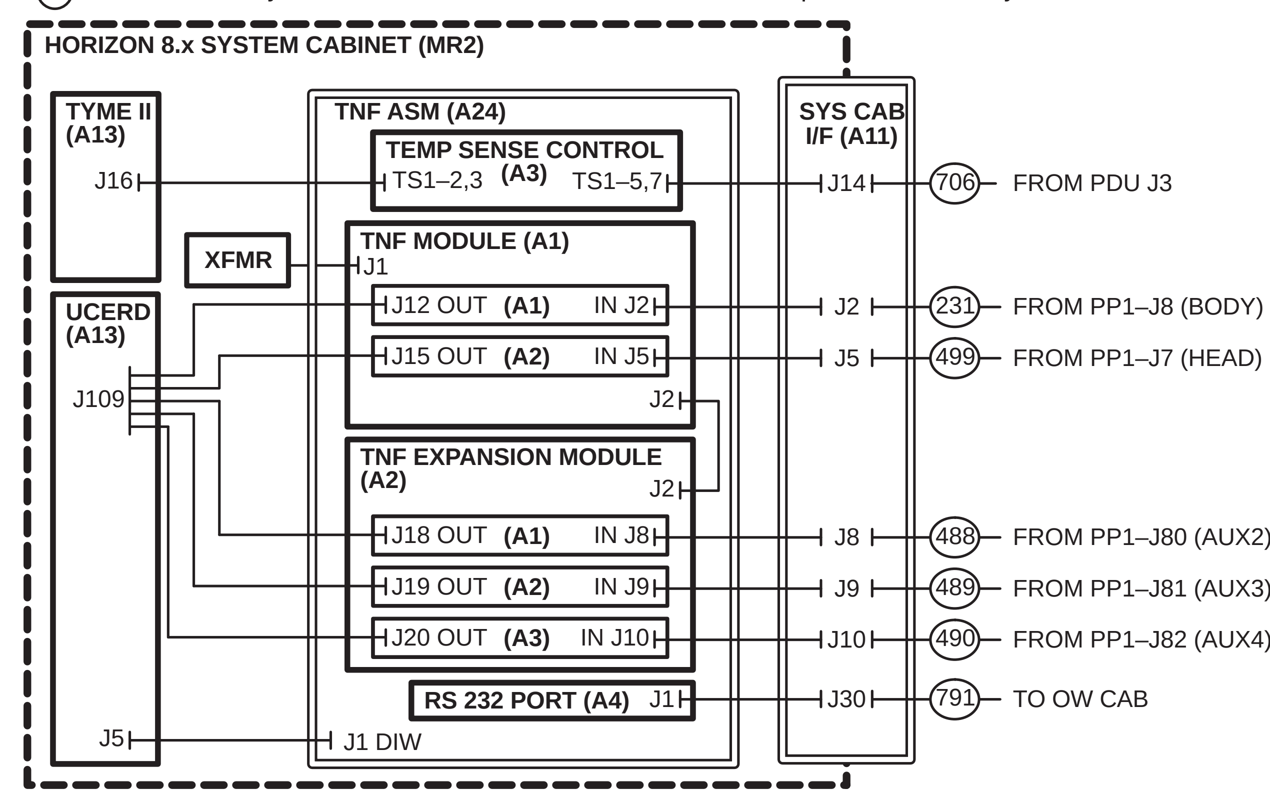
❑ B3 – TNF MODULE WIRING DIAGRAM

1. Connect new System Cabinet cables to TNF Module as indicated below.



❑ B4 – TNF MODULE WITH EXPANSION ASSEMBLY WIRING DIAGRAM

1. Connect new System Cabinet cables to TNF Module with Expansion Assembly as indicated below.



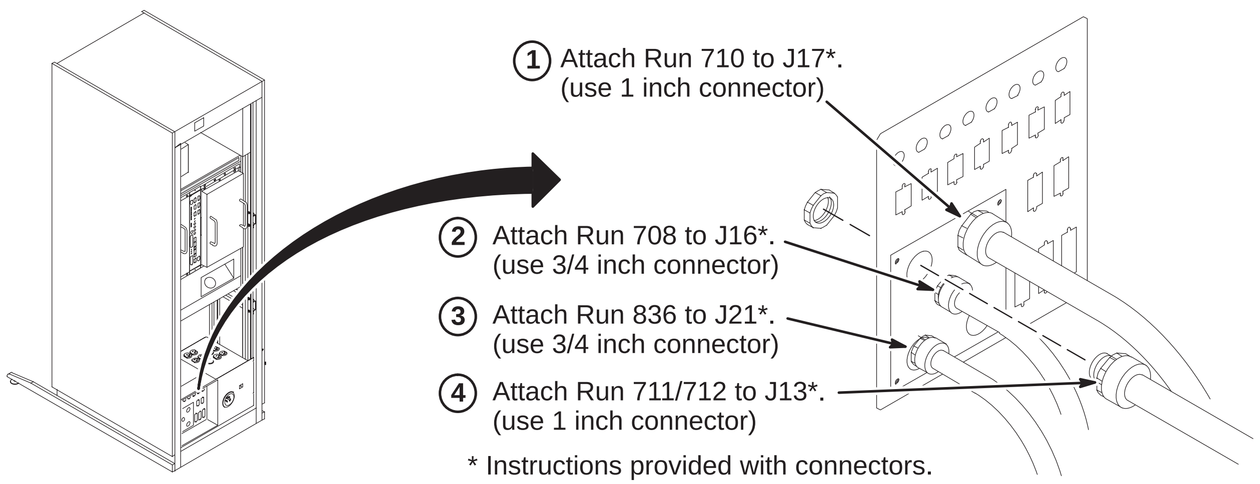
INSTALLATION STEPS SUMMARY

- ❑ C1 – Connect Fiber Optic Runs 708, 710, 711/712, and 836 to I/F Panel
- ❑ C2 – Route/Connect Fiber Optic Runs 836 to BIT3 Module
- ❑ C3 – Route/Connect Fiber Optic Runs 708, 710, and 711/712 to Tyme II
- ❑ C4 – Connect Power, Ground, and Data Cables

❑ C1 – CONNECT FIBER OPTIC RUNS 708, 710, 711/712, AND 836 TO I/F PANEL

CAUTION

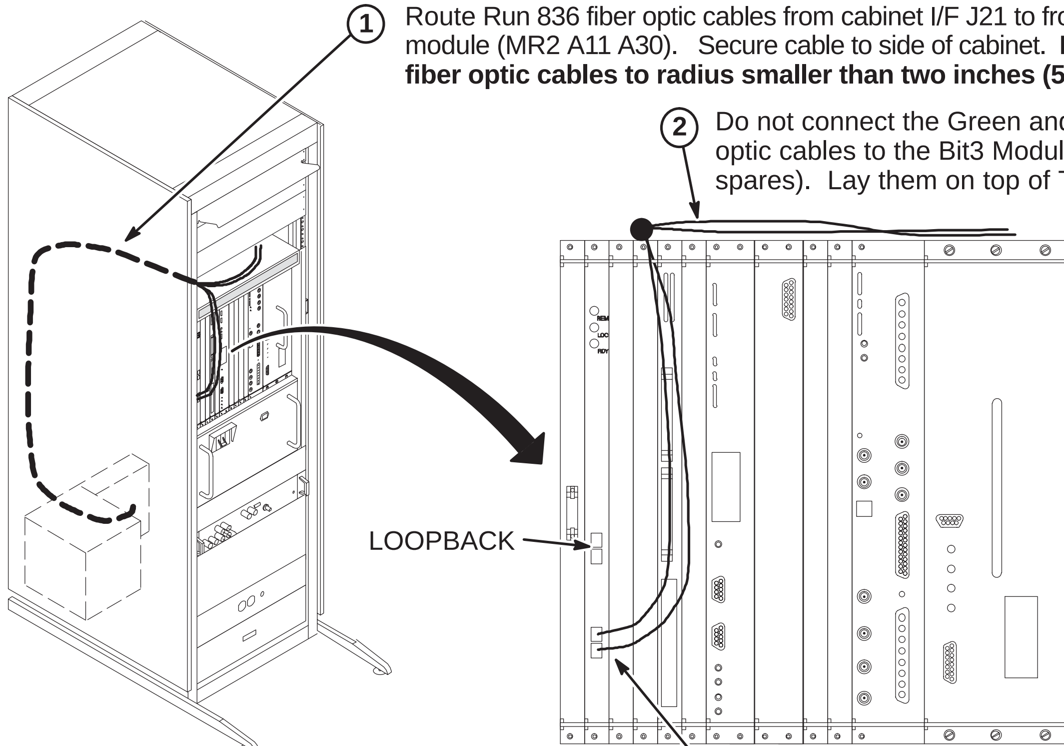
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



1. Attach Run 710 to J17*. (use 1 inch connector)
2. Attach Run 708 to J16*. (use 3/4 inch connector)
3. Attach Run 836 to J21*. (use 3/4 inch connector)
4. Attach Run 711/712 to J13*. (use 1 inch connector)

* Instructions provided with connectors.

❑ C2 – ROUTE/CONNECT FIBER OPTIC RUN 836 TO BIT3 MODULE

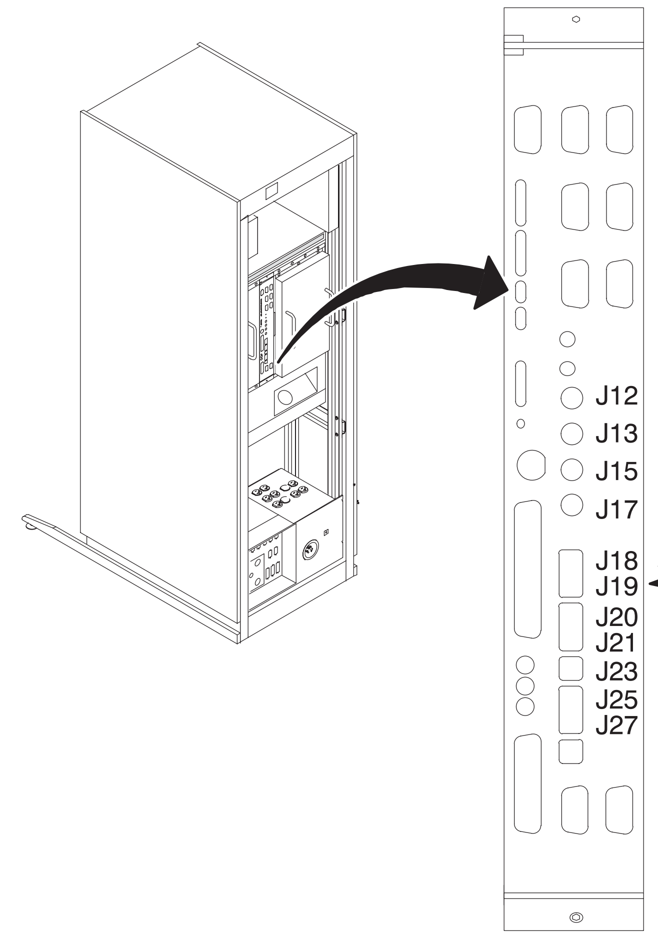


1. Route Run 836 fiber optic cables from cabinet I/F J21 to front of Bit3 module (MR2 A11 A30). Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**
2. Do not connect the Green and Brown fiber optic cables to the Bit3 Module (these are spares). Lay them on top of TPS/ISE Chassis.
3. Connect the Orange and Blue fiber optic cables to the Bit3 Module as indicated.

❑ C3 – ROUTE/CONNECT FIBER OPTIC RUNS 708, 710, AND 711/712 TO TYME II

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.

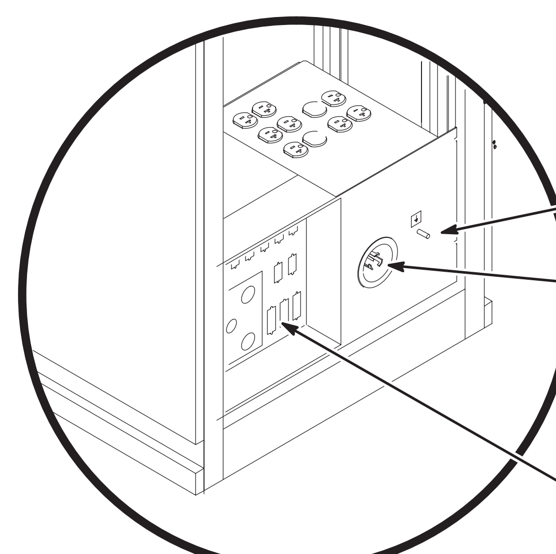


TYME II
MR2–A15–A23

Note:
Use tie-wraps to secure fiber optic cables to ISE Chassis cables.

1. Route/Connect as marked Run 710 fiber optic cables to J12, J13, J15, J17 and J18 on the TYME II Board.
2. Route/Connect as marked Run 708 fiber optic cable to J19 on the TYME II Board.
3. Route/Connect as marked Run 711/712 fiber optic cables to J20, J21, J23, J25, and J27 on the TYME II Board.

❑ C4 – CONNECT POWER, GROUND, AND DATA CABLES

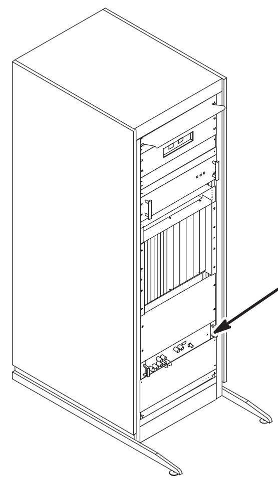


1. Connect Run 037 GND cable to ground stud.
2. Connect Run 030 cable plug to “J1”. Be sure to push and turn plug for secure connection.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.
4. Connect remaining cabinet cables as marked.

INSTALLATION STEPS SUMMARY

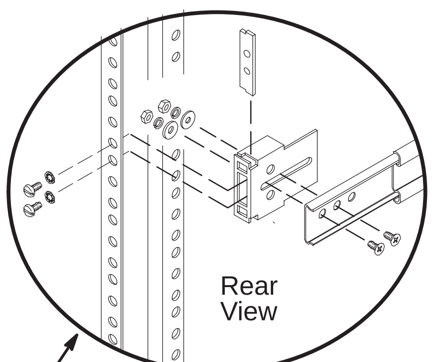
- ❑ **D1 – Remove Fast Receiver Module form Horizon System Cabinet**
- ❑ **D2 – Transfer and Install Fast Receiver Module**
- ❑ **D3 – Relabel and Connect Fast Receiver Cables**
- ❑ **D4 – Install Rating Plate**

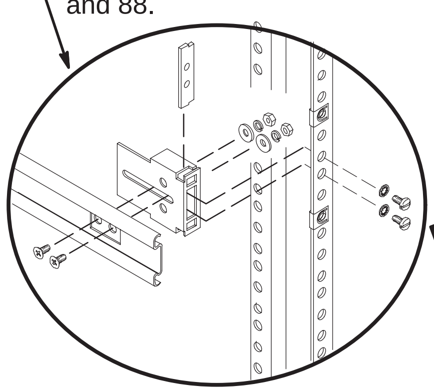
❑ D1 – REMOVE FAST RECEIVER MODULE FROM HORIZON SYSTEM CABINET

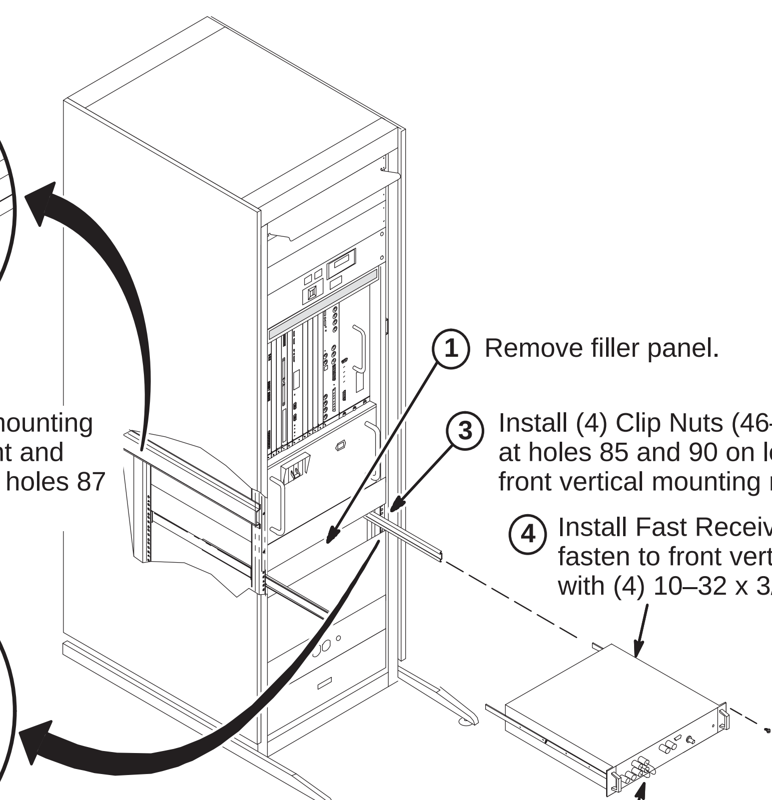


1. Remove Fast Receiver, rear and front vertical mounting rail brackets, mounting hardware, and slides from Horizon 5.x System Cabinet.

❑ D2 – TRANSFER AND INSTALL FAST RECEIVER MODULE



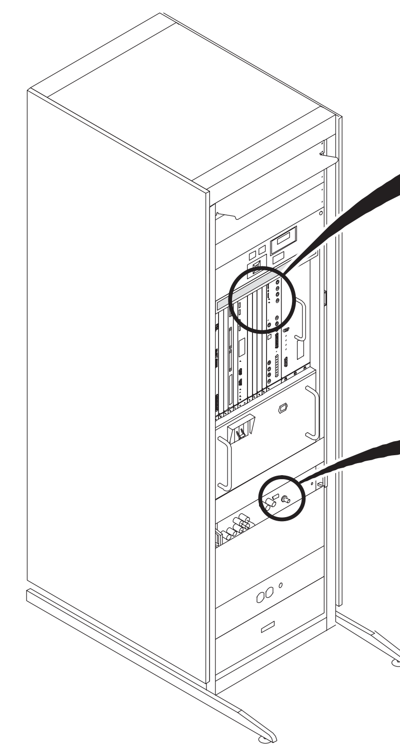


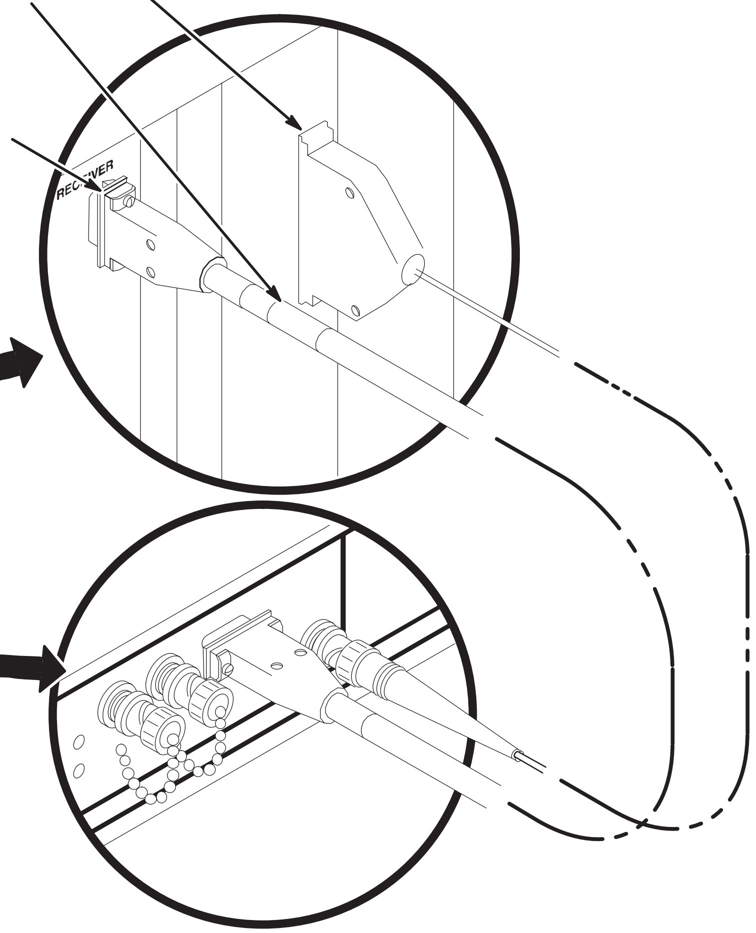


1. Remove filler panel.
2. Install slides, brackets, and mounting hardware for attaching to front and rear vertical mounting rails at holes 87 and 88.
3. Install (4) Clip Nuts (46–220288P1) at holes 85 and 90 on left and right front vertical mounting rails.
4. Install Fast Receiver and fasten to front vertical rails with (4) 10–32 x 3/4 Screws.
5. Make sure that (2) 5 inch BNC cables are connected to front panel of Fast Receiver as follows:
   "I A/D IN" to "I OUT" and
   "Q A/D IN" to "Q OUT"
6. Connect power cord to Power Module J9.

❑ D3 – RELABEL AND CONNECT FAST RECEIVER CABLES

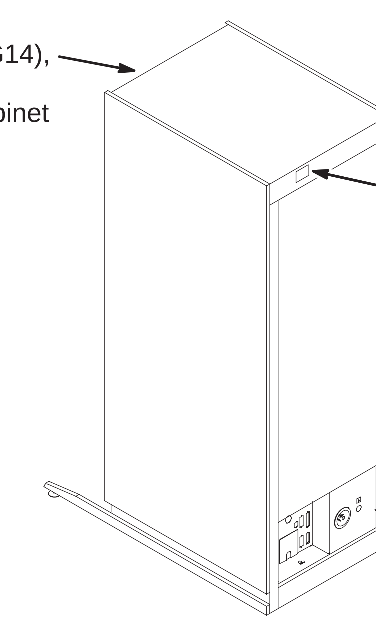
1. Connect Cable (2144527) from CERD J101 to Fast Receiver 2.5MHZ INPUT.
2. Relabel Digital Interface cable 2137306 with new label located in bag in rear of System cabinet. "MR2–A2–A9" added to label.
3. Connect Digital Interface Cable from Fast Receiver Board to Fast Receiver DIGITAL INTERFACE.





❑ D4 – INSTALL RATING PLATE

1. From the 8.0 8.x Insite Kit (46–301708G14), locate the "Systems" cabinet magnet (46–320095P5) and attach to font of cabinet as shown.
2. Apply Rating Plate (46–302200P12)

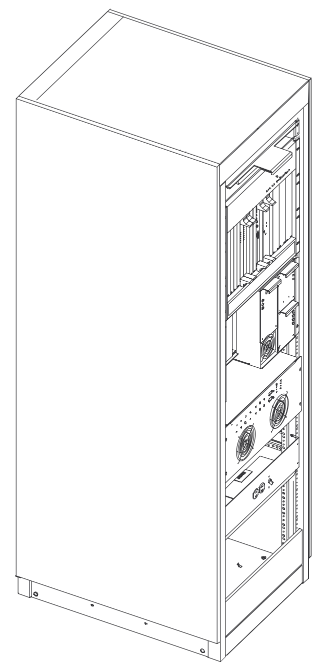


INSTALLATION STEPS SUMMARY

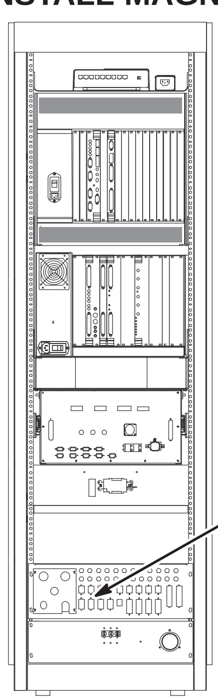
- ❑ **A – Position/Secure System Cabinet**
- ❑ **B – Install Magnet Monitoring Cables**
- ❑ **C – Connect Fiber Optic Runs 1033, 1034, 1044, and 1045 to I/F Panel**
- ❑ **D – Route/Connect Fiber Optic Run 1044 to APS Board**
- ❑ **E – Route/Connect Fiber Optic Runs1033, 1034, and 1045 to STIF Board**
- ❑ **F – Connect Power, Ground, and Data Cables**
- ❑ **G – Install InSite “SYSTEMS” Cabinet Magnet**

❑ A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.



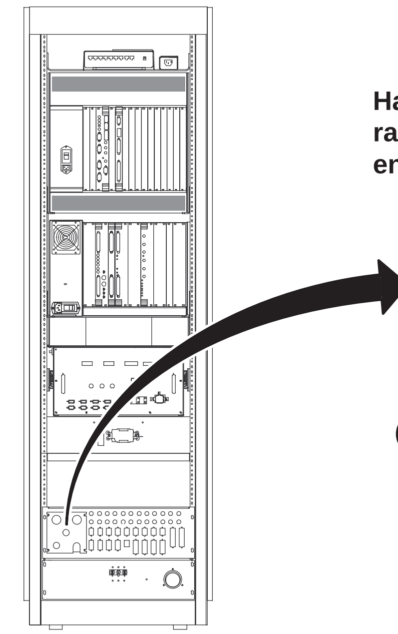
❑ B – INSTALL MAGNET MONITORING CABLES



Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual,* for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.

1. Connect Run 823 to J24.

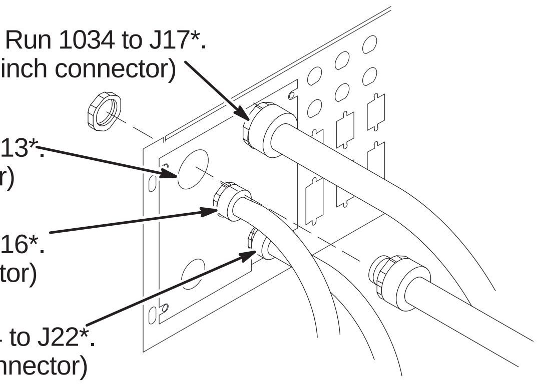
❑ C – CONNECT FIBER OPTIC RUNS 1033, 1034, 1044, AND 1045 TO I/F PANEL



CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

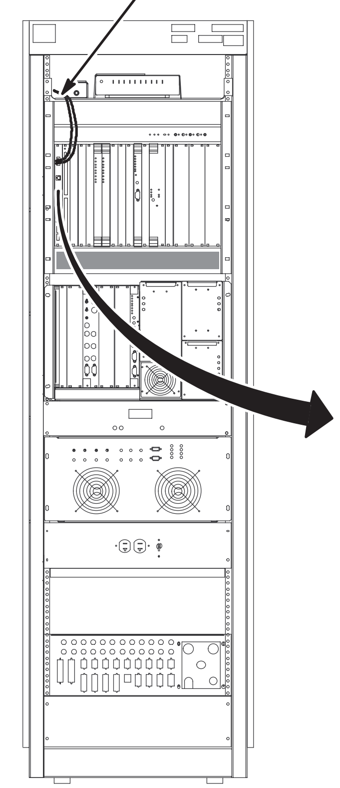
1. Attach Run 1034 to J17*. (use 1 inch connector)
2. Attach Run 1045 to J13*. (use 1 inch connector)
3. Attach Run 1033 to J16*. (use 3/4 inch connector)
4. Attach Run 1044 to J22*. (use 3/4 inch connector)



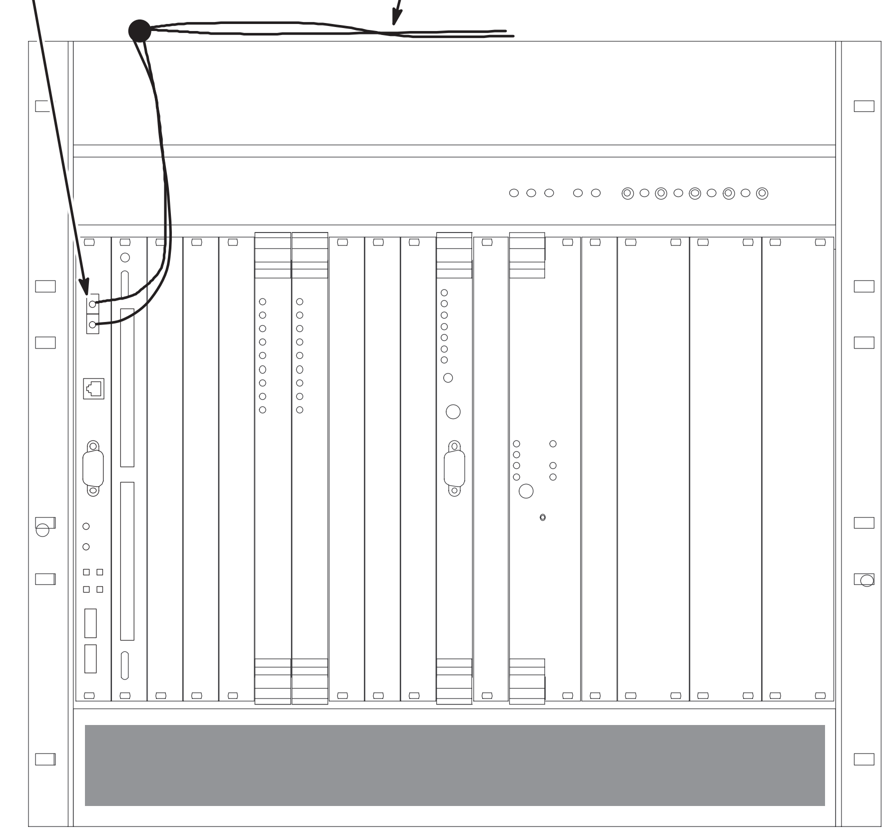
* Instructions provided with connectors.

❑ D – ROUTE/CONNECT FIBER OPTIC RUN 1044 TO APS BOARD

1. Route Run 1044 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**



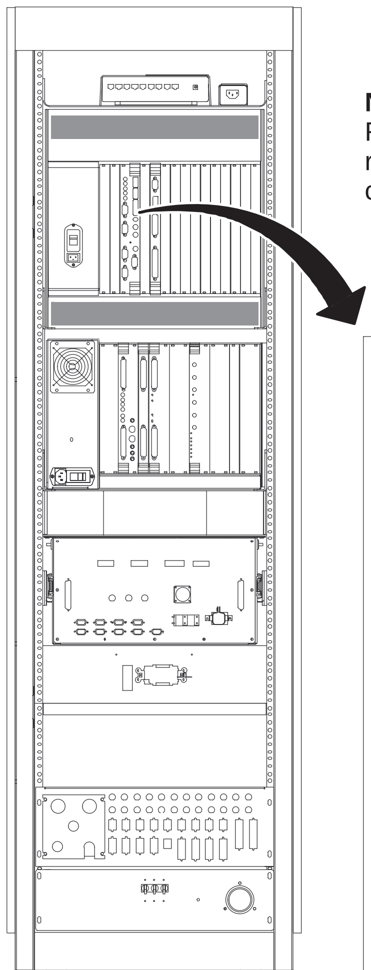
2. Connect the Orange and Blue fiber optic cables to the TI–MGD Daughter Board as indicated.
3. Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.



❑ E – ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD

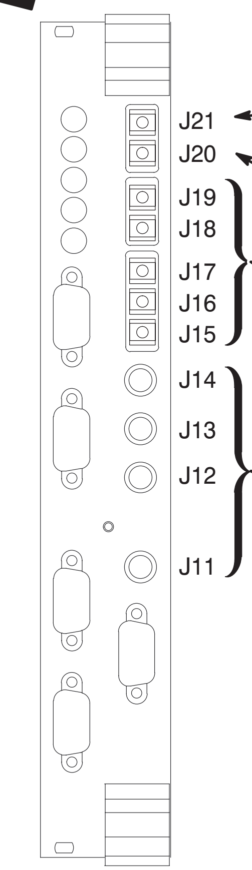
CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.



NOTE:
Refer to table at right for relabel information if existing cables are to be reused.

New Cables		Existing Cables	
RUN #	**10.x**	**RUN #**	**8.x/9.x**
1033–1	J21	708–1	J19
1034–1	J14	710–1	J12
1034–2	J13	710–2	J13
1034–3	J12	710–3	J15
1034–4	J11	710–4	J17
1034–5	J20	710–5	J18
1045–1	J17	711–2	J23
1045–2	J18,J19	711–3	J20, J21
1045–3	J15,J16	711–1	J25, J27



1. Relabel (if old cable re–used) and Route/Connect Run 1033 fiber optic cable to J21 on the STIF Board.

Part of Run 1034 below.

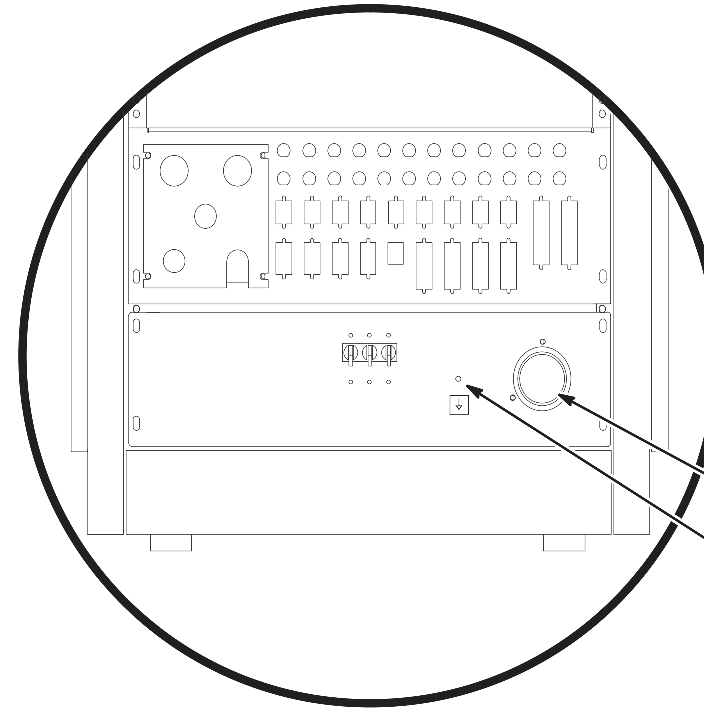
2. Relabel (if old cable re–used) and Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.

3. Relabel (if old cable re–used) and Route/Connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.

Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

❑ F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.

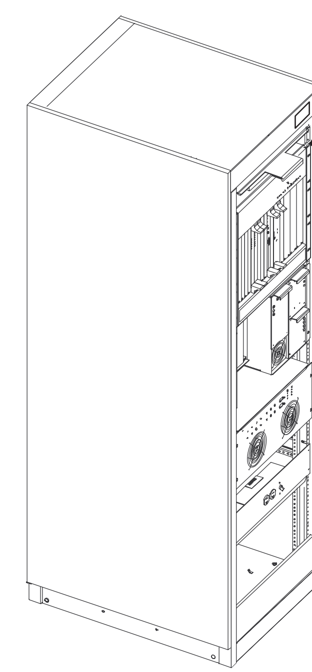


WARNING!

SYSTEM CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.

1. Connect Run 968 cable plug to “J1”. Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.
4. Connect remaining cabinet cables as marked.

❑ G – INSTALL INSITE “SYSTEMS” CABINET MAGNET



1. From the LX Insite Kit (46–301708G14), locate the ”Systems” cabinet magnet (46–320095P5) and attach to front of cabinet as shown.

ACCESSORIES AND OPTIONS INSTALLATION INSTRUCTIONS

2–1 Tab 11, Accessories & Options, contains installation instructions for the following:

- *DIRECTION 2123158* – Patient Transport Installation Instructions
- *DIRECTION 2231762* – Signa WideOpen Pneumatic Patient Alert Installation

Use only the applicable Installation Instruction Directions

PATIENT TRANSPORT INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

- ❑ **A – Patient Transport Installation**
- ❑ **B – Head Coil Holder**
- ❑ **C – Head Coil Adapters**

❑ A – PATIENT TRANSPORT INSTALLATION

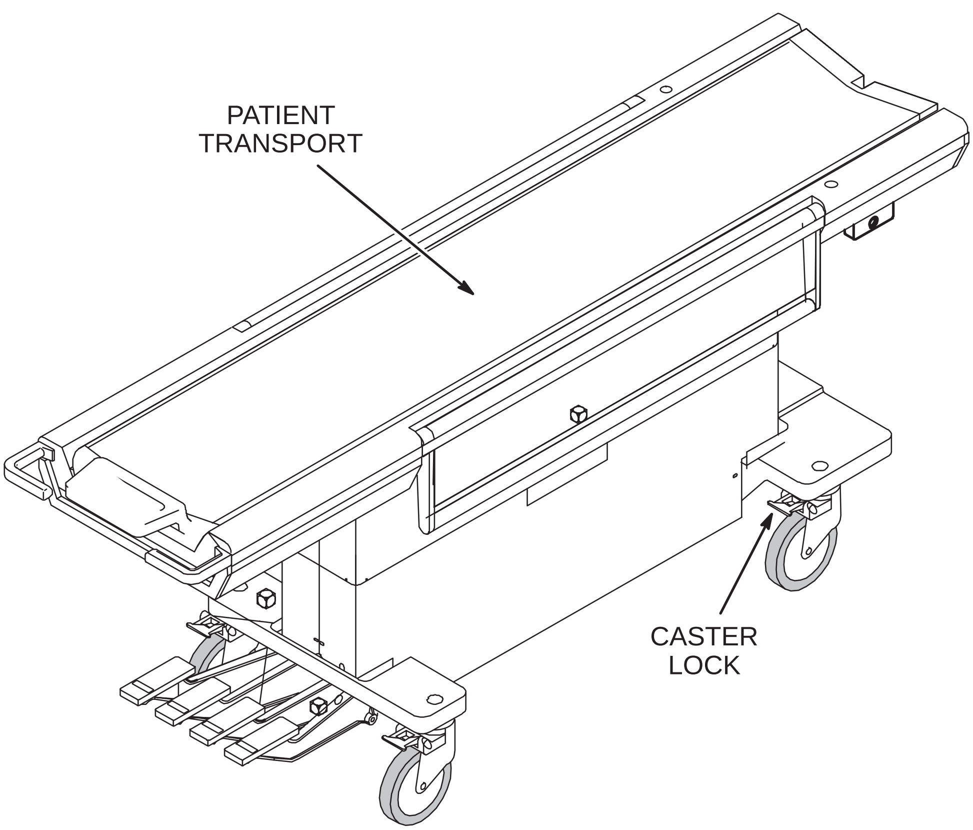
Patient Transport is shipped covered with protective cardboard which should remain in place until ready to be moved into the magnet room. Cover table with plastic previously removed when checks and adjustments are completed.

1. Unlock casters and move Patient Transport to magnet room for adjustments and function checks at the dock.
2. Remove protective plastic sheet from table.
3. Perform mechanial procedures in MR CD–ROM, Set up and Calibration: Patient Transport, LIGHTWEIGHT PATIENT TRANSPORT.

Note

If hardware fastening procedures for installation of the dock and support brackets were followed, the hardware used to fasten the dock and dock supports was left loose. The loose hardware permits a quick and easy dock centering adjustment with table docked.

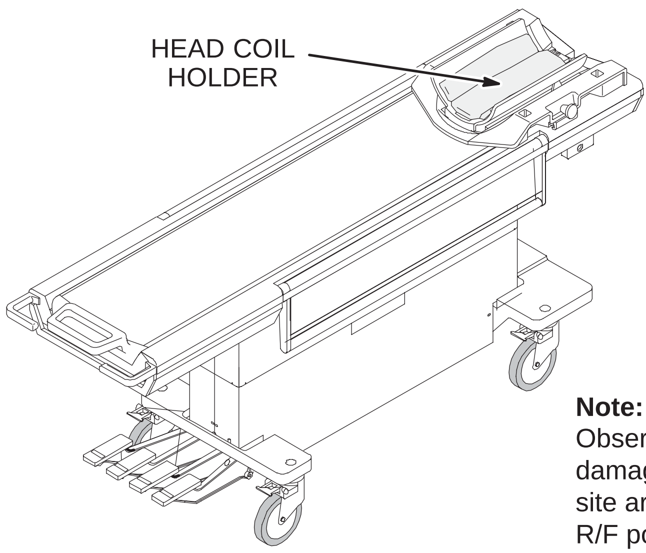
4. If additional Patient Transports are supplied, refer to *Direction 15477, Second Patient Transport Option Installation*, (delivered with M1020BD option).



PATIENT TRANSPORT INSTALLATION INSTRUCTIONS

❑ B – HEAD COIL HOLDER

1. Unpack Head Coil Holder and install Head Coil Holder on Patient Transport cradle.
2. With Patient Transport docked, check mechanical operation of Head Coil with holder.



Note:
Observe warning Label on head coil. To prevent head coil damage, be sure that all service personnel and users at the site are aware that the head coil must be removed before R/F power is applied to the body coil. This warning also applies to service procedures.

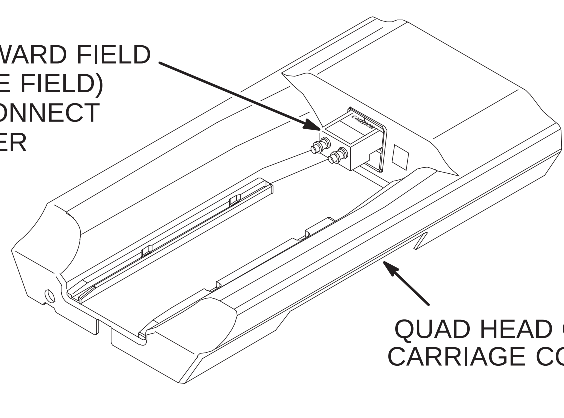
❑ C – HEAD COIL ADAPTERS

1. Unpack Head Coil, Coil Adapters (Head Coil Forward Field, Extremity Coil , Surface Coil), and adapter cables.
2. Install Head Coil Adapter (Head Coil Forward or Reverse Field) into upper coil adapter port on carriage cover.

Note

If magnet is configured with reverse polarity, "reverse field" Head and Surface Coil Adapters are required to be ordered. The "forward field" Coil Adapters must be returned. Since an unaware operator could use the wrong adapter, do not leave both adapters on site.

3. Remove head coil cables from bag attached to Head Coil. Place Head Coil on carriage cover and connect cables to verify connections. Connect blue cable to connector indicated with blue.
3. Set Head Coil with cables, holder, and adapters aside for safe storage (outside of magnet room) until needed during set–up calibrations, functional checks, and system tests.



INSTALLATION STEPS SUMMARY

- ❑ **A1 – Preinstallation Planning**
- ❑ **A2 – Squeeze Bulb Hanger Installation**
- ❑ **A3 – Squeeze Bulb Installation**
- ❑ **A4 – Magnet Room Routing of Tubing & Wave Guide Installation**
- ❑ **A5 – Determining Need of Extender Box**

❑ A1 – PREINSTALLATION PLANNING

The customer and users should be involved in the following decisions:

Routing of Pneumatic Tubing:

Location of Control Box

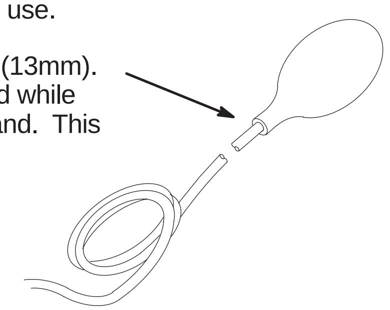
CAUTION

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

❑ A2 – SQUEEZE BULB INSTALLATION

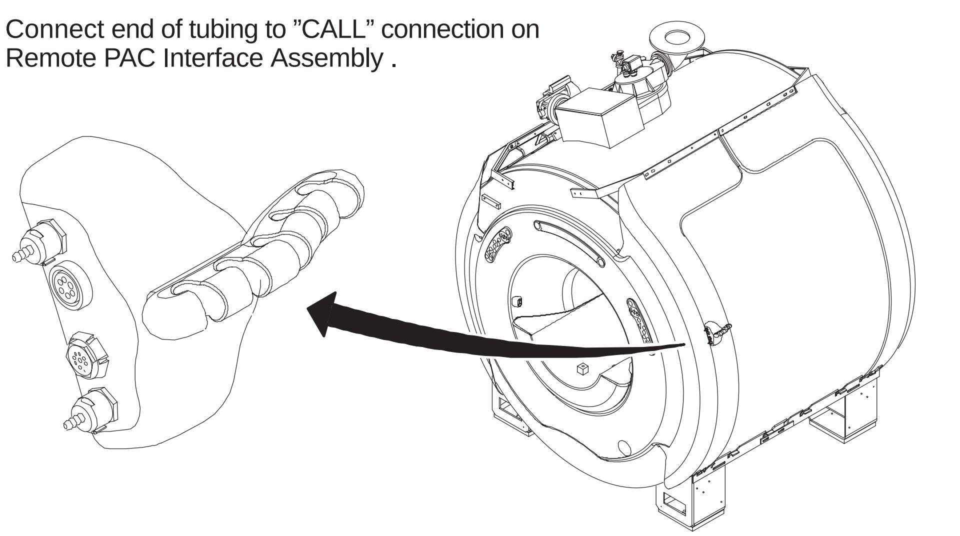
(2) Before cutting pneumatic tubing from roll, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the “CALL” connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.

(1) Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.



❑ A3 – CONNECT TUBE TO REMOTE PAC INTERFACE ASSEMBLY

(1) Connect end of tubing to ”CALL” connection on Remote PAC Interface Assembly .



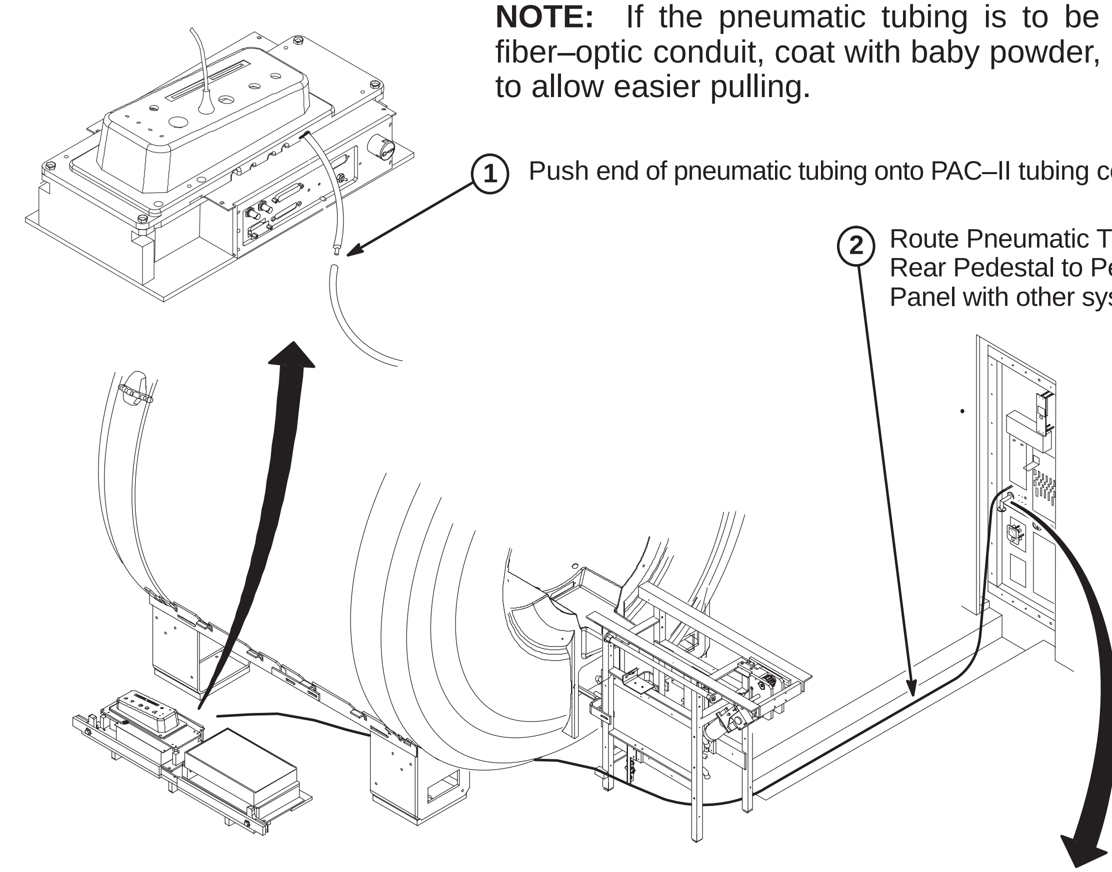
❑ A4 – MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION

CAUTION

Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

NOTE: Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

NOTE: If the pneumatic tubing is to be pulled thru fiber–optic conduit, coat with baby powder, not grease, to allow easier pulling.

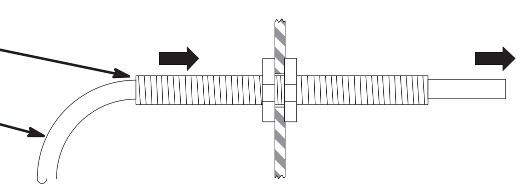


(1) Push end of pneumatic tubing onto PAC–II tubing connector.

(2) Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.

(3) Insert Wave Guide thru “ALERT AIR LINE” hole and fasten to Penetration Panel with nut on each side.

(4) Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.



❑ A5 – DETERMINING NEED OF EXTENDER BOX

Note: If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator’s Console), Extender Kit, 46–317758P2, must be ordered.

If the pneumatic tubing is long enough to reach control box:
The extender is not needed. Proceed to Step **B3**, CONTROL BOX INSTALLATION.

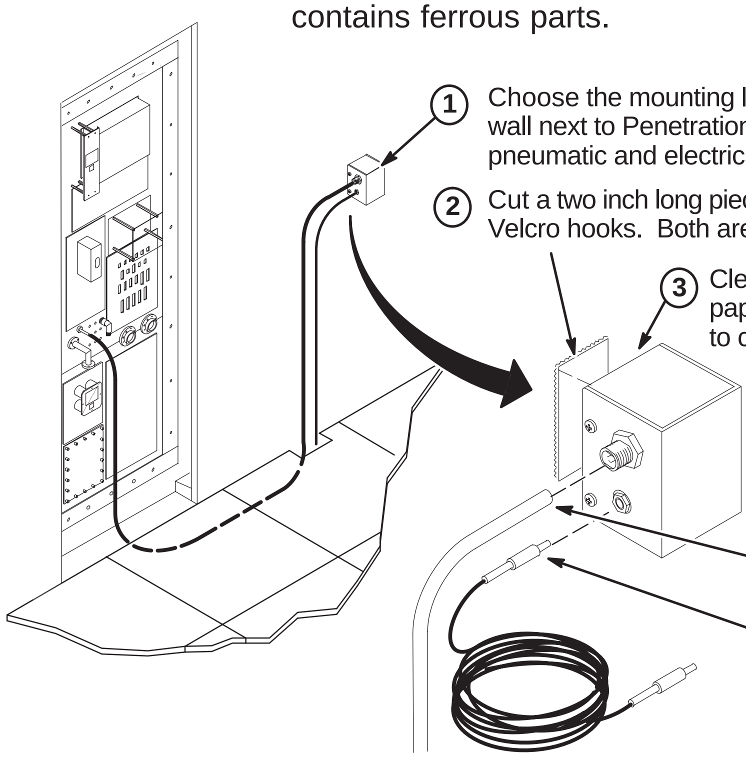
If the pneumatic tubing is not long enough to reach control box:
The extender is needed. Proceed to Step **B1**, INSTALLATION OF EXTENDER KIT.

INSTALLATION STEPS SUMMARY

- ❑ **B1 – Installation of Extender Kit (46–317758P2)**
- ❑ **B2 – Connection of Extender Wire to Control Box**
- ❑ **B3 – Control Box Installation**
- ❑ **B4 – Control Box Installation on Wall or Under Shelf**
- ❑ **B5 – Control Box Mounted with Velcro**

❑ B1 – INSTALLATION OF EXTENDER KIT (46–317758P2)

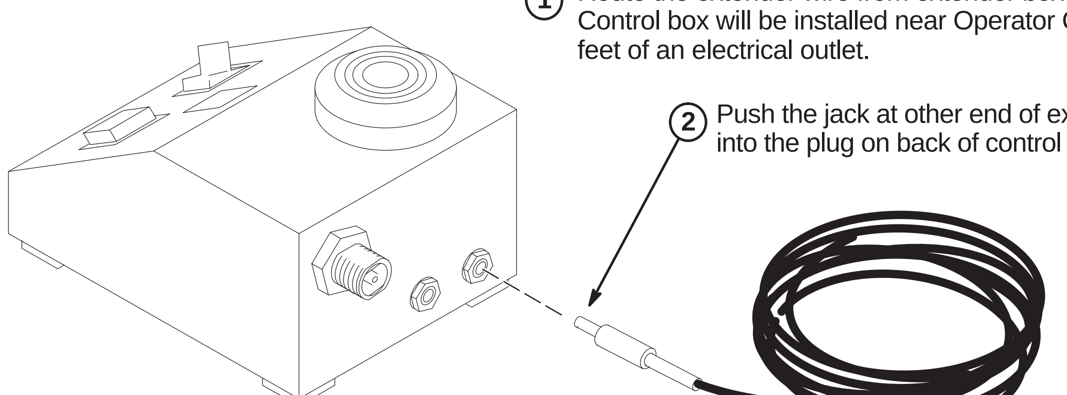
Note: Do not mount Extender Box in Magnet Room because the box contains ferrous parts.



1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed–thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.

❑ B2 – CONNECTION OF EXTENDER WIRE TO CONTROL BOX

1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.



❑ B3 – CONTROL BOX INSTALLATION

Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

Mount control box on a wall or other vertical surface per Step **B4**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.

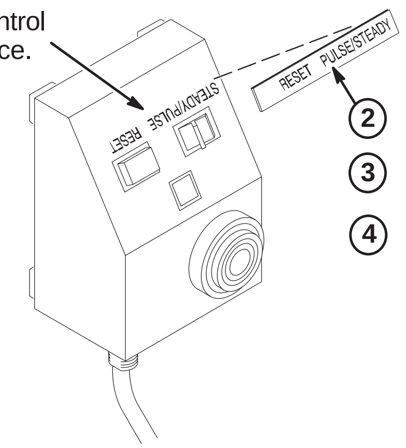
Mount control box under shelf per Step **B4**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.

Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

❑ B4 – CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF

1. Clean Control Box surface.



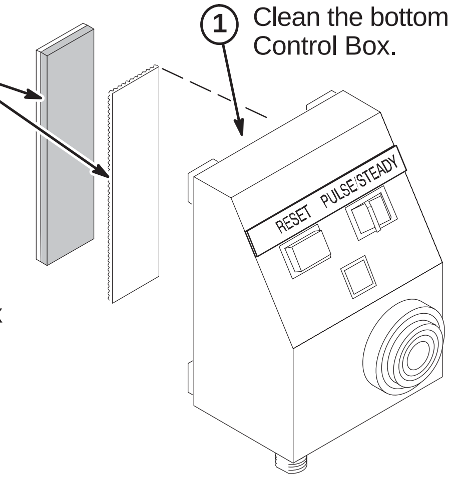
2. Remove adhesive backing from metalized nameplate.
3. Apply nameplate over silk screened lettering as shown.
4. **If control box will be mounted to WALL or SHELF with Velcro:** Proceed to Step **B5**.

   If control box will be mounted on WALL with screws: Proceed to Step **C1**.

   If control box will be mounted under SHELF with screws: Proceed to Step **C3**.

❑ B5 – CONTROL BOX MOUNTED WITH VELCRO

1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. Mount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.

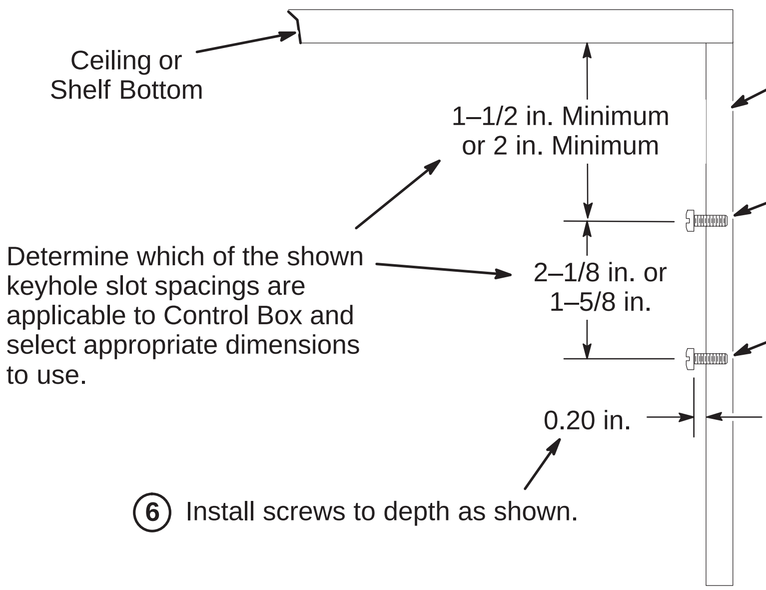


INSTALLATION STEPS SUMMARY

- ❑ **C1 – Installing Mounting Screws in Vertical Surface**
- ❑ **C2 – Installing Control Box on Vertical Surface**
- ❑ **C3 – Installing Mounting Screws Under Shelf**
- ❑ **C4 – Installing Control Box Under Shelf**
- ❑ **C5 – Final Connections to Control Box**

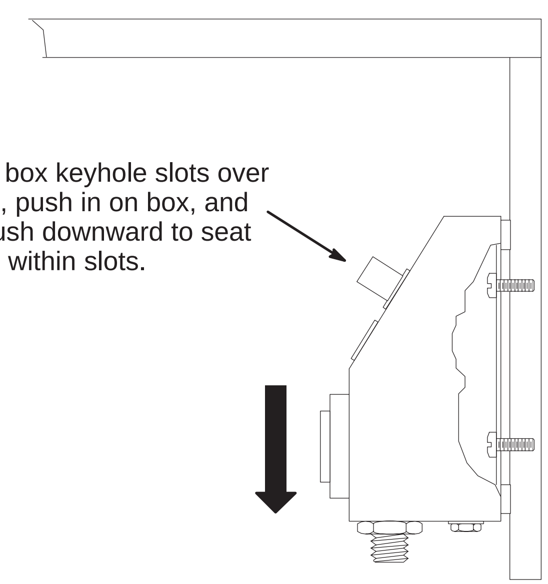
❑ C1 – INSTALLING MOUNTING SCREWS IN VERTICAL SURFACE

1. Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.



2. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
3. Mark top mounting hole position on wall.
4. Mark bottom mounting hole position on wall.
5. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
6. Install screws to depth as shown.

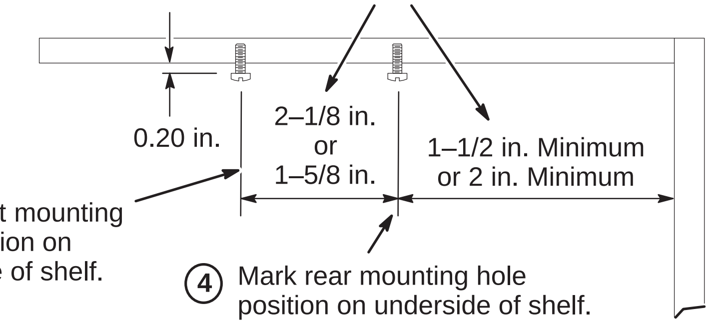
❑ C2 – INSTALLING CONTROL BOX ON VERTICAL SURFACE



1. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
2. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
3. Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

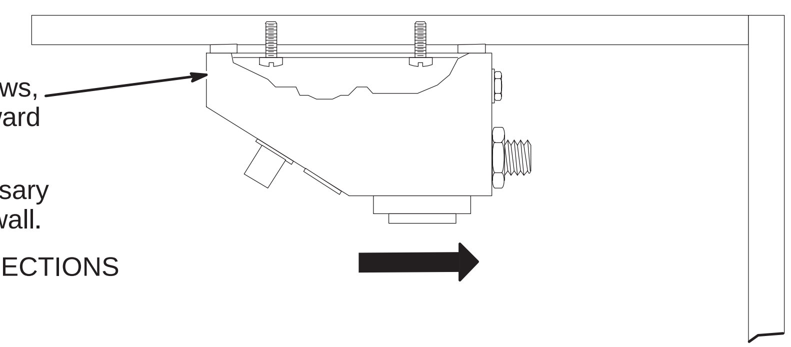
❑ C3 – INSTALLING MOUNTING SCREWS UNDER SHELF

1. Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.
2. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.



3. Mark front mounting hole position on underside of shelf.
4. Mark rear mounting hole position on underside of shelf.
5. Drill a 1/16 in. pilot hole at locations if furnished screws are used.

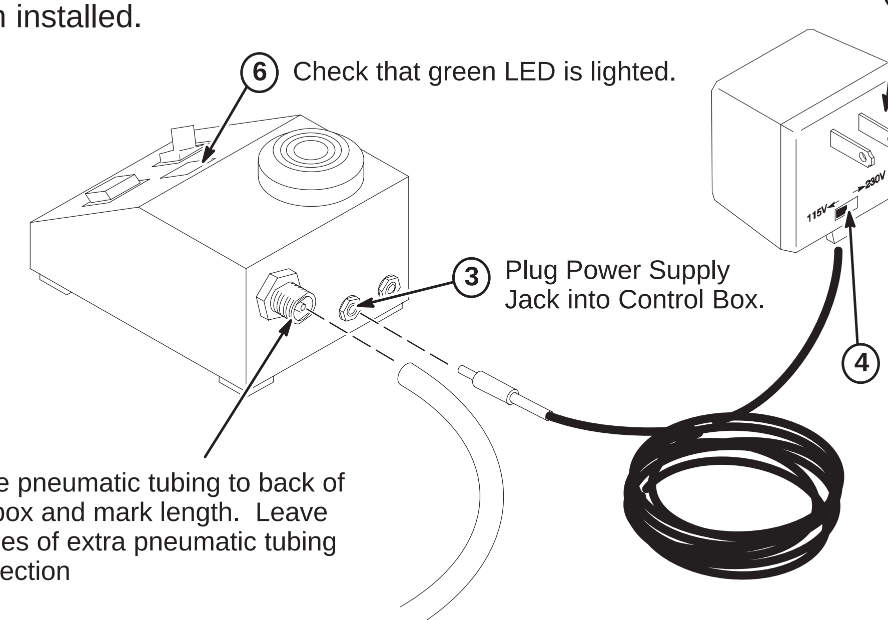
❑ C4 – INSTALLING CONTROL BOX UNDER SHELF



1. Center box keyhole slots over screws, push up on box, and then push inward to seat screws within slots.
2. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
3. Proceed to Step **C5**, FINAL CONNECTIONS TO CONTROL BOX.

❑ C5 – FINAL CONNECTIONS TO CONTROL BOX

Note: Steps 1 and 2 are not needed if Extender Box (46–317758P2) has been installed.



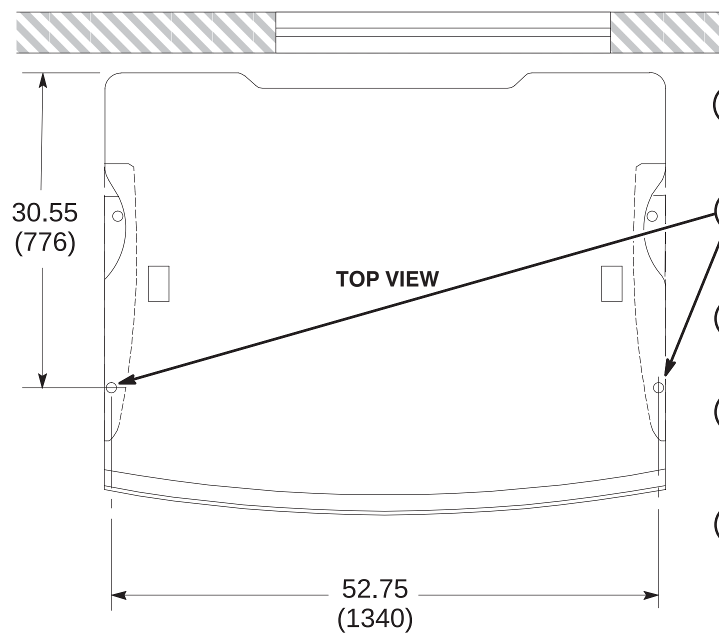
1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection
2. Cut tubing and push fully onto feed–thru connector on back of control box.
3. Plug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is lighted.

INSTALLATION STEPS SUMMARY

- ❑ **A1 – Operator Workspace Table Anchor Installation**
- ❑ **A2 – Securing Operator Workspace Table to Floor**
- ❑ **A3 – Attach Workspace Interface Module (2141978 series) to Table**
- ❑ **A4 – Attach Power Distribution Box (46–307830G1) to Table**

Note: The CERD Spike Noise Test Cable (2168505) is part of the 8.x Octane Operator Workspace Cable collector (2154392–2). Remove this cable from the collector and place it in the TPS RF Service Interface Kit (46–301927G1).

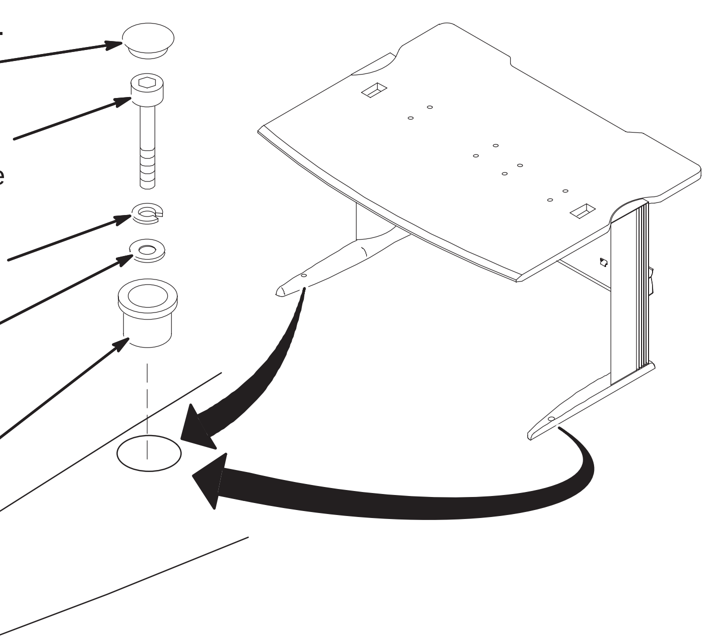
❑ A1 – OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION



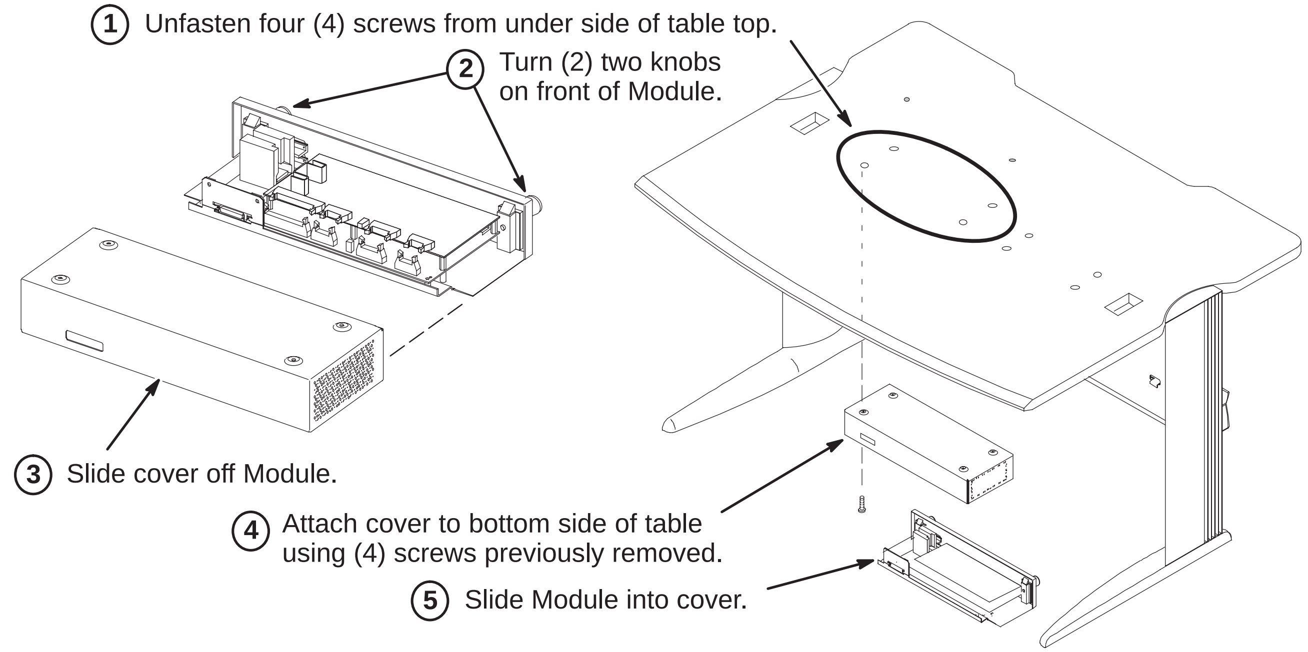
1. Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table (2135998–2).
2. Remove and discard the two (2) cover plugs. Locate and mark on floor the two hole locations at the front of the table legs.
3. Move table to prepare for installing the floor anchors in Steps 4 and 5.
4. At each hole marked in Step 2, drill a .38 inch hole and insert 1/4–20 x .375 diameter anchor (2168253).
5. Secure anchor in place with supplied Setting Tool (2168781).

❑ A2 – SECURING OPERATOR WORKSPACE TABLE TO FLOOR

6. Repeat anchoring process below for other leg.
5. Install black nylon Plug (2168254).
4. Depending upon depth of previously installed anchor, install either the 1–3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
3. Insert one Lock Washer (2168252) on top of Flat Washer(s).
2. Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
1. Insert Bushing (2151454) in table leg hole.

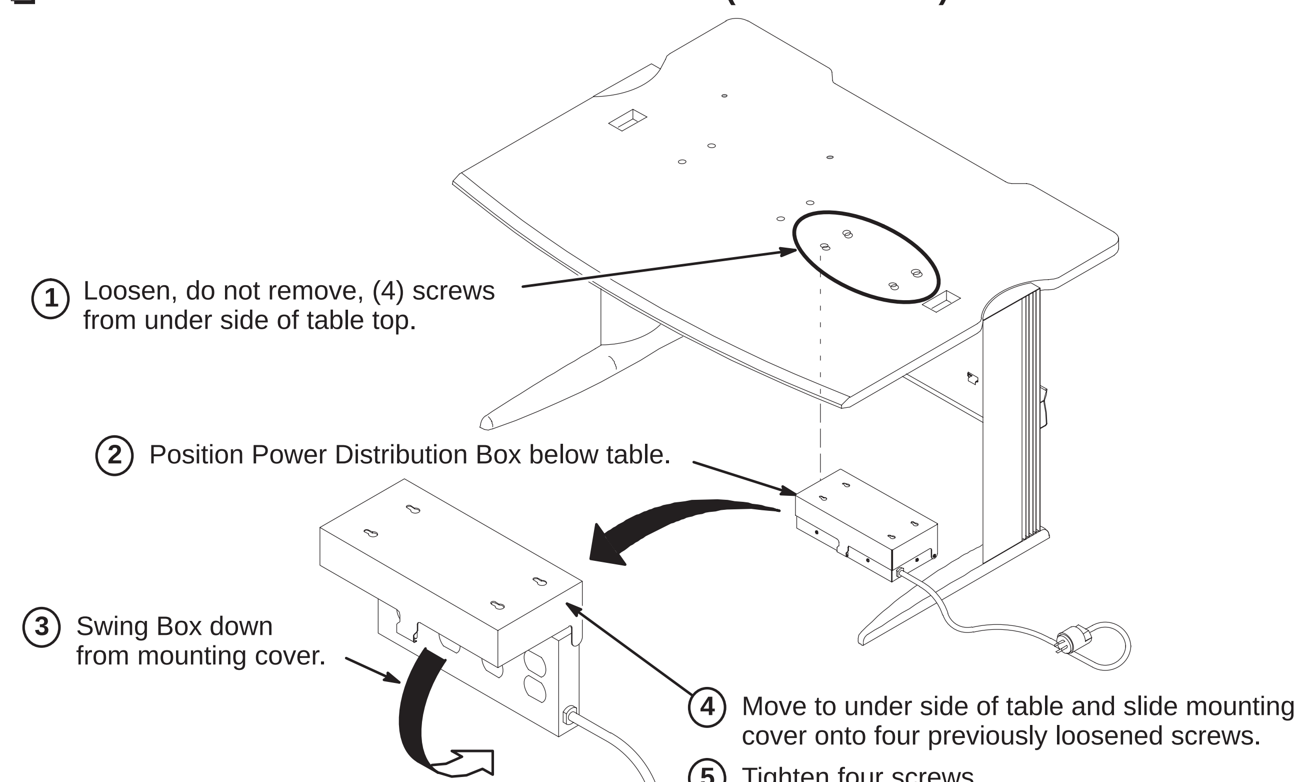


❑ A3 – ATTACH WORKSPACE INTERFACE MODULE (2141978 series) TO TABLE



1. Unfasten four (4) screws from under side of table top.
2. Turn (2) two knobs on front of Module.
3. Slide cover off Module.
4. Attach cover to bottom side of table using (4) screws previously removed.
5. Slide Module into cover.

❑ A4 – ATTACH POWER DISTRIBUTION BOX (46–307830G1) TO TABLE

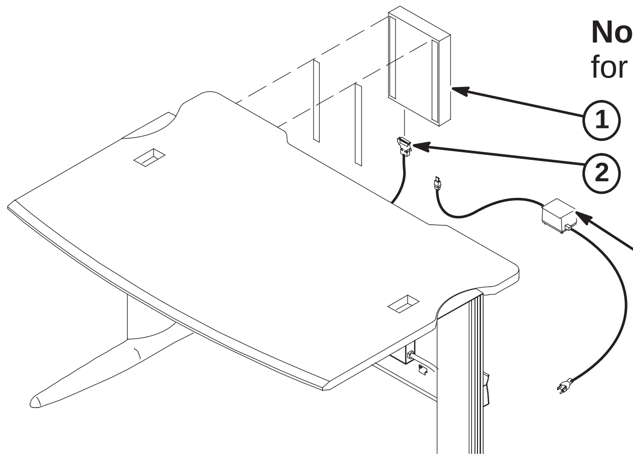


1. Loosen, do not remove, (4) screws from under side of table top.
2. Position Power Distribution Box below table.
3. Swing Box down from mounting cover.
4. Move to under side of table and slide mounting cover onto four previously loosened screws.
5. Tighten four screws.

INSTALLATION STEPS SUMMARY

- ❑ **B1 – Attach Modem to Table (May be replaced by ethernet connection)**
- ❑ **B2 – Attach DASM to Table**
- ❑ **B3 – Attach Serial Expansion Box Asm to Underside of Table Top**
- ❑ **B4 – Attach Serial Coverter to Underside of Table Top (MGD & Open*Speed* only)**

❑ B1 – ATTACH MODEM TO TABLE (May be replaced by ethernet connection)



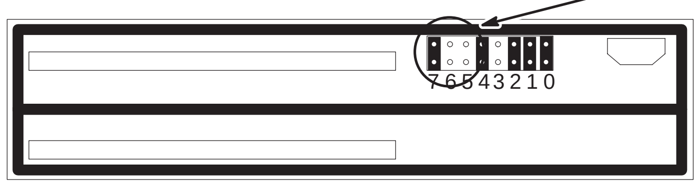
Note: Modem may be replaced by Ethernet cable for connection to InSite (See Procedure **E4**).

1. Attach to back plate of table.
2. Connect Run 812 (2188746) to Modem. Other end to be connected to Operator Workspace Cabinet SER 1 later.
3. Modem Power Adapter to be supplied by District office and connected to Power Distribution Box J7.

❑ B2 – ATTACH DASM TO TABLE

Note: The Analog and Digital DASMs are shipped with the jumper set to SCSI ID=0. This is the correct ID for an AW, but incorrect for Signa Horizon LX (SCSI ID=3).

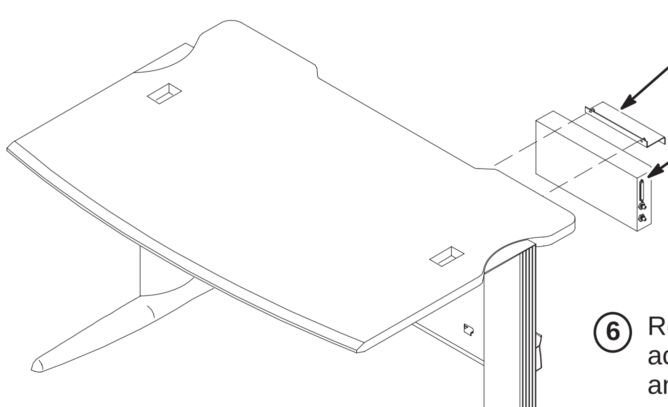
1. Unfasten DASM cover screw and remove cover.
2. For SCSI ID = 3, set Configuration Jumper J2 as follows:
   5=OFF, 6=OFF, 7=ON
   Note: A jumper in the OFF position enables the SCSI ID bit.



3. Re-attach DASM cover.

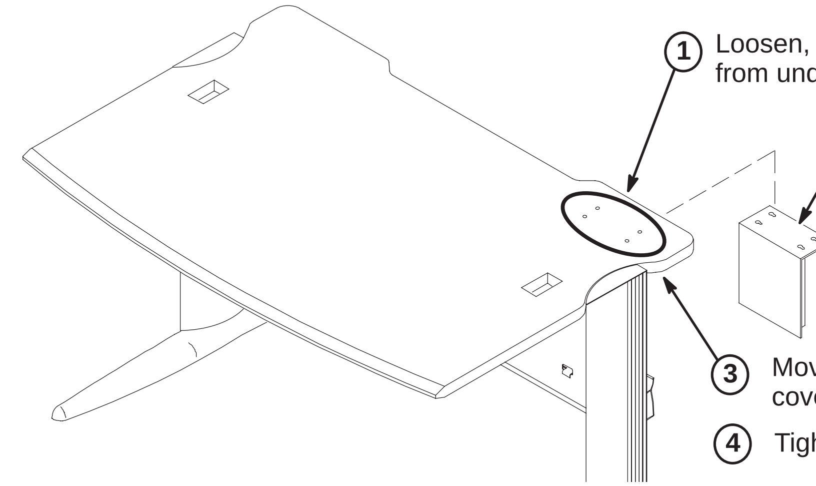
Note: Consult with Site customer for preference of DASM location. This procedure installs DASM in the cable trough. Customer may prefer DASM on top of table.

CAUTION: Longer SCSI cables cannot be substituted.



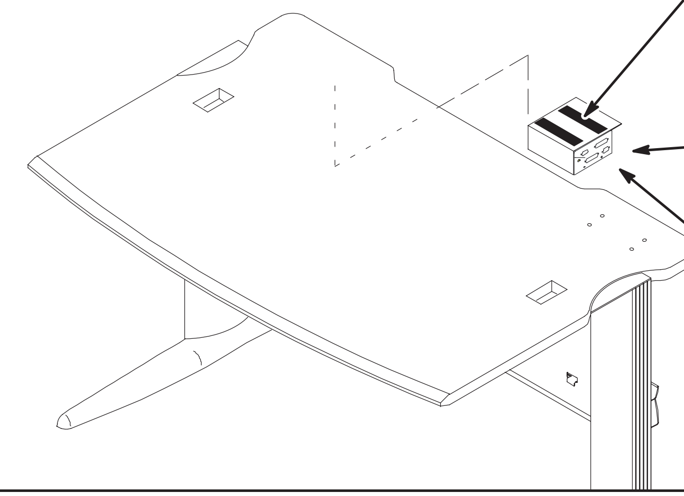
4. Unfasten DASM Bracket from back plate. Use only for 46–269566P3 or 2143388–2. Not used for 2251040 or 2251043.
5. Install DASM (Analog, 46–269566P3, or Digital, 2143388–2) in cable trough and fasten in place with DASM Bracket.
   OR
   Install DASM (Analog, 2251040, or Digital, 2251043) in cable trough. Refer to installation instructions that are shipped with DASM.
6. Route and connect DASM cables and accessories according to instructions provided with DASM Module and Cable Map in Overview Tab of this manual.
7. Connect power cable to Power Distribution Box J2.

❑ B3 – ATTACH SERIAL EXPANSION BOX ASM TO UNDERSIDE OF TABLE TOP



1. Loosen, do not remove, (4) screws from under side of table top.
2. Position Serial Expansion Box Asm (2204796) below table.
3. Move to under side of table and slide mounting cover onto four previously loosened screws.
4. Tighten four screws.

❑ B4 – ATTACH SERIAL CONVERTER TO BOTTOM OF TABLE (MGD & Open*Speed*)



1. Cut supplied velcro strip into two pieces and attach to module.
2. Attach Serial Converter Module to underside of table top.

3A. **FOR Open*Speed*:** Connect to module Runs 837 (J4), 838 (J3), 910 (J2), and 911 (J1).

OR

3B. **FOR MGD: If provided, c**onnect to module Runs 838 (J3) and 911 (J1). (future feature)

4. Connect Power cable to Power Distribution Box J8.

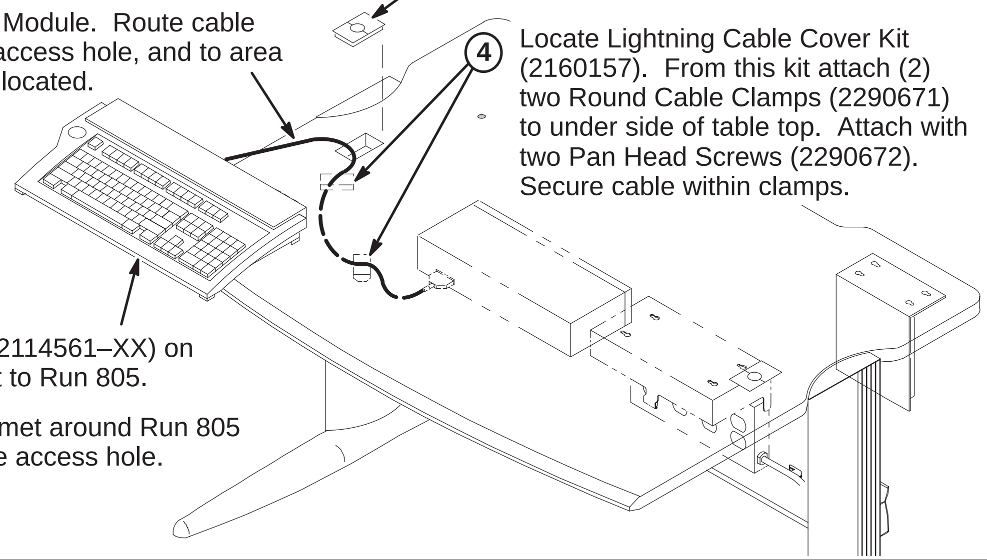
INSTALLATION STEPS SUMMARY

- ❑ **C1 – Install Keyboard on Table**
  OR
- ❑ **C2 – Install SCIM/Keyboard Data Cable on Table**
- ❑ **C3 – Install SCIM/Keyboard on Table**
- ❑ **C4 – Install Flat Screen Color Monitor on Table**

NOTE:
Consult with Site customer for exact location of Color Monitor on Operator Workspace Table.

❑ C1 – INSTALL KEYBOARD ON TABLE

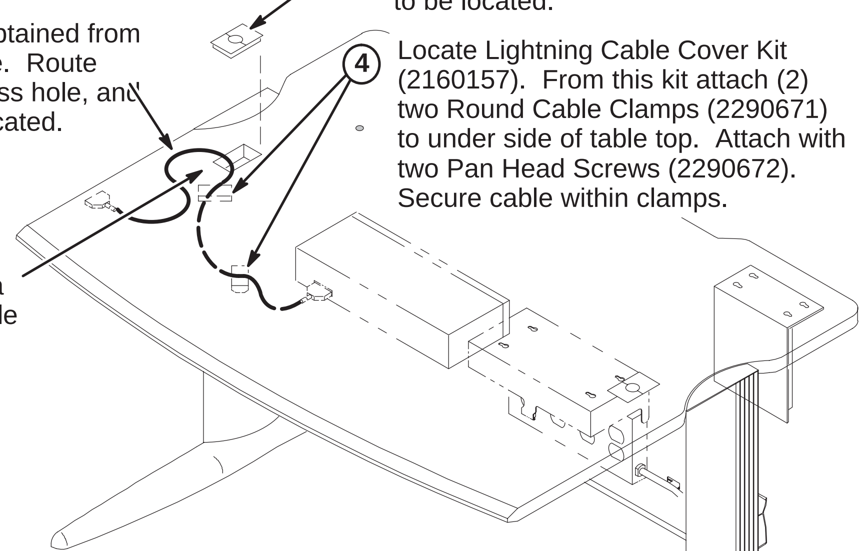
1. Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.
2. Remove cable grommet from side of table where Monitor and Keyboard are to be located.
3. Connect Run 805 to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.
4. Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
5. Place Keyboard (2114561–XX) on table and connect to Run 805.
6. Place cable grommet around Run 805 and insert in cable access hole.



OR

❑ C2 – INSTALL SCIM/KEYBOARD DATA CABLE ON TABLE

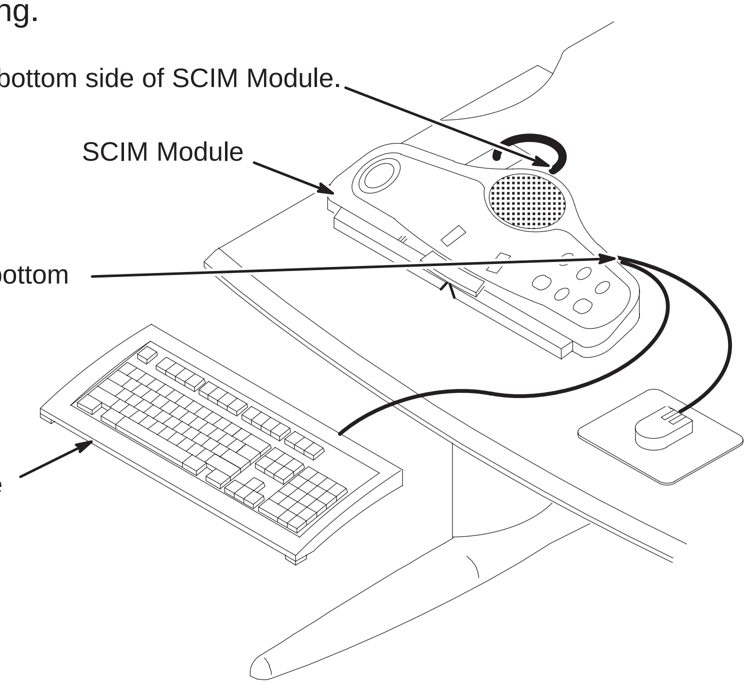
1. Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.
2. Remove cable grommet from side of table where Monitor and Keyboard are to be located.
3. Connect Data Cable (Run 805), obtained from SCIM Kit (2307175), to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.
4. Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
5. Place cable grommet around Data Cable (Run 805) and insert in cable access hole.



❑ C3 – INSTALL SCIM/KEYBOARD ON TABLE

NOTE: Refer to Installation document OW1INA3.DOC on Service Methods CD–ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

1. Connect Data Cable to bottom side of SCIM Module.
2. Connect Keyboard and Mouse cables to bottom side of SCIM Module as marked.
3. Place Keyboard on table in front of SCIM Module.

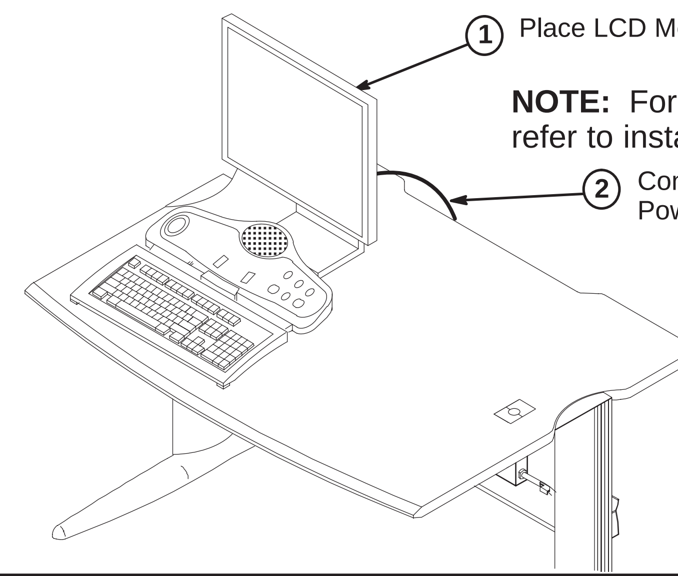


❑ C4 – INSTALL FLAT SCREEN COLOR MONITOR ON TABLE

1. Place LCD Monitor on table in back of SCIM.

NOTE: For fastening monitor to surface, refer to installation drawing 2204493DDW.

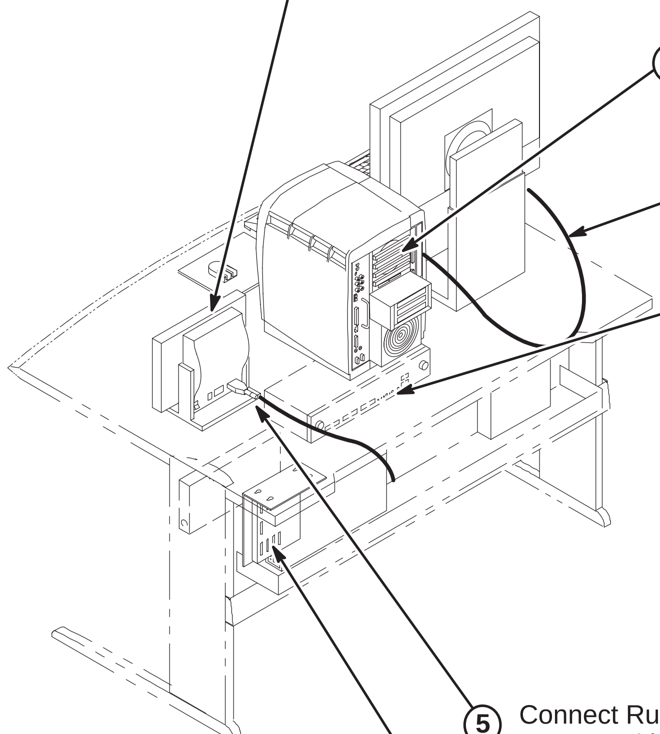
2. Connect power cable to Power Distribution Box J1.



INSTALLATION STEPS SUMMARY

- ❑ **D1 – Install LCD Display on Table and Position Octane Computer**
- ❑ **D2 – Connect Table Component Cables**
- ❑ **D3 – Serial Expansion Box Set Up**
- ❑ **D4 – SGI Octane Computer Cables Connections**

❑ D2 – CONNECT TABLE COMPONENT CABLES



1. Place LCD Display (2138599) on table. Connect power cable to Power Distribution Box J5.
2. Place Octane Computer on table.
   NOTE: Refer to Procedure **D4** for Run connections to SGI Octane Computer.
3. Connect Run 798/821 (2150533–2) to R IN, B IN, G IN on Monitor and route/connect to MONITOR on the Octane Computer module.
4. Connect the following cables from the OW I/F Module (OW A1 A4) for later connection to Operator Workspace Cabinet:

RUN #	CONNECTION	PART #
805	J6	2114561–38
810	J8	2141980–2
814	J9	2196990
817	J17	46–328000G958

Connect the following cables to OW I/F Module. These cables will be routed to the System Cabinet and Penetration Panel:

RUN #	CONNECTION	PART #
789	J18	46–317068G938
788	J7	46–328000G937

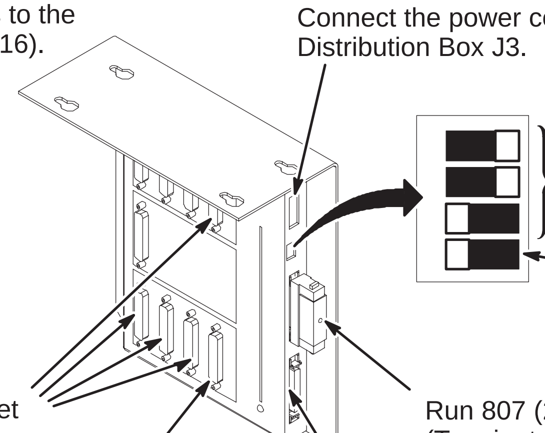
5. Connect Run 797 (2138599–2) to LCD Display. Other end is connected later to PC in the Operator Workspace Cabinet.

NOTE: Refer to Procedure **D3** for Run connections and DIP switch settings.

❑ D3 – SERIAL EXPANSION BOX SET UP FOR MGD (10.x)

1. Connect the following cables to the SCSI Expansion Box (OW A16).

Connect the power cord to Power Distribution Box J3.



2. Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled
NOTE: If DASM not installed, move swtich setting to disabled

Run 1046 from System cabinet to Ports 1, 6, 7, and 8.

Run 807 (2163886) from DASM. (Terminator shown)

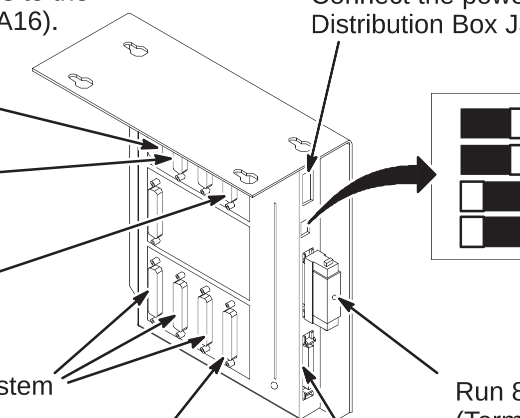
Run 911 from Serial Converter J1 to Port 5. (Future)

Run 815 (2198983) from Octane computer

❑ D3 – SERIAL EXPANSION BOX SET UP (9.x)

1. Connect the following cables to the SCSI Expansion Box (OW A16).

Connect the power cord to Power Distribution Box J3.



Run 1004 from TAC Cabinet J27 to Port 2. (Twin*Speed* option only)

Run 910 from Serial Converter J2 to Port 3. (Open*Speed* only)

Run 791 from System cabinet to Port 1.

Run 818 (2198993) from System cabinet to Ports 6, 7, and 8.

Run 911 from Serial Converter J1 to Port 5. (Open*Speed* only)

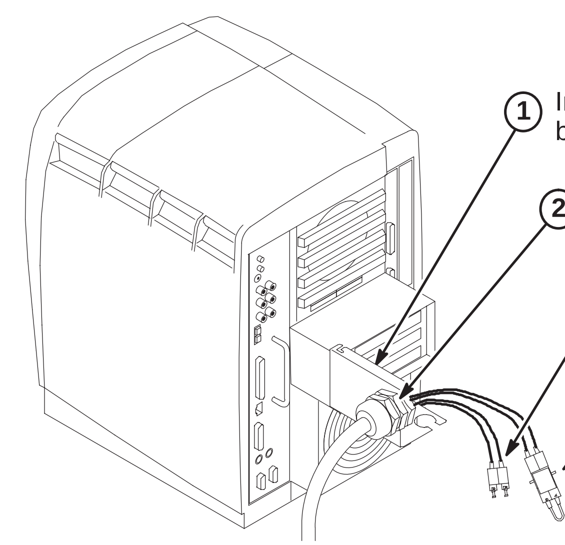
2. Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled
NOTE: If DASM not installed, move swtich setting to disabled

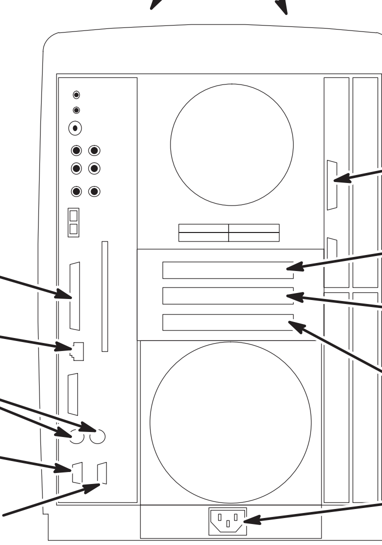
Run 807 (2163886) from DASM. (Terminator shown)

Run 815 (2198983) from Octane computer

❑ D4 – SGI OCTANE COMPUTER CABLES CONNECTIONS



1. Install fiber optic Strain Relief Bracket (2245994) to back of Octane Computer using existing screws.
2. Route and connect Bit3 Fiber Optic cable Run 836 (2241052) to bracket.
3. Connect Run 836 (or Run 1044, MGD) orange and blue fiber optic cables as indicated in Step 5 below.
4. Spare Fiber Optic cable and Loopback Assembly (2241052–6).
5. Connect the following cables to the SGI Octane Computer (OW A15) as marked.



Run 819

Run 811 (2153396–3)

Run 810 (2141980–2)

Run 813 (2188747)

Run 812 (2188746)

Run 798 or 821 (2150533–2)
OR
Run 799 or 820 (2150533)

Run 1031 (MGD)

Run 836 (2241052)
OR
Run 1044 (MGD)

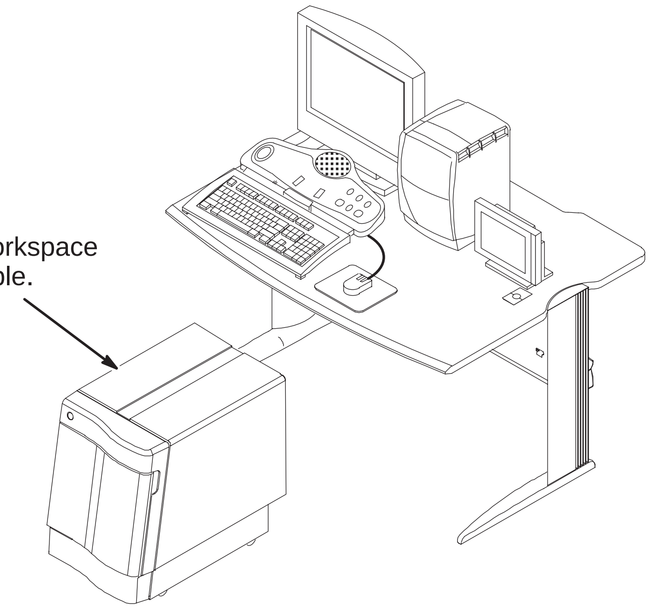
Run 815 (2198983)

Connect power cable to Power distribution Box J4.

INSTALLATION STEPS SUMMARY

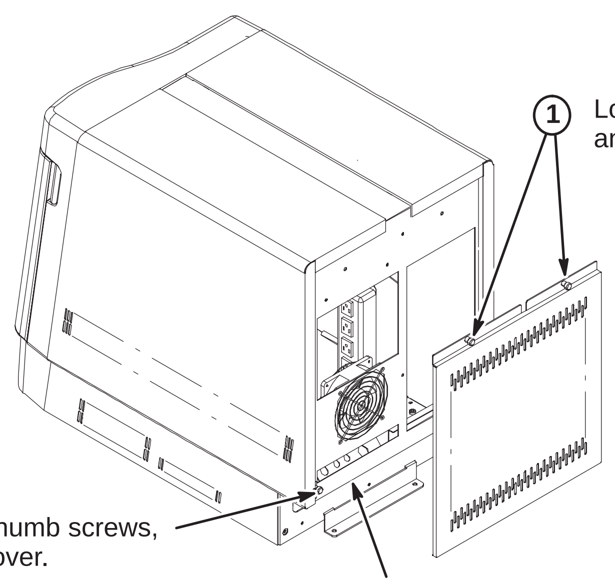
- ❑ **E1 – Locate Operator Workspace Cabinet in Front of Table**
- ❑ **E2 – Remove Operator Workspace Cabinet Rear Cover**
- ❑ **E3 – Remove Operator Workspace Cabinet Front And Side Covers**
- ❑ **E4 – Connect Ethernet Cables**
- ❑ **E5 – Route/Connect Runs 792, 797, 813, and 814 to PC Module**

❑ E1 – LOCATE OPERATOR WORKSPACE CABINET IN FRONT OF TABLE



1. Position Operator Workspace Cabinet in front of table.

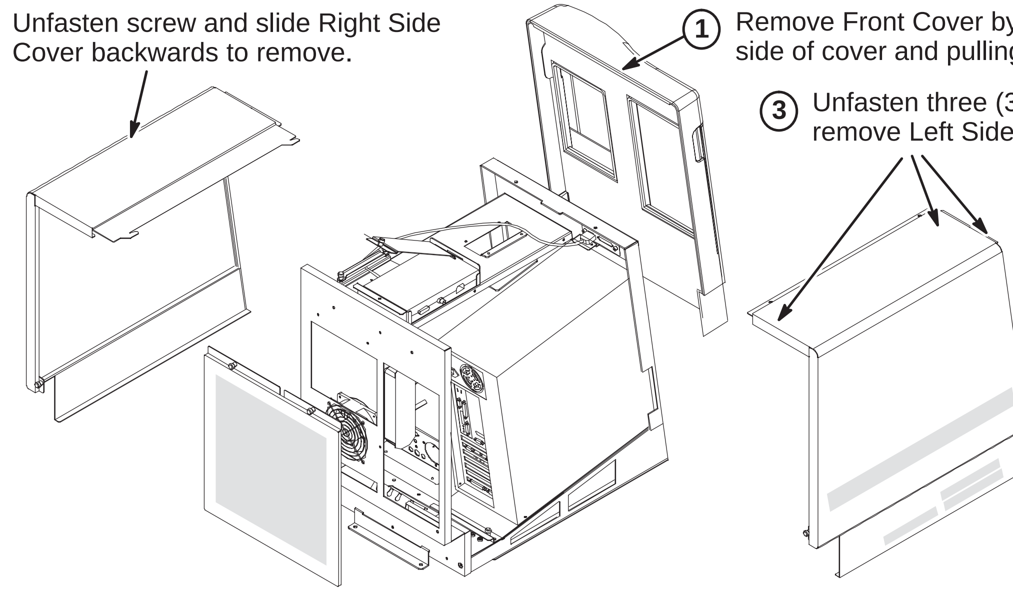
❑ E2 – REMOVE OPERATOR WORKSPACE CABINET REAR COVER



1. Loosen (2) two thumb screws and remove Rear Cover.
2. Loosen the (2) two thumb screws, one on each Side Cover.

Cable Access Hole

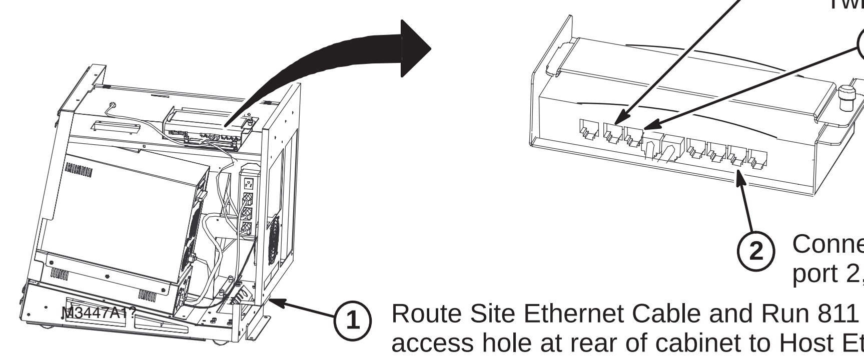
❑ E3 – REMOVE OPERATOR WORKSPACE CABINET FRONT AND SIDE COVERS



1. Remove Front Cover by grasping each side of cover and pulling forward.
2. Unfasten screw and slide Right Side Cover backwards to remove.
3. Unfasten three (3) screws and remove Left Side Cove.

❑ E4 – CONNECT ETHERNET CABLES

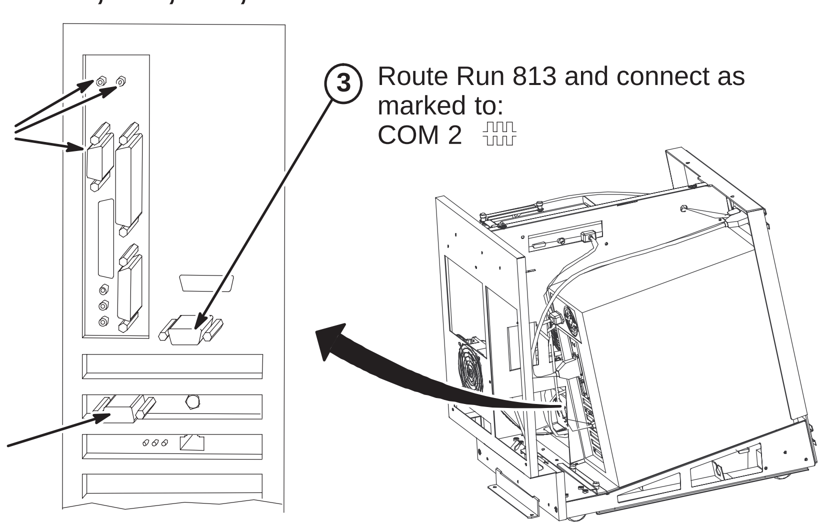
Note: Site Ethernet cable for InSite connection is supplied by customer. For Europe, connection to ISDN router is required. Refer to catalog required for country or region for specific equipment to be installed.



1. Route Site Ethernet Cable and Run 811 thru cable access hole at rear of cabinet to Host Ethernet Module.
2. Connect Run 811 from Octane Computer to port 2, FE laptop to port 3, and PC to port 4.
3. Connect Run 942 from Magnet Monitor J5 to port 7 (Open*Speed* and Twin*Speed*).
4. Site Ethernet cable is supplied by customer. If cable supplied is a Shielded Twisted Pair Cable, connect to port 8.

❑ E5 – ROUTE/CONNECT RUNS 792, 797, 813, AND 814 TO PC MODULE

Note: Route cables thru access hole at rear of cabinet.



1. Route Run 814 and connect as marked to:
   KEYBOARD
   MOUSE
   COM1
2. Route Run 797 and connect as marked to:
   VIDEO
3. Route Run 813 and connect as marked to:
   COM 2

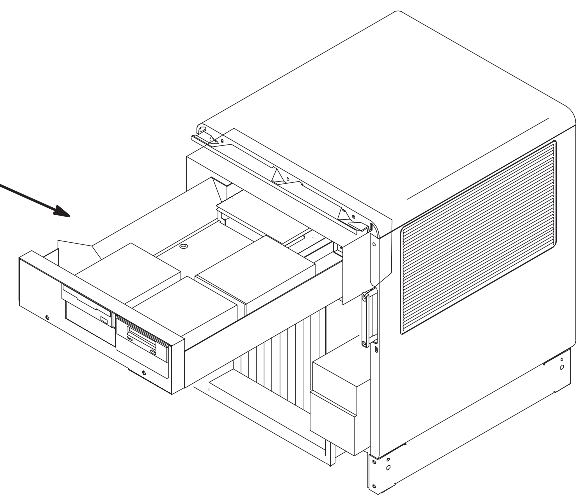
PROCEDURES F1 THRU F5 ARE USED ONLY WHEN UPGRADING FROM 5.x PLATFORMS, <u>DO NOT PERFORM PROCEDURES F1 THRU F5 FOR NEW OCTANE INSTALLATIONS</u>

<u>INSTALLATION STEPS SUMMARY</u>

- ❑ **F1 – Removal of DAT and MOD Drives from Genesis Computer**
- ❑ **F2 – Remove SCSI Expansion Box**
- ❑ **F3 – DAT Drive Settings**
- ❑ **F4 – MOD Drive Settings**
- ❑ **F5 – Install DAT and MOD Drives (Legacy Media) in SCSI Expansion Box**

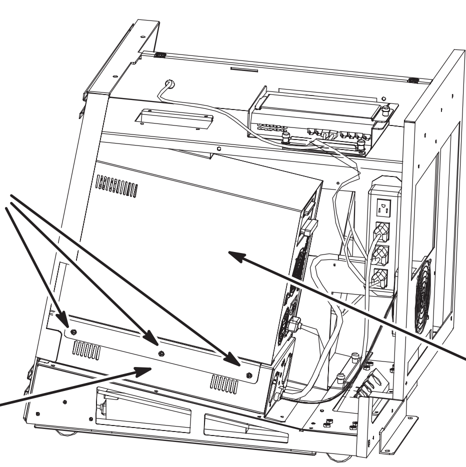
❑ F1 – REMOVAL OF DAT AND MOD DRIVES FROM GENESIS COMPUTER

UPGRADE FROM 5.x OPTIONS:
For transferring Legacy Media DAT and MOD Drives from Genesis Computer – refer to Tab 13, Directions 2174955 and 2174956, steps **A1** thru **A4** only for removal of existing drives.



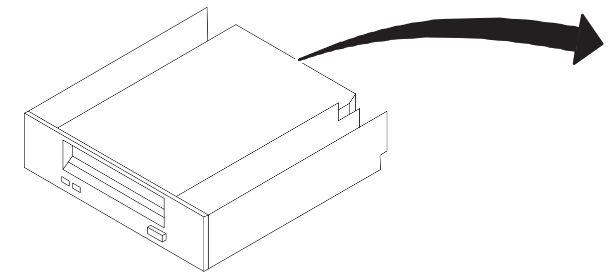
❑ F2 – REMOVE SCSI EXPANSION BOX

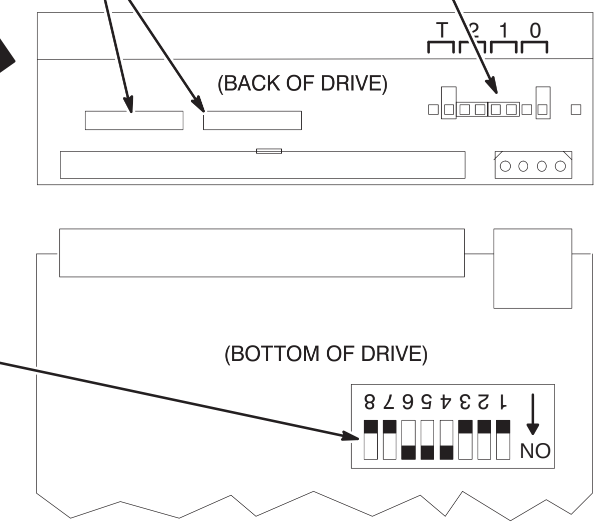
1. Unfasten three (3) screws from SCSI Expansion Box mounting bracket and remove Box and bracket from power module.
2. Unfasten six (6) screws that hold bracket to bottom of SCSI Expansion Box and remove bracket.
3. Remove SCSI Expansion Box cover.



❑ F3 – DAT DRIVE SETTINGS

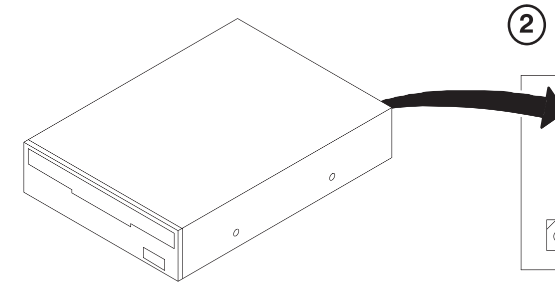
1. Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.
2. For Octane, set SCSI ID = 6
3. Set the swithches on the bottom of the drive as shown. This sets the drive to NOT perform “data disconnects” when there is no data available. This is required because of the odd byte count record packing done on GEMS tapes and SGI’s inability to handle odd byte “save pointer” SCSI requests.

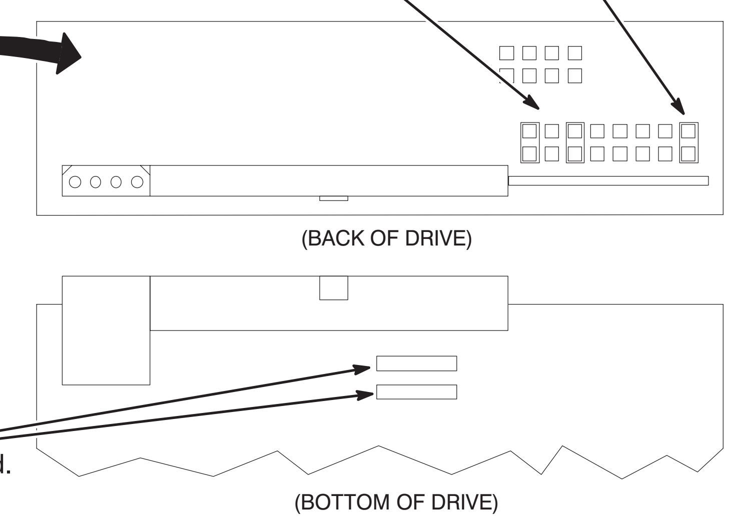




❑ F4 – MOD DRIVE SETTINGS

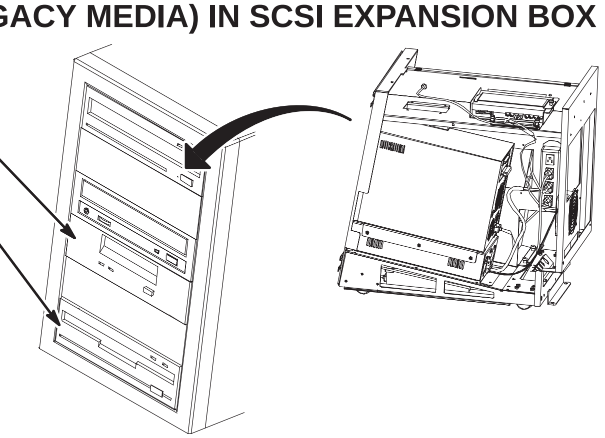
1. Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.
2. For Octane, SCSI ID = 5
3. Make sure jumper is installed on pins at position 8 for SCSI Optical Disk.





❑ F5 – INSTALL DAT AND MOD DRIVES (LEGACY MEDIA) IN SCSI EXPANSION BOX

1. Install DAT Drive in slot 4 with existing hardware.
2. Install MOD Drive in slot 6 with existing hardware.
3. Connect power cords to DAT and MOD Drives.
4. Connect ribbon cable to DAT and MOD Drives.
5. Install SCSI Expansion Box cover. Reattach bracket to bottom of Box. Reattach bracket and Box assembly to power module.

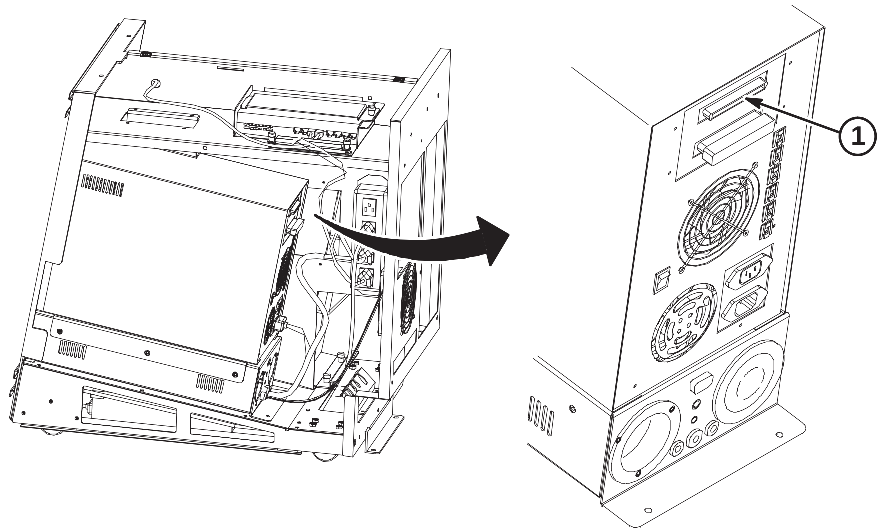


INSTALLATION STEPS SUMMARY

- ❑ **G1 – Route/Connect Run 819 to SCSI Expansion Box**
- ❑ **G2 – Route/Connect Runs 047, 048, 049/969, and 817 to Power Module**
- ❑ **G3 – Align Operator Workspace Cabinet Cables & Attach InSite "Computer" Magnet**
- ❑ **G4 – Bundle Operator Workspace Cabinet Cables**

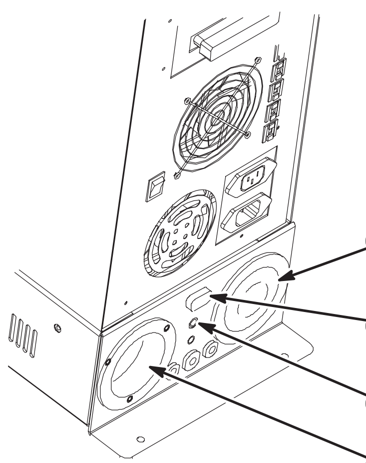
❑ G1 – ROUTE/CONNECT RUN 819 TO SCSI EXPANSION BOX

Note: Route all cables from SGI Octane Computer thru cable access hole at rear of cabinet.



1. Route/Connect Run 819 from SGI Octane Computer.

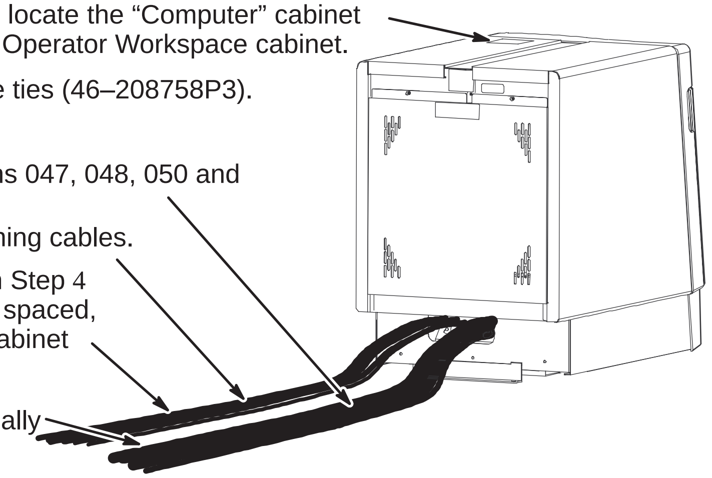
❑ G2 – ROUTE/CONNECT RUNS 047, 048, 049/969, AND 817 TO POWER MODULE



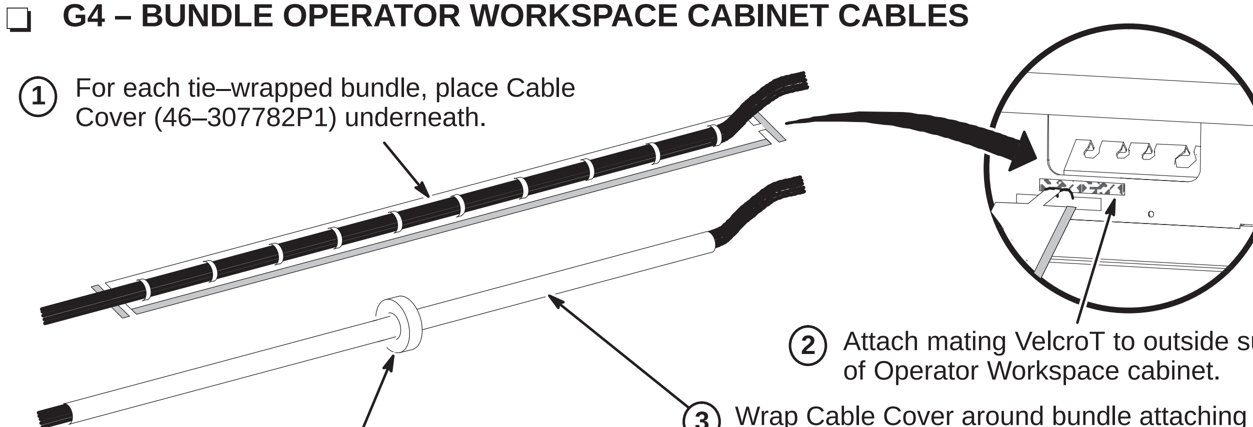
1. Route/Connect Run 049 or Run 969 Power Cable from Operator Workspace I/F to J2.
2. Route/Connect Run 817 Power Cable from Operator Workspace Power Module to J3.
3. Route/Connect Run 048 Ground Cable from PDU to ground screw.
4. Route/Connect Run 047 Power Cable from PDU to J1.

❑ G3 – ALIGN CABINET CABLES & ATTACH InSite "COMPUTER" MAGNET

1. Install Operator Workspace Cabinet right and left Side covers, Front cover and Rear cover.
2. From the 8.0 LX InSite Kit (46–301708G14), locate the "Computer" cabinet magnet (46–320095P1) and attach to top of Operator Workspace cabinet.
3. Locate the twenty (20) seven inch long cable ties (46–208758P3).
4. Seperate cables into two bundles;
   The first bundle will include power cable Runs 047, 048, 050 and fiber optic cable Run 790.
   The second bundle will include all the remaining cables.
5. Align the cables of the first bundle created in Step 4 and attach ten (10) of the cable ties, equally spaced, around the cable bundle between the OW Cabinet and the OW Table.
6. Attach the remaining ten (10) cable ties, equally spaced, around the second cable bundle.



❑ G4 – BUNDLE OPERATOR WORKSPACE CABINET CABLES



1. For each tie–wrapped bundle, place Cable Cover (46–307782P1) underneath.
2. Attach mating VelcroT to outside surface of Operator Workspace cabinet.
3. Wrap Cable Cover around bundle attaching the VelroT hooks and loops.
4. Attach cable bundle Clamp (46–307873P1) to center of Cable Cover.
5. Carefully roll OW Cabinet into final position. Make sure cable bundles are unobstructed and slide easily. Attach to mating Velcro on cabinet.
6. Repeat Step 5 as often as necessary.

INSTALLATION STEPS SUMMARY

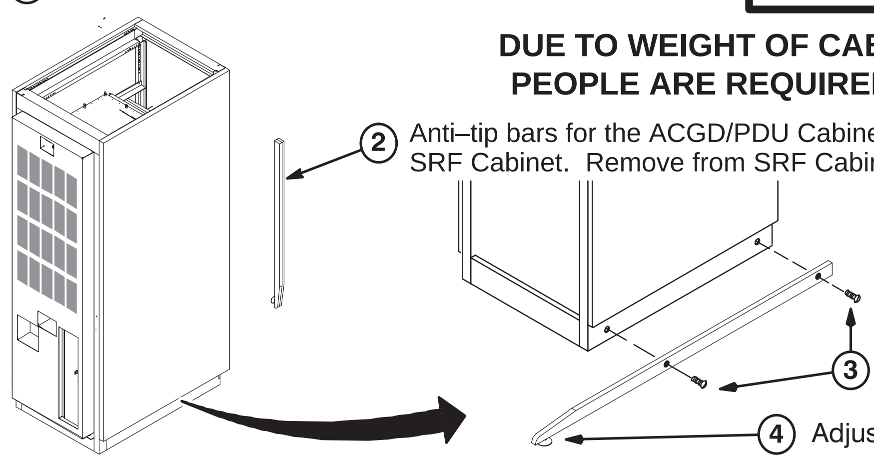
- ❑ A1 – Position ACGD/PDU Cabinet
- ❑ A2 – Install Fan Asembly and Attach InSite Cabinet Magnet
- ❑ A3 – Input Voltage Selection
- ❑ A4 – Circuit Breaker DIP Switch Settings

❑ A1 – POSITION ACGD/PDU CABINET

(1) Move cabinet to final position.

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

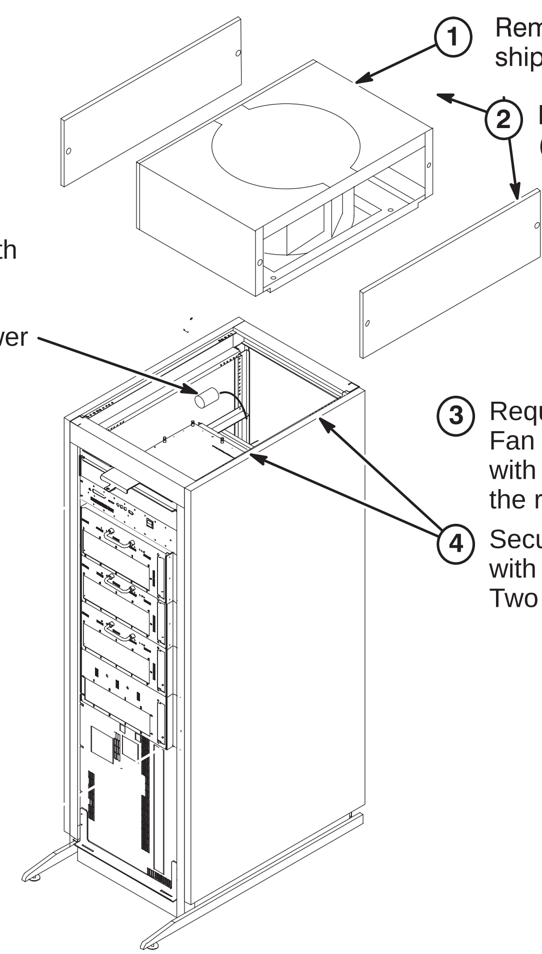


(2) Anti–tip bars for the ACGD/PDU Cabinet are shipped in the SRF Cabinet. Remove from SRF Cabinet for this procedure.

(3) Attach bars to cabinet.

(4) Adjust leveling pads to level cabinet.

❑ A2 – INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET



(1) Remove Fan Assembly from shipping container.

(2) Remove two side trim panels (two screws from each).

(3) Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.

(4) Secure Fan Assembly in place with (4) four screws provided. Two on each side of cabinet.

(5) Install two side trim panels with previously removed screws.

(6) Connect cable from SGA Power Supply to the fan connector.

DANGER!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

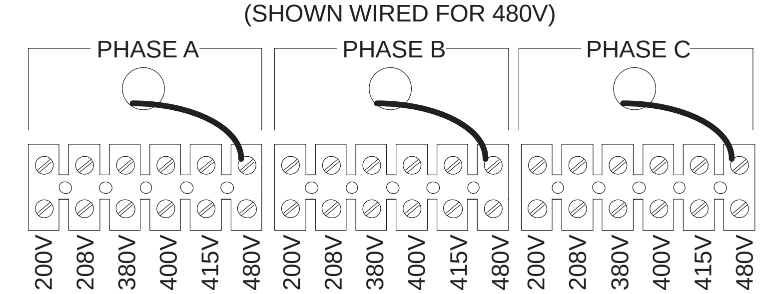
❑ A3 – INPUT VOLTAGE SELECTION

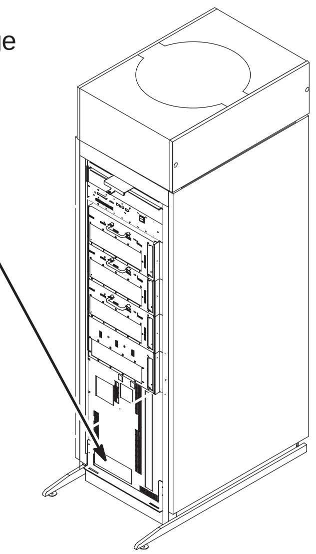
(1) Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.

(2) Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.

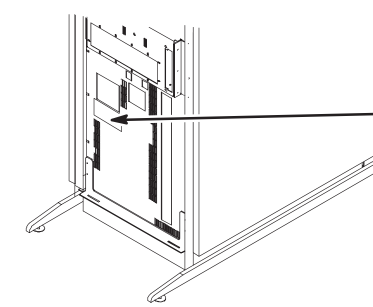
(3) For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.

(SHOWN WIRED FOR 480V)



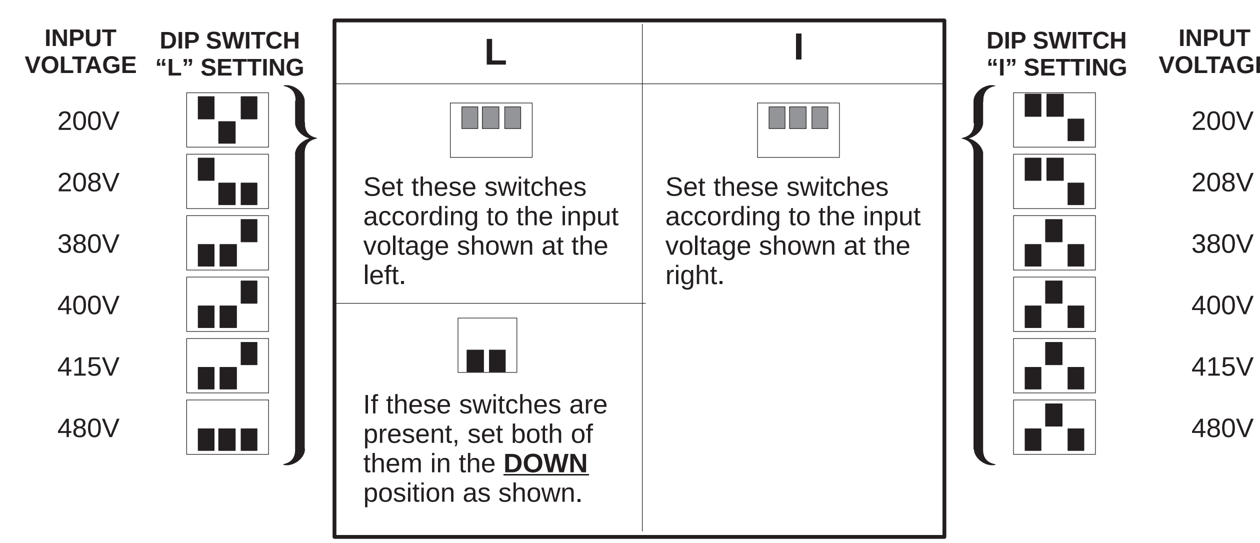


❑ A4 – CIRCUIT BREAKER DIP SWITCH SETTINGS



(1) Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.

(2) Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter "L" and "I" are the only ones that should be changed.



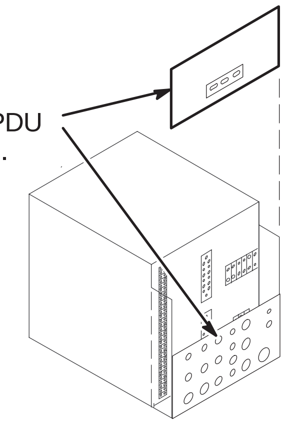
INSTALLATION STEPS SUMMARY

- ❑ **B1 – Facility Power and Ground Connention**
- ❑ **B2 – Route Power Cables Thru Interface Panel**
- ❑ **B3 – Run 967 Power Cable Alteration**
- ❑ **B4 – Connect Power Cables to PDU Module**

❑ B1 – FACILITY POWER AND GROUND CONNECTION

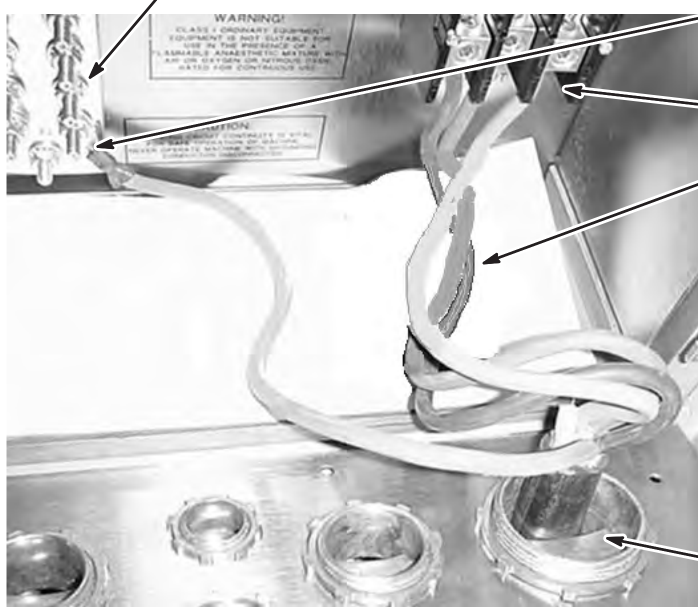
WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE ACGD/PDU CABINET.



1. Remove rear PDU module covers.

6. Connect RF screen room ground conductor to Ground Bus Bar.



5. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.
4. Connect L1–L3 conductors to terminal block.
3. **NOTE:** Air flow thru cabinet must exit thru top of cabinet. If not, fan at top of cabinet is rotating in wrong direction. To reverse rotation of fan, switch any two of the three incoming L1–L3 power wires.

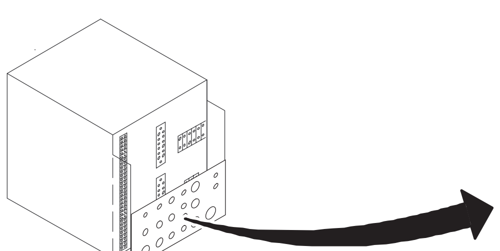
DANGER!!

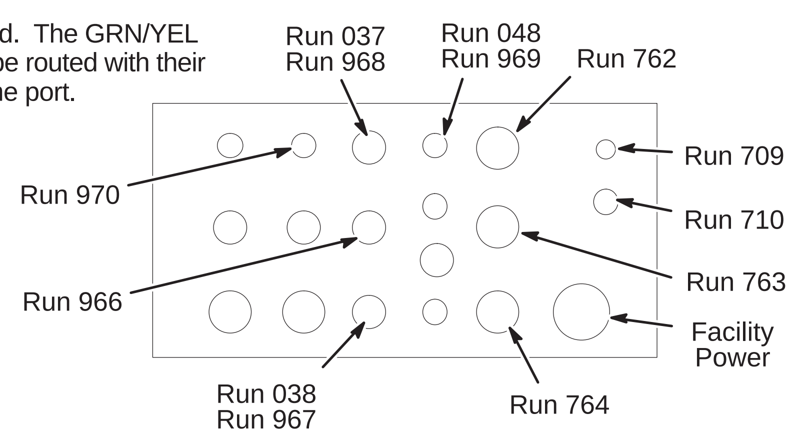
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE SWITCHING ANY ABOVE WIRES.

2. Route facility input power cable and ground wire thru interface panel.

❑ B2 – ROUTE POWER CABLES THRU INTERFACE PANEL

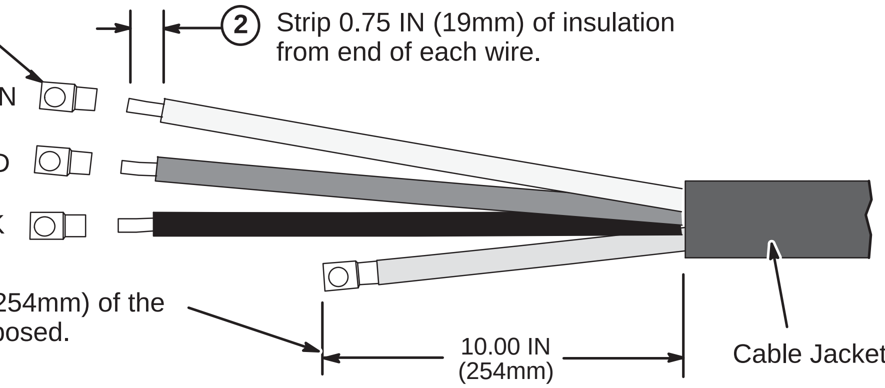
1. Route Runs thru interface panel as marked. The GRN/YEL ground wires (Runs 037 and 048) **MUST** be routed with their respective power cable Runs thru the same port.
2. Route any option Runs as marked.



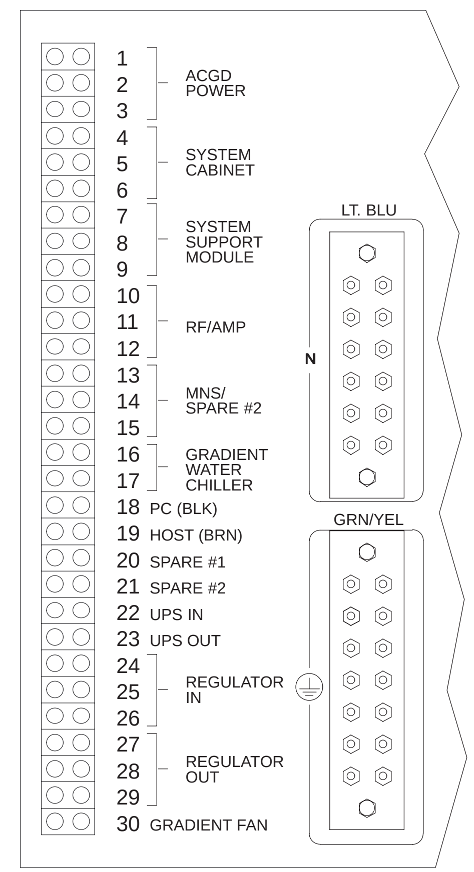


❑ B3 – RUN 967 POWER CABLE ALTERATION

1. Cut off ring terminal from ORN, RED, and BLK wires.
2. Strip 0.75 IN (19mm) of insulation from end of each wire.
3. Strip jacket until 10 Inches (254mm) of the GRN/YEL ground wire is exposed.

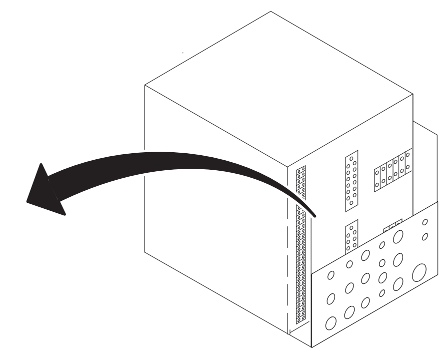


❑ B4 – CONNECT POWER CABLES TO PDU MODULE



WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.



1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
2. Trim and connect ground wire cables (Runs 037, 038, and 048) to ground bus connectors.
3. Tighten clamps of interface panel to secure cables in place.

CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	ϕ A, B, C	System Cabient 3ϕ, Neutral, Ground
7,8,9	966	ϕ A, B, C	SSM 3ϕ Neutral, Ground
10,11,12	967	ϕ A, B, C	RFI/AMP 3ϕ, Ground
16,17	970	ϕ A,B	Water Chiller Single ϕ, Ground
18,19	969	ϕ C,A	Operator Workspace Single ϕ, Neutral, GND
20	973	ϕ A	Heat Exchanger Single ϕ, Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire
GRN/YEL	048		Operator Workspace Ground Wire

COLOR CODE: for remainder of cables:
3ϕ= Black, Red, Orange   Neutral = Light blue   Gnd = Grn/Yel
Single ϕ = Brown, Black

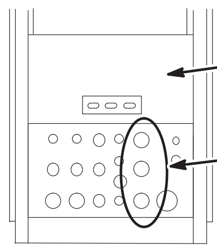
SIGNA ACGD/PDU CABINET INSTALLATION INSTRUCTIONS

INSTALLATION STEPS SUMMARY

- ❑ **C1 – Gradient Output Cable Interface Panel Access**
- ❑ **C2 – Gradient Output Cable Routing**
- ❑ **C3 – Route/Connect Fiber Optics Cables to GP Module**
- ❑ **C4 – Connect Runs 971 and 972 to PDU Module**
- ❑ **C5 – Atach InSite Cabinet Magnets**

❑ C1 – GRADIENT OUTPUT CABLE INTERFACE PANEL ACCESS

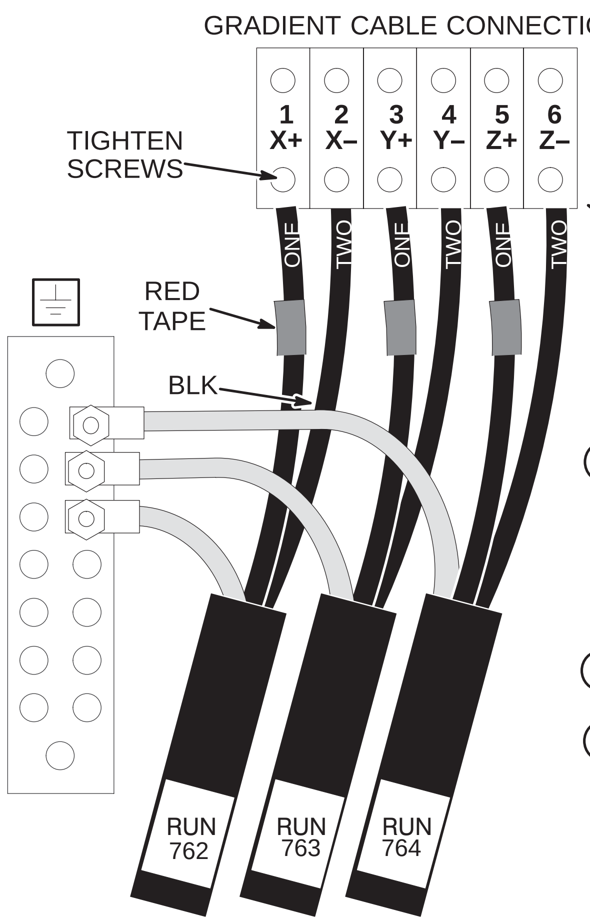
ACGD CABINET
(REAR VIEW)



NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.

1. Remove Rear Cover Plate by removing seven screws that hold it in place.
2. Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

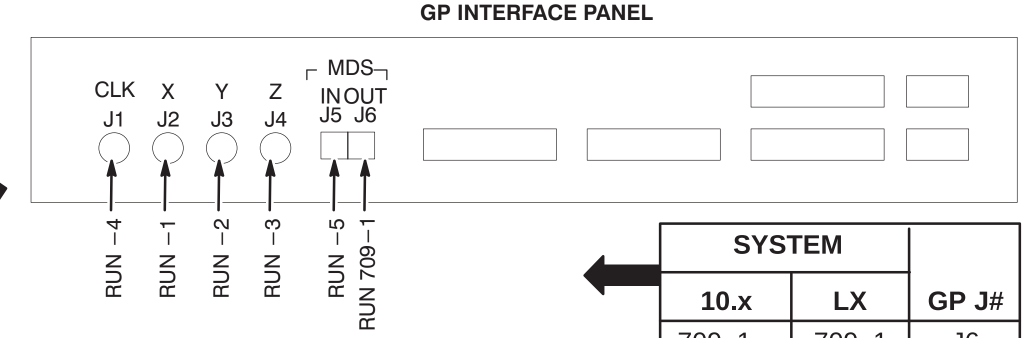
❑ C2 – GRADIENT OUTPUT CABLE ROUTING

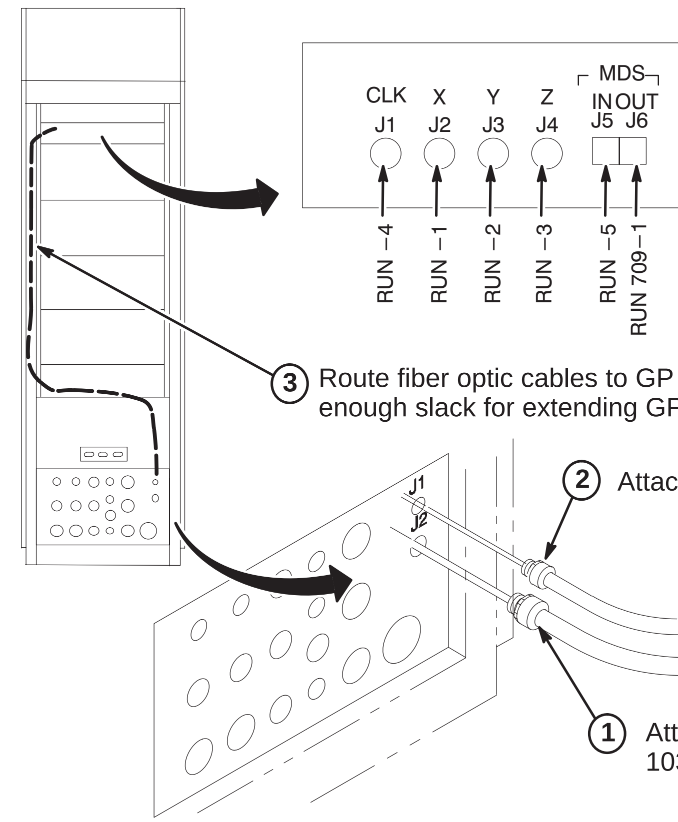


1. Connect gradient output cables with red–taped (+) #1 wires to + output terminals and black (–) #2 wires to – output terminals.

   NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.
2. Tighten screws to <u>52–65 inch–lbs (5,88–7,34 N–M).</u> This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
3. Connect the GND leads of each run to the Ground Bus Bar studs.
4. Tighten clamps on Interface Panel to secure gradient output cables.

❑ C3 – ROUTE/CONNECT FIBER OPTICS TO GP MODULE

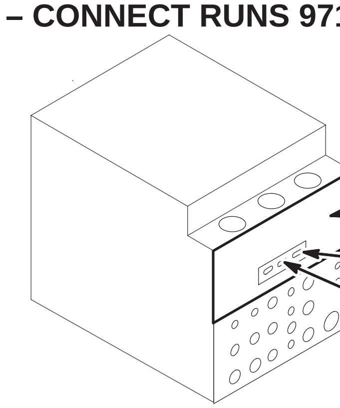




SYSTEM		
10.x	LX	GP J#
709–1	709–1	J6
1034–5	710–5	J5
1034–3	710–3	J4
1034–2	710–2	J3
1034–1	710–1	J2
1034–4	710–4	J1

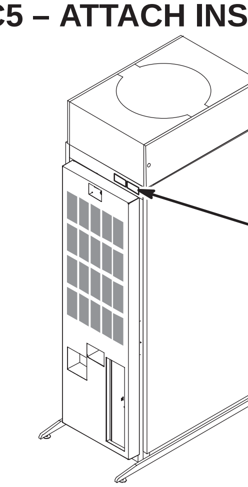
1. Attach Run 710 or Run 1034 to I/F J2
2. Attach Run 709 to I/F J1
3. Route fiber optic cables to GP Module. Leave enough slack for extending GP module forward.
4. Connect fiber optics to GIP module according to table.

❑ C4 – CONNECT RUNS 971 AND 972 TO PDU MODULE



1. Attach rear PDU module covers removed in Procedure **B1.**
2. Connect Run 972 to J3.
3. Connect Run 971 to J2.

❑ C5 – ATTACH INSITE CABINET MAGNETS



1. Attach front and rear covers.
2. From the 8.0 LX InSite Kit (46–301708G14), locate the "Gradient" cabinet magnet (46–320095P7) and the "PDU" cabinet magnet (46–320095P4). Attach to front of cabinet as shown.

SRF CABINET UPGRADE INSTRUCTIONS

Tab 16, SRF Cabinet, contains installation instructions for the following:

- *DIRECTION* 2286812 – Signa SRF Cabinet Installation Instructions
- *DIRECTION 2*346942 – Signa SRFD2 Cabinet Installation Instructions

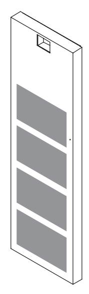
Use only the applicable SRF Cabinet Installation Instruction Direction for the site being upgraded

INSTALLATION STEPS SUMMARY

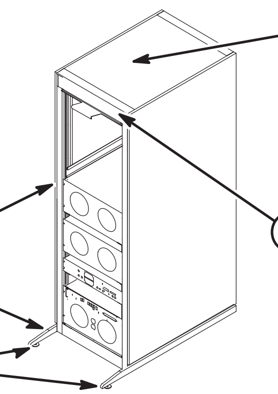
- ❑ **A – Install New SRF Cabinet**
- ❑ **B1 – Route/Connect Run 967 to Circuit Breaker**
  OR
- ❑ **B2 – Route/Connect Run 967 to Terminal Strip**
  } **Use the applicable procedure.**
- ❑ **C – Connect Runs 235/745 and 236/748 to RF Interface (RFI) Module**
- ❑ **D – Connect Fiber Optics and Run 229**
- ❑ **E – System Support Module Cable Connections**

❑ A – INSTALL NEW SRF CABINET

Note: The 1.5T SRF (Scaleable RF) cabinet is shown in these procedures. The 1.0T is the same except for the Heliax cables installed in Procedure **C** below.



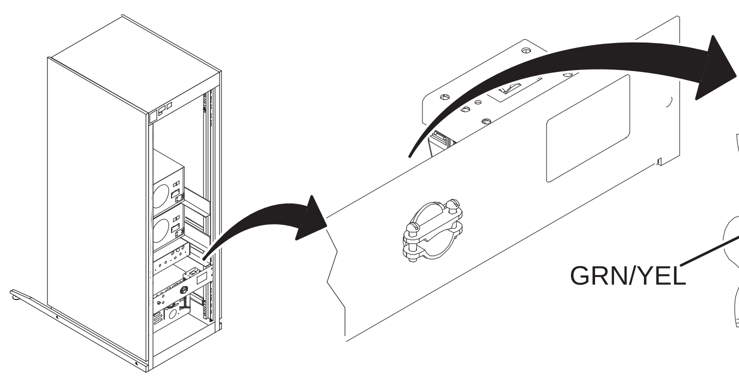
Note: Comply with UL requirements by either installing anti tip legs or securing the cabinet to the floor with local supplied brackets and floor anchors.

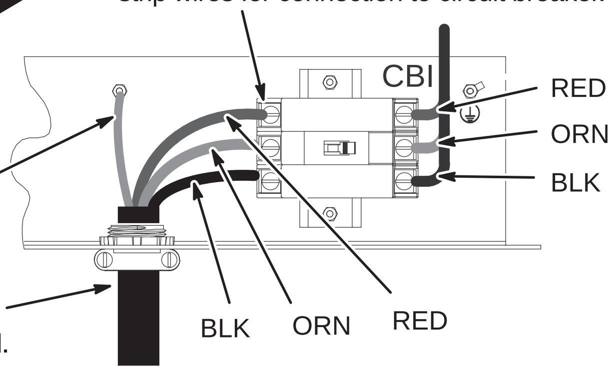


Note: UFI Label Kit (2146243) is attached to upper rail inside of cabinet.

1. Move new SRF Cabinet to final position.
2. Remove anti–tip bars from inside of cabinet and attach to each side.
3. Adjust leveling pads to level cabinet.
4. From the 8.0 LX InSite Kit (46–301708G14), locate the "RF/Penetration" cabinet magnet (46–320095P13). Attach to front of cabinet as shown.

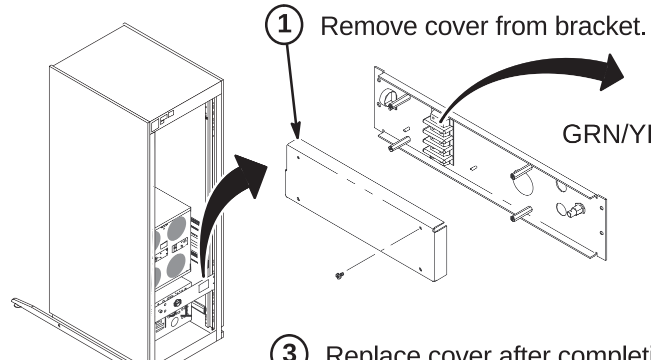
❑ B1 – ROUTE/CONNECT RUN 967 TO CIRCUIT BREAKER

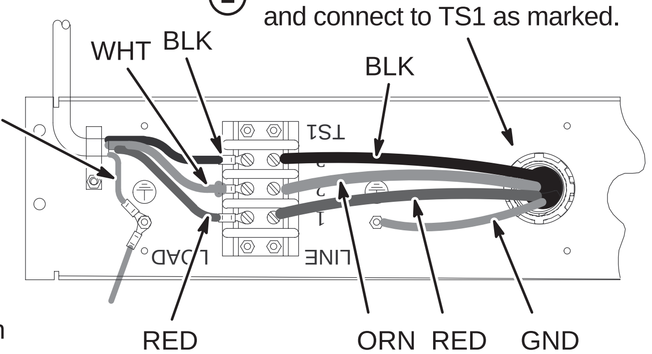




1. Route Run 967 to rear bracket and connect to CB1 as marked.
2. If power cable provided has ring terminals for RED, ORN, & BLK, remove rings and strip wires for connection to circuit breaker.

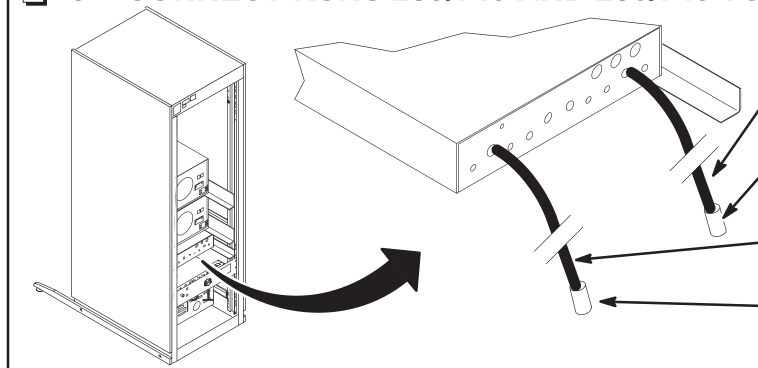
❑ B2 – ROUTE/CONNECT RUN 967 TO TERMINAL STRIP





1. Remove cover from bracket.
2. Route Run 967 to rear bracket and connect to TS1 as marked.
3. Replace cover after completion of power cable connection.

❑ C – CONNECT RUNS 235/745 AND 236/748 TO RF INTERFACE (RFI) MODULE

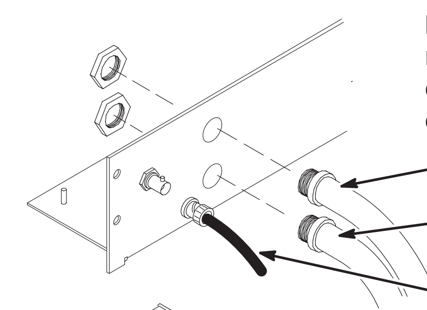


1. Connect Extension Cable with HN plugs (2105362) to J3 Head Output on RFI Module.
2. Connect Run 236 (1.5T) or Run 748 (1.0T) to above Extension cable using HN Bulkhead connector (46–221942P1).
3. Connect Extension Cable with N plugs (2105358) to J4 Body Output on RFI Module.
4. Connect Run 235 (1.5T) or Run 745 (1.0T) to above Extension cable using N Bulkhead connector (46–221865P1).

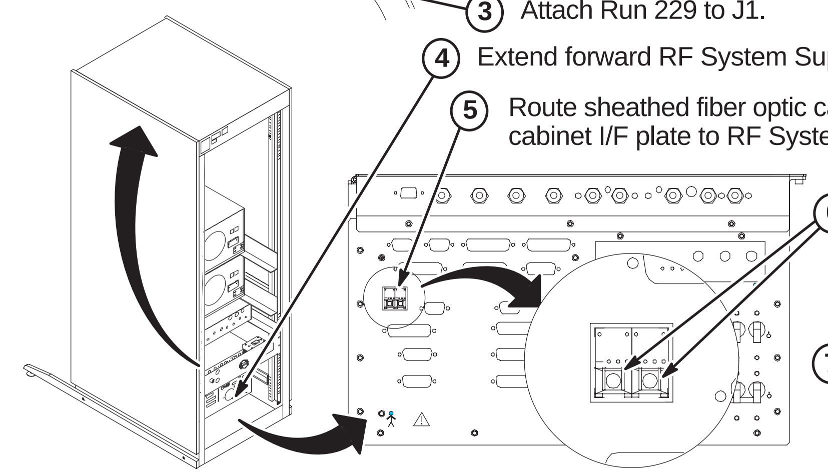
❑ D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



1. Attach Run 709 to I/F RX*.
2. Attach Run 708 (or 1033) to I/F TX*.
   * Instructions provided with connectors.
3. Attach Run 229 to J1.

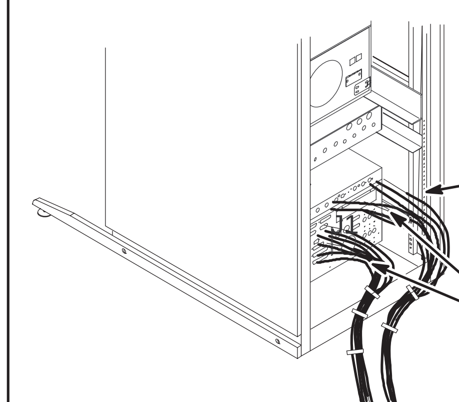


4. Extend forward RF System Support Module into servicing position.
5. Route sheathed fiber optic cables (Run 708 and Run 709) from cabinet I/F plate to RF System Support Module.
6. Connect fiber optic cables as marked to TR1 and TR2 at rear of Module:
   – Run 708 to Transmit (TX)
   – Run 709 to Receive (RX)
7. Dress and secure fiber optic cables along route from Module to I/F Panel. While checking Fiber Optic Cable routing, push RF System Support Module into cabinet.

❑ E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



1. Connect remainder of SRF Cabinet cables to System Support Module.
   Note:
   If needed, refer to Overview: Section 2 – System Cables, for Run descriptions and designator information located at end of each cable.
2. Dress and secure cables connected to System Support Module. Verify sufficient slack by moving Module completely forward and back again. Adjust as necessary.
3. Attach all front and rear covers.

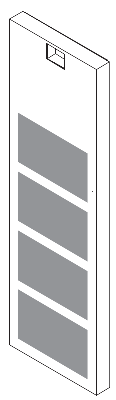
SIGNA SRFD2 CABINET INSTALLATION INSTRUCTIONS

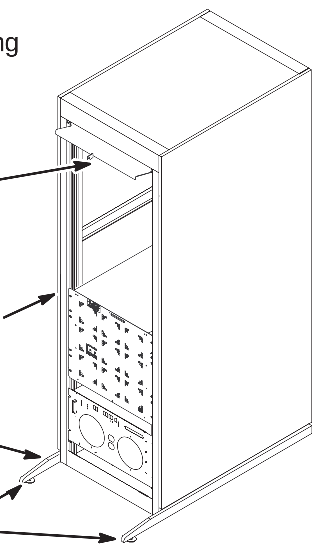
INSTALLATION STEPS SUMMARY

- ❑ A – Install New SRFD2 Cabinet
- ❑ B – Route/Connect Run 967
- ❑ C – Connect Runs 235 and 236 to RF Amp & Interface Module
- ❑ D – Connect Fiber Optics and Run 229
- ❑ E – System Support Module Cable Connections
- ❑ F – Attach InSite Cabinet Magnets

❑ A – INSTALL NEW SRFD2 CABINET

Note: Comply with UL requirements by either installing anti tip legs or securing the cabinet to the floor with local supplied brackets and floor anchors.

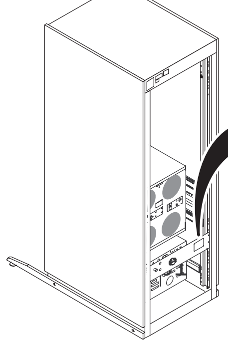


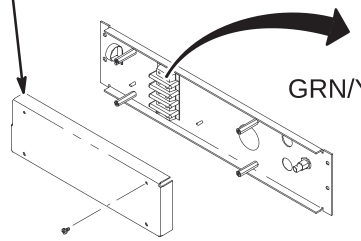


Note: UFI Label Kit (2146243) is attached to upper rail inside of cabinet.

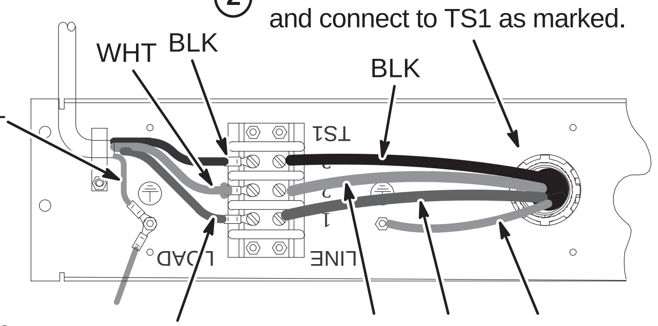
1. Move new SRFD2 Cabinet to final position.
2. Remove anti–tip bars from inside of cabinet and attach to each side.
3. Adjust leveling pads to level cabinet.

❑ B – ROUTE/CONNECT RUN 967

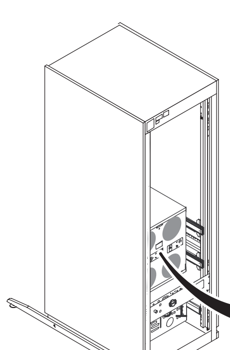


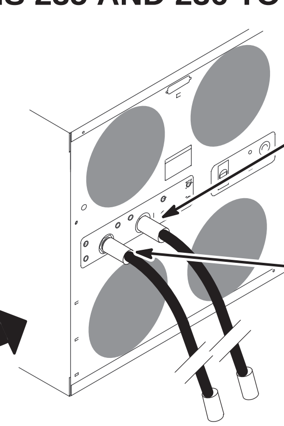


1. Remove cover from bracket.
2. Route Run 967 to rear bracket and connect to TS1 as marked.
3. Replace cover after completion of power cable connection.



❑ C – CONNECT RUNS 235 AND 236 TO RF AMP & INTERFACE MODULE



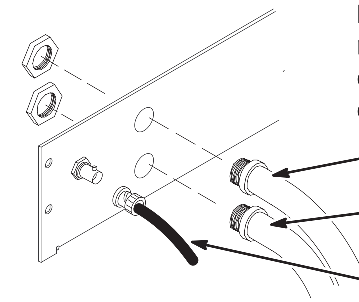


1. Connect Extension Cable with HN plugs (2105362) to J3 Head Output on RFI Module.
2. Connect Run 236 (1.5T) to above Extension cable using HN Bulkhead connector (46–221942P1).
3. Connect Extension Cable with N plugs (2105358) to J4 Body Output on RFI Module.
4. Connect Run 235 (1.5T) to above Extension cable using N Bulkhead connector (46–221865P1).

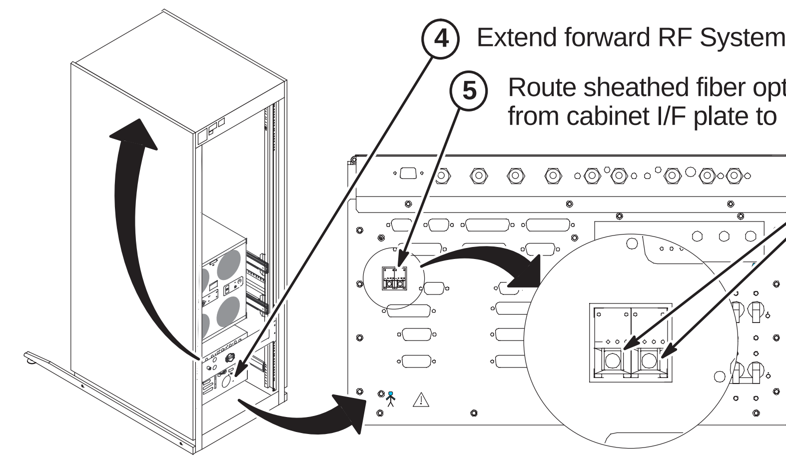
❑ D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



1. Attach Run 709 to I/F RX*.
2. Attach Run 708 (or1033) to I/F TX*.
   * Instructions provided with connectors.
3. Attach Run 229 to J1.
4. Extend forward RF System Support Module into servicing position.
5. Route sheathed fiber optic cables (Run 708/1033 and Run 709) from cabinet I/F plate to RF System Support Module.

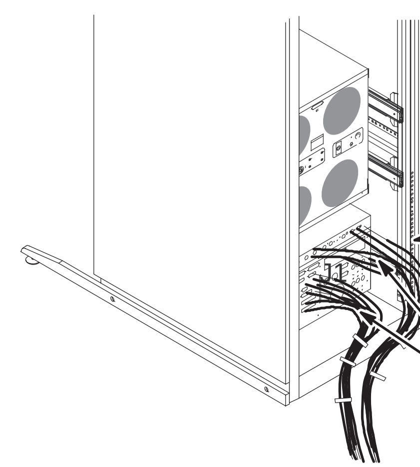


6. Connect fiber optic cables as marked to TR1 and TR2 at rear of Module:
   – Run 708/1033 to Transmit (TX)
   – Run 709 to Receive (RX)
7. Dress and secure fiber optic cables along route from Module to I/F Panel. While checking Fiber Optic Cable routing, push RF System Support Module into cabinet.

❑ E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

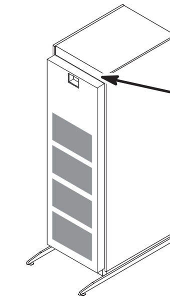
CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



1. Connect remainder of SRFD2 Cabinet cables to System Support Module.
   Note:
   If needed, refer to Overview: Section 2 – System Cables, for Run descriptions and designator information located at end of each cable.
2. Dress and secure cables connected to System Support Module. Verify sufficient slack for service loop by moving Module completely forward and back again. Adjust as necessary.

❑ F – ATTACH INSITE CABINET MAGNET



1. Attach all front and rear covers.
2. From the 8.0 LX InSite Kit (46–301708G14), locate the “RF/Penetration” cabinet magnet (46–320095P13). Attach to front of cabinet as shown.